

Additive Combinatorics

Volumes I-V

Version 1.10 / Formalization Revision R13

Canonical five-volume edition

14 September 2026

Additive Combinatorics
Volume I
Foundations of Additive and Analytic Combinatorics
Version 1.10

14 September 2026

Contents

1	Counting, inequalities, and graph language	1
2	Probability and energy language	9
2.1	Small sumsets: Ruzsa calculus and Plünnecke theory	11
2.2	Partial sumsets and the Balog–Szemerédi–Gowers theorem	15
3	Group theory from the required starting level	19
4	Elementary number theory	28
5	Characters and finite Fourier analysis	33
6	Foundational probabilistic and Fourier examples	39
7	Worked laboratories for the foundational toolkit	41

Chapter 1

Counting, inequalities, and graph language

The first recurring move in additive combinatorics is to convert a structural question into a count and then force that count to be large by an inequality. Double counting, Cauchy–Schwarz, concentration, and graph language are therefore not miscellaneous preliminaries: they are the engine behind energy arguments, dependent random choice, regularity, and the higher-order cube norms developed later in this book. We state the finite forms with explicit normalizations so later proofs can invoke them without changing conventions.

A useful mental model is that Cauchy–Schwarz trades *information* for *symmetry*. Each application duplicates a variable and creates a parallelogram or cube. The iterated version below is included early because the same geometry will reappear, at much greater depth, in Volumes III–V.

Theorem 1.1 (I.1.001: Double-counting principle in finite incidence structures.). *Let X, Y be finite sets and $I \subseteq X \times Y$. If $d_X(x) = |\{y : (x, y) \in I\}|$ and $d_Y(y) = |\{x : (x, y) \in I\}|$, then*

$$|I| = \sum_{x \in X} d_X(x) = \sum_{y \in Y} d_Y(y).$$

More generally, any finite incidence set may be counted by either coordinate.

Proof. Let I be the set of incidences. Counting the same finite set by its first coordinate gives $|I| = \sum_x d(x)$, and counting by its second coordinate gives $|I| = \sum_y d(y)$. Hence the two sums are equal. Every double-counting identity used later is this argument with a different incidence set. $\square$

Theorem 1.2 (I.1.002: Pigeonhole principle and quantitative pigeonhole variants.). *If N objects are partitioned into $m \geq 1$ classes, then some class has size at least N/m . Equivalently, if $w_1, \dots, w_m \geq 0$ and $\sum_j w_j = W$, then $\max_j w_j \geq W/m$.*

Proof. If N objects are partitioned among m boxes, then some box has at least N/m objects; otherwise the total would be $< N$. More generally, if nonnegative weights w_j sum to W , some j has $w_j \geq W/m$. The dyadic and density versions follow by applying this to the relevant partition. $\square$

Theorem 1.3 (I.1.003: Cauchy–Schwarz inequality in finite-sum, integral, and expectation forms.). *For complex numbers a_i, b_i ,*

$$\left| \sum_i a_i \bar{b}_i \right|^2 \leq \left(\sum_i |a_i|^2 \right) \left(\sum_i |b_i|^2 \right).$$

The same inequality holds for L^2 functions on a finite probability space, and hence in expectation form $|\mathbb{E}(XY)|^2 \leq \mathbb{E}|X|^2 \mathbb{E}|Y|^2$.

Proof. For finite sums, $0 \leq \sum_i |a_i - \lambda b_i|^2$ for every λ . Choosing $\lambda = \langle a, b \rangle / \|b\|_2^2$ yields $|\langle a, b \rangle|^2 \leq \|a\|_2^2 \|b\|_2^2$. The integral and expectation forms are identical in L^2 after replacing sums by integrals. $\square$

Box notation. Fix an integer $d \geq 1$ and finite probability spaces $X_1, \dots, X_d$. For each i let x_i^0, x_i^1 denote two independent samples from X_i . If $\omega = (\omega_1, \dots, \omega_d) \in \{0, 1\}^d$, write

$$x^\omega = (x_1^{\omega_1}, \dots, x_d^{\omega_d}), \quad |\omega| = \omega_1 + \dots + \omega_d,$$

and let $Cz = \bar{z}$, with C^m meaning the identity when m is even and complex conjugation when m is odd. For a family $\mathbf{F} = (F_\omega)_{\omega \in \{0,1\}^d}$ of functions $F_\omega : X_1 \times \dots \times X_d \rightarrow \mathbb{C}$, define its box multilinear form by

$$\Lambda_{\square^d}(\mathbf{F}) := \mathbb{E}_{x_1^0, x_1^1, \dots, x_d^0, x_d^1} \prod_{\omega \in \{0,1\}^d} C^{|\omega|} F_\omega(x^\omega).$$

For a single function $F : X_1 \times \dots \times X_d \rightarrow \mathbb{C}$, define

$$\|F\|_{\square^d(X_1, \dots, X_d)} := \left(\mathbb{E}_{x_1^0, x_1^1, \dots, x_d^0, x_d^1} \prod_{\omega \in \{0,1\}^d} C^{|\omega|} F(x^\omega) \right)^{1/2^d}.$$

The quantity inside the last parenthesis is real and nonnegative. Indeed, if y denotes all variables except x_d^0, x_d^1 and

$$A_y(t) := \prod_{\eta \in \{0,1\}^{d-1}} C^{|\eta|} F(x_1^{\eta_1}, \dots, x_{d-1}^{\eta_{d-1}}, t),$$

then that quantity equals

$$\mathbb{E}_y |\mathbb{E}_{t \in X_d} A_y(t)|^2 \geq 0.$$

Theorem 1.4 (I.1.004: Iterated Cauchy–Schwarz / box Cauchy–Schwarz lemma.). *Let $d \geq 1$, let $X_1, \dots, X_d$ be finite probability spaces, and let $F_\omega : X_1 \times \dots \times X_d \rightarrow \mathbb{C}$ for every $\omega \in \{0, 1\}^d$. Then*

$$|\Lambda_{\square^d}((F_\omega)_\omega)| \leq \prod_{\omega \in \{0,1\}^d} \|F_\omega\|_{\square^d(X_1, \dots, X_d)}.$$

Equivalently,

$$\left| \mathbb{E}_{x_1^0, x_1^1, \dots, x_d^0, x_d^1} \prod_{\omega \in \{0,1\}^d} C^{|\omega|} F_\omega(x^\omega) \right| \leq \prod_{\omega \in \{0,1\}^d} \left(\mathbb{E}_{x_1^0, x_1^1, \dots, x_d^0, x_d^1} \prod_{\eta \in \{0,1\}^d} C^{|\eta|} F_\omega(x^\eta) \right)^{1/2^d}.$$

Proof. For $1 \leq j \leq d$ and $b \in \{0, 1\}$, write $\omega^{(j \leftarrow b)}$ for the vertex obtained from ω by replacing its j th coordinate by b , and define the face-reflected family

$$F_\omega^{(j,b)} := F_{\omega^{(j \leftarrow b)}}.$$

We first prove the one-coordinate reflection inequality

$$|\Lambda_{\square^d}(\mathbf{F})|^2 \leq \Lambda_{\square^d}(\mathbf{F}^{(j,0)}) \Lambda_{\square^d}(\mathbf{F}^{(j,1)}).$$

Fix all variables except x_j^0, x_j^1 and denote the remaining collection by y . For $\eta = (\eta_1, \dots, \eta_{d-1}) \in \{0, 1\}^{d-1}$ and $b \in \{0, 1\}$, define the insertion map

$$\iota_j(\eta, b) := (\eta_1, \dots, \eta_{j-1}, b, \eta_j, \dots, \eta_{d-1}) \in \{0, 1\}^d.$$

Set

$$A_b(y) := \mathbb{E}_{t \in X_j} \prod_{\eta \in \{0, 1\}^{d-1}} \mathcal{C}^{|\eta|} F_{\iota_j(\eta, b)}(x_1^{\eta_1}, \dots, x_{j-1}^{\eta_{j-1}}, t, x_{j+1}^{\eta_j}, \dots, x_d^{\eta_d}).$$

Because the vertices on the j th upper face acquire one additional complex conjugation,

$$\Lambda_{\square^d}(\mathbf{F}) = \mathbb{E}_y A_0(y) \overline{A_1(y)}.$$

Ordinary Cauchy–Schwarz therefore gives

$$|\Lambda_{\square^d}(\mathbf{F})|^2 \leq \left(\mathbb{E}_y |A_0(y)|^2 \right) \left(\mathbb{E}_y |A_1(y)|^2 \right).$$

Expanding the two squares introduces independent copies x_j^0, x_j^1 and shows exactly that

$$\mathbb{E}_y |A_b(y)|^2 = \Lambda_{\square^d}(\mathbf{F}^{(j, b)}), \quad b \in \{0, 1\},$$

which proves the reflection inequality.

Apply the reflection inequality successively for $j = 1, 2, \dots, d$. At each step every current family splits into its two face reflections. After all d coordinates have been reflected, the 2^d terminal families are constant on the vertex set: for each $\beta \in \{0, 1\}^d$ every vertex carries the same function F_β . Consequently

$$|\Lambda_{\square^d}(\mathbf{F})|^{2^d} \leq \prod_{\beta \in \{0, 1\}^d} \Lambda_{\square^d}((F_\beta)_\omega) = \prod_{\beta \in \{0, 1\}^d} \|F_\beta\|_{\square^d}^{2^d}.$$

Taking the 2^d th root proves the boxed inequality.

For example, when $d = 2$ the theorem reads

$$\left| \mathbb{E}_{\substack{x^0, x^1 \in X_1 \\ y^0, y^1 \in X_2}} F_{00}(x^0, y^0) \overline{F_{01}(x^0, y^1)} \overline{F_{10}(x^1, y^0)} F_{11}(x^1, y^1) \right| \leq \prod_{a, b \in \{0, 1\}} \|F_{ab}\|_{\square^2(X_1, X_2)}.$$

Thus the phrase “duplicate one coordinate at a time” is not a substitute for the estimate: it is precisely the sequence of reflection inequalities above that proves the displayed box bound. $\square$

Theorem 1.5 (I.1.005: Hölder inequality and finite L^p interpolation consequences.). *If $1 \leq p, q \leq \infty$ are conjugate exponents, then for functions on a finite probability space*

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

If $1 \leq p < r < q \leq \infty$ and $1/r = \theta/p + (1 - \theta)/q$, then

$$\|f\|_r \leq \|f\|_p^\theta \|f\|_q^{1-\theta}.$$

Proof. For conjugate $p, q > 1$, Young’s scalar inequality $uv \leq u^p/p + v^q/q$, applied to $u = |f|/\|f\|_p$ and $v = |g|/\|g\|_q$, gives after summation

$$\sum_x |f(x)g(x)| \leq \|f\|_p \|g\|_q.$$

The endpoint cases $p = 1$ or $q = \infty$ are immediate from the definition, so this is Hölder in every finite case used below.

For interpolation, suppose $1 \leq p < r < q \leq \infty$ and choose $0 < \theta < 1$ from $1/r = \theta/p + (1 - \theta)/q$. For $q < \infty$ write

$$\sum_x |f(x)|^r = \sum_x |f(x)|^{\theta r} |f(x)|^{(1-\theta)r}$$

and apply Hölder with exponents $p/(\theta r)$ and $q/((1 - \theta)r)$; their reciprocals add to one. Taking the r th root gives $\|f\|_r \leq \|f\|_p^\theta \|f\|_q^{1-\theta}$. When $q = \infty$, use $|f|^{(1-\theta)r} \leq \|f\|_\infty^{(1-\theta)r}$ and the same calculation. Thus all finite L^p interpolation consequences used later follow directly from Hölder, with no operator-interpolation theorem hidden here. $\square$

Theorem 1.6 (I.1.006: Jensen inequality in the finite/probabilistic forms later used.). *Let ϕ be convex on an interval containing $x_1, \dots, x_m$, and let $\lambda_i \geq 0$ with $\sum_i \lambda_i = 1$. Then*

$$\phi\left(\sum_i \lambda_i x_i\right) \leq \sum_i \lambda_i \phi(x_i).$$

In particular, $\phi(\mathbb{E}X) \leq \mathbb{E}\phi(X)$ for every finite-valued random variable X for which the two sides are defined.

Proof. We use only the defining two-point inequality for convexity. The assertion for two points is exactly

$$\phi(tx + (1 - t)y) \leq t\phi(x) + (1 - t)\phi(y) \quad (0 \leq t \leq 1).$$

Assume the finite-weight statement is known for $m - 1$ points. If $\lambda_m = 1$ there is nothing to prove. Otherwise put $\Lambda = 1 - \lambda_m$ and

$$y = \frac{1}{\Lambda} \sum_{i < m} \lambda_i x_i.$$

The induction hypothesis gives $\phi(y) \leq \Lambda^{-1} \sum_{i < m} \lambda_i \phi(x_i)$. Applying two-point convexity to $\Lambda y + \lambda_m x_m$ yields

$$\phi\left(\sum_i \lambda_i x_i\right) \leq \Lambda \phi(y) + \lambda_m \phi(x_m) \leq \sum_i \lambda_i \phi(x_i).$$

For a finite-valued random variable, take the x_i to be its values and the λ_i to be their probabilities. $\square$

Theorem 1.7 (I.1.007: Markov and Chebyshev inequalities.). *If $X \geq 0$ and $t > 0$, then $\Pr(X \geq t) \leq \mathbb{E}X/t$. Consequently, for every square-integrable real random variable Y and $u > 0$,*

$$\Pr(|Y - \mathbb{E}Y| \geq u) \leq \frac{\text{Var}(Y)}{u^2}.$$

Proof. For $X \geq 0$, $X \geq t \mathbf{1}_{X \geq t}$, hence $\Pr(X \geq t) \leq \mathbb{E}X/t$. Apply this to $(Y - \mathbb{E}Y)^2$ with threshold t^2 to obtain Chebyshev. $\square$

Theorem 1.8 (I.1.008: Union bound and first-moment method.). *For events $E_1, \dots, E_m$, $\Pr(\bigcup_i E_i) \leq \sum_i \Pr(E_i)$. If X is a nonnegative integer-valued random variable, then $\Pr(X > 0) \leq \mathbb{E}X$; in particular, $\mathbb{E}X < 1$ guarantees the existence of an outcome with $X = 0$.*

Proof. The indicator of a union is pointwise at most the sum of the individual indicators. Taking expectations gives the union bound. The first-moment method is Markov at threshold 1: for an integer-valued $X \geq 0$, $\Pr(X > 0) \leq \mathbb{E}X$. $\square$

Theorem 1.9 (I.1.009: Second-moment method.). *For every nonnegative random variable $X \in L^2$,*

$$\Pr(X > 0) \geq \frac{(\mathbb{E}X)^2}{\mathbb{E}X^2}.$$

Hence if $\text{Var } X = o((\mathbb{E}X)^2)$ along a sequence of models with $\mathbb{E}X > 0$, then $\Pr(X > 0) = 1 - o(1)$.

Proof. For $X \geq 0$, Cauchy–Schwarz gives $(\mathbb{E}X)^2 = (\mathbb{E}[X\mathbf{1}_{X>0}])^2 \leq \mathbb{E}X^2 \Pr(X > 0)$. Thus $\Pr(X > 0) \geq (\mathbb{E}X)^2/\mathbb{E}X^2$. The usual concentration consequence follows by applying Chebyshev to X when $\text{Var } X = o((\mathbb{E}X)^2)$. $\square$

Theorem 1.10 (I.1.010: Basic Chernoff/Hoeffding concentration inequality for independent Bernoulli sums.). *Let $S = \sum_{i=1}^n X_i$, where the X_i are independent Bernoulli variables and $\mu = \mathbb{E}S$. Then for $\delta > 0$*

$$\Pr(S \geq (1 + \delta)\mu) \leq \exp\left(-\frac{\delta^2}{2 + \delta}\mu\right),$$

and for $0 < \delta < 1$

$$\Pr(S \leq (1 - \delta)\mu) \leq e^{-\delta^2\mu/2}.$$

More generally, if independent Y_i satisfy $a_i \leq Y_i \leq b_i$, then

$$\Pr\left(\sum_i (Y_i - \mathbb{E}Y_i) \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right).$$

Proof. For the Bernoulli upper tail, independence and $1 + x \leq e^x$ give, for $\lambda > 0$,

$$\mathbb{E}e^{\lambda S} = \prod_i (1 - p_i + p_i e^{\lambda}) \leq \exp(\mu(e^\lambda - 1)).$$

Markov's inequality therefore yields

$$\Pr(S \geq (1 + \delta)\mu) \leq \exp(\mu(e^\lambda - 1) - \lambda(1 + \delta)\mu).$$

The minimizing choice $e^\lambda = 1 + \delta$ gives

$$\exp\{-\mu((1 + \delta)\log(1 + \delta) - \delta)\} \leq e^{-\delta^2\mu/(2 + \delta)}.$$

The lower tail follows in the same way from $e^{-\lambda S}$ and $-\log(1 - \delta) - \delta \geq \delta^2/2$.

For completeness we also prove the bounded-variable step that was previously only alluded to. If $Y \in [a, b]$ almost surely, convexity of $e^{\lambda y}$ gives, after centering and rescaling to $[0, 1]$, Hoeffding's lemma

$$\mathbb{E}e^{\lambda(Y - \mathbb{E}Y)} \leq \exp\left(\frac{\lambda^2(b - a)^2}{8}\right).$$

One direct verification is to bound the exponential by the chord joining its endpoint values, take expectations, and maximize the resulting two-point expression over the mean; the maximum is at most $e^{\lambda^2(b-a)^2/8}$. Independence now gives

$$\mathbb{E} \exp\left(\lambda \sum_i (Y_i - \mathbb{E}Y_i)\right) \leq \exp\left(\frac{\lambda^2}{8} \sum_i (b_i - a_i)^2\right).$$

Markov's inequality and optimization at $\lambda = 4t/\sum_i (b_i - a_i)^2$ give the displayed Hoeffding bound. $\square$

Bipartite graph notation. A finite bipartite graph is a triple $G = (L, R, E)$ in which L and R are disjoint finite vertex sets and $E \subseteq L \times R$ is the edge set. For $x \in L$ and $y \in R$ define

$$d(x) := |\{y \in R : (x, y) \in E\}|, \quad d(y) := |\{x \in L : (x, y) \in E\}|.$$

Thus $d(v)$ is the degree of the vertex v . The *average left degree* is

$$d_L := \frac{1}{|L|} \sum_{x \in L} d(x).$$

When no confusion can arise, as in Theorem 1.12, we abbreviate d_L by d .

Theorem 1.11 (I.1.011: Bipartite graph degree-sum identities.). *If $G = (L, R, E)$ is a finite bipartite graph, then*

$$|E| = \sum_{x \in L} d(x) = \sum_{y \in R} d(y).$$

Proof. Every edge of a bipartite graph has exactly one endpoint in each part. Counting incident pairs (v, e) from the left gives $|E| = \sum_{x \in L} d(x)$ and from the right gives $|E| = \sum_{y \in R} d(y)$. $\square$

Common neighbourhoods and bad r -sets. For $x \in L$, write

$$N_R(x) := \{y \in R : (x, y) \in E\}.$$

More generally, if $S \subseteq L$, its *common neighbourhood in R* is

$$N_R(S) := \bigcap_{x \in S} N_R(x) = \{y \in R : (x, y) \in E \text{ for every } x \in S\}.$$

Thus $N_R(S)$ consists of the right-hand vertices adjacent to every vertex of S . Symmetrically, for $T \subseteq R$ define

$$N_L(T) := \{x \in L : (x, y) \in E \text{ for every } y \in T\}.$$

When $|S| = r$, we call S *bad (for the threshold q)* if

$$|N_R(S)| < q.$$

Here and below an “ r -set” means an unordered set of r distinct vertices; the proof does not use ordered r -tuples.

Theorem 1.12 (I.1.012: Dependent random choice lemma in a quantitative form sufficient for later dense graph arguments.). *Let $G = (L, R, E)$ be bipartite with $|L| = m$, $|R| = n$, and average left degree d . Let t, r, q be positive integers and let $a \geq 0$ be real. If*

$$m \left(\frac{d}{n} \right)^t - \binom{m}{r} \left(\frac{q}{n} \right)^t \geq a,$$

then there is $U \subseteq L$ with $|U| \geq a$ such that every r -element subset $S \subseteq U$ satisfies $|N_R(S)| \geq q$.

Worked example (common neighbourhoods and bad pairs). Let a bipartite graph have left vertex set $U = \{1, 2, 3, 4\}$, right vertex set $V = \{a, b, c\}$, and

$$N(1) = N(2) = \{a, b, c\}, \quad N(3) = \{a, b\}, \quad N(4) = \{b, c\}.$$

For a two-element set $S \subset U$, its common neighbourhood is

$$N(S) = \bigcap_{u \in S} N(u).$$

Taking “bad” to mean $|N(S)| < 2$, every pair containing 1 or 2 has at least two common neighbours, whereas

$$N(\{3, 4\}) = \{b\}.$$

Thus $\{3, 4\}$ is the unique bad unordered pair. If the theorem counts ordered pairs of distinct vertices, the bad pairs are $(3, 4)$ and $(4, 3)$; if repetitions are allowed, that convention must be stated separately. This finite calculation is exactly what the abstract bad- r -tuple definition records.

Proof. Choose $y_1, \dots, y_t \in R$ independently and uniformly, with repetition, and put

$$X := N_L(\{y_1, \dots, y_t\}) = \{x \in L : (x, y_j) \in E \text{ for every } 1 \leq j \leq t\}.$$

A fixed $x \in L$ lies in X precisely when all t samples land in $N_R(x)$, so

$$\Pr(x \in X) = \left(\frac{d(x)}{n}\right)^t.$$

Consequently

$$\mathbb{E}|X| = \sum_{x \in L} \left(\frac{d(x)}{n}\right)^t \geq m \left(\frac{d}{n}\right)^t,$$

where the last inequality is convexity of $u \mapsto u^t$ together with $m^{-1} \sum_{x \in L} d(x) = d$.

Now fix an r -element subset $S \subseteq L$. We have $S \subseteq X$ exactly when all y_j belong to the common neighbourhood $N_R(S)$; hence

$$\Pr(S \subseteq X) = \left(\frac{|N_R(S)|}{n}\right)^t.$$

If S is bad, this is $< (q/n)^t$, and therefore is certainly at most $(q/n)^t$. Let $B(X)$ denote the number of bad r -element subsets contained in X . By linearity of expectation,

$$\mathbb{E}B(X) \leq \binom{m}{r} \left(\frac{q}{n}\right)^t.$$

Thus

$$\mathbb{E}(|X| - B(X)) \geq m \left(\frac{d}{n}\right)^t - \binom{m}{r} \left(\frac{q}{n}\right)^t \geq a.$$

For at least one choice of $y_1, \dots, y_t$ we therefore have $|X| - B(X) \geq a$.

For that choice, select one vertex from each bad r -set contained in X and delete all selected vertices. At most $B(X)$ vertices are deleted, so the surviving set U has

$$|U| \geq |X| - B(X) \geq a.$$

No bad r -set can remain inside U : if $S \subseteq U$ were bad, then S was one of the bad r -sets of X , and the deletion procedure removed at least one of its vertices. Hence every r -element $S \subseteq U$ has $|N_R(S)| \geq q$, as claimed. $\square$

Definition 1.13 (Regular pair). For disjoint vertex sets A, B , write $d(A, B) = e(A, B)/(|A||B|)$. The pair (A, B) is ε -regular if

$$|d(A', B') - d(A, B)| \leq \varepsilon$$

whenever $|A'| \geq \varepsilon|A|$ and $|B'| \geq \varepsilon|B|$.

Theorem 1.14 (I.1.013: Tripartite regular-pair triangle counting lemma.). *Let A, B, C be the parts of a tripartite graph. Suppose each of $(A, B), (A, C), (B, C)$ is ε -regular and has density at least α , with $0 < 2\varepsilon < \alpha$. Then the number of triangles with one vertex in each part is at least*

$$(1 - 2\varepsilon)(\alpha - \varepsilon)^3|A||B||C|.$$

This is the elementary counting estimate required before the full graph/hypergraph regularity arguments later in the book; no graph-removal theorem is being smuggled into the present chapter.

Proof. Regularity of (A, B) implies that all but at most $\varepsilon|A|$ vertices $a \in A$ satisfy

$$|N_B(a)| \geq (d(A, B) - \varepsilon)|B| \geq (\alpha - \varepsilon)|B|.$$

Indeed, if more than $\varepsilon|A|$ vertices violated this inequality, taking those vertices as A' and taking $B' = B$ would contradict ε -regularity. The same argument for (A, C) leaves at least $(1 - 2\varepsilon)|A|$ vertices that are good simultaneously for B and C .

Fix such a good a . Since $\alpha - \varepsilon > \varepsilon$, both $N_B(a)$ and $N_C(a)$ are large enough for regularity of (B, C) to apply. Hence

$$e(N_B(a), N_C(a)) \geq (d(B, C) - \varepsilon)|N_B(a)||N_C(a)| \geq (\alpha - \varepsilon)^3|B||C|.$$

Every edge counted here forms a unique triangle with a . Summing over all simultaneously good a gives the result. $\square$

Chapter 2

Probability and energy language

Additive energy is the bridge between cardinality and probability. A representation function such as r_{A+B} may be normalized to a probability distribution; its collision probability is then exactly the quadratic statistic that additive combinatorics calls energy. This viewpoint makes Cauchy–Schwarz, popularity arguments, dyadic decomposition, and random sampling feel like parts of one language rather than isolated tricks.

The results in this chapter are deliberately elementary, but they will later sit inside Balog–Szemerédi–Gowers arguments, entropy proofs of PFR, and almost-periodicity. We therefore make the quantitative thresholds explicit.

Theorem 2.1 (I.2.001: Collision-probability identity for finite random variables.). *Let X, X' be independent copies of a random variable taking values in a finite set, with $p_x = \Pr(X = x)$. Then*

$$\Pr(X = X') = \sum_x p_x^2.$$

Proof. For independent copies X, X' , $\Pr(X = X') = \sum_x \Pr(X = x) \Pr(X' = x) = \sum_x p_x^2$. This is the squared ℓ^2 norm of the probability mass function. $\square$

Theorem 2.2 (I.2.002: Energy as an ℓ^2 norm of a representation function.). *Let A, B be finite subsets of an abelian group and set $r_{A+B}(x) = |\{(a, b) \in A \times B : a + b = x\}|$. Then*

$$E(A, B) := |\{(a_1, b_1, a_2, b_2) : a_1 + b_1 = a_2 + b_2\}| = \sum_x r_{A+B}(x)^2.$$

Proof. If $r_{A+B}(x) = |\{(a, b) : a + b = x\}|$, then $E(A, B) = |\{a_1 + b_1 = a_2 + b_2\}| = \sum_x r_{A+B}(x)^2 = \|1_A * 1_B\|_2^2$ under counting measure. $\square$

Theorem 2.3 (I.2.003: Additive-energy/Cauchy-Schwarz lower bound $E(A, B) \geq |A|^2 |B|^2 / |A+B|$). *For finite nonempty subsets A, B of an abelian group,*

$$E(A, B) \geq \frac{|A|^2 |B|^2}{|A+B|}.$$

Proof. Since $\sum_x r_{A+B}(x) = |A||B|$ and r is supported on $A+B$, Cauchy–Schwarz gives $|A|^2 |B|^2 \leq |A+B| \sum_x r(x)^2 = |A+B| E(A, B)$. $\square$

Worked Example 2.4 (Seeing additive energy as collision multiplicity). Take $A = B = \{0, 1, 3\} \subset \mathbb{Z}$. The nine ordered pairs in $A \times A$ produce the representation table

x	0	1	2	3	4	6
$r_{A+A}(x)$	1	2	1	2	2	1.

Thus

$$|A + A| = 6, \quad E(A) = 1^2 + 2^2 + 1^2 + 2^2 + 2^2 + 1^2 = 15.$$

The Cauchy–Schwarz lower bound gives

$$E(A) \geq \frac{|A|^4}{|A + A|} = \frac{81}{6} = 13.5,$$

so in this small example it already detects the repeated sums rather sharply. The three sums 1, 3, 4 are the “popular” fibres: each has two representations and hence contributes four energy quadruples rather than one.

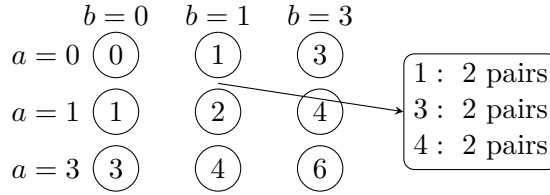


Figure 2.1: The sum map $A \times A \rightarrow A + A$ for $A = \{0, 1, 3\}$. Additive energy is the sum of the squared fibre sizes, so collisions in the sum map are exactly what energy measures.

Theorem 2.5 (I.2.004: Multiplicative-energy analogue in arbitrary finite groups.). *For finite nonempty subsets A, B of an arbitrary group, define $E_{\times}(A, B) = |\{(a_1, b_1, a_2, b_2) : a_1 b_1 = a_2 b_2\}|$. Then*

$$E_{\times}(A, B) = \sum_x r_{AB}(x)^2 \geq \frac{|A|^2 |B|^2}{|AB|}.$$

Proof. For a finite group set $r_{AB}(x) = |\{(a, b) : ab = x\}|$. Then $E_{\times}(A, B) = \sum_x r_{AB}(x)^2$, while $\sum_x r_{AB}(x) = |A||B|$. Cauchy–Schwarz on the support AB gives $E_{\times}(A, B) \geq |A|^2 |B|^2 / |AB|$. $\square$

Theorem 2.6 (I.2.005: Dyadic pigeonholing lemma.). *Let f be a nonnegative function on a finite set, with $f \leq M$ and $W = \sum_x f(x) > 0$. Among the dyadic level sets $L_j = \{x : 2^j \leq f(x) < 2^{j+1}\}$ there is one for which*

$$\sum_{x \in L_j} f(x) \geq \frac{W}{1 + \lceil \log_2 M \rceil},$$

with the obvious harmless modification when $M < 1$.

Proof. Partition positive values into dyadic classes $2^j \leq f < 2^{j+1}$. If $f \leq M$ and $\sum f = W$, at most $1 + \lceil \log_2 M \rceil$ classes occur; one class contributes at least $W/(1 + \lceil \log_2 M \rceil)$. This is the form used to freeze a multiplicity scale. $\square$

Theorem 2.7 (I.2.006: Popular-values/popular-differences lemma.). *Let r be a nonnegative function on a finite set, put $M = \sum_x r(x) > 0$ and $E = \sum_x r(x)^2$. Then the popular set*

$$P = \{x : r(x) \geq E/(2M)\}$$

carries at least half of the quadratic mass: $\sum_{x \in P} r(x)^2 \geq E/2$.

Proof. If nonnegative $r(x)$ has total mass M and $\sum r(x)^2 = E$, define $P = \{x : r(x) \geq E/(2M)\}$. The complement contributes at most $(E/(2M)) \sum r = E/2$ to $\sum r^2$, so $\sum_{x \in P} r(x)^2 \geq E/2$. Thus a positive fraction of the energy lies on popular values/differences. $\square$

Theorem 2.8 (I.2.007: Random-sampling preservation of density/energy at the level required later.). *Let A be a finite subset of an abelian group and let A_p be obtained by retaining each element independently with probability $0 < p \leq 1$. Then*

$$\mathbb{E}|A_p| = p|A|, \quad \mathbb{E}E(A_p) \geq p^4 E(A).$$

Moreover, if

$$|A|^3 e^{-p|A|/3} \leq \frac{1}{2} p^4 E(A),$$

there is a realization $A' \subseteq A$ with

$$|A'| \leq 2p|A|, \quad E(A') \geq \frac{1}{2} p^4 E(A).$$

Proof. Sample each element independently with probability p . Linearity of expectation gives $\mathbb{E}|A_p| = p|A|$. A fixed additive quadruple contains at most four distinct elements, so it survives with probability $p^d \geq p^4$, where $d \leq 4$ is the number of distinct entries. Summing over all additive quadruples gives $\mathbb{E}E(A_p) \geq p^4 E(A)$.

For the deterministic conclusion, Chernoff gives

$$\Pr(|A_p| > 2p|A|) \leq e^{-p|A|/3}.$$

Every subset of A has additive energy at most $|A|^3$, since three entries of a quadruple determine the fourth. If $\mathcal{B}$ is the event $|A_p| > 2p|A|$, then

$$\mathbb{E}(E(A_p)1_{\mathcal{B}}) \leq |A|^3 e^{-p|A|/3}.$$

Under the displayed hypothesis in the theorem,

$$\mathbb{E}(E(A_p)1_{\mathcal{B}^c}) \geq \frac{1}{2} p^4 E(A).$$

Hence some realization in $\mathcal{B}^c$ has energy at least that quantity and size at most $2p|A|$. $\square$

2.1 Small sumsets: Ruzsa calculus and Plünnecke theory

The preceding energy inequality is only the first consequence of small doubling. A second family of consequences is purely set-theoretic: one can compare many sumsets and difference sets by injections. These manipulations are usually called *Ruzsa calculus*. The deeper amplification step is Plünnecke theory. We prove both here because Volume II will use them as foundational input.

Throughout this section G is an abelian group and all sets are finite and nonempty. For an integer $m \geq 0$, write

$$mB := \underbrace{B + \cdots + B}_{m \text{ copies}}, \quad 0B := \{0\}.$$

Theorem 2.9 (I.2.008: Ruzsa triangle inequality.). *For finite nonempty $X, Y, Z \subseteq G$,*

$$\boxed{|Y - Z||X| \leq |Y - X||X - Z|}. \quad (2.8)$$

Equivalently,

$$|Y - Z| \leq \frac{|Y - X||X - Z|}{|X|}.$$

Proof. For each $d \in Y - Z$, choose once and for all one representation $d = y_d - z_d$ with $y_d \in Y$ and $z_d \in Z$. Define

$$\Phi : (Y - Z) \times X \longrightarrow (Y - X) \times (X - Z), \quad \Phi(d, x) := (y_d - x, x - z_d).$$

We claim that Φ is injective. From a pair $(u, v) = \Phi(d, x)$ we first recover

$$d = u + v = (y_d - x) + (x - z_d) = y_d - z_d.$$

Our chosen representation (y_d, z_d) is therefore known. We then recover $x = y_d - u$ (equivalently $x = v + z_d$). Thus (u, v) determines (d, x) , proving injectivity. Counting the domain and codomain gives (2.8). $\square$

Corollary 2.10 (Difference-set control from small doubling). *If $A \subseteq G$ and*

$$|A + A| \leq K|A|,$$

then

$$|A - A| \leq K^2|A|.$$

Proof. Apply Theorem 2.9 with $Y = A$, $Z = A$, and $X = -A$. Then $Y - X = A + A$ and $X - Z = -(A + A)$, so

$$|A - A| |A| \leq |A + A|^2 \leq K^2 |A|^2.$$

Divide by $|A|$. $\square$

Theorem 2.11 (I.2.009: Ruzsa covering lemma.). *Let $A, B \subseteq G$ and suppose*

$$|A + B| \leq K|B|.$$

Then there is a set $X \subseteq A$ with $|X| \leq K$ such that

$$\boxed{A \subseteq X + (B - B)}. \quad (2.9)$$

Proof. Choose $X \subseteq A$ maximal with the property that the translates $x + B$ ($x \in X$) are pairwise disjoint. Their union lies in $A + B$, hence

$$|X| |B| = \left| \bigsqcup_{x \in X} (x + B) \right| \leq |A + B| \leq K|B|,$$

so $|X| \leq K$.

Now let $a \in A$. By maximality, the translate $a + B$ cannot be disjoint from all $x + B$ with $x \in X$. Thus for some $x \in X$ there are $b_1, b_2 \in B$ with $a + b_1 = x + b_2$, and therefore

$$a = x + b_2 - b_1 \in x + (B - B).$$

Since a was arbitrary, (2.9) follows. $\square$

Lemma 2.12 (Petridis's minimal-growth lemma). *Let $X, B \subseteq G$ be finite and nonempty, and suppose X minimizes the ratio*

$$\frac{|Z + B|}{|Z|}$$

among all nonempty subsets $Z \subseteq X$. Put

$$\alpha := \frac{|X + B|}{|X|}.$$

Then for every finite set $C \subseteq G$,

$$\boxed{|X + B + C| \leq \alpha |X + C|}. \quad (2.10)$$

Proof. We induct on $|C|$. The case $C = \emptyset$ is trivial. Suppose C is nonempty, choose $c \in C$, and put $C' := C \setminus \{c\}$. Define

$$X_c := \{x \in X : x + c \in X + C'\}.$$

The new points contributed by the translate $X + c$ are exactly those coming from $X \setminus X_c$, so

$$|X + C| = |X + C'| + |X| - |X_c|. \quad (2.11)$$

Also

$$X + B + C = (X + B + C') \cup (X + B + c),$$

and the intersection contains $X_c + B + c$: if $x \in X_c$, then $x + c = x' + c'$ for some $x' \in X$, $c' \in C'$, hence $x + b + c = x' + b + c' \in X + B + C'$ for every $b \in B$. Therefore

$$|X + B + C| \leq |X + B + C'| + |X + B| - |X_c + B|.$$

If $X_c = \emptyset$, the final term is zero and the desired estimate follows from the induction hypothesis and $|X + B| = \alpha|X|$. If $X_c \neq \emptyset$, minimality of X gives $|X_c + B| \geq \alpha|X_c|$. In either case,

$$\begin{aligned} |X + B + C| &\leq \alpha|X + C'| + \alpha|X| - \alpha|X_c| \\ &= \alpha|X + C| \end{aligned}$$

by (2.11). This closes the induction. $\square$

Theorem 2.13 (I.2.010: Plünnecke–Petridis growth theorem.). *Let $A, B \subseteq G$ be finite and nonempty and suppose*

$$|A + B| \leq K|A|. \quad (2.12)$$

Then there is a nonempty subset $X \subseteq A$ such that, for every integer $m \geq 0$,

$$\boxed{|X + mB| \leq K^m|X|}. \quad (2.13)$$

Proof. Among the nonempty subsets $Z \subseteq A$, choose X minimizing $|Z + B|/|Z|$, and write

$$\alpha := \frac{|X + B|}{|X|}.$$

Since A itself is an admissible competitor,

$$\alpha \leq \frac{|A + B|}{|A|} \leq K.$$

Lemma 2.12, applied with $C = (m - 1)B$, gives for every $m \geq 1$

$$|X + mB| = |X + B + (m - 1)B| \leq \alpha|X + (m - 1)B|.$$

Iterating this inequality yields

$$|X + mB| \leq \alpha^m|X| \leq K^m|X|.$$

For $m = 0$ the assertion is equality. $\square$

Corollary 2.14 (I.2.011: Plünnecke–Ruzsa sum–difference inequalities.). *Under the hypotheses of Theorem 2.13, for all integers $m, n \geq 0$ with $m + n \geq 1$,*

$$\boxed{|mB - nB| \leq K^{m+n}|A|}. \quad (2.14)$$

In particular, if $B = A$ and $|A + A| \leq K|A|$, then

$$|mA - nA| \leq K^{m+n}|A|.$$

Proof. Let $X \subseteq A$ be supplied by Theorem 2.13. Apply the Ruzsa triangle inequality, Theorem 2.9, with the intermediate set $-X$, and with $Y = mB$, $Z = nB$. Since

$$|mB - (-X)| = |X + mB|, \quad |-X - nB| = |X + nB|,$$

we obtain

$$\begin{aligned} |mB - nB| &\leq \frac{|X + mB| |X + nB|}{|X|} \\ &\leq K^{m+n} |X| \\ &\leq K^{m+n} |A|. \end{aligned}$$

This proves the stated sum–difference bound. $\square$

Proposition 2.15 (I.2.012: The small-doubling implication package used from Volume II onward.). *Let $A \subseteq G$ be finite and nonempty and suppose*

$$|A + A| \leq K|A|.$$

Then all of the following conclusions hold:

$$E(A) \geq K^{-1}|A|^3 \tag{2.15a}$$

$$|A - A| \leq K^2|A| \tag{2.15b}$$

$$|mA - nA| \leq K^{m+n}|A| \quad (m, n \geq 0, m + n \geq 1) \tag{2.15c}$$

$$A \subseteq X + (A - A), \quad |X| \leq K \quad \text{for some } X \subseteq A. \tag{2.15d}$$

Here (2.15a) means that A has at least $K^{-1}|A|^3$ additive quadruples; (2.15b) and (2.15c) are the Ruzsa/Plünnecke growth consequences; and (2.15d) is the Ruzsa covering conclusion.

Proof. Equation (2.15a) is Theorem 2.3 with $B = A$. Equation (2.15b) is Corollary 2.10. Equation (2.15c) is Corollary 2.14 with $B = A$. Finally apply Theorem 2.11 with $B = A$ to obtain (2.15d). $\square$

Worked Example 2.16 (How the package reads for an interval). Let $A = \{0, 1, \dots, N-1\} \subset \mathbb{Z}$. Then

$$|A + A| = 2N - 1 < 2|A|,$$

so one may take $K = 2$. The general theory therefore gives

$$E(A) \geq \frac{1}{2}N^3, \quad |A - A| \leq 4N, \quad |mA - nA| \leq 2^{m+n}N.$$

Here the exact sum–difference set is the interval

$$mA - nA = \{-n(N-1), \dots, m(N-1)\},$$

so

$$|mA - nA| = (m+n)(N-1) + 1.$$

The Plünnecke–Ruzsa estimate is not sharp for this highly ordered example; its purpose is uniformity: it gives a bound depending only on the original doubling constant K , even when no interval structure is known in advance.

2.2 Partial sumsets and the Balog–Szemerédi–Gowers theorem

Small additive energy follows from a small complete sumset by Cauchy–Schwarz. In applications one often has a different kind of information: many pairs (a, b) are known to have their sums in a small set, but nothing is known about the remaining pairs. The Balog–Szemerédi–Gowers theorem is the mechanism that converts this *partial* additive control into an ordinary small-sumset statement on large subsets. We prove here the asymmetric graph form used later in Volumes II, IV, and V.

For finite sets A, B in an abelian group and a set of edges $\Gamma \subseteq A \times B$, define the *restricted sumset*

$$A +_{\Gamma} B := \{a + b : (a, b) \in \Gamma\}.$$

We regard Γ as a bipartite graph with left vertex class A and right vertex class B .

Lemma 2.17 (I.2.013: dense bipartite graphs have many two-step connections). *Let $\Gamma \subseteq A \times B$ satisfy*

$$|\Gamma| \geq \frac{|A||B|}{K}, \quad K \geq 1.$$

For every $0 < \varepsilon < 1$ there is a set $A_0 \subseteq A$ such that

$$|A_0| \geq \frac{|A|}{\sqrt{2} K},$$

and all but at most $\varepsilon|A_0|^2$ ordered pairs $(a, a') \in A_0^2$ have at least

$$\frac{\varepsilon}{2K^2}|B|$$

common neighbours in B .

Proof. For $a \in A$ and $b \in B$, write $N(a) \subseteq B$ and $N(b) \subseteq A$ for their neighbourhoods. Put

$$c(a, a') := |N(a) \cap N(a')|.$$

Double counting and Cauchy–Schwarz give

$$\begin{aligned} \frac{1}{|A|^2} \sum_{a, a' \in A} \frac{c(a, a')}{|B|} &= \frac{1}{|B|} \sum_{b \in B} \left(\frac{|N(b)|}{|A|} \right)^2 \\ &\geq \left(\frac{1}{|B|} \sum_{b \in B} \frac{|N(b)|}{|A|} \right)^2 \\ &= \left(\frac{|\Gamma|}{|A||B|} \right)^2 \geq \frac{1}{K^2}. \end{aligned}$$

Call (a, a') *bad* when

$$c(a, a') < \frac{\varepsilon}{2K^2}|B|.$$

Then

$$\frac{1}{|A|^2} \sum_{a, a' \in A} \left(1 - \frac{1}{\varepsilon} 1_{\text{bad}}(a, a') \right) \frac{c(a, a')}{|B|} \geq \frac{1}{2K^2}.$$

Rewrite the left side by summing first over the common neighbour b :

$$\frac{1}{|B|} \sum_{b \in B} \frac{1}{|A|^2} \sum_{a, a' \in N(b)} \left(1 - \frac{1}{\varepsilon} 1_{\text{bad}}(a, a') \right).$$

Hence some $b_0 \in B$ satisfies

$$\frac{1}{|A|^2} \sum_{a, a' \in N(b_0)} \left(1 - \frac{1}{\varepsilon} 1_{\text{bad}}(a, a')\right) \geq \frac{1}{2K^2}.$$

Set $A_0 = N(b_0)$. Since every summand is at most 1, $|A_0|^2/|A|^2 \geq 1/(2K^2)$, giving $|A_0| \geq |A|/(\sqrt{2}K)$. The same displayed inequality is positive, so

$$|A_0|^2 - \varepsilon^{-1} |\{(a, a') \in A_0^2 : (a, a') \text{ bad}\}| > 0.$$

Thus fewer than $\varepsilon|A_0|^2$ ordered pairs in A_0^2 are bad. $\square$

Lemma 2.18 (I.2.014: polynomially many three-step paths). *Under the hypotheses of Lemma 2.17, there are $A_1 \subseteq A$ and $B_1 \subseteq B$ such that*

$$|A_1| \geq \frac{|A|}{8\sqrt{2}K^2}, \quad |B_1| \geq \frac{|B|}{4K},$$

and every $(a, b) \in A_1 \times B_1$ is joined in the bipartite graph by at least

$$\frac{|A||B|}{2^{13}K^6}$$

paths of length three of the form

$$a \text{ --- } b' \text{ --- } a' \text{ --- } b.$$

Proof. Let

$$\tilde{A} := \left\{a \in A : |N(a)| \geq \frac{|B|}{2K}\right\}.$$

Deleting $A \setminus \tilde{A}$ removes fewer than $|A||B|/(2K)$ edges, so the induced graph on $\tilde{A} \times B$ has at least $|A||B|/(2K)$ edges. In particular

$$|\tilde{A}| \geq \frac{|A|}{2K}.$$

Relative to the vertex classes $\tilde{A}$ and B , this induced graph has edge density at least $1/(2K)$. Apply Lemma 2.17 with parameter $2K$ and

$$\varepsilon := \frac{1}{16K}.$$

We obtain $A_0 \subseteq \tilde{A}$ with

$$|A_0| \geq \frac{|\tilde{A}|}{2\sqrt{2}K} \geq \frac{|A|}{4\sqrt{2}K^2},$$

such that all but at most $\varepsilon|A_0|^2$ ordered pairs in A_0^2 have at least

$$\frac{\varepsilon}{2(2K)^2}|B| = \frac{|B|}{128K^3}$$

common neighbours.

Call those exceptional ordered pairs bad. By averaging, the set

$$A_1 := \{a \in A_0 : |\{a' \in A_0 : (a, a') \text{ bad}\}| \leq 2\varepsilon|A_0|\}$$

has size at least $|A_0|/2$, and therefore

$$|A_1| \geq \frac{|A|}{8\sqrt{2} K^2}.$$

Next define

$$B_1 := \left\{ b \in B : |N(b) \cap A_0| \geq \frac{|A_0|}{4K} \right\}.$$

Every $a \in A_0 \subseteq \tilde{A}$ has degree at least $|B|/(2K)$, so there are at least $|A_0||B|/(2K)$ edges between A_0 and B . Vertices outside B_1 account for fewer than $|A_0||B|/(4K)$ of these edges. Hence at least $|A_0||B|/(4K)$ edges meet B_1 , and consequently

$$|B_1| \geq \frac{|B|}{4K}.$$

Fix $a \in A_1$ and $b \in B_1$. There are at least $|A_0|/(4K)$ choices $a' \in A_0$ adjacent to b . At most $2\varepsilon|A_0| = |A_0|/(8K)$ of these form a bad pair with a . Thus at least $|A_0|/(8K)$ choices of a' are good, and each good pair (a, a') has at least $|B|/(128K^3)$ common neighbours b' . Therefore (a, b) is joined by at least

$$\frac{|A_0|}{8K} \frac{|B|}{128K^3} \geq \frac{|A||B|}{4096\sqrt{2} K^6} \geq \frac{|A||B|}{2^{13} K^6}$$

three-step paths, as claimed. $\square$

Theorem 2.19 (I.2.015: asymmetric Balog–Szemerédi–Gowers theorem). *Let A, B be finite nonempty subsets of an abelian group and let $\Gamma \subseteq A \times B$. Suppose*

$$|\Gamma| \geq \frac{|A||B|}{K_1}, \quad |A +_\Gamma B| \leq K_2(|A||B|)^{1/2},$$

where $K_1 \geq 1$ and $K_2 > 0$. Define

$$C_{\text{BSG}}(K_1, K_2) := 2^{16} K_1^8 \max(1, K_2)^3.$$

Then there are $A' \subseteq A$ and $B' \subseteq B$ satisfying

$$|A'| \geq C_{\text{BSG}}(K_1, K_2)^{-1}|A|, \quad |B'| \geq C_{\text{BSG}}(K_1, K_2)^{-1}|B|,$$

and

$$|A' + B'| \leq C_{\text{BSG}}(K_1, K_2)(|A'||B'|)^{1/2}.$$

Thus every dense partial sumset of size comparable with the geometric mean of $|A|$ and $|B|$ contains large vertex subsets whose complete sumset is polynomially controlled.

Proof. Put

$$S := A +_\Gamma B.$$

Apply Lemma 2.18 with $K = K_1$, obtaining $A_1 \subseteq A$ and $B_1 \subseteq B$. Fix $a \in A_1$ and $b \in B_1$. Every path

$$a \text{ --- } b' \text{ --- } a' \text{ --- } b$$

uses three edges of Γ , and therefore the three sums

$$x := a + b', \quad y := a' + b', \quad z := a' + b$$

all lie in S . They satisfy the identity

$$a + b = x - y + z.$$

For fixed a, b , the map from a path (a', b') to (x, y, z) is injective, because $b' = x - a$ and $a' = z - b$. Hence every fixed pair (a, b) has at least

$$L := \frac{|A||B|}{2^{13}K_1^6}$$

distinct representing triples $(x, y, z) \in S^3$.

For each $t \in A_1 + B_1$, choose one pair (a_t, b_t) with $a_t + b_t = t$. The corresponding sets of representing triples are disjoint for different t , since $x - y + z$ recovers t . Therefore

$$|A_1 + B_1|L \leq |S|^3,$$

and so

$$\begin{aligned} |A_1 + B_1| &\leq 2^{13}K_1^6 \frac{|S|^3}{|A||B|} \\ &\leq 2^{13}K_1^6 K_2^3 (|A||B|)^{1/2}. \end{aligned}$$

The size bounds in Lemma 2.18 give

$$|A| \leq 8\sqrt{2} K_1^2 |A_1|, \quad |B| \leq 4K_1 |B_1|.$$

Consequently

$$(|A||B|)^{1/2} \leq \sqrt{32\sqrt{2}} K_1^{3/2} (|A_1||B_1|)^{1/2} < 8K_1^{3/2} (|A_1||B_1|)^{1/2}.$$

Since $K_1 \geq 1$, we obtain

$$|A_1 + B_1| \leq 2^{16} K_1^8 \max(1, K_2)^3 (|A_1||B_1|)^{1/2}.$$

The same value of C_{BSG} is larger than both $8\sqrt{2}K_1^2$ and $4K_1$, so the stated lower bounds for $|A_1|$ and $|B_1|$ also hold. Take $A' = A_1$ and $B' = B_1$. $\square$

Chapter 3

Group theory from the required starting level

Later chapters and volumes move freely between finite abelian groups, vector spaces over finite fields, cyclic groups, and nilpotent groups. This chapter fixes the minimum algebraic infrastructure needed for those changes of setting. The emphasis is not on developing group theory for its own sake, but on identifying which quotient, stabilizer, commutator, or decomposition is actually being used in an additive argument.

For the student, the key structural theme is *passing to a quotient without losing the information one cares about*. The isomorphism theorems formalize that theme; the finite-abelian structure theorem makes it concrete; and the lower central series anticipates the nilpotent geometry of Volume V.

Theorem 3.1 (I.3.001: Lagrange theorem and coset partition theorem.). *If G is a finite group and $H \leq G$, the left cosets of H partition G and*

$$|G| = [G : H]|H|.$$

In particular $|H|$ divides $|G|$.

Proof. Left cosets of a subgroup $H \leq G$ are either equal or disjoint: if $gH \cap g'H \neq \emptyset$, then $g^{-1}g' \in H$. They partition finite G into sets of size $|H|$, so $|G| = [G : H]|H|$. $\square$

Theorem 3.2 (I.3.002: First isomorphism theorem.). *For a group homomorphism $\phi : G \rightarrow K$, the kernel is normal and the induced map*

$$G / \ker \phi \longrightarrow \text{im } \phi, \quad g \ker \phi \longmapsto \phi(g),$$

is an isomorphism.

Proof. First $\ker \phi$ is normal: if $h \in \ker \phi$ and $g \in G$, then

$$\phi(ghg^{-1}) = \phi(g)\phi(h)\phi(g)^{-1} = e,$$

so $ghg^{-1} \in \ker \phi$. Hence the quotient is defined. Put

$$\bar{\phi}(g \ker \phi) := \phi(g).$$

If $g \ker \phi = g' \ker \phi$, then $g'^{-1}g \in \ker \phi$, so $\phi(g) = \phi(g')$; thus $\bar{\phi}$ is well-defined. It is a homomorphism, its image is exactly $\text{im } \phi$, and $\bar{\phi}(g \ker \phi) = e$ implies $g \in \ker \phi$, so its kernel is trivial. Therefore $\bar{\phi}$ is an isomorphism onto $\text{im } \phi$. $\square$

Theorem 3.3 (I.3.003: Second and third isomorphism theorems in the forms later used.). *If $H \leq G$ and $N \triangleleft G$, then $H/(H \cap N) \cong HN/N$. If $N \leq H \triangleleft G$, then*

$$(G/N)/(H/N) \cong G/H.$$

Proof. Suppose first that $H \leq G$ and $N \triangleleft G$. Normality of N gives

$$(h_1 n_1)(h_2 n_2) = h_1 h_2 (h_2^{-1} n_1 h_2) n_2 \in HN,$$

and $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}(hn^{-1}h^{-1}) \in HN$, so HN is a subgroup. Also $H \cap N \triangleleft H$ because N is normal in G . The homomorphism

$$H \longrightarrow HN/N, \quad h \longmapsto hN,$$

has kernel $H \cap N$ and is onto: every coset $(hn)N$ equals hN . The first isomorphism theorem therefore gives $H/(H \cap N) \cong HN/N$.

Now suppose $N \leq H \triangleleft G$. The map

$$G/N \longrightarrow G/H, \quad gN \longmapsto gH,$$

is well-defined because $N \subseteq H$, is onto, and has kernel H/N . Applying the first isomorphism theorem once more gives $(G/N)/(H/N) \cong G/H$. $\square$

Worked Example 3.4 (The second isomorphism theorem in $\mathbb{Z}/12\mathbb{Z}$). Work additively in $G = \mathbb{Z}/12\mathbb{Z}$. Let

$$H = \langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}, \quad N = \langle 3 \rangle = \{0, 3, 6, 9\}.$$

Because G is abelian, $N \triangleleft G$. We have

$$H + N = G, \quad H \cap N = \{0, 6\} = \langle 6 \rangle.$$

Thus $H/(H \cap N)$ has $6/2 = 3$ elements. On the other side, G/N has $12/4 = 3$ elements, and because $H + N = G$ we have $(H + N)/N = G/N$. The second isomorphism theorem therefore gives

$$H/(H \cap N) \cong (H + N)/N \cong \mathbb{Z}/3\mathbb{Z}.$$

Concretely the cosets $\{0, 6\}, \{2, 8\}, \{4, 10\}$ map to $N, 2 + N, 4 + N$. The example shows what the quotient theorem does: directions already contained in N are collapsed, while the three genuinely different H -directions survive.

Group actions, orbits, and stabilizers. A (left) *action* of a group G on a set X is a map

$$G \times X \longrightarrow X, \quad (g, x) \longmapsto gx,$$

satisfying

$$ex = x, \quad (gh)x = g(hx)$$

for all $g, h \in G$ and $x \in X$, where e is the identity of G . For such an action and $x \in X$, the *orbit* of x is

$$Gx := \{gx : g \in G\},$$

and the *stabilizer* (or isotropy subgroup) of x is

$$G_x := \{g \in G : gx = x\}.$$

The set G_x is a subgroup of G : it contains the identity, is closed under products, and $gx = x$ implies $g^{-1}x = x$.

Theorem 3.5 (I.3.004: Orbit-stabilizer theorem.). *If a finite group G acts on a set X and $x \in X$, then the map*

$$\Phi : G/G_x \longrightarrow Gx, \quad gG_x \longmapsto gx,$$

is a bijection. Thus $|Gx| = [G : G_x]$.

Proof. We first check that Φ is well-defined. If $gG_x = hG_x$, then $h^{-1}g \in G_x$, so $(h^{-1}g)x = x$ and hence $gx = hx$.

The map is surjective by the definition of the orbit: every point of Gx has the form $gx = \Phi(gG_x)$. It is injective because

$$\Phi(gG_x) = \Phi(hG_x) \implies gx = hx \implies (h^{-1}g)x = x \implies h^{-1}g \in G_x,$$

which is equivalent to $gG_x = hG_x$. Thus Φ is a bijection. When G is finite, the number of left cosets of G_x is $[G : G_x]$, and therefore $|Gx| = [G : G_x]$. $\square$

Theorem 3.6 (I.3.005: Cauchy's theorem for finite groups.). *If a prime p divides the order of a finite group G , then G contains an element of order p .*

Proof. If a prime p divides $|G|$, let the cyclic group C_p act on $\Omega = \{(g_1, \dots, g_p) : g_1 \cdots g_p = e\}$ by rotation. Choosing the first $p-1$ coordinates freely determines the last, so $|\Omega| = |G|^{p-1}$ is divisible by p . Because p is prime, every C_p -orbit has size either 1 or p . Consequently $|\Omega|$ is congruent modulo p to the number of fixed points. A tuple is fixed precisely when it is $(g, \dots, g)$ with $g^p = e$. Hence the number of solutions of $g^p = e$ is divisible by p . The identity gives one such solution, so there is a nonidentity g with $g^p = e$; its order divides the prime p and is not 1, hence is exactly p . $\square$

Lemma 3.7 (Subgroups of $\mathbb{Z}^n$ are finitely generated free abelian). *Every subgroup $L \leq \mathbb{Z}^n$ is free abelian of rank at most n and admits a finite basis.*

Proof. We induct on n . The case $n = 0$ is trivial. Let $\pi : \mathbb{Z}^n \rightarrow \mathbb{Z}$ be projection onto the last coordinate. If $\pi(L) = 0$, then $L \leq \mathbb{Z}^{n-1}$ and induction applies. Otherwise $\pi(L) = d\mathbb{Z}$ for a unique $d \geq 1$. Choose $v \in L$ with last coordinate d , and put $K = L \cap \ker \pi$. By induction K has a finite basis $k_1, \dots, k_r$ with $r \leq n-1$.

For any $\ell \in L$, its last coordinate is md for some $m \in \mathbb{Z}$, so $\ell - mv \in K$. Hence

$$L = K \oplus \mathbb{Z}v.$$

Indeed, an element of K has last coordinate 0, whereas a nonzero multiple of v has nonzero last coordinate. Thus $k_1, \dots, k_r, v$ is a basis of L , and its size is at most n . $\square$

Integral row and column operations. An integer square matrix is *unimodular* if its determinant is ± 1 ; equivalently, it is invertible over $\mathbb{Z}$. The elementary integer row operations are: swapping two rows, multiplying a row by -1 , and adding an integer multiple of one row to another. Each is multiplication on the left by a unimodular matrix. The analogous column operations are multiplication on the right by a unimodular matrix. Hence all such operations are reversible over $\mathbb{Z}$ and preserve the isomorphism type of an integer cokernel.

Theorem 3.8 (I.3.006: Structure theorem for finitely generated abelian groups.). *Every finitely generated abelian group A is isomorphic to*

$$\mathbb{Z}^r \oplus \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_s\mathbb{Z}, \quad 1 < d_1 \mid d_2 \mid \cdots \mid d_s.$$

The integer r and the invariant factors d_i are uniquely determined by A .

Proof. Let A be generated by $a_1, \dots, a_n$. The map $\pi : \mathbb{Z}^n \rightarrow A$, $e_i \mapsto a_i$, is surjective, so $A \cong \mathbb{Z}^n/L$ with $L = \ker \pi$. Lemma 3.7 shows that L has a basis $\ell_1, \dots, \ell_m$. Put these basis vectors as the columns of an integer matrix M . Changing the basis of L performs a unimodular column operation, while changing the basis of $\mathbb{Z}^n$ performs a unimodular row operation; both preserve the isomorphism type of the quotient.

We diagonalize M by an explicit finite Smith-reduction procedure. Assume $M \neq 0$. Choose *any* nonzero entry, move it to the $(1, 1)$ position by a row swap and a column swap, and change the sign of the first row if necessary so that

$$d := M_{11} > 0.$$

There is no claim that this d is the smallest entry of the original matrix. Instead, the algorithm forces the pivot to decrease whenever the present pivot is not yet suitable.

First reduce the first column and first row by Euclidean division. For example, if $M_{i1} \neq 0$, write

$$M_{i1} = qd + r, \quad 0 \leq r < d.$$

Replace row i by row $i - q$ times row 1. If $r = 0$, the new $(i, 1)$ entry is zero. If $r > 0$, swap rows 1 and i ; the new upper-left entry is then r , which is a strictly smaller positive integer than the old pivot d . Apply the analogous column operation whenever some M_{1j} is nonzero. Whenever a nonzero remainder appears, restart the reduction with that smaller remainder as the new pivot. Because the positive upper-left entry decreases strictly at every restart, such restarts can occur only finitely many times. We therefore eventually reach a stage at which the first row and first column are zero off the diagonal:

$$M \sim \begin{pmatrix} d & 0 \\ 0 & B \end{pmatrix}, \quad d > 0. \quad (3.1)$$

We must still ensure that d divides every entry of the remaining block B . If it does, the first Smith step is complete. If not, choose an entry $b = B_{ij}$ with $d \nmid b$. Add the corresponding lower row to the first row. In the resulting matrix the first row contains both d in the first column and b in the column of that chosen entry. Write

$$b = qd + r, \quad 0 < r < d.$$

Subtract q times the first column from that column and then swap that column with the first. The new upper-left entry is r , a strictly smaller positive integer. Return to the preceding first-row/first-column reduction. Thus every failure of the divisibility condition also forces a strict decrease of the positive pivot. Again there can be only finitely many such decreases. Consequently the procedure terminates with

$$M \sim \begin{pmatrix} d_1 & 0 \\ 0 & M_1 \end{pmatrix}, \quad d_1 > 0, \quad d_1 \mid (M_1)_{ij} \quad \text{for every } i, j. \quad (3.2)$$

This d_1 is precisely a terminal minimal pivot for the above descent procedure: any obstruction to clearing the first row and column or to divisibility of the remaining block would create a smaller positive pivot and force another restart.

Now apply the same procedure recursively to M_1 , using only row and column operations supported on the remaining rows and columns. Since every entry of M_1 is divisible by d_1 , every integer row or column combination of those entries is still divisible by d_1 . Hence the next terminal pivot d_2 is divisible by d_1 . Repeating this argument gives

$$UMV = \text{diag}(d_1, \dots, d_t, 0, \dots, 0), \quad d_i > 0, \quad d_1 \mid d_2 \mid \dots \mid d_t,$$

with U, V unimodular. Therefore

$$A \cong \mathbb{Z}^{n-t} \oplus \bigoplus_{i=1}^t \mathbb{Z}/d_i\mathbb{Z}.$$

Factors with $d_i = 1$ are trivial and may be deleted.

It remains to prove that the free rank and invariant factors do not depend on the chosen presentation. For a prime p and a positive integer n , write

$$v_p(n) := \max\{e \geq 0 : p^e \mid n\}$$

for the p -adic valuation of n . For $k \geq 1$, the finite quotient $A/p^k A$ is intrinsic. If

$$A \cong \mathbb{Z}^r \oplus \bigoplus_{i=1}^s \mathbb{Z}/d_i\mathbb{Z}, \quad d_1 \mid \cdots \mid d_s,$$

then

$$\log_p |A/p^k A| = kr + \sum_{i=1}^s \min\{k, v_p(d_i)\}. \quad (3.3)$$

Hence

$$c_{p,k} := \log_p |A/p^k A| - \log_p |A/p^{k-1} A| = r + \#\{i : v_p(d_i) \geq k\}. \quad (3.4)$$

For k large these integers stabilize at r , so the free rank is intrinsic. After subtracting r , the sequence $c_{p,k} - r$ determines the multiset of positive p -adic valuations of the invariant factors. Because $d_1 \mid \cdots \mid d_s$, those valuations are nondecreasing in i . Let $s = \max_p(c_{p,1} - r)$, pad each prime's valuation list with leading zeros to length s , and combine the prime powers coordinatewise. This recovers every d_i uniquely. Thus both r and the invariant factors are uniquely determined by A . $\square$

Worked Example 3.9 (Smith reduction and an intrinsic check). Let $A = \mathbb{Z}^2/L$, where L is generated by $(2, 0)$ and $(1, 3)$. The relation matrix is

$$M = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

The gcd of its 1×1 minors is 1, while $|\det M| = 6$. Smith reduction therefore gives invariant factors 1, 6, so

$$A \cong \mathbb{Z}/6\mathbb{Z}.$$

The quotient test used in the uniqueness proof sees the same answer without knowing the presentation: $|A/2A| = 2$, $|A/3A| = 3$, and for higher powers the successive p -adic increments stabilize at free rank 0. The surviving 2- and 3-primary valuations are both 1, and combining them gives the single invariant factor 6. This small calculation is a useful model for the general uniqueness argument.

Theorem 3.10 (I.3.007: Classification of finite abelian groups.). *Every finite abelian group is a direct sum of cyclic groups of prime-power order; equivalently it has a unique invariant-factor decomposition $\bigoplus_i \mathbb{Z}/d_i\mathbb{Z}$ with $d_i \mid d_{i+1}$.*

Proof. Apply the preceding structure theorem to a finite abelian group. The free rank is zero, so

$$A \cong \bigoplus_i \mathbb{Z}/d_i\mathbb{Z}, \quad d_i \mid d_{i+1}.$$

It remains only to split each cyclic factor into its prime-power parts, and we prove the needed coprime case here rather than appeal forward to Chapter 4. If $d = ab$ with $\gcd(a, b) = 1$, consider

$$\psi : \mathbb{Z}/ab\mathbb{Z} \longrightarrow \mathbb{Z}/a\mathbb{Z} \oplus \mathbb{Z}/b\mathbb{Z}, \quad x \pmod{ab} \longmapsto (x \pmod{a}, x \pmod{b}).$$

If $\psi(x) = 0$, then both a and b divide x ; coprimality implies $ab \mid x$, so the kernel is trivial. The source and target both have ab elements, hence ψ is an isomorphism. Iterating this argument over the distinct prime divisors of d_i gives

$$\mathbb{Z}/d_i\mathbb{Z} \cong \bigoplus_{p \mid d_i} \mathbb{Z}/p^{v_p(d_i)}\mathbb{Z}.$$

Thus A is a direct sum of cyclic groups of prime-power order. Uniqueness of the invariant-factor decomposition was proved in Theorem 3.8; equivalently, for each prime p , the intrinsic numbers $\#\{i : v_p(d_i) \geq k\}$ from equation (3.4) recover the p -primary cyclic factors. $\square$

Theorem 3.11 (I.3.008: Basic structure and character groups of finite cyclic groups.). *Every subgroup of $\mathbb{Z}/n\mathbb{Z}$ is cyclic and corresponds to a divisor of n . Its character group is canonically parametrized by $k \in \mathbb{Z}/n\mathbb{Z}$ via*

$$\chi_k(x) = e^{2\pi i k x / n},$$

in particular $\widehat{\mathbb{Z}/n\mathbb{Z}} \cong \mathbb{Z}/n\mathbb{Z}$.

Proof. Every subgroup of $\mathbb{Z}/n\mathbb{Z}$ is generated by a divisor of n . A character is determined by the image of 1, which must be an n th root of unity; hence $\widehat{\mathbb{Z}/n\mathbb{Z}} \cong \mathbb{Z}/n\mathbb{Z}$ via $k \mapsto (x \mapsto e^{2\pi i k x / n})$. $\square$

Theorem 3.12 (I.3.009: Linear algebra over finite fields, including dimension formulas and quotient spaces.). *Let V be a finite-dimensional vector space over a finite field (indeed, over any field). Bases exist and may be extended; every linear map satisfies rank-nullity; and for every subspace $W \leq V$,*

$$\dim V = \dim W + \dim(V/W).$$

Proof. We record the argument because three later constructions use its individual steps. Start with a finite spanning list for V and repeatedly discard any vector lying in the span of its predecessors. The remaining list is linearly independent and still spans, hence is a basis. More generally, if $w_1, \dots, w_r$ is a basis of a subspace W , append any spanning list of V and perform the same deletion without ever deleting the w_i ; this extends the given basis to $w_1, \dots, w_r, v_{r+1}, \dots, v_n$.

Now let $T : V \rightarrow U$ be linear. Choose a basis $k_1, \dots, k_a$ of $\ker T$ and extend it to $k_1, \dots, k_a, v_{a+1}, \dots, v_n$ of V . The vectors $T(v_{a+1}), \dots, T(v_n)$ span $\operatorname{im} T$: subtracting the kernel part of any vector does not change its image. They are independent as well, because a relation among them would put the corresponding linear combination of the v_j in $\ker T$, contradicting independence of the chosen basis. Thus $\dim V = \dim \ker T + \dim \operatorname{im} T$.

Finally, for $W \leq V$, extend a basis $w_1, \dots, w_r$ of W to a basis $w_1, \dots, w_r, v_{r+1}, \dots, v_n$ of V . The cosets $v_{r+1} + W, \dots, v_n + W$ form a basis of V/W , so $\dim V = \dim W + \dim(V/W)$. Nothing in the proof uses finiteness of the field; finite fields are simply the case needed later. $\square$

Theorem 3.13 (I.3.010: Basic facts on S_n and A_n , including generation by transpositions/3-cycles.). *Every permutation in S_n is a product of transpositions. The sign map $S_n \rightarrow \{\pm 1\}$ has kernel A_n , and A_n is generated by 3-cycles.*

Proof. Every permutation is a product of transpositions by decomposing cycles, so transpositions generate S_n . The sign homomorphism has kernel A_n . Every even product of transpositions can be grouped in pairs, and $(ab)(bc) = (abc)$ while disjoint pairs $(ab)(cd) = (acb)(acd)$; hence 3-cycles generate A_n . $\square$

Theorem 3.14 (I.3.011: Basic matrix-group facts for GL_n and SL_n over fields.). *For a field F , $\det : \mathrm{GL}_n(F) \rightarrow F^\times$ is a homomorphism with kernel $\mathrm{SL}_n(F)$. For $n \geq 2$, elementary transvections generate $\mathrm{SL}_n(F)$. If $F = \mathbf{F}_q$, then*

$$|\mathrm{GL}_n(\mathbf{F}_q)| = \prod_{j=0}^{n-1} (q^n - q^j), \quad |\mathrm{SL}_n(\mathbf{F}_q)| = \frac{|\mathrm{GL}_n(\mathbf{F}_q)|}{q-1}.$$

Proof. Multiplicativity of the determinant makes $\det : \mathrm{GL}_n(F) \rightarrow F^\times$ a homomorphism, and its kernel is by definition $\mathrm{SL}_n(F)$.

Write $E_{ij}(t) = I + te_{ij}$ for an elementary transvection. Left multiplication by $E_{ij}(t)$ adds t times row j to row i . Gaussian elimination therefore reduces any $A \in \mathrm{SL}_n(F)$, using transvections, to a diagonal matrix of determinant one except possibly for the usual row interchanges and row rescalings. These operations themselves can be absorbed into transvections in 2×2 blocks. Indeed, for $a \in F^\times$, put

$$w(a) = E_{12}(a)E_{21}(-a^{-1})E_{12}(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}.$$

Then

$$w(a)w(-1) = \mathrm{diag}(a, a^{-1}),$$

so every determinant-one diagonal 2×2 block is a product of transvections. Since $w(-1)$ itself is such a product, the same calculation also realizes the needed row interchange with its compensating sign. Pairing diagonal entries and embedding these 2×2 identities in the (i, j) coordinates proves that the transvections generate $\mathrm{SL}_n(F)$.

For $F = \mathbf{F}_q$, choose the columns of an invertible matrix successively. The first has $q^n - 1$ choices; after j independent columns have been chosen, their span has q^j elements, so the next column has $q^n - q^j$ choices. Hence

$$|\mathrm{GL}_n(\mathbf{F}_q)| = \prod_{j=0}^{n-1} (q^n - q^j).$$

The determinant map is surjective because $\mathrm{diag}(a, 1, \dots, 1)$ has determinant a . All fibres of a surjective homomorphism have the size of its kernel, and therefore $|\mathrm{SL}_n(\mathbf{F}_q)| = |\mathrm{GL}_n(\mathbf{F}_q)|/(q-1)$. $\square$

Commutator notation. For elements x, y of a group G we use

$$[x, y] := xyx^{-1}y^{-1}.$$

If $A, B \leq G$ are subgroups, then $[A, B]$ denotes the subgroup generated by all $[a, b]$ with $a \in A$ and $b \in B$. Thus an expression such as $[G, [A, B]]$ has a precise subgroup meaning: it is generated by the commutators $[g, c]$ with $g \in G$ and $c \in [A, B]$.

Theorem 3.15 (I.3.012: Commutator identities and the precise quotient in which commutators are bilinear.). *Let G be a group. For all $x, y, z \in G$,*

$$[xy, z] = x[y, z]x^{-1}[x, z], \quad [x, yz] = [x, y]y[x, z]y^{-1}. \quad (3.12a)$$

Now let $A, B, N \triangleleft G$ be normal subgroups and suppose

$$[G, [A, B]] \leq N. \quad (3.12b)$$

Then, for $a_1, a_2 \in A$ and $b, b_1, b_2 \in B$,

$$[a_1 a_2, b]N = [a_1, b]N [a_2, b]N, \quad [a, b_1 b_2]N = [a, b_1]N [a, b_2]N \quad (3.12c)$$

in the quotient group G/N .

Equivalently: whenever the subgroup $[A, B]$ becomes central after passing to G/N , the commutator is multiplicative in each variable modulo N . This is the precise meaning of the phrase “the commutator is bilinear modulo higher commutators.”

Worked example (the graded commutator map in the Heisenberg group). Write

$$u(x, y, z) = \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}, \quad x, y, z \in \mathbb{Z}.$$

Matrix multiplication gives

$$u(x, y, z)u(x', y', z') = u(x + x', y + y', z + z' + xy').$$

With the convention $[g, h] = g^{-1}h^{-1}gh$ one obtains

$$[u(x, y, z), u(x', y', z')] = u(0, 0, xy' - x'y).$$

(With the opposite commutator convention the central coordinate is negated.) The commutator subgroup is the centre

$$\gamma_2(H) = \{u(0, 0, z) : z \in \mathbb{Z}\}, \quad \gamma_3(H) = \{1\}.$$

Consequently the induced map

$$H/\gamma_2(H) \times H/\gamma_2(H) \longrightarrow \gamma_2(H), \quad ((x, y), (x', y')) \longmapsto xy' - x'y,$$

is visibly additive in each variable. This is the concrete two-step model for the abstract bilinearity asserted on the associated graded group.

Proof. The two identities in (3.12a) are obtained by direct multiplication. For example,

$$\begin{aligned} x[y, z]x^{-1}[x, z] &= x(yzy^{-1}z^{-1})x^{-1}(xzx^{-1}z^{-1}) \\ &= xyzy^{-1}x^{-1}z^{-1} = [xy, z]. \end{aligned}$$

The second identity is checked in the same way.

Assume now (3.12b). Since $[a_2, b] \in [A, B]$, its conjugate by a_1 differs from $[a_2, b]$ by an element of $[G, [A, B]] \leq N$. Hence

$$a_1[a_2, b]a_1^{-1}N = [a_2, b]N.$$

Applying the first identity of (3.12a) gives

$$[a_1 a_2, b]N = [a_2, b]N [a_1, b]N.$$

The two factors lie in the image of $[A, B]$, which is central in G/N by (3.12b); therefore their order may be reversed, and the first formula in (3.12c) follows. The second formula is identical, using the second identity in (3.12a). Thus no unspecified “higher commutator” is being suppressed: the exact subgroup that must vanish is $[G, [A, B]]$ (or any normal subgroup containing it). $\square$

Theorem 3.16 (I.3.013: Lower central series elementary properties.). *Let $\gamma_1(G) = G$ and $\gamma_{i+1}(G) = [G, \gamma_i(G)]$. Then every $\gamma_i(G)$ is characteristic,*

$$[\gamma_i(G), \gamma_j(G)] \subseteq \gamma_{i+j}(G),$$

and $\gamma_i(G)/\gamma_{i+1}(G)$ is central in $G/\gamma_{i+1}(G)$.

Proof. Set $\gamma_i = \gamma_i(G)$. Characteristicity is immediate by induction: an automorphism carries commutators to commutators, so if it preserves γ_i it also preserves $[G, \gamma_i] = \gamma_{i+1}$.

We record the commutator containment used in the second assertion. For normal subgroups $A, B, C \triangleleft G$, the Hall–Witt identity implies

$$[[A, B], C] \subseteq [[B, C], A] [[C, A], B]. \quad (*)$$

To avoid a convention ambiguity, temporarily write $\langle x, y \rangle = x^{-1}y^{-1}xy$ and $w^g = g^{-1}wg$. The standard Hall–Witt identity in these conventions is

$$\langle \langle x, y^{-1} \rangle, z \rangle^y \langle \langle y, z^{-1} \rangle, x \rangle^z \langle \langle z, x^{-1} \rangle, y \rangle^x = e.$$

It is verified by expanding the three factors and cancelling. Passing to a quotient in which $[[B, C], A]$ and $[[C, A], B]$ vanish, the last two Hall–Witt factors vanish; the first then forces $[[A, B], C]$ to vanish as well. This proves (*). The subgroup commutators are unchanged if one switches between $xyx^{-1}y^{-1}$ and $x^{-1}y^{-1}xy$, because A, B, C are normal.

Now induct on i , simultaneously for every $j \geq 1$. For $i = 1$, $[\gamma_1, \gamma_j] = [G, \gamma_j] = \gamma_{j+1}$. Suppose $[\gamma_i, \gamma_j] \subseteq \gamma_{i+j}$ for every j . Applying (*) to $A = G$, $B = \gamma_i$, $C = \gamma_j$ gives

$$\begin{aligned} [\gamma_{i+1}, \gamma_j] &= [[G, \gamma_i], \gamma_j] \\ &\subseteq [[\gamma_i, \gamma_j], G] [[\gamma_j, G], \gamma_i] \\ &\subseteq [\gamma_{i+j}, G] [\gamma_{j+1}, \gamma_i] \subseteq \gamma_{i+j+1}. \end{aligned}$$

The last inclusion uses the induction hypothesis again (and symmetry of subgroup commutators) with $j + 1$ in place of j . Thus $[\gamma_i, \gamma_j] \subseteq \gamma_{i+j}$ for all i, j . Finally $[G, \gamma_i] = \gamma_{i+1}$, so γ_i/γ_{i+1} is central in G/γ_{i+1} . $\square$

Remark 3.17 (What “bilinear commutator” means for the lower central series). Theorem 3.16 now supplies the standard application of Theorem 3.15. If $A = \gamma_i(G)$, $B = \gamma_j(G)$ and $N = \gamma_{i+j+1}(G)$, then

$$[G, [A, B]] \subseteq [\gamma_1, \gamma_{i+j}] \subseteq \gamma_{i+j+1} = N.$$

Moreover $[\gamma_{i+1}, \gamma_j]$ and $[\gamma_i, \gamma_{j+1}]$ are both contained in γ_{i+j+1} . Therefore the rule

$$(x\gamma_{i+1}, y\gamma_{j+1}) \longmapsto [x, y]\gamma_{i+j+1}$$

is well defined and is a homomorphism in each variable

$$\gamma_i/\gamma_{i+1} \times \gamma_j/\gamma_{j+1} \longrightarrow \gamma_{i+j}/\gamma_{i+j+1}.$$

These quotients are abelian, so in additive notation this is literally a biadditive map. This is the form used later in nilpotent algebra.

Chapter 4

Elementary number theory

Prime moduli and congruence classes enter additive combinatorics long before the analytic theory of primes. We need a small amount of elementary number theory to manipulate those congruences, factor local conditions, and estimate harmless divisor weights. Nothing in this chapter uses the prime number theorem. Deeper prime-distribution and circle-method material is intentionally outside this book and belongs to the separate Analytic Number Theory project.

The Chinese remainder theorem is especially important conceptually: it explains why local obstructions factor prime by prime. Möbius inversion and Dirichlet convolution provide the corresponding multiplicative bookkeeping on arithmetic functions.

Theorem 4.1 (I.4.001: Euclidean algorithm and Bezout identity.). *For integers a, b , not both zero, the Euclidean algorithm terminates at $d = \gcd(a, b)$, and there exist $x, y \in \mathbb{Z}$ such that $d = ax + by$.*

Worked example (a local solution set and its Chinese-remainder factorization).

Let $F(n) = n(n + 2)$ and define

$$\Omega_F(q) = \{a \in \mathbb{Z}/q\mathbb{Z} : F(a) \equiv 0 \pmod{q}\}, \quad \beta_F(q) = \frac{|\Omega_F(q)|}{q}.$$

Modulo 3 the roots are 0 and 1, while modulo 5 they are 0 and 3. The Chinese remainder theorem therefore pairs these choices independently. The four resulting roots modulo 15 are

$$0, \quad 3, \quad 10, \quad 13.$$

Hence

$$|\Omega_F(15)| = 4 = |\Omega_F(3)| |\Omega_F(5)|, \quad \beta_F(15) = \frac{4}{15} = \frac{2}{3} \cdot \frac{2}{5}.$$

The calculation also isolates the logical content of multiplicativity: it is a statement about coprime moduli and the canonical CRT bijection. Determining $\Omega_F(p^k)$ for one prime power is a separate local problem; multiplicativity does not solve that problem for us.

Proof. The Euclidean algorithm repeatedly replaces (a, b) by $(b, a - qb)$ without changing the ideal they generate; it terminates at $d = \gcd(a, b)$. Back-substitution expresses $d = ax + by$. $\square$

Theorem 4.2 (I.4.002: Chinese remainder theorem.). *If $n_1, \dots, n_r$ are pairwise coprime and $N = \prod_i n_i$, then for every set of residues $a_i \pmod{n_i}$ there is a unique $x \pmod{N}$ satisfying $x \equiv a_i \pmod{n_i}$ for all i . Equivalently,*

$$\mathbb{Z}/N\mathbb{Z} \cong \prod_i \mathbb{Z}/n_i\mathbb{Z}$$

as rings.

Proof. For pairwise coprime n_i , let $N = \prod n_i$, $N_i = N/n_i$, and choose $u_i N_i \equiv 1 \pmod{n_i}$. Then $x = \sum a_i u_i N_i$ has the prescribed residues. Uniqueness modulo N follows because a difference divisible by every n_i is divisible by their product. This is a ring isomorphism $\mathbb{Z}/N\mathbb{Z} \cong \prod \mathbb{Z}/n_i\mathbb{Z}$. $\square$

Theorem 4.3 (I.4.003: Fermat and Euler congruence theorems.). *If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. More generally, if $(a, n) = 1$, then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Proof. If $p \nmid a$, multiplication by a permutes the nonzero residues mod p , so $(p-1)!a^{p-1} \equiv (p-1)! \pmod{p}$ and cancellation gives Fermat. For general n with $(a, n) = 1$, multiplication by a permutes $(\mathbb{Z}/n\mathbb{Z})^\times$; multiplying all units and cancelling gives $a^{\varphi(n)} \equiv 1 \pmod{n}$. $\square$

Definition 4.4 (Arithmetic functions, multiplicativity, and the basic examples). An *arithmetic function* is a function $f : \mathbb{N} \rightarrow \mathbb{C}$ (or into another commutative ring when explicitly stated). It is *multiplicative* if $f(1) = 1$ and

$$f(mn) = f(m)f(n) \quad \text{whenever } (m, n) = 1.$$

We write $1(n) = 1$ for the constant-one function. The *Möbius function* μ is defined by

$$\mu(n) = \begin{cases} 0, & p^2 \mid n \text{ for some prime } p, \\ (-1)^r, & n \text{ is a product of } r \text{ distinct primes,} \end{cases} \quad \mu(1) = 1.$$

For arithmetic functions f, g their *Dirichlet convolution* is

$$(f * g)(n) := \sum_{d \mid n} f(d)g(n/d).$$

The k -fold divisor function is $d_k := \underbrace{1 * \cdots * 1}_{k \text{ factors}}$; equivalently $d_k(n)$ counts ordered factorizations $n = n_1 \cdots n_k$ with $n_i \geq 1$. Finally

$$\sigma_s(n) := \sum_{d \mid n} d^s.$$

These definitions are placed here because the next theorem uses all of them.

Theorem 4.5 (I.4.004: Multiplicativity of phi and basic arithmetic functions.). *Euler's totient φ is multiplicative. More generally, the Dirichlet convolution of multiplicative arithmetic functions is multiplicative; hence the Möbius function μ , the divisor functions $d_k = 1^{*k}$, and the divisor sums $\sigma_s(n) = \sum_{d \mid n} d^s$ are multiplicative.*

Proof. CRT gives $(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$ when $(m, n) = 1$, hence $\varphi(mn) = \varphi(m)\varphi(n)$.

If f and g are multiplicative and $(m, n) = 1$, every divisor $d \mid mn$ factors uniquely as $d = d_1 d_2$ with $d_1 \mid m$, $d_2 \mid n$. Therefore

$$(f * g)(mn) = \sum_{d_1 \mid m} \sum_{d_2 \mid n} f(d_1 d_2) g(mn/(d_1 d_2)) = (f * g)(m)(f * g)(n).$$

Thus convolution preserves multiplicativity. The constant-one function and $n \mapsto n^s$ are multiplicative, so $d_k = 1^{*k}$ and $\sigma_s = 1 * (n \mapsto n^s)$ are multiplicative. For μ we argue directly from its definition, without using Möbius inversion before it has been proved. If $(m, n) = 1$, then mn is squarefree exactly when both m and n are squarefree. In that case, if m has r distinct prime factors and n has s , coprimality makes the prime factors of mn their disjoint union, so $\mu(mn) = (-1)^{r+s} = \mu(m)\mu(n)$; if either factor is not squarefree, both sides are zero. Hence μ is multiplicative. $\square$

Theorem 4.6 (I.4.005: Dirichlet convolution algebra of arithmetic functions.). *For arithmetic functions $f, g : \mathbb{N} \rightarrow \mathbb{C}$, define*

$$(f * g)(n) := \sum_{d|n} f(d)g(n/d).$$

Then $$ is associative and commutative, has identity ε given by $\varepsilon(1) = 1$ and $\varepsilon(n) = 0$ for $n > 1$, and every arithmetic function f with $f(1) \neq 0$ has a unique convolution inverse.*

Proof. For arithmetic functions define $(f * g)(n) = \sum_{d|n} f(d)g(n/d)$. Reindexing divisor factorizations proves associativity and commutativity; $\varepsilon(1) = 1$, $\varepsilon(n > 1) = 0$ is the identity. If $f(1) \neq 0$, recursively solving $(f * g)(n) = \varepsilon(n)$ gives a unique convolution inverse. $\square$

Theorem 4.7 (I.4.006: Mobius inversion formula.). *Let $F(n) = \sum_{d|n} f(d)$. Then*

$$f(n) = \sum_{d|n} \mu(d)F(n/d).$$

*Equivalently, $\mu * 1 = \varepsilon$ and $F = f * 1$ implies $f = F * \mu$.*

Proof. The function μ satisfies $\sum_{d|n} \mu(d) = \varepsilon(n)$, checked multiplicatively and on prime powers. Hence $\mu * 1 = \varepsilon$. If $F = f * 1$, convolving with μ gives $f = F * \mu$, i.e. $f(n) = \sum_{d|n} \mu(d)F(n/d)$. $\square$

Theorem 4.8 (I.4.007: Basic divisor-function bounds sufficient for early volumes.). *For every fixed $k \geq 2$ and every $\varepsilon > 0$, the k -fold divisor function satisfies*

$$d_k(n) \ll_{k,\varepsilon} n^\varepsilon.$$

Proof. If $n = \prod p_i^{a_i}$ then $d_k(n) = \prod_i \binom{a_i + k - 1}{k - 1}$. Fix $\varepsilon > 0$. For sufficiently large primes each factor is at most $p_i^{\varepsilon a_i/2}$; the finitely many small primes contribute at most a constant times $p_i^{\varepsilon a_i/2}$ because polynomial growth in a_i is dominated by an exponential. Multiplying yields $d_k(n) \ll_{k,\varepsilon} n^\varepsilon$. $\square$

Theorem 4.9 (I.4.008: Elementary prime-counting lower and upper bounds sufficient for this book.). *There are absolute constants $c, C > 0$ and x_0 such that for all $x \geq x_0$,*

$$c \frac{x}{\log x} \leq \pi(x) \leq C \frac{x}{\log x}.$$

No prime-number theorem is used.

Proof. We use Chebyshev's elementary central-binomial-coefficient argument. First,

$$\prod_{n < p \leq 2n} p \mid \binom{2n}{n} < 4^n,$$

so

$$(\pi(2n) - \pi(n)) \log n \leq 2n \log 2.$$

Summing this estimate over dyadic intervals gives $\pi(x) \ll x/\log x$.

For the lower bound put $\psi(x) = \sum_{p^a \leq x} \log p$. For every prime p , the p -adic valuation of $\binom{2n}{n}$ is at most the largest a with $p^a \leq 2n$; hence $\binom{2n}{n}$ divides $\text{lcm}(1, 2, \dots, 2n)$ and

$$\log \binom{2n}{n} \leq \psi(2n).$$

Since $\binom{2n}{n} \geq 4^n/(2n+1)$, this yields $\psi(x) \gg x$. On the other hand

$$\psi(x) = \vartheta(x) + \sum_{a \geq 2} \vartheta(x^{1/a}) = \vartheta(x) + O(\sqrt{x} \log^2 x)$$

by the trivial bound $\vartheta(y) \leq y \log y$. Thus $\vartheta(x) \gg x$ for all sufficiently large x , and because $\vartheta(x) \leq \pi(x) \log x$ we obtain $\pi(x) \gg x/\log x$. No analytic prime-number theorem has entered. $\square$

Lemma 4.10 (A prime in a fixed dilate). *There is an absolute constant $B > 1$ and x_1 such that every real $x \geq x_1$ has a prime p with*

$$x < p \leq Bx.$$

Consequently, for every $C_0 > 0$ and every integer

$$N \geq N_0(C_0) := \left\lceil \frac{x_1}{C_0} \right\rceil,$$

one may choose a prime p satisfying

$$C_0 N < p \leq B C_0 N.$$

Proof. Let c, C, x_0 be the constants from Theorem 4.9. Choose a fixed $B > 4C/c$. For sufficiently large x we have

$$\pi(Bx) - \pi(x) \geq c \frac{Bx}{\log(Bx)} - C \frac{x}{\log x}.$$

Since $\log(Bx)/\log x \rightarrow 1$, the first term is eventually at least $(cB/2)x/\log x > 2Cx/\log x$. Hence the difference is positive, and there is a prime in $(x, Bx]$. If $N \geq \lceil x_1/C_0 \rceil$, then $C_0 N \geq x_1$; applying the first statement with $x = C_0 N$ gives the second assertion. $\square$

Worked Example 4.11 (Why a fixed-dilate prime is enough). Later cyclic embeddings need a prime modulus larger than a prescribed constant multiple of the interval length, but they never need the sharp factor 2 from Bertrand's postulate. If, for example, a proof only requires $p > kN$ and $p = O_k(N)$, apply Lemma 4.10 with $C_0 = k$ to get

$$kN < p \leq BkN.$$

All density losses and multiplicity bounds remain functions of the fixed parameter k alone. Thus the elementary Chebyshev estimates already proved in this chapter supply exactly the prime-selection input used in this book.

Theorem 4.12 (I.4.009: Local solubility and factorization of CRT-compatible local densities.). *Fix integers $d \geq 1$, $r, s \geq 0$, polynomials*

$$F_1, \dots, F_r, G_1, \dots, G_s \in \mathbb{Z}[x_1, \dots, x_d],$$

and integers $a_1, \dots, a_r$. For each modulus $q \geq 1$, define the set of allowed residue classes

$$\Omega(q) := \left\{ \mathbf{n} \in (\mathbb{Z}/q\mathbb{Z})^d : \begin{array}{ll} F_i(\mathbf{n}) \equiv a_i \pmod{q} & (1 \leq i \leq r), \\ \gcd(G_j(\mathbf{n}), q) = 1 & (1 \leq j \leq s) \end{array} \right\}. \quad (4.9a)$$

We say that the system is locally soluble modulo q if $\Omega(q) \neq \emptyset$, and we define its normalized local density by

$$\beta(q) := \frac{|\Omega(q)|}{q^d}. \quad (4.9b)$$

If $(q_1, q_2) = 1$, the Chinese remainder isomorphism induces a bijection

$$\Omega(q_1 q_2) \cong \Omega(q_1) \times \Omega(q_2). \quad (4.9c)$$

Consequently

$$\beta(q_1 q_2) = \beta(q_1) \beta(q_2), \quad (4.9d)$$

and the system is locally soluble modulo $q_1 q_2$ if and only if it is locally soluble modulo both q_1 and q_2 .

In particular, if $q = \prod_p p^{\nu_p(q)}$, then

$$\Omega(q) \neq \emptyset \iff \Omega(p^{\nu_p(q)}) \neq \emptyset \text{ for every } p \mid q.$$

Thus a local obstruction at a composite modulus is already visible at one of its prime-power factors.

Proof. By Theorem 4.2, reduction modulo q_1 and q_2 gives a bijection

$$(\mathbb{Z}/q_1 q_2 \mathbb{Z})^d \longrightarrow (\mathbb{Z}/q_1 \mathbb{Z})^d \times (\mathbb{Z}/q_2 \mathbb{Z})^d.$$

Because integer polynomials commute with reduction modulo a modulus,

$$F_i(\mathbf{n}) \equiv a_i \pmod{q_1 q_2}$$

holds exactly when it holds modulo both q_1 and q_2 . Also, since $(q_1, q_2) = 1$,

$$\gcd(G_j(\mathbf{n}), q_1 q_2) = 1$$

holds exactly when $G_j(\mathbf{n})$ is coprime to each of q_1 and q_2 . Therefore the CRT bijection restricts to the bijection (4.9c).

Taking cardinalities gives $|\Omega(q_1 q_2)| = |\Omega(q_1)| |\Omega(q_2)|$. Dividing by $(q_1 q_2)^d = q_1^d q_2^d$ proves (4.9d). Nonemptiness of a Cartesian product is equivalent to nonemptiness of both factors, which proves the solubility assertion. Iterating over the prime-power factorization of q gives the final statement. $\square$

Worked Example 4.13 (The usual local obstruction for a family of affine forms). Let $L_1, \dots, L_t : \mathbb{Z}^d \rightarrow \mathbb{Z}$ be affine-linear forms and impose only the requirement that none of them vanish modulo any prime dividing q . Set

$$G(\mathbf{n}) := \prod_{i=1}^t L_i(\mathbf{n}).$$

Then Theorem 4.12 with $r = 0$, $s = 1$, and $G_1 = G$ gives

$$\Omega(q) = \{\mathbf{n} \pmod{q} : \gcd(G(\mathbf{n}), q) = 1\}.$$

For a prime p this is simply

$$\Omega(p) = \{\mathbf{n} \pmod{p} : L_i(\mathbf{n}) \not\equiv 0 \pmod{p} \text{ for every } i\}.$$

Hence a prime p is a local obstruction precisely when these forbidden hyperplanes cover all of $(\mathbb{Z}/p\mathbb{Z})^d$. For squarefree $q = p_1 \cdots p_m$, CRT gives the exact factorization

$$\frac{|\Omega(q)|}{q^d} = \prod_{j=1}^m \frac{|\Omega(p_j)|}{p_j^d}.$$

This is the elementary mechanism behind the Euler-product local factors that appear later; the analytic question of whether an infinite product converges is separate and is not asserted here.

Chapter 5

Characters and finite Fourier analysis

Finite Fourier analysis is the first genuinely structural microscope of the subject. Addition in the physical group becomes multiplication on the dual group; a large Fourier coefficient records correlation with a character; subgroups become annihilators; and additive energy becomes a fourth moment. These correspondences explain why Fourier methods are so effective for three-term progressions and small-doubling problems.

We use normalized counting measure throughout. Thus $\hat{f}(\chi) = \mathbb{E}_x f(x) \overline{\chi(x)}$, Fourier inversion has an unnormalized sum over $\hat{G}$, and Parseval reads $\mathbb{E}|f|^2 = \sum_{\chi} |\hat{f}(\chi)|^2$. Keeping this normalization fixed prevents hidden powers of $|G|$ later.

Theorem 5.1 (I.5.001: Character group of a finite abelian group.). *If G is a finite abelian group, its character group $\hat{G} = \text{Hom}(G, S^1)$ has cardinality $|G|$ and is (noncanonically) isomorphic to G .*

Proof. By the finite abelian classification, $G \cong \prod_j \mathbb{Z}/n_j\mathbb{Z}$. Characters of a product are products of characters, and the cyclic calculation gives $\hat{G} \cong \prod_j \mathbb{Z}/n_j\mathbb{Z}$ noncanonically. In particular $|\hat{G}| = |G|$. $\square$

Theorem 5.2 (I.5.002: Orthogonality relations for characters.). *For characters $\chi, \psi \in \hat{G}$,*

$$\sum_{x \in G} \chi(x) \overline{\psi(x)} = |G| 1_{\chi=\psi},$$

and dually

$$\sum_{\chi \in \hat{G}} \chi(x) \overline{\chi(y)} = |G| 1_{x=y}.$$

Proof. If χ is nontrivial, choose y with $\chi(y) \neq 1$. Translation invariance gives

$$\sum_x \chi(x) = \sum_x \chi(x+y) = \chi(y) \sum_x \chi(x),$$

hence the sum is zero. Applying this to the character $\chi \overline{\psi}$ proves the first relation.

For the dual relation put $z = x - y$. If $z = 0$, every summand is 1 and the sum is $|\hat{G}| = |G|$ by Theorem 5.1. If $z \neq 0$, the cyclic-factor description in the same theorem supplies a character ρ with $\rho(z) \neq 1$. Writing $S(z) = \sum_{\chi \in \hat{G}} \chi(z)$ and using that multiplication by ρ permutes $\hat{G}$, we get

$$S(z) = \sum_{\chi} (\rho\chi)(z) = \rho(z) S(z),$$

so $S(z) = 0$. This proves the second relation. $\square$

Theorem 5.3 (I.5.003: Pontryagin duality for finite abelian groups.). *For every finite abelian group G , the evaluation map*

$$G \longrightarrow \widehat{\widehat{G}}, \quad x \longmapsto (\chi \mapsto \chi(x)),$$

is an isomorphism.

Proof. Define $\iota : G \rightarrow \widehat{\widehat{G}}$ by $\iota(x)(\chi) = \chi(x)$. Characters separate points by the cyclic-factor description, so ι is injective. Both finite groups have order $|G|$, hence ι is an isomorphism. $\square$

Theorem 5.4 (I.5.004: Fourier inversion on finite abelian groups.). *For $f : G \rightarrow \mathbb{C}$ define $\widehat{f}(\chi) = \mathbb{E}_{x \in G} f(x) \overline{\chi(x)}$. Then*

$$f(y) = \sum_{\chi \in \widehat{G}} \widehat{f}(\chi) \chi(y).$$

Proof. With normalized transform $\widehat{f}(\chi) = \mathbb{E}_x f(x) \overline{\chi(x)}$, compute $\sum_{\chi} \widehat{f}(\chi) \chi(y) = \mathbb{E}_x f(x) \sum_{\chi} \chi(y - x)$. Dual orthogonality makes the inner sum $|G|$ when $x = y$ and 0 otherwise, giving $f(y)$. $\square$

Theorem 5.5 (I.5.005: Parseval identity.). *For functions $f, g : G \rightarrow \mathbb{C}$,*

$$\mathbb{E}_x f(x) \overline{g(x)} = \sum_{\chi \in \widehat{G}} \widehat{f}(\chi) \overline{\widehat{g}(\chi)}.$$

In particular $\mathbb{E}|f|^2 = \sum_{\chi} |\widehat{f}(\chi)|^2$.

Proof. Insert Fourier inversion for one function and use character orthogonality: $\mathbb{E}_x f(x) \overline{g(x)} = \sum_{\chi} \widehat{f}(\chi) \overline{\widehat{g}(\chi)}$. Taking $g = f$ gives Parseval. $\square$

Theorem 5.6 (I.5.006: Plancherel identity.). *Let G be a finite abelian group and define, for $f : G \rightarrow \mathbb{C}$,*

$$\widehat{f}(\chi) := \mathbb{E}_{x \in G} f(x) \overline{\chi(x)} \quad (\chi \in \widehat{G}).$$

Equip $L^2(G)$ with normalized counting measure and $\ell^2(\widehat{G})$ with counting measure. Then $f \mapsto \widehat{f}$ is a unitary isomorphism $L^2(G) \rightarrow \ell^2(\widehat{G})$; in particular $\|f\|_{L^2(G)} = \|\widehat{f}\|_{\ell^2(\widehat{G})}$.

Proof. Parseval says the normalized Fourier transform is an isometry from $L^2(G)$ to $\ell^2(\widehat{G})$ with the compatible normalization. Fourier inversion shows it is onto, hence unitary; this is Plancherel. $\square$

Theorem 5.7 (I.5.007: Convolution theorem.). *For normalized convolution $(f * g)(x) = \mathbb{E}_y f(y) g(x - y)$,*

$$\widehat{f * g}(\chi) = \widehat{f}(\chi) \widehat{g}(\chi).$$

Proof. For normalized convolution $(f * g)(x) = \mathbb{E}_y f(y) g(x - y)$, interchange finite sums: $\widehat{f * g}(\chi) = \mathbb{E}_{x,y} f(y) g(x - y) \overline{\chi(x)} = \widehat{f}(\chi) \widehat{g}(\chi)$. $\square$

Theorem 5.8 (I.5.008: Translation/modulation identities.). *Let $T_a f(x) = f(x - a)$ and $M_{\gamma} f(x) = \gamma(x) f(x)$ for $a \in G$, $\gamma \in \widehat{G}$. Then*

$$\widehat{T_a f}(\chi) = \overline{\chi(a)} \widehat{f}(\chi), \quad \widehat{M_{\gamma} f}(\chi) = \widehat{f}(\chi \overline{\gamma}).$$

Proof. For $T_a f(x) = f(x - a)$, substitution gives $\widehat{T_a f}(\chi) = \overline{\chi(a)} \widehat{f}(\chi)$. For modulation $M_{\gamma} f(x) = \gamma(x) f(x)$, $\widehat{M_{\gamma} f}(\chi) = \widehat{f}(\chi \overline{\gamma})$. $\square$

Theorem 5.9 (I.5.009: Fourier transform of subgroups and cosets.). *If $H \leq G$ and $H^\perp = \{\chi : \chi|_H = 1\}$, then*

$$\widehat{1_H}(\chi) = \frac{|H|}{|G|} 1_{H^\perp}(\chi).$$

For a coset $a + H$ the right side is multiplied by $\overline{\chi(a)}$.

Proof. If $H \leq G$, then $\widehat{1_H}(\chi) = |H|/|G|$ when $\chi|_H = 1$ and 0 otherwise by the nontrivial-character sum. A coset $a + H$ adds the phase $\overline{\chi(a)}$. $\square$

Theorem 5.10 (I.5.010: Fourier expression for additive energy.). *For finite $A, B \subseteq G$,*

$$E(A, B) = |G|^3 \sum_{\chi \in \widehat{G}} |\widehat{1_A}(\chi)|^2 |\widehat{1_B}(\chi)|^2.$$

Proof. Since $E(A, B) = |G|^3 \|1_A * 1_B\|_2^2$ under the normalized convention, Plancherel and the convolution theorem give $E(A, B) = |G|^3 \sum_{\chi} |\widehat{1_A}(\chi)|^2 |\widehat{1_B}(\chi)|^2$. $\square$

Theorem 5.11 (I.5.011: Fourier expression for 3-term progression counts.). *Assume G has odd order. For $f : G \rightarrow \mathbb{C}$,*

$$\mathbb{E}_{x,d} f(x) f(x+d) f(x+2d) = \sum_{\chi \in \widehat{G}} \widehat{f}(\chi)^2 \widehat{f}(\chi^{-2}).$$

Proof. For odd-order G , the normalized 3-AP count is $\mathbb{E}_{x,d} f(x) f(x+d) f(x+2d)$. Expand all three factors in Fourier series. Averaging in x, d forces the character constraints $\chi_1 \chi_2 \chi_3 = 1$ and $\chi_2 \chi_3^2 = 1$, hence $\chi_1 = \chi_3 = \chi$ and $\chi_2 = \chi^{-2}$. Thus the count is $\sum_{\chi} \widehat{f}(\chi)^2 \widehat{f}(\chi^{-2})$ (with the equivalent additive-frequency notation in cyclic groups). $\square$

Theorem 5.12 (I.5.012: Basic uncertainty principle for finite abelian groups.). *If $0 \neq f : G \rightarrow \mathbb{C}$, then*

$$|\text{supp } f| |\text{supp } \widehat{f}| \geq |G|.$$

Proof. If $f \not\equiv 0$, let $S = \text{supp } f$ and $T = \text{supp } \widehat{f}$. Fourier inversion gives $|f(x)| \leq \sum_{\chi \in T} |\widehat{f}(\chi)|$. Therefore $\|f\|_2^2 \leq |S| (\sum_T |\widehat{f}|)^2 / |G| \leq |S| |T| \sum_T |\widehat{f}|^2 / |G| = |S| |T| \|f\|_2^2 / |G|$ by Cauchy–Schwarz and Plancherel. Cancel to obtain $|S| |T| \geq |G|$. $\square$

Lemma 5.13 (Finite three-lines lemma). *Let F be continuous on the closed strip $0 \leq \Re z \leq 1$, analytic in its interior, and bounded there. If*

$$|F(it)| \leq M_0, \quad |F(1+it)| \leq M_1 \quad (t \in \mathbb{R}),$$

then, for $0 \leq \theta \leq 1$,

$$|F(\theta)| \leq M_0^{1-\theta} M_1^\theta.$$

Proof. The cases $M_0 M_1 = 0$ follow by applying the argument below with M_j replaced by $M_j + \varepsilon$ and then letting $\varepsilon \downarrow 0$, so assume $M_0, M_1 > 0$. Put

$$H(z) = F(z) M_0^{z-1} M_1^{-z}.$$

On the two vertical boundary lines $|H| \leq 1$. Fix $\varepsilon > 0$ and set

$$H_\varepsilon(z) = H(z) \exp(\varepsilon(z - \theta)^2).$$

Since H is bounded, on the horizontal sides of the rectangle $0 \leq \Re z \leq 1$, $|\Im z| \leq R$ the factor $\exp(\varepsilon((\Re z - \theta)^2 - R^2))$ tends uniformly to zero as $R \rightarrow \infty$. The maximum-modulus principle on this rectangle therefore gives

$$|H_\varepsilon(\theta)| \leq \max\{e^{\varepsilon\theta^2}, e^{\varepsilon(1-\theta)^2}, 1 + o_R(1)\}.$$

Let first $R \rightarrow \infty$ and then $\varepsilon \downarrow 0$. We obtain $|H(\theta)| \leq 1$, which is exactly the desired inequality. $\square$

Lemma 5.14 (Finite Riesz–Thorin interpolation). *Let X, Y be finite probability spaces and let $T : \mathbb{C}^X \rightarrow \mathbb{C}^Y$ be linear. Suppose*

$$\|Tf\|_{q_j} \leq M_j \|f\|_{p_j} \quad (j = 0, 1),$$

where $1 \leq p_j, q_j \leq \infty$. For $0 \leq \theta \leq 1$ define

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}.$$

Then

$$\|Tf\|_{q_\theta} \leq M_0^{1-\theta} M_1^\theta \|f\|_{p_\theta}.$$

Proof. It is enough, by finite-dimensional duality and approximation, to treat finite exponents and vectors f, g having no zero coordinates, normalized by $\|f\|_{p_\theta} = \|g\|_{q'_\theta} = 1$. Write $f(x) = |f(x)|e^{i\alpha_x}$ and $g(y) = |g(y)|e^{i\beta_y}$, and for $0 \leq \Re z \leq 1$ put

$$\begin{aligned} \frac{1}{p(z)} &= \frac{1-z}{p_0} + \frac{z}{p_1}, & \frac{1}{q'(z)} &= \frac{1-z}{q'_0} + \frac{z}{q'_1}, \\ f_z(x) &= |f(x)|^{p_\theta/p(z)} e^{i\alpha_x}, & g_z(y) &= |g(y)|^{q'_\theta/q'(z)} e^{i\beta_y}. \end{aligned}$$

The finite sum

$$\Phi(z) = \langle Tf_z, g_z \rangle_Y$$

is analytic in the open strip and bounded on its closure. On $\Re z = j$ the modulus of each complex power depends only on the real part of its exponent, so

$$\|f_{j+it}\|_{p_j} = 1, \quad \|g_{j+it}\|_{q'_j} = 1.$$

Hölder and the endpoint hypotheses therefore give $|\Phi(j+it)| \leq M_j$. By [theorem 5.13](#),

$$|\langle Tf, g \rangle| = |\Phi(\theta)| \leq M_0^{1-\theta} M_1^\theta.$$

Taking the supremum over all dual unit vectors g gives the asserted operator norm. If an exponent is infinite or a coordinate vanishes, approximate the exponent by finite ones and the vector by vectors with nonzero coordinates; all norms and the finite sum vary continuously, so the same inequality follows on passage to the limit. $\square$

Theorem 5.15 (I.5.013: Hausdorff–Young inequality in the finite setting). *For $1 \leq p \leq 2$ and conjugate exponent $p' = p/(p-1)$,*

$$\|\widehat{f}\|_{p'} \leq \|f\|_p$$

with normalized counting measure on G .

Proof. The normalized Fourier transform satisfies $\|\widehat{f}\|_\infty \leq \|f\|_1$, while Plancherel gives $\|\widehat{f}\|_2 = \|f\|_2$. Apply [theorem 5.14](#) between the operator bounds $L^1 \rightarrow L^\infty$ and $L^2 \rightarrow L^2$. If $1 \leq p \leq 2$ and $1/p' = 1 - 1/p$, the interpolated exponents are precisely (p, p') , and hence

$$\|\widehat{f}\|_{p'} \leq \|f\|_p.$$

□

Theorem 5.16 (I.5.014: Young convolution inequality in the finite setting.). *If $1 \leq p, q, r \leq \infty$ and*

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q},$$

then, with normalized counting measure,

$$\|f * g\|_r \leq \|f\|_p \|g\|_q.$$

Proof. First suppose $1 < p, q, r < \infty$. The exponent relation implies $p, q \leq r$. For each x , write

$$|f(y)g(x-y)| = (|f(y)|^{p/r} |g(x-y)|^{q/r}) |f(y)|^{1-p/r} |g(x-y)|^{1-q/r}.$$

Apply Hölder in y with reciprocal exponents

$$\frac{1}{r}, \quad \frac{r-p}{pr}, \quad \frac{r-q}{qr};$$

they sum to 1 because $1/p + 1/q - 1/r = 1$. Thus

$$|f * g(x)| \leq \left(\mathbb{E}_y |f(y)|^p |g(x-y)|^q \right)^{1/r} \|f\|_p^{1-p/r} \|g\|_q^{1-q/r}.$$

Raise to the r th power and average over x . Translation invariance gives

$$\mathbb{E}_{x,y} |f(y)|^p |g(x-y)|^q = \|f\|_p^p \|g\|_q^q,$$

so $\|f * g\|_r^r \leq \|f\|_p^r \|g\|_q^r$. If $p = 1$ or $q = 1$, the same conclusion is Minkowski's inequality. If $r = \infty$, then $1/p + 1/q = 1$ and the result is Hölder applied directly to the convolution sum. These cover all endpoints. □

Theorem 5.17 (I.5.015: Large-spectrum Parseval bound.). *If $\|f\|_2^2 \leq M$ and $\text{Spec}_\eta(f) = \{\chi : |\widehat{f}(\chi)| \geq \eta\}$, then*

$$|\text{Spec}_\eta(f)| \leq M/\eta^2.$$

Proof. If $\|f\|_2^2 \leq M$ and $\text{Spec}_\eta(f) = \{\chi : |\widehat{f}(\chi)| \geq \eta\}$, Parseval gives $M \geq \sum_{\text{Spec}} |\widehat{f}(\chi)|^2 \geq \eta^2 |\text{Spec}_\eta(f)|$. Hence the spectrum has size at most M/η^2 . □

Definition 5.18 (Fourier transform and convolution of probability masses). For an ordinary function $f : G \rightarrow \mathbb{C}$ we use the normalized transform $\widehat{f}(\chi) = |G|^{-1} \sum_{x \in G} f(x) \overline{\chi(x)}$. A probability mass μ instead satisfies $\sum_x \mu(x) = 1$, so for probability masses we use

$$\widehat{\mu}(\chi) := \sum_{x \in G} \mu(x) \overline{\chi(x)}, \quad (\mu *_{\text{m}} \nu)(x) := \sum_{y \in G} \mu(y) \nu(x-y).$$

Thus $\widehat{\mu}(1) = 1$. Equivalently, the density $m(x) = |G|\mu(x)$ with respect to normalized counting measure satisfies $\widehat{m}(\chi) = \widehat{\mu}(\chi)$, and normalized function convolution of the densities corresponds exactly to measure convolution of the probability masses.

Theorem 5.19 (I.5.016: Random-walk convolution interpretation and spectral diagonalization on finite abelian groups.). *Let μ be a probability mass on a finite abelian group G . The law of the n -step random walk with increment law μ is $\mu^{*\mathfrak{m}n}$. The convolution operator*

$$T_\mu f(x) = \sum_{y \in G} \mu(y) f(x - y)$$

has every character χ as an eigenfunction with eigenvalue $\widehat{\mu}(\chi)$, and

$$\widehat{\mu^{*\mathfrak{m}n}}(\chi) = \widehat{\mu}(\chi)^n.$$

Proof. Independence of the increments gives measure convolution: if S_n and Y_{n+1} have laws ν and μ , then

$$\Pr(S_n + Y_{n+1} = x) = \sum_y \mu(y) \nu(x - y) = (\mu *_{\mathfrak{m}} \nu)(x).$$

For a character,

$$T_\mu \chi(x) = \sum_y \mu(y) \chi(x - y) = \chi(x) \sum_y \mu(y) \overline{\chi(y)} = \widehat{\mu}(\chi) \chi(x).$$

Finally, interchanging two finite sums shows $\widehat{\mu *_{\mathfrak{m}} \nu} = \widehat{\mu} \widehat{\nu}$, and induction gives the stated n -step formula. The density reformulation in the preceding definition shows that this is perfectly consistent with the normalized function Fourier transform used elsewhere in the volume. $\square$

Chapter 6

Foundational probabilistic and Fourier examples

The three examples here are meant to show the preceding toolkit operating as a system. Random walks illustrate spectral diagonalization; a cyclic walk makes the Fourier calculation completely explicit; and a random subset shows how concentration and the union bound create Fourier-uniform objects.

These examples are prototypes, not side remarks. Later pseudorandom majorants, dense models, and probabilistic constructions use the same pattern: identify many observables, control each one, and then pay for the family by a union bound or an entropy estimate.

Definition 6.1 (Irreducible and aperiodic finite random walks). Let $X_{n+1} = X_n + Y_{n+1}$ be the random walk on a finite additive group G with independent increments of law μ . The walk is *irreducible* if every state can be reached from every other state with positive probability in some number of steps; equivalently, the subgroup generated by $\text{supp } \mu$ is all of G . Its *period* is the greatest common divisor of the positive return times n for which $\mu^{*n}(0) > 0$; it is *aperiodic* when that gcd is 1. On a finite abelian group, irreducibility together with aperiodicity is equivalent to

$$|\hat{\mu}(\chi)| < 1 \quad (\chi \neq 1).$$

The proof of this spectral equivalence is included as the first step of Theorem 6.2.

Theorem 6.2 (I.6.001: Mixing of random walk on a finite abelian group under the nondegeneracy hypothesis.). *Let μ be a probability measure on a finite abelian group G . Suppose the walk is irreducible and aperiodic; equivalently in this finite abelian setting,*

$$|\hat{\mu}(\chi)| < 1 \quad (\chi \neq 1).$$

*Then μ^{*n} converges exponentially to the uniform probability mass u_G in $\ell^2(G)$ and hence in total variation.*

Proof. We first justify the spectral equivalence rather than hiding it in the hypothesis. If $|\hat{\mu}(\chi)| = 1$, equality holds in the triangle inequality for the convex combination

$$\hat{\mu}(\chi) = \sum_x \mu(x) \overline{\chi(x)}.$$

Hence χ is constant, say c , on $\text{supp } \mu$. Every n -step return to 0 then gives $1 = \chi(0) = c^n$. Aperiodicity says the gcd of the return times is 1, so $c = 1$. Irreducibility says the support generates G , and therefore $\chi = 1$ on all of G . Thus every nontrivial character satisfies $|\hat{\mu}(\chi)| < 1$. Conversely, if irreducibility fails, a nontrivial character trivial on the subgroup

generated by the support has Fourier modulus one. It remains to treat periodicity without appealing to Markov-chain terminology. Assume the support S generates G and let $d > 1$ be the period. For $g \in G$, choose a word $s_1 + \cdots + s_m = g$ with $s_i \in S$ and define

$$\phi(g) = m \pmod{d}.$$

This is well defined: if another word of length m' also reaches g , choose a positive S -word of length r taking g back to 0; then both $m + r$ and $m' + r$ are return times, so $d \mid (m - m')$. Concatenating words shows that $\phi : G \rightarrow \mathbb{Z}/d\mathbb{Z}$ is a homomorphism, and every $s \in S$ satisfies $\phi(s) = 1$. Hence the nontrivial character $\chi(g) = e^{2\pi i \phi(g)/d}$ is constant on S , so $|\hat{\mu}(\chi)| = 1$. This proves the converse and hence the spectral equivalence.

Put $\rho = \max_{\chi \neq 1} |\hat{\mu}(\chi)| < 1$. For probability masses the counting-measure Parseval formula is

$$\sum_x |\nu(x)|^2 = \frac{1}{|G|} \sum_{\chi} |\hat{\nu}(\chi)|^2.$$

The uniform mass has Fourier transform 1 at the trivial character and 0 elsewhere, so

$$\|\mu^{*m} - u_G\|_{\ell^2}^2 = \frac{1}{|G|} \sum_{\chi \neq 1} |\hat{\mu}(\chi)|^{2m} \leq \rho^{2m} \|\delta_0 - u_G\|_{\ell^2}^2.$$

Finally, $\|\nu\|_{\text{TV}} \leq \frac{1}{2}|G|^{1/2}\|\nu\|_{\ell^2}$ by Cauchy–Schwarz, giving exponential total-variation convergence. $\square$

Theorem 6.3 (I.6.002: Fourier proof of equidistribution for aperiodic cyclic random walk.). *Let μ be an aperiodic increment law whose support generates $\mathbb{Z}/N\mathbb{Z}$. Then*

$$\mu^{*m}(x) = \frac{1}{N} \sum_{k \in \mathbb{Z}/N\mathbb{Z}} \hat{\mu}(k)^m e^{2\pi i k x / N}$$

and $\mu^{*m}(x) \rightarrow 1/N$ uniformly in x .

Proof. On $\mathbb{Z}/N\mathbb{Z}$, characters are $e_k(x) = e^{2\pi i k x / N}$. If the increment law is aperiodic, the previous argument gives $|\hat{\mu}(k)| < 1$ for $k \neq 0$. Fourier inversion for probability masses yields $\mu^{*m}(x) = N^{-1} \sum_k \hat{\mu}(k)^m e^{2\pi i k x / N}$, and every nonzero term tends uniformly to zero. Hence the walk equidistributes. $\square$

Theorem 6.4 (I.6.003: Basic Salem-Zygmund/random-Fourier-coefficient estimate in a finite model, at the level later used for examples.). *Let G be finite and form a random set $A \subseteq G$ by retaining each element independently with probability α . Put $f = 1_A - \alpha$. There is an absolute $c > 0$ such that for every $t > 0$ and every nontrivial character χ ,*

$$\Pr(|\hat{f}(\chi)| > t) \leq 4e^{-c|G|t^2}.$$

Consequently, with probability at least $1 - |G|^{-2}$,

$$\max_{\chi \neq 1} |\hat{f}(\chi)| \ll \sqrt{\frac{\log |G|}{|G|}}.$$

Proof. Let $A \subset G$ be formed by independent Bernoulli(α) choices and put $f = 1_A - \alpha$. For nontrivial χ , $\hat{f}(\chi) = |G|^{-1} \sum_x (X_x - \alpha)\chi(x)$. Apply Hoeffding separately to real and imaginary parts: $\Pr(|\hat{f}(\chi)| > t) \leq 4e^{-c|G|t^2}$. Union bound over at most $|G|$ characters shows that with probability at least $1 - |G|^{-2}$, $\max_{\chi \neq 1} |\hat{f}(\chi)| \ll \sqrt{\log |G| / |G|}$. This is the finite Salem–Zygmund estimate used for probabilistic examples. $\square$

Chapter 7

Worked laboratories for the foundational toolkit

The preceding chapters deliberately separated basic tools into small self-contained lemmas and propositions. We now put several of them together in examples that resemble the arguments used later.

Worked Example 7.1 (Dependent random choice in a dense bipartite graph). Let $G = (L, R, E)$ with $|L| = |R| = 1000$ and edge density $1/2$. Choose two vertices of R independently and let X be their common neighborhood. The DRC calculation gives

$$\mathbb{E}|X| = \sum_{x \in L} (d(x)/1000)^2 \geq 1000(1/2)^2 = 250.$$

Thus some pair on the right has at least 250 common neighbours. More generally, for fixed integers $r, q \geq 1$, if we subtract the expected number of bad r -sets in X —that is, r -element subsets $S \subseteq X$ with $|N_R(S)| < q$ —the same random experiment produces a large set on the left in which every r -element subset has at least q common neighbours. This is why DRC is a structural rather than merely existential pigeonhole argument.

Worked Example 7.2 (Energy of an interval). For $A = \{0, 1, \dots, n-1\} \subset \mathbb{Z}$, the representation function $r_{A-A}(t) = n - |t|$ for $|t| < n$. Hence

$$E(A) = \sum_t r_{A-A}(t)^2 = n^2 + 2 \sum_{j=1}^{n-1} j^2 = \frac{2}{3}n^3 + O(n^2).$$

The Cauchy–Schwarz lower bound $E(A) \geq |A|^4/|A-A|$ predicts only a constant multiple of n^3 , but it already detects the correct scale. Later inverse theorems ask when such large energy forces a set to resemble this structured model.

Worked Example 7.3 (Characters and CRT on $\mathbb{Z}/15\mathbb{Z}$). The Chinese remainder theorem identifies $\mathbb{Z}/15\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$. A character therefore factors as

$$x \longmapsto e^{2\pi i ax/3} e^{2\pi i bx/5}.$$

The Fourier transform on the product is the tensor product of the two transforms. This toy calculation is the finite-group prototype for separating local congruence conditions from global additive structure.

Worked Example 7.4 (A lazy random walk on $\mathbb{Z}/5\mathbb{Z}$, with an explicit mixing rate). Let the increment be 0 with probability $1/2$ and ± 1 with probability $1/4$ each. This walk is irreducible and aperiodic. For $\chi_r(x) = e^{2\pi i r x/5}$,

$$\widehat{\mu}(r) = \frac{1}{2} + \frac{1}{4}e^{-2\pi i r/5} + \frac{1}{4}e^{2\pi i r/5} = \frac{1}{2} + \frac{1}{2}\cos \frac{2\pi r}{5}.$$

The largest nontrivial modulus occurs at $r = 1, 4$ and equals

$$\rho = \frac{1}{2} + \frac{1}{2}\cos \frac{2\pi}{5} = \frac{3 + \sqrt{5}}{8} \approx 0.6545.$$

Thus, starting from 0,

$$\|\mu^{*n} - u_G\|_{\ell^2} \leq \rho^n \|\delta_0 - u_G\|_{\ell^2} = \rho^n \sqrt{\frac{4}{5}}.$$

Consequently

$$\|\mu^{*n} - u_G\|_{\text{TV}} \leq \frac{\sqrt{5}}{2} \sqrt{\frac{4}{5}} \rho^n = \rho^n.$$

For instance $\rho^{10} < 0.015$, so after ten steps the total-variation distance is already below 1.5%. The entire calculation is the spectral method in its smallest nontrivial form: characters diagonalize the walk, the trivial character survives, and the largest nontrivial Fourier eigenvalue controls mixing.

Additive Combinatorics
Volume II
Sumsets, Additive Energy, and Freiman Theory
Version 1.10

14 September 2026

Contents

Chapter 1

The Inverse Problem Becomes Geometric

1.1 What changes after Volume I?

Volume I did not merely record that a small sumset is “structured” in some vague sense. Volume I, Chapter 2, Section 2.1 proves the exact elementary package that will now be used repeatedly. If A is a finite nonempty subset of an abelian group and

$$|A + A| \leq K|A|, \tag{1.1}$$

then Volume I, Theorem I.2.003, the Ruzsa triangle and covering lemmas (Theorems I.2.008 and I.2.009), and the Plünnecke–Petridis theorem (Theorem I.2.010) give the concrete chain

$$\begin{array}{c} |A + A| \leq K|A| \\ \Downarrow \\ E(A) \geq K^{-1}|A|^3 = \#\{(a_1, a_2, a_3, a_4) \in A^4 : a_1 + a_2 = a_3 + a_4\}, \\ |A - A| \leq K^2|A|, \\ |mA - nA| \leq K^{m+n}|A| \quad (m, n \geq 0, m + n \geq 1), \\ A \subseteq X + (A - A) \quad \text{for some } X \subseteq A, |X| \leq K. \end{array} \tag{1.2}$$

The first line says that there are many additive relations. The next two lines are the core of Ruzsa calculus and Plünnecke theory: one small sumset controls all fixed iterated sum–difference sets. The final line says that A is covered by at most K translates of its difference set. Proposition I.2.012 collects these implications in one place in Volume I.

These conclusions are indispensable, but they are not yet a classification. They tell us that A has persistent additive growth control; they do not tell us what A *looks like*. The aim of inverse theory is to cross that gap.

Guiding Question 1.1 (The inverse problem). Let A be a finite nonempty subset of an abelian group G and suppose

$$|A + A| \leq K|A|.$$

Can one describe A , up to losses depending only on K , by a bounded-complexity algebraic or geometric object?

There are at least four inequivalent ways to make the word “describe” precise.

1. **Containment.** Find a structured set M with

$$A \subseteq M, \quad |M| \leq f(K)|A|.$$

2. **Covering.** Find a structured set M and a small set X such that

$$A \subseteq X + M, \quad |X| \leq f(K).$$

3. **Extraction.** Find $A' \subseteq A$ with

$$|A'| \geq f(K)^{-1}|A|$$

and A' structured.

4. **Control.** Find a structured set M for which both A and M can be covered by boundedly many translates of the other.

These formulations often imply one another after additional Ruzsa-covering arguments, but the quantitative losses matter. Modern polynomial Freiman–Ruzsa theory is largely about replacing exponential losses by polynomial ones.

Idea The guiding distinction

A direct theorem starts from an explicit progression, subgroup, or subgroup-plus-progression model and proves small doubling. An inverse theorem starts only from the number $|A + A|/|A|$ and reconstructs such geometry. The second direction is where almost all of the difficulty lies.

1.2 The model objects

The simplest torsion-free model is an arithmetic progression

$$P = a + \{0, 1, \dots, L - 1\}v.$$

Its doubling is at most 2. A rank- r generalized arithmetic progression is

$$P = a + \{n_1 v_1 + \dots + n_r v_r : 0 \leq n_i < L_i\}. \quad (1.1)$$

If the parameterization is injective, then

$$|P| = L_1 \cdots L_r, \quad |P + P| \leq \prod_{i=1}^r (2L_i - 1) < 2^r |P|.$$

Thus fixed rank implies bounded doubling.

In a group with torsion, subgroups give another source of zero growth:

$$H + H = H.$$

The correct hybrid object therefore combines a finite subgroup with progression geometry in the quotient. We now fix the terminology before using it in theorem statements.

Definition 1.2 (Progression terminology used from this point onward). A *generalized arithmetic progression* (GAP) of rank at most r is a set

$$a + \{n_1 v_1 + \dots + n_r v_r : 0 \leq n_i < L_i\}.$$

A displayed GAP is *proper* if two parameter vectors in the displayed box never represent the same group element. If $H \leq G$ is finite, a *coset progression* is a set $H + P$ whose image in G/H is a GAP; its rank is the rank of that quotient progression, and it is proper when the quotient parameterization is injective. Chapter ?? develops these notions carefully, including t -properness, symmetric and convex variants.

The central classical theorem can be stated schematically as follows.

Theorem 1.3 (Freiman–Green–Ruzsa, qualitative form). *For every $K \geq 1$ there exist integers $d = d(K)$ and $F = F(K)$ such that the following holds. If A is a finite nonempty subset of an abelian group and*

$$|A + A| \leq K|A|,$$

then A is contained in a coset progression M of dimension at most d and size at most

$$|M| \leq F|A|.$$

For $A \subset \mathbb{Z}$, the subgroup part is trivial and M may be taken to be a generalized arithmetic progression.

Remark 1.4 (Proof location). Chapter 9 proves the classical additive-combinatorial assembly. The geometry-of-numbers step that extracts a large progression from a simultaneous-character neighborhood is proved in Appendix ??, while bounded-rank properization is proved in Chapter ?. Together these results give a complete proof of Theorem ??.

We shall prove a quantitative classical version later in this volume. The point of stating it now is to identify the destination.

1.3 Why progressions rather than merely subgroups?

The set

$$A_N = \{1, 2, \dots, N\} \subset \mathbb{Z}$$

satisfies

$$|A_N + A_N| = 2N - 1,$$

so its doubling tends to 2. But the only finite subgroup of $\mathbb{Z}$ is $\{0\}$. Therefore no subgroup theorem can classify small doubling in torsion-free groups.

More generally, if

$$P = \{n_1 v_1 + \dots + n_r v_r : 0 \leq n_i < L_i\}$$

is proper and all L_i are large, then its doubling is asymptotic to 2^r . Thus the doubling constant remembers, at least approximately, a geometric dimension.

This observation suggests a rough dictionary:

doubling close to 1	subgroup-like
doubling close to 2	one-dimensional progression-like
doubling of order 2^r	possibly rank- r progression-like.

The dictionary is only heuristic: a set may occupy a sparse portion of a progression, and torsion can alter the picture drastically. Nevertheless, the idea that bounded doubling corresponds to bounded-dimensional additive geometry is correct.

Worked Example 1.5 (A sparse but still controlled set). Let

$$P = [0, L_1 - 1]v_1 + [0, L_2 - 1]v_2 \subset \mathbb{Z}$$

be a proper rank-two progression, and let $A \subseteq P$ have density $\delta > 0$ in P . Then

$$|A + A| \leq |P + P| < 4|P| = 4\delta^{-1}|A|.$$

So a set can have bounded doubling without resembling the whole progression pointwise. What survives is containment in a progression whose size is bounded by a constant multiple of $|A|$.

This is why the conclusion of Freiman’s theorem is naturally a containment theorem rather than a claim that A itself must be a progression.

1.4 Freiman homomorphisms preserve the relevant geometry

Inverse theory repeatedly moves a set from one ambient group to another. The map need not be a genuine group homomorphism on the whole ambient group. It only needs to preserve the additive relations visible inside the set.

Definition 1.6 (Freiman homomorphism). Let $A \subseteq G$, $A' \subseteq G'$ and $s \geq 2$. A map $\phi : A \rightarrow A'$ is a *Freiman s -homomorphism* if

$$a_1 + \cdots + a_s = b_1 + \cdots + b_s$$

with all $a_i, b_i \in A$ implies

$$\phi(a_1) + \cdots + \phi(a_s) = \phi(b_1) + \cdots + \phi(b_s).$$

It is a *Freiman s -isomorphism* if it is bijective and its inverse is also a Freiman s -homomorphism.

We introduce this notion here because it is the correct invariant for the modelling steps that follow. From this point onward it is structural rather than merely notational.

Proposition 1.7 (What a Freiman 2-isomorphism preserves). *If $\phi : A \rightarrow A'$ is a Freiman 2-isomorphism, then*

$$|A| = |A'|, \quad |A + A| = |A' + A'|, \quad E(A) = E(A').$$

More generally, the entire incidence pattern of additive quadruples in A is carried bijectively to that in A' .

Proof. The equality of cardinalities is immediate from bijectivity. Define

$$\Phi : A + A \longrightarrow A' + A'$$

by choosing a representation $x = a + b$ and setting

$$\Phi(x) = \phi(a) + \phi(b).$$

If $x = a + b = c + d$, the Freiman property gives

$$\phi(a) + \phi(b) = \phi(c) + \phi(d),$$

so Φ is well-defined. Applying the same construction to ϕ^{-1} shows that Φ is bijective. Hence the sumsets have equal cardinality.

For energy, the map

$$(a, b, c, d) \mapsto (\phi(a), \phi(b), \phi(c), \phi(d))$$

is a bijection between the solution sets of

$$a + b = c + d$$

and

$$a' + b' = c' + d'.$$

Therefore $E(A) = E(A')$. □

Idea Why modelling is legitimate

If a set can be Freiman-isomorphically moved into a finite cyclic group where it has positive density, then Fourier analysis becomes available without changing the additive relations we are trying to classify. Modelling is therefore not a cosmetic change of coordinates; it is a way of manufacturing an ambient space in which analytic tools are effective.

1.5 The classical four-step mechanism

A particularly influential route to Freiman's theorem can be summarized in four steps.

Proof architecture The classical inverse machine

Suppose A has doubling at most K .

Step 1: model. Replace A , or a large subset of it, by a Freiman-isomorphic set A' living densely inside a finite abelian group G' of size $K^{O_K(1)}|A|$.

Step 2: thicken. Use Fourier analysis to show that

$$2A' - 2A'$$

contains a Bohr neighborhood: a set on which a controlled family of characters is simultaneously close to 1.

Step 3: geometrize. Show that a sufficiently large Bohr set contains a proper low-dimensional generalized arithmetic progression or coset progression.

Step 4: pull back and cover. Transfer the progression through the Freiman isomorphism and use covering arguments to place all of A inside a bounded enlargement of that progression.

Each step solves a different obstruction. The model handles sparsity; Fourier analysis detects coherent frequencies; geometry of numbers converts frequency constraints into an actual progression; and covering upgrades local structure in $2A - 2A$ to global control of A .

1.6 A first worked model: a two-dimensional box

Consider

$$A = \{x + My : 0 \leq x < L, 0 \leq y < L\} \subset \mathbb{Z},$$

where $M > 2L$. The large separation ensures unique representation: if

$$x + My = x' + My',$$

then

$$M(y - y') = x' - x,$$

and the right side has absolute value less than M , forcing $y = y'$ and then $x = x'$.

Hence

$$|A| = L^2.$$

Likewise every element of $A + A$ has a unique parameter pair

$$u + Mv, \quad 0 \leq u, v \leq 2L - 2,$$

so

$$|A + A| = (2L - 1)^2.$$

Therefore

$$\frac{|A + A|}{|A|} = \left(2 - \frac{1}{L}\right)^2 \rightarrow 4.$$

Now reduce modulo a prime $p > 4ML$. No wraparound occurs on $2A - 2A$, so reduction modulo p is a Freiman 2-isomorphism on A . The set can therefore be studied inside $\mathbb{Z}/p\mathbb{Z}$ without changing any additive quadruple.

This toy calculation already contains the philosophy of modelling: choose the modulus large enough that the relations of interest do not wrap around, but small enough that the image has useful density.

1.7 The quantitative question

The qualitative theorem ?? hides two functions:

$$d(K), \quad F(K).$$

Their growth is not a technical afterthought. It measures how efficiently additive structure can be recovered from small doubling.

At one extreme, a theorem of the form

$$d(K) \leq \exp(\exp(K^{100})), \quad F(K) \leq \exp(\exp(\exp(K^{100})))$$

would still be a valid qualitative classification. At the other extreme, polynomial Freiman–Ruzsa philosophy asks for losses of polynomial size in K , with dimension essentially logarithmic in K when the model objects double like 2^d .

Two types of quantitative statement should be kept distinct.

- In groups of bounded torsion, the natural structured object is essentially a subgroup, and polynomial covering bounds are now theorems.
- In torsion-free groups such as $\mathbb{Z}$, the strongest full convex-progression statements remain subtler; the current best general bounds are very close to polynomial on a logarithmic scale but do not yet give the strongest conjectured polynomial form.

The last chapters of this volume explain this distinction carefully.

Research window Polynomial Freiman–Ruzsa and Marton’s conjecture

In $\mathbb{F}_2^n$, Marton’s polynomial Freiman–Ruzsa conjecture asks whether

$$|A + A| \leq K|A|$$

forces A to be coverable by $K^{O(1)}$ cosets of a subspace V with $|V| \leq |A|$. Gowers, Green, Manners and Tao proved this in 2023, using entropy and an induction on an entropic version of Ruzsa distance. Their argument was subsequently extended to all abelian groups of bounded torsion. The proof is conceptually quite different from the classical Bohr-set route, and we devote several chapters to building it from the ground up.

Proof deferred to a later volume Volume III: entropic PFR

The full Gowers–Green–Manners–Tao proof, including entropic Balog–Szemerédi–Gowers and the distance-decrement theorem, is given in Volume III. Chapters 11–13 of the present volume build and test the entropy language but mark the decrement inputs explicitly.

1.8 What will count as a proof in this volume?

There are three levels of results below.

1. **Internal theorems.** The proof is given completely, including auxiliary lemmas that are genuinely part of additive combinatorics.
2. **Companion prerequisites.** A standard theorem from probability, convex geometry, or information theory is stated in the precise form needed and either proved in an appendix or cross-referenced to an explicitly named prerequisite source. The present application is nevertheless proved in full.
3. **Later-volume theorems.** A substantial theorem not proved in the present volume is displayed with an explicit later-volume proof assignment. The surrounding chapter may explain its architecture, but the statement is never counted as proved until the designated later volume supplies the proof.

This distinction is especially important in the chapters on Sanders’ theorem and on the entropic PFR argument.

1.9 The volume in one diagram

The classical half follows

small doubling $\rightarrow$ finite model $\rightarrow$ large spectrum / Bohr set
 $\rightarrow$ progression $\rightarrow$ Freiman structure.

The modern half follows a different route:

small doubling $\rightarrow$ almost-periodicity or entropy
 $\rightarrow$ structured invariance $\rightarrow$ subgroup/coset control.

The point of studying both is not historical completeness. The two routes teach different ways of extracting structure. The classical method turns a Fourier spectrum into geometry. The modern method turns stability under convolution or entropy inequalities into approximate invariance. Both recur far beyond the abelian theory.

Chapter 2

Generalized Progressions and Approximate Additive Geometry

2.1 Boxes mapped into groups

A generalized arithmetic progression is best regarded as the image of an integer box under a homomorphism. This viewpoint separates the combinatorial geometry of the parameter set from the algebra of the ambient group.

Let

$$B = \prod_{i=1}^r [0, L_i - 1]_{\mathbb{Z}} \subset \mathbb{Z}^r,$$

where

$$[0, L - 1]_{\mathbb{Z}} = \{0, 1, \dots, L - 1\}.$$

Given $v_1, \dots, v_r \in G$, define

$$\phi : \mathbb{Z}^r \rightarrow G, \quad \phi(n_1, \dots, n_r) = n_1 v_1 + \dots + n_r v_r.$$

Then

$$P = a + \phi(B)$$

is a generalized arithmetic progression of rank at most r .

Definition 2.1 (Proper and t -proper progressions). The progression $P = a + \phi(B)$ is *proper* if ϕ is injective on B . For an integer $t \geq 1$, write

$$tB := \underbrace{B + \dots + B}_{t \text{ copies}} = \prod_{i=1}^r [0, t(L_i - 1)]_{\mathbb{Z}}.$$

We call this presentation *t -proper* if ϕ is injective on tB .

Thus t -properness says exactly that parameter vectors for sums of t elements do not wrap around through $\ker \phi$. Some authors instead use centered dilates, especially for symmetric progressions; whenever that stronger convention is needed, we shall state the relevant parameter region explicitly.

Warning 2.2 (Properness is not optional). The formal product $L_1 \cdots L_r$ counts parameters, not necessarily elements of P . If the directions satisfy short integer relations, many parameter points may map to the same group element. Any argument that identifies $|P|$ with a box volume must first verify properness.

Worked Example 2.3 (A nonproper rank-two description of an interval). Take $v_1 = 1$, $v_2 = 2$ in $\mathbb{Z}$ and

$$P = \{n_1 + 2n_2 : 0 \leq n_1, n_2 < L\}.$$

The parameter box has L^2 points, but

$$P = \{0, 1, \dots, 3L - 3\},$$

so $|P| = 3L - 2$. The apparent rank two is spurious: the relation

$$2v_1 - v_2 = 0$$

collapses one direction. A proper description has rank one.

2.2 The direct doubling calculation

Proposition 2.4 (Proper progressions have small doubling). *Let*

$$P = a + \left\{ \sum_{i=1}^r n_i v_i : 0 \leq n_i < L_i \right\}$$

be proper. Then

$$|P| = \prod_{i=1}^r L_i$$

and

$$|P + P| \leq \prod_{i=1}^r (2L_i - 1) < 2^r |P|.$$

If the progression is 2-proper, then equality holds in the first inequality:

$$|P + P| = \prod_{i=1}^r (2L_i - 1).$$

Proof. Properness gives the cardinality formula for P . Every sum of two elements of P has the form

$$2a + \sum_{i=1}^r m_i v_i, \quad 0 \leq m_i \leq 2L_i - 2,$$

so the number of possible parameter vectors is $\prod_i (2L_i - 1)$. This proves the upper bound. If the map is injective on the doubled parameter box, all these parameter vectors give distinct sums. Finally,

$$2L_i - 1 < 2L_i$$

for every i . □

The proposition explains the rough dimensional heuristic

$$K \approx 2^r.$$

One should not read this as a theorem saying $r \leq \log_2 K$. A subset may be sparse in its containing progression, and the progression itself may have very uneven side lengths. Nevertheless, logarithmic rank is the natural target in the strongest inverse conjectures.

2.3 Symmetric progressions

For structural arguments it is often better to center the progression at the origin.

Definition 2.5 (Symmetric GAP). A symmetric generalized arithmetic progression is a set

$$P = P(v; L) = \left\{ \sum_{i=1}^r n_i v_i : |n_i| \leq L_i \right\},$$

where L_i are nonnegative integers.

If P is proper then

$$|P| = \prod_{i=1}^r (2L_i + 1).$$

Moreover

$$|P + P| \leq \prod_{i=1}^r (4L_i + 1) < 2^r |P|.$$

The advantage of symmetry is algebraic stability:

$$0 \in P, \quad -P = P, \quad mP \subseteq P(v; mL).$$

Thus a symmetric progression already looks like an approximate subgroup of an abelian group.

Worked Example 2.6 (An anisotropic progression). Let

$$P = \{n_1 + 10^6 n_2 : |n_1| \leq 10^5, |n_2| \leq 10\} \subset \mathbb{Z}.$$

The representation is proper because

$$|n_1 - n'_1| < 10^6$$

while any nonzero change in n_2 changes the value by at least 10^6 . Hence

$$|P| = (2 \cdot 10^5 + 1)(21).$$

The set has rank two even though one direction is vastly longer than the other. Its doubling constant is still less than 4. This is a useful reminder that rank controls the number of independent additive directions, not the Euclidean aspect ratio of the parameter box.

2.4 Coset progressions

Torsion forces us to add subgroup directions which do not grow at all.

Definition 2.7 (Coset progression). Let $H \leq G$ be a finite subgroup and let $\pi : G \rightarrow G/H$ be the quotient map. A *coset progression* of rank r is a set

$$M = H + P,$$

where P is the lift to G of a generalized arithmetic progression in G/H . We call M proper if the progression in the quotient is proper.

This quotient formulation is cleaner than requiring a literal direct sum inside G . If

$$\pi(P) = \left\{ \sum_{i=1}^r n_i \pi(v_i) : 0 \leq n_i < L_i \right\}$$

is proper, then every coset of H represented by the parameter box is distinct and

$$|M| = |H| \prod_{i=1}^r L_i.$$

Proposition 2.8 (Doubling of a proper coset progression). *If $M = H + P$ is a proper rank- r coset progression, then*

$$|M + M| < 2^r |M|.$$

Proof. Since $H + H = H$,

$$M + M = H + (P + P).$$

Distinct cosets of H correspond exactly to distinct elements of the quotient progression. Apply ?? in G/H and multiply cardinalities by $|H|$. $\square$

Worked Example 2.9 (A progression times a subgroup). Let

$$G = \mathbb{Z} \times \mathbb{F}_2^m, \quad H = \{0\} \times \mathbb{F}_2^m,$$

and

$$M = \{0, 1, \dots, L-1\} \times \mathbb{F}_2^m.$$

Then

$$|M| = L2^m, \quad |M + M| = (2L-1)2^m.$$

The subgroup direction contributes no growth; all growth comes from the one torsion-free progression direction.

2.5 Convex progressions

Boxes are convenient but not canonical. Fourier arguments often produce regions defined by several simultaneous linear inequalities, and these are naturally convex rather than rectangular.

Definition 2.10 (Convex progression). Let $Q \subset \mathbb{R}^r$ be a compact convex set symmetric about the origin, let $\Gamma \leq \mathbb{R}^r$ be a full-rank lattice, and let

$$\phi : \Gamma \rightarrow G$$

be a homomorphism. The image

$$P = \phi(Q \cap \Gamma)$$

is a *symmetric convex progression*. It is proper if ϕ is injective on $Q \cap \Gamma$.

A box progression is the special case

$$Q = [-L_1, L_1] \times \dots \times [-L_r, L_r], \quad \Gamma = \mathbb{Z}^r.$$

Convex progressions are useful because dilations behave exactly at the geometric level:

$$m(Q \cap \Gamma) \subseteq mQ \cap \Gamma.$$

The remaining issue is to compare lattice-point counts in Q and mQ .

Context Geometry-of-numbers capsule

The few facts about lattices needed for this comparison are developed in Appendix ?? . The main principle is that if a centrally symmetric convex body Q contains sufficiently many lattice points, then one can choose lattice directions $w_1, \dots, w_d$ adapted to its successive minima. A box in those directions sits inside a controlled dilation of Q , while a controlled dilation of that box covers $Q \cap \Gamma$. This is the mechanism by which Bohr neighborhoods will later produce generalized arithmetic progressions.

2.6 Freiman dimension: geometry encoded by relations

A finite subset $A = \{a_1, \dots, a_n\}$ of an abelian group carries a family of additive relations

$$a_i + a_j = a_k + a_\ell.$$

These relations constrain every Freiman image of A . To see this, introduce formal basis vectors $e_1, \dots, e_n$ of $\mathbb{R}^n$ and let W_A be the linear span of all vectors

$$e_i + e_j - e_k - e_\ell$$

corresponding to relations

$$a_i + a_j = a_k + a_\ell.$$

The quotient

$$\mathbb{R}^n / W_A$$

records the degrees of freedom left after all additive quadruple relations are imposed.

There are always two obvious affine degrees of freedom. If x_i is a real realization of the relation pattern, then

$$x_i \mapsto \lambda x_i + t$$

preserves every relation $x_i + x_j = x_k + x_\ell$. This motivates the definition we shall use throughout the volume.

Definition 2.11 (Freiman dimension). For $A = \{a_1, \dots, a_n\}$, let

$$W_A := \text{span}_{\mathbb{R}}\{e_i + e_j - e_k - e_\ell : a_i + a_j = a_k + a_\ell\} \subseteq \mathbb{R}^n.$$

The *Freiman dimension* of A is

$$\dim_F(A) := n - 1 - \dim W_A.$$

The subtraction of 1 removes the unavoidable translation parameter. A nondegenerate arithmetic progression therefore has Freiman dimension one. Chapter ?? proves the equivalent, coordinate-free description in terms of real-valued Freiman homomorphisms and constructs the universal Euclidean model.

The key point is conceptual:

additive relations determine an affine geometry.

Freiman's theorem says that small doubling forces this geometry to have bounded complexity.

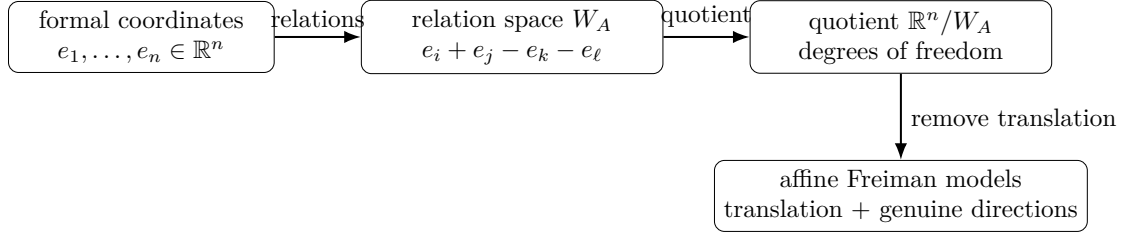


Figure 2.1: Freiman dimension as relation geometry. The ambient group may be highly torsion or otherwise complicated, but the additive quadruple relations cut out a linear relation space. The quotient records exactly the affine degrees of freedom available to a Freiman image.

Worked Example 2.12 (Why a four-point arithmetic progression has Freiman dimension one). Take $A = \{0, 1, 2, 3\}$. A Freiman homomorphism $\phi : A \rightarrow \mathbb{R}$ must respect

$$0 + 2 = 1 + 1, \quad 0 + 3 = 1 + 2.$$

Writing $x_j = \phi(j)$, these give

$$x_2 = 2x_1 - x_0, \quad x_3 = 3x_1 - 2x_0.$$

Thus every image has the form $x_j = t + jv$: one translation parameter t and one genuine direction v . After translation is removed, only one degree of freedom remains, so $\dim_F(A) = 1$. This is the relation-space version of the intuitive statement that a nondegenerate arithmetic progression is one-dimensional.

2.7 A rank-reduction phenomenon

Suppose a displayed rank- r progression is highly nonproper. Then its parameter map has a short kernel vector

$$q = (q_1, \dots, q_r) \neq 0, \quad q_1 v_1 + \dots + q_r v_r = 0.$$

This means one direction is algebraically dependent on the others. At a qualitative level, one can quotient the parameter lattice by the subgroup generated by q and obtain a lower-rank description.

The quantitative problem is harder: the new progression should not become enormously larger. This is why properization lemmas are nontrivial in high-quality versions of Freiman theory.

Proposition 2.13 (Qualitative rank reduction). *Let $P = \phi(B)$ be a finite generalized arithmetic progression of rank r . Then P is contained in a proper coset progression of rank at most r .*

Proof. Write the parameter homomorphism on the whole lattice as

$$\phi : \mathbb{Z}^r \rightarrow G, \quad (n_1, \dots, n_r) \mapsto \sum_{i=1}^r n_i v_i.$$

If the restriction of ϕ to the parameter box B is injective, then P is already proper (and is a coset progression with trivial subgroup part).

Otherwise there are distinct $x, y \in B$ with $\phi(x) = \phi(y)$, so

$$0 \neq x - y \in \ker \phi.$$

Hence $\ker \phi$ has positive rank. The quotient

$$\Lambda := \mathbb{Z}^r / \ker \phi$$

is a finitely generated abelian group whose free rank r' is at most $r - 1$. By the structure theorem,

$$\Lambda \cong T \oplus \mathbb{Z}^{r'}$$

for a finite abelian group T .

The homomorphism ϕ factors through an *injective* homomorphism

$$\bar{\phi} : \Lambda \hookrightarrow G,$$

because we have divided by the full kernel. Let $H = \bar{\phi}(T)$, a finite subgroup of G , and choose free generators $u_1, \dots, u_{r'}$ for the $\mathbb{Z}^{r'}$ part. The image of the finite box B in Λ is finite, so in the free coordinates it is contained in some integer box

$$[m_1, M_1] \times \cdots \times [m_{r'}, M_{r'}].$$

Translating the box if desired, we obtain a coset progression

$$Q = H + a_0 + \left\{ \sum_{j=1}^{r'} n_j \bar{\phi}(u_j) : 0 \leq n_j \leq M_j - m_j \right\}$$

containing P .

Because $T \oplus \mathbb{Z}^{r'}$ has unique coordinates and $\bar{\phi}$ is injective, the parameterization of Q modulo H is injective. Thus Q is proper. Its rank is $r' \leq r - 1$, and hence certainly at most r . $\square$

2.8 Worked example: a hidden rank drop

Let

$$P = \{n_1(1, 0) + n_2(0, 1) + n_3(1, 1) : 0 \leq n_i < L\} \subset \mathbb{Z}^2.$$

The displayed rank is three, but

$$(1, 1) = (1, 0) + (0, 1).$$

Every point is therefore

$$(n_1 + n_3, n_2 + n_3).$$

Consequently

$$P \subseteq [0, 2L - 2]^2,$$

a rank-two progression of size $(2L - 1)^2$.

The original parameter box has L^3 points, so for large L the average multiplicity of the rank-three parameterization is of order L . Nonproperness is not a small defect here; it completely changes the correct dimension.

The example is a useful warning for inverse theorems. A theorem should control the rank of a *proper* progression, not merely present A as the image of a high-dimensional box with massive collisions.

2.9 Worked example: a coset progression in a cyclic group

Let $N = qm$ and consider $G = \mathbb{Z}/N\mathbb{Z}$. The subgroup

$$H = q\mathbb{Z}/N\mathbb{Z}$$

has size m . Choose $v = 1$ and suppose $L < q/2$. Then

$$M = H + \{0, 1, \dots, L-1\}$$

consists of L distinct cosets of H , so

$$|M| = Lm.$$

Moreover

$$M + M = H + \{0, 1, \dots, 2L-2\},$$

and the assumption $2L-2 < q$ prevents wraparound in the quotient $G/H \cong \mathbb{Z}/q\mathbb{Z}$. Thus

$$|M + M| = (2L-1)m.$$

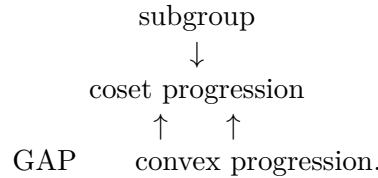
If L is allowed to exceed $q/2$, the quotient progression starts to wrap around and the doubling may become even smaller. In finite groups, small growth can come either from low-dimensional progression geometry or from saturation of a subgroup quotient. Coset progressions record both effects simultaneously.

Research window Convex progressions in modern Freiman theory

Modern quantitative inverse theorems are often stated using *convex coset progressions* rather than boxes. This is not cosmetic. Convex bodies interact efficiently with entropy, Fourier level sets, and geometry-of-numbers arguments, and they avoid some artificial losses caused by forcing every structured set into axis-parallel coordinates. Near-polynomial results for general abelian groups are naturally formulated in this language.

2.10 Summary of the geometric vocabulary

The objects introduced in this chapter form a hierarchy:



A subgroup has no growth. A rank- r proper progression has doubling at most about 2^r . A coset progression combines these two behaviors. Freiman theory seeks to prove that every finite set with bounded doubling is contained in, or efficiently covered by, an object from this hierarchy with bounded rank and controlled size.

Chapter 3

Freiman Relations, Universal Models, and Dimension

3.1 The relation space of a finite set

Let

$$A = \{a_1, \dots, a_n\}$$

be a finite subset of an abelian group. A Freiman homomorphism is determined not by the ambient group but by the additive relations among the a_i . It is therefore useful to package those relations algebraically.

Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{Z}^n$. Define the subgroup

$$R_2(A) \leq \mathbb{Z}^n$$

generated by all vectors

$$e_i + e_j - e_k - e_\ell$$

for which

$$a_i + a_j = a_k + a_\ell.$$

The quotient

$$\mathcal{F}_2(A) = \mathbb{Z}^n / R_2(A) \tag{3.1}$$

is the *universal Freiman 2-group* of A .

Let

$$\iota : A \rightarrow \mathcal{F}_2(A), \quad \iota(a_i) = e_i + R_2(A).$$

By construction, every additive quadruple relation in A is valid after applying ι .

Proposition 3.1 (Universal property). *Let H be an abelian group and let $f : A \rightarrow H$ be a Freiman 2-homomorphism. Then there is a unique group homomorphism*

$$\tilde{f} : \mathcal{F}_2(A) \rightarrow H$$

such that

$$f = \tilde{f} \circ \iota.$$

Proof. Define a homomorphism

$$F : \mathbb{Z}^n \rightarrow H$$

by

$$F(e_i) = f(a_i).$$

If

$$e_i + e_j - e_k - e_\ell$$

is one of the defining relation vectors, then

$$a_i + a_j = a_k + a_\ell,$$

so the Freiman property gives

$$F(e_i + e_j - e_k - e_\ell) = 0.$$

Hence $R_2(A) \subseteq \ker F$, and F factors uniquely through the quotient $(??)$. The factor map is $\tilde{f}$. $\square$

Idea Why this quotient is useful

The ambient group may contain irrelevant relations. The universal Freiman group remembers exactly the relations forced by the addition table of A and nothing else. It is the additive analogue of presenting an algebraic object by generators and relations.

3.2 Affine freedom and Freiman dimension

Suppose now that we look at real-valued Freiman homomorphisms

$$f : A \rightarrow \mathbb{R}.$$

The set of all such maps is a real vector space

$$\text{Hom}_{F_2}(A, \mathbb{R}).$$

Every constant map belongs to this space. More generally, if A is already realized in $\mathbb{R}^d$, then every affine functional

$$x \mapsto \lambda \cdot x + c$$

restricts to a Freiman homomorphism on A .

Recall Definition ???. The following coordinate-free description explains why that relation-space number is intrinsic.

Proposition 3.2 (Equivalent formulations of Freiman dimension). *For a finite set $A = \{a_1, \dots, a_n\}$,*

$$\dim_F(A) + 1 = \dim_{\mathbb{R}} \text{Hom}_{F_2}(A, \mathbb{R}) = \dim_{\mathbb{R}} (\mathcal{F}_2(A) \otimes \mathbb{R}).$$

Proof. Identify a function $f : A \rightarrow \mathbb{R}$ with the vector $(f(a_1), \dots, f(a_n)) \in \mathbb{R}^n$. It is a Freiman 2-homomorphism exactly when it annihilates every relation vector $e_i + e_j - e_k - e_\ell$, hence exactly when it lies in the annihilator $W_A^\perp$. Therefore

$$\dim_{\mathbb{R}} \text{Hom}_{F_2}(A, \mathbb{R}) = n - \dim W_A = \dim_F(A) + 1.$$

On the other hand, the universal property of $\mathcal{F}_2(A)$ gives $\text{Hom}_{F_2}(A, \mathbb{R}) \cong \text{Hom}(\mathcal{F}_2(A), \mathbb{R})$. Tensoring with $\mathbb{R}$ removes the torsion of $\mathcal{F}_2(A)$, and the last Hom-space has dimension $\dim_{\mathbb{R}}(\mathcal{F}_2(A) \otimes \mathbb{R})$. $\square$

Proposition 3.3 (Universal Euclidean realization). *If $d = \dim_F(A)$, then there is a Freiman 2-homomorphism*

$$\psi : A \rightarrow \mathbb{R}^d$$

whose image affinely spans $\mathbb{R}^d$. Moreover, every real-valued Freiman homomorphism on A is the restriction of an affine functional on $\psi(A)$.

Proof. Let

$$V = \text{Hom}_{F_2}(A, \mathbb{R})$$

and choose a basis

$$1, f_1, \dots, f_d,$$

where 1 denotes the constant function. Set

$$\psi(a) = (f_1(a), \dots, f_d(a)).$$

Each coordinate is a Freiman homomorphism, so ψ is a Freiman homomorphism.

If $\psi(A)$ were contained in an affine hyperplane

$$c_0 + c_1x_1 + \dots + c_dx_d = 0$$

with not all c_i zero, then

$$c_0 + c_1f_1 + \dots + c_df_d = 0$$

as functions on A , contradicting linear independence. Thus $\psi(A)$ affinely spans $\mathbb{R}^d$.

Finally any $f \in V$ has the form

$$f = c_0 + c_1f_1 + \dots + c_df_d,$$

which is exactly an affine functional of $\psi(a)$. □

The proposition converts an abstract addition table into Euclidean geometry. This is the precise meaning of the informal phrase “the additive relations determine an affine geometry.”

3.3 Worked example: an arithmetic progression

Let

$$A = \{0, 1, \dots, N-1\} \subset \mathbb{Z}, \quad N \geq 3.$$

If $f : A \rightarrow \mathbb{R}$ is a Freiman homomorphism, fix $1 \leq j \leq N-2$ and apply the defining relation to the four elements

$$a = j+1, \quad b = 0, \quad c = j, \quad d = 1.$$

Since $a + b = c + d$ in $\mathbb{Z}$, we obtain

$$f(j+1) + f(0) = f(j) + f(1).$$

Equivalently,

$$f(j+1) - f(j) = f(1) - f(0)$$

for every j . Therefore

$$f(j) = f(0) + j(f(1) - f(0)).$$

So every Freiman homomorphism is affine in the integer coordinate. The space of such maps has dimension 2, and therefore

$$\dim_F(A) = 1.$$

The calculation shows explicitly how the numerous additive relations of an interval collapse the possible images to a one-dimensional family.

3.4 Worked example: a proper rank-two box

Let

$$A = \{i + Mj : 0 \leq i < L_1, 0 \leq j < L_2\} \subset \mathbb{Z}$$

with $M > 2L_1$ and $L_1, L_2 \geq 2$. The map

$$(i, j) \mapsto i + Mj$$

is a Freiman 2-isomorphism from the rectangular grid

$$B = [0, L_1 - 1]_{\mathbb{Z}} \times [0, L_2 - 1]_{\mathbb{Z}}$$

onto A .

Let $f : B \rightarrow \mathbb{R}$ be a Freiman homomorphism. Horizontal parallelogram relations give

$$f(i + 1, j) - f(i, j)$$

independent of i and j , while vertical parallelograms give the same conclusion in the j direction. Consequently

$$f(i, j) = c + \alpha i + \beta j.$$

Thus

$$\dim_F(A) = 2.$$

This is the first clean example in which Freiman dimension recovers the rank of a proper progression.

3.5 Freiman's dimension lemma

The next theorem is the bridge from small doubling to bounded affine dimension.

Theorem 3.4 (Freiman's dimension lemma). *Let $A \subset \mathbb{R}^d$ be finite and suppose that its affine span has dimension d . Then*

$$|A + A| \geq (d + 1)|A| - \frac{d(d + 1)}{2}. \quad (3.2)$$

The bound is sharp for every d and arbitrarily large $|A|$.

Remark 3.5 (Proof location). Appendix ?? explains the shelling mechanism behind (??); Appendix ?? gives the complete pulling-triangulation, simplicial-decomposition, and translate-counting proof. No later argument in this volume requires a stronger geometric statement.

Even before reading that appendix, the consequence for small doubling is immediate.

Corollary 3.6 (Small doubling bounds Euclidean Freiman dimension). *Let A be Freiman 2-isomorphic to a finite set $A' \subset \mathbb{R}^d$ whose affine span has dimension d , and write $|A| = n$. If*

$$|A + A| \leq K|A|,$$

then

$$d + 1 - \frac{d(d + 1)}{2n} \leq K.$$

In particular, if $n \geq d(d + 1)$, then

$$d \leq K - \frac{1}{2},$$

and hence

$$d \leq \left\lfloor K - \frac{1}{2} \right\rfloor.$$

If K is an integer this simplifies to $d \leq K - 1$.

Proof. A Freiman 2-isomorphism preserves both cardinality and the equality pattern of pair sums, and hence

$$|A' + A'| = |A + A|, \quad |A'| = |A| = n.$$

Apply ?? to A' :

$$Kn \geq (d+1)n - \frac{d(d+1)}{2}.$$

Dividing by n gives the first assertion. If $n \geq d(d+1)$, then the correction term is at most $1/2$, so

$$K \geq d + \frac{1}{2}.$$

Thus $d \leq K - 1/2$, equivalently $d \leq \lfloor K - 1/2 \rfloor$. When K is an integer this is $d \leq K - 1$. $\square$

Warning 3.7 (Why we do not apply this corollary to arbitrary torsion sets). The universal Euclidean map in ?? need not be injective: tensoring the universal Freiman group with $\mathbb{R}$ kills torsion. Thus one cannot silently replace an arbitrary set in a torsion group by a Freiman-isomorphic subset of $\mathbb{R}^d$. The corollary is deliberately stated only when such a Euclidean Freiman model is known. In general abelian groups, subgroup directions must be retained, and this is exactly why the correct structural objects are coset progressions rather than ordinary Euclidean progressions.

3.6 Sharpness of the dimension lemma

Let $N \geq 1$ and consider

$$A = \{0, e_1, \dots, e_{d-1}\} + \{0, 1, \dots, N-1\}e_d \subset \mathbb{R}^d.$$

This is the union of d parallel arithmetic progressions of length N , so

$$|A| = dN.$$

A direct layer count gives the correct leading term in (??). The sums split according to the first $d-1$ coordinates. There are:

- one layer corresponding to $0 + 0$;
- $d-1$ layers corresponding to $0 + e_i$;
- $d-1$ layers corresponding to $e_i + e_i$;
- $\binom{d-1}{2}$ layers corresponding to $e_i + e_j$ with $i < j$.

Every layer is an interval in the e_d direction of length $2N-1$. Distinct first-coordinate types are disjoint. Hence

$$|A + A| = \left(1 + 2(d-1) + \binom{d-1}{2}\right)(2N-1).$$

This particular box-of-lines model is not the most economical sharp example for every finite N , but it shows why the leading coefficient $d+1$ is unavoidable. Exact equality examples are given by suitable stacked configurations; Appendix ?? records and counts one standard construction.

3.7 The universal model as a finitely generated abelian group

The group $\mathcal{F}_2(A)$ is finitely generated, hence by the structure theorem

$$\mathcal{F}_2(A) \cong \mathbb{Z}^d \oplus T$$

for some finite abelian group T . The free rank is

$$d = \dim_F(A) + 1$$

up to the one affine coordinate convention. Thus every finite addition table has a canonical ambient group consisting of a free part and a torsion part.

This decomposition anticipates the eventual form of Freiman's theorem:

$$\boxed{\text{free directions} \rightsquigarrow \text{progression coordinates}}$$

and

$$\boxed{\text{torsion directions} \rightsquigarrow \text{subgroup coordinates.}}$$

3.8 Worked example: torsion and free directions together

Let

$$A = \{0, 1, \dots, L-1\} \times H \subset \mathbb{Z} \times H,$$

where H is a finite abelian group. Any real-valued Freiman homomorphism must be constant on each H -coset direction, because every $h \in H$ has finite order m and repeated Freiman relations force

$$m(f(x, h) - f(x, 0)) = 0$$

in $\mathbb{R}$. Hence the torsion direction contributes no real Freiman dimension.

Along the integer coordinate the map must be affine, as in the interval example. Therefore

$$\dim_F(A) = 1,$$

although A may have arbitrarily large cardinality in the H direction.

This is exactly the geometry of the coset progression

$$H + [0, L-1](1, 0).$$

3.9 Relation complexity versus ambient dimension

A recurring mistake is to identify the dimension of a set with the dimension of the group in which it happens to be written. For example, a long arithmetic progression can be embedded in $\mathbb{Z}^{1000}$ along a single line. Its ambient dimension is 1000, but its Freiman dimension is 1.

Inverse theory is concerned with the latter. The addition table sees intrinsic affine degrees of freedom, not unused coordinates.

Research window Dimension lemmas beyond abelian groups

In torsion-free nilpotent groups there is a noncommutative analogue of the Freiman-dimension philosophy: small approximate subgroups can be covered by boundedly many cosets of a nilpotent subgroup of bounded rank. Breuillard, Green and Tao developed an explicit “nilpotent Freiman dimension lemma” of this type. The necessary

noncommutative structure theory is outside the scope of this book and is treated in the separate Bourgain–Gamburd project. The conceptual ancestor is already visible here: failure of growth restricts the number of genuinely independent directions.

Scope boundary. The noncommutative analogue is contextual only: no theorem below in Volumes I–V depends on it. Product growth, approximate groups, nilprogressions, and expansion are maintained in the separate Bourgain–Gamburd project rather than as future volumes of this book.

3.10 What dimension does and does not give us

The dimension lemma is an important first inverse statement:

$$|A + A| \leq K|A| \implies \dim_F(A) = O(K)$$

in the torsion-free component.

But bounded dimension alone is not enough. A set of N points in a line can have enormous diameter and very irregular spacing. We still need a theorem saying that small doubling controls not only dimension but also the *scale and occupancy* of a progression containing A .

That is the role of modelling and the Bohr-set argument.

Chapter 4

Critical Pairs: Kneser, Vosper, and the $3k - 4$ Theorem

4.1 Why critical-pair theory belongs here

The preceding chapters introduced Freiman maps and dimension. Before turning to analytic modelling, it is useful to close a classical circle of inverse questions that was deliberately left open in Volume I. These are the extremal and near-extremal cases of small sumsets.

For subsets of the integers, the elementary inequality

$$|A + B| \geq |A| + |B| - 1$$

is sharp for arithmetic progressions. In a finite abelian group, however, subgroups and periodicity create additional extremizers. Kneser's theorem is the correct replacement for the integer lower bound. In a cyclic group of prime order, periodicity disappears and Vosper's theorem identifies the equality cases. Finally, Freiman's $3k - 4$ theorem shows that in the integers a whole interval of *near* equality is still one-dimensional.

The three proofs below are included because they are genuine inverse-additive mechanisms. In particular, no later theorem in this volume will be allowed to hide Kneser or $3k - 4$ behind the phrase “standard critical-pair theory.”

4.2 Kneser's theorem

For a finite set S in an abelian group G , define its stabilizer

$$\text{Stab}(S) = \{g \in G : S + g = S\}.$$

It is a subgroup of G . If $H = \text{Stab}(S)$, then S is a union of H -cosets.

Theorem 4.1 (Kneser's addition theorem). *Let A, B be finite nonempty subsets of an abelian group G , and put*

$$H = \text{Stab}(A + B).$$

Then

$$|A + B| \geq |A + H| + |B + H| - |H|.$$

Equivalently, after passing to G/H one has

$$|\overline{A} + \overline{B}| \geq |\overline{A}| + |\overline{B}| - 1.$$

The proof below is a streamlined intersection–union proof, following the strategy of DeVos [3]. The central idea is worth isolating.

Idea Search for the smallest periodic witness

In the aperiodic case $\text{Stab}(A + B) = \{0\}$, one tries to prove the integer-looking inequality $|A + B| \geq |A| + |B| - 1$. Intersections and unions replace (A, B) by a smaller pair without changing the sum $|A| + |B|$. If this does not already prove the inequality, the obstruction is a periodic subset $C \subseteq A + B$. Choosing such a witness with the smallest possible stabilizer forces a contradiction.

Proof. We induct on

$$\mathcal{N}(A, B) = |A + B| + |A|.$$

The assertion is immediate when $|A| = 1$, so the induction starts.

Suppose first that $H \neq \{0\}$. Let

$$\phi : G \longrightarrow G/H$$

be the quotient map. The sumset $\phi(A) + \phi(B)$ is aperiodic in G/H : indeed, any nonzero stabilizer there would lift to a subgroup strictly larger than H stabilizing $A + B$. The induction parameter decreases after quotienting, because $A + B$ is a finite union of nontrivial H -cosets. Applying the inductive hypothesis in G/H gives

$$|\phi(A) + \phi(B)| \geq |\phi(A)| + |\phi(B)| - 1.$$

Multiplying by $|H|$ yields

$$|A + B| = |H| |\phi(A) + \phi(B)| \geq |A + H| + |B + H| - |H|,$$

as required.

It remains to prove the aperiodic case

$$\text{Stab}(A + B) = \{0\}.$$

If $|A| = 1$ there is nothing to prove, so assume $|A| > 1$. Choose distinct $a, a' \in A$. Since

$$\text{Stab}(B) \leq \text{Stab}(A + B) = \{0\},$$

the nonzero element $a' - a$ does not stabilize B . Thus some $b \in B$ satisfies

$$b + a' - a \notin B.$$

Replace B by the translate $B - b + a$. Translation does not alter any cardinality in the theorem, and after this replacement

$$\emptyset \neq A \cap B \neq A.$$

For a set $C \subseteq A + B$ write $K_C = \text{Stab}(C)$. Call C *convergent* if

$$|C| + |K_C| \geq |A \cap B| + |(A \cup B) + K_C|. \quad (4.1)$$

Set

$$C_0 = (A \cap B) + (A \cup B).$$

Because $A \cap B$ is nonempty and strictly smaller than A , the induction hypothesis applies to the pair $(A \cap B, A \cup B)$. If $K_0 = \text{Stab}(C_0)$, it gives

$$|C_0| + |K_0| \geq |(A \cap B) + K_0| + |(A \cup B) + K_0| \geq |A \cap B| + |(A \cup B) + K_0|.$$

Hence a convergent set exists.

Choose a convergent $C \subseteq A + B$ for which

$$H_0 := \text{Stab}(C)$$

has minimum cardinality. If $H_0 = \{0\}$, then

$$|A + B| \geq |C| \geq |A \cap B| + |A \cup B| - 1 = |A| + |B| - 1,$$

and the aperiodic case is proved. We therefore assume, for contradiction, that $H_0 \neq \{0\}$.

Since $A + B$ is aperiodic whereas C is H_0 -periodic, there exist $a \in A$ and $b \in B$ such that

$$a + b + H_0 \not\subseteq A + B.$$

Define

$$A_1 = A \cap (a + H_0), \quad B_1 = B \cap (b + H_0),$$

$$A_2 = A \cap (b + H_0), \quad B_2 = B \cap (a + H_0).$$

The sets A_1, B_1 are nonempty. Whenever A_i, B_i are nonempty, put

$$C_i = C \cup (A_i + B_i), \quad H_i = \text{Stab}(C_i + B_i).$$

The coset $a + b + H_0$ cannot be contained in C , since $C \subseteq A + B$. Consequently C is disjoint from that entire H_0 -coset, while $A_i + B_i$ lies inside it. It follows that

$$\text{Stab}(C_i) = H_i < H_0.$$

Indeed $H_i \leq H_0$ and stabilizes both pieces; conversely a stabilizer of C_i must preserve the unique partially occupied H_0 -coset and hence stabilize $A_i + B_i$. The inclusion is strict because $A_i + B_i$ does not fill the whole H_0 -coset.

By the minimal choice of H_0 , the set C_i is not convergent. Subtracting its failed convergence inequality from the convergence inequality for C gives

$$\begin{aligned} |(A \cup B) + H_0| - |(A \cup B) + H_i| &< |H_0| - |A_i + B_i| - |H_i| \\ &\leq |H_0| - |A_i + H_i| - |B_i + H_i|, \end{aligned} \quad (4.2)$$

where the last step is the inductive Kneser inequality applied to A_i, B_i . This has so far been obtained for $i = 1$ and also for $i = 2$ whenever A_2, B_2 are nonempty.

We now show that A_2 and B_2 are indeed nonempty. If $B_2 = \emptyset$, then the part of the H_0 -coset $a + H_0$ missing from $A_1 + H_1$ contributes at least

$$|H_0| - |A_1 + H_1|$$

to the difference

$$|(A \cup B) + H_0| - |(A \cup B) + H_1|,$$

contradicting (??). The case $A_2 = \emptyset$ is symmetric. Thus (??) holds for $i = 1, 2$.

The two cosets $a + H_0$ and $b + H_0$ are distinct. Otherwise $A_1 = A_2$ and $B_1 = B_2$, and the missing part of that one coset would give

$$|(A \cup B) + H_0| - |(A \cup B) + H_1| \geq |H_0| - |A_1 + H_1| - |B_1 + H_1|,$$

again contradicting the strict inequality (??).

The left side of (??) is nonnegative. All terms on the right are multiples of $|H_i|$. Hence the strict inequality forces a gap of at least one H_i -coset:

$$|H_0| \geq |A_i| + |B_i| + |H_i|, \quad i = 1, 2. \quad (4.3)$$

Finally let

$$S = (a + H_0) \setminus (A_1 \cup B_2), \quad T = (b + H_0) \setminus (A_2 \cup B_1).$$

These sets are disjoint. Since $A + B$ itself is not convergent—otherwise its stabilizer $\{0\}$ would contradict the minimality of H_0 —and since the inductive Kneser inequality for $A_i + B_i$ gives

$$|A_i + B_i| \geq |A_i + H_i| + |B_i + H_i| - |H_i| > |A_i| + |B_i| - |H_i|,$$

the following count holds for $i = 1, 2$:

$$\begin{aligned} |H_0| &\geq |(A \cup B) + H_0| + |A \cap B| - |C| \\ &\geq |S| + |T| + |A \cup B| + |A \cap B| - |A + B| + |A_i + B_i| \\ &> |S| + |T| + |A_i| + |B_i| - |H_i|. \end{aligned} \quad (4.4)$$

The first line is (??); in the second, the two holes S, T are added to $A \cup B$, while C is replaced by the larger ambient set $A + B$ and the missing piece $A_i + B_i$ is restored. The strict final step uses non-convergence of $A + B$ together with the inductive estimate just displayed.

Add (??) and (??) for $i = 1, 2$ and divide by two. We obtain

$$2|H_0| > |A_1| + |B_2| + |S| + |A_2| + |B_1| + |T|.$$

But

$$a + H_0 = S \sqcup A_1 \sqcup B_2, \quad b + H_0 = T \sqcup A_2 \sqcup B_1,$$

so the right-hand side is exactly $2|H_0|$. This contradiction proves the aperiodic case and hence the theorem. $\square$

Corollary 4.2 (The deficient case). *If*

$$|A + B| \leq |A| + |B| - 2,$$

then $A + B$ has a nontrivial stabilizer.

Proof. If the stabilizer were trivial, Kneser's theorem would give $|A + B| \geq |A| + |B| - 1$. $\square$

Corollary 4.3 (Kneser's equality form in the deficient range). *If*

$$|A + B| < |A| + |B|$$

and $H = \text{Stab}(A + B)$, then

$$|A + B| + |H| = |A + H| + |B + H|.$$

Proof. Pass to G/H . The quotient sumset is aperiodic, so Kneser's theorem there gives

$$|\bar{A} + \bar{B}| \geq |\bar{A}| + |\bar{B}| - 1.$$

If the inequality were strict, integrality would give

$$|\bar{A} + \bar{B}| \geq |\bar{A}| + |\bar{B}|,$$

and multiplication by $|H|$ would imply

$$|A + B| \geq |A + H| + |B + H| \geq |A| + |B|,$$

contrary to the hypothesis. Hence equality holds in the quotient, and multiplying by $|H|$ gives the claimed identity. $\square$

4.3 Vosper's theorem

We now specialize to $G = \mathbb{Z}/p\mathbb{Z}$, where p is prime. Every nonempty proper subset is aperiodic, so Cauchy–Davenport gives the universal lower bound and Vosper identifies equality. We prove the form used in Volume I.

For $S \subseteq G$ write $\overline{S} = G \setminus S$.

Lemma 4.4 (One-step progression criterion). *Let $X \subseteq G$, $|X| \geq 2$, and $d \neq 0$. If*

$$|X \cup (X + d)| \leq |X| + 1,$$

then X is an arithmetic progression with common difference d .

Proof. After scaling, take $d = 1$. Regard the residues cyclically. The quantity

$$|(X + 1) \setminus X|$$

is the number of transitions from an occupied residue to an unoccupied residue as one walks around the cycle. The hypothesis says this number is at most one. It cannot be zero because X is a nonempty proper subset whenever the displayed cardinality is $|X| + 1$; if $X = G$ the conclusion is also trivial. Thus there is exactly one occupied block, so X consists of consecutive residues. $\square$

Lemma 4.5 (The e -transform). *Let $A, B \subseteq G$ be nonempty, suppose $A + B \neq G$, and translate so that $0 \in B$. Put*

$$C = -\overline{A + B}, \quad r = |A + B| - |A| - |B| + 1,$$

and

$$E = (A + 2B) \setminus (A + B).$$

For $e \in E$ define

$$B_e = B \cap (e + C), \quad B^e = B \setminus B_e.$$

Then $0 \in B_e$, $B^e \neq \emptyset$, and

$$|A + B| - |B| \geq |A + B_e| - |B_e|. \tag{4.5}$$

If moreover $B_e = \{0\}$ for every $e \in E$, then

$$A + 2B \neq G \implies |B| \leq r + 2,$$

and

$$A + 2B = G \implies |C| \leq r + 1.$$

Proof. Because $e \notin A + B$, we have $-e \in C$, hence $0 \in e + C$ and therefore $0 \in B_e$. Since $e \in A + 2B$, write $e = a + b_1 + b_2$. If both b_1, b_2 lay in B_e , then $e - b_1 = a + b_2 \in A + B$, contradicting the definition of B_e . Thus B^e is nonempty.

If $b \in B^e$, then $b \notin e + C$, so $e - b \notin -C = \overline{A + B}$ and hence $e - b \in A + B$. Therefore

$$e - B^e \subseteq A + B.$$

Also $A + B_e \subseteq A + B$. These two subsets are disjoint: if $a + b_1 = e - b_2$ with $b_1 \in B_e$ and $b_2 \in B^e$, then $e - b_1 = a + b_2 \in A + B$, whereas $b_1 \in e + C$ says precisely that $e - b_1 \notin A + B$. Hence

$$|A + B| \geq |A + B_e| + |B^e| = |A + B_e| + |B| - |B_e|,$$

which is (??).

Now assume every $B_e = \{0\}$. Put $B^\times = B \setminus \{0\}$. For every $e \in E$ the preceding argument gives

$$e - B^\times \subseteq (A + B) \setminus A.$$

Moreover the whole set $E - B^\times$ is disjoint from A : if $e - b \in A$, then $e = (e - b) + b \in A + B$, impossible. Thus

$$|A + B| \geq |A| + |E - B^\times|.$$

Cauchy–Davenport gives

$$|E - B^\times| \geq |E| + |B| - 2;$$

the saturated alternative cannot occur because this set is disjoint from the nonempty set A . Consequently

$$|E| \leq r + 1.$$

If $A + 2B \neq G$, Cauchy–Davenport applied to $(A + B) + B$ gives

$$|E| = |A + 2B| - |A + B| \geq |B| - 1,$$

so $|B| \leq r + 2$. If $A + 2B = G$, then

$$E = G \setminus (A + B),$$

so $|C| = |E| \leq r + 1$. □

Theorem 4.6 (Vosper). *Let $A, B \subseteq \mathbb{Z}/p\mathbb{Z}$ satisfy*

$$|A|, |B| \geq 2, \quad |A + B| = |A| + |B| - 1 \leq p - 2.$$

Then A and B are arithmetic progressions with the same nonzero common difference.

Proof. Assume a counterexample exists and choose one with $|B|$ minimal. Translate B so that $0 \in B$. Set $C = -\overline{A} + \overline{B}$. The hypothesis $|A + B| \leq p - 2$ gives $|C| \geq 2$, and here the excess parameter in ?? is $r = 0$.

If $|B| = 2$, write $B = \{0, d\}$. Then

$$|A \cup (A + d)| = |A + B| = |A| + 1,$$

so ?? makes A an arithmetic progression of difference d . Thus a minimal counterexample has $|B| \geq 3$.

The set $E = (A + 2B) \setminus (A + B)$ is nonempty. Indeed, if $A + 2B = A + B$, then every $b \in B$ stabilizes the proper nonempty set $A + B$; in a group of prime order this would force $B = \{0\}$, contrary to $|B| \geq 2$.

Fix $e \in E$. By (??),

$$|A + B_e| - |B_e| \leq |A| - 1.$$

Cauchy–Davenport gives the reverse inequality, and saturation is impossible because $A + B_e \subseteq A + B \neq G$. Hence

$$|A + B_e| = |A| + |B_e| - 1.$$

If $|B_e| \geq 2$, then (A, B_e) satisfies the same Vosper hypotheses and has a strictly smaller second set. By minimality it is not a counterexample, so A is an arithmetic progression. We now show this already forces the original B to have the same common difference, contradicting the choice of counterexample.

Indeed, after an affine change of variables write

$$A = \{0, 1, \dots, m-1\}.$$

If $B \cup (B+1)$ had size at least $|B| + 2$, then Cauchy–Davenport applied to

$$(B + \{0, 1\}) + \{0, 1, \dots, m-2\} = A + B$$

would give

$$|A + B| \geq |B| + m,$$

because the right-hand side is at most $p-1$ under the hypothesis $|A + B| \leq p-2$. This contradicts equality. Thus

$$|B \cup (B+1)| = |B| + 1,$$

and ?? shows that B is an arithmetic progression of the same difference.

We conclude that every B_e must be a singleton. Since $0 \in B_e$, this means $B_e = \{0\}$ for all $e \in E$. Apply the final part of ?? with $r = 0$. If $A + 2B = G$, it gives $|C| \leq 1$, contradicting $|C| \geq 2$. If $A + 2B \neq G$, it gives $|B| \leq 2$, contradicting $|B| \geq 3$. No counterexample exists. $\square$

The proof is in the critical-pair tradition of Vosper [12]: one shrinks one summand while preserving exact Cauchy–Davenport equality until a two-point set forces an arithmetic progression.

4.4 Freiman’s $3k - 4$ theorem

The integer theorem is not merely an equality classification. It says that the entire first range above the minimum still forces one-dimensional structure. We shall prove it through a two-set reduced-length inequality due to Lev–Smeliansky [8]. Kneser’s theorem is the only group addition theorem used in the proof.

For a finite set $A \subset \mathbb{Z}$, define its *reduced length* by

$$\ell_{\text{red}}(A) = \frac{\max A - \min A}{\gcd\{a - a' : a, a' \in A\}}.$$

Thus after translating and dividing by the common difference, a set of reduced length q lies in $[0, q]$, contains both endpoints, and has $\gcd 1$.

Lemma 4.7 (Lev–Smeliansky reduced-length inequality). *Let*

$$A = \{0 = a_1 < a_2 < \dots < a_k = q\}, \quad B = \{0 = b_1 < \dots < b_\ell \leq q\},$$

with $\gcd(a_1, \dots, a_k) = 1$. Let

$$\delta = \begin{cases} 1, & b_\ell = q, \\ 0, & b_\ell < q. \end{cases}$$

Then

$$|A + B| \geq \min\{q, k + \ell - 2 - \delta\} + \ell.$$

Proof. If

$$|A + B| > k + 2\ell - 3 - \delta,$$

then, since cardinalities are integers,

$$|A + B| \geq k + 2\ell - 2 - \delta \geq \min\{q, k + \ell - 2 - \delta\} + \ell,$$

so there is nothing to prove. We may therefore assume

$$|A + B| \leq k + 2\ell - 3 - \delta \quad (4.6)$$

and prove the stronger inequality

$$|A + B| \geq q + \ell. \quad (4.7)$$

Let

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}/q\mathbb{Z}$$

be reduction modulo q , and write

$$\bar{A} = \phi(A), \quad \bar{B} = \phi(B), \quad \bar{S} = \bar{A} + \bar{B} = \phi(A + B).$$

Because 0 and q coalesce,

$$|\bar{A}| = k - 1, \quad |\bar{B}| = \ell - \delta.$$

We first claim

$$|\bar{S}| \leq |A + B| - \ell. \quad (4.8)$$

If $b_\ell < q$, the 2ℓ sums

$$0 + b_i, \quad q + b_i \quad (1 \leq i \leq \ell)$$

are pairwise distinct integers but occur in ℓ equal pairs modulo q . If $b_\ell = q$, then for $2 \leq i \leq \ell - 1$ the sums b_i and $q + b_i$ coalesce, while the residue 0 has the three distinct lifts $0, q, 2q$. Again at least ℓ collisions are lost on reduction. This proves (??).

Combining (??) and (??) gives

$$|\bar{S}| \leq |\bar{A}| + |\bar{B}| - 2.$$

By ??, $\bar{S}$ is periodic. Let

$$H = \text{Stab}(\bar{S}).$$

Every subgroup of $\mathbb{Z}/q\mathbb{Z}$ has the form $d\mathbb{Z}/q\mathbb{Z}$ with $d \mid q$. If $H = \mathbb{Z}/q\mathbb{Z}$, then $|\bar{S}| = q$, and (??) immediately yields

$$|A + B| \geq q + \ell,$$

which is (??). We therefore assume that H is a proper nontrivial subgroup and derive a contradiction.

Let

$$\sigma : \mathbb{Z}/q\mathbb{Z} \rightarrow (\mathbb{Z}/q\mathbb{Z})/H$$

be the quotient map and put

$$\tilde{A} = \sigma(\bar{A}), \quad \tilde{B} = \sigma(\bar{B}).$$

Kneser's theorem is exact here:

$$|\tilde{A} + \tilde{B}| = |\tilde{A}| + |\tilde{B}| - 1. \quad (4.9)$$

Indeed, the deficient estimate above gives $|\bar{S}| < |\bar{A}| + |\bar{B}|$. Therefore Corollary ??, applied to $\bar{A}, \bar{B}$, gives

$$|\bar{S}| + |H| = |\bar{A} + H| + |\bar{B} + H|.$$

Dividing by $|H|$ is exactly (??). The same equality also gives the hole identity

$$|(\bar{A} + H) \setminus \bar{A}| + |(\bar{B} + H) \setminus \bar{B}| = |H| - (|\bar{A}| + |\bar{B}| - |\bar{S}|). \quad (4.10)$$

We now count integer sums according to their image under $\sigma\phi$. First consider quotient classes belonging to $\tilde{B}$. Since $\bar{B} + H \subseteq \bar{A} + \bar{B} = \bar{S}$ (because $0 \in \bar{A}$), every residue in $\bar{B} + H$ occurs among the reductions of integer sums. In addition, the ℓ integers $b_1, \dots, b_\ell$ supply the endpoint-lift multiplicity lost modulo q . Consequently

$$\#\{c \in A + B : \sigma\phi(c) \in \tilde{B}\} \geq \ell + |\tilde{B}| |H|. \quad (4.11)$$

This is exactly the same endpoint-lift count as in (??), now restricted to the union of H -cosets above $\tilde{B}$.

Next fix

$$\tilde{c} \in (\tilde{A} + \tilde{B}) \setminus \tilde{B}$$

and choose

$$\tilde{c} = \tilde{a} + \tilde{b}, \quad \tilde{a} \in \tilde{A}, \quad \tilde{b} \in \tilde{B}.$$

The fibers

$$A_{\tilde{a}} = A \cap \phi^{-1}\sigma^{-1}(\tilde{a}), \quad B_{\tilde{b}} = B \cap \phi^{-1}\sigma^{-1}(\tilde{b})$$

are finite subsets of $\mathbb{Z}$. The elementary integer sumset inequality gives

$$|A_{\tilde{a}} + B_{\tilde{b}}| \geq |A_{\tilde{a}}| + |B_{\tilde{b}}| - 1.$$

Using (??), each selected pair of fibers has total size at least

$$2|H| - |(\bar{A} + H) \setminus \bar{A}| - |(\bar{B} + H) \setminus \bar{B}|,$$

so

$$\#\{c \in A + B : \sigma\phi(c) = \tilde{c}\} \geq |H| - 1 + |\bar{A}| + |\bar{B}| - |\bar{S}|. \quad (4.12)$$

There are, by (??), exactly $|\tilde{A}| - 1$ quotient classes of the second type. Summing (??) and (??) yields

$$|A + B| \geq \ell + |\bar{S}| + (|\tilde{A}| - 1)(|\bar{A}| + |\bar{B}| - |\bar{S}| - 1). \quad (4.13)$$

Set

$$x = |\bar{A}| + |\bar{B}| - |\bar{S}|.$$

We already know $x \geq 2$. Rewriting (??) and comparing with (??), namely

$$|A + B| \leq |\bar{A}| + |\bar{B}| + \ell - 2,$$

gives

$$1 + (|\tilde{A}| - 2)(x - 1) \leq 0.$$

Since $x - 1 \geq 1$, this forces

$$|\tilde{A}| = 1.$$

Thus every element of $\bar{A}$ lies in $H = d\mathbb{Z}/q\mathbb{Z}$. Because all representatives a_i lie between 0 and q , this says that d divides every a_i . The subgroup H was proper, so $d > 1$, contradicting $\gcd(a_1, \dots, a_k) = 1$.

The proper-periodic case is impossible. Hence H must have been the whole cyclic group, and (??) follows. This completes the proof. $\square$

Remark 4.8 (Where the proof uses one-dimensional order). The modular step is group-theoretic, but the fiber estimate

$$|X + Y| \geq |X| + |Y| - 1$$

and the endpoint-lift count use the linear order of the integers. This is why $3k - 4$ is a genuinely one-dimensional theorem, not merely a consequence of abstract small-doubling theory.

Theorem 4.9 (Freiman's $3k - 4$ theorem). *Let $A \subset \mathbb{Z}$ be finite with $|A| = k \geq 3$. If*

$$|A + A| \leq 3k - 4,$$

then A is contained in an arithmetic progression of length at most

$$|A + A| - |A| + 1.$$

Proof. Translate A so that its minimum is 0, and divide by the gcd of all its differences. These operations preserve cardinalities of sumsets and turn A into a set

$$A' = \{0 = a_1 < \dots < a_k = q\}$$

with gcd 1. Its reduced length q is exactly one less than the length of the shortest arithmetic progression, in the original common difference, containing A .

Apply ?? with $B = A'$. Here $\ell = k$ and $\delta = 1$, so

$$|2A'| \geq \min\{q, 2k - 3\} + k.$$

If the minimum were $2k - 3$, then

$$|2A'| \geq 3k - 3,$$

contrary to the hypothesis. Therefore the minimum is q , and

$$q \leq |2A'| - k.$$

Hence A' lies in the interval $[0, q]$ of length

$$q + 1 \leq |2A'| - k + 1.$$

Undoing the translation and scaling gives an arithmetic progression containing A of length at most

$$|A + A| - |A| + 1.$$

□

Worked Example 4.10 (The theorem is sharp at the threshold). Take

$$A = \{0, 1, \dots, k - 2\} \cup \{2k - 3\}.$$

Then A has k elements and its containing interval has length $2k - 2$. Moreover

$$[0, k - 2] + [0, k - 2] = [0, 2k - 4],$$

while adding the outlier gives

$$(2k - 3) + [0, k - 2] = [2k - 3, 3k - 5]$$

and the outlier with itself gives the isolated sum $4k - 6$. The first two intervals are adjacent, so

$$2A = [0, 3k - 5] \cup \{4k - 6\}, \quad |2A| = 3k - 3.$$

Thus increasing the hypothesis from $3k - 4$ to $3k - 3$ already permits a set whose shortest containing progression has length $2k - 2$. This explicit calculation shows why the constant $3k - 4$ marks the end of the first completely rigid one-dimensional regime.

4.5 How these theorems fit the later inverse machine

Kneser, Vosper, and $3k - 4$ occupy three different levels of inverse theory:

Kneser	periodicity explains deficient growth,
Vosper	exact equality in prime cyclic groups is progression-like,
$3k - 4$	near equality in $\mathbb{Z}$ is still one-dimensional.

Freiman's general theorem, proved later in this volume in its classical parameter-volume form, is what replaces one-dimensional progression structure when the doubling constant is an arbitrary fixed K .

Chapter 5

Modelling Small-Doubling Sets

5.1 Why one needs a model

Fourier analysis is strongest when a set occupies a noticeable proportion of a finite ambient group. A finite set of integers, however, has density zero in $\mathbb{Z}$, and reducing it modulo an enormous prime may leave it with negligible density. The modelling problem is therefore:

Guiding Question 5.1 (Good finite model). Given a finite set A with

$$|A + A| \leq K|A|,$$

can one move A , without changing its short additive relations, into a finite abelian group G' whose order is at most $F(K)|A|$?

If the answer is yes, then the image of A has density at least $F(K)^{-1}$ in G' . Bogolyubov-type Fourier arguments can then produce a large structured neighborhood inside $2A - 2A$.

5.2 Finite sets always have finite models

We first ignore quantitative efficiency.

Lemma 5.2 (Existence of a finite Freiman model). *Let A be a finite subset of an abelian group G and let $s \geq 2$. Then A is Freiman s -isomorphic to a subset of some finite abelian group.*

Proof. Let $\langle A \rangle$ denote the subgroup generated by A . It is finitely generated, so by the structure theorem for finitely generated abelian groups

$$\langle A \rangle \cong H \oplus \mathbb{Z}^d$$

with H finite. Under this identification, A lies inside

$$H \times [-M, M]^d$$

for some M .

Choose an integer

$$q > 4sM.$$

Project the free coordinates modulo q :

$$\pi : H \oplus \mathbb{Z}^d \longrightarrow H \oplus (\mathbb{Z}/q\mathbb{Z})^d.$$

Suppose

$$\pi(a_1) + \cdots + \pi(a_s) = \pi(b_1) + \cdots + \pi(b_s)$$

with $a_i, b_i \in A$. In each free coordinate the integer difference

$$a_1 + \cdots + a_s - b_1 - \cdots - b_s$$

has absolute value at most $2sM < q/2$, yet is divisible by q . Hence it is zero. The torsion coordinates are unchanged. Therefore the original s -term relation holds in $\langle A \rangle$.

The forward preservation of relations is automatic because π is a homomorphism. Thus $\pi|_A$ is a Freiman s -isomorphism. $\square$

The lemma is deliberately wasteful: q depends on the diameter M , which may be arbitrarily large compared with $|A|$. A good model theorem must exploit small doubling to remove that dependence.

5.3 Minimal models

Among all finite s -models of A , choose one with smallest ambient group. Call it

$$(A', G')$$

and suppress the prime from the notation. The philosophy is simple:

> if A occupies too tiny a fraction of a minimal model, then some nontrivial character of G' must almost fail to distinguish points of $A - A$; that near-constancy allows us to compress one cyclic direction and obtain a smaller model, a contradiction.

The compression step itself is elementary and worth proving before the analytic input that produces the character.

5.4 Compressing a cyclic direction

Let

$$\psi : G \rightarrow \mathbb{Z}/q\mathbb{Z}$$

be a homomorphism. We identify intervals in $\mathbb{Z}/q\mathbb{Z}$ with sets of consecutive residues that do not wrap around.

Lemma 5.3 (Short-arc compression). *Let $s \geq 2$. Suppose*

$$\psi(A) \subseteq \{0, 1, \dots, L\} \subset \mathbb{Z}/q\mathbb{Z}$$

with

$$2sL < q - 1.$$

Assume ψ is onto. Then A has a Freiman s -model in an abelian group of cardinality

$$|G| \frac{q-1}{q} < |G|.$$

Proof. Let

$$H = \ker \psi.$$

Choose $z \in G$ with $\psi(z) = 1$. Every $g \in G$ has a unique residue $\lambda \in \{0, 1, \dots, q-1\}$ determined by $\psi(g) = \lambda$. With that λ fixed,

$$h = g - \lambda z \in H$$

is uniquely determined, so

$$g = h + \lambda z, \quad h \in H, \quad 0 \leq \lambda < q.$$

Define

$$G' = H \times \mathbb{Z}/(q-1)\mathbb{Z}$$

and, for $a \in A$ with $\psi(a) = \lambda \in [0, L]$, write

$$a = h_a + \lambda z$$

and set

$$\theta(a) = (h_a, \lambda \bmod q-1).$$

Suppose

$$a_1 + \cdots + a_s = b_1 + \cdots + b_s.$$

Applying ψ gives

$$\lambda_1 + \cdots + \lambda_s \equiv \mu_1 + \cdots + \mu_s \pmod{q}.$$

Both sides lie between 0 and $sL < q/2$, so they are equal as integers. Hence the H coordinates also sum equally, and therefore

$$\theta(a_1) + \cdots + \theta(a_s) = \theta(b_1) + \cdots + \theta(b_s).$$

Conversely, suppose the equality holds after applying θ . The two integer coordinate sums lie in $[0, sL]$, and their difference is divisible by $q-1$. Since

$$sL < (q-1)/2,$$

the difference must be zero. The H -coordinate equality then implies equality of the original sums in G . Thus θ is a Freiman s -isomorphism onto its image.

Finally

$$|G'| = |H|(q-1) = |G|\frac{q-1}{q} < |G|,$$

contradicting minimality whenever (A, G) was a minimal model. □

Idea What compression needs

To contradict minimality it is enough to find a nontrivial cyclic quotient in which A lies in a very short arc. The analytic part of the good-model theorem exists only to produce such a quotient.

5.5 From a nearly constant character to a short arc

Suppose $\gamma : G \rightarrow S^1$ is a character of order q . After identifying the q th roots of unity with $\mathbb{Z}/q\mathbb{Z}$, γ corresponds to a quotient homomorphism

$$\psi : G \rightarrow \mathbb{Z}/q\mathbb{Z}.$$

If the normalized character average of a set S satisfies

$$\left| \frac{1}{|S|} \sum_{s \in S} \gamma(s) \right| \approx 1,$$

then the phases $\gamma(s)$, $s \in S$, point almost in the same direction. Most of S must therefore lie in a short arc.

The following elementary estimate makes that statement quantitative.

Lemma 5.4 (Phase concentration). *Let S be finite, let $z_s \in S^1$, and suppose*

$$\left| \frac{1}{|S|} \sum_{s \in S} z_s \right| \geq 1 - \eta.$$

Fix $0 < \delta \leq 1/2$. Then there is an arc of angular radius $2\pi\delta$ containing all but at most

$$\frac{\eta}{1 - \cos(2\pi\delta)} |S|$$

of the points z_s .

Proof. Rotate all phases so that the average is nonnegative real. Then

$$\frac{1}{|S|} \sum_s \Re z_s \geq 1 - \eta.$$

If z_s lies outside the arc $|\arg z| < 2\pi\delta$, then

$$\Re z_s \leq \cos(2\pi\delta).$$

If M points lie outside, then

$$1 - \eta \leq 1 - \frac{M}{|S|} (1 - \cos(2\pi\delta)).$$

Rearranging gives the claim. □

For small δ ,

$$1 - \cos(2\pi\delta) \asymp \delta^2,$$

so an error of order η in Fourier magnitude forces all but $O(\eta\delta^{-2})$ of the set into an arc of width $O(\delta)$.

5.6 A geometric upgrade from $A - A$ to A

If most of $A - A$ lies in a short arc under ψ , then in fact all of A lies in a comparably short arc. The reason is that every translate of a large subset of A must intersect that popular part of $A - A$.

Lemma 5.5 (Difference concentration rectifies the set). *Let $A \subseteq G$ be finite and let*

$$\psi : G \rightarrow \mathbb{Z}/q\mathbb{Z}$$

be a homomorphism. Suppose that an interval $I \subset \mathbb{Z}/q\mathbb{Z}$ of length $L < q/3$ has the property that all but fewer than $|A|/2$ distinct elements $d \in D := A - A$ satisfy $\psi(d) \in I$. Then $\psi(A)$ is contained in a translate of $I - I$, hence in an interval of length at most $2L$.

Proof. Fix $a_* \in A$. Among the $|A|$ differences

$$a - a_* \quad (a \in A),$$

at most $|A|/2$ can belong to the exceptional set

$$D \setminus \psi^{-1}(I).$$

Consequently the subset

$$A_0 := \{a \in A : \psi(a - a_*) \in I\}$$

has cardinality $|A_0| > |A|/2$, and

$$\psi(A_0) \subseteq \psi(a_*) + I.$$

Now fix an arbitrary $a \in A$. The set of differences

$$A_0 - a = \{a_0 - a : a_0 \in A_0\}$$

has cardinality $|A_0| > |A|/2$. Since fewer than $|A|/2$ elements of the whole difference set are exceptional, at least one $a_0 \in A_0$ satisfies

$$\psi(a_0 - a) \in I.$$

For that a_0 ,

$$\psi(a) = \psi(a_0) - \psi(a_0 - a) \in \psi(a_*) + I - I.$$

As a was arbitrary,

$$\psi(A) \subseteq \psi(a_*) + I - I.$$

Because $L < q/3$, the difference $I - I$ may be represented without wrapping as an interval of length at most $2L$. This proves the claim. $\square$

5.7 The analytic compression theorem, proved

The remaining step is analytic but elementary once the correct covering statement has been extracted from small doubling. We give the complete argument, following the Green–Ruzsa rectification method [6].

5.7.1 A covering lemma for fourfold differences

Lemma 5.6 (Green–Ruzsa covering lemma). *Let A, B_1, B_2 be finite subsets of an abelian group and suppose*

$$|A + B_i| = K_i |A| \quad (i = 1, 2).$$

Then there is a set $T \subseteq B_1 + B_2$ with

$$|T| \leq 2K_1 K_2 - 1$$

such that

$$B_1 - B_1 + B_2 - B_2 \subseteq A - A + T - T.$$

Proof. By the two-summand Plünnecke inequality proved in Volume I, there is a nonempty $A' \subseteq A$ such that

$$|A' + B_1 + B_2| \leq K_1 K_2 |A'|.$$

Choose $t_1, t_2, \dots, t_m \in B_1 + B_2$ greedily, for as long as possible, so that

$$\left| (A' + t_j) \setminus \bigcup_{i < j} (A' + t_i) \right| > \frac{|A'|}{2}.$$

Then

$$\left| \bigcup_{i=1}^m (A' + t_i) \right| > \left(1 + \frac{m}{2}\right) |A'|,$$

while the union is contained in $A' + B_1 + B_2$. Hence

$$m \leq 2K_1K_2 - 1.$$

Put $T = \{t_1, \dots, t_m\}$.

Take $u_1, u_2 \in B_1 + B_2$. Maximality says that for each $j = 1, 2$ more than half of the elements $a \in A'$ satisfy

$$a + u_j \in A' + T.$$

The two corresponding subsets of A' therefore intersect. Choose a common $a \in A'$ and write

$$a + u_j = a_j + t_{i_j} \quad (a_j \in A', t_{i_j} \in T).$$

Subtracting gives

$$u_1 - u_2 = a_1 - a_2 + t_{i_1} - t_{i_2} \in A - A + T - T.$$

Since every element of $B_1 - B_1 + B_2 - B_2$ is $u_1 - u_2$ for suitable $u_1, u_2 \in B_1 + B_2$, the result follows. $\square$

Corollary 5.7 (The difference set is finitely covering). *Suppose*

$$\min\{|2A|, |A - A|\} \leq K|A|$$

and put $D = A - A$. Then there is a set T with

$$|T| \leq 2K^2 - 1$$

such that

$$2D \subseteq D + T - T.$$

Consequently, for every integer $m \geq 1$,

$$(m + 1)D \subseteq D + m(T - T).$$

Proof. If $|2A| \leq K|A|$, apply ?? with $B_1 = B_2 = A$. If $|A - A| \leq K|A|$, apply it with $B_1 = B_2 = -A$. In either case the left side is $2(A - A) = 2D$. The iterated inclusion follows by induction: add D to the m th inclusion and use $2D \subseteq D + T - T$. $\square$

5.7.2 Counting the iterated covering set

For integers $k, m \geq 1$, let $J(k, m)$ be the number of integer vectors $x = (x_1, \dots, x_k)$ satisfying

$$\sum_i x_i^+ = \sum_i x_i^- \leq m,$$

where $x_i^+ = \max(x_i, 0)$ and $x_i^- = \max(-x_i, 0)$.

Lemma 5.8 (Coefficient-vector count). *For $m \geq k$,*

$$J(k, m) < \left(\frac{14m}{k}\right)^k.$$

Proof. Every admissible vector satisfies

$$\sum_i |x_i| \leq 2m.$$

Choose the signs of the k coordinates in at most 2^k ways and then choose nonnegative integers $y_i = |x_i|$ with $\sum y_i \leq 2m$. Stars and bars gives at most

$$2^k \binom{2m+k}{k}$$

possibilities. Using

$$\binom{2m+k}{k} \leq \frac{(2m+k)^{2m+k}}{k^k (2m)^{2m}} \leq \left(\frac{7m}{k}\right)^k$$

for $m \geq k$ gives the claim. For the last inequality, put $k = tm$, $0 < t \leq 1$; it is equivalent to

$$(2+t)^{2+t} \leq 4 \cdot 7^t,$$

which holds at $t = 0, 1$ and throughout the interval by convexity of $x \mapsto x \log x$. $\square$

Lemma 5.9 (Polynomial growth from finite covering). *Let B be finite and suppose*

$$2B \subseteq B + T - T, \quad |T| = k.$$

Then for $m \geq k$,

$$|(m+1)B| < \left(\frac{14m}{k}\right)^k |B|.$$

Proof. The covering inclusion iterates to

$$(m+1)B \subseteq B + m(T - T).$$

Writing $T = \{t_1, \dots, t_k\}$, every element of $m(T - T)$ is

$$\sum_{i=1}^k x_i t_i$$

with $\sum x_i^+ = \sum x_i^- \leq m$. Hence

$$|m(T - T)| \leq J(k, m),$$

and ?? gives the result. $\square$

5.7.3 High Fourier moments

For this subsection use the unnormalized Fourier transform

$$\widehat{1_B}(\gamma) = \sum_{b \in B} \gamma(b)$$

on a finite abelian group G of order N .

Lemma 5.10 (Sparse finite-covering sets have a near-maximal coefficient). *Let $B \subseteq G$ have density $\beta = |B|/N$. Suppose*

$$2B \subseteq B + T - T, \quad |T| = k,$$

and

$$\beta \leq 14^{-k-1}.$$

Then there is a nontrivial character γ for which

$$|\widehat{1_B}(\gamma)| \geq (1 - \eta)|B|, \quad \eta = 18 \frac{\beta^{1/k} \log(1/\beta)}{k}.$$

Proof. Fix $m \geq k$ and let

$$R = |(m+1)B|.$$

Let $r_{m+1}(x)$ count representations

$$x = b_1 + \cdots + b_{m+1}, \quad b_i \in B.$$

Then

$$\sum_x r_{m+1}(x) = |B|^{m+1}.$$

Since r_{m+1} is supported on a set of size R , Cauchy–Schwarz yields

$$\sum_x r_{m+1}(x)^2 \geq \frac{|B|^{2m+2}}{R}.$$

Parseval therefore gives

$$\sum_{\gamma \in \widehat{G}} |\widehat{1_B}(\gamma)|^{2m+2} = N \sum_x r_{m+1}(x)^2 \geq \frac{N}{R} |B|^{2m+2}.$$

Subtract the trivial character, whose Fourier coefficient is $|B|$:

$$\sum_{\gamma \neq 1} |\widehat{1_B}(\gamma)|^{2m+2} \geq \left(\frac{N}{R} - 1 \right) |B|^{2m+2}.$$

On the other hand Parseval at the second moment gives

$$\sum_{\gamma \neq 1} |\widehat{1_B}(\gamma)|^2 \leq N|B|.$$

If

$$M = \max_{\gamma \neq 1} |\widehat{1_B}(\gamma)|,$$

then

$$M^{2m} N |B| \geq \left(\frac{N}{R} - 1 \right) |B|^{2m+2},$$

so

$$M^{2m} \geq \left(\frac{1}{R} - \frac{1}{N} \right) |B|^{2m+1}. \tag{5.1}$$

By ??,

$$R < \left(\frac{14m}{k} \right)^k |B|.$$

Substitution into (5.1) gives

$$M \geq \left(\left(\frac{k}{14m} \right)^k - \beta \right)^{1/(2m)} |B|.$$

Choose

$$m = \left\lfloor \frac{k(2\beta)^{-1/k}}{14} \right\rfloor.$$

The density hypothesis ensures $m \geq k$. Since this m is no larger than the quantity inside the floor,

$$\left(\frac{k}{14m} \right)^k \geq 2\beta,$$

and hence

$$M \geq \beta^{1/(2m)} |B|.$$

Also the same density hypothesis gives

$$m \geq \frac{k\beta^{-1/k}}{36}.$$

Therefore

$$\beta^{1/(2m)} = \exp\left(-\frac{\log(1/\beta)}{2m}\right) > 1 - 18 \frac{\beta^{1/k} \log(1/\beta)}{k},$$

using $e^{-x} > 1 - x$. This proves the lemma. $\square$

Theorem 5.11 (Green–Ruzsa analytic compression). *Let $A \subseteq G$ be finite, $|G| = N$, and suppose*

$$|A + A| \leq K|A|.$$

Put

$$D = A - A, \quad \tau = |D|/N.$$

If

$$\tau \leq 14^{-2K^2},$$

then there is a nontrivial character γ such that

$$\left| \frac{1}{|D|} \sum_{d \in D} \gamma(d) \right| \geq 1 - \eta,$$

where, with a harmless adjustment of absolute constants,

$$\eta \leq 10K^{-2} \tau^{1/(2K^2)} \log(1/\tau).$$

Proof. By ??, the set D is k -covering for some

$$k \leq 2K^2 - 1.$$

Apply ?? to D . Under the displayed sparsity condition, the function

$$x \mapsto \frac{\tau^{1/x}}{x}$$

is increasing for $1 \leq x \leq 2K^2$ because $\log(1/\tau) \geq 2K^2 \log 14 > x$. Thus the worst bound occurs at $x = 2K^2$ and

$$18 \frac{\tau^{1/k} \log(1/\tau)}{k} \leq 9K^{-2} \tau^{1/(2K^2)} \log(1/\tau).$$

Replacing 9 by 10 absorbs the integral rounding in the covering number. Dividing the resulting Fourier estimate by $|D|$ gives the theorem. $\square$

5.8 The good-model theorem

Theorem 5.12 (Good finite models). *For every $K \geq 1$ and integer $s \geq 2$ there is a constant $F_s(K)$ such that every finite set A in an abelian group with*

$$|A + A| \leq K|A|$$

is Freiman s -isomorphic to a subset of a finite abelian group G' satisfying

$$|G'| \leq F_s(K)|A|.$$

One may take a classical bound of the shape

$$F_s(K) \leq (CsK)^{CK^2}$$

for an absolute constant C .

Proof architecture Proof of the good-model theorem

1. By ??, choose a finite s -model of minimal ambient size.
2. Suppose its density α is smaller than $F_s(K)^{-1}$.
3. Apply the analytic compression theorem to $D = A - A$ and obtain a nontrivial character almost constant on D .
4. By ??, almost all of D lies in a short arc of the corresponding cyclic quotient.
5. Since $|D| \leq K^2|A|$ by Ruzsa calculus, choose the parameters so that the exceptional part has size less than $|A|/2$.
6. Apply ??: all of A lies in an arc of length less than $q/(2s)$.
7. Apply ?? and obtain a smaller s -model, contradicting minimality.

Proof. Let (A, G) be a minimal finite s -model and write

$$\alpha = \frac{|A|}{|G|}.$$

Suppose, toward a contradiction, that

$$|G| > F_s(K)|A|,$$

where $F_s(K) = (CsK)^{CK^2}$ and the absolute constant C will be chosen large. Let

$$D = A - A, \quad \tau = \frac{|D|}{|G|}.$$

Volume I gives

$$|D| \leq K^2|A|,$$

so

$$\tau \leq K^2\alpha < K^2F_s(K)^{-1}.$$

For C sufficiently large this implies both

$$\tau \leq 14^{-2K^2}$$

and, by the elementary bound $t^a \log(1/t) \rightarrow 0$ as $t \rightarrow 0$,

$$10K^{-2}\tau^{1/(2K^2)} \log(1/\tau) \leq c_0s^{-2}K^{-2} \tag{5.2}$$

for an absolute constant $c_0 > 0$ as small as we wish. (Substituting $\tau \leq K^2(CsK)^{-CK^2}$ shows directly that the left side is bounded by a negative power of CsK , times $O(K^2 \log(CsK))$.)

By ??, there is a nontrivial character γ such that

$$\left| \frac{1}{|D|} \sum_{d \in D} \gamma(d) \right| \geq 1 - \eta, \quad \eta \leq c_0 s^{-2} K^{-2}.$$

Choose

$$\delta = \frac{1}{100s}.$$

Since $1 - \cos(2\pi\delta) \gg \delta^2$, ?? shows that all but

$$O(\eta s^2)|D|$$

elements of D lie in an arc of angular radius $2\pi\delta$. Using $|D| \leq K^2|A|$ and (??), the exceptional number is $O(c_0)|A|$. Taking c_0 small enough makes it less than $|A|/2$.

Let

$$\psi : G \rightarrow \mathbb{Z}/q\mathbb{Z}$$

be the cyclic quotient determined by γ . The phase arc corresponds to an interval I of length less than $q/(40s)$ after harmless rounding. Therefore ?? places $\psi(A)$ in a translate of $I - I$, an interval of length $L < q/(20s)$. Translate A by a preimage of the left endpoint of this interval; translation preserves Freiman s -relations, so we may assume $\psi(A) \subseteq \{0, 1, \dots, L\}$. If $L = 0$, then A lies in the proper subgroup $\ker \psi$, which is already a smaller ambient s -model and contradicts minimality. If $L \geq 1$, then $q > 20s$ and

$$2sL < q/10 < q - 1.$$

Thus the hypothesis of ?? holds, and that lemma again produces an s -model in a strictly smaller ambient group, contradicting minimality.

Hence the density of a minimal model is at least $F_s(K)^{-1}$, which is equivalent to

$$|G| \leq F_s(K)|A|.$$

□

5.9 Worked example: why a huge modulus is useless

Let

$$A = \{0, 1, \dots, N-1\} \cup \{10^{100N}\}.$$

This set does not have bounded doubling independent of N because the isolated point creates many new sums; nevertheless it illustrates the modelling issue. Reducing modulo a prime larger than 10^{100N} preserves all two-term relations, but the density of the image is roughly

$$10^{-100N},$$

far too small for a density-based Fourier theorem to produce a useful neighborhood.

For a genuine small-doubling set, the good-model theorem says that such a gigantic diameter is an artifact of coordinates: there is some Freiman-equivalent finite ambient group whose size is controlled by K and $|A|$, not by the largest integer appearing in A .

5.10 Worked example: modelling a separated box

Let

$$A = \{i + Mj : 0 \leq i, j < L\}$$

with $M \gg L$. Although the diameter is of order ML , the set is Freiman-isomorphic to

$$B = [0, L - 1]^2 \subset \mathbb{Z}^2.$$

Reducing each coordinate of B modulo a prime $p > 4sL$ gives an s -model inside

$$(\mathbb{Z}/p\mathbb{Z})^2$$

of size

$$p^2 = O_s(L^2) = O_s(|A|).$$

Thus in this example the good model is visible by inspection: first remove the artificial scale separation M by passing to intrinsic progression coordinates, then reduce those coordinates modulo a small prime.

The general theorem performs an analogous compression without being given the progression coordinates in advance.

Research window How small can a model be?

A tempting hope is that every K -doubling set has a Freiman model in a group of size $K^{O(1)}|A|$. This is false in complete generality: torsion patterns can force any model to retain many different element orders. Green and Ruzsa constructed examples showing that the minimal model size may be exponentially larger than $|A|$ as a function of K . The lesson is that modelling and structural covering are not the same quantitative problem. Polynomial Freiman–Ruzsa theorems can hold even when a single globally faithful finite model cannot always be polynomially small.

Proof deferred to a later volume Volume III / later research extensions: quantitative modelling obstructions

The Green–Ruzsa lower-bound examples for globally faithful model size are contextual research-window material here; they are not used by the arguments in Volumes I–V. They are contextual here and are not used in the good-model theorem proved above.

5.11 Why order eight will later appear

The classical proof will find a progression inside

$$2A - 2A.$$

To pull additive relations inside that set back through a model, one needs more than a Freiman 2-isomorphism on A . A relation among two elements of $2A - 2A$ expands into a relation involving up to eight elements of A . Thus it is convenient to build a Freiman 8-model at the outset.

This is a typical bookkeeping principle:

Idea Choose the model order from the future algebra

If a later argument manipulates elements of $kA - \ell A$ and must preserve two-term relations among them, then the original model should preserve relations involving at least $2(k + \ell)$ elements of A .

We shall therefore use ?? mainly with $s = 8$.

Chapter 6

Bohr Sets: Approximate Annihilators in Physical Space

6.1 Characters as additive coordinates

Let G be a finite abelian group and let $\widehat{G}$ be its character group. We use normalized Fourier transform

$$\widehat{f}(\gamma) = \mathbb{E}_{x \in G} f(x) \overline{\gamma(x)}.$$

Volume I proved inversion, Plancherel, and the convolution identity. Here we use characters not merely as Fourier basis functions but as approximate coordinates on G .

Write every $z \in S^1$ as

$$z = e^{2\pi i \theta}, \quad \theta \in \mathbb{R}/\mathbb{Z},$$

and define

$$\|z\|_{\mathbb{T}} = \min_{m \in \mathbb{Z}} |\theta - m|.$$

If $\gamma \in \widehat{G}$, abbreviate

$$\|\gamma(x)\|_{\mathbb{T}}$$

for the distance of its phase from 1.

Definition 6.1 (Bohr set). Let $\Gamma \subseteq \widehat{G}$ be finite and let $\rho > 0$. The Bohr set with frequency set Γ and radius ρ is

$$B(\Gamma, \rho) = \{x \in G : \|\gamma(x)\|_{\mathbb{T}} \leq \rho \text{ for every } \gamma \in \Gamma\}.$$

Its rank is at most $|\Gamma|$. For $\rho \geq 1/2$ the condition is vacuous and the Bohr set is all of G ; the interesting range is therefore $0 < \rho \leq 1/2$.

The phrase “at most” matters because the characters in Γ may satisfy dependencies.

Idea Approximate annihilator

The exact annihilator of Γ is

$$\Gamma^\perp = \{x : \gamma(x) = 1 \ \forall \gamma \in \Gamma\},$$

a subgroup. A Bohr set relaxes equality to approximate equality. It is therefore an approximate subgroup defined by Fourier constraints.

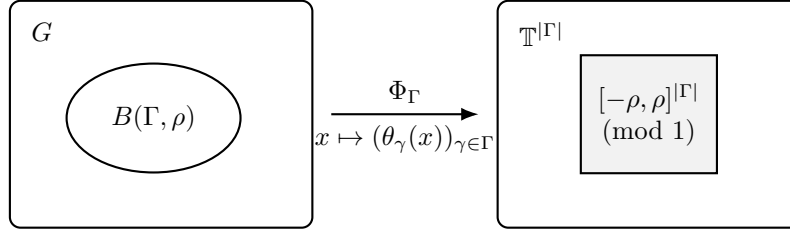


Figure 6.1: A Bohr set is the inverse image of a small coordinate box under the character map: $B(\Gamma, \rho) = \Phi_\Gamma^{-1}([- \rho, \rho]^{|\Gamma|})$. This picture explains both the “approximate annihilator” language and the pseudometric d_Γ .

6.2 Elementary geometry of Bohr sets

Proposition 6.2 (Symmetry, nesting, and addition). *For any finite $\Gamma \subseteq \widehat{G}$:*

1. $0 \in B(\Gamma, \rho)$ and $-B(\Gamma, \rho) = B(\Gamma, \rho)$;
2. if $0 < \rho \leq \rho'$, then

$$B(\Gamma, \rho) \subseteq B(\Gamma, \rho');$$

3. for $\rho + \rho' \leq 1/2$,

$$B(\Gamma, \rho) + B(\Gamma, \rho') \subseteq B(\Gamma, \rho + \rho').$$

Proof. The first two statements are immediate. For the third, if x and y belong to the two Bohr sets, then for every $\gamma \in \Gamma$,

$$\gamma(x + y) = \gamma(x)\gamma(y).$$

On $\mathbb{R}/\mathbb{Z}$, phase distance satisfies the triangle inequality, so

$$\|\gamma(x + y)\|_{\mathbb{T}} \leq \|\gamma(x)\|_{\mathbb{T}} + \|\gamma(y)\|_{\mathbb{T}} \leq \rho + \rho'.$$

□

Thus a Bohr set behaves like a ball for the pseudometric

$$d_\Gamma(x, y) = \max_{\gamma \in \Gamma} \|\gamma(x - y)\|_{\mathbb{T}}.$$

6.3 Bohr sets cannot be too small

Proposition 6.3 (Elementary size lower bound). *Let $d = |\Gamma|$ and $0 < \rho \leq 1/2$. Then*

$$|B(\Gamma, \rho)| \geq \left(\frac{\rho}{2}\right)^d |G|.$$

Proof. Consider the map

$$\Phi : G \rightarrow (\mathbb{R}/\mathbb{Z})^d, \quad x \mapsto (\theta_\gamma(x))_{\gamma \in \Gamma},$$

where $\gamma(x) = e^{2\pi i \theta_\gamma(x)}$.

Let

$$m = \lceil 1/\rho \rceil.$$

Partition the d -torus into m^d half-open cubes of side length $1/m$. By the pigeonhole principle, one cube contains at least

$$|G|/m^d$$

points of $\Phi(G)$ counted with multiplicity from G . Let $S \subseteq G$ be the corresponding preimage and fix $x_0 \in S$. For every $x \in S$, the points $\Phi(x)$ and $\Phi(x_0)$ lie in the same cube, so each coordinate of

$$\Phi(x - x_0)$$

has phase distance at most $1/m \leq \rho$. Hence

$$S - x_0 \subseteq B(\Gamma, \rho).$$

Translation is injective, so

$$|B(\Gamma, \rho)| \geq |S| \geq |G|m^{-d}.$$

Since $\rho \leq 1/2$,

$$m \leq \frac{1}{\rho} + 1 \leq \frac{2}{\rho},$$

which gives the stated bound. $\square$

The estimate is crude but already explains why bounded-rank, fixed-radius Bohr sets occupy positive density.

Worked Example 6.4 (A single frequency in a cyclic group). Take $G = \mathbb{Z}/N\mathbb{Z}$ and

$$\gamma_k(x) = e^{2\pi i k x / N}.$$

If $\gcd(k, N) = 1$, then γ_k permutes the N th roots of unity. The Bohr set $B(\{\gamma_k\}, \rho)$ is therefore the inverse image of a short arc around 1. After multiplying by k^{-1} modulo N , it becomes a short interval centred at 0.

For a completely explicit instance, take $N = 16$, $k = 1$ and $\rho = 1/8$. The condition is

$$\left\| \frac{x}{16} \right\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{1}{8},$$

so, using representatives in $\{-8, -7, \dots, 7\}$,

$$B(\{\gamma_1\}, 1/8) = \{-2, -1, 0, 1, 2\} \pmod{16}.$$

At radius $1/16$ we similarly obtain $\{-1, 0, 1\}$. Hence the addition law is visible without asymptotic notation:

$$B(\{\gamma_1\}, 1/16) + B(\{\gamma_1\}, 1/16) = \{-2, -1, 0, 1, 2\} = B(\{\gamma_1\}, 1/8).$$

Thus a rank-one Bohr set is literally a modular arithmetic progression at small radius, and the abstract inclusion $B(\rho) + B(\rho') \subseteq B(\rho + \rho')$ is just interval addition in phase coordinates.

6.4 Dependent frequencies

If Γ contains many algebraically dependent characters, the rank $|\Gamma|$ overstates the true complexity.

For example, in $G = \mathbb{F}_2^n$, every character has the form

$$\gamma_\xi(x) = (-1)^{\xi \cdot x}.$$

If $0 < \rho < 1/2$, then

$$\|\gamma_\xi(x)\|_{\mathbb{T}} \leq \rho$$

forces

$$\gamma_\xi(x) = 1,$$

so

$$B(\Gamma, \rho) = \{x : \xi \cdot x = 0 \text{ for all } \xi \in \Gamma\}.$$

This is the orthogonal complement of the linear span of Γ . Its codimension is

$$\dim \text{span}(\Gamma),$$

not $|\Gamma|$.

This finite-field example foreshadows Chang's theorem: a large Fourier spectrum may contain many frequencies, but only a controlled number can be genuinely independent.

6.5 Regular radii

Later arguments convolve with normalized indicators of Bohr sets. To do so efficiently, we want small changes in radius to cause only small changes in cardinality. The first ingredient is a genuine doubling estimate for the Bohr metric; unlike the crude lower bound above, it compares two nearby radii and is therefore independent of the ambient density of the Bohr set.

Lemma 6.5 (Bohr doubling). *Let $|\Gamma| = d$ and $0 < r \leq 1/4$. Then*

$$|B(\Gamma, 2r)| \leq 4^d |B(\Gamma, r)|.$$

Proof. For each $x \in B(\Gamma, 2r)$ choose the unique phase representatives

$$\theta_\gamma(x) \in [-2r, 2r] \quad (\gamma \in \Gamma)$$

with $\gamma(x) = e^{2\pi i \theta_\gamma(x)}$. Partition the cube $[-2r, 2r]^d$ into at most 4^d half-open boxes of side length at most r . From each occupied box choose one element $x_j \in B(\Gamma, 2r)$ whose phase vector lies in that box.

If x has its phase vector in the same box as x_j , then for every $\gamma \in \Gamma$ the difference

$$\theta_\gamma(x) - \theta_\gamma(x_j)$$

has absolute value at most r . Since both phases lie in $[-1/2, 1/2]$, this difference represents the phase of $\gamma(x - x_j)$ without any ambiguity that can increase its distance from 0 modulo 1. Hence

$$x - x_j \in B(\Gamma, r).$$

Thus $B(\Gamma, 2r)$ is covered by at most 4^d translates of $B(\Gamma, r)$, proving the claim. $\square$

Definition 6.6 (Regular Bohr set). Let $B(\Gamma, \rho)$ have rank d . We call the radius ρ *regular* if for every real η with

$$|\eta| \leq \frac{1}{1000d},$$

one has the explicit logarithmic estimate

$$\left| \log \frac{|B(\Gamma, (1 + \eta)\rho)|}{|B(\Gamma, \rho)|} \right| \leq 100d|\eta|.$$

The numerical constants are conventional. Since $100d|\eta| \leq 1/10$, this implies the more familiar multiplicative form

$$|B(\Gamma, (1 + \eta)\rho)| = (1 + O(d|\eta|))|B(\Gamma, \rho)|.$$

The logarithmic formulation is simply cleaner for the existence proof.

Proposition 6.7 (Regular radii exist). *For every $0 < \rho \leq 1/2$ and finite Γ of size $d \geq 1$, there is*

$$\rho' \in [\rho/2, \rho]$$

such that $B(\Gamma, \rho')$ is regular.

Proof. We give the standard monotone-selection argument because it is easy to misuse if one bounds Bohr-set sizes only against $|G|$.

Define

$$F(t) = \log |B(\Gamma, e^t \rho)|, \quad -\log 2 \leq t \leq 0.$$

The function F is nondecreasing. By ??, applied with $r = \rho/2 \leq 1/4$,

$$F(0) - F(-\log 2) = \log \frac{|B(\Gamma, \rho)|}{|B(\Gamma, \rho/2)|} \leq d \log 4. \quad (5.1)$$

This $O(d)$ bound, rather than a bound involving $\log(1/\rho)$, is the crucial point.

Put

$$J = \left[-\log 2 + \frac{1}{200d}, -\frac{1}{200d} \right].$$

For $d \geq 1$, J has length bounded below by an absolute positive constant. Suppose, for contradiction, that no $e^t \rho$ with $t \in J$ is regular. Then for every $t \in J$ there exists η_t , with

$$0 < |\eta_t| \leq \frac{1}{1000d},$$

such that, writing $s_t = \log(1 + \eta_t)$,

$$|F(t + s_t) - F(t)| > 100d|\eta_t|. \quad (5.2)$$

This is exactly the negation of the explicit regularity inequality. Since $|\eta_t| \leq 1/100$, we have

$$\frac{1}{2}|\eta_t| \leq |s_t| \leq 2|\eta_t|,$$

so (5.2) implies

$$|F(t + s_t) - F(t)| > 50d|s_t|. \quad (5.3)$$

The choice of J ensures that both endpoints t and $t + s_t$ remain in $[-\log 2, 0]$.

Associate to t the interval I_t having endpoints t and $t + s_t$. These intervals cover J in the sense that every $t \in J$ is an endpoint of its own I_t . The one-dimensional $5r$ covering lemma gives a pairwise disjoint subcollection I_{t_j} whose fivefold concentric dilates cover the union of the original intervals, hence cover J . Consequently

$$\sum_j |I_{t_j}| \geq \frac{1}{5}|J|. \quad (5.4)$$

For completeness, the $5r$ lemma is obtained by selecting intervals greedily in decreasing dyadic length classes: every rejected interval meets a selected interval at least as long, and is therefore contained in its fivefold dilate.

Because F is nondecreasing and the selected intervals are disjoint, the sum of its increments over them is at most the total variation in (5.1). On the other hand, (5.3) and (5.4) give

$$d \log 4 \geq \sum_j |F(t_j + s_{t_j}) - F(t_j)| > 50d \sum_j |I_{t_j}| \geq 10d|J|.$$

But $|J| = \log 2 - O(1/d)$, and even for $d = 1$ the last lower bound is larger than $d \log 4$, a contradiction. Thus some $t \in J$ satisfies

$$\left| \log \frac{|B(\Gamma, (1+\eta)e^t \rho)|}{|B(\Gamma, e^t \rho)|} \right| \leq 100d|\eta| \quad \left(|\eta| \leq \frac{1}{1000d} \right).$$

Taking $\rho' = e^t \rho$ completes the proof. $\square$

6.6 Bogolyubov's lemma in a general finite abelian group

We now reach the first major use of Bohr sets.

Theorem 6.8 (Bogolyubov). *Let $A \subseteq G$ have density*

$$\alpha = \frac{|A|}{|G|} > 0.$$

Then $2A - 2A$ contains a Bohr set

$$B(\Gamma, 1/6)$$

with

$$|\Gamma| \leq 2\alpha^{-2}.$$

Consequently

$$|B(\Gamma, 1/6)| \geq (1/12)^{2\alpha^{-2}} |G|.$$

Proof architecture Why four copies of A appear

The convolution

$$1_A * 1_A * 1_{-A} * 1_{-A}$$

is supported on $2A - 2A$. Its Fourier transform is

$$|\widehat{1_A}(\gamma)|^4,$$

which is nonnegative. We split the spectrum into large and small coefficients. Small coefficients have small total fourth-power mass; on a Bohr set generated by the large coefficients, all the surviving Fourier phases point roughly in the positive direction. The convolution is therefore positive there.

Proof. Use normalized convolution

$$(f * g)(x) = \mathbb{E}_y f(y)g(x - y).$$

Let

$$F = 1_A * 1_A * 1_{-A} * 1_{-A}.$$

Then $F(x) > 0$ implies $x \in 2A - 2A$.

Fourier inversion gives

$$F(x) = \sum_{\gamma \in \widehat{G}} |\widehat{1_A}(\gamma)|^4 \gamma(x).$$

The trivial character contributes

$$\alpha^4.$$

Set

$$\tau = \frac{\alpha^{3/2}}{\sqrt{2}}$$

and define the large spectrum

$$\Gamma = \{\gamma : |\widehat{1_A}(\gamma)| \geq \tau\}.$$

By Parseval,

$$\sum_{\gamma} |\widehat{1_A}(\gamma)|^2 = \mathbb{E} 1_A^2 = \alpha,$$

so

$$|\Gamma| \tau^2 \leq \alpha.$$

Since $\tau^2 = \alpha^3/2$,

$$|\Gamma| \leq 2\alpha^{-2}.$$

Now let $x \in B(\Gamma, 1/6)$. For every $\gamma \in \Gamma$,

$$\Re \gamma(x) \geq \cos(\pi/3) = \frac{1}{2}.$$

In particular, the large-spectrum terms have nonnegative real part.

For the complementary spectrum,

$$\begin{aligned} \sum_{\gamma \notin \Gamma} |\widehat{1_A}(\gamma)|^4 &\leq \tau^2 \sum_{\gamma} |\widehat{1_A}(\gamma)|^2 \\ &= \tau^2 \alpha = \frac{\alpha^4}{2}. \end{aligned}$$

Therefore

$$\Re F(x) \geq \alpha^4 - \frac{\alpha^4}{2} > 0.$$

Since F is real and nonnegative by its convolution definition, in fact $F(x) > 0$. Hence

$$x \in 2A - 2A.$$

This proves the containment. The size estimate follows from ??.

□

6.7 Why Bogolyubov is an inverse theorem

The conclusion does not say that A itself is a Bohr set. It says that after four additive operations, a neighborhood of the identity emerges:

$$B(\Gamma, \rho) \subseteq 2A - 2A.$$

This is a form of smoothing. The irregular boundary of A disappears under repeated convolution, while the frequencies at which A has large Fourier mass determine the surviving structured core.

The argument is a prototype for a general scheme:

$$\boxed{\text{large spectrum}} \longrightarrow \boxed{\text{approximate annihilator}} \longrightarrow \boxed{\text{physical-space structure}}.$$

6.8 Worked example: a subgroup

Let $A = H \leq G$ be a subgroup of density α . Then

$$\widehat{1_H}(\gamma) = \begin{cases} \alpha, & \gamma \in H^\perp, \\ 0, & \gamma \notin H^\perp. \end{cases}$$

Thus the large spectrum is exactly the annihilator $H^\perp$ for every threshold below α .

If $\rho < 1/2$ in a group of exponent two, or if one uses radius zero in general, the corresponding Bohr set is

$$(H^\perp)^\perp = H.$$

Bogolyubov's theorem then recovers the exact identity

$$2H - 2H = H.$$

So the theorem is genuinely an approximate version of subgroup duality.

6.9 Worked example: an interval modulo a prime

Let

$$G = \mathbb{Z}/p\mathbb{Z}, \quad A = \{-L, \dots, L\}$$

with $4L < p$. Then

$$2A - 2A = \{-4L, \dots, 4L\}$$

without wraparound. The low-frequency characters

$$\gamma_k(x) = e^{2\pi i kx/p}$$

with small $|k|$ have large Fourier coefficients on A . The Bohr set generated by these characters is a short interval around 0, exactly the kind of structure already visible by inspection.

For a general dense set A , the theorem finds an analogous low-complexity neighborhood without knowing in advance which frequencies play the role of the low integers k .

6.10 The bottleneck: too many large frequencies

The rank bound

$$|\Gamma| \leq 2\alpha^{-2}$$

is enough for a qualitative theorem, but it is poor when α is small. In applications arising from a K -doubling set, a good model may have density exponentially small in a polynomial of K , so an α^{-2} rank is disastrous.

The key refinement is that the large spectrum cannot contain too many *independent* characters. This is the content of Chang's theorem, to which we now turn.

Chapter 7

Dissociated Frequencies and Chang's Theorem

7.1 The defect in the naive spectrum bound

Bogolyubov's lemma used Parseval to bound the *number* of Fourier coefficients above a threshold. But a large spectrum can contain many frequencies generated by only a few basic ones. Since a Bohr set only needs constraints on generators, cardinality is the wrong complexity measure.

The appropriate notion is dissociativity.

Definition 7.1 (Dissociated set of characters). A finite set $\Lambda \subseteq \widehat{G}$ is *dissociated* if the only relation

$$\prod_{\lambda \in \Lambda} \lambda^{\varepsilon_\lambda} = 1, \quad \varepsilon_\lambda \in \{-1, 0, 1\},$$

is the trivial relation $\varepsilon_\lambda = 0$ for every λ .

Define the $\{-1, 0, 1\}$ -span

$$\langle \Lambda \rangle = \left\{ \prod_{\lambda \in \Lambda} \lambda^{\varepsilon_\lambda} : \varepsilon_\lambda \in \{-1, 0, 1\} \right\}.$$

A maximal dissociated subset of a set Γ spans all of Γ in this sense.

Lemma 7.2 (Maximal dissociated sets span). *If $\Lambda \subseteq \Gamma \subseteq \widehat{G}$ is maximal subject to being dissociated, then*

$$\Gamma \subseteq \langle \Lambda \rangle.$$

Proof. If $\gamma \in \Gamma \setminus \langle \Lambda \rangle$, then any relation

$$\gamma^\varepsilon \prod_{\lambda \in \Lambda} \lambda^{\varepsilon_\lambda} = 1, \quad \varepsilon, \varepsilon_\lambda \in \{-1, 0, 1\},$$

with $\varepsilon \neq 0$ would express γ or γ^{-1} as an element of $\langle \Lambda \rangle$. Therefore $\varepsilon = 0$, and dissociativity of Λ forces all other coefficients to vanish. Hence $\Lambda \cup \{\gamma\}$ is dissociated, contradicting maximality. $\square$

Thus it suffices to control the size of dissociated subsets of the large spectrum.

7.2 Rudin's inequality

The analytic input is a Khintchine-type inequality for dissociated characters.

Theorem 7.3 (Rudin inequality). *There is an absolute constant C such that if $\Lambda \subseteq \widehat{G}$ is dissociated, $p \geq 2$, and $(a_\lambda)_{\lambda \in \Lambda}$ are complex coefficients, then*

$$\left\| \sum_{\lambda \in \Lambda} a_\lambda \lambda \right\|_{L^p(G)} \leq C \sqrt{p} \left(\sum_{\lambda \in \Lambda} |a_\lambda|^2 \right)^{1/2}.$$

The L^p norm uses normalized counting measure on G .

Context Why this is a harmonic-analysis prerequisite rather than a black box

Rudin's inequality is the character analogue of the Khintchine inequality for independent random signs. Dissociated characters are not probabilistically independent in every high moment, but Riesz-product estimates make them behave as if they were subgaussian. Appendix ?? proves the precise form above, beginning from a Riesz product and an exponential-moment bound. We use no stronger theorem from harmonic analysis.

A useful dual form follows immediately.

Corollary 7.4 (Dual Rudin bound). *Let $p \geq 2$ and $p' = p/(p-1)$. If $f : G \rightarrow \mathbb{C}$ and Λ is dissociated, then*

$$\left(\sum_{\lambda \in \Lambda} |\widehat{f}(\lambda)|^2 \right)^{1/2} \leq C \sqrt{p} \|f\|_{L^{p'}(G)}.$$

Proof. Put

$$S = \left(\sum_{\lambda \in \Lambda} |\widehat{f}(\lambda)|^2 \right)^{1/2}.$$

If $S = 0$ there is nothing to prove. Otherwise set

$$a_\lambda = \frac{\widehat{f}(\lambda)}{S}.$$

Then $\sum_{\lambda \in \Lambda} |a_\lambda|^2 = 1$. With the normalized inner product

$$\langle u, v \rangle = \mathbb{E}_x u(x) \overline{v(x)},$$

Fourier duality gives

$$\begin{aligned} \left\langle f, \sum_{\lambda \in \Lambda} a_\lambda \lambda \right\rangle &= \sum_{\lambda \in \Lambda} \overline{a_\lambda} \widehat{f}(\lambda) \\ &= \frac{1}{S} \sum_{\lambda \in \Lambda} |\widehat{f}(\lambda)|^2 = S. \end{aligned}$$

Therefore Hölder and ?? yield

$$S \leq \|f\|_{p'} \left\| \sum_{\lambda \in \Lambda} a_\lambda \lambda \right\|_p \leq C \sqrt{p} \|f\|_{p'}.$$

□

7.3 Chang's large-spectrum theorem

For $A \subseteq G$ of density α and $0 < \rho \leq 1$, define

$$\text{Spec}_\rho(A) = \{\gamma \in \widehat{G} : |\widehat{1_A}(\gamma)| \geq \rho\alpha\}.$$

Theorem 7.5 (Chang). *Let $A \subseteq G$ have density $0 < \alpha \leq 1/2$, and let $0 < \rho \leq 1$. Then there is a dissociated set*

$$\Lambda \subseteq \text{Spec}_\rho(A)$$

with

$$|\Lambda| \leq C\rho^{-2} \log(1/\alpha)$$

such that

$$\text{Spec}_\rho(A) \subseteq \langle \Lambda \rangle.$$

Here C is an absolute constant.

Proof architecture The proof in one sentence

Take a maximal dissociated subset Λ of the large spectrum. Every coefficient on Λ is at least $\rho\alpha$, so their ℓ^2 mass is at least $\rho\alpha\sqrt{|\Lambda|}$. Dual Rudin bounds the same mass by $O(\sqrt{p}\alpha^{1-1/p})$. Choosing $p \asymp \log(1/\alpha)$ makes the density loss constant and forces $|\Lambda| = O(\rho^{-2} \log(1/\alpha))$.

Proof. Let Λ be a maximal dissociated subset of $\text{Spec}_\rho(A)$. By ??, it remains only to bound $|\Lambda|$.

Since every $\lambda \in \Lambda$ satisfies

$$|\widehat{1_A}(\lambda)| \geq \rho\alpha,$$

we have

$$\left(\sum_{\lambda \in \Lambda} |\widehat{1_A}(\lambda)|^2 \right)^{1/2} \geq \rho\alpha\sqrt{|\Lambda|}.$$

Apply ?? to $f = 1_A$. Since

$$\|1_A\|_{p'} = \alpha^{1/p'},$$

we obtain

$$\rho\alpha\sqrt{|\Lambda|} \leq C\sqrt{p}\alpha^{1/p'}.$$

Now

$$\frac{1}{p'} = 1 - \frac{1}{p},$$

so after dividing by α ,

$$\rho\sqrt{|\Lambda|} \leq C\sqrt{p}\alpha^{-1/p}.$$

Choose

$$p = \max\{2, \log(1/\alpha)\}.$$

Then

$$\alpha^{-1/p} = \exp\left(\frac{\log(1/\alpha)}{p}\right) \leq e.$$

Therefore

$$|\Lambda| \leq C'\rho^{-2}p \leq C''\rho^{-2}\log(1/\alpha),$$

after enlarging the constant to handle bounded $\log(1/\alpha)$. □

7.4 What Chang's theorem really says

Parseval alone gives

$$|\text{Spec}_\rho(A)| \leq \rho^{-2} \alpha^{-1}.$$

Chang's theorem replaces the factor α^{-1} by $\log(1/\alpha)$, at the price of measuring *dimension* rather than cardinality.

This is a dramatic improvement when A is sparse. The large spectrum may contain exponentially many characters, but they are generated by only

$$O((\rho^{-2} \log(1/\alpha)))$$

basic frequencies.

Worked Example 7.6 (A subspace in $\mathbb{F}_2^n$). Let $A = V \leq \mathbb{F}_2^n$ have codimension d , so $\alpha = 2^{-d}$. Its Fourier transform is supported exactly on the annihilator

$$V^\perp,$$

a d -dimensional subspace of the dual group. Thus the spectrum has $2^d = \alpha^{-1}$ elements but can be generated by $d = \log_2(1/\alpha)$ independent characters.

Chang's logarithmic dependence is therefore qualitatively sharp.

7.5 From a spanning spectrum to a Bohr set

If

$$\Gamma \subseteq \langle \Lambda \rangle$$

and $|\Lambda| = d$, then making every $\lambda \in \Lambda$ very close to 1 makes every $\gamma \in \Gamma$ moderately close to 1.

Lemma 7.7 (Bohr control passes to the span). *If $\Gamma \subseteq \langle \Lambda \rangle$ and $d = |\Lambda|$, then*

$$B\left(\Lambda, \frac{\rho}{d}\right) \subseteq B(\Gamma, \rho)$$

for every $0 < \rho \leq 1/2$.

Proof. Write

$$\gamma = \prod_{\lambda \in \Lambda} \lambda^{\varepsilon_\lambda}, \quad \varepsilon_\lambda \in \{-1, 0, 1\}.$$

If $x \in B(\Lambda, \rho/d)$, the phase triangle inequality gives

$$\|\gamma(x)\|_{\mathbb{T}} \leq \sum_{\lambda \in \Lambda} |\varepsilon_\lambda| \|\lambda(x)\|_{\mathbb{T}} \leq d \frac{\rho}{d} = \rho.$$

□

7.6 Bogolyubov–Chang for small-doubling sets

We now combine energy, Chang's theorem, and the Bohr argument in a form tailored to Freiman theory.

Theorem 7.8 (Bogolyubov–Chang). *Let $A \subseteq G$ have density α and suppose*

$$|A + A| \leq K|A|.$$

Then $2A - 2A$ contains a Bohr set

$$B(\Lambda, c/d)$$

where

$$d = |\Lambda| \leq CK \log(1/\alpha)$$

and $c, C > 0$ are absolute constants.

Proof architecture Why small doubling improves the spectral threshold

The basic Bogolyubov proof compared all Fourier mass with the trivial coefficient α^4 . Small doubling gives much larger additive energy:

$$E(A) \geq K^{-1}|A|^3.$$

After normalization, the total fourth Fourier mass is at least α^3/K . We may therefore call a coefficient “large” already at scale

$$\alpha K^{-1/2},$$

not $\alpha^{3/2}$. Chang’s theorem then costs only $O(K \log(1/\alpha))$ independent frequencies.

Proof. Let

$$F = 1_A * 1_A * 1_{-A} * 1_{-A}.$$

At the origin,

$$F(0) = \sum_{\gamma} |\widehat{1_A}(\gamma)|^4 = \frac{E(A)}{|G|^3} \geq \frac{\alpha^3}{K}.$$

Set

$$\tau = \frac{\alpha}{2\sqrt{K}}$$

and

$$\Gamma = \{\gamma : |\widehat{1_A}(\gamma)| \geq \tau\} = \text{Spec}_{1/(2\sqrt{K})}(A).$$

For the complementary spectrum,

$$\begin{aligned} \sum_{\gamma \notin \Gamma} |\widehat{1_A}(\gamma)|^4 &\leq \tau^2 \sum_{\gamma} |\widehat{1_A}(\gamma)|^2 \\ &= \frac{\alpha^2}{4K} \alpha = \frac{\alpha^3}{4K} \leq \frac{1}{4} F(0). \end{aligned}$$

By Chang’s theorem, there is a dissociated $\Lambda \subseteq \Gamma$ with

$$\Gamma \subseteq \langle \Lambda \rangle$$

and

$$d := |\Lambda| \leq CK \log(1/\alpha).$$

Let

$$x \in B\left(\Lambda, \frac{1}{6d}\right).$$

By ??, every $\gamma \in \Gamma$ satisfies

$$\|\gamma(x)\|_{\mathbb{T}} \leq 1/6,$$

so

$$\Re \gamma(x) \geq 1/2.$$

Therefore

$$\begin{aligned} \Re F(x) &\geq \frac{1}{2} \sum_{\gamma \in \Gamma} |\widehat{1_A}(\gamma)|^4 - \sum_{\gamma \notin \Gamma} |\widehat{1_A}(\gamma)|^4 \\ &\geq \frac{1}{2} \left(F(0) - \frac{1}{4} F(0) \right) - \frac{1}{4} F(0) \\ &= \frac{1}{8} F(0) > 0. \end{aligned}$$

Hence $F(x) > 0$ and $x \in 2A - 2A$. □

This is the spectral heart of the classical Freiman theorem.

7.7 Worked example: the gain over naive Bogolyubov

Suppose a good finite model has density

$$\alpha = e^{-K^2}.$$

The naive Bogolyubov rank bound is of order

$$\alpha^{-2} = e^{2K^2},$$

which is enormous. Bogolyubov–Chang gives instead

$$d = O(K \log(1/\alpha)) = O(K^3).$$

The radius is about $1/d$, so the subsequent geometry-of-numbers step loses exponentially in a polynomial of K , rather than doubly exponentially in K through the rank itself.

The numerical example explains why Chang’s theorem transformed quantitative Freiman theory.

7.8 Worked example: spectrum of a product set

Let

$$G = \mathbb{F}_2^d \times \mathbb{F}_2^m$$

and

$$A = \mathbb{F}_2^d \times S,$$

where $S \subseteq \mathbb{F}_2^m$ has density β . Then A has density $\alpha = \beta$, independent of d . Its Fourier transform factors:

$$\widehat{1_A}(\xi, \eta) = \widehat{1_{\mathbb{F}_2^d}}(\xi) \widehat{1_S}(\eta).$$

The first factor vanishes unless $\xi = 0$. Hence every large Fourier coefficient lies in the m -dimensional dual slice

$$\{0\} \times \widehat{\mathbb{F}_2^m}.$$

The ambient dimension $d + m$ may be huge, but the spectrum ignores the entire subgroup direction on which A is already invariant. This is the Fourier analogue of separating subgroup structure from progression structure.

Research window Large spectra as an inverse principle

Chang's theorem is an early example of a recurring modern phenomenon: a set of large correlations may be huge in cardinality but low-dimensional in an appropriate additive sense. Higher-order Fourier analysis replaces characters by polynomial phases or by nilsequences (observables sampled along polynomial orbits on nilpotent homogeneous spaces, defined precisely in Volume V), and analogous “large spectrum has structure” principles become inverse theorems for Gowers norms (the higher-order cube-averaging uniformity norms defined in Volume III). The linear theory developed here is the model case for that much deeper story.

Chapter 8

From Bohr Sets to Proper Progressions

8.1 The remaining geometric step

Chapters ??–?? have produced the following situation. Inside a finite abelian group G we know that

$$B(\Gamma, \rho) \subseteq 2A - 2A,$$

where $d = |\Gamma|$ is controlled and ρ is not too small. To obtain a Freiman theorem we must replace the implicit Fourier description

$$\|\gamma(x)\|_{\mathbb{T}} \leq \rho \quad (\gamma \in \Gamma)$$

by an explicit low-dimensional progression.

This is where geometry of numbers enters.

Idea The frequency map turns a Bohr set into lattice points in a box

Map

$$x \mapsto (\theta_1(x), \dots, \theta_d(x)) \in \mathbb{T}^d,$$

where $\gamma_j(x) = e^{2\pi i \theta_j(x)}$. The image of G is a finite subgroup of the torus. Lifting that subgroup to $\mathbb{R}^d$ produces a lattice containing $\mathbb{Z}^d$. The Bohr set is precisely the preimage of lattice points in a small cube. Successive minima then find independent short lattice vectors, and integer combinations of those vectors form the desired progression.

8.2 From a finite subgroup of a torus to a lattice

Let

$$\Phi : G \rightarrow \mathbb{T}^d, \quad \Phi(x) = (\theta_1(x), \dots, \theta_d(x))$$

be the homomorphism determined by $\Gamma = \{\gamma_1, \dots, \gamma_d\}$. Let

$$K = \ker \Phi = \bigcap_{j=1}^d \ker \gamma_j.$$

Then

$$H := \Phi(G) \cong G/K$$

is a finite subgroup of $\mathbb{T}^d$.

Let

$$\pi : \mathbb{R}^d \rightarrow \mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$$

be the quotient map and define

$$\Lambda = \pi^{-1}(H).$$

Then Λ is a full-rank lattice containing $\mathbb{Z}^d$.

Lemma 8.1 (Covolume of the lifted lattice). *Let G be a finite abelian group, let $\Phi : G \rightarrow \mathbb{T}^d$ be a homomorphism, put $H := \Phi(G)$, let $\pi : \mathbb{R}^d \rightarrow \mathbb{T}^d$ be the quotient map, and set $\Lambda := \pi^{-1}(H)$. Then Λ is a full-rank lattice containing $\mathbb{Z}^d$ and*

$$\det \Lambda = |H|^{-1}.$$

Proof. Because $\mathbb{Z}^d \subseteq \Lambda$, the index

$$[\Lambda : \mathbb{Z}^d]$$

is finite. Reduction modulo $\mathbb{Z}^d$ identifies this quotient with H , so the index is $|H|$.

For lattices $L_1 \subseteq L_2$ of the same rank,

$$\det L_1 = [L_2 : L_1] \det L_2.$$

Since $\det \mathbb{Z}^d = 1$, we obtain

$$1 = |H| \det \Lambda.$$

□

8.3 The cube corresponding to the Bohr set

Let

$$Q_\rho = [-\rho, \rho]^d \subset \mathbb{R}^d.$$

If $0 < \rho < 1/2$, every point of H whose torus coordinates have distance at most ρ from zero has a unique lift to Q_ρ . Consequently

$$B(\Gamma, \rho)/K \leftrightarrow \Lambda \cap Q_\rho.$$

The subgroup K is already a structured component. We therefore need only find a proper progression in the quotient H generated by some lattice points of Q_ρ .

8.4 Successive minima

For a centrally symmetric convex body Q and a full-rank lattice Λ , the i th successive minimum is

$$\lambda_i(Q, \Lambda) = \inf\{\lambda > 0 : \lambda Q \text{ contains } i \text{ linearly independent points of } \Lambda\}.$$

Thus

$$0 < \lambda_1 \leq \dots \leq \lambda_d.$$

Choose linearly independent lattice vectors

$$v_i \in \lambda_i Q.$$

We use the following form of Minkowski's second theorem.

Theorem 8.2 (Minkowski’s second theorem, upper half). *For a centrally symmetric convex body $Q \subset \mathbb{R}^d$ with nonempty interior and a full-rank lattice Λ ,*

$$\lambda_1 \cdots \lambda_d \operatorname{vol}(Q) \leq 2^d \det \Lambda.$$

Remark 8.3 (Proof location). A complete proof of Minkowski’s second theorem is given in Appendix ?? . The present chapter uses only its upper bound, which converts Bohr volume into the product of successive-minimum scales.

Appendix ?? explains the theorem and the elementary lattice facts surrounding it; Appendix ?? gives the complete proof of Minkowski’s second theorem. No deeper geometry of numbers is used in the main text.

For the cube Q_ρ ,

$$\operatorname{vol}(Q_\rho) = (2\rho)^d.$$

Combining this with ?? gives

$$\prod_{i=1}^d \lambda_i \leq \frac{1}{\rho^d |H|}. \quad (8.1)$$

8.5 Building the progression

Theorem 8.4 (A Bohr set contains a large proper coset progression). *Let G be a finite abelian group, let $\Gamma \subseteq \widehat{G}$ have cardinality $d \geq 1$, and let*

$$0 < \rho \leq \frac{1}{4}.$$

Then $B(\Gamma, \rho)$ contains a proper symmetric coset progression M of rank at most d satisfying

$$|M| \geq \left(\frac{\rho}{2d} \right)^d |G|. \quad (8.2)$$

Proof architecture The proof

1. Quotient by the exact annihilator K of Γ .
2. Lift the finite torus subgroup G/K to a lattice $\Lambda \subset \mathbb{R}^d$ of covolume $1/|G/K|$.
3. Take lattice vectors v_i at the successive-minimum scales λ_i .
4. Keep only directions with $\lambda_i \leq 1/(2d)$ and allow roughly $1/(d\lambda_i)$ integer steps in direction v_i .
5. The triangle inequality for the gauge of the cube keeps every such combination inside Q_ρ .
6. Minkowski’s product bound guarantees that the product of the side lengths is large.
7. Because $2Q_\rho$ does not contain a nonzero point of $\mathbb{Z}^d$ when $\rho \leq 1/4$, the progression is proper modulo the torus lattice.

Proof. Use the notation of the preceding sections. Let

$$\lambda_1 \leq \cdots \leq \lambda_d$$

be the successive minima of Q_ρ with respect to Λ , and choose linearly independent

$$v_i \in \lambda_i Q_\rho \cap \Lambda.$$

Let r be the largest index for which

$$\lambda_r \leq \frac{1}{2d};$$

if no such index exists, set $r = 0$. For $1 \leq i \leq r$ define

$$L_i = \left\lfloor \frac{1}{d\lambda_i} \right\rfloor.$$

Since $1/(d\lambda_i) \geq 2$,

$$L_i \geq \frac{1}{2d\lambda_i}.$$

Consider

$$P_\Lambda = \left\{ \sum_{i=1}^r n_i v_i : |n_i| \leq L_i \right\}.$$

The Minkowski functional of Q_ρ satisfies

$$\|v_i\|_{Q_\rho} \leq \lambda_i.$$

Hence for $x = \sum n_i v_i$ in P_Λ ,

$$\|x\|_{Q_\rho} \leq \sum_{i=1}^r |n_i| \lambda_i \leq \sum_{i=1}^r \frac{1}{d} \leq 1.$$

Thus

$$P_\Lambda \subseteq Q_\rho \cap \Lambda.$$

Project P_Λ to $H = \Lambda/\mathbb{Z}^d$. We claim the projection is injective. If two elements $x, y \in P_\Lambda$ have the same image, then

$$x - y \in \mathbb{Z}^d.$$

But $x - y \in 2Q_\rho \subseteq [-1/2, 1/2]^d$. Every nonzero vector of $\mathbb{Z}^d$ has at least one coordinate of absolute value at least 1, so the only integer vector in this cube is 0. Thus $x = y$. Since the v_i are linearly independent, all parameters agree. The projected progression is proper.

Its cardinality is

$$|P_H| = \prod_{i=1}^r (2L_i + 1) \geq \prod_{i=1}^r L_i \geq \prod_{i=1}^r \frac{1}{2d\lambda_i}.$$

For $i > r$ we have

$$\frac{1}{2d\lambda_i} < 1.$$

Therefore

$$\prod_{i=1}^r \frac{1}{2d\lambda_i} \geq \prod_{i=1}^d \frac{1}{2d\lambda_i}.$$

By (??),

$$\prod_{i=1}^d \frac{1}{\lambda_i} \geq \rho^d |H|.$$

Hence

$$|P_H| \geq \left(\frac{\rho}{2d} \right)^d |H|.$$

Finally lift P_H to G and add the kernel K . This gives a coset progression

$$M = K + P$$

contained in $B(\Gamma, \rho)$, of the same rank r , with

$$|M| = |K| |P_H| \geq |K| \left(\frac{\rho}{2d} \right)^d |H| = \left(\frac{\rho}{2d} \right)^d |G|.$$

□

8.6 Worked example: rank one recovered exactly

Let $G = \mathbb{Z}/N\mathbb{Z}$ and let $\Gamma = \{\gamma_1\}$ with

$$\gamma_1(x) = e^{2\pi i x/N}.$$

Then

$$B(\Gamma, \rho) = \{-\lfloor \rho N \rfloor, \dots, \lfloor \rho N \rfloor\}.$$

The lifted lattice is

$$\Lambda = N^{-1}\mathbb{Z} \subset \mathbb{R}$$

modulo $\mathbb{Z}$, with determinant $1/N$. The successive minimum of $[-\rho, \rho]$ is

$$\lambda_1 = \frac{1}{\rho N}$$

up to the endpoint convention. The construction chooses

$$L_1 \asymp \rho N,$$

which recovers a progression of the same order of magnitude as the whole Bohr set.

Thus the general geometry-of-numbers argument is a high-dimensional version of the obvious fact that a short arc in a cyclic group is an arithmetic progression.

8.7 Worked example: exponent two

Take $G = \mathbb{F}_2^n$ and $\rho < 1/2$. As observed earlier,

$$B(\Gamma, \rho) = \Gamma^\perp$$

is already a subgroup. In the lattice construction, all nontrivial progression directions disappear into the kernel K . The theorem returns a rank-zero coset progression, namely a subgroup.

This is exactly what inverse theory should do: in bounded-exponent groups, subgroup structure replaces Euclidean progression directions.

8.8 Combining with Bogolyubov–Chang

Suppose $A \subseteq G$ has density α and doubling K . By ??,

$$2A - 2A$$

contains a Bohr set of rank

$$d \ll K \log(1/\alpha)$$

and radius

$$\rho \gg d^{-1}.$$

Applying ?? gives a proper coset progression $M \subseteq 2A - 2A$ with rank at most d and

$$|M| \geq \exp(-O(d \log d))|G|.$$

Since $d \ll K \log(1/\alpha)$, this is

$$|M| \geq \exp(-O(K \log(1/\alpha) \log(K \log(1/\alpha))))|G|.$$

For a good model, $|G|$ is at most $F(K)|A|$, so α is bounded below by a function of K . Therefore the progression has size at least

$$c(K)|A|.$$

This is exactly the scale needed for Ruzsa covering to control all of A .

8.9 Why a large progression inside $2A - 2A$ is enough

Suppose

$$M \subseteq 2A - 2A$$

is a proper symmetric coset progression with

$$|M| \geq c(K)|A|.$$

Then

$$A + M \subseteq 3A - 2A.$$

Plünnecke–Ruzsa gives

$$|3A - 2A| \leq K^5|A|.$$

Therefore

$$|A + M| \leq K^5 c(K)^{-1} |M|.$$

Ruzsa’s covering lemma then gives a set X with

$$|X| \leq K^5 c(K)^{-1}$$

such that

$$A \subseteq X + (M - M).$$

Since M is symmetric,

$$M - M = 2M,$$

which is another coset progression of the same rank with only an exponential-in-rank size inflation.

The remaining issue is to absorb the finitely many translates X into one progression. A crude but sufficient method adds each translate as a new side direction of length two. This increases the rank by at most $|X|$, still a function only of K .

The next chapter makes this assembly precise and completes the classical theorem.

Research window Why convex progressions are natural

The lattice proof actually constructs a progression from a convex body. Forcing the result into a rectangular GAP loses factors such as d^d . Modern near-polynomial Freiman theorems retain convex progressions for as long as possible, because entropy and Fourier estimates interact more efficiently with convex geometry than with an

arbitrarily chosen coordinate box.

Chapter 9

Freiman's Theorem: Assembling the Classical Machine

9.1 What remains to be done

We now possess all the conceptual pieces of the classical proof of Freiman's theorem in an arbitrary abelian group:

$\begin{aligned} \text{small doubling} &\longrightarrow \text{good finite model} \longrightarrow \text{large spectrum} \\ &\longrightarrow \text{Bohr set} \longrightarrow \text{coset progression.} \end{aligned}$

The purpose of this chapter is not merely to concatenate the preceding theorems. Two points deserve to be made completely explicit:

1. how a progression found in a model is transported back without losing its additive geometry;
2. how a large structured set inside $2A - 2A$ is upgraded to a structural description of all of A .

Those are precisely the places at which a proof can look convincing while silently using more Freiman-isomorphism order than has actually been constructed.

9.2 The classical structural theorem

Theorem 9.1 (Freiman–Green–Ruzsa, qualitative quantitative form). *For every $K \geq 1$ there are finite constants $r(K)$ and $F(K)$ with the following property. If A is a finite nonempty subset of an abelian group G and*

$$|A + A| \leq K|A|,$$

then A is contained in a proper coset progression Q of rank at most $r(K)$ and

$$|Q| \leq F(K)|A|.$$

One may take $r(K)$ and $\log F(K)$ to be polynomial in K up to logarithmic factors by the classical Green–Ruzsa argument.

Remark 9.2 (Proof organization). The additive-combinatorial assembly to a large structured progression plus bounded covering is proved in this chapter. The two geometric refinements needed to obtain a single *proper* containing coset progression are proved in Appendix ?? (Minkowski's second theorem) and Chapter ?? (bounded-rank t -properization). Combining those ingredients proves the stated proper-containment form.

The precise classical estimates are less important here than the architecture. Later chapters will explain why modern methods replace polynomial-in- K rank by polylogarithmic rank, and in bounded torsion by genuinely polynomial Freiman–Ruzsa bounds.

9.3 Extending a Freiman map to a difference set

Let $\phi : A \rightarrow B$ be a Freiman s -isomorphism. If s is sufficiently large, then ϕ induces maps on iterated sum-difference sets.

Lemma 9.3 (Transport on $2A - 2A$). *Suppose $\phi : A \rightarrow B$ is a Freiman 8-isomorphism. Define, for*

$$x = a_1 + a_2 - a_3 - a_4 \in 2A - 2A,$$

$$\Phi(x) = \phi(a_1) + \phi(a_2) - \phi(a_3) - \phi(a_4).$$

Then Φ is a well-defined Freiman 2-isomorphism

$$2A - 2A \longrightarrow 2B - 2B.$$

Proof. First suppose

$$a_1 + a_2 - a_3 - a_4 = a'_1 + a'_2 - a'_3 - a'_4.$$

Rearranging gives the four-term equality

$$a_1 + a_2 + a'_3 + a'_4 = a'_1 + a'_2 + a_3 + a_4.$$

A Freiman 8-isomorphism certainly preserves and reflects this equality, so the displayed formula for Φ is independent of the chosen representation. Applying the same argument to ϕ^{-1} shows that Φ is bijective.

It remains to check the Freiman 2 property. Write each $x_i \in 2A - 2A$ as a sum of two elements of A minus two elements of A . An equality

$$x_1 + x_2 = x_3 + x_4$$

can be rearranged into an equality of two sums of eight elements of A : the positive terms from x_1, x_2 and the negative terms from x_3, x_4 lie on one side, and the remaining eight terms lie on the other. Hence the equality is preserved and reflected by the Freiman 8-isomorphism. This is exactly the Freiman 2 property for Φ . $\square$

9.4 Freiman images of coset progressions

Lemma 9.4 (A Freiman 2-isomorphism preserves progression geometry). *Let S, S' be subsets of abelian groups and let $f : S \rightarrow S'$ be a Freiman 2-isomorphism. Suppose S contains a proper symmetric coset progression*

$$M = H + \left\{ \sum_{i=1}^r n_i v_i : |n_i| \leq L_i \right\}$$

centred at 0. Then $f(M)$ is a translate of a proper coset progression of rank at most r and of the same cardinality.

Proof. Normalize by setting

$$g(x) = f(x) - f(0).$$

For $h_1, h_2 \in H$, the relation

$$h_1 + h_2 = (h_1 + h_2) + 0$$

consists entirely of points of $H \subset M$. Thus

$$g(h_1 + h_2) = g(h_1) + g(h_2).$$

Hence $g(H)$ is a subgroup; bijectivity of f on H shows that $g|_H$ is a group isomorphism onto this subgroup.

For a progression direction v_i , write $w_i = g(v_i)$ whenever $L_i \geq 1$. We claim that

$$g\left(h + \sum_i n_i v_i\right) = g(h) + \sum_i n_i w_i.$$

Starting at h , move one unit at a time in each coordinate. Whenever both x and $x + v_i$ lie in the parameter box, the parallelogram relation

$$(x + v_i) + 0 = x + v_i$$

gives

$$g(x + v_i) - g(x) = g(v_i).$$

The same argument with $-v_i$ handles negative steps. Induction over the coordinate steps proves the formula.

Therefore

$$f(M) = f(0) + g(H) + \left\{ \sum_i n_i w_i : |n_i| \leq L_i \right\}.$$

Because f is injective on M , the displayed coset progression is proper and has $|M|$ elements. $\square$

9.5 The central assembly theorem

We now prove a slightly weaker but already fully quantitative structural conclusion. A finite-rank properization lemma, recorded and explained in Appendix ??, converts it into ??.

Theorem 9.5 (Large progression plus bounded covering). *For each $K \geq 1$ there are constants $c(K) > 0$ and integers $r(K), L(K)$ such that, whenever*

$$|A + A| \leq K|A|,$$

there is a set

$$M \subseteq 2A - 2A$$

which is a translate of a proper symmetric coset progression P of rank at most $r(K)$, satisfying

$$|M| = |P| \geq c(K)|A|,$$

and A is covered by at most $L(K)$ translates of

$$P - P = 2P = M - M.$$

Proof. Apply the good-model theorem, ??, with $s = 8$. Thus A is Freiman 8-isomorphic to a set B in a finite abelian group G' with

$$|G'| \leq F_8(K)|A|.$$

Since the Freiman map preserves pair-sum equality, it preserves doubling:

$$|B + B| = |A + A| \leq K|B|.$$

Write

$$\alpha = \frac{|B|}{|G'|} \geq F_8(K)^{-1}.$$

By the Bogolyubov–Chang theorem of Chapter ??, $2B - 2B$ contains a Bohr set of rank

$$d \ll K \log(1/\alpha) = O_K(1)$$

and radius at least a positive function of K . By ??, this Bohr set contains a proper symmetric coset progression M' of rank at most d with

$$|M'| \geq c_1(K)|G'|.$$

Since $|G'| \geq |B| = |A|$,

$$|M'| \geq c_1(K)|A|.$$

Let $\phi : A \rightarrow B$ be the Freiman 8-isomorphism. By ??, its induced map

$$\Phi : 2A - 2A \rightarrow 2B - 2B$$

is a Freiman 2-isomorphism. Set

$$M = \Phi^{-1}(M').$$

By ??, M is a translate of a proper symmetric coset progression of the same rank and cardinality. Write

$$M = m_0 + P,$$

where P is centred at 0 and symmetric. In particular

$$|M| = |P| = |M'| \geq c_1(K)|A|.$$

We next cover all of A . The companion Volume I proves the Plünnecke–Ruzsa inequality in the form

$$|mA - nA| \leq K^{m+n}|A| \quad (m, n \geq 0, m + n \geq 1).$$

Since $M \subseteq 2A - 2A$,

$$A + M \subseteq 3A - 2A,$$

and therefore

$$|A + M| \leq K^5|A| \leq \frac{K^5}{c_1(K)}|M|.$$

Ruzsa’s covering lemma, applied to A and M , produces a set $X \subseteq A$ with

$$|X| \leq \frac{K^5}{c_1(K)}$$

such that

$$A \subseteq X + (M - M).$$

Since $M = m_0 + P$ with P symmetric and centred at 0,

$$M - M = P - P = 2P.$$

Thus A is covered by at most

$$L(K) := \left\lceil K^5/c_1(K) \right\rceil$$

translates of $2P$ (equivalently, of $M - M$). □

Idea What the proof really did

The Fourier and geometry-of-numbers stages did *not* directly describe A . They found a structured chunk of the difference geometry of A . Plünnecke theory then showed that adding A to this chunk still creates only bounded growth, and Ruzsa covering converted that numerical fact into a cover of all of A . The final theorem is therefore an interaction of three languages:

$$\text{Fourier structure} \quad + \text{additive growth} \quad + \text{geometric covering}.$$

9.6 Absorbing finitely many translates

Suppose

$$A \subseteq X + M, \quad |X| = t,$$

where $M = H + P$ is a rank- r coset progression. Fix $x_0 \in X$ and write

$$X \subseteq x_0 + \{0, q_1\} + \cdots + \{0, q_{t-1}\}.$$

Then

$$A \subseteq x_0 + H + P + \{0, q_1\} + \cdots + \{0, q_{t-1}\}.$$

The right-hand side is a coset progression of rank at most $r + t - 1$, although it need not be proper. Its parameter-box size is bounded by 2^{t-1} times that of P .

Appendix ?? records the following technical form of properization and its rank-reduction mechanism. Its complete t -proper form is proved internally as Theorem II.4.005 in Chapter ??.

Proposition 9.6 (Bounded-rank properization). *For every integer r there is a constant C_r such that every finite rank- r coset progression P is contained in a proper coset progression Q of rank at most r and*

$$|Q| \leq C_r |P|_{\text{par}},$$

where $|P|_{\text{par}}$ is the cardinality of its parameter box times the size of its subgroup part.

Remark 9.7 (Proof location). The proposition is proved in Chapter ?? by induction on rank, using a shortest nontrivial parameter relation and a rank-reduction step; the same chapter proves the stronger t -proper form.

Applying the internally proved properization proposition to the progression obtained by absorbing the $L(K)$ translates in ?? gives a proper coset progression containing A , of rank $O_K(1)$ and size $O_K(|A|)$. This completes the proper form ??; the arbitrary-abelian-group theorem in this form is due to Green and Ruzsa [7].

9.7 Worked example: why “large structure inside $2A - 2A$ ” is enough

Let

$$A = P \cup (P + N^3), \quad P = \{0, 1, \dots, N - 1\}.$$

Then A is not itself one interval, but

$$|A| = 2N,$$

and $A + A$ is the union of at most three intervals of length $2N - 1$, so

$$|A + A| \leq 6N = 3|A|.$$

Meanwhile

$$P - P = \{-(N - 1), \dots, N - 1\} \subseteq A - A.$$

Thus the difference set already contains a progression whose size is comparable with $|A|$. Covering recovers the global geometry: A is the union of two translates of that progression scale.

The example is deliberately elementary, but it mirrors the general proof. One should not insist that the first structured set discovered be all of A ; a large structured object inside an iterated difference set is enough if growth estimates let us cover A by boundedly many of its translates.

9.8 Worked example: a torsion component survives modelling

Let

$$G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$$

and

$$A = (\mathbb{Z}/2\mathbb{Z}) \times \{0, 1, \dots, N - 1\}.$$

Then

$$|A| = 2N, \quad |A + A| = 2(2N - 1),$$

so the doubling constant tends to 2. Any correct inverse theorem must remember both pieces of structure:

$$\underbrace{\mathbb{Z}/2\mathbb{Z}}_{\text{subgroup direction}} \times \underbrace{\{0, \dots, N - 1\}}_{\text{progression direction}}.$$

A purely Euclidean progression forgets the first factor. A coset progression records it exactly.

Research window The quantitative story

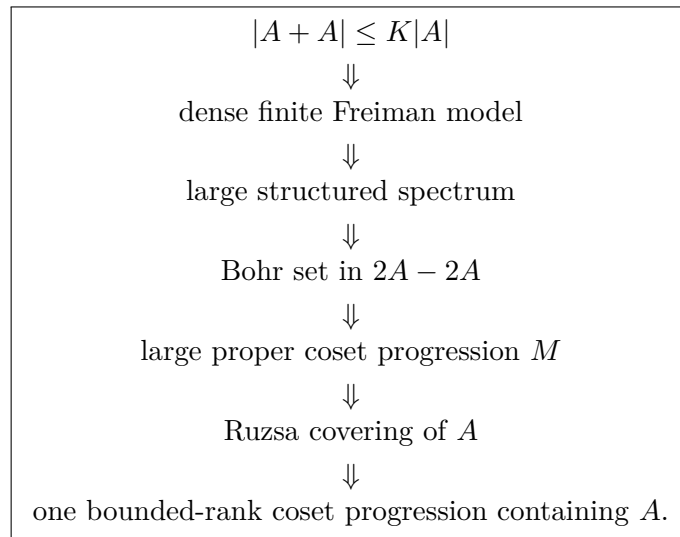
The original general abelian-group theorem of Green and Ruzsa gives explicit bounds with rank polynomial in K (up to logarithms) and size exponential in a polynomial of K . Sanders' Bogolyubov–Ruzsa method later reduced the structural complexity to polylogarithmic functions of K . In bounded-torsion groups, the polynomial Freiman–Ruzsa conjecture is now a theorem: Gowers, Green, Manners and Tao proved polynomial covering by subgroup cosets. The final chapters explain the entropy method behind that breakthrough and place it beside the classical model–Bohr–progression proof.

Proof deferred to a later volume Volume III: polynomial Freiman–Ruzsa

The Sanders and polynomial Freiman–Ruzsa quantitative results mentioned in this window are proved in Volume III. The classical Green–Ruzsa mechanism used in the present volume is proved here, subject only to the explicitly scheduled geometry-of-numbers and properization steps.

9.9 What should be remembered

The classical theorem is not a mysterious single leap from small doubling to a progression. It is a pipeline:



The later theory changes the efficiency of several arrows, but this architecture remains one of the basic mental models of inverse additive theory.

Chapter 10

Almost-Periodicity and the Croot–Sisask Method

10.1 Why almost-periods matter

A subgroup H has exact periods:

$$1_H(x + h) = 1_H(x) \quad (h \in H).$$

A progression does not, but convolution smooths sets. If

$$f = \mu_A * 1_B$$

is an averaged translate of 1_B , then one may hope that many shifts t satisfy

$$f(x + t) \approx f(x).$$

Croot and Sisask discovered that small product or sum growth produces a surprisingly large set of such *almost-periods* [2]. The proof is probabilistic and elementary in spirit: approximate a convolution by a small random sample, then use the fact that small growth forces many translated samples to collide.

This chapter develops that mechanism in full detail.

10.2 Normalized notation

Let G be a finite abelian group. We use normalized averaging

$$\mathbb{E}_x f(x) = \frac{1}{|G|} \sum_{x \in G} f(x)$$

and normalized convolution

$$(f * g)(x) = \mathbb{E}_y f(y)g(x - y).$$

For a nonempty set A of density

$$\alpha = |A|/|G|,$$

write

$$\mu_A = \frac{1_A}{\alpha}.$$

Then $\mathbb{E}\mu_A = 1$, and

$$(\mu_A * 1_B)(x) = \frac{1}{|A|} \sum_{a \in A} 1_B(x - a).$$

Thus $\mu_A * 1_B$ is literally the fraction of elements $a \in A$ for which $x - a \in B$.

For $t \in G$ define translation by

$$(\tau_t f)(x) = f(x - t).$$

Translations preserve every normalized L^p norm.

10.3 The sampling lemma

The probabilistic engine is a standard moment inequality. We state it in the precise form needed here and prove the reduction to it; Appendix ?? gives a proof from symmetrization and the Khintchine inequality.

Lemma 10.1 (Vector-valued sampling bound). *Let $p \geq 2$, let Ω be a probability space, and let $Y_1, \dots, Y_k$ be independent random functions on Ω with a common mean $F = \mathbb{E}Y_i$. Then*

$$\mathbb{E} \left\| \frac{1}{k} \sum_{i=1}^k (Y_i - F) \right\|_{L^p(\Omega)}^p \leq \left(C \sqrt{\frac{p}{k}} \right)^p \mathbb{E} \|Y_1\|_{L^p(\Omega)}^p$$

whenever all the Y_i are identically distributed. Here C is an absolute constant.

The point of this formulation is that it does not ask us to estimate a high moment of an empirical variance. Symmetrization creates random signs, Khintchine controls those signs pointwise, and the elementary inequality

$$\left(\sum_{i=1}^k u_i^2 \right)^{p/2} \leq k^{p/2-1} \sum_{i=1}^k |u_i|^p$$

finishes the argument. Appendix ?? gives every step.

Proposition 10.2 (Random sampling approximates a convolution). *Let $A, B \subseteq G$ be nonempty, let $p \geq 2$, and choose independent uniform random elements*

$$a_1, \dots, a_k \in A.$$

Set

$$g_{\mathbf{a}}(x) = \frac{1}{k} \sum_{i=1}^k 1_B(x - a_i), \quad f = \mu_A * 1_B,$$

and let $\beta = |B|/|G|$. Then

$$\mathbb{E}_{\mathbf{a}} \|g_{\mathbf{a}} - f\|_p^p \leq \left(C \sqrt{\frac{p}{k}} \right)^p \beta.$$

Consequently, if

$$k \geq C_1 p \varepsilon^{-2},$$

then with probability at least $1/2$,

$$\|g_{\mathbf{a}} - f\|_p \leq \frac{\varepsilon}{2} \beta^{1/p}.$$

Proof. For a uniform random $a \in A$, let

$$Y_a(x) = 1_B(x - a).$$

Then

$$\mathbb{E}_a Y_a(x) = f(x),$$

and translation invariance gives

$$\|Y_a\|_p^p = \|1_B\|_p^p = \beta.$$

Apply ?? to the independent random functions $Y_{a_1}, \dots, Y_{a_k}$. Since

$$g_{\mathbf{a}} = \frac{1}{k} \sum_{i=1}^k Y_{a_i},$$

we obtain

$$\mathbb{E}_{\mathbf{a}} \|g_{\mathbf{a}} - f\|_p^p \leq \left(C \sqrt{\frac{p}{k}}\right)^p \beta.$$

If $k \geq C_1 p \varepsilon^{-2}$ and C_1 is a sufficiently large absolute constant, the last expression is at most

$$\frac{1}{2} \left(\frac{\varepsilon}{2}\right)^p \beta.$$

Markov's inequality therefore gives, with probability at least $1/2$,

$$\|g_{\mathbf{a}} - f\|_p^p \leq \left(\frac{\varepsilon}{2}\right)^p \beta,$$

which is the asserted bound after taking p th roots. $\square$

10.4 Small growth forces collisions of samples

Suppose now that $S + A$ is small. Take a good sample

$$\mathbf{a} = (a_1, \dots, a_k) \in A^k$$

and translate every coordinate by $s \in S$:

$$s + \mathbf{a} = (s + a_1, \dots, s + a_k) \in (S + A)^k.$$

There are $|S||A|^k$ possible input pairs $(s, \mathbf{a})$ but only $|S + A|^k$ possible translated tuples. If

$$|S + A| \leq K|A|,$$

many inputs must collide.

The crucial observation is that a collision between translated samples gives an almost-period of the *true* convolution.

Theorem 10.3 (Croot–Sisask almost-periodicity, basic form). *Let A, B, S be finite nonempty subsets of a finite abelian group G , and suppose*

$$|A + S| \leq K|A|.$$

Let $p \geq 2$ and $0 < \varepsilon \leq 1$. Then there exists $T \subseteq S$ with

$$|T| \geq \frac{|S|}{2K^k}, \quad k \leq Cp\varepsilon^{-2},$$

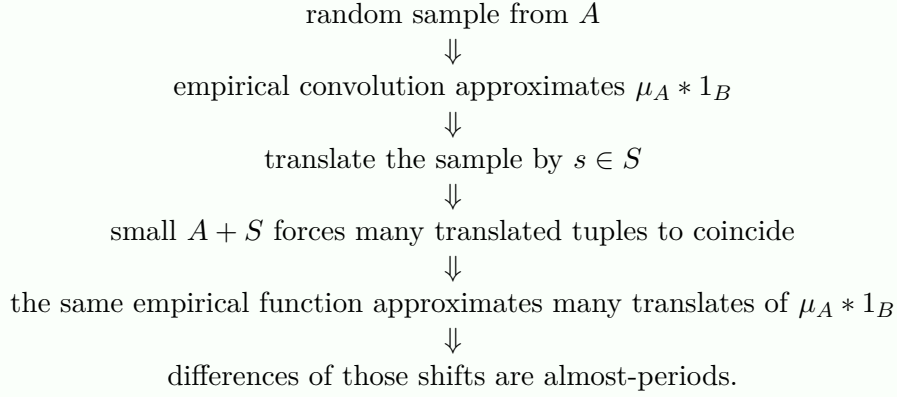
such that every $t \in T$ satisfies

$$\|\tau_t(\mu_A * 1_B) - \mu_A * 1_B\|_p \leq \varepsilon \|1_B\|_p. \tag{10.1}$$

Equivalently,

$$|T| \geq (2K)^{-Cp/\varepsilon^2} |S|.$$

Proof architecture The proof in one picture



Proof. Let

$$f = \mu_A * 1_B, \quad \beta = |B|/|G|,$$

and choose

$$k = \lceil C_1 p \varepsilon^{-2} \rceil$$

large enough that ?? gives

$$\|g_{\mathbf{a}} - f\|_p \leq \frac{\varepsilon}{2} \beta^{1/p}$$

for at least half of all tuples $\mathbf{a} \in A^k$. Call these tuples good.

Consider all pairs

$$(s, \mathbf{a}) \in S \times A^k$$

with $\mathbf{a}$ good, and map them to

$$\Theta(s, \mathbf{a}) = s + \mathbf{a} = (s + a_1, \dots, s + a_k) \in (A + S)^k.$$

There are at least

$$\frac{1}{2} |S| |A|^k$$

good input pairs, while the image has cardinality at most

$$|A + S|^k \leq K^k |A|^k.$$

Hence some tuple $\mathbf{z} \in (A + S)^k$ has at least

$$\frac{|S|}{2K^k}$$

good preimages. Let $T \subseteq S$ be the set of first coordinates of those preimages. For a fixed s and $\mathbf{z}$ the tuple $\mathbf{a} = \mathbf{z} - s$ is uniquely determined, so

$$|T| \geq \frac{|S|}{2K^k}.$$

For each $s \in T$, let $\mathbf{a}^{(s)}$ be the corresponding good tuple. Since

$$\mathbf{z} = s + \mathbf{a}^{(s)},$$

the empirical function attached to $\mathbf{z}$ is

$$g_{\mathbf{z}} = \tau_s g_{\mathbf{a}^{(s)}}.$$

Translation invariance of L^p therefore gives

$$\|g_{\mathbf{z}} - \tau_s f\|_p = \|g_{\mathbf{a}(s)} - f\|_p \leq \frac{\varepsilon}{2} \beta^{1/p}.$$

If $s, u \in T$, the triangle inequality yields

$$\|\tau_s f - \tau_u f\|_p \leq \varepsilon \beta^{1/p}.$$

Translate by $-u$ and write $t = s - u$. Then

$$\|\tau_t f - f\|_p \leq \varepsilon \beta^{1/p} = \varepsilon \|1_B\|_p.$$

Every $t \in T - T$ arises in this way. □

10.5 Why the theorem is stronger than it first looks

The theorem gives almost-periods not merely as an unstructured set but as a *difference set* $T - T$, where T itself is large. This matters because additive combinatorics has powerful tools for extracting further structure from large difference sets. One can iterate the estimate, use Plünnecke theory to control sumsets of T , or feed T into Fourier/Chang machinery.

A typical strategy is

$$\begin{aligned} \text{small doubling} &\rightarrow \text{large almost-period set } T - T \\ &\rightarrow \text{structured subset of } kT - kT \rightarrow \text{Bohr/progression structure.} \end{aligned}$$

This is the bridge from the probabilistic Croot–Sisask lemma to the quasipolynomial Bogolyubov–Ruzsa theorem.

10.6 Worked example: intervals

Let $G = \mathbb{Z}/N\mathbb{Z}$ with $N \gg L$, and take

$$A = B = \{0, 1, \dots, L-1\}.$$

Then

$$f(x) = (\mu_A * 1_A)(x)$$

is the normalized triangular function

$$f(x) = \frac{1}{L} |A \cap (x - A)|.$$

Away from wraparound,

$$f(x) = \begin{cases} (x+1)/L, & 0 \leq x \leq L-1, \\ (2L-1-x)/L, & L \leq x \leq 2L-2, \\ 0, & \text{otherwise.} \end{cases}$$

A translation by one changes f by at most $1/L$ pointwise except at the ends. Thus an interval has many obvious almost-periods at small scales.

The Croot–Sisask theorem is interesting because it finds an analogous phenomenon without knowing in advance that A resembles an interval.

10.7 Worked example: an almost-period can be invisible on the original set

Let $A \subseteq G$ be a set for which 1_A itself is highly irregular. The convolution

$$\mu_A * 1_A$$

counts normalized representations of x as a sum of two elements of A . Even if shifting A by t changes a large fraction of its points, the representation counts may vary only slightly after averaging over one copy of A .

This is the key smoothing effect. Croot–Sisask does not usually assert

$$1_A(x+t) \approx 1_A(x).$$

It asserts that an *averaged additive statistic* is stable. Modern additive combinatorics repeatedly exploits the fact that convolution is much smoother than the underlying indicator functions.

10.8 The noncommutative version

The sampling proof used almost no commutativity. In a finite group, one replaces $A + S$ by a product set such as SA , chooses a consistent left/right translation convention, and obtains almost-periods in $T^{-1}T$. What changes is the bookkeeping of factor order, not the probabilistic heart of the argument.

This is one reason Croot–Sisask belongs naturally in a series that eventually reaches approximate groups and expansion: its mechanism is fundamentally about convolution and small growth, not about coordinates in an abelian group.

Research window From almost-periodicity to modern Roth bounds

Croot–Sisask almost-periodicity became one of the central tools in the modern quantitative theory of three-term arithmetic progressions. Bloom and Sisask used increasingly refined almost-periodicity and density-increment arguments to break long-standing barriers; Kelley and Meka introduced a powerful sifting architecture; more recent work continues to combine these ideas. The methods are developed systematically in Volume III, which now contains the complete modern Roth and PFR proofs. Here the point is structural: a probabilistic sampling lemma created a new route from small doubling to highly organized additive information.

Proof deferred to a later volume Volume III: modern Roth theory

The modern quantitative Roth applications of almost-periodicity are proved in Volume III. The Croot–Sisask theorem itself is proved in full in the present volume.

10.9 A reusable proof pattern

The method is worth remembering independently of the theorem:

1. replace a complicated average by a small random empirical average;
2. prove the empirical object is accurate with positive probability;
3. move the sample through a set of transformations;

4. use small growth to force many transformed samples to collide;
5. compare the true objects through the common empirical approximation.

This pattern has applications far beyond the particular convolution studied in this chapter.

Chapter 11

The Bogolyubov–Ruzsa Lemma and Sanders’ Quasipolynomial Upgrade

11.1 The quantitative question

The classical proof of Freiman’s theorem found a large coset progression inside $2A - 2A$, but its rank was a relatively poor function of the doubling constant K . The modern goal is much sharper:

Guiding Question 11.1. If

$$|A + A| \leq K|A|,$$

how complicated must a large structured subset of

$$2A - 2A$$

really be?

Sanders’ answer was a decisive improvement: the relevant dimension can be taken polylogarithmic in K . This chapter first shows, completely, how Croot–Sisask almost-periodicity creates *actual elements* of $2A - 2A$. We then isolate the technically deeper bootstrapping step that turns this first gain into the quasipolynomial theorem.

11.2 From almost-periods to exact containment

Let $f : G \rightarrow [0, \infty)$ and let

$$\langle f, g \rangle = \mathbb{E}_x f(x) \overline{g(x)}.$$

If f is supported on a set S , then

$$\langle \tau_t f, f \rangle > 0$$

forces

$$t \in S - S.$$

Indeed some x must satisfy both $f(x - t) > 0$ and $f(x) > 0$.

Lemma 11.2 (An L^2 almost-period becomes a support difference). *Let $f \geq 0$ be nonzero. If*

$$\|\tau_t f - f\|_2 < \sqrt{2} \|f\|_2,$$

then

$$\langle \tau_t f, f \rangle > 0.$$

Consequently, if $\text{supp } f \subseteq S$, then $t \in S - S$.

Proof. Translation invariance gives

$$\begin{aligned}\|\tau_t f - f\|_2^2 &= \|\tau_t f\|_2^2 + \|f\|_2^2 - 2\Re\langle \tau_t f, f \rangle \\ &= 2\|f\|_2^2 - 2\Re\langle \tau_t f, f \rangle.\end{aligned}$$

If the left side is strictly less than $2\|f\|_2^2$, then the real part of the inner product is positive. Since f is real and nonnegative, the inner product itself is nonnegative real. $\square$

11.3 A first Bogolyubov–Ruzsa consequence of Croot–Sisask

Take

$$f = \mu_A * 1_A.$$

Then

$$\text{supp } f = A + A.$$

If $|A + A| \leq K|A|$, we can estimate its L^2 norm from below.

Lemma 11.3 (Energy lower bound in convolution form). *Let $A \subseteq G$ have density α and doubling at most K . Then*

$$\|\mu_A * 1_A\|_2 \geq \sqrt{\frac{\alpha}{K}}.$$

Proof. The function $f = \mu_A * 1_A$ has mean

$$\mathbb{E}f = \alpha$$

and is supported on $A + A$, whose density is at most $K\alpha$. Cauchy–Schwarz on the support gives

$$\alpha^2 = (\mathbb{E}f)^2 \leq \frac{|A + A|}{|G|} \|f\|_2^2 \leq K\alpha \|f\|_2^2.$$

Cancel $\alpha > 0$. $\square$

Now apply ?? with

$$B = A, \quad S = A, \quad p = 2,$$

and choose

$$\varepsilon = \frac{1}{2\sqrt{K}}.$$

Since

$$\|1_A\|_2 = \sqrt{\alpha},$$

every $t \in T - T$ satisfies

$$\|\tau_t f - f\|_2 \leq \frac{1}{2\sqrt{K}} \sqrt{\alpha} \leq \frac{1}{2} \|f\|_2.$$

Thus ?? implies

$$T - T \subseteq (A + A) - (A + A) = 2A - 2A.$$

The almost-periodicity theorem also gives

$$|T| \geq K^{-O(K)} |A|.$$

We have proved:

Proposition 11.4 (A first structured difference set). *If $|A + A| \leq K|A|$, then there is $T \subseteq A$ with*

$$|T| \geq K^{-O(K)}|A|$$

such that

$$T - T \subseteq 2A - 2A.$$

Proof. This is exactly the conclusion of the preceding calculation: Croot–Sisask with $\varepsilon \asymp K^{-1/2}$ gives $|T| \geq K^{-O(K)}|A|$, and Lemma ?? applied to the displayed L^2 estimate gives $T - T \subseteq 2A - 2A$. $\square$

This is not yet Sanders’ theorem, but it illustrates the decisive new feature: almost-periodicity has produced an *exact containment* inside $2A - 2A$.

11.4 Why the naive argument is quantitatively wasteful

The loss came from choosing

$$\varepsilon \asymp K^{-1/2}.$$

The Croot–Sisask sample size is proportional to ε^{-2} , so this costs a factor of order K in the exponent. If we now apply ordinary Bogolyubov–Chang to T , the resulting rank is still far from polylogarithmic in K .

The modern insight is that one should not insist on approximating the rough two-fold convolution at a scale dictated by its worst possible L^2 norm. Instead one:

1. works in a suitable L^p space with p of order $\log K$;
2. passes to more smoothed convolutions;
3. uses almost-periods repeatedly, each time improving the effective regularity;
4. converts the final large almost-period set into spectral structure by Chang-type arguments.

The proof is an iterative regularization scheme.

11.5 A useful amplification lemma

The next elementary observation explains why controlling a generating set of almost-periods controls many-fold sums of that set.

Lemma 11.5 (Triangle amplification). *Let $1 \leq p \leq \infty$, let $f : G \rightarrow \mathbb{C}$, and suppose*

$$\|\tau_t f - f\|_p \leq \eta$$

for every $t \in T$. Then for every

$$u = t_1 + \cdots + t_m, \quad t_i \in T,$$

one has

$$\|\tau_u f - f\|_p \leq m\eta.$$

The same holds for signed sums from $mT - mT$ with $2m\eta$ on the right.

Proof. Write

$$\tau_{t_1+\cdots+t_m} f - f = \sum_{j=1}^m \tau_{t_1+\cdots+t_{j-1}} (\tau_{t_j} f - f).$$

Take L^p norms, use translation invariance and the triangle inequality. The signed version is identical, noting that $-T$ has the same almost-period bound because

$$\|\tau_{-t}f - f\|_p = \|\tau_tf - f\|_p.$$

□

The lemma seems to lose a factor m , but if the original error is chosen as $1/m$, a large iterated sumset of almost-periods can still consist entirely of useful shifts. The art is to choose m and p so that Croot–Sisask produces enough almost-periods without paying too much.

11.6 The quasipolynomial Bogolyubov–Ruzsa theorem

We now state the deep quantitative result in a form sufficient for the rest of this book.

Theorem 11.6 (Sanders’ Bogolyubov–Ruzsa theorem). *There are absolute constants $C_1, C_2 > 0$ such that the following holds. Let A be a finite nonempty subset of an abelian group and suppose*

$$|A + A| \leq K|A|, \quad K \geq 1.$$

Then $2A - 2A$ contains a proper symmetric convex coset progression M satisfying

$$\dim M \leq C_1 \log^6(2K)$$

and

$$|M| \geq \exp(-C_2 \log^7(2K))|A|.$$

This is the quantitative form used in the later structural arguments.

Proof deferred to a later volume Proof in Volume III: Sanders Bogolyubov–Ruzsa theorem

The theorem itself, including the regular-Bohr bookkeeping and the sharp local Fourier iteration, is proved in Volume III. The material below proves the Croot–Sisask component and explains the remaining mechanism, but is not presented as a substitute for that later proof.

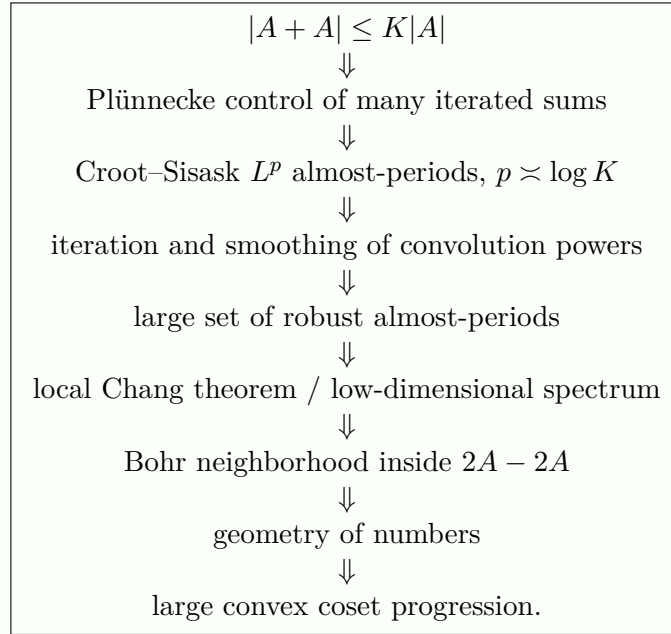
Proof deferred to a later volume Proof in Volume III: Sanders local-Fourier bootstrap

The technically deepest stage is a local Fourier/Chang bootstrap: from a small-doubling set one can construct, through repeated Croot–Sisask almost-periodicity, a regular Bohr-type neighborhood B contained in $2A - 2A$ whose rank and logarithmic reciprocal density are both polylogarithmic in K . The construction uses $p \asymp \log K$, carefully chosen convolution powers, and a local version of Chang’s spectral-dimension argument.

This is no longer a forward reference: Volume III contains the complete local Bohr-set regularity, local Chang theorem, and the full Sanders bootstrap line by line. The present chapter remains an architectural preview; the load-bearing complete proof is Volume III. What matters in the present volume is to understand exactly how the pieces interact.

Proof architecture Sanders' mechanism

A useful schematic form is



11.7 From Sanders to a quasipolynomial Freiman theorem

The same covering step used in Chapter ?? now has much better parameters.

Let $M \subseteq 2A - 2A$ be supplied by ??. Write

$$d = O(\log^6(2K)), \quad |M| \geq e^{-O(\log^7(2K))}|A|.$$

Since

$$A + M \subseteq 3A - 2A,$$

Plünnecke–Ruzsa gives

$$|A + M| \leq K^5|A|.$$

Therefore

$$\frac{|A + M|}{|M|} \leq \exp(O(\log^7(2K))).$$

Ruzsa covering covers A by

$$\exp(O(\log^7(2K)))$$

translates of $M - M$. This is the natural quasipolynomial *covering* form of a Freiman–Ruzsa theorem: a polylogarithmic-dimensional structured set, together with only quasipolynomially many translates, controls all of A .

One should not silently turn this into a comparably efficient single containing progression. The naive absorption trick from Chapter ?? adds one binary direction for essentially every translate, and therefore destroys the polylogarithmic rank when the covering number is quasipolynomial. This distinction between *covering by* a structured object and *containment in* one is quantitatively important, and it is exactly the covering formulation that later polynomial Freiman–Ruzsa theorems optimize.

The important advance is qualitative as well as numerical: the dimension of the controlling structured piece is no longer a power of K but a power of $\log K$.

11.8 Worked example: why smoothing can improve the error scale

Suppose f is the normalized triangular convolution of an interval from Chapter ?? . Its first convolution is piecewise linear. Convolving once more produces a piecewise quadratic profile with smaller discrete derivatives relative to its bulk size. Repeated convolution continues this smoothing.

This elementary picture is a good model for the modern argument. A rough indicator may require a very small shift to remain close in L^p ; a sufficiently smoothed convolution can tolerate a much larger family of shifts. Croot–Sisask finds those shifts without any explicit coordinate description of the original set.

11.9 Worked example: exponent-two groups

In $G = \mathbb{F}_2^n$, every Bohr set of radius below $1/2$ is a subspace. Hence the Bohr-set stage of Sanders’ argument immediately becomes a subspace theorem:

$$2A - 2A = 4A$$

contains a subspace of codimension polylogarithmic in K after passing to a suitable finite model. This bounded-exponent simplification is one reason the polynomial Freiman–Ruzsa problem was especially natural in vector spaces over finite fields.

But a polylogarithmic codimension is still not the polynomial covering statement predicted by Marton. Closing that gap required a new idea: entropy.

Research window Why this theorem was a turning point

Sanders’ result showed that the classical Fourier/Bohr framework could be pushed dramatically further once combined with probabilistic almost-periodicity. For more than a decade it supplied the strongest general quantitative route toward polynomial Freiman–Ruzsa phenomena. The later entropy proof of Marton’s conjecture does not make these techniques obsolete: almost-periodicity remains central in modern progression bounds, and Fourier/entropy hybrids now give the strongest general abelian-group Freiman–Ruzsa estimates known beyond bounded torsion.

Chapter 12

Entropy as Additive Geometry

12.1 Why introduce random variables?

Classical inverse theory begins with a uniform random point of a set A . Entropy allows us to forget the uniformity requirement and retain only the effective amount of information carried by a distribution. This added flexibility turns out to be decisive.

For a finitely supported random variable X with probabilities $p(x)$, define its Shannon entropy by

$$H(X) = - \sum_x p(x) \log p(x),$$

with the convention $0 \log 0 = 0$. We use natural logarithms. If X is uniform on a set A , then

$$H(X) = \log |A|.$$

Thus entropy is exactly the logarithm of cardinality for uniform distributions.

Appendix ?? develops the elementary entropy identities used below. The main ones are the chain rule, conditioning reduces entropy, and submodularity. We nevertheless prove every additive consequence in the present chapter.

12.2 Entropy of sums and differences

Let X, Y be independent group-valued random variables. Conditioning on Y gives

$$H(X - Y \mid Y) = H(X),$$

because for each fixed y , translation $x \mapsto x - y$ is a bijection. Since conditioning cannot increase entropy,

$$H(X - Y) \geq H(X).$$

By symmetry,

$$H(X - Y) \geq H(Y).$$

This is the entropy analogue of the elementary cardinality inequalities

$$|A - B| \geq |A|, \quad |A - B| \geq |B|.$$

12.3 Entropic Ruzsa distance

Definition 12.1 (Entropic Ruzsa distance). Let X, Y be finitely supported random variables in an abelian group. Let X', Y' be independent variables with the same respective laws as X, Y . Define

$$d_{\text{ent}}(X, Y) = H(X' - Y') - \frac{1}{2}H(X') - \frac{1}{2}H(Y').$$

The quantity depends only on the laws of X and Y .

The elementary entropy-of-differences inequality immediately yields

$$d_{\text{ent}}(X, Y) \geq 0.$$

It is symmetric because

$$H(X' - Y') = H(Y' - X').$$

Worked Example 12.2 (Uniform distributions on subgroups). Let $H \leq G$ be finite and let U_H be uniform on H . The difference of two independent uniform elements of H is again uniform on H . Hence

$$H(U_H - U_H) = \log |H| = H(U_H),$$

and therefore

$$d_{\text{ent}}(U_H, U_H) = 0.$$

Thus subgroups are zero-distance objects for entropic Ruzsa geometry.

12.4 The entropic triangle inequality

The set-theoretic Ruzsa triangle inequality was proved in Volume I by an injection. Entropy has its own triangle inequality, but the proof uses submodularity rather than representatives.

Lemma 12.3 (Functional submodularity). *For discrete random variables A, B, C ,*

$$H(A, B, C) + H(C) \leq H(A, C) + H(B, C).$$

This standard entropy inequality is proved in Appendix ??; it is equivalent to nonnegativity of conditional mutual information.

Theorem 12.4 (Entropic Ruzsa triangle inequality). *For any finitely supported G -valued random variables X, Y, Z ,*

$$d_{\text{ent}}(X, Z) \leq d_{\text{ent}}(X, Y) + d_{\text{ent}}(Y, Z).$$

Proof. Since the distance depends only on marginal laws, take Y independent of the pair (X, Z) ; X and Z need not be independent for the intermediate inequality. It is enough to prove

$$H(X - Z) \leq H(X - Y) + H(Y - Z) - H(Y). \quad (12.1)$$

Indeed, afterwards take X, Z independent as well and subtract

$$\frac{1}{2}H(X) + \frac{1}{2}H(Z)$$

from both sides.

Apply ?? with

$$A = X - Y, \quad B = Z, \quad C = X - Z.$$

The triple (A, B, C) determines (X, Y, Z) and conversely: from $B = Z$ and $C = X - Z$ recover X , then from $A = X - Y$ recover Y . Hence

$$H(A, B, C) = H(X, Y, Z) = H(X, Z) + H(Y),$$

where the last equality uses independence of Y from (X, Z) .

Also $(A, C) = (X - Y, X - Z)$ determines

$$Y - Z = (X - Z) - (X - Y) = C - A,$$

so

$$H(A, C) = H(X - Y, Y - Z) \leq H(X - Y) + H(Y - Z).$$

Finally $(B, C) = (Z, X - Z)$ is bijectively equivalent to (X, Z) , hence

$$H(B, C) = H(X, Z).$$

Submodularity gives

$$H(X, Z) + H(Y) + H(X - Z) \leq H(X - Y) + H(Y - Z) + H(X, Z).$$

Cancel $H(X, Z)$ to obtain (??). □

Idea Why entropy improves the geometry

The combinatorial Ruzsa distance uses the cardinality of an entire difference set, so a tiny exceptional part of a set can be expensive. Entropy weights points by probability. One may therefore condition, project, or discard low-probability behavior while keeping quantitative control. This flexibility is precisely what the polynomial Freiman–Ruzsa proof exploits.

12.5 Homomorphisms contract entropic distance

A second decisive advantage over cardinality is monotonicity under quotient maps.

Theorem 12.5 (Contraction under homomorphisms). *Let*

$$\pi : G \rightarrow H$$

be a group homomorphism. Then

$$d_{\text{ent}}(\pi X, \pi Y) \leq d_{\text{ent}}(X, Y).$$

Proof. Take X, Y independent and put

$$U = \pi X, \quad V = \pi Y.$$

Since $U - V = \pi(X - Y)$,

$$H(X - Y) = H(U - V) + H(X - Y \mid U - V),$$

while

$$H(X) = H(U) + H(X \mid U), \quad H(Y) = H(V) + H(Y \mid V).$$

Thus it suffices to prove

$$H(X - Y \mid U - V) \geq \frac{1}{2}H(X \mid U) + \frac{1}{2}H(Y \mid V).$$

Conditioning further can only reduce entropy, so

$$H(X - Y \mid U - V) \geq H(X - Y \mid U, V).$$

For fixed values $U = u, V = v$, the conditional variables $X \mid U = u$ and $Y \mid V = v$ remain independent. Therefore

$$H(X - Y \mid U = u, V = v) \geq \max\{H(X \mid U = u), H(Y \mid V = v)\}.$$

The maximum is at least the average. Average over (u, v) to obtain the desired inequality. $\square$

For ordinary doubling, the image of a small-doubling set under a homomorphism may have awkward fibers and no monotone doubling constant. Entropic distance behaves cleanly. That is not a cosmetic reformulation; it is a new induction tool.

12.6 From sets to entropy

Let U_A denote a uniform random element of a finite nonempty set A .

If $A \subseteq \mathbb{F}_2^n$, subtraction equals addition, and

$$d_{\text{ent}}(U_A, U_A) = H(U_A + U'_A) - \log |A|.$$

Since the sum is supported on $A + A$,

$$H(U_A + U'_A) \leq \log |A + A|.$$

Therefore

$$d_{\text{ent}}(U_A, U_A) \leq \log \frac{|A + A|}{|A|}. \quad (12.2)$$

In a general abelian group the natural same-set quantity is

$$d_{\text{ent}}(U_A, -U_A) = H(U_A + U'_A) - \log |A|,$$

so the identical bound holds.

Thus

$$|A + A| \leq K|A| \implies d_{\text{ent}} \leq \log K.$$

The parameter $\log K$, rather than K , is the natural scale in entropy space.

12.7 Worked example: an interval

Let

$$A = \{0, 1, \dots, N-1\} \subset \mathbb{Z}$$

and X, X' be independent uniform variables on A . The distribution of $X + X'$ is triangular, not uniform on $A + A$. Therefore

$$H(X + X') < \log(2N - 1).$$

Consequently

$$d_{\text{ent}}(X, -X) < \log(2N - 1) - \log N \rightarrow \log 2.$$

Entropy sees the same bounded-doubling phenomenon as cardinality, but with a strictly smaller effective support because central sums are more likely than endpoint sums.

This simple example already shows that entropy is not merely “log cardinality with new notation.”

12.8 Zero entropic doubling characterizes coset-uniformity

The analogue of exact doubling one also has a clean entropy formulation.

Proposition 12.6 (Zero distance means a coset of a subgroup). *Let X be finitely supported. Then*

$$d_{\text{ent}}(X, X) = 0$$

if and only if X is uniform on a coset of a finite subgroup.

Proof. The “if” direction follows from translation invariance and the subgroup example.

Conversely, let X' be an independent copy of X . We always have

$$H(X - X') \geq H(X - X' \mid X') = H(X).$$

Equality of entropic distance with zero forces equality here. Hence

$$I(X - X'; X') = 0,$$

so $X - X'$ and X' are independent.

Let S be the support of X . For every $y \in S$, conditioning on $X' = y$ shows that the law of $X - y$ is the same common law of $X - X'$. Thus the translates of the probability mass function satisfy

$$p_X(z + y) = p_X(z + y')$$

whenever $y, y' \in S$ and the terms are interpreted on the ambient group. In particular, translation by every difference $y - y'$ preserves the law of X . Let

$$H = \langle S - S \rangle.$$

Then the law is invariant under H .

Choose $x_0 \in S$. Since $S - S \subseteq H$, we have $S \subseteq x_0 + H$. Conversely, invariance under every $h \in H$ gives

$$p_X(x_0 + h) = p_X(x_0) > 0,$$

so $x_0 + H \subseteq S$. Because S is finite, this inclusion also shows that H is finite. Hence $S = x_0 + H$, and invariance makes the mass constant on this finite coset. Therefore X is uniform there. $\square$

This exact theorem is the entropy analogue of

$$|A + A| = |A| \iff A \text{ is a coset of a subgroup.}$$

The polynomial Freiman–Ruzsa problem asks for a quantitative stability theorem: if entropic distance is merely small, or moderately large, how close is the distribution to a subgroup-uniform law?

12.9 Distance to a subgroup and quotient entropy

Let $V \leq G$ be finite and let U_V be uniform on V , independent of X . Let

$$\pi : G \rightarrow G/V$$

be the quotient map. Then $X - U_V$ is uniform inside the random coset selected by $\pi(X)$. Therefore

$$H(X - U_V) = H(\pi X) + \log |V|. \tag{12.3}$$

It follows that

$$d_{\text{ent}}(X, U_V) = H(\pi X) + \frac{1}{2}(\log |V| - H(X)). \quad (12.4)$$

This formula makes the geometry transparent. Small distance to U_V simultaneously says:

- the entropy scale of V is close to that of X ;
- the quotient variable πX has small entropy, so most of the mass of X lies in relatively few V -cosets.

12.10 Worked example: a union of cosets

Suppose A is the disjoint union of r cosets of a subgroup V , all of equal size $|V|$, and $X = U_A$. Then πX is uniform on r quotient points, so

$$H(\pi X) = \log r, \quad H(X) = \log r + \log |V|.$$

Substituting into (??),

$$d_{\text{ent}}(X, U_V) = \frac{1}{2} \log r.$$

Thus covering by r subgroup cosets corresponds naturally to entropic distance of order $\log r$.

This is precisely the logarithmic conversion needed for a polynomial covering theorem.

12.11 Entropy concentration from small entropy

We will need one elementary probabilistic lemma in the next chapter.

Lemma 12.7 (A low-entropy variable is mostly supported on few atoms). *Let Z be a discrete random variable with*

$$H(Z) \leq L.$$

For every $0 < \delta < 1$, there is a set S of at most

$$\exp(L/\delta)$$

values such that

$$\mathbb{P}(Z \in S) \geq 1 - \delta.$$

Proof. For a realized value z , define its information content

$$I(z) = \log \frac{1}{\mathbb{P}(Z = z)}.$$

Then

$$\mathbb{E}I(Z) = H(Z) \leq L.$$

By Markov's inequality,

$$\mathbb{P}(I(Z) > L/\delta) \leq \delta.$$

Let

$$S = \{z : I(z) \leq L/\delta\}.$$

Every $z \in S$ has probability at least $e^{-L/\delta}$, so there can be at most $e^{L/\delta}$ such atoms. The Markov estimate gives the mass conclusion. $\square$

Research window Entropy before PFR

Entropy analogues of sumset inequalities were developed by Ruzsa, Tao, Madiman, Kontoyiannis and others. Green, Manners and Tao subsequently emphasized that entropy doubling contracts under homomorphisms and proved strong “very small doubling” stability results. Those features became central in the Gowers–Green–Manners–Tao proof of Marton’s conjecture. The next chapter turns this language into a polynomial covering theorem.

Proof deferred to a later volume Volume III: entropy stability and PFR

The entropy-doubling stability theorems and their use in the Gowers–Green–Manners–Tao proof are proved in Volume III. This chapter proves only the entropy identities and inequalities it uses internally.

Chapter 13

Polynomial Freiman–Ruzsa in Characteristic Two

13.1 Marton’s conjecture

Let $A \subseteq \mathbb{F}_2^n$ and suppose

$$|A + A| \leq K|A|.$$

In this setting, generalized arithmetic progressions are not the natural final objects: bounded-exponent additive geometry is governed by subspaces. Marton’s conjecture asked for a polynomial-strength structural theorem.

Theorem 13.1 (Polynomial Freiman–Ruzsa in $\mathbb{F}_2^n$). *There is an absolute constant C such that every nonempty*

$$A \subseteq \mathbb{F}_2^n, \quad |A + A| \leq K|A|,$$

can be covered by at most

$$2K^C$$

cosets of a subspace $H \leq \mathbb{F}_2^n$ satisfying

$$|H| \leq |A|.$$

The Gowers–Green–Manners–Tao proof gives $C = 12$ through its basic conversion [5], and a refinement of Liao gives $C = 9$ [9].

Proof deferred to a later volume Volume III: polynomial Freiman–Ruzsa

The set-theoretic theorem is deduced below from entropic PFR; its load-bearing entropic PFR owner is Volume III. The present chapter is retained as an explanatory precursor rather than a competing complete proof.

The striking feature is the word *polynomial*. Sanders’ earlier bounds were quasipolynomial in K ; Marton’s prediction says that the logarithmic complexity of the structure is only $O(\log K)$.

13.2 The entropic theorem

The proof is most naturally stated for arbitrary probability distributions.

Theorem 13.2 (Gowers–Green–Manners–Tao entropic PFR). *Let X_1^0, X_2^0 be random variables taking values in $\mathbb{F}_2^n$. There is a subspace $H \leq \mathbb{F}_2^n$ such that*

$$d_{\text{ent}}(X_1^0, U_H) + d_{\text{ent}}(X_2^0, U_H) \leq 11d_{\text{ent}}(X_1^0, X_2^0). \quad (13.1)$$

Moreover each individual distance on the left is at most

$$6d_{\text{ent}}(X_1^0, X_2^0).$$

Proof ownership. The complete load-bearing proof is in Volume III, Chapter “Bridge Lemmas for Quantitative Additive Combinatorics,” under the theorem “Entropic polynomial Freiman–Ruzsa in characteristic two.” The present chapter uses that theorem as a cross-volume dependency and proves the entropy-to-combinatorics conversion below.

Proof deferred to a later volume Volume III: entropic PFR

The complete proof, including the distance-decrement proposition and entropic BSG, is proved in Volume III, which develops the modern entropy/PFR machinery without compression. The present chapter proves the entropy identities and the conversion from the entropic theorem to polynomial coset covering, and gives the proof architecture of the decrement argument.

We first show in full detail how this theorem implies the combinatorial PFR statement. We then explain the proof of ?? itself.

13.3 Entropy to combinatorics: the complete conversion

Let $X = U_A$. Because the group has characteristic two,

$$X - X' = X + X'.$$

By (??),

$$d_{\text{ent}}(X, X) \leq \log K.$$

Apply ?? with $X_1^0 = X_2^0 = X$. There is a subspace H for which

$$d_{\text{ent}}(X, U_H) \leq \frac{11}{2} \log K. \quad (13.2)$$

We temporarily replace 11 by a general constant C' to make the parameter conversion visible.

Lemma 13.3 (Entropy scales of nearby variables). *For any finitely supported X, Y ,*

$$|H(X) - H(Y)| \leq 2d_{\text{ent}}(X, Y).$$

Proof. Since X', Y' are independent,

$$H(X' - Y') \geq \max\{H(X), H(Y)\}.$$

Hence

$$\begin{aligned} d_{\text{ent}}(X, Y) &\geq \max\{H(X), H(Y)\} - \frac{1}{2}H(X) - \frac{1}{2}H(Y) \\ &= \frac{1}{2}|H(X) - H(Y)|. \end{aligned}$$

□

Thus

$$|\log |H| - \log |A|| \leq C' \log K. \quad (13.3)$$

Next, the definition of entropic distance gives

$$H(X - U_H) \leq \frac{1}{2} \log |A| + \frac{1}{2} \log |H| + \frac{C'}{2} \log K.$$

Every discrete random variable Z has an atom of probability at least $e^{-H(Z)}$: indeed

$$H(Z) = \mathbb{E} \log \frac{1}{p_Z(Z)} \geq \log \frac{1}{\max_z p_Z(z)}.$$

Therefore there exists x_0 such that

$$\mathbb{P}(X - U_H = x_0) \geq |A|^{-1/2} |H|^{-1/2} K^{-C'/2}.$$

But

$$\mathbb{P}(X - U_H = x_0) = \frac{|A \cap (x_0 + H)|}{|A||H|}.$$

Hence, with

$$B = A \cap (x_0 + H),$$

we obtain

$$|B| \geq K^{-C'/2} |A|^{1/2} |H|^{1/2}. \quad (13.4)$$

Since $B \subseteq A$,

$$|A + B| \leq |A + A| \leq K|A|.$$

Ruzsa's covering lemma covers A by at most

$$\frac{|A + B|}{|B|} \leq K^{C'/2+1} \left(\frac{|A|}{|H|} \right)^{1/2}$$

translates of

$$B - B \subseteq H.$$

This already solves the *covering-number* problem. Indeed, if $|H| > |A|$, the square-root factor is < 1 , so the displayed estimate is already at most $K^{C'/2+1}$. No smaller subspace is needed to obtain a polynomially small cover.

There remains, however, a separate normalization required by the statement of combinatorial PFR: the structural subspace is required to have size at most $|A|$. If $|H| \leq |A|$, then no modification is needed, and (??) gives

$$\left(\frac{|A|}{|H|} \right)^{1/2} \leq K^{C'/2},$$

so A is covered by at most $K^{C'+1}$ cosets of H .

Only when $|H| > |A|$ do we shrink H , and we do so solely to enforce this size normalization. Choose a subspace $H' \leq H$ with

$$\frac{|A|}{2} < |H'| \leq |A|.$$

Such a subspace exists because subspace sizes are powers of two. Each H -coset is a union of

$$[H : H'] < \frac{2|H|}{|A|}$$

cosets of H' . Refining the already obtained H -coset cover therefore gives at most

$$2K^{C'/2+1} \left(\frac{|H|}{|A|} \right)^{1/2} \leq 2K^{C'+1}$$

cosets of H' , where the last step uses (??).

We have proved:

Proposition 13.4 (Entropic PFR implies combinatorial PFR). *If ?? holds with C' in place of 11, then every set of doubling at most K in $\mathbb{F}_2^n$ is covered by at most*

$$2K^{C'+1}$$

cosets of a subspace of size at most $|A|$.

Proof. The proof is the calculation immediately preceding the statement. The entropic theorem supplies a subgroup H and a translate meeting A in the required proportion, and Ruzsa covering already gives a polynomially small H -coset cover in both size regimes. If $|H| \leq |A|$, this is the required structural subspace. If $|H| > |A|$, the sole purpose of replacing H by a subspace $H' \leq H$ with $|A|/2 < |H'| \leq |A|$ is to enforce the normalization $|H'| \leq |A|$; refining each H -coset into H' -cosets multiplies the covering number by less than $2|H|/|A|$. Equation (??) then gives the final bound $2K^{C'+1}$. $\square$

For $C' = 11$ this gives $2K^{12}$.

Idea The conversion has only three moves

The proof above is worth remembering independently:

small entropic distance to $U_H \rightarrow$ large intersection with one H -coset
 $\rightarrow$ Ruzsa covering $\rightarrow$ polynomially many H -cosets.

The entropy theorem is therefore exactly the right strength: distance $O(\log K)$ becomes $K^{O(1)}$ covering.

13.4 How the entropic theorem is proved

The proof of ?? is an induction on Ruzsa distance disguised as an optimization argument. The key innovation is to penalize moving too far from the original variables.

Set

$$\eta = \frac{1}{9}$$

and, for candidate variables X_1, X_2 , define the potential

$$\tau(X_1, X_2) = d_{\text{ent}}(X_1, X_2) + \eta d_{\text{ent}}(X_1^0, X_1) + \eta d_{\text{ent}}(X_2^0, X_2). \quad (13.5)$$

The first term rewards making the two variables more subgroup-like relative to each other. The penalty terms prevent us from winning by replacing them with unrelated variables.

Proof deferred to a later volume Volume III: GGMT multidistance decrement

Let $G = \mathbb{F}_2^n$. If

$$d_{\text{ent}}(X_1, X_2) > 0,$$

then there exist G -valued variables X'_1, X'_2 such that

$$\tau(X'_1, X'_2) < \tau(X_1, X_2).$$

Once this proposition is known, the global theorem is short and completely transparent.

Proof of ?? from ??. The space of probability distributions on the finite set G is compact, so the continuous functional τ attains a minimum at some pair (X_1, X_2) . By the contrapositive of the decrement proposition,

$$d_{\text{ent}}(X_1, X_2) = 0.$$

The triangle inequality first gives

$$d_{\text{ent}}(X_1, X_1) \leq 2d_{\text{ent}}(X_1, X_2) = 0.$$

Hence ?? identifies the law of X_1 with the uniform law on a coset of some subspace H ; translating that coset does not change entropic Ruzsa distance, so we may write the corresponding reference law as U_H . A second use of the triangle inequality then gives

$$d_{\text{ent}}(X_2, U_H) \leq d_{\text{ent}}(X_2, X_1) + d_{\text{ent}}(X_1, U_H) = 0.$$

Thus

$$d_{\text{ent}}(X_1, U_H) = d_{\text{ent}}(X_2, U_H) = 0.$$

Therefore

$$\begin{aligned} & \eta(d_{\text{ent}}(X_1^0, U_H) + d_{\text{ent}}(X_2^0, U_H)) \\ &= \eta(d_{\text{ent}}(X_1^0, X_1) + d_{\text{ent}}(X_2^0, X_2)) \\ &= \tau(X_1, X_2) \\ &\leq \tau(X_1^0, X_2^0) \\ &= (1 + 2\eta)d_{\text{ent}}(X_1^0, X_2^0). \end{aligned}$$

With $\eta = 1/9$,

$$\frac{1 + 2\eta}{\eta} = 11,$$

which gives (??).

Finally the triangle inequality implies

$$\left| d_{\text{ent}}(X_1^0, U_H) - d_{\text{ent}}(X_2^0, U_H) \right| \leq d_{\text{ent}}(X_1^0, X_2^0).$$

Together with the sum bound $11d_{\text{ent}}(X_1^0, X_2^0)$, this shows that each term is at most $6d_{\text{ent}}(X_1^0, X_2^0)$. $\square$

13.5 Inside the distance-decrement proposition

The decrement proposition is the mathematical core. It is not a black-box Fourier estimate; it is a carefully organized choice between *sums* and *fibres*.

Take independent copies

$$X_1, \tilde{X}_1 \sim X_1, \quad X_2, \tilde{X}_2 \sim X_2.$$

The first candidate replacements are sums:

$$(X_1 + \tilde{X}_2, X_2 + \tilde{X}_1)$$

or

$$(X_1 + \tilde{X}_1, X_2 + \tilde{X}_2).$$

The second candidates are fibres, such as

$$X_1 \mid X_1 + \tilde{X}_2 = g, \quad X_2 \mid X_2 + \tilde{X}_1 = g'.$$

Why these choices? Addition smooths a distribution; conditioning on a sum moves in the opposite direction, selecting a fibre. The entropy chain rule forces a tradeoff between the two. If the sum choices do not decrease the potential enough, the relevant fibre choices become favorable unless a set of entropy inequalities is nearly sharp.

Proof architecture The four-stage core argument

1. **Fibring inequalities.** Compare the Ruzsa distances of the sum variables with the average distances of their fibres. Contraction under homomorphisms is the conceptual source of these inequalities.
2. **Try the obvious sum/fibre candidates.** If any candidate gives sufficient improvement, the decrement proposition is finished.
3. **Near-equality forces conditional near-independence.** If every easy candidate fails, several conditional mutual informations must be small. Thus three particular sums are almost pairwise independent after conditioning on their total.
4. **Endgame.** Three pairwise independent variables whose sum is constant satisfy a remarkable identity: the sum of their three pairwise Ruzsa distances is zero. With small rather than zero mutual information, an entropic Balog–Szemerédi–Gowers lemma gives a pair with sufficiently small distance, producing the required decrement.

We now verify the exact zero-information model, because it explains the endgame better than pages of bookkeeping.

Lemma 13.5 (The exact endgame). *Suppose T_1, T_2, T_3 are pairwise independent $\mathbb{F}_2^n$ -valued variables and*

$$T_1 + T_2 + T_3 = c$$

almost surely for a constant c . Then

$$d_{\text{ent}}(T_1, T_2) + d_{\text{ent}}(T_2, T_3) + d_{\text{ent}}(T_3, T_1) = 0.$$

In particular every one of the three distances is zero.

Proof. Because T_1, T_2 are independent and

$$T_1 + T_2 = c - T_3,$$

we have

$$d_{\text{ent}}(T_1, T_2) = H(T_3) - \frac{1}{2}H(T_1) - \frac{1}{2}H(T_2).$$

The two cyclic analogues are

$$d_{\text{ent}}(T_2, T_3) = H(T_1) - \frac{1}{2}H(T_2) - \frac{1}{2}H(T_3),$$

and

$$d_{\text{ent}}(T_3, T_1) = H(T_2) - \frac{1}{2}H(T_3) - \frac{1}{2}H(T_1).$$

Adding gives zero. Each Ruzsa distance is nonnegative, so all three vanish. $\square$

This identity is the conceptual heart of the endgame. The actual proof shows that if the conditional mutual informations are merely small, a quantitative replacement of this conclusion survives.

13.6 The entropic Balog–Szemerédi–Gowers step

The robust endgame uses an information-theoretic version of BSG. In schematic form:

Proof deferred to a later volume Volume III: entropic Balog–Szemerédi–Gowers

If two random variables R_1, R_2 have small mutual information and their sum has entropy not much larger than the average of their individual entropies, then after conditioning on suitable fibres one obtains variables R'_1, R'_2 with small entropic Ruzsa distance. Quantitatively, the loss is linear in the mutual information and the entropy defect.

This is the entropy counterpart of the classical statement

$$\text{many additive coincidences} \implies \text{large subset with small doubling.}$$

The difference is that conditioning replaces subset extraction, and entropy keeps precise track of how expensive that conditioning was.

13.7 Why characteristic two matters in this proof

Several identities simplify because

$$x = -x.$$

More importantly, the exact endgame and the sum/fibre transformations interact particularly cleanly with subspaces in $\mathbb{F}_2^n$. Discrete Gaussian distributions, which create an obstruction in torsion-free groups, do not appear in the same way.

The bounded-torsion extension therefore requires a more symmetric object than pairwise Ruzsa distance: *multidistance*. That is the subject of the next chapter.

13.8 Worked example: a random dense subset of a subspace

Let $H \leq \mathbb{F}_2^n$ and let $A \subseteq H$ be a random subset of density about $1/K$. Heuristically,

$$|A| \approx |H|/K,$$

while for a sufficiently dense random set,

$$A + A \approx H.$$

Thus the doubling constant is about K .

The correct structural conclusion is not that A itself contains a huge subspace. Rather, A is contained in one coset of the subspace H , whose size is about $K|A|$. If the theorem insists on a subgroup of size at most $|A|$, one passes to a subspace of H of index about K and uses about K cosets.

This model explains why polynomial *covering* is the right conclusion and why the factor of two in the theorem is natural near powers-of-two size thresholds.

Research window Quantitative refinement

The original characteristic-two argument yields the covering exponent 12 through the entropic constant 11 and the conversion in ???. Jyun-Jie Liao refined both pieces of the argument and obtained exponent 9. The conceptual architecture is unchanged: entropy distance, potential descent, sum/fibre alternatives, and an endgame controlled by mutual information.

Proof deferred to a later volume Volume III: entropic PFR complete proof

The characteristic-two bounded-torsion formulation is proved in Volume III together with the GGMT descent.

Chapter 14

Bounded Torsion and the Current Freiman–Ruzsa Frontier

14.1 From characteristic two to exponent m

Let G be an abelian group of torsion m , meaning

$$mg = 0 \quad \text{for every } g \in G.$$

The characteristic-two argument cannot simply be copied: the three-variable identity used in the endgame is special, and signs no longer disappear. Gowers, Green, Manners and Tao overcame this by replacing pairwise Ruzsa distance with an m -variable quantity.

Theorem 14.1 (Marton’s conjecture for bounded torsion). *Let G be an abelian group of torsion $m \geq 2$, and let $A \subseteq G$ be finite and nonempty with*

$$|A + A| \leq K|A|.$$

Then A can be covered by at most

$$(2K)^{O(m^3)}$$

cosets of a subgroup $H \leq G$ satisfying

$$|H| \leq |A|.$$

For fixed m , the dependence on K is polynomial.

Proof deferred to a later volume Proof in Volume III: bounded-torsion PFR

The full Gowers–Green–Manners–Tao proof is proved in Volume III, including the multidistance decrement, the $m \times m$ entropy array, and the stability endgame. This chapter establishes the multidistance calculus used by that proof and records the iteration architecture, but no earlier theorem depends on the bounded-torsion conclusion.

This is the full bounded-torsion form of Marton’s conjecture.

14.2 Multidistance

Definition 14.2 (Multidistance). Let I be a nonempty finite index set and let $(X_i)_{i \in I}$ be finitely supported G -valued random variables. Take independent copies $\tilde{X}_i$ and define

$$D[(X_i)_{i \in I}] = H\left(\sum_{i \in I} \tilde{X}_i\right) - \frac{1}{|I|} \sum_{i \in I} H(\tilde{X}_i).$$

The quantity is nonnegative. Indeed the entropy of a sum of independent variables is at least the entropy of any one summand, so it is at least their average.

When $|I| = 2$,

$$D[X_1, X_2] = d_{\text{ent}}(X_1, -X_2).$$

Thus multidistance is genuinely an extension of entropic Ruzsa distance.

Why take exactly m variables in a group of torsion m ? Because the relation

$$mx = 0$$

permits a cyclic array of sums whose total collapses. The m -fold symmetry replaces the characteristic-two cancellation $x + x = 0$.

14.3 The multidistance chain rule

A remarkable feature is that multidistance decomposes cleanly across a homomorphism.

Theorem 14.3 (Multidistance chain rule). *Let $\pi : G \rightarrow H$ be a homomorphism and suppose the X_i are independent. Then*

$$\begin{aligned} D[(X_i)_{i \in I}] &= D[(X_i)_{i \in I} \mid (\pi X_i)_{i \in I}] \\ &\quad + D[(\pi X_i)_{i \in I}] \\ &\quad + I \left(\sum_i X_i : (\pi X_i)_{i \in I} \mid \pi \left(\sum_i X_i \right) \right). \end{aligned}$$

Here the first term means the average multidistance of the conditional fibre distributions.

Proof. Write

$$S = \sum_i X_i.$$

By the chain rule for entropy,

$$H(S) = H(S \mid (\pi X_i)_i) + H(\pi S) + I(S : (\pi X_i)_i \mid \pi S).$$

For each i ,

$$H(X_i) = H(X_i \mid \pi X_i) + H(\pi X_i).$$

Average the latter identities over i and subtract them from the first. The three resulting differences are precisely the three terms in ?? $\square$

The last term is a mutual information and hence nonnegative. Therefore the chain rule simultaneously controls the multidistance of the quotient and the average multidistance inside fibres. This is the many-variable form of the sum/fibre tradeoff from Chapter ??.

14.4 The decrement theorem

For $I = \{1, \dots, m\}$, the bounded-torsion proof is driven by the following quantitative descent.

Proof deferred to a later volume Volume III: GGMT multidistance decrement

There is an absolute $c > 0$ such that, with

$$\eta = c/m^3,$$

whenever

$$D[(X_i)_{i=1}^m] > 0,$$

one can construct variables $X'_1, \dots, X'_m$ satisfying

$$D[(X'_i)] \leq (1 - \eta)D[(X_i)] - \eta \sum_{i=1}^m d_{\text{ent}}(X_i, X'_i). \quad (14.1)$$

If the original variables are supported in a symmetric set S containing 0, the new variables may be chosen inside $m^3 S$.

The proof occupies the main technical body of the bounded-torsion argument. Its architecture is worth understanding.

Proof architecture Why an $m \times m$ array appears

Take independent variables

$$X_{i,j}, \quad i, j \in \mathbb{Z}/m\mathbb{Z},$$

so that each column is a permutation of the original m laws. Compare the column sums

$$C_j = \sum_i X_{i,j}$$

with the row sums

$$R_i = \sum_j X_{i,j}.$$

Because the group has torsion m , carefully shifted choices of the array make the total relations among rows and columns compatible with the identity $mx = 0$.

Repeated application of the multidistance chain rule gives a dichotomy:

1. a quotient or a family of fibres produces a definite decrement; or
2. all the inequalities are close to equality, forcing the row and column statistics to have small conditional mutual information.

In the second case an m -variable endgame, together with an entropic BSG argument, constructs the improved tuple. The bookkeeping costs a factor of order m^3 , which is the source of the exponent in ??.

14.5 Iteration to the small-distance regime

The global part of the bounded-torsion proof is elementary once the decrement theorem is available.

Set

$$k_0 = D[(X_i^{(0)})].$$

Iterate (??). Ignoring the nonpositive movement penalty gives

$$D[(X_i^{(t)})] \leq (1 - \eta)^t k_0.$$

After

$$t = O(m^3 \log(2 + k_0))$$

steps, the multidistance is below an absolute small constant.

The required base case is a stability theorem:

Proof deferred to a later volume Proof in Volume III: GGMT stability theorem

If

$$D[(X_i)_{i=1}^m] \leq c_0$$

for a sufficiently small absolute c_0 , then there is a subgroup H for which

$$\sum_{i=1}^m d_{\text{ent}}(X_i, U_H) \ll m D[(X_i)].$$

If all variables are supported in a symmetric set S containing 0, one may take $H \subseteq 6S$.

Sum the movement penalties in (??) and use the Ruzsa triangle inequality to compare the terminal variables with the initial ones. This yields

$$\sum_{i=1}^m d_{\text{ent}}(X_i^{(0)}, U_H) \ll m^3 D[(X_i^{(0)})]. \quad (14.2)$$

The same set-to-entropy and covering conversion as in Chapter ?? now gives ??.

Idea The conceptual progression from 2 to m

Characteristic two used a pairwise distance and a three-variable endgame. Bounded torsion uses an m -variable distance and an $m \times m$ chain-rule machine. What survives unchanged is the meta-principle:

not subgroup-like
 $\implies$ construct a new law with smaller entropy potential.

The subgroup appears as the unique stable endpoint of the descent.

14.6 Worked example: $\mathbb{F}_3^n$

Let $G = \mathbb{F}_3^n$. Here $m = 3$, so for independent X_1, X_2, X_3 one considers

$$D[X_1, X_2, X_3] = H(X_1 + X_2 + X_3) - \frac{1}{3}(H(X_1) + H(X_2) + H(X_3)).$$

If all three are uniform on the same subspace V , then their sum is uniform on V , so

$$D = 0.$$

If they are uniform on three cosets of V , the same is true after translating the sum. Thus multidistance correctly regards affine subspace-uniform distributions as exact structured objects.

The bounded-torsion theorem says that every small-doubling set in $\mathbb{F}_3^n$ is polynomially covered by cosets of a subspace, with constants depending only on the exponent 3. This is a special case of the bounded-torsion theorem of Gowers–Green–Manners–Tao [4].

14.7 Why this does not solve general PFR

The hypothesis “torsion m ” means that every element has uniformly bounded order. It excludes groups such as

$$\mathbb{Z}, \quad \mathbb{Z}/N\mathbb{Z} \text{ with unbounded } N,$$

and arbitrary products with large cyclic factors.

In torsion-free groups, approximate structure has genuine progression directions rather than only subgroup directions. Entropy descent encounters distributions analogous to discrete Gaussians, which have bounded entropy doubling but are not close to uniform measures on finite subgroups. A general theory must therefore accommodate both:

$$\text{subgroup structure} \quad \text{and} \quad \text{convex/progression geometry}.$$

Thus the bounded-torsion argument does not by itself resolve the unbounded-torsion or torsion-free quantitative problem; progression geometry has to be incorporated as an additional mechanism.

14.8 The general abelian frontier

The classical Green–Ruzsa theorem gives K -dependent coset progressions; Sanders and Konyagin brought the dependence to quasipolynomial scale. Recent entropy–Fourier methods have moved much closer to the polynomial threshold.

A useful modern formulation asks for a function $f(K)$ such that every A of doubling at most K can be covered by at most

$$\exp(f(K))$$

translates of a convex coset progression of dimension at most $f(K)$ and size at most

$$\exp(f(K))|A|.$$

Polynomial Freiman–Ruzsa corresponds to

$$f(K) = O(\log K).$$

Raghavan’s February 2026 revision proves the following genuinely stronger near-polynomial statement: for every fixed $\varepsilon > 0$ there is a constant C_ε such that one may take

$$f_\varepsilon(K) = C_\varepsilon \log(2K)^{1+\varepsilon}.$$

Equivalently, the covering number, progression dimension, and relative size loss are bounded by $\exp(C_\varepsilon \log(2K)^{1+\varepsilon})$. This improves the older regime in which the power of $\log K$ was larger than 3, but it does *not* prove an $O(\log K \log \log K)$ bound. The method combines entropy and Fourier analysis rather than belonging purely to either tradition [11].

Research window What the current “near polynomial” bound means

For fixed $\varepsilon > 0$,

$$\exp(C_\varepsilon \log(2K)^{1+\varepsilon}) = (2K)^{C_\varepsilon \log(2K)^\varepsilon}.$$

Thus the covering number is still super-polynomial in K : its effective polynomial exponent grows like a small power of $\log K$. But because ε may be chosen arbitrarily small (with a corresponding change in C_ε), the theorem gets arbitrarily close to the conjectural $K^{O(1)}$ scale in the exponent-of-log K sense. This is the precise meaning of “near polynomial” used here.

14.9 Algorithmic PFR

A structural theorem can ask only whether an appropriate subspace exists. An algorithmic theorem asks whether it can actually be found from access to the set.

A 2026 preprint of Castro–Silva, Briët, Arunachalam, Dutt and Gur gives an algorithmic version of PFR in $\mathbb{F}_2^n$ [1]: from sample and membership-query access to a small-doubling set, one can efficiently output a basis for a subspace of size at most $|A|$ whose polynomially many translates cover A . Their framework also produces algorithmic versions of related polynomial inverse theorems and makes a striking connection between quadratic Fourier analysis and finite symplectic geometry.

This development is important conceptually. The Gowers–Green–Manners–Tao proof is organized around an entropy-minimization/decrement argument whose clean existence form does not by itself supply an efficient recovery procedure. Algorithmic PFR asks for the hidden additive structure in a form that can actually be recovered.

14.10 A 2026 robustness direction

An August 2026 preprint of Cheng Peng asks what happens when the structured set is not observed exactly [10]. Its model gives access to a set B that is adversarially close in symmetric difference to an unknown small-doubling set A , and asks for subspace structure that remains valid for compatible hidden sets. This is a very recent, unrefereed research-frontier result; it is recorded only as a direction and is not used anywhere in the proof architecture of this book.

The point of recording these developments is not to turn this volume into a news survey. It is to show that the structural language developed here—energy, covering, Freiman models, entropy distance and subgroup approximation—is now being converted into computational and robust forms.

Proof deferred to a later volume Proof in Volume III for the stated PFR theorem; further variants are research extensions

The broader general-abelian entropy–Fourier refinements discussed here remain a research window outside the frozen scope of this book; they are not part of the arguments in Volumes I–V. Any future treatment should be maintained separately and rechecked against the final published versions of results that are currently preprints.

14.11 Volume II synthesis

The volume began with the classical inverse question

$$|A + A| \leq K|A| \implies \text{what is } A?$$

We have now seen three generations of answers.

Generation I: geometric inverse theory

Freiman relations $\rightarrow$ models $\rightarrow$ Bohr sets $\rightarrow$ coset progressions.

Generation II: almost-periodicity

random sampling $\rightarrow$ Croot–Sisask almost-periods
 $\rightarrow$ Sanders’ polylogarithmic-dimensional structure.

Generation III: entropy descent

entropic Ruzsa distance $\rightarrow$ sum/fibre descent
 $\rightarrow$ subgroup-uniform endpoints
 $\rightarrow$ polynomial PFR in bounded torsion.

The mathematics changes, but the central philosophy from Volume I survives intact:

failure of growth is evidence of hidden algebraic geometry.

Volumes III–V turn from inverse structure to quantitative Fourier methods, additive patterns, Gowers uniformity, and higher-order inverse theory. Circle-method and prime-distribution material belongs to the separate Analytic Number Theory project; noncommutative growth and expansion belong to the separate Bourgain–Gamburd project. They are not future volumes of this book.

Chapter 15

Auxiliary Results for the Classical Freiman Machinery

This chapter collects the auxiliary results needed to complete the classical Freiman-theoretic arguments of Volume II. Kneser's theorem, Vosper's theorem, Ruzsa calculus, Plünnecke–Ruzsa, the asymmetric Balog–Szemerédi–Gowers theorem (Theorem I.2.015 of Volume I), the Green–Ruzsa modelling argument, Minkowski's theorems, and Freiman's dimension lemma were established earlier; we record here the remaining critical-pair, robust-BSG, geometry-of-numbers, properization, and dimension-reduction consequences.

15.1 Critical-pair consequences

Theorem 15.1 (II.1.003: Scherk–Kemperman inequality). *Let A, B be finite nonempty subsets of an abelian group. Suppose some $c \in A + B$ has a unique representation $c = a + b$. Then*

$$|A + B| \geq |A| + |B| - 1.$$

Proof. Translate so that $c = 0$ and its unique representation is $0 = 0 + 0$; thus $0 \in A \cap B$ and $A \cap (-B) = \{0\}$. Let $H = \text{Stab}(A + B)$. Kneser's theorem gives

$$|A + B| \geq |A + H| + |B + H| - |H|.$$

Write $a = |A|$, $b = |B|$, $h = |H|$ and let r, s be the numbers of H -cosets meeting A, B . Thus $|A + H| = rh$ and $|B + H| = sh$. If both A and B filled all the H -cosets they meet, then the cosets containing 0 would contain H in both A and B , and every $x \in H$ would give $0 = x + (-x)$, contrary to uniqueness unless $H = \{0\}$. More generally, in the two zero cosets the subsets $A \cap H$ and $-B \cap H$ intersect only at 0, so

$$|A \cap H| + |B \cap H| \leq h + 1.$$

All other occupied cosets contain at most h points. Consequently

$$a + b \leq (r + s - 1)h + 1 = |A + H| + |B + H| - h + 1.$$

Combining with Kneser gives $|A + B| \geq a + b - 1$. □

Theorem 15.2 (II.1.006: Aperiodic reduction and the prime-cyclic equality case). *Let A, B be finite nonempty subsets of an abelian group G , let $H = \text{Stab}(A + B)$, and write $\bar{A}, \bar{B}$ for the images of A, B in G/H . Then:*

(i) $\text{Stab}(\bar{A} + \bar{B}) = \{0\}$ and

$$|\bar{A} + \bar{B}| \geq |\bar{A}| + |\bar{B}| - 1.$$

(ii) If $G = \mathbb{Z}/p\mathbb{Z}$ for a prime p , $|A|, |B| \geq 2$, and

$$|A + B| = |A| + |B| - 1 \leq p - 2,$$

then A and B are arithmetic progressions with a common nonzero difference.

Proof. For (i), $A + B$ is a union of H -cosets and the quotient sumset has trivial stabilizer by maximality of H . Kneser's theorem in the quotient gives the displayed aperiodic bound. Part (ii) is exactly the already-proved Vosper theorem. $\square$

15.2 A robust BSG form

Theorem 15.3 (II.3.007: robust/noisy BSG). *For every $K \geq 1$ and $0 \leq \eta < 1/2$ there is $C = C(K, \eta)$ with the following property. Let A, B be finite sets and let $\Gamma \subseteq A \times B$ have at least $|A||B|/K$ edges. Suppose a subset $\Gamma_0 \subseteq \Gamma$ with $|\Gamma \setminus \Gamma_0| \leq \eta|\Gamma|$ satisfies*

$$|A +_{\Gamma_0} B| \leq K(|A||B|)^{1/2}.$$

Then there are $A' \subseteq A$, $B' \subseteq B$ with

$$|A'| \geq C^{-1}|A|, \quad |B'| \geq C^{-1}|B|, \quad |A' + B'| \leq C(|A'||B'|)^{1/2}.$$

With $C_{\text{BSG}}(K_1, K_2)$ as defined in the asymmetric BSG theorem, Theorem I.2.015 of Volume I, one may take

$$C = C_{\text{BSG}}\left(\frac{K}{1-\eta}, K\right).$$

In particular, the polynomial form of that theorem gives polynomial dependence on $K/(1-\eta)$.

Proof. We have $|\Gamma_0| \geq (1-\eta)|\Gamma| \geq (1-\eta)|A||B|/K$. Apply Theorem I.2.015 of Volume I (the asymmetric BSG theorem) to the graph Γ_0 , with density parameter $K/(1-\eta)$ and restricted-sumset parameter K . Its conclusion is precisely the displayed one, with polynomial dependence on those parameters. This is exactly the constant displayed in the statement. Thus deleting a controlled fraction of noisy edges changes only the BSG parameters. $\square$

15.3 Geometry of numbers and properization

Proposition 15.4 (II.4.004: lattice points from successive minima). *Let $K \subset \mathbb{R}^d$ be a symmetric convex body, Λ a full lattice, and $\lambda_1 \leq \dots \leq \lambda_d$ the successive minima of K with respect to Λ . Then*

$$|K \cap \Lambda| \leq 3^d \prod_{i=1}^d \max(1, \lambda_i^{-1}).$$

Proof. Let m be maximal with $\lambda_m \leq 1$; if none exists then $K \cap \Lambda = \{0\}$. By definition of successive minima, $K \cap \Lambda$ lies in an m -dimensional subspace V , and $\Lambda_V = \Lambda \cap V$ is a rank- m lattice. Apply Blichfeldt's packing argument in V to the half body $\frac{1}{2}K \cap V$: every translate of a fundamental parallelepiped contains at most one point from a maximal $(\frac{1}{2}K)$ -separated subset of $K \cap \Lambda$, while the translates of $\frac{1}{2}K$ around that subset lie in $\frac{3}{2}K$. Hence

$$|K \cap \Lambda| \leq \frac{\text{vol}_V(\frac{3}{2}K)}{\det \Lambda_V} + 1 \leq 3^m \frac{\text{vol}_V(\frac{1}{2}K)}{\det \Lambda_V} + 1.$$

Minkowski's second theorem in V gives

$$\text{vol}_V(K) \leq 2^m \det \Lambda_V (\lambda_1 \cdots \lambda_m)^{-1}.$$

Substitution and a harmless enlargement of the constant yield $|K \cap \Lambda| \leq 3^m \prod_{i \leq m} \lambda_i^{-1}$, and inserting factors 1 for $i > m$ proves the claim. $\square$

Theorem 15.5 (II.4.005: bounded-rank properization). *For every rank r and $t \geq 1$ there is $C_{r,t}$ such that every symmetric rank- r generalized arithmetic progression*

$$P = \{n_1 v_1 + \cdots + n_r v_r : |n_i| \leq N_i\}$$

in an abelian group is contained in a symmetric t -proper progression Q of rank at most r satisfying

$$|Q|_{\text{par}} \leq C_{r,t} |P|_{\text{par}}.$$

The same statement holds for coset progressions with the subgroup part unchanged.

Proof. We induct on the rank. The assertion is trivial in rank 0, and if P is already t -proper there is nothing to change. Suppose it is not t -proper. Write $B = \prod_{i=1}^r [-N_i, N_i] \subset \mathbb{Z}^r$ and $\phi(n) = \sum_i n_i v_i$. Non- t -properness gives distinct $n, n' \in tB$ with $\phi(n) = \phi(n')$. After dividing $a = n - n'$ by the gcd of its coordinates we have a primitive nonzero vector

$$a \in \ker \phi, \quad |a_i| \leq 2tN_i. \quad (2.21)$$

Let $\pi : \mathbb{Z}^r \rightarrow \Lambda := \mathbb{Z}^r / \mathbb{Z}a$. Since a is primitive, Λ is a free lattice of rank $r - 1$, and ϕ factors through a homomorphism $\bar{\phi} : \Lambda \rightarrow G$. Hence it is enough to contain $\pi(B)$ efficiently by a rank- $(r - 1)$ progression in the quotient lattice.

We record the quantitative point. Identify the real quotient with $a^\perp$ by orthogonal projection and put $K = \pi_{\mathbb{R}}(\prod [-N_i, N_i])$. The projected lattice has covolume $\det \Lambda = \|a\|_2^{-1}$. Cauchy's projection formula for a rectangular box gives

$$\text{vol}_{r-1}(K) = \sum_{i=1}^r \frac{|a_i|}{\|a\|_2} \prod_{j \neq i} (2N_j).$$

Consequently

$$\frac{\text{vol}_{r-1}(K)}{\det \Lambda} = \sum_i |a_i| \prod_{j \neq i} (2N_j) \leq C_{r,t} \prod_i (2N_i + 1) = C_{r,t} |B|_{\text{par}}, \quad (2.22)$$

where (2.21) was used in the last step. The lattice-point estimate II.4.004, applied in the quotient space, therefore gives $|K \cap \Lambda| \leq C'_{r,t} |B|_{\text{par}}$.

Apply the convex-lattice approximation theorem II.4.008 below to K and Λ . It gives a proper symmetric quotient progression R of rank at most $r - 1$ such that $\pi(B) \subseteq K \cap \Lambda \subseteq R_{C_r}$ and

$$|R_{C_r}|_{\text{par}} \leq C''_{r,t} |B|_{\text{par}}.$$

Its image $P_1 = \bar{\phi}(R_{C_r})$ is a symmetric progression in G , contains P , has rank at most $r - 1$, and has parameter-box size at most $C''_{r,t} |P|_{\text{par}}$. It need not yet be t -proper, but its rank has strictly decreased. Apply the induction hypothesis to P_1 . At most r rank reductions occur, so the product of the rank-dependent enlargement constants is still a constant $C_{r,t}$. A subgroup part in a coset progression is unaffected throughout.

The proof of Theorem II.4.008 uses only the adapted-basis and successive-minima lemmas, so its use here is independent of the properization theorem. $\square$

Theorem 15.6 (II.4.008: convex-lattice progression approximation). *Let $K \subset \mathbb{R}^d$ be a symmetric convex body and Λ a lattice. There is a proper symmetric GAP $P \subseteq K \cap \Lambda$ of rank at most d such that*

$$K \cap \Lambda \subseteq P_{C_d} \quad \text{and} \quad |P| \asymp_d |K \cap \Lambda|.$$

Here P_{C_d} denotes dilation of the parameter box by a dimension-dependent constant.

Proof. Let $V = \text{span}(K \cap \Lambda)$ and $m = \dim V$. The adapted-basis lemma proved in Appendix ?? gives a basis $b_1, \dots, b_m$ of the lattice Λ_V such that, with λ_i the successive minima,

$$c_m \lambda_i^{-1} \leq M_i := \max\{M \geq 0 : [-M, M]b_i \subseteq C_m K\} \leq C_m \lambda_i^{-1}.$$

After shrinking the M_i by a fixed m -dependent factor, convexity gives the proper progression

$$P = \left\{ \sum_{i=1}^m n_i b_i : |n_i| \leq c'_m \lambda_i^{-1} \right\} \subseteq K \cap \Lambda.$$

Conversely the same adapted-coordinate bounds put every point of $K \cap \Lambda$ in a C_m -dilate of this box, proving the covering inclusion. Finally

$$|P| \asymp_m \prod_i \max(1, \lambda_i^{-1}),$$

and the preceding lattice-point proposition gives the matching upper bound for $|K \cap \Lambda|$, while $P \subseteq K \cap \Lambda$ gives the lower bound. This is the John-type form required later; no sharper convex-geometric constant is used. $\square$

15.4 The remaining Freiman closures

Proposition 15.7 (II.5.005: quantitative dimension reduction). *Let A be a finite set Freiman-isomorphic to a full-dimensional subset of $\mathbb{R}^d$ and suppose $|A + A| \leq K|A|$. Then*

$$d \leq K - 1 + \frac{\binom{d+1}{2}}{|A|}.$$

In particular, if $|A| \geq 2K^2$, then $d \leq K$, while if $|A| < 2K^2$ the set has cardinality bounded solely in terms of K and may be absorbed into a rank- $O_K(1)$ progression directly.

Proof. The Freiman–Matolcsi–Ruzsa dimension lemma already proved in the classical core gives

$$|A + A| \geq (d+1)|A| - \binom{d+1}{2}.$$

Combining this with $|A + A| \leq K|A|$ gives the displayed inequality

$$d \leq K - 1 + \frac{d(d+1)}{2|A|}. \tag{2.31}$$

Assume now $M := |A| \geq 2K^2$. Because A affinely spans dimension d , we also have $d \leq M - 1$. Rearranging (2.31),

$$F(d) := d^2 + (1 - 2M)d + 2M(K - 1) \geq 0. \tag{2.32}$$

The quadratic F is convex. Directly,

$$F(K) = K(K + 1 - 2M) + 2M(K - 1) = K^2 + K - 2M < 0$$

for every $K \geq 1$ and $M \geq 2K^2$, since $K^2 + K - 2M \leq K - 3K^2 < 0$. On the other hand the second root of F is larger than $M - 1$: indeed

$$F(M - 1) = -(M - 1)^2 + 2M(K - 1) < 0$$

under $M \geq 2K^2$, while $F(d) \rightarrow +\infty$ only after passing beyond the midpoint $M - \frac{1}{2}$. Hence the only integers $0 \leq d \leq M - 1$ satisfying (2.32) lie at or below the first root, which is $< K + 1$. Therefore $d \leq K$.

If $M < 2K^2$, translate one point of A to 0 and take the remaining at most $M - 1$ elements as generators. The box with coefficients in $\{0, 1\}$ contains the translated set, so A is contained in a progression of rank at most $M - 1 < 2K^2$. This is the bounded-cardinality alternative needed in the Freiman assembly. $\square$

Theorem 15.8 (II.5.006: classical finite-field Freiman theorem). *Fix a prime p . There is $C_p > 0$ such that whenever $A \subset \mathbb{F}_p^n$ is nonempty and*

$$|A + A| \leq K|A|,$$

there is an affine subspace $x + V$ containing A with

$$|V| \leq p^{K^{C_p}} |A|.$$

Thus the dependence is exponential in a polynomial of K , the classical level required in this volume.

Proof. Apply the quantitative Green–Ruzsa Freiman theorem already assembled in this volume (whose geometric inputs are now proved in II.4.004–II.4.008). It gives a proper coset progression

$$P = H + \{n_1 v_1 + \cdots + n_r v_r : |n_i| \leq N_i\}$$

containing A , with $r \leq K^C$ and $|P| \leq \exp(K^C)|A|$ for an absolute classical exponent C (we do not optimize it here).

Because the ambient group has exponent p , both H and every v_i lie in an elementary abelian p -group. Let

$$V = H + \text{span}_{\mathbb{F}_p}(v_1, \dots, v_r).$$

Then $P \subseteq V$, hence after undoing the initial translation A lies in an affine subspace. Moreover

$$|V| \leq p^r |H| \leq p^r |P| \leq p^{K^C} \exp(K^C) |A|.$$

Since p is fixed, enlarge the exponent to K^{C_p} to absorb the exponential factor. This proves the stated classical finite-field Freiman bound without importing polynomial Freiman–Ruzsa. $\square$

Theorem 15.9 (II.5.008: torsion-sensitive modelling). *Let A be a finite subset of an abelian group of exponent dividing m with $|A + A| \leq K|A|$. There is a Freiman 2-isomorphic model A' inside a finite abelian group G' of exponent dividing m such that $G' = \langle A' \rangle$ and*

$$|G'| \leq m^{K^C} \exp(K^C) |A|$$

for an absolute classical exponent C . In particular, for fixed torsion exponent the model has size $\exp(K^{O(1)})|A|$ and retains the original torsion bound.

Proof. Translate A so that $0 \in A$ and apply the quantitative Green–Ruzsa theorem in the same form as above. Thus A is contained in a proper coset progression

$$P = H + \{n_1 v_1 + \cdots + n_r v_r : |n_i| \leq N_i\},$$

where $r \leq K^C$ and $|P| \leq \exp(K^C)|A|$. Put

$$G' = H + \langle v_1, \dots, v_r \rangle.$$

This is a subgroup of the original ambient group, so its exponent still divides m , and the identity map on A is certainly a Freiman 2-isomorphism onto its image $A' \subseteq G'$. Since each added generator has order dividing m ,

$$|G'| \leq m^r |H| \leq m^{K^C} |P| \leq m^{K^C} \exp(K^C) |A|.$$

Thus the modelling step preserves bounded torsion with an explicit classical bound. The point is that no passage to a large cyclic group of unrelated exponent is necessary. $\square$

Chapter 16

Worked structure laboratories

Worked Example 16.1 (Kneser equality from a subgroup obstruction). Let $H \leq G$ be finite and take $A = H \cup (a + H)$ with $a \notin H$. Then A is H -periodic and

$$A + A = H \cup (a + H) \cup (2a + H),$$

unless two of these cosets coincide. In the generic case $|A| = 2|H|$ and $|A + A| = 3|H| = 2|A| - |H|$. Kneser's correction term is therefore not an artifact of the proof: it measures the actual periodic obstruction to the aperiodic lower bound $|A + B| \geq |A| + |B| - 1$.

Worked Example 16.2 (A one-frequency Bohr set). In $\mathbb{Z}/N\mathbb{Z}$ fix $r \neq 0$ and let

$$B = \{x : \|rx/N\|_{\mathbb{R}/\mathbb{Z}} \leq \rho\}.$$

Since multiplication by r permutes the group, B is simply the pullback of an arc of length about $2\rho N$. It therefore contains a progression of length $\asymp \rho N$. In several frequencies a Bohr set is the simultaneous version of this picture; geometry of numbers replaces the obvious one-dimensional ordering.

Worked Example 16.3 (Entropy distance vanishes on a common coset model). Let $H \leq G$ and let X, Y be independent uniform random variables on the same coset $a + H$. Then $X - Y$ is uniform on H and

$$H(X) = H(Y) = H(X - Y) = \log |H|.$$

After the normalization used in the entropic Ruzsa distance, this is the zero-distance model. The GGMT decrement argument can be read as a quantitative attempt to move a small-doubling distribution toward precisely this subgroup/coset geometry.

Appendix A

Probability and Riesz-Product Tools

This appendix supplies the two analytic-probabilistic tools that were used in the main text but are not themselves part of the additive-combinatorial story: the sampling moment estimate behind Croot–Sisask and Rudin’s inequality for dissociated characters. The proofs are included because both tools are short enough that treating them as black boxes would hide exactly the mechanisms that later reappear in more sophisticated forms.

A.1 Symmetrization

Lemma A.1 (Symmetrization). *Let $Z_1, \dots, Z_k$ be independent random variables taking values in a normed vector space, let $Z'_1, \dots, Z'_k$ be an independent copy, and let $p \geq 1$. Then*

$$\mathbb{E} \left\| \sum_{i=1}^k (Z_i - \mathbb{E} Z_i) \right\|^p \leq \mathbb{E} \left\| \sum_{i=1}^k (Z_i - Z'_i) \right\|^p.$$

Proof. Condition on $Z_1, \dots, Z_k$. Since

$$\sum_{i=1}^k (Z_i - \mathbb{E} Z_i) = \mathbb{E}_{Z'} \sum_{i=1}^k (Z_i - Z'_i),$$

convexity of $v \mapsto \|v\|^p$ and Jensen’s inequality give

$$\left\| \sum_i (Z_i - \mathbb{E} Z_i) \right\|^p \leq \mathbb{E}_{Z'} \left\| \sum_i (Z_i - Z'_i) \right\|^p.$$

Now average over Z . □

A.2 Khintchine’s inequality from an exponential moment

Let $\varepsilon_1, \dots, \varepsilon_k$ be independent random signs, each taking values ± 1 with probability $1/2$.

Lemma A.2 (Subgaussian moment of a Rademacher sum). *For $p \geq 2$ and complex numbers $c_1, \dots, c_k$,*

$$\left(\mathbb{E}_{\varepsilon} \left| \sum_{i=1}^k \varepsilon_i c_i \right|^p \right)^{1/p} \leq C \sqrt{p} \left(\sum_{i=1}^k |c_i|^2 \right)^{1/2}$$

for an absolute constant C .

Proof. For real c_i , independence and $\cosh u \leq e^{u^2/2}$ give

$$\mathbb{E}_\varepsilon e^{t \sum_i \varepsilon_i c_i} = \prod_i \cosh(tc_i) \leq \exp\left(\frac{t^2}{2} \sum_i c_i^2\right).$$

Chernoff's bound therefore yields

$$\mathbb{P}\left(\left|\sum_i \varepsilon_i c_i\right| \geq u\right) \leq 2 \exp\left(-\frac{u^2}{2 \sum_i c_i^2}\right).$$

Using

$$\mathbb{E}|X|^p = p \int_0^\infty u^{p-1} \mathbb{P}(|X| \geq u) du$$

and the elementary estimate $\Gamma(p/2 + 1)^{1/p} \ll \sqrt{p}$ gives the real case. For complex coefficients, apply the real estimate separately to the real and imaginary parts and use $|z| \leq |\Re z| + |\Im z|$. $\square$

A.3 The sampling estimate used by Croot–Sisask

Theorem A.3 (Vector-valued sampling estimate). *Let (Ω, ν) be a probability space, $p \geq 2$, and let $Y_1, \dots, Y_k$ be independent identically distributed random functions in $L^p(\Omega)$. Write $F = \mathbb{E}Y_1$. Then*

$$\mathbb{E} \left\| \frac{1}{k} \sum_{i=1}^k (Y_i - F) \right\|_{L^p(\Omega)}^p \leq \left(C \sqrt{\frac{p}{k}} \right)^p \mathbb{E} \|Y_1\|_{L^p(\Omega)}^p.$$

Proof. Let $Y'_1, \dots, Y'_k$ be an independent copy. By ??,

$$\mathbb{E} \left\| \sum_i (Y_i - F) \right\|_p^p \leq \mathbb{E} \left\| \sum_i (Y_i - Y'_i) \right\|_p^p.$$

The independent differences $D_i = Y_i - Y'_i$ are symmetric, so their joint law is unchanged after multiplication by independent random signs ε_i . Hence the last expression equals

$$\mathbb{E}_{Y, Y'} \mathbb{E}_\varepsilon \int_\Omega \left| \sum_i \varepsilon_i D_i(x) \right|^p d\nu(x).$$

By ??, pointwise in x this is at most

$$(C\sqrt{p})^p \mathbb{E}_{Y, Y'} \int_\Omega \left(\sum_i |D_i(x)|^2 \right)^{p/2} d\nu(x).$$

For $p \geq 2$,

$$\left(\sum_i |d_i|^2 \right)^{p/2} \leq k^{p/2-1} \sum_i |d_i|^p.$$

Also

$$|Y_i - Y'_i|^p \leq 2^{p-1} (|Y_i|^p + |Y'_i|^p).$$

Consequently

$$\mathbb{E} \left\| \sum_i (Y_i - F) \right\|_p^p \leq (C\sqrt{p})^p k^{p/2} \mathbb{E} \|Y_1\|_p^p,$$

after absorbing the harmless factor 2^p into C^p . Divide by k^p . $\square$

This is exactly ??. Notice that the proof uses only symmetrization, random signs, and the convexity inequality comparing ℓ^2 and ℓ^p norms. No concentration theorem is hidden in the argument.

A.4 Riesz products and dissociated characters

Let G be a finite abelian group with normalized counting measure and let $\Lambda \subseteq \widehat{G}$ be dissociated. If $0 \leq u_\lambda \leq 1/2$ and $\omega_\lambda \in \mathbb{C}$ have modulus one, define the Riesz product

$$R(x) = \prod_{\lambda \in \Lambda} (1 + u_\lambda \Re(\omega_\lambda \lambda(x))).$$

Each factor is nonnegative.

Lemma A.4 (The constant term of a dissociated Riesz product). *Let G be a finite abelian group, let $\Lambda \subseteq \widehat{G}$ be dissociated, let $0 \leq u_\lambda \leq 1/2$, and let $|\omega_\lambda| = 1$. Define*

$$R(x) := \prod_{\lambda \in \Lambda} (1 + u_\lambda \Re(\omega_\lambda \lambda(x))).$$

Then

$$\mathbb{E}_{x \in G} R(x) = 1.$$

Proof. Expand every factor using

$$\Re(\omega_\lambda \lambda) = \frac{1}{2}(\omega_\lambda \lambda + \overline{\omega_\lambda} \lambda^{-1}).$$

A nonconstant term in the expansion has a character of the form

$$\prod_{\lambda \in \Lambda} \lambda^{\varepsilon_\lambda}, \quad \varepsilon_\lambda \in \{-1, 0, 1\},$$

with not all ε_λ equal to zero. Dissociativity says that this character is nontrivial, and the average of a nontrivial character is zero. Only the constant term 1 survives. $\square$

Lemma A.5 (Subgaussian exponential moment for a dissociated sum). *Let*

$$F(x) = \sum_{\lambda \in \Lambda} a_\lambda \lambda(x), \quad \sum_{\lambda} |a_\lambda|^2 = 1.$$

Then for every $t \geq 0$ and every complex ω of modulus one,

$$\mathbb{E}_x \exp(t \Re(\omega F(x))) \leq e^{4t^2}.$$

Proof. Fix $t > 0$ and split

$$\Lambda = L \sqcup S, \quad L = \{\lambda : t|a_\lambda| > 1/2\}.$$

Since $\sum |a_\lambda|^2 = 1$,

$$|L| < 4t^2, \quad \sum_{\lambda \in L} |a_\lambda| \leq \sqrt{|L|} \leq 2t.$$

Thus the large-coefficient part obeys the pointwise bound

$$t \Re \left(\omega \sum_{\lambda \in L} a_\lambda \lambda(x) \right) \leq 2t^2.$$

For $\lambda \in S$ set

$$u_\lambda = t|a_\lambda| \leq \frac{1}{2}, \quad z_\lambda(x) = \Re \left(\omega \frac{a_\lambda}{|a_\lambda|} \lambda(x) \right),$$

with an arbitrary phase if $a_\lambda = 0$. For $|w| \leq 1/2$,

$$\log(1+w) \geq w - 2w^2,$$

so

$$u_\lambda z_\lambda \leq \log(1 + u_\lambda z_\lambda) + 2u_\lambda^2.$$

Exponentiating and multiplying over S gives

$$\exp\left(t\Re\left(\omega \sum_{\lambda \in S} a_\lambda \lambda(x)\right)\right) \leq e^{2t^2 \sum_S |a_\lambda|^2} \prod_{\lambda \in S} (1 + u_\lambda z_\lambda(x)).$$

Average over x and use ???. The small part contributes at most e^{2t^2} , and the large part at most e^{2t^2} . Their product is at most e^{4t^2} . $\square$

Theorem A.6 (Rudin's inequality). *There is an absolute C such that, for every $p \geq 2$,*

$$\left\| \sum_{\lambda \in \Lambda} a_\lambda \lambda \right\|_p \leq C\sqrt{p} \left(\sum_{\lambda \in \Lambda} |a_\lambda|^2 \right)^{1/2}.$$

Proof. By homogeneity normalize the ℓ^2 norm of the coefficients to one. From ??? and Chernoff's bound,

$$\mathbb{P}(\Re(\omega F) \geq u) \leq \inf_{t>0} e^{4t^2 - tu} \leq e^{-u^2/16}.$$

For every complex number z , one of the four numbers $z, -z, iz, -iz$ has real part at least $|z|/\sqrt{2}$. Therefore

$$\mathbb{P}(|F| \geq u) \leq 4e^{-u^2/32}.$$

Integrating this tail as in the proof of Khintchine's inequality gives

$$\|F\|_p \leq C\sqrt{p}.$$

Rescale the coefficients. This proves ???. $\square$

Appendix B

Entropy Identities Used in Additive Geometry

This appendix develops the finite-valued entropy facts used in Chapters ??–??. We use base-2 logarithms. All random variables below take values in finite sets.

B.1 Entropy, conditional entropy, and relative entropy

For a random variable X with law $p(x)$,

$$H(X) = - \sum_x p(x) \log p(x),$$

where $0 \log 0$ is interpreted as 0. If (X, Y) has joint law $p(x, y)$, define

$$H(X | Y) = \sum_y p(y) H(X | Y = y).$$

For probability distributions p, q on the same finite set, with p supported where $q > 0$, define the relative entropy

$$D(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

Lemma B.1 (Log-sum inequality). *For nonnegative a_i, b_i with $A = \sum_i a_i > 0$ and $B = \sum_i b_i > 0$,*

$$\sum_i a_i \log \frac{a_i}{b_i} \geq A \log \frac{A}{B}.$$

Equality holds exactly when a_i/b_i is constant on indices with $a_i > 0$.

Proof. Put $p_i = a_i/A$ and $q_i = b_i/B$. Since $-\log$ is convex,

$$\sum_i p_i \log \frac{q_i}{p_i} \leq \log \left(\sum_i p_i \frac{q_i}{p_i} \right) = 0.$$

Multiply by A and add $A \log(A/B)$. □

Corollary B.2 (Gibbs inequality). *$D(p||q) \geq 0$, with equality if and only if $p = q$.*

Proof. Apply the log-sum inequality with $a_i = p_i$ and $b_i = q_i$. Since both totals are 1, the right-hand side is 0. Equality in log-sum requires p_i/q_i to be constant on the support; normalization forces that constant to be 1, hence $p = q$. □

B.2 The chain rule

Theorem B.3 (Chain rule). *For finite-valued X, Y ,*

$$H(X, Y) = H(Y) + H(X | Y) = H(X) + H(Y | X).$$

Proof. Write $p(x, y) = p(y)p(x | y)$ wherever $p(y) > 0$. Then

$$\begin{aligned} H(X, Y) &= - \sum_{x,y} p(x, y) \log p(x, y) \\ &= - \sum_{x,y} p(x, y) \log p(y) - \sum_{x,y} p(x, y) \log p(x | y) \\ &= H(Y) + H(X | Y). \end{aligned}$$

The other identity is symmetric. □

Repeatedly applying the chain rule gives

$$H(X_1, \dots, X_m) = \sum_{j=1}^m H(X_j | X_1, \dots, X_{j-1}).$$

B.3 Mutual information and conditioning

Define

$$I(X; Y) = H(X) + H(Y) - H(X, Y).$$

By the chain rule,

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X).$$

Proposition B.4 (Mutual information is nonnegative).

$$I(X; Y) \geq 0.$$

Consequently,

$$H(X | Y) \leq H(X).$$

Proof. If p_{XY} is the joint law and $p_X p_Y$ the product of the marginals, then direct expansion gives

$$I(X; Y) = D(p_{XY} \| p_X p_Y).$$

Apply ?? □

Conditionally on a third random variable Z , define

$$I(X; Y | Z) = H(X | Z) + H(Y | Z) - H(X, Y | Z).$$

It is the average, over $Z = z$, of the mutual information computed under the conditional law. Hence

$$I(X; Y | Z) \geq 0.$$

Corollary B.5 (Submodularity). *For finite-valued X, Y, Z ,*

$$H(X, Z) + H(Y, Z) \geq H(Z) + H(X, Y, Z).$$

Equivalently,

$$H(X | Z) + H(Y | Z) \geq H(X, Y | Z).$$

Proof. The difference between the two sides is $I(X; Y | Z)$. □

This is the precise entropy inequality used in the proof of the entropic Ruzsa triangle inequality in Chapter ??.

B.4 Functions cannot increase entropy

Proposition B.6 (Deterministic data processing). *If $Y = f(X)$ is a deterministic function of X , then*

$$H(Y) \leq H(X).$$

More generally, if (Y, Z) is a deterministic function of X , then

$$H(Y, Z) \leq H(X).$$

Proof. Since $H(Y | X) = 0$, the chain rule gives

$$H(X) = H(X, Y) = H(Y) + H(X | Y) \geq H(Y).$$

The second assertion is identical. □

B.5 Entropy and support size

Proposition B.7 (Maximum entropy on a finite support). *If X is supported on a finite set S , then*

$$H(X) \leq \log |S|.$$

Equality holds if and only if X is uniform on S .

Proof. Let u be the uniform distribution on S . Gibbs' inequality gives

$$0 \leq D(p_X \| u) = \sum_{x \in S} p_X(x) \log p_X(x) + \log |S| = \log |S| - H(X).$$

Equality is equivalent to $p_X = u$. □

A complementary elementary observation is useful when converting entropy back to sets.

Lemma B.8 (A heavy atom from entropy). *For every finite-valued X ,*

$$\max_x \mathbb{P}(X = x) \geq e^{-H(X)}.$$

Proof. If $M = \max_x p(x)$, then $-\log p(x) \geq -\log M$ for every x in the support. Hence

$$H(X) = \sum_x p(x)(-\log p(x)) \geq -\log M.$$

Rearrange. □

B.6 Two equality cases worth remembering

If X and Y are independent, then

$$H(X, Y) = H(X) + H(Y).$$

Conversely, equality is equivalent to independence because it is $D(p_{XY} \| p_X p_Y) = 0$.

If X is uniform on a coset $x + H$ of a finite subgroup H , then

$$H(X) = \log |H|,$$

and sums or differences of independent such variables remain uniform on appropriate cosets of H . This is why uniform subgroup measures are the zero-distance objects for the entropic Ruzsa geometry developed in the main text.

Appendix C

Geometry-of-Numbers Capsules and Properization

Several classical geometric facts enter the additive-combinatorial machine in a very specific way. Reproducing a full course in convex and discrete geometry here would violate the editorial policy of the book: prerequisites already developed in the companion mathematics books are introduced in context, while the additive consequences are derived in full. This appendix therefore does three things:

1. it proves the elementary lattice facts used in the Bohr-set lift;
2. it states precisely the two classical geometric inputs—Freiman’s shelling lemma and Minkowski’s second theorem—and explains the mechanism of their proofs;
3. it derives the forms needed in Chapters ??, ??, and ??, including the sharp model for the dimension lemma and the bounded-rank properization principle.

Nothing stronger is used elsewhere in Volume II.

C.1 Lattices and covolume

A full-rank lattice in $\mathbb{R}^d$ is a subgroup

$$\Lambda = B\mathbb{Z}^d$$

for an invertible real matrix B . Its determinant, or covolume, is

$$\det \Lambda = |\det B|.$$

This is the volume of a fundamental parallelepiped.

Lemma C.1 (Index and covolume). *If $\Lambda' \leq \Lambda$ are full-rank lattices of finite index, then*

$$\det \Lambda' = [\Lambda : \Lambda'] \det \Lambda.$$

Proof. A fundamental domain for Λ' is, up to measure-zero boundaries, the disjoint union of $[\Lambda : \Lambda']$ translates of a fundamental domain for Λ . Taking volumes gives the formula. $\square$

Lemma C.2 (Covolume of a finite torus lift). *Let H be a finite subgroup of $\mathbb{R}^d/\mathbb{Z}^d$, and let*

$$\Lambda = \{x \in \mathbb{R}^d : x + \mathbb{Z}^d \in H\}.$$

Then $\mathbb{Z}^d \leq \Lambda$, $\Lambda/\mathbb{Z}^d \cong H$, and

$$\det \Lambda = |H|^{-1}.$$

Proof. The quotient map $\Lambda \rightarrow H$ has kernel $\mathbb{Z}^d$, so the first two claims follow from the first isomorphism theorem. Apply ?? to $\mathbb{Z}^d \leq \Lambda$; equivalently, $[\Lambda : \mathbb{Z}^d] = |H|$ and $\det \mathbb{Z}^d = 1$. $\square$

C.2 Successive minima and Minkowski's second theorem

Let $Q \subset \mathbb{R}^d$ be centrally symmetric, convex, bounded, and with nonempty interior. For a full-rank lattice Λ , the successive minima are

$$\lambda_i(Q, \Lambda) = \inf\{\lambda > 0 : \lambda Q \text{ contains } i \text{ linearly independent lattice vectors}\}.$$

Theorem C.3 (Minkowski's second theorem). *For every centrally symmetric convex body $Q \subset \mathbb{R}^d$ with nonempty interior and every full-rank lattice $\Lambda \subset \mathbb{R}^d$,*

$$\frac{2^d}{d!} \det \Lambda \leq \lambda_1(Q, \Lambda) \cdots \lambda_d(Q, \Lambda) \operatorname{vol}(Q) \leq 2^d \det \Lambda.$$

Proof. Choose linearly independent vectors $v_i \in \lambda_i(Q, \Lambda)Q$; the usual limiting argument permits this choice with the successive minima themselves. By symmetry and convexity, Q contains the cross-polytope with vertices $\pm v_i / \lambda_i(Q, \Lambda)$, whose volume equals

$$\frac{2^d}{d!} \frac{|\det(v_1, \dots, v_d)|}{\lambda_1(Q, \Lambda) \cdots \lambda_d(Q, \Lambda)}.$$

The vectors v_i generate a full-rank sublattice of Λ , so $|\det(v_1, \dots, v_d)| \geq \det \Lambda$. This proves the lower bound.

For the upper bound, apply the complete internal proof of Minkowski's second theorem, Theorem ?? in Appendix ??, with $K = Q$. Its hypotheses are exactly the symmetry, convexity, compactness (after replacing Q by its closure if necessary), nonempty interior, and full-rank lattice hypotheses stated here. $\square$

Theorem ?? is precisely the upper half of Theorem ??.

For the cube

$$Q_\rho = [-\rho, \rho]^d,$$

we have $\operatorname{vol}(Q_\rho) = (2\rho)^d$, and hence

$$\lambda_1 \cdots \lambda_d \leq \frac{\det \Lambda}{\rho^d}. \quad (\text{C.1})$$

When Λ is the lift of a finite torus subgroup H , this becomes

$$\lambda_1 \cdots \lambda_d \leq \frac{1}{\rho^d |H|},$$

which is exactly the estimate used in (??).

C.3 Why a small cube gives properness

The following elementary observation is the part of the Bohr-to-progression argument most easily obscured by notation.

Lemma C.4 (No wrap-around in a quarter cube). *If $0 < \rho \leq 1/4$, then*

$$2[-\rho, \rho]^d \cap \mathbb{Z}^d = \{0\}.$$

Consequently, if two points $x, y \in [-\rho, \rho]^d$ are congruent modulo $\mathbb{Z}^d$, then $x = y$.

Proof. Every coordinate of a point in the doubled cube has absolute value at most $2\rho \leq 1/2$. The only integer belonging to $[-1/2, 1/2]$ is 0. Therefore the only integer vector in the doubled cube is the zero vector. If $x, y \in [-\rho, \rho]^d$ are congruent modulo $\mathbb{Z}^d$, then $x - y \in 2[-\rho, \rho]^d \cap \mathbb{Z}^d = \{0\}$, and hence $x = y$. $\square$

C.4 Freiman's dimension lemma from the shelling input

The exact discrete-geometric fact behind ?? is the following shelling statement.

Remark C.5 (Precise shelling input). The exact geometric statements used here are the connected simplicial decomposition Lemma ?? and the simplex-counting Lemma ?? in Appendix ?. Their combination gives the fully stated Freiman dimension theorem there and hence

$$|A + A| \geq (d + 1)|A| - \frac{d(d + 1)}{2}$$

for every finite affinely spanning $A \subset \mathbb{R}^d$.

The triangulation is shelled: each new simplex meets the union of the preceding simplices in a union of facets. Translating A by the vertices of the exposed simplex and ordering points according to the shelling ensures that overlaps are charged only to already exposed facets. A d -simplex has $d + 1$ vertices, and the cumulative unavoidable overlap is exactly

$$1 + 2 + \cdots + d = \frac{d(d + 1)}{2}.$$

This produces the displayed count. The remaining issue is the shellability of a suitable pulling triangulation. That geometric step is now proved internally in Appendix ?, together with the connected simplicial decomposition and the simplex counting lemma. Thus ?? no longer has an external geometric proof dependency.

A sharp stacked configuration

The numerical constant can be checked without any triangulation theory. Let

$$L = \{0, e_d, 2e_d, \dots, Ne_d\}$$

and set

$$A = L \cup \{e_1, \dots, e_{d-1}\} \subset \mathbb{R}^d.$$

Then

$$|A| = N + d.$$

The sumset is a disjoint union of the following types:

- $L + L$, of size $2N + 1$;
- $e_i + L$ for $1 \leq i \leq d - 1$, each of size $N + 1$;
- $e_i + e_j$ for $1 \leq i \leq j \leq d - 1$, giving $d(d - 1)/2$ further points.

Therefore

$$\begin{aligned} |A + A| &= (2N + 1) + (d - 1)(N + 1) + \frac{d(d - 1)}{2} \\ &= (d + 1)N + \frac{d(d + 1)}{2} \\ &= (d + 1)|A| - \frac{d(d + 1)}{2}. \end{aligned}$$

So the dimension lemma is sharp for every d and arbitrarily large cardinality.

C.5 Bounded-rank properization

A generalized progression may have collisions in its parameter box. The standard cure is to use a shortest nontrivial relation to eliminate a direction, then repeat. The quantitative statement needed in Chapter ?? is as follows.

Proposition C.6 (Bounded-rank properization). *For every integer $r \geq 0$ there is a constant $C_r < \infty$ such that the following holds. If*

$$P = H + \{n_1 v_1 + \cdots + n_r v_r : |n_i| \leq L_i\}$$

is a finite symmetric coset progression, then P is contained in a proper symmetric coset progression Q of rank at most r and

$$|Q| \leq C_r |H| \prod_{i=1}^r (2\lfloor L_i \rfloor + 1).$$

Proof. Apply Theorem II.4.005 in Chapter ?? with $t = 1$. Its coset-progression form keeps the subgroup H unchanged and produces a proper symmetric coset progression $Q \supseteq P$ of rank at most r with

$$|Q|_{\text{par}} \leq C_{r,1} |P|_{\text{par}} = C_{r,1} |H| \prod_{i=1}^r (2\lfloor L_i \rfloor + 1).$$

Since a proper coset progression has cardinality equal to its parameter-box cardinality, the asserted bound follows with $C_r = C_{r,1}$. $\square$

This proposition immediately implies ??.

Appendix D

Complete Geometry Proof Owners for Volume II

This appendix gives the complete geometry-of-numbers and discrete-geometry proofs used in Volume II. In particular, it proves Minkowski's second theorem and Freiman's sharp dimension lemma before those results are invoked through their cross-references.

D.1 Lattices, convex bodies, and successive minima

Let $\Lambda \subset \mathbb{R}^d$ be a full-rank lattice with covolume $\det \Lambda$, and let $K \subset \mathbb{R}^d$ be a compact, convex, centrally symmetric body with nonempty interior. Its successive minima are

$$\lambda_i(K, \Lambda) = \inf\{\lambda > 0 : \dim \text{span}(\Lambda \cap \lambda K) \geq i\}.$$

Lemma D.1 (Blichfeldt principle). *If $E \subset \mathbb{R}^d$ is measurable and $\text{vol}(E) > \det \Lambda$, then there are distinct $x, y \in E$ with $x - y \in \Lambda$.*

Proof. Let F be a half-open fundamental parallelepiped for Λ . Translate the pieces $E \cap (F + \lambda)$, $\lambda \in \Lambda$, back into F . If no two points of E differ by a lattice vector, these translated pieces are disjoint up to null boundaries, so their total measure is at most $\text{vol}(F) = \det \Lambda$. This contradicts the assumed volume of E . $\square$

Theorem D.2 (Minkowski's convex body theorem). *If*

$$\text{vol}(K) > 2^d \det \Lambda,$$

then K contains a nonzero point of Λ .

Proof. Apply ?? to $\frac{1}{2}K$. Distinct $x, y \in \frac{1}{2}K$ with $x - y \in \Lambda$ give a nonzero lattice vector $x - y \in K$ because K is symmetric and convex. $\square$

D.2 An adapted lattice flag

Lemma D.3 (Basis adapted to successive-minimum subspaces). *After an invertible linear change of variables sending Λ to $\mathbb{Z}^d$, one may arrange that, for $1 \leq i \leq d$,*

$$\text{int}(\lambda_i K) \cap \mathbb{Z}^d \subset L_{i-1} := \text{span}(e_1, \dots, e_{i-1}).$$

Proof. After a preliminary linear change we may assume $\Lambda = \mathbb{Z}^d$. For each i put

$$W_i = \text{span}_{\mathbb{R}}(\text{int}(\lambda_i K) \cap \mathbb{Z}^d).$$

By the definition of the i -th minimum, $\dim W_i \leq i - 1$: otherwise i independent lattice points would lie in the open set $\text{int}(\lambda_i K)$, and a slightly smaller homothetic copy of K would still contain them, contradicting minimality of λ_i . The spaces W_i are rational and nested because the dilates are nested.

We use the following elementary flag-extension fact. If

$$W_1 \subset W_2 \subset \cdots \subset W_d$$

is a nested chain of rational subspaces with $\dim W_i \leq i - 1$, then there is a complete rational flag

$$0 = L_0 \subset L_1 \subset \cdots \subset L_d = \mathbb{R}^d, \quad \dim L_i = i,$$

with $W_{i+1} \subset L_i$. To prove it, choose a lattice basis adapted to the chain $W_i \cap \mathbb{Z}^d$: first choose a basis of the smallest nonzero subgroup, extend it to a basis of the next, and continue; primitiveness follows from rationality. Order the resulting basis vectors by the first W_i in which they occur. The inequality $\dim W_i \leq i - 1$ guarantees that by position $i - 1$ all vectors needed for W_i have appeared. Fill any unused positions with further lattice basis vectors. The spans of the first i vectors give the required L_i .

Apply this fact to the W_i . The nested primitive subgroups $L_i \cap \mathbb{Z}^d$ admit an adapted lattice basis $b_1, \dots, b_d$ with

$$L_i = \text{span}(b_1, \dots, b_i).$$

The unimodular map carrying b_i to e_i preserves the lattice and sends the chain to the coordinate flag. Since $W_i \subset L_{i-1}$, the desired containment follows. $\square$

D.3 Minkowski's second theorem

Theorem D.4 (Minkowski's second theorem). *Let $K \subset \mathbb{R}^d$ be a symmetric convex body with nonempty interior, let $\Lambda \subset \mathbb{R}^d$ be a full-rank lattice, and let $0 < \lambda_1 \leq \cdots \leq \lambda_d$ be the successive minima of K with respect to Λ . Then*

$$\frac{2^d}{d!} \det \Lambda \leq \lambda_1 \cdots \lambda_d \text{vol}(K) \leq 2^d \det \Lambda.$$

Proof. The lower bound is immediate from a cross-polytope. Choose independent $v_i \in \Lambda \cap \lambda_i K$. Then $u_i = v_i / \lambda_i \in K$, so

$$C = \text{conv}(\pm u_1, \dots, \pm u_d) \subset K.$$

Its volume is

$$\text{vol}(C) = \frac{2^d}{d!} \frac{|\det(v_1, \dots, v_d)|}{\lambda_1 \cdots \lambda_d}.$$

The lattice generated by the v_i is a sublattice of Λ , hence its covolume is at least $\det \Lambda$. Rearranging gives the lower inequality.

For the upper bound, apply a linear map and scale volume so that $\Lambda = \mathbb{Z}^d$; at the end we restore the factor $\det \Lambda$. Use ???. Put

$$K_i = \frac{\lambda_i}{2} K, \quad M_q^d = \{z \in \mathbb{Z}^d : |z_j| \leq q\}, \quad M_q^i = M_q^d \cap L_i.$$

Since K_d is bounded, there is a constant γ depending only on K_d such that

$$\text{vol}(M_q^d + K_d) \leq (2q + \gamma)^d. \tag{D.1}$$

Distinct translates $z + \text{int } K_1$, $z \in \mathbb{Z}^d$, are disjoint: an intersection would give a nonzero lattice point in $\text{int}(K_1 - K_1) = \text{int}(\lambda_1 K)$, contradicting the definition of λ_1 . Consequently

$$\text{vol}(M_q^d + K_1) = (2q+1)^d \text{vol}(K_1) = (2q+1)^d \left(\frac{\lambda_1}{2}\right)^d \text{vol}(K). \quad (\text{D.2})$$

We next prove, for $1 \leq i < d$,

$$\text{vol}(M_q^d + K_{i+1}) \geq \left(\frac{\lambda_{i+1}}{\lambda_i}\right)^{d-i} \text{vol}(M_q^d + K_i). \quad (\text{D.3})$$

If $\lambda_{i+1} = \lambda_i$ there is nothing to prove, so suppose the ratio $r > 1$.

First, translates of $\text{int } K_{i+1}$ indexed by lattice points with different last $d-i$ coordinates are disjoint. Otherwise their difference would be a lattice point outside L_i lying in

$$\text{int}(K_{i+1} - K_{i+1}) = \text{int}(\lambda_{i+1} K),$$

contradicting ???. Therefore Fubini over the last $d-i$ coordinate directions gives

$$\text{vol}(M_q^d + K_j) = (2q+1)^{d-i} \text{vol}(M_q^i + K_j), \quad j = i, i+1.$$

It remains to compare the two lower-dimensional unions.

Let T_1 multiply the first i coordinates by r and fix the remaining coordinates; let T_2 fix the first i coordinates and multiply the last $d-i$ coordinates by r . Since $T_2 T_1 = rI$ and T_2 fixes M_q^i pointwise,

$$M_q^i + K_{i+1} = T_2(M_q^i + T_1 K_i).$$

Hence

$$\text{vol}(M_q^i + K_{i+1}) = r^{d-i} \text{vol}(M_q^i + T_1 K_i).$$

We claim

$$\text{vol}(M_q^i + T_1 K_i) \geq \text{vol}(M_q^i + K_i).$$

To see this, fix a point x in the orthogonal complement of L_i and consider the fiber

$$C_x = \{u \in L_i : u + x \in K_i\}.$$

Whenever C_x is nonempty, choose $c_x \in C_x$. Convexity gives

$$C_x + (r-1)c_x \subset rC_x,$$

because $u + (r-1)c_x = r((1/r)u + (1-1/r)c_x)$ and the point in parentheses belongs to C_x . Thus the i -dimensional measure of $M_q^i + C_x$ is at most that of $M_q^i + rC_x$. Integrating these fiber inequalities over x proves the claim. This establishes (??).

Iterating (??) and using (??) yields

$$\text{vol}(M_q^d + K_d) \geq \frac{\lambda_1 \cdots \lambda_d}{2^d} (2q+1)^d \text{vol}(K).$$

Together with (??),

$$\lambda_1 \cdots \lambda_d \text{vol}(K) \leq 2^d \left(\frac{2q+\gamma}{2q+1}\right)^d.$$

Letting $q \rightarrow \infty$ proves the upper bound for $\mathbb{Z}^d$. Undoing the initial linear change multiplies volume by $\det \Lambda$, giving the stated inequality. $\square$

D.4 A simplicial decomposition

We now prove Freiman's sharp dimension lemma. The geometric decomposition follows the simplicial convex-hull method of Matolcsi–Ruzsa, but every needed step is included.

Lemma D.5 (Connected simplicial decomposition). *Let $B \subset \mathbb{R}^d$ be finite and affinely spanning. Then $\text{conv}(B)$ can be written as a union of d -simplices $S_1, \dots, S_m$ such that:*

1. *every vertex of every S_i belongs to B , and $S_i \cap B$ is exactly its vertex set;*
2. *two simplices meet, if at all, in a common face;*
3. *for every $i > 1$, the simplex S_i shares a facet with some earlier S_j .*

Proof. Induct simultaneously on the affine dimension and on $|B|$. The case of a simplex is immediate. Choose an extreme point $b \in B$ and set $B' = B \setminus \{b\}$.

If B' is still d -dimensional, triangulate $\text{conv}(B')$ inductively. A boundary facet is visible from b when the open segment from b to a relative interior point of that facet does not enter $\text{conv}(B')$. Cone each visible facet to b . These cones fill exactly $\text{conv}(B) \setminus \text{conv}(B')$, their mutual intersections are common faces, and no point of B' lies in the interior of a new cone. Each cone shares its base facet with an old simplex.

If B' has dimension $d - 1$, triangulate it in its affine hull and cone every $(d - 1)$ -simplex to b . The same face-intersection properties hold. In either case the facet-adjacency graph of the resulting simplices is connected: a generic line segment joining interior points of two simplices crosses only facets, producing a path in that graph. Reorder the simplices along a spanning tree rooted at the first one to obtain property (3). $\square$

D.5 The simplex counting lemma

Lemma D.6 (Addition by the vertices of a simplex). *Let $B = \{b_0, \dots, b_d\}$ be affinely independent and let $A \subset \text{conv}(B)$ be finite, $|A| = m$. Put $m_1 = |A \cap B|$. Then*

$$|A + B| = (m - m_1)(d + 1) + \binom{d + 2}{2} - \binom{d - m_1 + 2}{2}.$$

In particular,

$$|A + B| \geq (d + 1)m - \binom{d + 1}{2},$$

and if $m_1 \leq 1$, then $|A + B| = (d + 1)m$.

Proof. Every $a \in \text{conv}(B)$ has unique barycentric coordinates

$$a = \sum_{i=0}^d \alpha_i b_i, \quad \alpha_i \geq 0, \quad \sum_i \alpha_i = 1.$$

Suppose $a + b_i = a' + b_j$. Comparison of barycentric coordinates shows that if either a or a' is not a vertex, then the only possibility is $a = a'$ and $i = j$: otherwise two coordinate vectors in the simplex would differ by $e_j - e_i$, which forces one of them to be a vertex. Thus the translates $a + B$ for $a \in A \setminus B$ are pairwise disjoint and avoid $(A \cap B) + B$. They contribute $(m - m_1)(d + 1)$ points.

After relabeling, let $A \cap B = \{b_0, \dots, b_{m_1-1}\}$. The set $(A \cap B) + B$ consists of unordered two-vertex sums with at least one index below m_1 . Affine independence makes different unordered pairs give different sums. Hence its size is

$$\binom{d + 2}{2} - \binom{d - m_1 + 2}{2}.$$

This proves the exact formula. The lower bound follows by minimizing in m_1 , which occurs at $m_1 = d + 1$; if $m_1 \leq 1$ the formula simplifies to $(d + 1)m$. $\square$

Theorem D.7 (Freiman's dimension lemma). *Let $A \subset \mathbb{R}^d$ be finite and affinely spanning. Then*

$$|A + A| \geq (d + 1)|A| - \frac{d(d + 1)}{2}.$$

Proof. Apply ?? to $B = A$, with simplices S_i and vertex sets B_i . Partition A by

$$A_1 = A \cap S_1, \quad A_i = (A \cap S_i) \setminus \bigcup_{j < i} S_j \quad (i > 1).$$

These sets are disjoint and cover A .

We claim the sumsets $A_i + B_i$ are pairwise disjoint. Suppose $i < j$ and

$$a + b = a' + b', \quad a \in A_i, \quad b \in B_i, \quad a' \in A_j, \quad b' \in B_j.$$

Then $(a + b)/2$ belongs to both S_i and S_j , hence to their common face. In the unique barycentric representation inside S_j , the point $(a' + b')/2$ is the average of the coordinate vector of a' and a vertex vector. Since all coordinates are nonnegative, every coordinate outside the common face must vanish separately. Therefore a' belongs to that common face and hence to S_i , contradicting the definition of A_j .

For $i > 1$, property (3) gives an earlier simplex sharing a facet with S_i . The d vertices of that facet already belong to the earlier union, so at most the opposite vertex can lie in $A_i \cap B_i$. By ??,

$$|A_i + B_i| = (d + 1)|A_i| \quad (i > 1).$$

For $i = 1$,

$$|A_1 + B_1| \geq (d + 1)|A_1| - \binom{d + 1}{2}.$$

Every $B_i \subset A$, so the disjoint union of these partial sumsets lies inside $A + A$. Summing gives

$$|A + A| \geq (d + 1)|A| - \binom{d + 1}{2},$$

as required. $\square$

References

- [1] D. Castro-Silva, J. Briët, S. Arunachalam, A. Dutt, and T. Gur. *An algorithmic Polynomial Freiman–Ruzsa theorem*. 2026. arXiv: [2604.04547](#).
- [2] Ernie Croot and Olof Sisask. “A probabilistic technique for finding almost-periods of convolutions”. In: *Geometric and Functional Analysis* 20 (2010), pp. 1367–1396. arXiv: [1003.2978](#).
- [3] Matt DeVos. “A short proof of Kneser’s addition theorem for abelian groups”. In: *arXiv preprint arXiv:1303.3539* (2013).
- [4] W. T. Gowers, Ben Green, Freddie Manners, and Terence Tao. *Marton’s conjecture in abelian groups with bounded torsion*. 2024. arXiv: [2404.02244](#).
- [5] W. T. Gowers, Ben Green, Freddie Manners, and Terence Tao. “On a conjecture of Marton”. In: *Annals of Mathematics* 201.2 (2025), pp. 515–549.
- [6] Ben Green and Imre Z. Ruzsa. *Sets with small sumset and rectification*. 2004. arXiv: [math/0403338](#).
- [7] Ben Green and Imre Z. Ruzsa. “Freiman’s theorem in an arbitrary abelian group”. In: *Journal of the London Mathematical Society* 75 (2007), pp. 163–175. arXiv: [math/0505198](#).
- [8] Vsevolod F. Lev and Pavel Y. Smeliansky. “On addition of two distinct sets of integers”. In: *Acta Arithmetica* 70.1 (1995), pp. 85–91.
- [9] J.-J. Liao. *Improved exponent for Marton’s conjecture in $\mathbb{F}_2^n$* . 2024. arXiv: [2404.09639](#).
- [10] C. Peng. *Robust Polynomial Freiman–Ruzsa from Corrupted Set Observations via a Sharp Persistent-Subset BSG Compiler*. Version 2, 6 August 2026. 2026. arXiv: [2608.00451](#).
- [11] R. Raghavan. *Improved bounds for the Freiman–Ruzsa theorem*. Version 2, 26 February 2026. 2026. arXiv: [2512.11217](#).
- [12] A. G. Vosper. “The critical pairs of subsets of a group of prime order”. In: *Journal of the London Mathematical Society*. s1 31.2 (1956), pp. 200–205. DOI: [10.1112/jlms/s1-31.2.200](#).

Additive Combinatorics
Volume III
Fourier Methods and Quantitative Additive Combinatorics
Version 1.10

14 September 2026

Contents

1	Counting Progressions: Correlation, Fourier Structure, and the Inverse Question	1
1.1	The first counting operator	1
1.2	Fourier diagonalization	1
1.3	The direct and inverse viewpoints	2
1.4	Why four terms are already different	3
1.5	Degenerate and nondegenerate configurations	3
1.6	A recurring proof architecture	3
2	Roth’s Theorem by a Local Density Increment	4
2.1	Why locality matters	4
2.2	Relative balancing inside an interval	4
2.3	The interval has many progressions	5
2.4	Linearizing one frequency	5
2.5	Iteration	6
2.6	What is quantitatively wasteful	7
3	Bohr Sets and Local Fourier Analysis	8
3.1	Why progressions are too rigid	8
3.2	Regular radii	8
3.3	Approximate translation invariance	9
3.4	Freezing a generated spectrum	9
3.5	The local spectral density increment, proved	10
3.6	Chang’s theorem as a rank bound	12
3.7	Bourgain’s change of ambient object	12
4	Gowers Uniformity Norms and Cube Calculus	13
4.1	Multiplicative derivatives	13
4.2	Reading the cube	13
4.3	Polynomial phases act as gauge symmetries	14
4.4	U^2 is ordinary Fourier analysis	14
4.5	Gowers–Cauchy–Schwarz	15
4.6	Polynomial phases	16
4.7	Why the hierarchy is nested conceptually	16
4.8	The derivative philosophy	16
4.9	Norms on intervals	17
5	Generalized von Neumann Inequalities and Complexity	18
5.1	Cauchy–Schwarz as an elimination algorithm	18
5.2	Warm-up: three-term progressions from Fourier analysis	18

5.3	The elimination lemma	19
5.4	Cauchy–Schwarz complexity	19
5.5	Worked complexity calculations	21
5.6	True complexity	21
5.7	Nondegeneracy and characteristic	21
5.8	What Cauchy–Schwarz forgets	22
6	From Uniformity to Szemerédi’s Theorem	23
6.1	The theorem	23
6.2	The density-increment template	23
6.3	An abstract iteration lemma	25
6.4	Varnavides supersaturation	25
6.5	Regularity versus density increment	27
6.6	A four-term obstruction	27
6.7	Where quantitative bounds are lost	27
7	Arithmetic Regularity: Structure, Uniformity, and Error	28
7.1	The U^2 model	28
7.2	An abstract inverse theorem is enough	29
7.3	Where the L^2 -small term comes from	30
7.4	Torus prototype	31
7.5	What remains genuinely difficult	31
8	Dense Models and Transference	32
8.1	The sparse problem	32
8.2	A finite dense-model theorem	32
8.3	Counting after modelling	33
8.4	Pseudorandomness is theorem-dependent	33
9	Modern Roth Theory: Spectra and Almost-Periodicity	34
9.1	Past the one-frequency argument	34
9.2	Energy of spectra	34
9.3	Croot–Sisask almost-periodicity	34
9.4	Smoothing or structure	35
9.5	Quantitative milestones	35
9.6	The methodological change	36
10	The Kelley–Meka Paradigm: Spreadness, High Moments, and Potentials	37
10.1	Spreadness	37
10.2	High moments detect rare concentration	37
10.3	Structure versus pseudorandomness	37
10.4	Why multiplicative increments are decisive	38
10.5	A potential-function viewpoint	38
10.6	A research lesson	38
11	Systems of Linear Forms and True Complexity	39
11.1	Beyond one progression	39
11.2	Tensor powers	39
11.3	Why true complexity matters for primes	40
11.4	Cauchy–Schwarz versus true complexity	40
11.5	The tensor criterion in coordinates	40

11.6	A phase obstruction explains necessity	41
11.7	Cauchy–Schwarz complexity can be wasteful	41
11.8	Affine forms and constant terms	42
11.9	What the theorem does not say	42
12	Polynomial Configurations and PET Induction	43
12.1	From linear to polynomial patterns	43
12.2	A degree-lowering calculation	43
12.3	PET ordering	43
12.4	Why inverse theory still appears	44
12.5	The van der Corput identity on a finite group	44
12.6	A complete quadratic differencing example	44
12.7	PET equivalence classes	45
12.8	The PET termination theorem	45
12.9	Termination is finite but not cheap	47
12.10	Polynomial systems with collisions	47
12.11	PET and the inverse theorem play different roles	47
13	Quantitative Szemerédi Theory: Where Bounds Are Lost	48
13.1	A proof as a parameter machine	48
13.2	Three bottlenecks	48
13.3	Behrend’s lower bound	48
13.4	A methodological comparison	49
13.5	Optimizing the Behrend parameters	49
13.6	Why the sphere is progression-free	50
13.7	Upper and lower bounds measure different difficulties	50
13.8	Tracking parameters through an iteration	50
14	The Polynomial-Method Contrast: Cap Sets and Slice Rank	51
14.1	Cap sets	51
14.2	The diagonal-tensor idea	51
14.3	Why the method is so different	51
14.4	Slice rank	52
14.5	The cap-set tensor	53
14.6	Estimating the monomial count	53
14.7	What exactly made the argument possible?	54
15	Local Fourier Analysis on Bohr Sets	55
15.1	Local convolution	55
15.2	Local characters	55
15.3	Nested scales	55
15.4	Regularity as a boundary theorem	56
15.5	A local Parseval calculation	56
15.6	Frequency growth versus radius loss	57
16	Three Proof Architectures for Szemerédi-Type Theorems	58
16.1	Density increment	58
16.2	Regularity and counting	58
16.3	Transference	58
16.4	Choosing a proof	58

17 Research Outlook: From Uniformity to Higher-Order Structure	60
17.1 What Volume III has established	60
17.2 Quantitative frontiers	60
17.3 The conceptual transition	60
 I Modern Roth Theory and Polynomial Freiman–Ruzsa	 61
18 Bridge Lemmas for Quantitative Additive Combinatorics	62
18.1 Purpose and dependencies	62
18.2 Density-increment iteration	62
18.3 Regular Bohr sets as local approximate groups	63
18.4 Bohr sets as Bourgain systems	65
18.5 Dissociation and spectral bookkeeping	66
18.6 Almost-periodicity for indicator convolutions	67
18.7 The entropic Balog–Szemerédi–Gowers lemma	67
18.8 Boundary of this bridge chapter	68
18.9 Entropic stability and the fibring calculus	69
18.10 The characteristic-two distance decrement	71
 19 The Local Chang–Sanders Chain	 77
19.1 The local Chang–Sanders argument	77
19.2 Two covering lemmas needed by the bootstrap	81
19.3 From polynomial growth to a controlled Bohr envelope	81
19.4 Low-dimensional progressions from one-scale Bohr growth	83
19.5 The quantitative Sanders bootstrap	84
 20 The Kelley–Meka Regularization Mechanism	 89
20.1 Why a separate potential is needed	89
20.2 Local L^∞ almost-periodicity on Bohr sets	93
20.3 The localized integer mechanism	95
 21 Bounded-Torsion Polynomial Freiman–Ruzsa	 104
21.1 Scope and proof architecture	104
21.2 A self-contained very-small-distance theorem	104
21.3 Multidistance and its chain rule	107
21.4 The bounded-torsion decrement	108
21.5 Iteration and the entropic theorem	112
21.6 From entropy to Marton’s theorem	113
 22 Bloom–Sisask: spectral boosting and the logarithmic barrier	 115
22.1 The two-strength density-increment engine	115
22.2 Hereditary energy versus relative dimension	116
22.2.1 From a spectrum to a low-dimensional weighted piece	120
22.3 The additive-framework structure theorem: referee expansion	123
22.4 From few progressions to a robust non-smoothing spectrum	134
22.4.1 Constructing the additive framework from Bohr spectra	134
22.4.2 A progression deficit creates weighted spectral mass	136
22.4.3 Two spectral conversion lemmas	138
22.4.4 The exact spectrum alternative used by the structure theorem	142
22.5 Structure of a non-smoothing spectrum	144

22.6	Spectral boosting	145
22.7	A baseline local Roth theorem from almost-periodicity	147
22.8	Iteration and the historical theorem	151
23	Roadmap and model calculations for the modern quantitative arc	153
23.1	Croot–Sisask: why random translates create almost-periods	153
23.2	Kelley–Meka: spreadness as a stopping rule	153
23.3	GGMT: distance decrement and the subgroup fixed point	153
A	Probability and Convexity Tools Used in Volume III	154
A.1	Markov, Chebyshev, and high moments	154
A.2	Finite-dimensional separation	154
A.3	Polynomial approximation	154
B	Normalization Dictionary	155
C	A Gowers–Cauchy–Schwarz Proof Laboratory	156
C.1	The 2-cube	156
C.2	The inductive cube form	156
C.3	The triangle inequality	157
C.4	A generalized von Neumann calculation for 3-APs	157
C.5	What information is lost?	158
D	Progression Counts, Degeneracies, and Supersaturation	159
D.1	Ordered versus unordered progressions	159
D.2	The exact three-term Fourier identity	159
D.3	Balanced expansion	160
D.4	Varnavides averaging without boundary errors	160
D.5	Why a positive count is the stable formulation	161
D.6	Boundary conventions on intervals	161
E	Hypergraph Removal and a Complete Proof of Szemerédi’s Theorem	162
E.1	Why this appendix is here	162
E.2	Hypergraph systems and face sigma-algebras	162
E.3	The one-step energy increment	163
E.4	Regularity at one level	164
E.5	Randomness versus structure at one level	164
E.6	Preliminary strong regularity	165
E.7	A finite domination principle for growth functions	166
E.8	Multilevel regularity	166
E.9	Good atoms	168
E.10	The counting lemma	168
E.11	The removal lemma	170
E.12	Encoding arithmetic progressions as simplices	171
E.13	From the cyclic theorem to intervals	172
E.14	What this proof contributes to the rest of the book	172
	References	173

Chapter 1

Counting Progressions: Correlation, Fourier Structure, and the Inverse Question

1.1 The first counting operator

Let G be a finite abelian group in which multiplication by 2 is invertible. For functions $f_0, f_1, f_2 : G \rightarrow \mathbb{C}$ define

$$\Lambda_3(f_0, f_1, f_2) := \mathbb{E}_{x,d \in G} f_0(x) f_1(x+d) f_2(x+2d).$$

If $A \subseteq G$, then $|G|^2 \Lambda_3(1_A, 1_A, 1_A)$ is the number of ordered three-term arithmetic progressions in A .

The first lesson of this volume is that a configuration count should be regarded as a multilinear correlation. Once written in this way, two questions become unavoidable:

1. what norm controls the correlation when one input is pseudorandom;
2. what structure is forced if the norm is large?

For three terms the answers are ordinary Fourier analysis and characters. For longer configurations they are Gowers norms and higher-order phases.

1.2 Fourier diagonalization

With the normalized Fourier transform

$$\widehat{f}(\gamma) = \mathbb{E}_{x \in G} f(x) \overline{\gamma(x)},$$

one has the exact identity

Proposition 1.1 (Three-term Fourier formula).

$$\Lambda_3(f_0, f_1, f_2) = \sum_{\gamma \in \widehat{G}} \widehat{f}_0(\gamma) \widehat{f}_1(-2\gamma) \widehat{f}_2(\gamma).$$

Worked example (why U^3 detects quadratic phase). Let p be an odd prime, write $e_p(t) = \exp(2\pi i t/p)$, and take

$$f(x) = e_p(ax^2 + bx + c), \quad x \in \mathbb{F}_p,$$

with $a \neq 0$. Using the multiplicative derivative

$$\Delta_h f(x) = f(x+h) \overline{f(x)},$$

three successive derivatives annihilate the quadratic polynomial, so

$$\Delta_{h_1} \Delta_{h_2} \Delta_{h_3} f(x) = 1 \quad \text{for every } x, h_1, h_2, h_3.$$

By the normalized cube formula this gives $\|f\|_{U^3} = 1$.

The U^2 norm sees something different. With normalized Fourier transform

$$\widehat{f}(\xi) = \mathbb{E}_{x \in \mathbb{F}_p} f(x) e_p(-\xi x),$$

the quadratic Gauss-sum evaluation gives $|\widehat{f}(\xi)| = p^{-1/2}$ for every ξ . Therefore

$$\|f\|_{U^2}^4 = \sum_{\xi \in \mathbb{F}_p} |\widehat{f}(\xi)|^4 = p \cdot p^{-2} = p^{-1}, \quad \|f\|_{U^2} = p^{-1/4}.$$

Thus a genuine quadratic phase is maximally structured for U^3 but becomes asymptotically invisible to U^2 . This calculation is the simplest model for the degree-lowering principle behind higher-order inverse theory.

Proof. Expand all three functions by Fourier inversion. The average in x kills every triple of frequencies except those whose sum is zero. The average in d imposes the second relation coming from the coefficients 0, 1, 2. Solving the two relations leaves one free frequency, giving the displayed formula. $\square$

Worked Example 1.2 (Balanced three-term count). Let $A \subseteq G$ have density α and put $f = 1_A - \alpha$. Since $\widehat{f}(0) = 0$,

$$\Lambda_3(1_A, 1_A, 1_A) = \alpha^3 + \Lambda_3(f, f, f).$$

Hence a substantial deficit from the random prediction α^3 forces

$$|\Lambda_3(f, f, f)| \gg \alpha^3.$$

By the Fourier formula and Parseval,

$$|\Lambda_3(f, f, f)| \leq \|\widehat{f}\|_\infty \sum_{\gamma} |\widehat{f}(\gamma)|^2 \leq \alpha \|\widehat{f}\|_\infty.$$

Thus some nonzero character satisfies $|\widehat{f}(\gamma)| \gg \alpha^2$. This is the analytic seed of Roth's density-increment argument.

1.3 The direct and inverse viewpoints

A character γ is a direct source of additive structure: if 1_A correlates with γ , then A has a density bias on regions where the phase is nearly constant. The inverse problem reverses this observation:

$$\text{abnormal progression count} \Rightarrow \text{large Fourier correlation} \Rightarrow \text{spatial concentration}.$$

The last arrow is the density-increment step and must be proved in a form stable under iteration.

1.4 Why four terms are already different

For four terms define

$$\Lambda_4(f_0, f_1, f_2, f_3) = \mathbb{E}_{x,d} \prod_{j=0}^3 f_j(x + jd).$$

No one-dimensional Fourier diagonalization controls this expression. The reason is structural rather than technical.

Worked Example 1.3 (Quadratic obstruction). Let $G = \mathbb{F}_p$ with $p > 3$ and $f(x) = e_p(ax^2)$, $a \neq 0$. Completing the square gives

$$|\widehat{f}(\xi)| = p^{-1/2}$$

for every ξ , so all ordinary Fourier coefficients tend to zero as $p \rightarrow \infty$. Nevertheless the third additive derivative of ax^2 vanishes, and therefore f has perfect U^3 structure. Products of f along four-term progressions can retain coherent phase even though ordinary Fourier analysis sees no large coefficient.

The correct hierarchy is therefore

$$\text{characters} \leftrightarrow U^2, \quad \text{quadratic phases} \leftrightarrow U^3, \quad \text{higher-degree/nilpotent phases} \leftrightarrow U^{s+1}.$$

The remainder of Volumes III–V makes this hierarchy precise.

1.5 Degenerate and nondegenerate configurations

When counting progressions one must distinguish genuine patterns from degenerate solutions. In $[N]$, a three-term progression $(x, x + d, x + 2d)$ is nontrivial when $d \neq 0$. In a finite group with torsion, $d \neq 0$ need not imply three distinct points. This is why hypotheses such as invertibility of 2 and 3 appear in clean finite-group statements. Throughout the volume, every use of division by an integer will be stated explicitly.

1.6 A recurring proof architecture

Most of the theorems in this arc fit the diagram

$$\begin{aligned} \text{configuration deficit} &\longrightarrow \text{large uniformity norm} \\ &\longrightarrow \text{inverse structure} \\ &\longrightarrow \text{density increment or counting lemma.} \end{aligned}$$

The first arrow is usually Cauchy–Schwarz. The second is inverse theory. The third is a geometric or probabilistic argument converting correlation into usable density. The quantitative strength of a theorem is often determined not by the first arrow but by the efficiency of the last two.

Chapter 2

Roth's Theorem by a Local Density Increment

2.1 Why locality matters

An iterative proof of Roth's theorem must be closed under its own output. If the proof finds an arithmetic progression $P \subseteq [N]$ on which the density increases, then after identifying P affinely with a shorter interval, the same argument must apply again. We therefore prove the density increment directly for subsets of intervals rather than silently iterating a theorem valid only in one cyclic group.

Theorem 2.1 (A quantitative Roth theorem). *There are absolute constants $C > 0$ and $N_0 \geq 3$ such that, for every integer $N \geq N_0$, every set $A \subseteq [N]$ containing no nontrivial three-term arithmetic progression satisfies*

$$\frac{|A|}{N} \leq \frac{C}{\sqrt{\log \log N}}.$$

The bound is intentionally crude. It is strong enough to display every logical step of a valid density-increment iteration.

2.2 Relative balancing inside an interval

Choose a prime p with $4N < p < 8N$ and place $I = [N]$ inside $G = \mathbb{Z}/p\mathbb{Z}$. Write

$$\mu = \frac{N}{p}, \quad \frac{1}{8} < \mu < \frac{1}{4}.$$

A congruence $x + z \equiv 2y \pmod{p}$ with $x, y, z \in I$ is an equality in the integers because $|x + z - 2y| < 2N < p$.

If $A \subseteq I$ has relative density α , define

$$f = 1_A - \alpha 1_I.$$

Then $\mathbb{E}_G f = 0$. Balancing against $\alpha 1_I$, rather than a constant on all of G , is what makes the argument local.

2.3 The interval has many progressions

Let $T(I)$ be the number of ordered (x, d) with $x, x + d, x + 2d \in I$. Directly,

$$T(I) = N + 2 \sum_{1 \leq d \leq (N-1)/2} (N - 2d) \geq \frac{N^2}{3}.$$

Hence

$$\Lambda_3(1_I, 1_I, 1_I) \geq \frac{\mu^2}{3}.$$

If A is progression-free, only $d = 0$ remains and

$$\Lambda_3(1_A, 1_A, 1_A) = \frac{|A|}{p^2} = \frac{\alpha\mu}{p}.$$

Lemma 2.2 (Fourier obstruction on an interval). *If $N \geq C\alpha^{-2}$, then some nonzero frequency $r \in \mathbb{Z}/p\mathbb{Z}$ satisfies*

$$|\widehat{f}(r)| \geq c\alpha^3\mu,$$

or equivalently

$$|\mathbb{E}_{x \in I} f(x) e(-rx/p)| \geq c\alpha^3.$$

Proof. Expand $1_A = \alpha 1_I + f$ in the trilinear form. The main term is at least $\alpha^3\mu^2/3$. Every one of the seven remaining terms contains f . Put $M = \|\widehat{f}\|_\infty$. In the Fourier formula, place one copy of f in L^∞ and use Cauchy–Schwarz and Parseval on the other two factors. Because

$$\|1_I\|_2^2 = \mu, \quad \|f\|_2^2 = \mu\alpha(1 - \alpha) \leq \mu,$$

every mixed term has magnitude at most $M\mu$. Therefore

$$\frac{\alpha^3\mu^2}{3} \leq \frac{\alpha\mu}{p} + 7M\mu.$$

For $N \geq C\alpha^{-2}$ the first term on the right is at most half the main term, giving $M \geq c\alpha^3\mu$. Since $\widehat{f}(0) = 0$, the frequency is nonzero. Finally f is supported on I , so dividing by $\mu = N/p$ gives the interval-average formulation. $\square$

2.4 Linearizing one frequency

We shall use the following elementary form of simultaneous approximation only for one frequency. Its proof is included because it is exactly the scale-producing step in the local Roth iteration.

Lemma 2.3 (Dirichlet approximation, one frequency). *For every $\theta \in \mathbb{R}/\mathbb{Z}$ and every integer $Q \geq 1$ there is an integer $1 \leq q \leq Q$ such that*

$$\|q\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{1}{Q}.$$

Proof. Place the $Q + 1$ points

$$0, \theta, 2\theta, \dots, Q\theta$$

on the circle $\mathbb{R}/\mathbb{Z}$ and divide the circle into Q half-open intervals of length $1/Q$. Two of the points lie in the same interval. Their difference is $q\theta$ for some $1 \leq q \leq Q$, and its distance to the nearest integer is at most $1/Q$. $\square$

Lemma 2.4 (Local density increment). *Suppose*

$$|\mathbb{E}_{x \in I} f(x) e(-rx/p)| \geq \eta$$

for nonzero r , and $\eta\sqrt{N}$ is larger than an absolute constant. Then I contains an arithmetic progression P with

$$|P| \geq c\eta\sqrt{N}, \quad \text{dens}_P(A) \geq \alpha + c\eta.$$

Proof. Apply Lemma 2.3 to $\theta = r/p$ with $Q = \lfloor \sqrt{p} \rfloor$. We obtain $1 \leq q \leq \sqrt{p}$ and

$$\left\| \frac{rq}{p} \right\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{1}{\lfloor \sqrt{p} \rfloor} \leq 2p^{-1/2},$$

for $p \geq 4$. The harmless factor 2 is absorbed into the absolute constants below. Partition I by residues modulo q , and split each residue class into consecutive blocks of length

$$L = \lfloor c_1 \eta \sqrt{N} \rfloor.$$

The union E of terminal incomplete blocks has size at most $qL \leq Cc_1\eta N$. On a full block the character varies by $O(L/\sqrt{p}) = O(c_1\eta)$.

For the full blocks P_j set

$$w_j = \frac{|P_j|}{N}, \quad m_j = \mathbb{E}_{P_j} f.$$

Approximating the character by a constant on each block and choosing c_1 small gives

$$\sum_j w_j |m_j| \geq \eta/2.$$

Since $\mathbb{E}_I f = 0$, the weighted positive and negative masses differ by at most $|E|/N \leq \eta/8$. Thus the positive mass is at least $3\eta/16$, and some full block has $m_j \geq 3\eta/16$. On that progression

$$\text{dens}_{P_j}(A) = \alpha + m_j \geq \alpha + c\eta.$$

□

2.5 Iteration

Apply the two lemmas with $\eta \gg \alpha^3$. After affine reparametrization of the progression produced at each step,

$$\alpha_{j+1} \geq \alpha_j + c\alpha_j^3, \quad N_{j+1} \geq c\alpha_j^3 \sqrt{N_j}.$$

The density recurrence permits only $O(\alpha^{-2})$ stages: for $b \geq a + c\alpha^3$, monotonicity in b gives

$$\frac{1}{a^2} - \frac{1}{b^2} \geq \frac{1}{a^2} - \frac{1}{(a + c\alpha^3)^2} \geq c'$$

for $0 < a \leq 1$. On the other hand

$$\log N_{j+1} \geq \frac{1}{2} \log N_j - C \log(1/\alpha).$$

After $t = O(\alpha^{-2})$ steps,

$$\log N_t \geq 2^{-t} \log N - C \log(1/\alpha).$$

Thus all local lemmas remain applicable when $\log \log N \gg \alpha^{-2}$, while the density cannot increase forever. This proves [theorem 2.1](#).

2.6 What is quantitatively wasteful

There are two major losses. Relative balancing on an interval loses one power of α in the Fourier obstruction. Then one-frequency localization loses a square root of the ambient length. Bourgain's Bohr-set method and modern almost-periodicity/spreadness arguments address precisely these losses. This suggests a recurring quantitative strategy: identify not only which estimates are valid, but also exactly where scale or density is lost.

Chapter 3

Bohr Sets and Local Fourier Analysis

3.1 Why progressions are too rigid

The density increment in the preceding chapter freezes one frequency on a long arithmetic progression. If many Fourier frequencies are simultaneously relevant, restricting one at a time destroys too much scale. A Bohr set freezes several characters at once.

For a finite abelian group G , a finite frequency set $\Gamma \subseteq \widehat{G}$, and $0 < \rho \leq 2$, define

$$B(\Gamma, \rho) = \{x \in G : |1 - \gamma(x)| \leq \rho \text{ for every } \gamma \in \Gamma\}.$$

Its rank is $|\Gamma|$. Bohr sets are approximate subgroups: if $x \in B(\Gamma, \rho)$ and $y \in B(\Gamma, \rho')$, then

$$x + y \in B(\Gamma, \rho + \rho'),$$

because $|1 - zw| \leq |1 - z| + |1 - w|$ on the unit circle.

3.2 Regular radii

A radius is useful only if a small dilation changes the size of the Bohr set predictably. We call $B(\Gamma, \rho)$ regular if, for $|\kappa| \leq c/|\Gamma|$,

$$|B(\Gamma, (1 + \kappa)\rho)| = (1 + O(|\Gamma||\kappa|))|B(\Gamma, \rho)|.$$

As proved in Volume II, every Bohr set has a nearby regular dilation. That result is a routine piece of Bohr-set geometry rather than an inverse theorem, so we use it here without repetition.

For a finite nonempty set $S \subseteq G$ write

$$\mu_S(x) := \frac{|G|}{|S|} 1_S(x),$$

so that $\mathbb{E}_G \mu_S = 1$. We use the normalized convolution

$$(f * g)(x) = \mathbb{E}_{y \in G} f(y)g(x - y).$$

Then

$$(1_A * \mu_S)(x) = \frac{|A \cap (x - S)|}{|S|},$$

which is literally the density of A on the translate $x - S$.

3.3 Approximate translation invariance

Proposition 3.1 (Regular Bohr translation). *Let $B = B(\Gamma, \rho)$ be regular and let $t \in B(\Gamma, \rho')$ with*

$$\rho' \leq c\rho/|\Gamma|.$$

Then

$$\|\tau_t \mu_B - \mu_B\|_1 \ll |\Gamma| \frac{\rho'}{\rho}.$$

Proof. If $x \in B$ but $x + t \notin B$, then for some $\gamma \in \Gamma$ the value $\gamma(x)$ lies within $O(\rho')$ of the boundary of the arc defining B . Equivalently,

$$(B + t) \triangle B \subseteq B(\Gamma, \rho + \rho') \setminus B(\Gamma, \rho - \rho').$$

Regularity therefore gives

$$|(B + t) \triangle B| \ll |\Gamma| \frac{\rho'}{\rho} |B|.$$

Since μ_B has the constant value $|G|/|B|$ on B , normalized $L^1(G)$ converts the last estimate exactly into the stated bound. $\square$

A useful set of centers is the *interior*

$$B^- := B(\Gamma, \rho - \rho').$$

For every $x \in B^-$ and every $s \in B(\Gamma, \rho')$ we have $x - s \in B$. Thus translates centered in B^- stay inside the old ambient Bohr set.

3.4 Freezing a generated spectrum

For a finite set $\Lambda \subseteq \widehat{G}$, write

$$\text{span}(\Lambda) = \left\{ \prod_{\lambda \in \Lambda} \lambda^{\varepsilon_\lambda} : \varepsilon_\lambda \in \{-1, 0, 1\} \right\}.$$

Lemma 3.2 (A Bohr set freezes the whole span). *Let $S = B(\Lambda, \sigma)$. If $\gamma \in \text{span}(\Lambda)$, then for every $x \in S$,*

$$|1 - \gamma(x)| \leq |\Lambda| \sigma.$$

Consequently, if $|\Lambda| \sigma \leq 1/4$, then

$$|\widehat{\mu_S}(\gamma)| \geq 3/4.$$

Proof. Write $\gamma = \lambda_1^{\varepsilon_1} \cdots \lambda_r^{\varepsilon_r}$ with $r \leq |\Lambda|$ and $\varepsilon_j \in \{-1, 1\}$. For unit complex numbers,

$$|1 - z_1 \cdots z_r| \leq \sum_{j=1}^r |1 - z_j|,$$

and $|1 - \bar{z}| = |1 - z|$. Hence the first assertion follows. Averaging over S gives

$$|1 - \widehat{\mu_S}(\gamma)| = \left| \mathbb{E}_{x \in S} (1 - \overline{\gamma(x)}) \right| \leq 1/4,$$

which implies the second assertion. $\square$

3.5 The local spectral density increment, proved

We now prove the mechanism that is frequently quoted as a “local Fourier density increment”. The statement is deliberately arranged so that every boundary term is visible.

Theorem 3.3 (Local spectral density increment). *Let $B = B(\Gamma, \rho)$ be a regular Bohr set, let $A \subseteq B$, and write*

$$\alpha = \frac{|A|}{|B|}, \quad \beta = \frac{|B|}{|G|}, \quad f = 1_A - \alpha 1_B.$$

Let $\Delta \subseteq \widehat{G}$ and suppose

$$\sum_{\gamma \in \Delta} |\widehat{f}(\gamma)|^2 \geq \nu \alpha^2 \beta \tag{3.1}$$

for some $0 < \nu \leq 1$. Assume that $\Delta \subseteq \text{span}(\Lambda)$ for a set Λ of size r .

There are absolute constants $c, c_0 > 0$ with the following property. Define

$$\rho_0 := \min \left\{ \frac{1}{2}, c_0 \rho \min \left(\frac{\nu \alpha^2}{|\Gamma| + 1}, \frac{1}{r + 1} \right) \right\}.$$

Then there is a regular radius $\rho' \in [\rho_0/2, \rho_0]$ and, with

$$B' = B(\Gamma \cup \Lambda, \rho'),$$

a translate $x - B' \subseteq B$ such that

$$\frac{|A \cap (x - B')|}{|B'|} \geq \alpha + c\nu \alpha^2.$$

Proof. We divide the proof into the three operations that recur throughout local Fourier analysis.

Step 1: choose an interior scale and freeze the spectrum. Choose $c_0 > 0$ in the theorem statement smaller than the reciprocal of the absolute constants required in the freezing and boundary estimates below. By the regular-radius lemma from Volume II there is a regular $\rho' \in [\rho_0/2, \rho_0]$. In particular,

$$\rho' \leq c_0 \rho \min \left\{ \frac{\nu \alpha^2}{|\Gamma| + 1}, \frac{1}{r + 1} \right\}. \tag{3.2}$$

Because B' freezes every character in Λ , Lemma 3.2 gives

$$|\widehat{\mu_{B'}}(\gamma)| \geq \frac{3}{4} \quad (\gamma \in \Delta). \tag{3.3}$$

Let

$$B^- := B(\Gamma, \rho - \rho').$$

Then $x - B' \subseteq B$ whenever $x \in B^-$. Regularity and (3.2) give

$$|(B + B') \setminus B^-| \leq C |\Gamma| \frac{\rho'}{\rho} |B| \leq c_0 \nu \alpha^2 |B|, \tag{3.4}$$

after decreasing the absolute c_0 in the theorem statement if necessary.

Step 2: spectral energy gives a large L^2 fluctuation. Set

$$g := f * \mu_{B'}.$$

Fourier inversion and Parseval give

$$\|g\|_2^2 = \sum_{\gamma \in \widehat{G}} |\widehat{f}(\gamma)|^2 |\widehat{\mu_{B'}}(\gamma)|^2.$$

Using (3.1) and (3.3),

$$\|g\|_2^2 \geq \frac{9}{16} \nu \alpha^2 \beta. \quad (3.5)$$

Also

$$\mathbb{E}g = (\mathbb{E}f)(\mathbb{E}\mu_{B'}) = 0$$

because $\mathbb{E}f = |A|/|G| - \alpha|B|/|G| = 0$. Finally, g is supported on $B + B'$ and satisfies $|g| \leq 1$: indeed

$$g(x) = (1_A * \mu_{B'})(x) - \alpha(1_B * \mu_{B'})(x),$$

and both convolutions lie in $[0, 1]$.

Step 3: a large L^2 fluctuation cannot hide entirely in the negative part or the boundary. On B^- we have $1_B * \mu_{B'} = 1$, so

$$g(x) = \frac{|A \cap (x - B')|}{|B'|} - \alpha \quad (x \in B^-). \quad (3.6)$$

Suppose, for contradiction, that

$$g(x) < \delta_0 := \frac{\nu \alpha^2}{16} \quad (x \in B^-). \quad (3.7)$$

Write $g = g_+ - g_-$ with $g_{\pm} \geq 0$. Since $\mathbb{E}g = 0$,

$$\mathbb{E}g_+ = \mathbb{E}g_- =: M.$$

The positive mass on B^- is at most $\delta_0|B|/|G| = \delta_0\beta$ by (3.7); outside B^- , the support of g is contained in the boundary set from (3.4) and $g_+ \leq 1$. Therefore

$$M \leq \delta_0\beta + c_0\nu\alpha^2\beta.$$

Since $|g| \leq 1$,

$$\|g\|_2^2 \leq \mathbb{E}|g| = 2M \leq 2(1/16 + c_0)\nu\alpha^2\beta.$$

Taking $c_0 \leq 1/32$ makes the right-hand side at most

$$\frac{3}{16}\nu\alpha^2\beta,$$

contradicting (3.5).

Thus (3.7) fails. There exists $x \in B^-$ with

$$g(x) \geq \frac{\nu \alpha^2}{16}.$$

Using (3.6),

$$\frac{|A \cap (x - B')|}{|B'|} \geq \alpha + \frac{\nu \alpha^2}{16}.$$

This proves the theorem with $c = 1/16$. □

Proof architecture What the proof actually did

The theorem is a clean composition of three elementary facts:

generated spectrum $\implies$ Fourier multiplier close to one,

spectral energy $\xRightarrow{\text{Parseval}}$ large spatial variance,

and

mean zero + small boundary $\implies$ a positive interior fluctuation.

The first line is why Chang's theorem controls rank; the second is Fourier analysis; the third is the density increment. No hidden inverse theorem occurs in this passage.

3.6 Chang's theorem as a rank bound

The large-spectrum theorem from Volume II says that if A has density α , then

$$\text{Spec}_\eta(A) = \{\gamma : |\widehat{1_A}(\gamma)| \geq \eta\alpha\}$$

is contained in the $\{-1, 0, 1\}$ -span of

$$O((\eta^{-2} \log(1/\alpha)))$$

characters. The key point is conceptual: a large cloud of coefficients may be frozen at a cost determined by its additive dimension, not its cardinality.

Worked Example 3.4 (Two hyperplanes at once). Let $G = \mathbb{F}_p^n$ and let H be the intersection of two independent hyperplanes. The Fourier support of 1_H is the two-dimensional orthogonal space $H^\perp$. Freezing one frequency sees only one hyperplane and raises density from p^{-2} to p^{-1} . Freezing a basis of $H^\perp$ simultaneously identifies the full coset of H and raises density to 1. Bohr-set localization is the cyclic analogue of this simultaneous restriction.

3.7 Bourgain's change of ambient object

The crucial conceptual shift is that the iteration no longer moves from interval to a much shorter interval. It moves from one approximate group to a slightly more complicated approximate group. Rank measures how many frequencies have been frozen; radius measures spatial scale. Quantitative Roth theory is largely the art of balancing these two costs.

Chapter 4

Gowers Uniformity Norms and Cube Calculus

4.1 Multiplicative derivatives

For $f : G \rightarrow \mathbb{C}$ define

$$\Delta_h f(x) = f(x+h) \overline{f(x)}.$$

A linear phase has constant derivative; a quadratic phase has linear derivative. Repeated differencing therefore lowers algebraic degree.

Definition 4.1 (Gowers norm). For $d \geq 1$,

$$\|f\|_{U^d(G)}^{2^d} = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{\omega \in \{0,1\}^d} \mathcal{C}^{|\omega|} f(x + \omega \cdot h).$$

The recursive identity

$$\|f\|_{U^d}^{2^d} = \mathbb{E}_h \|\Delta_h f\|_{U^{d-1}}^{2^{d-1}}$$

is immediate by grouping the cube vertices according to the last coordinate.

4.2 Reading the cube

For $d = 1$ the definition gives

$$\|f\|_{U^1}^2 = \mathbb{E}_{x, h} f(x) \overline{f(x+h)} = |\mathbb{E} f|^2,$$

so U^1 measures only the mean. For $d = 2$ the four vertices are

$$x, \quad x + h_1, \quad x + h_2, \quad x + h_1 + h_2,$$

and

$$\|f\|_{U^2}^4 = \mathbb{E}_{x, h_1, h_2} f(x) \overline{f(x+h_1) f(x+h_2) f(x+h_1+h_2)}.$$

Thus U^2 measures the extent to which multiplicative derivatives in one direction remain correlated after a second shift. The d -dimensional definition is not an arbitrary collection of 2^d terms: it is exactly the result of applying d commuting multiplicative derivatives and averaging the surviving phase.

For a phase $f = e(P)$ of modulus one,

$$\prod_{\omega} \mathcal{C}^{|\omega|} f(x + \omega \cdot h) = e\left((-1)^d \partial_{h_1} \cdots \partial_{h_d} P(x)\right),$$

up to the harmless sign convention for additive derivatives. Consequently the Gowers hierarchy is designed to measure polynomial degree by repeated differencing.

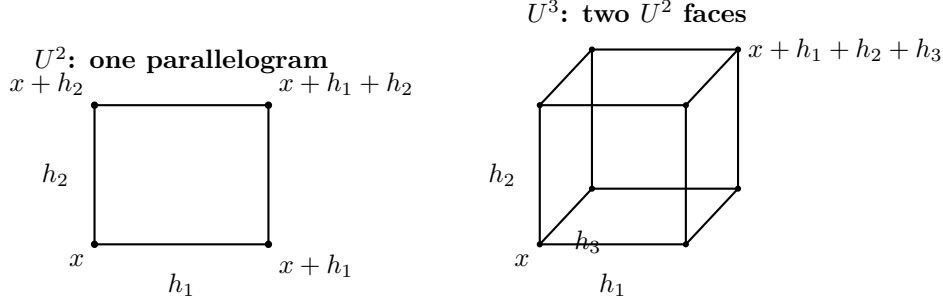


Figure 4.1: The cube geometry behind the Gowers norms. Splitting a d -cube according to its last binary coordinate produces two parallel $(d-1)$ -cubes; this is the geometric content of the recursive identity for U^d .

Worked Example 4.2 (The parallelogram identity). If $f = 1_A$, then

$$\|1_A\|_{U^2}^4 = \mathbb{P}_{x,h,k}[x, x+h, x+k, x+h+k \in A].$$

Equivalently, after writing $a = x$, $b = x+h$, $c = x+k$, $d = x+h+k$, it is the normalized number of additive quadruples

$$a + d = b + c.$$

Thus the U^2 norm is another normalization of additive energy. This is the first direct bridge between the sumset theory of Volumes I–II and the cube language developed in Volumes III–V.

4.3 Polynomial phases act as gauge symmetries

Suppose $P : G \rightarrow \mathbb{R}/\mathbb{Z}$ has additive degree at most $d-1$, so its d -fold derivative vanishes on every d -cube. Then for every f ,

$$\|f e(P)\|_{U^d} = \|f\|_{U^d}.$$

Indeed, in the cube product the extra phase contributed by $e(P)$ is exactly the exponential of its d -fold derivative, hence equals one. Therefore U^d deliberately regards multiplication by degree- $(d-1)$ phases as invisible. This is why an inverse theorem for U^d should recover precisely degree- $(d-1)$ structure rather than ordinary Fourier characters alone.

4.4 U^2 is ordinary Fourier analysis

Theorem 4.3.

$$\|f\|_{U^2}^4 = \sum_{\gamma \in \widehat{G}} |\widehat{f}(\gamma)|^4.$$

Proof. Set

$$C(h) = \mathbb{E}_x f(x+h) \overline{f(x)}.$$

Grouping the parallelogram by the second direction gives

$$\|f\|_{U^2}^4 = \mathbb{E}_h |C(h)|^2.$$

With the normalized Fourier convention of Volume I,

$$f(x) = \sum_{\gamma \in \widehat{G}} \widehat{f}(\gamma) \gamma(x).$$

Substitution and character orthogonality give

$$C(h) = \sum_{\gamma} |\hat{f}(\gamma)|^2 \gamma(h).$$

Parseval in the variable h therefore yields

$$\mathbb{E}_h |C(h)|^2 = \sum_{\gamma} |\hat{f}(\gamma)|^4,$$

which is the stated identity. $\square$

Corollary 4.4 (The U^2 inverse theorem). *If $|f| \leq 1$ and $\|f\|_{U^2} \geq \delta$, then some character satisfies*

$$|\hat{f}(\gamma)| \geq \delta^2.$$

Proof. By Parseval, $\sum |\hat{f}|^2 \leq 1$, so

$$\delta^4 \leq \sum |\hat{f}|^4 \leq \|\hat{f}\|_{\infty}^2 \sum |\hat{f}|^2.$$

$\square$

The exponent δ^2 is worth noticing. The inverse statement follows from the elementary interpolation

$$\|\hat{f}\|_4^4 \leq \|\hat{f}\|_{\infty}^2 \|\hat{f}\|_2^2.$$

In higher order there is no comparable linear transform whose point masses are the structured objects. Replacing this one-line extraction is the central difficulty of higher-order inverse theory.

Worked Example 4.5 (A random-sign benchmark). Let $f : G \rightarrow \{-1, 1\}$ be chosen independently with mean zero. For a fixed nontrivial cube with distinct vertices, the expected cube product is zero, while degenerate cubes contribute only a lower-order fraction when $|G|$ is large. Thus a typical random sign function has small U^d for every fixed d . Polynomial phases behave in exactly the opposite way at their natural order. The contrast gives concrete content to the words “uniform” and “structured”.

4.5 Gowers–Cauchy–Schwarz

For a family (f_{ω}) on the cube, define the multilinear cube form by replacing the same f at each vertex with f_{ω} .

Theorem 4.6 (Gowers–Cauchy–Schwarz).

$$|\Lambda_{\square}((f_{\omega}))| \leq \prod_{\omega \in \{0,1\}^d} \|f_{\omega}\|_{U^d}.$$

Proof. The full induction is written out in Appendix C; we record the mechanism here. Split the last cube coordinate from the first $d-1$. Cauchy–Schwarz duplicates the last shift, and after replacing the two copies by their difference, every vertex function occurs through a multiplicative derivative $\Delta_u f_{\omega}$. The remaining variables form a $(d-1)$ -cube. Applying the induction hypothesis to that cube gives factors

$$\left(\mathbb{E}_u \|\Delta_u f_{\omega}\|_{U^{d-1}}^{2^{d-1}} \right)^{1/2^d} = \|f_{\omega}\|_{U^d}.$$

Multiplying these factors over the vertices proves the inequality. $\square$

This inequality gives the triangle inequality for U^d , $d \geq 2$, by multilinearly expanding the cube expression for $f + g$.

4.6 Polynomial phases

If $G = \mathbb{F}_p^n$, $p > d$, and P has degree at most $d - 1$, then its d -fold additive derivative vanishes. Hence

$$\|e_p(P)\|_{U^d} = 1.$$

Worked Example 4.7 (Quadratic versus ordinary Fourier). For $f(x) = e_p(ax^2)$ on $\mathbb{F}_p$, $a \neq 0$, Gauss sums give

$$|\widehat{f}(\xi)| = p^{-1/2}, \quad \|f\|_{U^2} = p^{-1/4},$$

while the third derivative of ax^2 vanishes and therefore $\|f\|_{U^3} = 1$.

Worked Example 4.8 (A cubic phase). Let $p > 3$ and $f(x) = e_p(x^3)$. Then $\|f\|_{U^4} = 1$, while

$$\partial_a \partial_b \partial_c (x^3) = 6abc.$$

Thus

$$\|f\|_{U^3}^8 = \mathbb{E}_{a,b,c} e_p(-6abc) = \mathbb{P}(ab = 0) = \frac{2p-1}{p^2}.$$

The hierarchy distinguishes algebraic degree one level at a time.

4.7 Why the hierarchy is nested conceptually

The recursion

$$\|f\|_{U^d}^{2^d} = \mathbb{E}_h \|\Delta_h f\|_{U^{d-1}}^{2^{d-1}}$$

should be read as an induction principle. If $\|f\|_{U^d} \geq \delta$ and $|f| \leq 1$, then a positive proportion of increments h have

$$\|\Delta_h f\|_{U^{d-1}} \gg_\delta 1.$$

Indeed the averaged nonnegative quantity on the right is at least δ^{2^d} and at most one. Thus a degree- d inverse problem is automatically converted into many degree- $(d-1)$ inverse problems. What the recursion does *not* provide is compatibility between the structures obtained for different h .

Warning 4.9 (Individual derivative structure is insufficient). For each good h one may choose a structured model of $\Delta_h f$. Independent choices can vary chaotically with h . The higher-dimensional cube identities are needed again to prove that these models satisfy many additive relations. Volume IV begins exactly at this coherence problem for U^3 .

4.8 The derivative philosophy

Large U^d implies that many derivatives have large U^{d-1} . An inverse proof therefore tries to run

$$\text{large } U^d(f) \Rightarrow \text{structured } \Delta_h f \text{ for many } h \Rightarrow \text{coherence in } h \Rightarrow \text{integrated structure of } f.$$

The difficult steps are coherence and integration. For U^3 they become linearization and symmetry of a derivative-frequency map. In general they become nilpotent geometry.

4.9 Norms on intervals

For $f : [N] \rightarrow \mathbb{C}$, extend by zero to $\mathbb{Z}/\tilde{N}\mathbb{Z}$ with $\tilde{N} > 2^d N$ and set

$$\|f\|_{U^d[N]} := \frac{\|f1_{[N]}\|_{U^d(\mathbb{Z}/\tilde{N}\mathbb{Z})}}{\|1_{[N]}\|_{U^d(\mathbb{Z}/\tilde{N}\mathbb{Z})}}.$$

Because no relevant cube wraps around, numerator and denominator acquire the same ambient normalization factor, so the ratio is independent of the chosen sufficiently large auxiliary group.

Chapter 5

Generalized von Neumann Inequalities and Complexity

5.1 Cauchy–Schwarz as an elimination algorithm

Fix a finite field $\mathbb{F}$ and a finite-dimensional $\mathbb{F}$ -vector space V . An affine-linear form in t vector variables is a map

$$L(\mathbf{x}) = b + \sum_{r=1}^t a_r x_r, \quad \mathbf{x} = (x_1, \dots, x_t) \in V^t,$$

with $b \in V$ and coefficient vector $a = (a_1, \dots, a_t) \in \mathbb{F}^t$. For a system $\mathcal{L} = (L_1, \dots, L_m)$ and functions $f_j : V \rightarrow \mathbb{C}$, define

$$\Lambda_{\mathcal{L}}(f_1, \dots, f_m) = \mathbb{E}_{\mathbf{x} \in V^t} \prod_{j=1}^m f_j(L_j(\mathbf{x})).$$

Repeated Cauchy–Schwarz can eliminate bounded factors until only a cube built from one chosen function remains.

The word “eliminate” has a precise meaning. If, after a linear change of variables, one nuisance factor is independent of a coordinate u , then applying Cauchy–Schwarz in all the other variables removes that nuisance factor using only its L^∞ bound. The target factor is duplicated at two u -values. Repeating this operation manufactures the vertices of a cube.

5.2 Warm-up: three-term progressions from Fourier analysis

Before using repeated Cauchy–Schwarz, it is useful to see the smallest case exactly. Assume multiplication by 2 is invertible on G . Fourier expansion gives

$$\Lambda_3(f_0, f_1, f_2) := \mathbb{E}_{x,d} f_0(x) f_1(x+d) f_2(x+2d) = \sum_{\gamma} \widehat{f_0}(\gamma) \widehat{f_1}(\gamma^{-2}) \widehat{f_2}(\gamma),$$

with the evident multiplicative notation for characters. If $|f_1|, |f_2| \leq 1$, then Cauchy–Schwarz and Parseval give

$$|\Lambda_3| \leq \|\widehat{f_0}\|_\infty \|\widehat{f_1}\|_2 \|\widehat{f_2}\|_2 \leq \|\widehat{f_0}\|_\infty.$$

Since

$$\|\widehat{f_0}\|_\infty \leq \left(\sum_{\gamma} |\widehat{f_0}(\gamma)|^4 \right)^{1/4} = \|f_0\|_{U^2},$$

we obtain

$$|\Lambda_3(f_0, f_1, f_2)| \leq \|f_0\|_{U^2}.$$

By symmetry the same holds with any input as target. This is the generalized von Neumann inequality behind Roth's theorem, and it explains why U^2 —ordinary Fourier uniformity—is exactly the right norm for three-term progressions.

Theorem 5.1 (Four-term generalized von Neumann). *Let G be a finite abelian group on which 2 and 3 are invertible. If $|f_i| \leq 1$, then*

$$\left| \mathbb{E}_{x,d} \prod_{j=0}^3 f_j(x + jd) \right| \leq \min_j \|f_j\|_{U^3(G)}.$$

Proof. Fix one factor, say f_0 , as the target. Apply Cauchy–Schwarz three times, each time after an invertible linear change of variables chosen so that one non-target form is independent of the variable being doubled. Because the non-target functions are 1-bounded they disappear. The eight surviving copies of f_0 occupy the vertices of a three-dimensional affine cube with the conjugation pattern of the U^3 norm. Thus the eighth power of the original average is at most $\|f_0\|_{U^3}^8$. $\square$

The same procedure gives U^{k-1} control for k -term progressions under the corresponding invertibility hypotheses.

5.3 The elimination lemma

Lemma 5.2. *Let $L_0, L_1, \dots, L_r$ be linear forms on $\mathbb{F}^t$ and suppose L_0 is not in the span of $L_1, \dots, L_r$. Then there is an invertible change of variables with a coordinate u such that L_0 depends nontrivially on u while every L_j , $j \geq 1$, is independent of u .*

Proof. Let W be the span of $L_1, \dots, L_r$. Since $L_0 \notin W$, there is $v \in \ker W$ with $L_0(v) \neq 0$; otherwise L_0 would vanish on $\ker W$ and belong to $(\ker W)^\perp = W$. Extend v to a basis and use its coefficient as u . $\square$

5.4 Cauchy–Schwarz complexity

Definition 5.3 (Cauchy–Schwarz complexity at a form). Write $a_j \in \mathbb{F}^t$ for the coefficient vector of L_j . The system $\mathcal{L}$ has *Cauchy–Schwarz complexity at most s at L_i* if the index set $\{1, \dots, m\} \setminus \{i\}$ can be partitioned into $s + 1$ classes

$$C_1 \sqcup \dots \sqcup C_{s+1}$$

such that, for every r ,

$$a_i \notin \text{span}_{\mathbb{F}}\{a_j : j \in C_r\}.$$

Thus the nondegeneracy needed below is part of the definition rather than an unspoken extra hypothesis.

Theorem 5.4 (Generalized von Neumann principle). *Let $\mathbb{F}$ be a finite field, let V be a finite-dimensional $\mathbb{F}$ -vector space, and let $\mathcal{L} = (L_1, \dots, L_m)$ be affine-linear forms $L_j : V^t \rightarrow V$ with coefficient vectors $a_j \in \mathbb{F}^t$. Suppose $\mathcal{L}$ has Cauchy–Schwarz complexity at most s at L_i in the sense of Definition 5.3, and let $f_1, \dots, f_m : V \rightarrow \mathbb{C}$ satisfy $|f_j| \leq 1$. Then*

$$|\Lambda_{\mathcal{L}}(f_1, \dots, f_m)| \leq \|f_i\|_{U^{s+1}(V)}.$$

No additional coefficient-invertibility condition is required in this vector-space formulation: every nonzero scalar in the field $\mathbb{F}$ is invertible.

Proof. Choose a witnessing partition $C_1, \dots, C_{s+1}$. By [theorem 5.2](#), applied to the coefficient vectors in C_r , there is for each r a direction $v_r \in \mathbb{F}^t$ such that

$$a_j \cdot v_r = 0 \quad (j \in C_r), \quad a_i \cdot v_r \neq 0.$$

Rescaling v_r by the nonzero scalar $(a_i \cdot v_r)^{-1}$, we may and do arrange

$$a_i \cdot v_r = 1. \tag{5.1}$$

Here $\mathbf{x} + hv_r$ means $(x_1 + h(v_r)_1, \dots, x_t + h(v_r)_t)$ for $h \in V$.

We now perform the Cauchy–Schwarz eliminations explicitly. At the first stage choose a linear complement to the one-dimensional direction $\mathbb{F}v_1$ in $\mathbb{F}^t$ and write the variable $\mathbf{x}$ as $\mathbf{y} + uv_1$ with $u \in V$. Every factor indexed by C_1 is independent of u , because $a_j \cdot v_1 = 0$. Treating the product of those factors as a 1-bounded function of $\mathbf{y}$ and applying Cauchy–Schwarz in $\mathbf{y}$ gives

$$|\Lambda_{\mathcal{L}}|^2 \leq \mathbb{E}_{\mathbf{x}, h_1} \prod_{j \notin C_1} f_j(L_j(\mathbf{x})) \overline{f_j(L_j(\mathbf{x} + h_1 v_1))},$$

up to harmless conjugations of the factors. The important point is that every C_1 factor has disappeared, whereas each surviving factor has acquired the two shifts 0 and $h_1 v_1$.

At stage r , all copies of a factor with index $j \in C_r$ are still independent of the new coordinate in direction v_r : a previous shift changes $\mathbf{x}$ by a linear combination of $v_1, \dots, v_{r-1}$, but differentiation in the fresh direction v_r sees only the coefficient $a_j \cdot v_r = 0$. Cauchy–Schwarz therefore eliminates simultaneously every copy of every factor in C_r and doubles all remaining copies. Iterating through $r = 1, \dots, s+1$ yields

$$|\Lambda_{\mathcal{L}}|^{2^{s+1}} \leq \mathbb{E}_{\mathbf{x}, h_1, \dots, h_{s+1}} \prod_{\omega \in \{0,1\}^{s+1}} \mathcal{C}^{|\omega|} f_i \left(L_i(\mathbf{x}) + \sum_{r=1}^{s+1} \omega_r h_r \right).$$

Indeed, (5.1) says exactly that shifting $\mathbf{x}$ by $h_r v_r$ shifts the target form by h_r ; its affine constant is unchanged. The right-hand side is $\|f_i\|_{U^{s+1}(V)}^{2^{s+1}}$ by the definition of the Gowers norm. Taking the $2^{-(s+1)}$ power proves the claim. $\square$

For the k forms $L_j(x, d) = x + jd$, fixing L_i and putting each of the other $k-1$ forms in its own class gives $s+1 = k-1$, hence complexity at most $k-2$ and U^{k-1} control whenever the relevant coefficient vectors are distinct up to scalar multiple.

Worked Example 5.5 (The three elimination directions for a four-term progression). Let $\mathbb{F}$ have characteristic greater than 3 and take

$$L_j(x, d) = x + jd \quad (0 \leq j \leq 3).$$

Target L_0 . Its coefficient vector is $a_0 = (1, 0)$, while $a_j = (1, j)$ for $j = 1, 2, 3$. Put the three non-target forms into the singleton classes $C_j = \{L_j\}$. Since no $(1, j)$ with $j \neq 0$ is a scalar multiple of $(1, 0)$, the system has Cauchy–Schwarz complexity at most 2 at L_0 .

The elimination directions can be written down explicitly:

$$v_j = (1, -j^{-1}) \in \mathbb{F}^2.$$

Indeed,

$$a_j \cdot v_j = 1 - jj^{-1} = 0, \quad a_0 \cdot v_j = 1.$$

Thus the three Cauchy–Schwarz steps shift the target by independent increments h_1, h_2, h_3 while eliminating, in turn, the L_1, L_2, L_3 factors. The eight target copies are therefore

$$f_0(x + \omega_1 h_1 + \omega_2 h_2 + \omega_3 h_3), \quad \omega \in \{0, 1\}^3,$$

with the alternating conjugation pattern, giving exactly $\|f_0\|_{U^3}^8$. This calculation also pinpoints the characteristic restriction: the displayed elimination directions require 1, 2, 3 to be nonzero and hence invertible.

5.5 Worked complexity calculations

Worked Example 5.6 (A corner). On G^2 consider

$$(x, y), \quad (x + d, y), \quad (x, y + d).$$

For the corner-counting average, two Cauchy–Schwarz steps isolate any chosen input and create a two-dimensional parallelogram. Thus U^2 -type box uniformity is sufficient. The configuration has three points, but its geometry is genuinely two-dimensional; the relevant cube is dictated by the dependencies of the forms, not by the number of points.

Worked Example 5.7 (Why four-term progressions need one more derivative). Take $f_j(x) = e_p(c_j x^2)$ on $\mathbb{F}_p$, $p > 3$, and choose coefficients $c_j = (-1)^j \binom{3}{j}$. Because a quadratic has vanishing third finite difference,

$$\prod_{j=0}^3 f_j(x + jd) = 1$$

for every (x, d) . Yet each nonconstant quadratic phase has Fourier coefficients of size $p^{-1/2}$ and hence U^2 norm $p^{-1/4}$. Therefore no inequality controlling the four-term progression average by the U^2 norm of one arbitrary input can hold uniformly in p . The obstruction disappears at U^3 , where every quadratic phase has norm one.

5.6 True complexity

Cauchy–Schwarz complexity measures one proof strategy, not always the intrinsic uniformity level. Gowers and Wolf showed in finite-field settings that the correct level is governed by linear independence of suitable tensor powers of the coefficient vectors [4].

Worked Example 5.8 (Same number of forms, different complexity). The parallelogram system

$$x, \quad x + h, \quad x + k, \quad x + h + k$$

is naturally a two-dimensional cube and is controlled at the U^2 level after conjugation. The four-term progression

$$x, x + d, x + 2d, x + 3d$$

has four forms as well, but quadratic phases survive and force U^3 . Counting the forms tells us little; their algebraic dependencies determine the uniformity degree.

5.7 Nondegeneracy and characteristic

In the vector-space theorem above, the relevant nondegeneracy is encoded exactly by [theorem 5.3](#): for each elimination class, the target coefficient vector must lie outside the span of that class. In other ambient groups additional coefficient conditions can reappear. For example, the progression argument in [theorem 5.1](#) uses scalar changes of variables involving 2 and 3; over a cyclic group they must act invertibly. These are not cosmetic technicalities: in small characteristic or torsion settings distinct-looking forms can acquire new coincidences and an intended change of variables can cease to be bijective.

A safe reading practice is therefore to separate three questions:

1. Which forms are genuinely distinct up to scalar multiple?
2. Which coefficients must be invertible for the proposed changes of variables?
3. What cube dimension is produced after the eliminations?

Only after these checks should one quote a uniformity norm.

5.8 What Cauchy–Schwarz forgets

Every Cauchy–Schwarz step removes phase information from some inputs. This is why generalized von Neumann inequalities are robust but inverse theorems are difficult. When the final cube norm is large, the inverse theorem must reconstruct coherent algebra from information that repeated squaring has deliberately discarded.

Chapter 6

From Uniformity to Szemerédi's Theorem

6.1 The theorem

Theorem 6.1 (Szemerédi). *For every $k \geq 3$ and $\delta > 0$ there is $N_0(k, \delta)$ such that every $A \subseteq [N]$ with $|A| \geq \delta N$ and $N \geq N_0(k, \delta)$ contains a nontrivial k -term arithmetic progression.*

Proof location. A complete proof is given later under the heading “From the cyclic theorem to intervals.” The present statement is the precise dependency interface used before that proof.

For this book, the important issue is not merely existence. We want to understand how a configuration deficit is turned into a structured obstruction and then into an increment.

6.2 The density-increment template

Let $f = 1_A - \alpha$. Expanding the k -term progression count gives the random term α^k plus error terms involving f . Generalized von Neumann inequalities show that if every relevant balanced term is small in U^{k-1} , then the progression count is close to random. Therefore

$$\text{too few } k\text{-APs} \Rightarrow \|f\|_{U^{k-1}} \text{ is large.}$$

The remaining step is inverse theory: large uniformity must provide a structured region on which density increases.

Definition 6.2 (Nilsequence preview for this chapter). A *polynomial nilsequence* is a sequence

$$n \mapsto F(g(n)\Gamma),$$

where G/Γ is a compact nilmanifold, $g : \mathbb{Z} \rightarrow G$ is a polynomial sequence adapted to a finite filtration of G , and $F : G/\Gamma \rightarrow \mathbb{C}$ is bounded and Lipschitz. Its *step* is the length of the filtration. In this chapter, a bound on its *complexity* means simultaneous bounds on the dimension and step of G , the rational/Mal’cev data of G/Γ , the polynomial-coordinate data of g , and the Lipschitz norm of F . Volume V constructs these notions carefully. This preview is enough to make the conditional reduction below a fully stated assertion rather than a promise involving undefined terms.

Warning 6.3 (Two routes, and which one proves the theorem here). A common exposition says “apply an inverse theorem and localize the resulting nilsequence.” The localization step for general nilsequences is itself substantial, so we do not use that sentence as a hidden proof.

Appendix E gives a complete proof of Szemerédi's theorem through hypergraph regularity, counting, and removal. The density-increment argument below is retained as a second structural route: it explains how higher-order inverse theory would produce an iterative proof once a suitable localization theorem is available. It is not used to certify Theorem 6.1.

Proposition 6.4 (Structural reduction to inverse theory and localization). *Fix $k \geq 3$ and put $s = k - 2$. Suppose the following two statements have been proved for every $\eta > 0$.*

- (I) *If $g : [M] \rightarrow \mathbb{C}$ is 1-bounded and $\|g\|_{U^{s+1}[M]} \geq \eta$, then g correlates by at least $c_s(\eta) > 0$ with an s -step nilsequence of complexity at most $C_s(\eta)$.*
- (L) *Every nilsequence of complexity at most C can, for every $\varepsilon > 0$, be partitioned into $O_{C,\varepsilon}(M^{1-c_{C,\varepsilon}})$ arithmetic progressions, apart from $O_{C,\varepsilon}(1)$ discarded points, so that it oscillates by at most ε on each progression and every non-singleton progression has length at least $M^{c_{C,\varepsilon}}$.*

Then the density-increment hypothesis of Lemma 6.5 holds, and hence Szemerédi's theorem follows.

Proof. Let $A \subseteq [M]$ have density α and contain no nontrivial k -term progression. Put $f = 1_A - \alpha$. The generalized von Neumann theorem of Chapter 5, applied with the normalized zero-extension definition from Section 4.9 and the boundary convention recorded in Section D.6, shows that a deficit of the k -AP count from its random main term forces

$$\|f\|_{U^{k-1}[M]} \geq \eta_k(\alpha) > 0.$$

Indeed, the constant term contributes a positive quantity comparable with α^k , while every mixed term is controlled by the U^{k-1} norm of one balanced factor. Degenerate progressions contribute only $O_k(M^{-1})$, which is absorbed once M exceeds a threshold depending on (k, α) .

Apply (I). There is a 1-bounded structured sequence ψ of complexity $C_k(\alpha)$ such that, after multiplying by a constant phase,

$$\Re \mathbb{E}_{n \leq M} f(n) \overline{\psi(n)} \geq c_k(\alpha).$$

Apply (L) with $\varepsilon = c_k(\alpha)/8$. Let $P_1, \dots, P_J$ be the resulting non-singleton progressions and choose $n_j \in P_j$. On P_j ,

$$\psi(n) = \psi(n_j) + O(\varepsilon).$$

The total contribution of the oscillation error to the correlation is at most ε , and the discarded points contribute $o_{k,\alpha}(1)$. Therefore, for M sufficiently large,

$$\left| \sum_j \frac{|P_j|}{M} \overline{\psi(n_j)} \left(\frac{1}{|P_j|} \sum_{n \in P_j} f(n) \right) \right| \geq \frac{1}{2} c_k(\alpha).$$

Put

$$d_j := \frac{1}{|P_j|} \sum_{n \in P_j} f(n).$$

Since $|\psi(n_j)| \leq 1$, the triangle inequality gives

$$\sum_j \frac{|P_j|}{M} |d_j| \geq \frac{1}{2} c_k(\alpha).$$

On the other hand, the progressions partition all but $o(M)$ points and $\sum_{n \leq M} f(n) = 0$, so

$$\sum_j \frac{|P_j|}{M} d_j = o_{k,\alpha}(1).$$

Writing $d_j = d_j^+ - d_j^-$ and comparing the positive and negative masses, we obtain, for large M ,

$$\sum_j \frac{|P_j|}{M} d_j^+ \geq \frac{1}{8} c_k(\alpha).$$

Hence at least one j has

$$d_j \geq \frac{1}{8} c_k(\alpha).$$

Consequently

$$\text{dens}_{P_j}(A) \geq \alpha + \frac{1}{8} c_k(\alpha).$$

By (L), $|P_j| \geq M^{c'_k(\alpha)}$. Thus Lemma 6.5 applies with

$$\rho(\alpha) = \frac{1}{8} c_k(\alpha), \quad c(\alpha) = c'_k(\alpha),$$

and proves the theorem. $\square$

6.3 An abstract iteration lemma

Lemma 6.5. *Suppose that for every density $\alpha > 0$, every progression-free A of density α in a progression P of length $M \geq M_*(\alpha)$ has a subprogression P' satisfying*

$$|P'| \geq M^{c(\alpha)}, \quad \text{dens}_{P'}(A) \geq \alpha + \rho(\alpha).$$

Assume that for every $\alpha_0 > 0$,

$$\inf_{\alpha_0 \leq u \leq 1} c(u) > 0, \quad \inf_{\alpha_0 \leq u \leq 1} \rho(u) > 0, \quad \sup_{\alpha_0 \leq u \leq 1} M_*(u) < \infty.$$

Then for every $\alpha_0 > 0$ there is an integer $M_0 = M_0(\alpha_0)$ such that, whenever P is an arithmetic progression of length $M \geq M_0$, every progression-free set $A \subseteq P$ satisfies $\text{dens}_P(A) < \alpha_0$.

Proof. Starting from α_0 , put

$$c_* := \inf_{u \in [\alpha_0, 1]} c(u) > 0, \quad \rho_* := \inf_{u \in [\alpha_0, 1]} \rho(u) > 0.$$

At most $t = \lceil (1 - \alpha_0)/\rho_* \rceil + 1$ increments are possible. After $j \leq t$ stages the ambient length is at least $M^{c_*^j}$. Choose M so large that $M^{c_*^t}$ exceeds the uniform threshold $\sup M_*(u)$. Then all t increments are legal, and the next one would force density above 1. $\square$

6.4 Varnavides supersaturation

For transference, one needs not merely existence but many progressions.

Proposition 6.6 (Varnavides). *Fix $k \geq 3$ and $\delta > 0$. There are $c = c(k, \delta) > 0$ and N_0 such that every $A \subseteq [N]$ of density at least δ contains at least cN^2 ordered nonconstant k -term arithmetic progressions for $N \geq N_0$.*

Proof. We give an exact averaging argument that avoids boundary bookkeeping. Let $B > 1$ be the absolute constant in the fixed-dilate prime corollary of the elementary Chebyshev bounds proved in Volume I. Choose $M = M(k, \delta)$ so large that every subset of $[M]$ of density at least $\delta/(4B)$ contains a nonconstant k -term progression. Increase N_0 so that $N \geq N_0$ implies $M < 2N$.

For all sufficiently large N , that fixed-dilate prime corollary gives a prime p with

$$2N < p \leq 2BN.$$

Embed $[N]$, and hence A , in the cyclic group $\mathbb{Z}/p\mathbb{Z}$. Put

$$\beta := \frac{|A|}{p}.$$

Since $|A| \geq \delta N$ and $p \leq 2BN$,

$$\beta \geq \frac{\delta}{2B}.$$

For $a \in \mathbb{Z}/p\mathbb{Z}$ and $q \in (\mathbb{Z}/p\mathbb{Z})^\times$, consider the affine copy

$$P_{a,q} := \{a + iq : 0 \leq i < M\}.$$

Because $M < p$, the M points are distinct. Averaging over all $p(p-1)$ pairs (a, q) gives the exact identity

$$\mathbb{E}_{a,q \neq 0} \frac{|A \cap P_{a,q}|}{M} = \beta.$$

Indeed, for each fixed i and $q \neq 0$, the point $a + iq$ is uniformly distributed as a varies.

Let θ be the proportion of pairs (a, q) for which

$$\frac{|A \cap P_{a,q}|}{M} \geq \frac{\beta}{2}.$$

Since every relative density is at most 1,

$$\beta \leq \theta \cdot 1 + (1 - \theta) \frac{\beta}{2},$$

whence

$$\theta \geq \frac{\beta/2}{1 - \beta/2} \geq \frac{\beta}{2} \geq \frac{\delta}{4B}.$$

Thus at least $(\delta/(4B))p(p-1)$ affine copies are good. For each good copy, the index set

$$B_{a,q} := \{0 \leq i < M : a + iq \in A\}$$

has density at least $\beta/2 \geq \delta/(4B)$ in $[M]$. By the choice of M , it contains indices

$$i_0, i_0 + r, \dots, i_0 + (k-1)r, \quad r \geq 1.$$

Their images form a nonconstant modular k -term progression in A .

Because every image lies in the integer interval $[N]$ and $p > 2N$, this modular progression is in fact an ordinary integer progression. To see this, if its consecutive terms are $x_j \in [N]$, then

$$x_j - 2x_{j+1} + x_{j+2} \equiv 0 \pmod{p},$$

while the absolute value of the left-hand side is strictly smaller than $2N < p$; hence it is zero as an integer.

It remains only to control multiplicity. Fix an ordered nonconstant integer k -AP

$$x, x + d, \dots, x + (k-1)d \subseteq A.$$

If it arises from $P_{a,q}$ through the index progression $i_0, i_0 + r, \dots, i_0 + (k-1)r$, then

$$rq \equiv d \pmod{p}.$$

For each admissible pair (i_0, r) in $[M]$ there is at most one q , because $1 \leq r < M < p$, and then $a \equiv x - i_0 q \pmod{p}$ is determined. There are at most M^2 possible pairs (i_0, r) . Therefore each output progression is counted at most M^2 times.

Consequently the number of distinct ordered nonconstant k -APs in A is at least

$$\frac{(\delta/(4B))p(p-1)}{M^2} \gg_{k,\delta} N^2.$$

This is the required estimate. □

Idea Why supersaturation matters

A dense-model or regularity argument changes a counting average by a small error. The assertion “there is at least one progression” is not stable under such perturbation. A positive cN^2 main term is stable. This is why existence theorems are routinely upgraded to counting theorems before transference.

6.5 Regularity versus density increment

There are two major ways to use inverse structure. Density increment localizes to a region where the set is denser and iterates. Regularity decomposes

$$f = f_{\text{str}} + f_{\text{sml}} + f_{\text{unf}}$$

and counts directly on the structured component. Both are manifestations of the same dichotomy: either the balanced function is uniform enough to count, or it has a structured obstruction.

6.6 A four-term obstruction

Worked Example 6.7. Let $G = \mathbb{F}_p^n$, $p > 3$, and $f(x) = e_p(Q(x))$ for a quadratic form Q . Then $\|f\|_{U^3} = 1$. Along a four-term progression the third finite difference of Q vanishes. Thus ordinary Fourier coefficients may all be small while the four-term count remains highly non-random. This is the concrete reason Roth’s one-frequency proof does not extend by cosmetic modification to $k = 4$.

6.7 Where quantitative bounds are lost

The abstract lemma hides two functions: the increment size $\rho(\alpha)$ and the scale exponent $c(\alpha)$. If $c(\alpha)$ is tower-small, the final N_0 is tower-large. Quantitative Szemerédi theory can therefore be read as a forensic analysis of the inverse theorem: which step introduces the worst loss, and can it be reorganized rather than merely estimated more carefully?

Chapter 7

Arithmetic Regularity: Structure, Uniformity, and Error

7.1 The U^2 model

Before nilsequences enter, the energy-increment mechanism can be proved with ordinary characters. Let $\mathcal{B}$ be a finite factor generated by discretized character values. Define

$$\mathcal{E}_f(\mathcal{B}) = \|\mathbb{E}(f \mid \mathcal{B})\|_2^2.$$

Refinement increases energy because conditional expectation is an orthogonal projection.

Lemma 7.1 (One coefficient gives energy). *Let $g = f - \mathbb{E}(f \mid \mathcal{B})$ and suppose $|\widehat{g}(\gamma)| \geq \eta$. Refine $\mathcal{B}$ by quantizing γ to mesh $M \gg \eta^{-1}$. Then the energy increases by at least $c\eta^2$.*

Proof. The refined factor contains a measurable approximation $\tilde{\gamma}$ with

$$\|\gamma - \tilde{\gamma}\|_\infty = O(M^{-1}).$$

Taking M large enough gives

$$|\langle g, \tilde{\gamma} \rangle| \geq \eta/2.$$

The orthogonal projection of g onto the new measurable subspace therefore has L^2 norm at least $\eta/2$. Pythagoras gives an energy increment at least $\eta^2/4$. $\square$

Theorem 7.2 (U^2 regularity). *For every $\eta > 0$ and every 1-bounded $f : G \rightarrow \mathbb{C}$, there is a factor generated by $O(\eta^{-2})$ characters such that*

$$f = f_{\text{str}} + f_{\text{unf}}, \quad f_{\text{str}} = \mathbb{E}(f \mid \mathcal{B}), \quad \|f_{\text{unf}}\|_{U^2} \leq C\eta^{1/2}.$$

Proof. Set $\mathcal{B}_0$ to be the trivial factor. At stage j let

$$g_j = f - \mathbb{E}(f \mid \mathcal{B}_j).$$

If $\|g_j\|_{U^2} \leq C\eta^{1/2}$, stop. Otherwise the U^2 inverse theorem from Chapter 4 gives a Fourier coefficient of magnitude at least η ; Lemma 7.1 then produces a refinement whose energy rises by at least $c\eta^2$.

The energy is at most $\|f\|_2^2 \leq 1$, so this can happen only $O(\eta^{-2})$ times. At termination the stated decomposition holds. $\square$

7.2 An abstract inverse theorem is enough

The higher-order regularity mechanism is not a second mysterious theorem. Once an inverse theorem has been proved, regularity follows from the same Hilbert-space argument as above. We formulate and prove the abstract statement now; Volume IV verifies its hypothesis for Gowers norms by proving the nilsequence inverse theorem.

Let X be a finite probability space. A *structured observable of complexity at most M* is simply a 1-bounded function belonging to a prescribed class $\mathcal{S}_M$. We assume the classes are nested in M .

Definition 7.3 (Inverse property). A seminorm $\|\cdot\|_{\mathcal{U}}$ has the (c, M) inverse property at threshold η if every 1-bounded g with

$$\|g\|_{\mathcal{U}} > \eta$$

correlates with some $\phi \in \mathcal{S}_{M(\eta)}$:

$$|\langle g, \phi \rangle| \geq c(\eta) > 0.$$

The first issue is that a structured observable need not itself be measurable with respect to a finite factor. Quantization resolves this without losing much correlation.

Lemma 7.4 (Quantizing an observable). *Let $\phi : X \rightarrow \mathbb{C}$ satisfy $|\phi| \leq 1$ and let $m \geq 1$. There is a finite factor $\mathcal{P}(\phi, m)$ with $O(m^2)$ atoms and a $\mathcal{P}(\phi, m)$ -measurable function $\tilde{\phi}$ such that*

$$\|\phi - \tilde{\phi}\|_{\infty} \leq \frac{3}{m}.$$

Proof. Partition the square $[-1, 1]^2 \subset \mathbb{R}^2 \cong \mathbb{C}$ into squares of side $1/m$. Pull this partition back by $\tilde{\phi}$. On each nonempty atom choose the center of the corresponding square as the value of $\tilde{\phi}$. Euclidean distance is at most $\sqrt{2}/m$, and the harmless constant $3/m$ avoids boundary conventions. $\square$

Lemma 7.5 (Correlation forces energy increment). *Let $\mathcal{B}$ be a factor, put*

$$g = f - \mathbb{E}(f \mid \mathcal{B}),$$

and suppose $|\langle g, \phi \rangle| \geq c$ for a 1-bounded ϕ . Let $m \geq 12/c$ and refine $\mathcal{B}$ by $\mathcal{P}(\phi, m)$ to obtain $\mathcal{B}'$. Then

$$\|\mathbb{E}(f \mid \mathcal{B}')\|_2^2 - \|\mathbb{E}(f \mid \mathcal{B})\|_2^2 \geq \frac{c^2}{16}.$$

Proof. Let $\tilde{\phi}$ be the approximation from Lemma 7.4. Since $\|g\|_1 \leq 2$,

$$|\langle g, \tilde{\phi} \rangle| \geq c - \|g\|_1 \|\phi - \tilde{\phi}\|_{\infty} \geq c/2.$$

Write

$$h = \mathbb{E}(f \mid \mathcal{B}') - \mathbb{E}(f \mid \mathcal{B}).$$

Because $\tilde{\phi}$ is $\mathcal{B}'$ -measurable and g is orthogonal to every $\mathcal{B}$ -measurable function,

$$\langle g, \tilde{\phi} \rangle = \langle \mathbb{E}(g \mid \mathcal{B}'), \tilde{\phi} \rangle = \langle h, \tilde{\phi} \rangle.$$

As $\|\tilde{\phi}\|_2 \leq 1 + 3/m \leq 2$, Cauchy–Schwarz gives

$$\|h\|_2 \geq c/4.$$

Finally h is orthogonal to $\mathbb{E}(f \mid \mathcal{B})$, because conditional expectation onto $\mathcal{B}$ of h is zero. Hence Pythagoras gives the displayed energy increment. $\square$

Theorem 7.6 (Abstract arithmetic regularity from an inverse theorem). *Assume the seminorm $\|\cdot\|_{\mathcal{U}}$ has the inverse property of Definition 7.3 at threshold η . Then every 1-bounded f admits a decomposition*

$$f = f_{\text{str}} + f_{\text{unf}}$$

such that

$$\|f_{\text{unf}}\|_{\mathcal{U}} \leq \eta,$$

and f_{str} is measurable with respect to a factor generated by at most

$$O(c(\eta)^{-2})$$

quantized observables from $\mathcal{S}_{M(\eta)}$.

Proof. Start with the trivial factor $\mathcal{B}_0$. Given $\mathcal{B}_j$, put

$$g_j = f - \mathbb{E}(f \mid \mathcal{B}_j).$$

If $\|g_j\|_{\mathcal{U}} \leq \eta$, stop. Otherwise the inverse property provides $\phi_j \in \mathcal{S}_{M(\eta)}$ with

$$|\langle g_j, \phi_j \rangle| \geq c(\eta).$$

Refine by its quantization. Lemma 7.5 increases the energy by at least $c(\eta)^2/16$.

Since

$$0 \leq \|\mathbb{E}(f \mid \mathcal{B}_j)\|_2^2 \leq \|f\|_2^2 \leq 1,$$

there are at most $16c(\eta)^{-2}$ nonterminal stages. At the final stage take

$$f_{\text{str}} = \mathbb{E}(f \mid \mathcal{B}_j), \quad f_{\text{unf}} = f - f_{\text{str}}.$$

□

7.3 Where the L^2 -small term comes from

Applications often need

$$f = f_{\text{str}} + f_{\text{sml}} + f_{\text{unf}}.$$

This is not an additional inverse theorem. It is a stability refinement of Theorem 7.6.

Proposition 7.7 (Three-term regularity by energy stabilization). *Fix $\eta, \varepsilon > 0$ and suppose the inverse property holds at threshold η . Then one may choose two nested structured factors $\mathcal{B} \subseteq \mathcal{B}'$ so that*

$$f_{\text{str}} := \mathbb{E}(f \mid \mathcal{B}), \quad f_{\text{sml}} := \mathbb{E}(f \mid \mathcal{B}') - \mathbb{E}(f \mid \mathcal{B}), \quad f_{\text{unf}} := f - \mathbb{E}(f \mid \mathcal{B}')$$

satisfy

$$\|f_{\text{sml}}\|_2 \leq \varepsilon, \quad \|f_{\text{unf}}\|_{\mathcal{U}} \leq \eta.$$

The number of structured generators is finite as a function of (η, ε) and the inverse-theorem parameters.

Proof. Run the refinement algorithm of Theorem 7.6 repeatedly in blocks. Record the bounded nondecreasing energies

$$E_j = \|\mathbb{E}(f \mid \mathcal{B}_j)\|_2^2 \in [0, 1].$$

Among any more than ε^{-2} successive blocks there are two consecutive block endpoints, say $i < j$, with

$$E_j - E_i \leq \varepsilon^2;$$

otherwise the total increase would exceed 1.

Choose j to be the endpoint of a completed regularity block, so the residual at $\mathcal{B}_j$ is U -uniform. Set $\mathcal{B} = \mathcal{B}_i$ and $\mathcal{B}' = \mathcal{B}_j$. Orthogonality of conditional expectations gives

$$\|f_{\text{sml}}\|_2^2 = E_j - E_i \leq \varepsilon^2.$$

The final residual is uniform by construction. $\square$

In nilsequence applications one makes one further approximation: a step function of finitely many nilsequence coordinates is smoothed away from the quantization boundaries. The discarded boundary set has arbitrarily small measure and therefore contributes to f_{sml} ; the smoothed observable is a Lipschitz function on a finite product nilmanifold and is again a nilsequence. This is a compactness/approximation step, not a new structural theorem.

7.4 Torus prototype

On an interval, a U^2 structured function may be written

$$f_{\text{str}}(n) = F(n\theta \bmod 1), \quad F : \mathbb{T}^d \rightarrow \mathbb{C}.$$

If $k \cdot \theta$ is irrational for every nonzero $k \in \mathbb{Z}^d$, then Weyl's criterion gives

$$\frac{1}{N} \sum_{n \leq N} F(n\theta) \rightarrow \int_{\mathbb{T}^d} F.$$

This is the abelian prototype of the nilsequence factorization-and-counting theorem: separate rational resonances, keep an equidistributed orbit, and replace a long average by a geometric integral.

Worked Example 7.8 (Structured does not mean periodic). Take

$$f(n) = 1_{[0,1/3]}(\{n\sqrt{2}\}).$$

It is controlled by one irrational rotation and is highly structured, but it is not periodic. Fourier regularity should therefore be understood as low-complexity phase dependence, not as finite periodicity.

7.5 What remains genuinely difficult

The proof above shows that *regularity itself is elementary once the correct inverse theorem is available*. The hard theorem in higher order is therefore the inverse theorem: identifying the structured observables that correlate with a function of large U^{s+1} norm. Volume IV proves the U^3 case in full and then develops the general nilsequence inverse theorem. This separation prevents the energy-increment argument from being credited with work actually done by inverse theory.

Chapter 8

Dense Models and Transference

8.1 The sparse problem

A dense theorem assumes $0 \leq f \leq 1$. Prime-number weights and other sparse objects are typically unbounded after normalization. Transference begins with a majorant

$$0 \leq f \leq \nu,$$

where ν has mean approximately one and looks like the constant function 1 to every bounded-complexity test generated by the counting argument.

8.2 A finite dense-model theorem

Let X be finite and let $\mathcal{F}$ be a finite family of functions $\phi : X \rightarrow [-1, 1]$, closed under negation. Write

$$\|h\|_{\mathcal{F}} = \sup_{\phi \in \mathcal{F}} |\mathbb{E}h\phi|.$$

Let $\mathcal{F}^{(\leq m)}$ be the products of at most m tests.

Theorem 8.1 (Finite dense model). *For every $\epsilon > 0$ there are m and $\eta > 0$ such that the following holds. Suppose $\nu \geq 0$, $\mathbb{E}\nu = 1$, and*

$$|\mathbb{E}(\nu - 1)\Phi| \leq \eta \quad (\Phi \in \mathcal{F}^{(\leq m)}).$$

If $0 \leq f \leq \nu$ and $\mathbb{E}f = \alpha \leq 1$, then there is $g : X \rightarrow [0, 1]$ with

$$\mathbb{E}g = \alpha, \quad \|f - g\|_{\mathcal{F}} \leq \epsilon.$$

Proof. Assume no model exists. Map h to the vector of correlations $(\mathbb{E}h\phi)_{\phi \in \mathcal{F}}$. Finite-dimensional hyperplane separation gives

$$\psi = \sum_j c_j \phi_j, \quad \sum_j |c_j| = 1,$$

with $\|\psi\|_{\infty} \leq 1$ and

$$\mathbb{E}f\psi \geq \sup_{0 \leq g \leq 1, \mathbb{E}g = \alpha} \mathbb{E}g\psi + \epsilon.$$

The maximizing dense function fills the points with largest ψ first, so for some threshold t it is 1 above t , 0 below t , and fractional only on the level set. Call it g_0 . Since $\mathbb{E}(f - g_0) = 0$,

$$\mathbb{E}(f - g_0)(\psi - t) \geq \epsilon.$$

Let $q_t(u) = (u - t)_+$. Pointwise,

$$g_0(\psi - t) = q_t(\psi), \quad f(\psi - t) \leq \nu q_t(\psi).$$

Hence

$$\mathbb{E}(\nu - 1)q_t(\psi) \geq \epsilon.$$

The family q_t , $t \in [-1, 1]$, is compact in $C([-1, 1])$. By Weierstrass and a finite-net argument, for a chosen κ every q_t has a polynomial approximation

$$P_t(u) = \sum_{j=0}^m a_j(t)u^j$$

of uniform error at most κ , with m and the coefficient sum bounded uniformly in t . Taking $\kappa \ll \epsilon$ yields

$$\mathbb{E}(\nu - 1)P_t(\psi) \gg \epsilon.$$

But ψ^j expands into products of j members of $\mathcal{F}$, with total absolute coefficient mass at most one. The pseudorandomness hypothesis therefore bounds the last correlation by $C_\epsilon \eta$, a contradiction when η is small enough. $\square$

8.3 Counting after modelling

The test family must be chosen so that the configuration operator is continuous in the induced norm. If

$$|\Lambda(f, \dots, f) - \Lambda(g, \dots, g)| \ll \|f - g\|_{\mathcal{F}},$$

then a supersaturated dense theorem for g transfers to f .

Worked Example 8.2 (Random sparse majorant). Let $S \subset X$ be a random set of density ρ and $\nu = \rho^{-1}1_S$. For any fixed finite family of bounded test products, concentration gives

$$\mathbb{E}(\nu - 1)\Phi = o(1)$$

with high probability. Thus a random sparse set behaves densely for bounded-complexity tests after rescaling. The primes are not literally random because of congruence obstructions. Removing those obstructions with analytic number theory, the W -trick, sieve majorants, and prime-specific transference is intentionally left to the separate Analytic Number Theory project.

8.4 Pseudorandomness is theorem-dependent

There is no universal pseudorandom measure. The test family must be rich enough to include everything generated by the Cauchy–Schwarz proof, but weak enough that the number theory can verify its correlations. Over-specifying pseudorandomness may make the majorant impossible to construct; under-specifying it invalidates transference.

Chapter 9

Modern Roth Theory: Spectra and Almost-Periodicity

9.1 Past the one-frequency argument

The classical density increment loses too much scale. Modern arguments exploit entire families of moderate Fourier coefficients and, increasingly, almost-periods of convolutions rather than individual exact frequencies.

If A has density α and too few three-term progressions, the Fourier identity forces substantial cubic spectral mass. Dyadic pigeonholing produces a level η and an annular spectrum

$$\Delta = \{\gamma : \eta\alpha \leq |\widehat{1_A}(\gamma)| < 2\eta\alpha\}$$

carrying a large part of that mass. The question becomes geometric: what additive structure does Δ possess inside the dual group?

9.2 Energy of spectra

Worked Example 9.1 (A toy spectral extraction). Suppose

$$E(\Delta) \geq \frac{|\Delta|^3}{K}.$$

Balog–Szemerédi–Gowers gives a large $\Delta' \subseteq \Delta$ with

$$|\Delta' + \Delta'| \leq K^{O(1)}|\Delta'|.$$

Freiman theory then controls Δ' by a low-complexity progression or coset progression. Freezing the corresponding generators in physical space creates a density increment. The best Roth arguments cannot afford the losses in this toy pipeline, but it displays the strategy: turn a progression deficit into additive structure of the spectrum, then push that structure back to the original set.

9.3 Croot–Sisask almost-periodicity

For a normalized measure μ_A , convolution

$$F = f * \mu_A$$

is an average of translates of f . Sampling $a_1, \dots, a_m \in A$ gives

$$F_{\mathbf{a}}(x) = \frac{1}{m} \sum_{j=1}^m f(x - a_j).$$

A vector-valued moment estimate shows that for $m \asymp p\epsilon^{-2}$, many samples approximate F in L^p . If two good samples differ by a common translation t , then

$$\|\tau_t F - F\|_p \ll \epsilon \|f\|_p.$$

Additive covering arguments turn the set of good samples into many such t . Thus almost-periodicity is a law-of-large-numbers phenomenon converted into additive structure.

Idea Why almost-periods beat frequency-by-frequency localization

A convolution may be the sum of many medium Fourier modes, none individually strong enough to justify an expensive new Bohr frequency. Sampling can find translations that preserve the aggregate convolution without paying rank for every mode. The cost depends on the requested L^p accuracy and small-doubling parameters rather than directly on the raw spectral cardinality.

9.4 Smoothing or structure

Repeated convolution normally lowers peaks. If smoothing succeeds, error terms in the progression count become small. If smoothing fails, the measure retains concentration under addition, which itself signals additive structure. This dichotomy

$$\text{smoothing} \quad \text{or} \quad \text{failure of smoothing forces structure}$$

is a commutative precursor of the flattening dichotomy in Bourgain–Gamburd expansion theory.

9.5 Quantitative milestones

Bloom and Sisask broke the classical logarithmic barrier by combining almost-periodicity with spectral boosting [1]. Kelley and Meka later obtained a stretched-exponential-in-a-power-of- $\log N$ bound using spreadness and high moments [5]. A 2026 preprint of Raghavan gives a further improvement of the form

$$r_3(N) \leq N \exp\left(-c \frac{(\log N)^{1/6}}{\log \log N}\right)$$

for an absolute $c > 0$ [8]. We treat this final statement as a current preprint result and do not use it as an input elsewhere.

Proof deferred to a later volume Volume III supplement: modern Roth theory

The full Bloom–Sisask and Kelley–Meka arguments, including local almost-periodicity, spectral boosting, sifting, spreadness, and potential bookkeeping, are proved in the Volume III supplement. The 2026 Raghavan result remains a research-frontier preprint and is not used as an input in Volumes I–V.

9.6 The methodological change

The classical question “which Fourier coefficient is large?” has become “what collective geometry is carried by a distribution of coefficients, level sets, convolutions, and almost-periods?” This is a recurring feature of mature inverse theory: the first obstruction is an individual structured object; the strongest bounds often require understanding the organization of many such obstructions at once.

Chapter 10

The Kelley–Meka Paradigm: Spreadness, High Moments, and Potentials

10.1 Spreadness

A set $A \subseteq \mathbb{F}_q^n$ of density α is (γ, r) -spread if no affine subspace of codimension at most r has relative density exceeding $\gamma\alpha$. Failure of spreadness is already a multiplicative density increment.

The key change in the Kelley–Meka method is that density is not the only monotone quantity. Auxiliary sets may become sparser while a convolution concentration statistic becomes stronger.

10.2 High moments detect rare concentration

Normalize $\mu_A = \alpha^{-1}1_A$. A random-like set should have $\mu_A * \mu_A \approx 1$. High L^p norms detect exceptional peaks.

Worked Example 10.1 (A rare peak). Suppose a nonnegative F of mean one equals $1 + \delta^{-1/2}$ on a set of measure δ and is adjusted slightly below one elsewhere. For $p \gg \log(1/\delta)$,

$$\|F\|_p \gtrsim \delta^{1/p}(1 + \delta^{-1/2}) \asymp \delta^{-1/2}.$$

A rare concentration invisible to low moments becomes dominant at high moments. In additive combinatorics, such concentration represents popular sums or differences.

10.3 Structure versus pseudorandomness

Research window Kelley–Meka sifting principle: preview

In finite-field and localized integer settings, an abnormally large high-moment convolution norm can be sifted into one of two outcomes: a multiplicative density increment on a low-complexity structured region, or auxiliary sets for which a stronger convolution concentration parameter is obtained. A bounded potential prevents the second alternative from continuing indefinitely [5].

Proof deferred to a later volume Volume III supplement: Kelley–Meka

The sifting principle above is proved in the Volume III supplement. This chapter only develops the motivating high-moment and potential calculations, and no theorem in Volumes III–IV uses the unproved preview.

10.4 Why multiplicative increments are decisive

An additive increment

$$\alpha \mapsto \alpha + c\alpha^2$$

can require polynomially many stages in $1/\alpha$. A multiplicative increment

$$\alpha \mapsto (1 + c)\alpha$$

requires only $O(\log(1/\alpha))$ stages. If each stage costs polylogarithmic codimension, this difference is the distinction between very weak and quasipolynomial-type bounds.

Worked Example 10.2 (Codimension bookkeeping). Suppose every failure of spreadness gives density at least $(1 + \epsilon)\alpha$ on a subspace whose codimension increases by at most r . There can be at most

$$\frac{\log(1/\alpha)}{\log(1 + \epsilon)} = O((1 + \epsilon^{-1}) \log(1/\alpha))$$

such increments. The total codimension is therefore

$$O(r\epsilon^{-1} \log(1/\alpha)).$$

This elementary calculation explains why keeping r polynomial in $\log(1/\alpha)$ is so valuable.

10.5 A potential-function viewpoint

For a high moment p , consider schematically

$$\Phi_p(A) = \|\mu_A * \mu_A\|_p.$$

Since $\mu_A * \mu_A \leq \alpha^{-1}$ pointwise,

$$\Phi_p(A) \leq \alpha^{-1+1/p}.$$

If a sifting operation increases a suitable localized version of Φ_p by a fixed factor, it can happen only $O(\log(1/\alpha))$ times. The actual proof uses more carefully designed convolutions and local measures, but its termination mechanism is this bounded-potential argument.

10.6 A research lesson

When the obvious monotone quantity cannot improve efficiently, search for a different potential whose increase records the same obstruction. This principle extends well beyond Roth’s theorem: entropy increments, energy increments, flattening norms, and spectral gaps are all examples of choosing the right bounded quantity to drive an iteration.

Chapter 11

Systems of Linear Forms and True Complexity

11.1 Beyond one progression

Let V be a vector space over $\mathbb{F}_p$ and let

$$L_i(x_1, \dots, x_d) = \lambda_{i1}x_1 + \dots + \lambda_{id}x_d$$

be linear forms. The counting operator

$$\Lambda_{\mathcal{L}}(f_1, \dots, f_m) = \mathbb{E}_{\mathbf{x}} \prod_i f_i(L_i(\mathbf{x}))$$

contains corners, parallelepipeds, constellations, and arithmetic progressions as special cases.

11.2 Tensor powers

Write $\lambda_i = (\lambda_{i1}, \dots, \lambda_{id})$. The heuristic behind true complexity is that a degree- s phase inserted in one form creates a top multilinear contribution controlled by the $(s+1)$ -fold tensor

$$\lambda_i^{\otimes(s+1)}.$$

If these tensors are linearly independent, no nontrivial degree- s phase can cancel across the system in the dangerous way.

Research window Gowers–Wolf true-complexity theorem: preview

Over finite fields of sufficiently large characteristic, the least Gowers-uniformity degree needed to control a nondegenerate system is governed by the first tensor power at which the coefficient vectors become linearly independent in the appropriate sense [4]. Precise hypotheses exclude proportional forms and characteristic pathologies.

Proof deferred to a later volume Research window: true-complexity theory is not a theorem used in Volumes I–V

The general true-complexity theorem is retained here as a research window rather than a theorem used in Volumes I–V; it is not incorporated as a load-bearing theorem in this book. The calculations below prove the tensor criterion on the principal progression examples and explain necessity, but the general theorem is not used as a black box in

Worked Example 11.1 (Four terms on a line). For

$$L_j(x, d) = x + jd, \quad j = 0, 1, 2, 3,$$

the coefficient vectors are $(1, j)$. Ordinary Fourier structure corresponds to first tensor powers and is not enough; quadratic relations survive. At the next level the relevant tensor data separate the forms, matching the fact that U^3 controls four-term progressions.

11.3 Why true complexity matters for primes

A linear-forms condition for a pseudorandom majorant becomes more expensive as the required Gowers degree increases. Determining the actual complexity of a system therefore tells us which uniformity theorem for the von Mangoldt function is necessary. For prime applications the abstract algebra of coefficient tensors becomes a concrete analytic-number-theory question. That prime-specific theory is treated in the separate Analytic Number Theory project, not in later volumes of this book.

11.4 Cauchy–Schwarz versus true complexity

Cauchy–Schwarz complexity asks how a particular elimination algorithm can isolate one form. True complexity asks what norm is intrinsically necessary for all bounded inputs. The first is constructive and robust; the second can be smaller and requires more algebra. This distinction is a model of a broader phenomenon: a proof complexity need not equal the complexity of the underlying object.

11.5 The tensor criterion in coordinates

Let $L_i(\mathbf{x}) = \lambda_i \cdot \mathbf{x}$ with $\lambda_i \in \mathbb{F}_p^d$. A homogeneous polynomial phase of degree r may be written, at top order, as a symmetric r -linear form T evaluated diagonally:

$$P(v) = \frac{1}{r!} T(v, \dots, v), \quad p > r.$$

Hence

$$P(L_i(\mathbf{x})) = \frac{1}{r!} T(\lambda_i \cdot \mathbf{x}, \dots, \lambda_i \cdot \mathbf{x}).$$

The coefficient of the top multilinear part is encoded by $\lambda_i^{\otimes r}$. If a combination

$$\sum_i c_i P_i(L_i(\mathbf{x}))$$

can cancel at degree r , then the corresponding tensor coefficients satisfy a linear relation. This is the origin of the tensor-power criterion; it is not a mysterious algebraic decoration added after the fact.

Worked Example 11.2 (Three-term progressions have true complexity one). For the forms

$$L_0(x, d) = x, \quad L_1(x, d) = x + d, \quad L_2(x, d) = x + 2d,$$

the coefficient vectors are

$$(1, 0), \quad (1, 1), \quad (1, 2).$$

Their first tensor powers lie in a two-dimensional space and are therefore dependent. Their symmetric squares may be represented by

$$(1, 0, 0), \quad (1, 1, 1), \quad (1, 2, 4).$$

The determinant is

$$\det \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix} = 2,$$

which is nonzero when $p \neq 2$. Thus degree-one uniformity, namely U^2 , is the first level capable of controlling the system. This reproduces Roth's Fourier theory from the tensor viewpoint.

Worked Example 11.3 (Four-term progressions have true complexity two). For $L_j(x, d) = x + jd$, $j = 0, 1, 2, 3$, the symmetric squares lie in a three-dimensional space, so four such tensors are necessarily dependent. Hence U^2 cannot control the system. At cubic level, identify

$$(1, j)_{\text{sym}}^{\otimes 3} \longleftrightarrow (1, j, j^2, j^3).$$

The four vectors form a Vandermonde matrix with determinant

$$\prod_{0 \leq i < j \leq 3} (j - i).$$

This is nonzero when the characteristic exceeds 3. Thus the true complexity is 2, and U^3 is the correct controlling norm.

11.6 A phase obstruction explains necessity

Why does tensor dependence obstruct a lower Gowers norm? Suppose coefficients c_i , not all zero, satisfy

$$\sum_i c_i \lambda_i^{\otimes(s+1)} = 0.$$

Choose a high-rank homogeneous polynomial P of degree $s + 1$ and insert phases

$$f_i(t) = e_p(c_i P(t)).$$

Then the top-degree part of

$$\sum_i c_i P(L_i(\mathbf{x}))$$

cancels. The product of the $f_i(L_i)$ therefore has lower degree and can have a large average even though each individual f_i may have very small U^{s+1} . This constructs the basic counterexample showing that a lower uniformity norm cannot suffice.

The phrase “high rank” matters. A low-rank polynomial may accidentally correlate with lower-degree phases, which would fail to separate the norms. True-complexity proofs therefore combine tensor algebra with a rank/bias principle.

11.7 Cauchy–Schwarz complexity can be wasteful

Consider a system with many forms but with coefficient tensors becoming independent at a low degree. The class-partition definition may require many Cauchy–Schwarz rounds simply because it eliminates forms one class at a time. True complexity recognizes that a single high-rank phase cannot create a global cancellation unless a tensor relation exists. In applications this can lower the required Gowers order dramatically.

Idea Algebra replaces repeated squaring

Cauchy–Schwarz is universal because it ignores phase. True-complexity theory is sharper because it keeps enough algebraic information to determine whether the dangerous phases can actually coexist across the system.

11.8 Affine forms and constant terms

For affine forms

$$L_i(\mathbf{x}) = \lambda_i \cdot \mathbf{x} + b_i,$$

constant terms affect lower-degree pieces of $P(L_i)$ but not the top tensor $\lambda_i^{\otimes r}$. Thus the tensor criterion is naturally projective in the linear coefficients. Degeneracies such as proportional linear parts must still be excluded: two proportional forms can coincide on large sets and create correlations unrelated to higher-order uniformity.

11.9 What the theorem does not say

For the nondegenerate large-characteristic linear systems covered by the Gowers–Wolf tensor criterion described at the start of this chapter, true complexity identifies the least controlling Gowers-uniformity *order*. Even in that precise scope it does not automatically provide:

- a quantitatively efficient inverse theorem at that order;
- a sharp dependence on the density or error threshold;
- a sparse transference theorem for the same system;
- control in small characteristic where classical tensor powers can collapse.

These are separate layers of the problem. The dense higher-order aspects reappear in Volumes IV–V; sparse prime applications are outside this book and belong to the separate Analytic Number Theory project.

Chapter 12

Polynomial Configurations and PET Induction

12.1 From linear to polynomial patterns

Consider a pattern

$$x, \quad x + P_1(d), \dots, x + P_m(d)$$

with integer polynomials $P_i(0) = 0$. Linear changes of variables no longer reduce the counting problem to a fixed system of linear forms. The basic replacement is van der Corput differencing: compare the pattern at d and $d + h$ so that top-degree terms cancel.

12.2 A degree-lowering calculation

If $P(d) = ad^r + \dots$, then

$$P(d + h) - P(d) = arh d^{r-1} + \text{lower-degree terms in } d.$$

Thus one Cauchy–Schwarz/van der Corput step lowers polynomial degree by one, at the cost of introducing extra shifts and extra polynomials.

Worked Example 12.1 (Quadratic three-term pattern). For

$$x, \quad x + d^2, \quad x + 2d^2,$$

compare the expression at d and $d + h$. The shift difference

$$(d + h)^2 - d^2 = 2hd + h^2$$

is linear in d . After one differencing step the problem has become a family of linear configurations parameterized by h . This is the simplest instance of polynomial exhaustion.

12.3 PET ordering

Polynomial exhaustion technique, or PET induction, organizes a family by a weight vector recording how many essentially different leading coefficients occur at each degree. One chooses a polynomial of maximal degree, performs a van der Corput step relative to it, and verifies that the new family has smaller PET weight. Since the weight is well ordered, the process terminates at a linear system.

Idea What PET is really doing

Repeated Cauchy–Schwarz does not merely increase the number of variables. It trades polynomial degree for combinatorial complexity. The PET weight is the bookkeeping device proving that this trade eventually reaches degree one.

12.4 Why inverse theory still appears

Once polynomial differencing reduces a problem to a high-complexity linear system, a Gowers norm controls the count. The inverse theorem then describes the obstruction. Polynomial Szemerédi theory therefore combines two inductions: PET lowers the degree of the configuration, while higher-order Fourier analysis describes the structured functions that survive the resulting uniformity norm.

12.5 The van der Corput identity on a finite group

The mechanism behind PET can be written without boundary terms on a finite abelian group G . If $a : G \rightarrow \mathbb{C}$, then

$$|\mathbb{E}_n a(n)|^2 = \mathbb{E}_h \mathbb{E}_n a(n+h) \overline{a(n)}.$$

Indeed,

$$\mathbb{E}_h \mathbb{E}_n a(n+h) \overline{a(n)} = \mathbb{E}_{m,n} a(m) \overline{a(n)} = |\mathbb{E}_n a(n)|^2,$$

using the bijection $(n, h) \leftrightarrow (n, m = n + h)$.

For interval averages one cannot average all shifts equally without leaving the interval. A truncated version gives, for $1 \leq H \leq N$ and $|a_n| \leq 1$,

$$\left| \frac{1}{N} \sum_{n \leq N} a_n \right|^2 \ll \frac{1}{H} + \frac{1}{H} \sum_{h=1}^H \left| \frac{1}{N} \sum_{n \leq N-h} a_{n+h} \overline{a_n} \right|.$$

This is the form used in polynomial exhaustion and in Weyl differencing.

12.6 A complete quadratic differencing example

Let $G = \mathbb{F}_p$ with p odd and consider

$$\Lambda = \mathbb{E}_{x,d} f_0(x) f_1(x+d) f_2(x+d^2), \quad |f_i| \leq 1.$$

We show how the quadratic term becomes linear.

Apply Cauchy–Schwarz in x after regarding the average as $\mathbb{E}_x f_0(x) B(x)$:

$$|\Lambda|^2 \leq \mathbb{E}_x |B(x)|^2.$$

Expanding gives

$$\mathbb{E}_{x,d,d'} f_1(x+d) \overline{f_1(x+d')} f_2(x+d^2) \overline{f_2(x+d'^2)}.$$

Set $h = d' - d$. Then

$$d'^2 - d^2 = 2hd + h^2.$$

After translating x and relabeling variables, the arguments of the surviving f_2 factors differ by an affine-linear function of d . A further Cauchy–Schwarz step now produces an ordinary affine cube. Consequently the original average is controlled by a finite-order Gowers norm of one of the inputs. The exact order is less important here than the structural fact that one differencing step removed the quadratic degree.

Warning 12.2 (Do not say “differencing lowers every polynomial degree”). If one subtracts two polynomials with the same leading term, the degree drops. If their leading coefficients are different, it may not. PET induction therefore groups polynomials by their leading-term equivalence classes and chooses the pivot carefully. This is why a weight vector is necessary.

12.7 PET equivalence classes

For polynomials P, Q of the same positive degree, write

$$P \sim Q$$

if $P - Q$ has smaller degree. Thus P and Q have the same leading term. A polynomial family is summarized by the number of equivalence classes at each degree. Starting from the highest degree, these numbers form a lexicographically ordered weight vector.

Suppose P_* is a pivot of minimal degree among a suitable maximal-degree class. A van der Corput step replaces each P_i by combinations such as

$$P_i(n + h) - P_*(n) \quad \text{and} \quad P_i(n) - P_*(n).$$

The pivot itself produces

$$P_*(n + h) - P_*(n),$$

which has lower degree. Polynomials sharing the pivot’s leading term also drop degree. The remaining top-degree classes are unchanged. With the correct pivot choice, the weight vector decreases lexicographically.

12.8 The PET termination theorem

We now make the preceding bookkeeping precise. Constants can always be discarded from a configuration because they can be absorbed into the base variable or into the corresponding bounded function. We therefore consider only nonconstant polynomials.

For a finite family $\mathcal{P}$ and $1 \leq j \leq D$, let $w_j(\mathcal{P})$ be the number of equivalence classes of degree- j polynomials under

$$P \sim Q \iff \deg(P - Q) < j.$$

Write

$$w(\mathcal{P}) = (w_D, w_{D-1}, \dots, w_1)$$

and order these vectors lexicographically from the highest degree downward.

Fix $P_* \in \mathcal{P}$ and a shift parameter h . The associated PET transform is the family obtained from

$$P(n + h) - P_*(n), \quad P(n) - P_*(n) \quad (P \in \mathcal{P}),$$

after deleting constants and duplicate polynomials. We denote it by $\partial_{P_*,h}\mathcal{P}$.

Theorem 12.3 (PET weight descent). *Let $\mathcal{P}$ be a finite family of distinct nonconstant polynomials over a field of characteristic 0, or of characteristic larger than all degrees under consideration. Choose $P_* \in \mathcal{P}$ of minimal positive degree d . For every shift h ,*

$$w(\partial_{P_*,h}\mathcal{P}) < w(\mathcal{P})$$

in the preceding lexicographic order. Consequently, after finitely many PET transforms every polynomial family is reduced to a family of degree at most one.

Proof. Write $a(P)$ for the leading coefficient of a polynomial P of degree j . For every P of degree strictly larger than d , subtraction of the lower-degree polynomial P_* does not change its leading term. Moreover

$$P(n+h) - P_*(n) \quad \text{and} \quad P(n) - P_*(n)$$

have the same degree and the same leading coefficient. Thus each old equivalence class of degree $j > d$ produces exactly one degree- j equivalence class after duplicate classes are identified. No new degree- j class is created. Hence

$$w_j(\partial_{P_*,h}\mathcal{P}) = w_j(\mathcal{P}), \quad j > d.$$

Now consider degree d . The pivot contributes

$$P_*(n+h) - P_*(n),$$

whose degree is at most $d-1$, while $P_*(n) - P_*(n)$ is zero and is discarded. If P lies in the same degree- d equivalence class as P_* , then $P - P_*$ already has degree below d . Translation does not change the leading coefficient, so the degree- d terms also cancel in $P(n+h) - P_*(n)$ for every h ; any further cancellation only lowers the degree still more.

If P belongs to a different degree- d class, then its leading coefficient differs from that of P_* . Both

$$P(n+h) - P_*(n), \quad P(n) - P_*(n)$$

have degree d and leading coefficient $a(P) - a(P_*)$, so they lie in one common new degree- d class. Distinct old classes remain distinct after subtracting the same leading coefficient $a(P_*)$. Therefore every degree- d class other than the pivot class survives exactly once, while the pivot class disappears. Thus

$$w_d(\partial_{P_*,h}\mathcal{P}) = w_d(\mathcal{P}) - 1.$$

The higher entries of the weight vector are unchanged, so the new weight is strictly smaller lexicographically.

The set of vectors in $\mathbb{N}^D$ is well ordered by this finite lexicographic order: one cannot have an infinite strictly descending sequence because the first coordinate can decrease only finitely often; after it stabilizes the same is true of the second, and so on. Repeated PET transforms must therefore terminate. The only terminal families have no polynomial of degree at least two, hence are linear. This proves both claims. $\square$

Corollary 12.4 (Qualitative polynomial exhaustion). *For every fixed finite polynomial system there is a finite number of van der Corput/Cauchy–Schwarz steps, depending only on the system, after which its counting average is reduced to finitely many averages attached to linear systems. Over $G = \mathbb{F}_p$ with p tending to infinity, the finitely many shift parameters for which distinct transformed polynomials collide or a nonconstant lower-degree term degenerates contribute $o(1)$.*

Proof. At each van der Corput step use the identity from the preceding section and choose the minimal-degree pivot supplied by Theorem 12.3. Cauchy–Schwarz removes the pivot function and replaces the remaining polynomial arguments by the PET transform. The theorem says that only finitely many transformations occur before all arguments are linear. At each fixed step, failure of the expected degree pattern is described by the vanishing of one of finitely many nonzero polynomial expressions in the shift parameters. Over $G = \mathbb{F}_p$ with characteristic exceeding the relevant degrees, each such one-variable polynomial has only $O(1)$ roots, so the union of all exceptional choices has density $O(1/p)$ at that stage. Since the number of stages is fixed, their total contribution is $o(1)$ for bounded functions. $\square$

12.9 Termination is finite but not cheap

The preceding theorem proves finite termination, but it says nothing pleasant about the size of the resulting system. The number of forms can roughly double at each van der Corput step, and the required Gowers order can grow rapidly. PET is therefore a qualitative termination machine before it is a quantitative one.

Worked Example 12.5 (Cubes created by two differencing steps). For a cubic phase $P(n) = \alpha n^3$, the first difference is

$$\Delta_h P(n) = 3\alpha h n^2 + 3\alpha h^2 n + \alpha h^3.$$

A second difference gives

$$\Delta_k \Delta_h P(n) = 6\alpha h k n + 3\alpha h k (h + k),$$

and a third gives the constant

$$\Delta_\ell \Delta_k \Delta_h P(n) = 6\alpha h k \ell.$$

The successive differencing parameters (h, k, ℓ) are precisely the edge directions of the Gowers cube. PET induction and Gowers norms are therefore two views of the same repeated-difference geometry.

12.10 Polynomial systems with collisions

Exceptional parameter values can cause two derived polynomials to coincide or become constant. For example, in characteristic p , the derivative of n^p is constant because of Frobenius. One must either assume characteristic larger than the degrees involved or use a low-characteristic polynomial formalism. This is the polynomial analogue of the non-classical phases that appear in Volume IV.

12.11 PET and the inverse theorem play different roles

PET answers:

Which uniformity norm is sufficient to control this polynomial configuration?

The inverse theorem answers:

What does it mean for a function to fail that uniformity test?

Keeping these questions distinct helps organize polynomial Szemerédi arguments. Differencing analyzes the *configuration*; inverse theory analyzes the *function*.

Chapter 13

Quantitative Szemerédi Theory: Where Bounds Are Lost

13.1 A proof as a parameter machine

A qualitative proof can hide enormous losses. For quantitative work, every stage should be recorded as a map between parameters:

$$(\alpha, N) \mapsto (\eta, M) \mapsto (\text{structure complexity}) \mapsto (\alpha', N').$$

The worst map usually determines the final extremal bound.

13.2 Three bottlenecks

For longer progressions, three costs dominate:

1. the generalized von Neumann step determines the required Gowers order;
2. the inverse theorem determines the complexity of the structured obstruction;
3. the conversion of correlation to density increment determines how much ambient scale survives.

A better exponential-sum estimate cannot repair a tower loss introduced by an inverse theorem, and a sharp inverse theorem can still be squandered by inefficient localization.

13.3 Behrend's lower bound

Any upper bound for $r_3(N)$ must confront Behrend's construction. Take the integer box

$$\{0, 1, \dots, m-1\}^d$$

and choose a sphere layer

$$S_R = \{x : \|x\|_2^2 = R\}.$$

Strict convexity implies that a nontrivial three-term progression cannot lie on one Euclidean sphere. Since the possible squared radii number only $O(dm^2)$, some sphere contains at least $m^d/O(dm^2)$ points.

Worked Example 13.1 (Carry-free encoding). Define

$$\phi(x_0, \dots, x_{d-1}) = \sum_{j=0}^{d-1} x_j B^j$$

with $B > 2m$. If

$$\phi(x) + \phi(z) = 2\phi(y),$$

no base- B carry occurs, so equality holds coordinatewise:

$$x_j + z_j = 2y_j.$$

Thus a progression-free sphere layer gives a progression-free subset of the integers. Optimizing d and m yields the classical scale

$$N \exp(-C\sqrt{\log N}).$$

13.4 A methodological comparison

Density-increment proofs exploit universal additive structure and adapt well to sparse settings. Behrend's construction exploits Euclidean geometry. Polynomial/slice-rank arguments exploit finite-field algebra. Quantitative additive combinatorics advances fastest when one distinguishes what is universal from what is special to the ambient model.

13.5 Optimizing the Behrend parameters

We now perform the calculation suppressed in the first presentation. Let the digit set be $\{0, \dots, m-1\}^d$ and take a base $B \geq 2m$. The encoded integers lie below

$$B^d.$$

A sphere layer has size at least

$$\frac{m^d}{dm^2} = \frac{m^{d-2}}{d},$$

because $\|x\|_2^2$ takes at most $d(m-1)^2 + 1 \leq dm^2$ values.

Set $N = B^d$ and, for simplicity, $B = 2m$. The logarithmic density loss is

$$\log \frac{N}{|A|} \leq d \log 2 + 2 \log m + \log d + O(1).$$

Write $L = \log N = d \log(2m)$ and $t = \log m$. Then

$$d = \frac{L}{t + \log 2},$$

so the two dominant losses are

$$\frac{L \log 2}{t + \log 2} + 2t.$$

Balancing them gives $t \asymp \sqrt{L}$ and $d \asymp \sqrt{L}$. Consequently

$$|A| \geq N \exp(-C\sqrt{\log N})$$

for an absolute constant C .

The calculation is as important as the construction: it shows exactly where the square root in the exponent comes from. Dimension costs 2^{-d} , while reserving room for a carry-free digit alphabet costs roughly m^{-2} . The optimal choice balances these two losses.

13.6 Why the sphere is progression-free

Suppose $x, z \in \mathbb{R}^d$ are distinct and $y = (x + z)/2$. The parallelogram identity gives

$$\|x\|_2^2 + \|z\|_2^2 = 2\|y\|_2^2 + \frac{1}{2}\|x - z\|_2^2.$$

If x, y, z all lie on the same sphere, the left side is $2R$ and the first term on the right is also $2R$, forcing $x = z$. Thus a Euclidean sphere contains no nontrivial three-term progression. Strict convexity is the geometric core of Behrend's argument.

13.7 Upper and lower bounds measure different difficulties

An upper-bound proof must show that *every* sufficiently large set has structure. A lower-bound construction needs only one carefully designed family that avoids the configuration. There is no reason the mechanisms should resemble each other. For Roth's problem the upper theory is Fourier/additive, while the lower construction is geometric.

Research window The quantitative gap

Even very strong modern upper bounds remain far from the Behrend-type lower-bound scale. This gap warns against interpreting an improved exponent in an upper bound as evidence that the extremal problem is close to solved. Quantitative Roth theory is still constrained by a major structural gap between known constructions and known universal arguments.

13.8 Tracking parameters through an iteration

For any iterative additive-combinatorial proof, record a table with four columns:

stage	density	ambient scale	structural complexity
initial	α	N	0
inverse	α	N	$M(\alpha)$
localize	$\alpha + \rho(\alpha)$	$N^{c(\alpha)}$	$M(\alpha)$
iterate	$\dots$	$\dots$	$\dots$

The final extremal estimate is obtained by asking when the scale becomes too small before the density exceeds one. Writing this table often reveals immediately which lemma is quantitatively dominant.

Chapter 14

The Polynomial-Method Contrast: Cap Sets and Slice Rank

14.1 Cap sets

In $\mathbb{F}_3^n$, the equation

$$x + z = 2y$$

is equivalent to

$$x + y + z = 0,$$

because $2 = -1$. If two variables coincide, then all three do. Thus nontrivial three-term progressions are exactly affine lines of three distinct points.

Fourier density increments prove that cap sets have density tending to zero, but the polynomial method gives exponentially stronger decay.

14.2 The diagonal-tensor idea

Suppose $A \subseteq \mathbb{F}_3^n$ contains no three distinct x, y, z with $x + y + z = 0$. A polynomial indicator of the equation can be arranged so that when restricted to A^3 , the resulting tensor is supported essentially on the diagonal $x = y = z$. A diagonal tensor has slice rank $|A|$. On the other hand, expanding the polynomial shows that its slice rank is bounded by the number of monomials below a certain degree threshold.

The remarkable point is that this outline can be completed with elementary polynomial algebra and one tensor-rank lemma. We do so below, obtaining the classical Ellenberg–Gijswijt exponential bound [3, 2]. We do not pursue the polynomial method as a separate research programme later in this book, and no unproved cap-set estimate is used here.

14.3 Why the method is so different

Gowers norms ask whether a function correlates with a low-complexity structured phase. Slice rank asks whether an algebraic tensor representing the forbidden equation can have high rank on the set. The latter uses special polynomial identities in finite characteristic and is not a general replacement for higher-order Fourier analysis.

Research window A useful warning

A universal theory can be structurally correct and quantitatively far from optimal in a special ambient category. Cap sets are the cleanest example. One should therefore learn higher-order Fourier analysis as a robust language, not as a dogma that every additive problem must be solved through Gowers norms.

14.4 Slice rank

For a tensor $T : X \times Y \times Z \rightarrow \mathbb{F}$, an x -slice has the form

$$a(x)b(y, z),$$

and similarly for y - and z -slices. The *slice rank* is the smallest number of slices of all three types whose sum is T .

For matrices, a rank-one term separates the two variables. Slice rank is the three-variable analogue that allows one variable to separate from the other two.

Lemma 14.1 (Diagonal slice-rank lemma). *Let $D : X^3 \rightarrow \mathbb{F}$ vanish off the diagonal and suppose exactly m diagonal entries $D(x, x, x)$ are nonzero. Then the slice rank of D is m .*

Proof. The upper bound is immediate by writing one x -slice for each nonzero diagonal point. For the lower bound, suppose

$$D = \sum_{i=1}^r a_i(x)b_i(y, z) + \sum_{j=1}^s c_j(y)d_j(x, z) + \sum_{k=1}^t e_k(z)g_k(x, y).$$

Let S be the m points supporting the diagonal. Consider functions $h : S \rightarrow \mathbb{F}$ satisfying

$$\sum_{x \in S} h(x)a_i(x) = 0 \quad (1 \leq i \leq r).$$

This solution space has dimension at least $m - r$. A d -dimensional subspace of $\mathbb{F}^S$ contains a vector with at least d nonzero coordinates. Indeed, choose a basis in reduced row-echelon form and add all d basis vectors: at each of the d pivot coordinates the sum is exactly 1. Thus choose h with at least $m - r$ nonzero values.

Contract the tensor in the x variable:

$$M(y, z) = \sum_{x \in S} h(x)D(x, y, z).$$

Since D is diagonal, M is a diagonal matrix with at least $m - r$ nonzero diagonal entries, hence

$$\text{rank } M \geq m - r.$$

The x -slice terms vanish by the choice of h . Every y -slice contracts to a rank-one matrix in (y, z) , as does every z -slice. Therefore

$$\text{rank } M \leq s + t.$$

Hence $m - r \leq s + t$, or $m \leq r + s + t$. Since the decomposition was arbitrary, the slice rank is at least m . $\square$

14.5 The cap-set tensor

For $x, y, z \in \mathbb{F}_3^n$, define

$$T(x, y, z) = \prod_{j=1}^n \left(1 - (x_j + y_j + z_j)^2\right).$$

In $\mathbb{F}_3$, the factor $1 - u^2$ equals 1 if $u = 0$ and 0 otherwise. Hence

$$T(x, y, z) = 1_{\{x+y+z=0\}}.$$

Expand the product. Every monomial has degree at most 2 in each coordinate and total degree at most $2n$. Split its variables into three blocks x, y, z . At least one block has degree at most $2n/3$. Group every monomial according to one such low-degree block. This gives a slice decomposition with at most

$$3m_{2n/3}$$

slices, where

$$m_d = \# \left\{ a \in \{0, 1, 2\}^n : \sum_j a_j \leq d \right\}.$$

Therefore

$$\text{srnk}(T) \leq 3m_{2n/3}.$$

Now restrict T to A^3 , where A is a cap set. If $x + y + z = 0$ and two of x, y, z are equal, then all three are equal. Since A contains no three distinct collinear points, the restricted tensor is exactly diagonal:

$$T|_{A^3}(x, y, z) = 1_{\{x=y=z\}}.$$

By Lemma 14.1, its slice rank is $|A|$. Restriction cannot increase slice rank, so

$$|A| \leq 3m_{2n/3}.$$

This is the essential Ellenberg–Gijswijt bound.

14.6 Estimating the monomial count

For any $0 < t < 1$,

$$m_d t^d \leq \sum_{a \in \{0,1,2\}^n} t^{\sum a_j} = (1 + t + t^2)^n.$$

Hence

$$m_d \leq t^{-d} (1 + t + t^2)^n.$$

With $d = 2n/3$,

$$|A| \leq 3 \left[t^{-2/3} (1 + t + t^2) \right]^n.$$

Minimizing the bracket over $0 < t < 1$ gives a constant strictly smaller than 2.756. Thus

$$|A| \ll 2.756^n,$$

exponentially smaller than 3^n .

14.7 What exactly made the argument possible?

Three special facts align:

1. the equation $x + y + z = 0$ has a low-degree polynomial delta function over $\mathbb{F}_3$;
2. the forbidden-configuration hypothesis makes that tensor diagonal on A^3 ;
3. the degree bound forces one variable block of every monomial to have small degree.

None of these has a direct analogue for arbitrary subsets of $[N]$. The polynomial method is therefore spectacularly strong in the finite-field model without superseding the general higher-order Fourier framework.

Chapter 15

Local Fourier Analysis on Bohr Sets

15.1 Local convolution

Let B be regular and use the probability density

$$\beta_B = \frac{|G|}{|B|} 1_B.$$

For f supported on a controlled dilation of B , convolution with $\beta_{B'}$ for a much smaller B' behaves like averaging over a local translation group.

The approximate invariance estimate

$$\|\tau_t \beta_B - \beta_B\|_1 \ll \varepsilon \quad (t \in B')$$

allows global Fourier identities to be reproduced with $O(\varepsilon)$ boundary errors.

15.2 Local characters

A global character may be nearly constant on B' , but in higher-order arguments one also meets phases defined only locally. A local linear phase is a function whose additive derivative is constant whenever the relevant parallelogram stays inside the ambient Bohr set. The need for such local objects is already visible in bracket polynomials and becomes unavoidable in the U^3 inverse theorem on cyclic groups.

Worked Example 15.1 (Why boundary control is part of the theorem). Suppose $B = [-M, M] \subset \mathbb{Z}$ in an informal interval model and $B' = [-m, m]$ with $m \ll M$. For x far from the boundary, averaging $f(x+t)$ over $t \in B'$ is translation invariant. Near $\pm M$ part of the translate leaves B . A regular Bohr set is the group-theoretic analogue of an interval whose boundary layer under small shifts is negligible. Ignoring this layer invalidates local Parseval and local density computations.

15.3 Nested scales

A robust local proof rarely uses one Bohr set. It uses a chain

$$B \supset B_1 \supset B_2 \supset \cdots$$

with radii separated enough that each smaller set acts almost invariantly on the previous measure. Frequencies are added only when necessary; radii are shrunk only enough to preserve regularity. This multiscale bookkeeping is one reason modern Roth proofs are technically longer than their high-level diagrams suggest.

15.4 Regularity as a boundary theorem

For a Bohr set

$$B(\Gamma, \rho) = \{x : |1 - \gamma(x)| \leq \rho \text{ for all } \gamma \in \Gamma\},$$

regularity says that a small multiplicative change in radius produces only a proportionally small change in cardinality:

$$|B(\Gamma, (1 + \eta)\rho)| = (1 + O(|\Gamma||\eta|))|B(\Gamma, \rho)|$$

for $|\eta| \ll |\Gamma|^{-1}$. The point is not a smoothness property of the indicator function; it is a quantitative statement that the boundary shell is small.

If $t \in B(\Gamma, \eta\rho)$, then triangle inequalities for characters give

$$B(\Gamma, (1 - \eta)\rho) \subseteq B(\Gamma, \rho) \cap (B(\Gamma, \rho) - t)$$

and

$$B(\Gamma, \rho) \cup (B(\Gamma, \rho) - t) \subseteq B(\Gamma, (1 + \eta)\rho).$$

For a regular radius this yields

$$|B \Delta (B - t)| \ll |\Gamma|\eta|B|,$$

which is exactly the L^1 approximate-translation-invariance estimate for β_B .

15.5 A local Parseval calculation

Let f be supported on B and let B' act almost invariantly on B . Define

$$F(x) = f * \beta_{B'}(x).$$

Then

$$\mathbb{E}_{x \in B} |F(x)|^2$$

can be expanded as an average of local correlations

$$\mathbb{E}_{t, t' \in B'} \mathbb{E}_{x \in B} f(x - t) \overline{f(x - t')}.$$

Changing variables would be exact on a subgroup. On a regular Bohr set, the symmetric-difference estimate shows that the error is bounded by the relative boundary size. This is the model calculation behind essentially every “local Fourier identity” in the subject.

Warning 15.2 (Normalization matters). The density

$$\beta_B = \frac{|G|}{|B|} 1_B$$

has global average 1. The probability measure on B is instead $|B|^{-1} 1_B$ when sums are unnormalized. Mixing these two conventions changes factors of $|G|$ and can silently corrupt local Parseval identities. This volume consistently uses normalized ambient expectations.

15.6 Frequency growth versus radius loss

Adding one character to Γ increases rank by one but may force a radius reduction. If this is done r times naively, the radius may shrink exponentially in r . Modern arguments therefore try to add *structured families* of frequencies at once, or to use almost-periodicity without adjoining every visible spectral frequency.

This explains a recurring quantitative tradeoff:

$$\text{more Fourier information} \quad \Longleftrightarrow \quad \text{less physical-space scale.}$$

Good proofs spend this resource only when the expected density or energy gain compensates for the lost radius.

Chapter 16

Three Proof Architectures for Szemerédi-Type Theorems

16.1 Density increment

The density-increment architecture is

$$\text{deficit} \rightarrow \text{large norm} \rightarrow \text{structured region} \rightarrow \text{higher density} \rightarrow \text{iterate}.$$

It is local and often yields explicit extremal bounds. Its weakness is cumulative scale loss.

16.2 Regularity and counting

The regularity architecture is

$$f = f_{\text{str}} + f_{\text{sml}} + f_{\text{unf}}$$

followed by a counting lemma for f_{str} . It separates the structural theorem from the application and is ideal for counting many configurations at once. Its weakness is potentially enormous structured complexity.

16.3 Transference

The transference architecture first constructs a pseudorandom ambient majorant, models the sparse function densely, and then applies a dense theorem. It is indispensable for primes and other sparse sequences. Its weakness is that the pseudorandomness conditions must exactly match the Cauchy–Schwarz tests generated by the target configuration.

Worked Example 16.1 (One theorem, three uses of structure). For three-term progressions, a large Fourier coefficient can be used to restrict to a progression and increase density; it can be added to a Fourier factor and increase energy; or it can be included in a test family against which a sparse majorant must look random. The same character appears in three proof architectures, but its operational role is different.

16.4 Choosing a proof

A research problem should not be attacked by asking only “which theorem is strongest?” Ask instead:

- Is the target dense or sparse?

- Is one configuration needed or an asymptotic count for many systems?
- Are quantitative bounds central?
- Does localization preserve the ambient category?
- Can the structured objects be counted explicitly?

The answer determines which architecture is appropriate.

Chapter 17

Research Outlook: From Uniformity to Higher-Order Structure

17.1 What Volume III has established

The forward theory is now in place:

1. multilinear configuration counts are controlled by Gowers norms through Cauchy–Schwarz;
2. Roth’s theorem illustrates how non-uniformity produces density increments;
3. Bohr sets and almost-periodicity make localization more efficient;
4. regularity and dense models turn structural information into counting theorems;
5. complexity theory identifies which uniformity degree a general linear system requires.

The unresolved higher-order inverse question is completed in Volume V; Volume IV first develops the Gowers/Szemer’edi route without depending forward on that inverse theory.

17.2 Quantitative frontiers

For three-term progressions, modern work has moved far beyond one-frequency Fourier increments. Kelley–Meka proved a strong stretched-exponential-type bound [5], and the 2026 Raghavan preprint improves the exponent further [8]. Full proofs of the mature Bloom–Sisask/Kelley–Meka line are proved in the Volume III supplement; the newer Raghavan statement remains a clearly labelled research-frontier item and is not used in the dependency chain of Volumes I–V. For longer progressions, improvements in quantitative inverse theory and nilsequence equidistribution feed directly into improved Szemerédi bounds [7, 6].

17.3 The conceptual transition

For U^2 , the inverse object is a character and the dual group is again abelian. For U^3 , derivatives have character structure, but those characters depend on the increment h . Coherence forces the frequency map to behave linearly, symmetry forces a bilinear form, and integration produces a quadratic phase. For higher U^d , this derivative-integration pattern continues but the integrated object is no longer conveniently described by an ordinary polynomial on the integers.

Volume V develops the nilpotent language that makes this statement precise; Volume IV supplies the intervening Gowers-uniformity and Szemer’edi theory.

Part I

Modern Roth Theory and Polynomial Freiman–Ruzsa

Chapter 18

Bridge Lemmas for Quantitative Additive Combinatorics

18.1 Purpose and dependencies

The modern quantitative results in this volume are connected by several short but logically essential bridge lemmas. We collect those bridges here so that the deep inputs are not joined by unexplained phrases such as “standard localization” or “the usual entropy argument.” The local Chang–Sanders chain and the characteristic-two Gowers–Green–Manners–Tao stability/decrement chain are developed below; Bloom–Sisask, Kelley–Meka, and the bounded-torsion multidistance argument have their own full proof chapters later in the volume. This organization makes the dependency architecture readable before the parameter-heavy arguments begin.

Throughout, G is a finite abelian group, $\widehat{G}$ its character group, and

$$\widehat{f}(\gamma) = \mathbb{E}_{x \in G} f(x) \overline{\gamma(x)}.$$

For a nonempty set $B \subseteq G$ of density β , write $\mu_B = 1_B/\beta$.

18.2 Density-increment iteration

Lemma 18.1 (A general reciprocal-potential iteration bound). *Let $0 < \alpha_0 \leq 1$, let $r > 1$, and suppose a finite sequence satisfies*

$$\alpha_{j+1} \geq \alpha_j + c\alpha_j^r$$

for an absolute $c > 0$. Then the number t of strict increment steps is at most

$$t \leq C_{c,r} \alpha_0^{1-r}.$$

Proof. Put $\Phi(x) = x^{1-r}$. Since $\Phi'(x) = -(r-1)x^{-r}$, the mean-value theorem gives, for $0 < a \leq 1$ and $b \geq a + ca^r$,

$$\Phi(a) - \Phi(b) = (r-1)\xi^{-r}(b-a) \geq (r-1)b^{-r}ca^r.$$

Because $b \leq a + ca^r \leq (1+c)a$, the last quantity is at least the positive constant

$$c(r-1)(1+c)^{-r}.$$

Thus $\Phi(\alpha_j)$ decreases by a fixed positive amount at every step. Since it is nonnegative, there are at most $C_{c,r}\Phi(\alpha_0) = C_{c,r}\alpha_0^{1-r}$ steps. $\square$

Corollary 18.2 (Closing the classical Roth iteration). *Suppose a local Roth step on an interval of length N_j gives*

$$\alpha_{j+1} \geq \alpha_j + c\alpha_j^3, \quad N_{j+1} \geq c\alpha_j^3 N_j^{1/2}.$$

Then there are $O(\alpha_0^{-2})$ steps, and if $\log \log N_0 \gg \alpha_0^{-2}$ all the local steps remain available until the density iteration terminates. Consequently the local density-increment proof given earlier yields a quantitative Roth bound of the form

$$\alpha_0 \ll (\log \log N_0)^{-1/2}.$$

Proof. The first assertion is Lemma 18.1 with $r = 3$. Since $\alpha_j \geq \alpha_0$,

$$\log N_{j+1} \geq \frac{1}{2} \log N_j - C \log(1/\alpha_0).$$

Iterating for $t = O(\alpha_0^{-2})$ steps gives

$$\log N_t \geq 2^{-t} \log N_0 - C \log(1/\alpha_0).$$

If $\log \log N_0$ is a sufficiently large multiple of α_0^{-2} , the right-hand side remains above the fixed threshold required by the local Fourier lemma at every stage. The process therefore cannot stop through loss of scale before the density potential forces termination. Rearranging gives the stated bound. $\square$

18.3 Regular Bohr sets as local approximate groups

Let

$$B = B(\Gamma, \rho) = \{x : \|\gamma(x)\|_{\mathbb{T}} \leq \rho \ \forall \gamma \in \Gamma\}$$

with $d = |\Gamma|$. For $\lambda > 0$ write $B_\lambda = B(\Gamma, \lambda\rho)$ whenever the radius remains in the non-vacuous range.

Lemma 18.3 (Regular-radius stability). *Assume B is regular in the logarithmic sense*

$$\left| \log \frac{|B_{1+\eta}|}{|B|} \right| \leq 100d|\eta| \quad (|\eta| \leq (1000d)^{-1}).$$

Then, for $0 \leq \eta \leq (1000d)^{-1}$,

$$|B_{1+\eta} \triangle B| \leq Cd\eta |B| \quad \text{and} \quad |B \triangle B_{1-\eta}| \leq Cd\eta |B|.$$

In particular, for $y \in B_\eta$,

$$|(B + y) \triangle B| \leq Cd\eta |B|.$$

Proof. For $\eta \geq 0$, nesting gives $B \subseteq B_{1+\eta}$. The regularity estimate and $e^u - 1 \ll u$ for $0 \leq u \leq 1/10$ imply

$$|B_{1+\eta} \setminus B| = |B_{1+\eta}| - |B| \leq Cd\eta |B|.$$

The shrinking estimate is identical. If $y \in B_\eta$, the Bohr triangle inequality gives

$$B_{1-\eta} + y \subseteq B.$$

Since $B_{1-\eta} \subseteq B$, the same translated set also lies in $B + y$; hence

$$|B \cap (B + y)| \geq |B_{1-\eta}|.$$

Because $|B + y| = |B|$,

$$|(B + y) \triangle B| = 2(|B| - |B \cap (B + y)|) \leq 2(|B| - |B_{1-\eta}|) \ll d\eta |B|,$$

by the shrinking estimate from the first part. $\square$

Lemma 18.4 (Bohr refinement covering and volume). *Let*

$$B = B(\Gamma, \rho), \quad d = |\Gamma|,$$

and let $0 < \lambda, \sigma \leq 1$. If Λ is a set of r characters, then

$$B_\lambda \cap B(\Lambda, \sigma)$$

has size at least

$$\left(\frac{\lambda}{4}\right)^d \left(\frac{\sigma}{4}\right)^r |B|. \quad (\text{BV})$$

In particular

$$|B_\lambda| \geq (\lambda/4)^d |B|.$$

Proof. For $x \in B$, record the phase vector

$$\Phi(x) = ((\|\gamma(x)\|_{\mathbb{T}})_{\gamma \in \Gamma}, (\|\xi(x)\|_{\mathbb{T}})_{\xi \in \Lambda}),$$

using signed representatives in $[-1/2, 1/2)$. Partition each of the first d coordinates of the box $[-\rho, \rho]$ into intervals of length at most $\lambda\rho$, and partition each of the last r coordinates of $[-1/2, 1/2)$ into intervals of length at most σ . At most

$$(4/\lambda)^d (4/\sigma)^r$$

boxes are needed. In every occupied box choose one representative $x_j \in B$. If x and x_j lie in the same box, then for every $\gamma \in \Gamma$

$$\|\gamma(x - x_j)\|_{\mathbb{T}} \leq \lambda\rho,$$

and for every $\xi \in \Lambda$

$$\|\xi(x - x_j)\|_{\mathbb{T}} \leq \sigma.$$

Thus

$$x - x_j \in B_\lambda \cap B(\Lambda, \sigma).$$

Consequently B is covered by at most $(4/\lambda)^d (4/\sigma)^r$ translates of the refinement. Comparing cardinalities proves (BV). Taking $\Lambda = \emptyset$ gives the last assertion. $\square$

Lemma 18.5 (Approximate annihilation). *If $x \in B(\Gamma, \rho)$ and $\gamma \in \Gamma$, then*

$$|1 - \gamma(x)| \leq 2\pi\rho.$$

If $x \in B(\Gamma, \eta\rho)$, the right-hand side is $2\pi\eta\rho$.

Proof. Write $\gamma(x) = e^{2\pi i\theta}$ with $\|\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \rho$. Then

$$|e^{2\pi i\theta} - 1| \leq 2\pi|\theta| \leq 2\pi\rho.$$

$\square$

Proposition 18.6 (Local translation/convolution stability). *Let B be a regular Bohr set of rank d and let $y \in B_\eta$ with $0 < \eta \leq (1000d)^{-1}$. Then*

$$\|\tau_y \mu_B - \mu_B\|_1 \leq Cd\eta.$$

Consequently, for every $1 \leq p \leq \infty$ and every $f : G \rightarrow \mathbb{C}$,

$$\|\tau_y(f * \mu_B) - f * \mu_B\|_p \leq Cd\eta \|f\|_p.$$

Proof. Because $\mu_B = 1_B/\beta$ with $\beta = |B|/|G|$,

$$\|\tau_y \mu_B - \mu_B\|_1 = \frac{|(B+y) \triangle B|}{|B|}.$$

Lemma 18.3 gives the first assertion. For the second, use

$$\tau_y(f * \mu_B) - f * \mu_B = f * (\tau_y \mu_B - \mu_B)$$

and Young's inequality $\|f * g\|_p \leq \|f\|_p \|g\|_1$ in normalized counting measure. $\square$

Proposition 18.7 (Local Parseval and Bessel in a Bohr window). *Let $B \subseteq G$ have density $\beta > 0$ and define the localized transform*

$$\widehat{f}_B(\gamma) = \mathbb{E}_x f(x) \mu_B(x) \overline{\gamma(x)}.$$

Then

$$\sum_{\gamma \in \widehat{G}} |\widehat{f}_B(\gamma)|^2 = \beta^{-1} \mathbb{E}_{x \in B} |f(x)|^2.$$

Hence, if

$$\text{Spec}_{B,\lambda}(f) = \left\{ \gamma : |\widehat{f}_B(\gamma)| \geq \lambda \right\},$$

then

$$|\text{Spec}_{B,\lambda}(f)| \lambda^2 \leq \beta^{-1} \mathbb{E}_{x \in B} |f(x)|^2.$$

Proof. Apply global Parseval to the function $f \mu_B$. Since $\mu_B = \beta^{-1} 1_B$,

$$\mathbb{E} |f \mu_B|^2 = \beta^{-2} \mathbb{E} 1_B |f|^2 = \beta^{-1} \mathbb{E}_{x \in B} |f(x)|^2.$$

The second assertion follows by retaining only the large-spectrum terms in the sum. $\square$

Lemma 18.8 (Frequency adjoining costs one rank). *For a finite frequency set Γ and a character γ , the Bohr set*

$$B(\Gamma \cup \{\gamma\}, \rho')$$

has rank at most $|\Gamma| + 1$ and is contained in $B(\Gamma, \rho')$. After adjoining m frequencies, rank grows by at most m .

Proof. This is immediate from the defining system of inequalities. Each new character contributes at most one new scalar constraint, and adding constraints only shrinks the set. $\square$

18.4 Bohr sets as Bourgain systems

Definition 18.9 (Bourgain system in the form used here). A one-parameter family $\mathcal{B} = (B_\lambda)_{0 < \lambda \leq 2}$ is a Bourgain system of dimension at most D if $0 \in B_\lambda$, $B_\lambda = -B_\lambda$, the family is nested, $B_\lambda + B_{\lambda'} \subseteq B_{\lambda+\lambda'}$ when $\lambda + \lambda' \leq 2$, and

$$|B_{2\lambda}| \leq 2^D |B_\lambda|$$

whenever $2\lambda \leq 2$.

Proposition 18.10 (Bohr-to-Bourgain construction). *Fix $B(\Gamma, \rho)$ with $d = |\Gamma|$ and $\rho \leq 1/4$, and set*

$$B_\lambda = B(\Gamma, \lambda \rho) \quad (0 < \lambda \leq 2).$$

Then (B_λ) is a Bourgain system of dimension at most $2d$.

Proof. The origin, symmetry, nesting and addition axioms follow directly from the Bohr pseudometric triangle inequality. The Bohr doubling lemma proved earlier gives

$$|B_{2\lambda}| \leq 4^d |B_\lambda| = 2^{2d} |B_\lambda|,$$

which is the dimension bound. $\square$

Proposition 18.11 (Regularization of a Bourgain system arising from a Bohr set). *Let $\Gamma \subseteq \widehat{G}$ have $|\Gamma| = d$, let $0 < \rho \leq 1/4$, and set*

$$B_\lambda := B(\Gamma, \lambda\rho) \quad (0 < \lambda \leq 2).$$

Then there are absolute constants $c, C > 0$ and a number $\lambda_0 \in [1/2, 1]$ such that, for

$$B'_\lambda := B_{\lambda_0\lambda},$$

one has

$$\left| \frac{|B'_{1+\eta}|}{|B'_1|} - 1 \right| \leq Cd|\eta| \quad (|\eta| \leq c/d).$$

Proof. Apply the regular-radius existence theorem proved earlier to the radius ρ . It gives $\rho' = \lambda_0\rho$ with $\lambda_0 \in [1/2, 1]$ and the logarithmic regularity estimate. Exponentiating that estimate in the range $|\eta| \leq c/d$ gives the stated multiplicative form. $\square$

18.5 Dissociation and spectral bookkeeping

Lemma 18.12 (Subset-sum uniqueness for a dissociated set). *Let $\Lambda = \{\lambda_1, \dots, \lambda_m\}$ be dissociated in an abelian group, meaning that*

$$\sum_{j=1}^m \varepsilon_j \lambda_j = 0, \quad \varepsilon_j \in \{-1, 0, 1\},$$

forces all $\varepsilon_j = 0$. Then the 2^m subset sums

$$\sum_{j \in J} \lambda_j \quad (J \subseteq \{1, \dots, m\})$$

are distinct. In particular, if Λ lies in a finite group H , then

$$m \leq \log_2 |H|.$$

Proof. If two subset sums agree, subtract them. The resulting relation has coefficients in $\{-1, 0, 1\}$, so dissociation forces the two subsets to coincide. There are therefore 2^m distinct group elements, giving $2^m \leq |H|$. $\square$

Proposition 18.13 (Large-spectrum energy consequence). *There is an absolute constant $C > 0$ with the following property. Let G be a finite abelian group, let $A \subseteq G$ have density $\alpha \in (0, 1]$, and let $0 < \tau \leq 1$. Every dissociated set*

$$\Lambda \subseteq \text{Spec}_\tau(A) := \{\gamma \in \widehat{G} : |\widehat{1_A}(\gamma)| \geq \tau\alpha\}$$

satisfies

$$|\Lambda| \leq C\tau^{-2} \log(2/\alpha).$$

In particular, if Λ_0 is maximal dissociated in $\text{Spec}_\tau(A)$, then

$$|\Lambda_0| \leq C\tau^{-2} \log(2/\alpha), \quad \text{Spec}_\tau(A) \subseteq \text{span}_{\{-1, 0, 1\}}(\Lambda_0).$$

Proof. The quantitative dissociated-spectrum estimate is the Chang–Rudin bound proved earlier: one applies Rudin’s L^p inequality to a phase choice on Λ , takes $p \asymp \log(2/\alpha)$, pairs the resulting trigonometric polynomial with 1_A , and optimizes. This gives the displayed cardinality estimate with an absolute constant. If a character $\gamma \in \text{Spec}_\tau(A)$ were outside the signed span of a maximal dissociated set Λ_0 , then $\Lambda_0 \cup \{\gamma\}$ would still be dissociated, contradicting maximality. Hence the signed-span inclusion follows. $\square$

18.6 Almost-periodicity for indicator convolutions

Corollary 18.14 (Indicator-convolution form of Croot–Sisask). *Assume the hypotheses of the Croot–Sisask theorem proved earlier:*

$$|A + S| \leq K|A|, \quad p \geq 2, \quad 0 < \varepsilon \leq 1.$$

Then there is $T \subseteq S$ with

$$|T| \geq (2K)^{-Cp/\varepsilon^2} |S|$$

such that every $t \in T - T$ satisfies

$$\|\tau_t(\mu_A * 1_B) - \mu_A * 1_B\|_p \leq \varepsilon \|1_B\|_p.$$

More generally, if h is bounded and supported on B , the same sampling proof gives

$$\|\tau_t(\mu_A * h) - \mu_A * h\|_p \leq \varepsilon \|h\|_p$$

with the same parameter shape after replacing each sampled translate of 1_B by the corresponding translate of h .

Proof. The indicator statement is exactly the Croot–Sisask theorem proved earlier. For the stated extension, in the random-sampling proof use $Y_a(x) = h(x - a)$ instead of $1_B(x - a)$. Translation invariance gives $\|Y_a\|_p = \|h\|_p$, so the vector-valued sampling bound and the collision argument are unchanged. $\square$

18.7 The entropic Balog–Szemerédi–Gowers lemma

Theorem 18.15 (Entropic BSG in the conditional-sum form). *Let (A, B) be a jointly distributed pair of random variables taking values in an abelian group, and put $Z = A + B$. Then*

$$\mathbb{E}_z d_{\text{ent}}(A \mid Z = z, B \mid Z = z) \leq 3I(A; B) + 2H(Z) - H(A) - H(B).$$

Here, inside each conditional Ruzsa distance, independent copies of the two conditional laws are used, as required by the definition of d_{ent} .

Proof. Construct two conditionally independent trials (A_1, B_1) and (A_2, B_2) of the pair (A, B) over the common value of Z . Thus

$$A_1 + B_1 = A_2 + B_2 = Z,$$

and, conditional on $Z = z$, the two pairs are independent copies of $(A, B) \mid Z = z$. In particular, conditional on $Z = z$, A_1 and B_2 are independent. Therefore the average conditional Ruzsa distance on the left is

$$H(A_1 - B_2 \mid Z) - \frac{1}{2}H(A_1 \mid Z) - \frac{1}{2}H(B_2 \mid Z). \quad (*)$$

We first bound the first term. Submodularity, applied to the variables

$$A_1 - B_2, \quad A_1, \quad B_1,$$

gives

$$H(A_1 - B_2) + H(A_1 - B_2, A_1, B_1) \leq H(A_1 - B_2, A_1) + H(A_1 - B_2, B_1).$$

The tuple $(A_1 - B_2, A_1, B_1)$ determines both conditional trials: from A_1, B_1 we know Z , then $B_2 = A_1 - (A_1 - B_2)$ and $A_2 = Z - B_2$. Hence conditional independence over Z gives

$$H(A_1 - B_2, A_1, B_1) = H(A_1, B_1, A_2, B_2) = 2H(A, B) - H(Z).$$

Next,

$$H(A_1 - B_2, A_1) = H(B_2, A_1) \leq H(A) + H(B).$$

Using $A_1 + B_1 = A_2 + B_2$ we have $A_1 - B_2 = A_2 - B_1$, and therefore

$$H(A_1 - B_2, B_1) = H(A_2, B_1) \leq H(A) + H(B).$$

It follows that

$$H(A_1 - B_2) \leq 2I(A; B) + H(Z).$$

Conditioning cannot increase entropy, so the same upper bound holds for $H(A_1 - B_2 \mid Z)$.

Finally, because (A_1, B_1) has the original joint law and $Z = A_1 + B_1$,

$$H(A_1 \mid Z) = H(A, B) - H(Z) = H(A) + H(B) - I(A; B) - H(Z).$$

The identical formula holds for $H(B_2 \mid Z)$. Substituting these three estimates into $(*)$ yields

$$\mathbb{E}_z d_{\text{ent}}(A \mid Z = z, B \mid Z = z) \leq 3I(A; B) + 2H(Z) - H(A) - H(B),$$

as claimed. $\square$

Remark 18.16 (What the lemma does in the PFR endgame). When A and B are almost independent and their joint sum has unusually small entropy, the right-hand side is small. Averaging then produces a fibre $Z = z$ on which the two conditional laws have small entropic Ruzsa distance. This is the entropy analogue of extracting a high-energy fibre in the classical Balog–Szemerédi–Gowers argument.

18.8 Boundary of this bridge chapter

The lemmas above establish only the short logical bridges. The following deeper theorems are proved in dedicated chapters later in this volume:

- the local Chang theorem at the strength required by the Sanders iteration;
- the Sanders local-Fourier bootstrap and the resulting polylogarithmic Bogolyubov–Ruzsa theorem;
- the Bloom–Sisask spectral-boosting density-increment theorem;
- the Kelley–Meka spreadness, sifting, and high-moment boosting mechanism;
- the GGMT distance-decrement theorem, its stability/decrement auxiliaries, or the bounded-characteristic multidistance extension.

Each item is proved in a later dedicated chapter of Volume III; the bridge lemmas above are used only to connect these results.

18.9 Entropic stability and the fibring calculus

The next package is the exact stability layer needed before the distance–decrement argument. We spell it out because none of these estimates may be hidden inside the eventual polynomial Freiman–Ruzsa theorem.

Lemma 18.17 (Entropy increment under an independent summand). *Let X, Y, Z be independent random variables taking values in an abelian group. Then*

$$H(X + Y + Z) - H(X + Y) \leq H(Y + Z) - H(Y).$$

Proof. Conditional mutual information is nonnegative, so

$$I(X; Z \mid X + Y + Z) \geq 0.$$

Expanding it gives

$$\begin{aligned} I(X; Z \mid X + Y + Z) &= H(X, X + Y + Z) + H(Z, X + Y + Z) \\ &\quad - H(X, Z, X + Y + Z) - H(X + Y + Z). \end{aligned}$$

The indicated changes of variables are bijective and, using independence,

$$\begin{aligned} H(X, X + Y + Z) &= H(X, Y + Z) = H(X) + H(Y + Z), \\ H(Z, X + Y + Z) &= H(Z, X + Y) = H(Z) + H(X + Y), \\ H(X, Z, X + Y + Z) &= H(X, Y, Z) = H(X) + H(Y) + H(Z). \end{aligned}$$

Substitution and cancellation turn $I(X; Z \mid X + Y + Z) \geq 0$ exactly into the asserted inequality. $\square$

Lemma 18.18 (Conditioning cost for entropic Ruzsa distance). *Let (X, Z) and (Y, W) be random variables, with X, Y taking values in an abelian group. Define*

$$d_{\text{ent}}(X \mid Z, Y \mid W) = \sum_{z, w} \mathbb{P}(Z = z) \mathbb{P}(W = w) d_{\text{ent}}(X \mid Z = z, Y \mid W = w).$$

Then

$$d_{\text{ent}}(X \mid Z, Y \mid W) \leq d_{\text{ent}}(X, Y) + \frac{1}{2}I(X; Z) + \frac{1}{2}I(Y; W).$$

Proof. Take independent copies (X', Z') and (Y', W') of the two pairs. By the definition of conditional entropic Ruzsa distance,

$$\begin{aligned} d_{\text{ent}}(X \mid Z, Y \mid W) &= H(X' - Y' \mid Z', W') - \frac{1}{2}H(X' \mid Z') - \frac{1}{2}H(Y' \mid W') \\ &\leq H(X' - Y') - \frac{1}{2}H(X' \mid Z') - \frac{1}{2}H(Y' \mid W') \\ &= d_{\text{ent}}(X, Y) + \frac{1}{2}I(X; Z) + \frac{1}{2}I(Y; W), \end{aligned}$$

where only “conditioning reduces entropy” is used in the inequality. $\square$

Lemma 18.19 (Perturbation by an independent summand). *Let X, Y, Z be random variables in an abelian group, with Y, Z independent. Then*

$$d_{\text{ent}}(X, Y - Z) - d_{\text{ent}}(X, Y) \leq \frac{1}{2}(H(Y - Z) - H(Y)), \quad (18.1)$$

$$d_{\text{ent}}(X, Y \mid Y - Z) - d_{\text{ent}}(X, Y) \leq \frac{1}{2}(H(Y - Z) - H(Z)). \quad (18.2)$$

Equivalently, since Y, Z are independent,

$$\begin{aligned}\frac{1}{2}(H(Y - Z) - H(Y)) &= \frac{1}{2}d_{\text{ent}}(Y, Z) + \frac{1}{4}H(Z) - \frac{1}{4}H(Y), \\ \frac{1}{2}(H(Y - Z) - H(Z)) &= \frac{1}{2}d_{\text{ent}}(Y, Z) + \frac{1}{4}H(Y) - \frac{1}{4}H(Z).\end{aligned}$$

Proof. In the definition of $d_{\text{ent}}(X, \cdot)$ we may replace X by an independent copy, so assume X is independent of Y, Z . Then

$$\begin{aligned}d_{\text{ent}}(X, Y - Z) - d_{\text{ent}}(X, Y) \\ = H(X - Y + Z) - H(X - Y) - \frac{1}{2}H(Y - Z) + \frac{1}{2}H(Y).\end{aligned}$$

Lemma 18.17, with Y replaced by $-Y$, bounds the first entropy difference by $H(Y - Z) - H(Y)$. This proves (18.1).

For the conditional form, Lemma 18.18 with conditioning variable $Y - Z$ gives

$$d_{\text{ent}}(X, Y \mid Y - Z) - d_{\text{ent}}(X, Y) \leq \frac{1}{2}I(Y; Y - Z).$$

Because $(Y, Y - Z)$ is a bijective transform of (Y, Z) and Y, Z are independent,

$$I(Y; Y - Z) = H(Y) + H(Y - Z) - H(Y, Z) = H(Y - Z) - H(Z),$$

which gives (18.2). The displayed equivalent forms follow immediately from

$$d_{\text{ent}}(Y, Z) = H(Y - Z) - \frac{1}{2}H(Y) - \frac{1}{2}H(Z).$$

□

Proposition 18.20 (Fibering lemma with exact defect). *Let $\pi : H \rightarrow H'$ be a homomorphism of abelian groups and let Z_1, Z_2 be H -valued random variables. Then*

$$d_{\text{ent}}(Z_1, Z_2) \geq d_{\text{ent}}(\pi Z_1, \pi Z_2) + d_{\text{ent}}(Z_1 \mid \pi Z_1, Z_2 \mid \pi Z_2).$$

If independent copies are used in the definition of the left-hand distance, the defect between the two sides is exactly

$$I(Z_1 - Z_2; (\pi Z_1, \pi Z_2) \mid \pi(Z_1 - Z_2)).$$

Proof. Take Z_1, Z_2 independent. The conditional-distance term equals

$$H(Z_1 - Z_2 \mid \pi Z_1, \pi Z_2) - \frac{1}{2}H(Z_1 \mid \pi Z_1) - \frac{1}{2}H(Z_2 \mid \pi Z_2).$$

Since $(\pi Z_1, \pi Z_2)$ determines $\pi(Z_1 - Z_2)$,

$$H(Z_1 - Z_2 \mid \pi Z_1, \pi Z_2) \leq H(Z_1 - Z_2 \mid \pi(Z_1 - Z_2)).$$

Also

$$H(Z_i \mid \pi Z_i) = H(Z_i) - H(\pi Z_i)$$

and

$$H(Z_1 - Z_2 \mid \pi(Z_1 - Z_2)) = H(Z_1 - Z_2) - H(\pi(Z_1 - Z_2)).$$

Substitution yields the inequality. The difference between the two sides is

$$\begin{aligned}H(Z_1 - Z_2 \mid \pi(Z_1 - Z_2)) - H(Z_1 - Z_2 \mid \pi Z_1, \pi Z_2) \\ = I(Z_1 - Z_2; (\pi Z_1, \pi Z_2) \mid \pi(Z_1 - Z_2)),\end{aligned}$$

by the identity $H(A \mid B) - H(A \mid B, C) = I(A; C \mid B)$. □

Corollary 18.21 (Four-variable fibring identity). *Let Y_1, Y_2, Y_3, Y_4 be independent random variables in an abelian group. Then*

$$\begin{aligned} & d_{\text{ent}}(Y_1 - Y_3, Y_2 - Y_4) + d_{\text{ent}}(Y_1 \mid Y_1 - Y_3, Y_2 \mid Y_2 - Y_4) \\ & \quad + I(Y_1 - Y_2; Y_2 - Y_4 \mid Y_1 - Y_2 - Y_3 + Y_4) \\ & = d_{\text{ent}}(Y_1, Y_2) + d_{\text{ent}}(Y_3, Y_4). \end{aligned}$$

Proof. Apply Proposition 18.20 to $H = G \times G$, to the subtraction map $\pi(x, y) = x - y$, and to

$$Z_1 = (Y_1, Y_3), \quad Z_2 = (Y_2, Y_4).$$

Independence makes entropic Ruzsa distance additive on product distributions:

$$d_{\text{ent}}(Z_1, Z_2) = d_{\text{ent}}(Y_1, Y_2) + d_{\text{ent}}(Y_3, Y_4).$$

Writing out the exact-defect term gives precisely the stated conditional mutual information. $\square$

Proof architecture Exact stability bookkeeping used by the characteristic-two decrement

Fix $G = \mathbb{F}_2^n$ and independent $X_1, X_2, \tilde{X}_1, \tilde{X}_2$, with $\tilde{X}_i \stackrel{d}{=} X_i$. No additional “stability theorem” is used below. Every substitution is one of the following already-proved identities or inequalities:

1. for an independent-sum replacement, the distance increment is bounded by

$$d_{\text{ent}}(X, Y - Z) - d_{\text{ent}}(X, Y) \leq \frac{1}{2}(H(Y - Z) - H(Y))$$

from (18.1);

2. for the corresponding fibre replacement,

$$d_{\text{ent}}(X, Y \mid Y - Z) - d_{\text{ent}}(X, Y) \leq \frac{1}{2}(H(Y - Z) - H(Z))$$

by (18.2), and general conditioning is bounded by Lemma 18.18;

3. for the four-variable sum/fibre decomposition, Corollary 18.21 is an exact identity and its third term is the nonnegative conditional mutual information

$$I(Y_1 - Y_2; Y_2 - Y_4 \mid Y_1 - Y_2 - Y_3 + Y_4) \geq 0;$$

4. for the endgame variables

$$U = X_1 + X_2, \quad V = \tilde{X}_1 + X_2, \quad W = X_1 + \tilde{X}_1,$$

conditioning on $S = X_1 + X_2 + \tilde{X}_1 + \tilde{X}_2$ is again governed by Lemma 18.18.

Thus every numerical loss subsequently used in the decrement proof is one of the displayed entropy or mutual-information terms above; this box is a bookkeeping map, not a new theorem.

18.10 The characteristic-two distance decrement

We now prove the full characteristic-two decrement rather than citing polynomial Freiman–Ruzsa. The proof is organized so that every numerical loss has an owner in the preceding stability package.

Theorem 18.22 (Characteristic-two distance decrement). *Let $G = \mathbb{F}_2^n$, let X_1^0, X_2^0 be fixed reference variables, and put $\eta = 1/9$. For G -valued variables X_1, X_2 define*

$$\tau(X_1, X_2) = d_{\text{ent}}(X_1, X_2) + \eta d_{\text{ent}}(X_1^0, X_1) + \eta d_{\text{ent}}(X_2^0, X_2).$$

If $k = d_{\text{ent}}(X_1, X_2) > 0$, then there exist G -valued X'_1, X'_2 such that

$$\tau(X'_1, X'_2) < \tau(X_1, X_2).$$

Proof. Assume to the contrary that no such pair exists. Then for every pair Y_1, Y_2 we have

$$d_{\text{ent}}(Y_1, Y_2) \geq k - \eta(d_{\text{ent}}(X_1^0, Y_1) - d_{\text{ent}}(X_1^0, X_1)) - \eta(d_{\text{ent}}(X_2^0, Y_2) - d_{\text{ent}}(X_2^0, X_2)). \quad (18.3)$$

Averaging the same inequality over fibres gives its conditional form: if (Y_1, Z_1) and (Y_2, Z_2) are arbitrary, then

$$\begin{aligned} d_{\text{ent}}(Y_1 \mid Z_1, Y_2 \mid Z_2) &\geq k - \eta(d_{\text{ent}}(X_1^0, Y_1 \mid Z_1) - d_{\text{ent}}(X_1^0, X_1)) \\ &\quad - \eta(d_{\text{ent}}(X_2^0, Y_2 \mid Z_2) - d_{\text{ent}}(X_2^0, X_2)). \end{aligned} \quad (18.4)$$

Indeed, apply (18.3) on each pair of fibres and average.

Let $\tilde{X}_1, \tilde{X}_2$ be independent copies of X_1, X_2 , all four variables independent, and set

$$S = X_1 + X_2 + \tilde{X}_1 + \tilde{X}_2.$$

We first extract the two mutual-information estimates that drive the endgame.

First estimate. Apply Corollary 18.21 in characteristic 2 to

$$(Y_1, Y_2, Y_3, Y_4) = (X_1, X_2, \tilde{X}_2, \tilde{X}_1).$$

Writing

$$I_1 = I(X_1 + X_2; \tilde{X}_1 + X_2 \mid S),$$

we obtain the exact identity

$$d_{\text{ent}}(X_1 + \tilde{X}_2, X_2 + \tilde{X}_1) + d_{\text{ent}}(X_1 \mid X_1 + \tilde{X}_2, X_2 \mid X_2 + \tilde{X}_1) + I_1 = 2k. \quad (18.5)$$

Apply (18.3) to the first distance and (18.4) to the second. The four reference-distance changes that appear are bounded by Lemma 18.19. Pairing the two bounds involving X_1^0 and X_2^0 makes the entropy-asymmetry terms cancel, and their total is at most $2k$. Comparing with (18.5) therefore gives

$$I_1 \leq 2\eta k. \quad (18.6)$$

The same argument, retaining only the lower bound for the fibre term in (18.5), gives

$$d_{\text{ent}}(X_1 + \tilde{X}_2, X_2 + \tilde{X}_1) \leq (1 + \eta)k - I_1.$$

Since each of $X_1 + \tilde{X}_2$ and $X_2 + \tilde{X}_1$ has entropy

$$k + \frac{1}{2}H(X_1) + \frac{1}{2}H(X_2),$$

and their difference is S , this is equivalent to

$$H(S) \leq \frac{1}{2}H(X_1) + \frac{1}{2}H(X_2) + (2 + \eta)k - I_1. \quad (18.7)$$

Second estimate. Apply the four-variable fibring identity instead to

$$(Y_1, Y_2, Y_3, Y_4) = (X_2, X_1, \tilde{X}_2, \tilde{X}_1).$$

Set

$$I_2 = I(X_1 + X_2; X_1 + \tilde{X}_1 \mid S).$$

We obtain

$$d_{\text{ent}}(X_1 + \tilde{X}_1, X_2 + \tilde{X}_2) + d_{\text{ent}}(X_1 \mid X_1 + \tilde{X}_1, X_2 \mid X_2 + \tilde{X}_2) + I_2 = 2k.$$

Put

$$D = d_{\text{ent}}(X_1, X_1) + d_{\text{ent}}(X_2, X_2).$$

The self-sum instances of Lemma 18.19 bound each pair of reference penalties by $D/2$. Hence (18.3) and its conditional version imply

$$I_2 \leq \eta D \tag{18.8}$$

and

$$d_{\text{ent}}(X_1 + \tilde{X}_1, X_2 + \tilde{X}_2) \geq k - \frac{\eta}{2} D. \tag{18.9}$$

On the other hand, by the definition of D ,

$$\begin{aligned} d_{\text{ent}}(X_1 + \tilde{X}_1, X_2 + \tilde{X}_2) \\ = H(S) - \frac{1}{2} H(X_1) - \frac{1}{2} H(X_2) - \frac{1}{2} D. \end{aligned}$$

Using (18.7) gives the upper bound

$$d_{\text{ent}}(X_1 + \tilde{X}_1, X_2 + \tilde{X}_2) \leq (2 + \eta)k - I_1 - \frac{1}{2} D.$$

Together with (18.9) this yields

$$D \leq 2k + \frac{2(2\eta k - I_1)}{1 - \eta}. \tag{18.10}$$

Consequently

$$I_2 \leq 2\eta k + \frac{2\eta(2\eta k - I_1)}{1 - \eta}. \tag{18.11}$$

By symmetry the same bound holds for

$$I_3 = I(\tilde{X}_1 + X_2; X_1 + \tilde{X}_1 \mid S).$$

Three-variable endgame. Define

$$U = X_1 + X_2, \quad V = \tilde{X}_1 + X_2, \quad W = X_1 + \tilde{X}_1.$$

In characteristic 2 we have the exact identity

$$U + V + W = 0.$$

Let

$$\tilde{\delta} = I(U; V \mid S) + I(V; W \mid S) + I(W; U \mid S).$$

By (18.6) and (18.11),

$$\tilde{\delta} \leq 6\eta k - \frac{1-5\eta}{1-\eta}(2\eta k - I_1). \quad (18.12)$$

We need one final lemma, proved immediately below as Lemma 18.23: if $T_1 + T_2 + T_3 = 0$ identically and

$$\delta = I(T_1; T_2) + I(T_2; T_3) + I(T_3; T_1),$$

then there are T'_1, T'_2 with

$$\begin{aligned} & d_{\text{ent}}(T'_1, T'_2) + \eta(d_{\text{ent}}(X_1^0, T'_1) - d_{\text{ent}}(X_1^0, X_1)) \\ & + \eta(d_{\text{ent}}(X_2^0, T'_2) - d_{\text{ent}}(X_2^0, X_2)) \\ & \leq \delta + \frac{\eta}{3} \left(\delta + \sum_{i=1}^2 \sum_{j=1}^3 (d_{\text{ent}}(X_i^0, T_j) - d_{\text{ent}}(X_i^0, X_i)) \right). \end{aligned} \quad (18.13)$$

Apply this to the conditional triple

$$(T_1, T_2, T_3) = (U \mid S = s, V \mid S = s, W \mid S = s)$$

and average over s . The no-decrement assumption (18.3) applied to the output of the lemma implies

$$k \leq \tilde{\delta} + \frac{\eta}{3} \left(\tilde{\delta} + \sum_{i=1}^2 \sum_{A \in \{U, V, W\}} (d_{\text{ent}}(X_i^0, A \mid S) - d_{\text{ent}}(X_i^0, X_i)) \right). \quad (18.14)$$

It remains to bound the six reference terms. Combining the two perturbation estimates in Lemma 18.19 gives, for independent Y, Z, Z' ,

$$\begin{aligned} d_{\text{ent}}(X, Y - Z \mid Y - Z - Z') - d_{\text{ent}}(X, Y) & \leq \frac{1}{2}(H(Y - Z - Z') + H(Y - Z)) \\ & \quad - \frac{1}{2}(H(Y) + H(Z')). \end{aligned}$$

Use this with the obvious permutations of $X_1, X_2, \tilde{X}_1, \tilde{X}_2$. Since the intermediate two-sum always has the same law as a sum of X_1 and X_2 , the six inequalities add to

$$\begin{aligned} & \sum_{i=1}^2 \sum_{A \in \{U, V, W\}} (d_{\text{ent}}(X_i^0, A \mid S) - d_{\text{ent}}(X_i^0, X_i)) \\ & \leq 3H(S) - \frac{3}{2}H(X_1) - \frac{3}{2}H(X_2) \\ & \leq (6 - 3\eta)k + 3(2\eta k - I_1), \end{aligned} \quad (18.15)$$

where the last line is (18.7) rewritten so that the nonnegative gap $2\eta k - I_1$ is visible.

Substituting (18.12) and (18.15) into (18.14) gives

$$\begin{aligned} k & \leq (8\eta + \eta^2)k \\ & \quad - \left(\frac{1-5\eta}{1-\eta} \left(1 + \frac{\eta}{3} \right) - \eta \right) (2\eta k - I_1). \end{aligned}$$

For $\eta = 1/9$, the coefficient multiplying $2\eta k - I_1$ is nonnegative, while

$$8\eta + \eta^2 = \frac{73}{81} < 1.$$

Since $2\eta k - I_1 \geq 0$ by (18.6), we obtain $k \leq 73k/81$, hence $k = 0$, contradicting the assumption $k > 0$. The theorem follows. $\square$

Lemma 18.23 (Ternary endgame from entropic BSG). *Let T_1, T_2, T_3 be $\mathbb{F}_2^n$ -valued random variables satisfying $T_1 + T_2 + T_3 = 0$ identically, and set*

$$\delta = I(T_1; T_2) + I(T_2; T_3) + I(T_3; T_1).$$

For fixed reference variables X_1^0, X_2^0 and fixed comparison variables X_1, X_2 , there exist T'_1, T'_2 satisfying (18.13).

Proof. Apply Theorem 18.15 to the joint pair (T_1, T_2) and its sum T_3 . Because any two of T_1, T_2, T_3 determine the third, all three pair-entropies are equal to $H(T_1, T_2, T_3)$. A direct expansion therefore shows that the right-hand side of the entropic BSG estimate is exactly

$$I(T_1; T_2) + I(T_2; T_3) + I(T_3; T_1) = \delta.$$

Thus

$$\mathbb{E}_{t_3} d_{\text{ent}}(T_1 \mid T_3 = t_3, T_2 \mid T_3 = t_3) \leq \delta. \quad (18.16)$$

For each $i = 1, 2$, Lemma 18.18 gives

$$d_{\text{ent}}(X_i^0, T_1 \mid T_3) - d_{\text{ent}}(X_i^0, X_i) \leq d_{\text{ent}}(X_i^0, T_1) - d_{\text{ent}}(X_i^0, X_i) + \frac{1}{2}I(T_1; T_3),$$

and similarly for T_2 . Combine these inequalities with (18.16). Choosing a fibre t_3 with value no larger than the average produces a pair of conditional laws and an estimate involving the ordered pair (T_1, T_2) .

Now repeat the same construction for all six permutations $(T_\alpha, T_\beta, T_\gamma)$ of (T_1, T_2, T_3) and average the six bounds. Each reference-distance term involving a fixed T_j occurs equally often, and the half-mutual-information penalties average to $\delta/3$. Therefore some permutation gives

$$\begin{aligned} & d_{\text{ent}}(T'_1, T'_2) + \eta(d_{\text{ent}}(X_1^0, T'_1) - d_{\text{ent}}(X_1^0, X_1)) + \eta(d_{\text{ent}}(X_2^0, T'_2) - d_{\text{ent}}(X_2^0, X_2)) \\ & \leq \delta + \frac{\eta}{3} \left(\delta + \sum_{i=1}^2 \sum_{j=1}^3 (d_{\text{ent}}(X_i^0, T_j) - d_{\text{ent}}(X_i^0, X_i)) \right), \end{aligned}$$

which is (18.13). □

Theorem 18.24 (Entropic polynomial Freiman–Ruzsa in characteristic two). *Let $G = \mathbb{F}_2^n$ and let X_1^0, X_2^0 be G -valued random variables. There exists a subgroup $H \leq G$ such that*

$$d_{\text{ent}}(X_1^0, U_H) + d_{\text{ent}}(X_2^0, U_H) \leq 11 d_{\text{ent}}(X_1^0, X_2^0).$$

Moreover each of the two distances on the left is at most

$$6 d_{\text{ent}}(X_1^0, X_2^0).$$

Proof. The space of probability distributions on the finite group G is compact, and entropy is continuous there. Hence the functional τ in Theorem 18.22 attains a minimum at some pair (X_1, X_2) . The distance decrement theorem forces

$$d_{\text{ent}}(X_1, X_2) = 0.$$

The zero-distance theorem proved earlier for entropic Ruzsa distance then supplies a subgroup $H \leq G$ such that

$$d_{\text{ent}}(X_1, U_H) = d_{\text{ent}}(X_2, U_H) = 0.$$

By minimality,

$$\begin{aligned}\eta(d_{\text{ent}}(X_1^0, U_H) + d_{\text{ent}}(X_2^0, U_H)) &= \tau(X_1, X_2) \\ &\leq \tau(X_2^0, X_1^0) \\ &= (1 + 2\eta)d_{\text{ent}}(X_1^0, X_2^0).\end{aligned}$$

With $\eta = 1/9$, $(1 + 2\eta)/\eta = 11$, proving the sum bound. The triangle inequality gives

$$|d_{\text{ent}}(X_1^0, U_H) - d_{\text{ent}}(X_2^0, U_H)| \leq d_{\text{ent}}(X_1^0, X_2^0).$$

Combining this with the sum bound shows that each term is at most $6d_{\text{ent}}(X_1^0, X_2^0)$. $\square$

Corollary 18.25 (Polynomial Freiman–Ruzsa in $\mathbb{F}_2^n$). *There is an absolute constant C such that, whenever $A \subseteq \mathbb{F}_2^n$ is nonempty and*

$$|A + A| \leq K|A|,$$

there is a subgroup $H \leq \mathbb{F}_2^n$ with $|H| \leq |A|$ such that A is covered by at most $2K^C$ cosets of H .

Proof. Apply Theorem 18.24 with both reference variables equal to the uniform distribution U_A . Since

$$d_{\text{ent}}(U_A, U_A) = H(U_A - U_A) - H(U_A) \leq \log |A - A| - \log |A| \leq \log K$$

in characteristic 2, the entropic theorem gives a subgroup law U_H at entropic distance $O(\log K)$ from U_A . The entropy-to-combinatorial PFR proposition proved earlier converts precisely such a bound into a polynomial number of cosets of a subgroup of size at most $|A|$. Its proof is internal in the preceding entropy chapter and uses only entropy support bounds and a covering argument; inserting the constant 11 above yields $2K^C$ for an absolute C . $\square$

Remark 18.26 (Characteristic two versus bounded torsion). The theorem above treats characteristic two. The bounded-characteristic theorem requires a genuinely different multi-distance argument, which is proved in Chapter 21.

Chapter 19

The Local Chang–Sanders Chain

19.1 The local Chang–Sanders argument

The Croot–Sisask sampling theorem and the classical Chang theorem were established earlier. We now prove the local Chang estimate and the quantitative Sanders bootstrap. The inputs are Plünnecke–Ruzsa inequalities, the Croot–Sisask L^p theorem, regularization of Bohr sets, Ruzsa–Chang covering, and the geometry-of-numbers lemma that extracts a proper progression from a convex lattice body.

For a probability measure ν on a finite abelian group G and $f \in L^1(\nu)$ put

$$\text{Spec}_\varepsilon(f; \nu) = \left\{ \gamma \in \widehat{G} : \left| \int f \bar{\gamma} d\nu \right| \geq \varepsilon \|f\|_{L^1(\nu)} \right\}.$$

If $\Lambda \subseteq \widehat{G}$ and $\omega : \Lambda \rightarrow \{z : |z| \leq 1\}$, set

$$p_{\omega, \Lambda}(x) = \prod_{\lambda \in \Lambda} (1 + \Re(\omega(\lambda)\lambda(x))).$$

For $K \geq 0$ we say that Λ is K -dissociated relative to ν when

$$\int p_{\omega, \Lambda} d\nu \leq e^K \quad \text{for every } \omega.$$

The (K, ν) -relative entropy of a set of characters is the largest cardinality of a K -dissociated subset. Thus the case $K = 1$ is the notion used in the local Chang lemma below.

Lemma 19.1 (Subgaussian estimate for a relative Riesz product). *Let Λ be 1-dissociated relative to a probability measure ν , let $m = |\Lambda|$, and choose phases $c_\lambda \in \mathbb{S}^1$. If $2 \leq p \leq m$, then*

$$\left\| \sum_{\lambda \in \Lambda} c_\lambda \lambda \right\|_{L^p(\nu)} \leq C \sqrt{pm}.$$

Proof. It suffices to control the real and imaginary parts. Write

$$S(x) = \sum_{\lambda \in \Lambda} \Re(c_\lambda \lambda(x)).$$

For $|t| \leq 1$ and $u \in [-1, 1]$ the elementary inequality

$$e^{tu} \leq (1 + tu)e^{Ct^2}$$

holds with an absolute C . Multiplying it over Λ and using relative dissociativity with $\omega(\lambda) = tc_\lambda$ gives

$$\int e^{tS} d\nu \leq \exp(1 + Ct^2m).$$

The same estimate holds for $-S$. Chernoff's inequality therefore yields

$$\nu\{|S| \geq u\} \leq 2e \exp\left(-\frac{u^2}{4Cm}\right) \quad (0 \leq u \leq 2Cm).$$

Since $|S| \leq m$ and $p \leq m$, integration of the distribution function,

$$\|S\|_p^p = p \int_0^m u^{p-1} \nu\{|S| > u\} du,$$

gives $\|S\|_p \leq C\sqrt{pm}$. Apply the same argument to the imaginary part and use $|z| \leq |\Re z| + |\Im z|$. $\square$

Theorem 19.2 (Local Chang lemma). *Let ν be a probability measure on a finite abelian group and let $0 \neq f \in L^2(\nu)$. Put*

$$L_f = \frac{\|f\|_{L^2(\nu)}}{\|f\|_{L^1(\nu)}} \geq 1.$$

Then $\text{Spec}_\varepsilon(f; \nu)$ has $(1, \nu)$ -relative entropy at most

$$C\varepsilon^{-2} \log(2L_f).$$

Proof. Let Λ be a relatively 1-dissociated subset of the spectrum and write $m = |\Lambda|$. For every $\lambda \in \Lambda$ choose a phase c_λ so that

$$c_\lambda \int f \bar{\lambda} d\nu = \left| \int f \bar{\lambda} d\nu \right|.$$

Let

$$F = \sum_{\lambda \in \Lambda} \bar{c}_\lambda \lambda.$$

For any $p \geq 2$, with conjugate exponent $q = p/(p-1)$, Hölder gives

$$\begin{aligned} \varepsilon m \|f\|_1 &\leq \sum_{\lambda \in \Lambda} \left| \int f \bar{\lambda} d\nu \right| \\ &= \left| \int f \bar{F} d\nu \right| \leq \|f\|_q \|F\|_p. \end{aligned}$$

Interpolation between L^1 and L^2 gives

$$\|f\|_q \leq \|f\|_1^{1-2/p} \|f\|_2^{2/p} = \|f\|_1 L_f^{2/p}.$$

Choose

$$p = \max\{2, \lceil 4 \log(2L_f) \rceil\}.$$

If $m < p$ there is nothing to prove. Otherwise Lemma 19.1 applies and yields

$$\varepsilon m \leq CL_f^{2/p} \sqrt{pm} \leq C' \sqrt{pm},$$

because $L_f^{2/p} \leq e^{1/2}$. Squaring and cancelling m gives

$$m \leq C\varepsilon^{-2} p \leq C'\varepsilon^{-2} \log(2L_f).$$

Taking the largest such Λ proves the theorem. $\square$

Lemma 19.3 (Entropy regularization for Bohr scales). *Let B_0 be a Bohr set and let $0 < a < b$ with $b/a \geq 2$. Put*

$$H_0 = \log \frac{|(B_0)_b|}{|(B_0)_a|}.$$

There is $\lambda \in [a, b]$, staying an absolute multiplicative distance from the two endpoints, such that $B = (B_0)_\lambda$ satisfies

$$\left| \log \frac{|B_{1+u}|}{|B|} \right| \leq C(1 + H_0)|u| \quad \left(|u| \leq \frac{c}{1 + H_0} \right).$$

Moreover, if $2\lambda \leq b$ and $\lambda/2 \geq a$, then

$$h(B) := \log \frac{|B_2|}{|B_{1/2}|} \leq H_0.$$

Proof. Write

$$F(t) = \log |(B_0)_{e^t}|, \quad \log a \leq t \leq \log b.$$

This is nondecreasing and its total variation is H_0 . Restrict to a closed subinterval J obtained by deleting an absolute fraction of the length at both ends. Suppose every $t \in J$ violates the asserted local regularity. Then for each t there is s_t , with $0 < |s_t| \leq c/(1 + H_0)$, such that the interval joining t to $t + s_t$ remains in $[\log a, \log b]$ and

$$|F(t + s_t) - F(t)| > C_1(1 + H_0)|s_t|.$$

Choose, by the one-dimensional $5r$ covering lemma, a pairwise disjoint subfamily of these intervals whose fivefold dilates cover J . Their total length is therefore at least $|J|/5$. Monotonicity of F and disjointness give

$$H_0 \geq \sum_j |F(t_j + s_{t_j}) - F(t_j)| > \frac{C_1}{5}(1 + H_0)|J|,$$

which is impossible once the absolute constant C_1 is chosen large relative to the fixed positive length of J . Hence some t is regular. Replacing e^t by λ proves the first assertion. The last assertion is immediate from nesting:

$$|B_2| = |(B_0)_{2\lambda}| \leq |(B_0)_b|, \quad |B_{1/2}| = |(B_0)_{\lambda/2}| \geq |(B_0)_a|.$$

□

Lemma 19.4 (Entropy-regular approximate annihilator). *Let B satisfy the regularity conclusion of Lemma 19.3 with parameter $H \geq h(B)$, and let β be its uniform probability measure. If $|\hat{\beta}(\chi)| \geq \theta > 0$, then for $y \in B_u$, $0 < u \leq c/(1 + H)$,*

$$|1 - \chi(y)| \leq C\theta^{-1}(1 + H)u.$$

Proof. Translation gives

$$|1 - \chi(y)| |\hat{\beta}(\chi)| = \left| \int \chi(x) d\beta(x) - \int \chi(x + y) d\beta(x) \right| \leq \|\tau_y \beta - \beta\|_1.$$

As $B_{1-u} + y \subseteq B \subseteq B_{1+u}$, the symmetric difference between B and $B + y$ has relative size $O((1 + H)u)$ by entropy regularity. Dividing by θ proves the claim. □

Lemma 19.5 (Annihilating a low-relative-entropy spectrum). *Let B be an entropy-regular Bohr set, let β be uniform on B , and write $H = 1 + h(B)$. Suppose $\Delta \subseteq \widehat{G}$ has (η, β) -relative entropy at most k , where $0 < \eta \leq 1$. Then there are a set $\Lambda \subseteq \Delta$, $|\Lambda| \leq k$, and a regular dilate B_ρ with*

$$\rho \geq \frac{c\eta}{H(k + \log(2/\eta) + 1)}$$

such that, for every $\gamma \in \Delta$ and all $\rho', \nu > 0$ in the regularity range,

$$|1 - \gamma(x)| \leq C(k\nu + (1 + h(B_\rho))\rho')$$

whenever

$$x \in (B_\rho)_{\rho'} \cap B(\Lambda, \nu).$$

Here $B(\Lambda, \nu) = \{x : |1 - \lambda(x)| \leq \nu \ \forall \lambda \in \Lambda\}$.

Proof. Let

$$L = \lceil C_0(k + \log(2/\eta) + 1) \rceil.$$

Choose $\rho \asymp \eta/(HL)$, decreasing it by at most an absolute factor so that B_ρ is entropy-regular. Put

$$\beta^+ = \beta_{1+L\rho} * \beta_\rho^{*L}.$$

For $x \in B$, every decomposition used by the convolution stays inside $B_{1+L\rho}$, hence

$$\beta(x) \leq \frac{|B_{1+L\rho}|}{|B|} \beta^+(x) \leq (1 + \eta/3) \beta^+(x)$$

by regularity and the choice of ρ . Consequently every set which is $\eta/2$ -dissociated relative to β^+ is η -dissociated relative to β and therefore has size at most k .

Set $\eta_i = i\eta/(2(k+1))$. Starting with the empty set, adjoin a character from Δ whenever the enlarged set is η_{i+1} -dissociated relative to β^+ . The process must stop at some $i \leq k$; otherwise it would produce an $\eta/2$ -dissociated set of size $k+1$. Let the terminal set be Λ . Thus for every $\gamma \in \Delta \setminus \Lambda$ there are coefficients $\omega : \Lambda \rightarrow \{z : |z| \leq 1\}$ and a phase v , $|v| \leq 1$, for which

$$\int p_{\omega, \Lambda}(1 + \Re(v\gamma)) d\beta^+ > e^{\eta_{i+1}}, \quad \int p_{\omega, \Lambda} d\beta^+ \leq e^{\eta_i}.$$

Rotating v if necessary and subtracting the two bounds gives

$$\left| \int p_{\omega, \Lambda} \gamma d\beta^+ \right| \geq c \frac{\eta}{k+1}. \quad (*)$$

The Fourier expansion of a Riesz product is supported on $\text{Span}(\Lambda)$ and the sum of the absolute values of its coefficients is at most 2^k . Also

$$\widehat{\beta^+}(\xi) = \widehat{\beta_{1+L\rho}}(\xi) \widehat{\beta_\rho}(\xi)^L.$$

Applying Fourier inversion to (*) therefore shows that for some $\lambda \in \text{Span}(\Lambda)$,

$$|\widehat{\beta_\rho}(\gamma - \lambda)|^L \geq c 2^{-k} \frac{\eta}{k+1}.$$

The definition of L makes the right side at least 2^{-L} , so

$$|\widehat{\beta_\rho}(\gamma - \lambda)| \geq \frac{1}{2}.$$

By Lemma 19.4, applied to the regular Bohr set B_ρ , the residual character $\gamma - \lambda$ is $O((1 + h(B_\rho))\rho')$ -constant on $(B_\rho)_{\rho'}$. On $B(\Lambda, \nu)$ every $\lambda \in \text{Span}(\Lambda)$ is $O(k\nu)$ -constant, by the triangle inequality on the unit circle. Combining the two estimates proves the displayed conclusion. Characters already in Λ satisfy it without the residual term. $\square$

19.2 Two covering lemmas needed by the bootstrap

For a finite set T write

$$\text{Span}(T) = \left\{ \sum_{t \in T} \epsilon_t t : \epsilon_t \in \{-1, 0, 1\} \right\}.$$

Lemma 19.6 (Finite-scale Chang covering). *Let A, S be finite subsets of an abelian group. If*

$$|rA + S| < 2^r |S|$$

for an integer $r \geq 1$, then there is $T \subseteq A$, $|T| < r$, such that

$$A \subseteq \text{Span}(T) + S - S.$$

Proof. Choose a maximal subset $T \subseteq A$ such that the translates

$$\epsilon \cdot T + S, \quad \epsilon \in \{0, 1\}^T,$$

are pairwise disjoint. Maximality implies the covering assertion: if $a \in A \setminus T$, then adjoining a destroys disjointness, so two binary sums, one using a and one not using it, differ by an element of $S - S$; solving for a places it in $\text{Span}(T) + S - S$. If $|T| \geq r$, choose $T_0 \subseteq T$ with $|T_0| = r$. The 2^r translates $\epsilon \cdot T_0 + S$ are disjoint and all lie in $rA + S$, contradicting the hypothesis. $\square$

Lemma 19.7 (Finite-scale growth implies polynomial growth). *Let $X = -X$ contain 0. If for some integer $r \geq 1$*

$$|(3r + 1)X| < 2^r |X|,$$

then there is $T \subseteq 3X$ with $|T| < r$ such that

$$3X \subseteq \text{Span}(T) + 2X,$$

and consequently, for every $n \geq 1$,

$$|(n + 2)X| \leq (2n + 1)^r |2X|.$$

Proof. Apply Lemma 19.6 with $A = 3X$ and $S = X$. The hypothesis is exactly

$$|r(3X) + X| < 2^r |X|.$$

Thus $3X \subseteq \text{Span}(T) + X - X = \text{Span}(T) + 2X$. Induction gives

$$(n + 2)X \subseteq n \text{Span}(T) + 2X.$$

Every element of $n \text{Span}(T)$ has a coefficient vector in $[-n, n]^T$, hence there are at most $(2n + 1)^{|T|} \leq (2n + 1)^r$ possibilities. $\square$

19.3 From polynomial growth to a controlled Bohr envelope

We use the multiplicative-circle form of Bohr sets in this section. Thus $B(\Gamma, \rho)$ consists of those x for which $|1 - \gamma(x)| \leq \rho$ for every $\gamma \in \Gamma$, and B_λ denotes width $\lambda\rho$.

Lemma 19.8 (Near-maximal Fourier coefficients annihilate differences). *Let $A, S \subseteq G$ and assume $|A + S| \leq K|A|$. If*

$$|\widehat{1_{A+S}}(\gamma)| \geq (1 - \delta)|A + S|$$

for the unnormalised Fourier transform, then

$$|1 - \gamma(s_0 - s_1)| \leq C\sqrt{K\delta} \quad (s_0, s_1 \in S).$$

Proof. Choose $\omega \in \mathbb{S}^1$ so that

$$\sum_{x \in A+S} \Re(\omega\gamma(x)) \geq (1 - \delta)|A + S|.$$

Since $|1 - z|^2 = 2(1 - \Re z)$ on the unit circle,

$$\sum_{x \in A+S} |1 - \omega\gamma(x)|^2 \leq 2\delta|A + S|.$$

For $s \in S$, $A + s \subseteq A + S$, hence

$$\sum_{a \in A} |1 - \omega\gamma(a + s)|^2 \leq 2\delta|A + S|.$$

For $s_0, s_1 \in S$, the triangle inequality in $L^2(A)$ gives

$$|1 - \gamma(s_0 - s_1)|\sqrt{|A|} \leq 2\sqrt{2\delta|A + S|},$$

which is the asserted estimate. $\square$

Proposition 19.9 (Polynomial-growth Bohr envelope). *Let G be finite abelian and let $X = -X \subseteq G$ contain 0. Suppose*

$$|nX| \leq n^d|X| \quad (n \geq 1),$$

with $d \geq 1$. For $0 < \kappa \leq 1/2$ there is an entropy-regular Bohr set B such that

$$X - X \subseteq B_\kappa, \quad |B_2| \leq \exp(Cd \log(2d/\kappa))|X|,$$

and, in addition,

$$h(B) \leq Cd \log(2d/\kappa).$$

Proof. Use normalized Haar measure μ_G throughout and put $\alpha = \mu_G(\ell X)$. Choose $\ell = O(d \log(2d))$ so that

$$|\ell X| \leq 2|(\ell - 1)X|;$$

otherwise multiplying the successive growth ratios up to this scale would contradict $|nX| \leq n^d|X|$. Set $\delta = c_0\kappa^2$ and

$$\Gamma = \left\{ \gamma : |\widehat{1_{\ell X}}(\gamma)| \geq (1 - \delta)\alpha \right\}.$$

Let B' have frequency set Γ and constant multiplicative width $1/2$. Lemma 19.8, applied to $A = (\ell - 1)X$ and $S = X$, gives

$$X - X \subseteq B'_{\kappa/4}$$

after fixing c_0 sufficiently small.

We next bound B' . Take

$$q \asymp d\kappa^{-2} \log(2d/\kappa).$$

For the q -fold convolution $F = 1_{\ell X}^{(*q)}$, Cauchy–Schwarz on its support gives

$$\|F\|_2^2 \geq \frac{\alpha^{2q}}{\mu_G(q\ell X)} \geq \frac{\alpha^{2q}}{(q\ell)^d \mu_G(X)}. \quad (1)$$

Outside Γ we have $|\widehat{1_{\ell X}}| \leq (1 - \delta)\alpha$. Parseval and $\sum_{\gamma} |\widehat{1_{\ell X}}(\gamma)|^2 = \alpha$ therefore give

$$\sum_{\gamma \notin \Gamma} |\widehat{1_{\ell X}}(\gamma)|^{2q} \leq e^{-c\delta q} \alpha^{2q-1}.$$

Since $X \subseteq \ell X$, we have $\alpha \geq \mu_G(X)$. Thus the ratio of this error term to the lower bound in (1) is at most $e^{-c\delta q}(q\ell)^d$, which is at most $1/2$ by the choice of q . Hence at least half the $2q$ -moment lies on Γ .

Let β' be the uniform probability measure on B' . Since every $\gamma \in \Gamma$ varies by at most an absolute constant on B' , shrinking the defining width once if necessary gives $|\widehat{\beta'}(\gamma)| \geq 1/2$. Thus Parseval and Young's inequality imply

$$\begin{aligned} c \frac{\alpha^{2q}}{(q\ell)^d \mu_G(X)} &\leq \sum_{\gamma \in \Gamma} |\widehat{1_{\ell X}}(\gamma)|^{2q} \\ &\leq C \|F * \beta'\|_2^2 \\ &\leq C \|F * \tilde{F}\|_1 \|\beta' * \tilde{\beta'}\|_\infty = C \alpha^{2q} \mu_G(B')^{-1}. \end{aligned}$$

Therefore

$$\mu_G(B') \leq C(q\ell)^d \mu_G(X) \leq \exp(Cd \log(2d/\kappa)) \mu_G(X). \quad (2)$$

Apply Lemma 19.3 to the scale interval $[1/4, 1/2]$ of B' . Its total logarithmic growth is at most the right-hand side of (2), because $B'_{1/4} \supseteq X - X$ and $B'_{1/2} \subseteq B'$. Let $\lambda \in [1/4, 1/2]$ be the resulting regular scale and put $B = B'_\lambda$. Then $B_2 \subseteq B'$ and, since $\lambda \geq 1/4$, $B_\kappa \supseteq B'_{\kappa/4} \supseteq X - X$. Moreover $B_{1/2} \supseteq B'_{1/8} \supseteq B'_{\kappa/4} \supseteq X - X$. Combining this with (2) gives both the asserted size bound and

$$h(B) = \log \frac{|B_2|}{|B_{1/2}|} \leq Cd \log(2d/\kappa).$$

□

19.4 Low-dimensional progressions from one-scale Bohr growth

Proposition 19.10 (One-scale Bohr growth gives a progression). *Let B be a Bohr set in a finite abelian group. Suppose that, for some integer $r \geq 1$ and $0 < \rho < 1/(4(3r+1))$,*

$$|B_{(3r+1)\rho}| < 2^r |B_\rho|.$$

Then B_ρ contains a proper symmetric coset progression P of rank at most r such that

$$|P| \geq \exp(-Cr \log(2r)) |B_\rho|.$$

Proof. Let Γ be the frequency set of B and quotient by $H = \bigcap_{\gamma \in \Gamma} \ker \gamma$. At the end we lift the resulting progression and add H , so assume $H = \{0\}$. For $x \in B_{(3r+1)\rho}$ choose the unique principal arguments $\theta_\gamma(x) \in (-1/2, 1/2)$ with $\gamma(x) = e^{2\pi i \theta_\gamma(x)}$. Because $(3r+1)\rho < 1/4$, no wrap-around occurs in sums of at most $3r+1$ elements. Thus

$$R(x) = (\theta_\gamma(x))_{\gamma \in \Gamma}$$

is a Freiman homomorphism of the required order on $B_{(3r+1)\rho}$, and it is injective by the trivial-kernel assumption.

Apply Lemma 19.6 in the additive image with $A = 3R(B_\rho)$ and $S = R(B_\rho)$. The growth hypothesis gives a set T of fewer than r image-points such that

$$3R(B_\rho) \subseteq \text{Span}(T) + 2R(B_\rho).$$

Let V be the real span of T . Induction gives $nR(B_\rho) \subseteq V + 2R(B_\rho)$ for every n . If $v \in 2R(B_\rho)$, then $nv = w_n + u_n$ with $w_n \in V$ and $u_n \in 2R(B_\rho)$; after division by n , boundedness of u_n shows $\text{dist}(v, V) = 0$. Hence $R(B_\rho) \subseteq V$ and $\dim V \leq r$.

The lift of the finite subgroup $R(G)$ together with the integer lattice inside the ambient argument space induces a full lattice Λ in V . The set $R(B_\rho)$ is exactly the lattice points of Λ in the symmetric cube section

$$K = V \cap [-c\rho, c\rho]^\Gamma.$$

The convex-lattice progression lemma proved in Volume II (from successive minima and Minkowski's second theorem) supplies a proper rank- $\leq r$ progression $Q \subseteq K \cap \Lambda$ with

$$|Q| \geq e^{-Cr \log(2r)} |K \cap \Lambda|.$$

Because R and its inverse are Freiman homomorphisms of order at least two on the relevant cube, $R^{-1}(Q)$ is a proper progression of the same rank and size inside B_ρ . Restoring the common kernel turns it into a coset progression and proves the proposition. $\square$

19.5 The quantitative Sanders bootstrap

Lemma 19.11 (Croo–Sisask correlation package). *Let $A, S, T \subseteq G$ be finite nonempty sets with $|A + S| \leq K|A|$ and $|T + S| \leq L|S|$. Let $q \geq 1$ and $0 < \varepsilon \leq 1$. Then there is a symmetric $X \ni 0$ satisfying*

$$|X| \geq \exp(-C\varepsilon^{-2}q^2 \log(2K) \log(2L)) |T|$$

and

$$|\mu_{-A} * 1_{A+S} * \mu_{-S}(x) - 1| \leq \varepsilon \quad (x \in qX).$$

Proof. Apply the already-proved Croo–Sisask L^p theorem to $f = 1_{A+S}$ and the translating set T relative to S , with $p = 2 + \lceil \log K \rceil$ and one-step error $\varepsilon/(eq)$. It produces a set X_0 of the displayed size up to a change of the absolute constant and such that

$$\|\tau_x(1_{A+S} * \mu_{-S}) - 1_{A+S} * \mu_{-S}\|_p \leq \varepsilon e^{-1} \|1_{A+S}\|_p$$

for $x \in q(X_0 - X_0)$, by triangle amplification. Replace $X_0 - X_0$ by a symmetric translate-containing subset $X \ni 0$ with the same exponential lower bound.

Convolving with μ_{-A} and evaluating at 0, Hölder gives

$$|\mu_{-A} * 1_{A+S} * \mu_{-S}(x) - \mu_{-A} * 1_{A+S} * \mu_{-S}(0)| \leq \varepsilon K^{1/p} / e \leq \varepsilon.$$

The López–Ross identity gives the value at 0 equal to 1, since every pair $(a, s) \in A \times S$ contributes inside $A + S$. This proves the claim. $\square$

Lemma 19.12 (Fourier correlation from an annihilating set). *There is an absolute $c > 0$ with the following property. Suppose $A, S, S', X \subseteq G$, $S' \subseteq S$, satisfy*

$$|A + S| \leq K \min(|A|, |S|), \quad |S'| \geq |S|/K^2,$$

and for an integer $q \geq \log(4K)$,

$$|\mu_{-A} * 1_{A+S'} * \mu_{-S'}(x) - 1| \leq 1/64 \quad (x \in qX).$$

If a set M satisfies

$$|1 - \gamma(x)| \leq c/K^2 \quad (x \in M, \gamma \in \text{Spec}_c(\mu_X)),$$

then every probability measure σ supported on M obeys

$$\|1_{A+S'} * \sigma\|_\infty \geq 15/16.$$

Proof. Average the first displayed estimate over the q -fold convolution of μ_X . Fourier inversion shows that the resulting correlation differs from 1 by at most $1/64$. Split the Fourier sum into the large spectrum $\text{Spec}_c(\mu_X)$ and its complement. On the complement, $|\widehat{\mu_X}|^q \leq c^q$, while Cauchy–Schwarz and Parseval give

$$\sum_{\gamma} |\widehat{1_{A+S'}}(\gamma) \widehat{\mu_A}(\gamma) \widehat{\mu_{S'}}(\gamma)| \leq K^2.$$

Thus the complementary contribution is at most $1/64$ after decreasing c . On the large spectrum, the hypothesis on M implies $|1 - \widehat{\sigma}(\gamma)| \leq c/K^2$, so inserting $\widehat{\sigma}$ changes the Fourier sum by at most another $1/64$. Altogether the correlation of $1_{A+S'} * \sigma$ against the probability measure $\mu_A * \mu_{S'} * \mu_X^{(*q)} * \sigma$ is at least $61/64 > 15/16$. Since the latter is a probability measure, the asserted L^∞ bound follows. $\square$

Theorem 19.13 (Sanders–Bogolyubov–Ruzsa, selected quantitative form). *Let A be a finite nonempty subset of an abelian group and suppose*

$$|A + A| \leq K|A|, \quad K \geq 2.$$

Then $2A - 2A$ contains a proper symmetric convex coset progression M with

$$\text{rank } M \leq C \log^6(2K)$$

and

$$|M| \geq \exp(-C \log^6(2K) \log \log(4K)) |A|.$$

In particular the coarser size bound $|M| \geq \exp(-C \log^7(2K)) |A|$ holds.

Proof. By the finite modelling theorem proved in Volume II it suffices to work in a finite abelian group; all relations used below have bounded order and pull back under the model.

Step 1: prepare a smoothed almost-period set. Plünnecke’s inequality supplies $S' \subseteq A$, $S' \neq \emptyset$, with

$$|2A + S'| \leq K^2 |S'|.$$

Since $|2A + S'| \geq |A|$, this also gives $|S'| \geq |A|/K^2$. A second Plünnecke application gives

$$|(A + S') + (A + S')| \leq K^4 |A + S'|.$$

Choose

$$q = C_0 \log(2K)$$

with C_0 a sufficiently large absolute constant. Applying Lemma 19.11 with fixed small error, $T = A$, produces a symmetric set $X \ni 0$ such that

$$|X| \geq \exp(-Cq^2 \log^2(2K)) |A| \tag{19.1}$$

and

$$|\mu_{-A} * 1_{A+S'} * \mu_{-S'}(x) - 1| \leq 1/64 \quad (x \in qX). \quad (19.2)$$

In particular the left side of the convolution is positive on qX , so

$$qX \subseteq (A + S') - (A + S').$$

Step 2: force polynomial growth. The Plünnecke–Ruzsa inequalities applied to $A + S'$ give, for every $m \geq 1$,

$$|m((A + S') - (A + S'))| \leq K^{8m}|A + S'|.$$

Together with (19.1), this implies

$$|4\ell qX| \leq \exp(C\ell \log(2K) + Cq^2 \log^2(2K))|X|.$$

Take $\ell = C_1 q^2 \log(2K)$ and put $r = \ell q$. Since $q = C_0 \log(2K)$ with C_0 chosen first large enough, the right side is strictly smaller than $2^r |X|$. Therefore

$$|(3r + 1)X| < 2^r |X|.$$

Lemma 19.7 now yields

$$|nX| \leq n^{d_0} |2X| \quad (n \geq 2), \quad d_0 = O(r) = O(\log^4(2K)). \quad (19.3)$$

Absorbing the harmless factor $|2X|/|X| \leq 2^{O(r)}$ into the exponent lets us use Proposition 19.9 with $d = O(\log^4(2K))$.

Step 3: a local Fourier model. Choose $\kappa = K^{-C_2}$, with C_2 a sufficiently large absolute constant. Proposition 19.9 gives a regular Bohr set B such that

$$X - X \subseteq B_\kappa, \quad |B_2| \leq e^{d_1} |X|, \quad d_1 = O(d \log(2dK)) = O(\log^5(2K)).$$

Let β be uniform measure on B . Since $0 \in X$ and X is symmetric, $X \subseteq X - X \subseteq B_\kappa \subseteq B$; hence $\beta(X) \geq e^{-O(d_1)}$. Apply the local Chang theorem, Theorem 19.2, to 1_X relative to β at a fixed threshold c . Its large spectrum

$$\Delta = \text{Spec}_c(1_X; \beta) = \text{Spec}_c(\mu_X)$$

has relative entropy

$$s = O(\log(2\beta(X)^{-1})) = O(d_1).$$

Since $X - X \subseteq B_\kappa$ and $|B_2| \leq e^{d_1} |X|$, while $|X - X| \geq |X|$, we have

$$h(B) \leq d_1.$$

Apply Lemma 19.5 with $\eta = 1$. It gives a set $\Lambda \subseteq \Delta$, $|\Lambda| \leq s$, and a regular scale

$$\rho \geq c(1 + d_1)^{-1}(s + 1)^{-1} \geq d_1^{-O(1)}$$

such that every $\gamma \in \Delta$ obeys

$$|1 - \gamma(x)| \leq C(s\nu + (1 + h(B_\rho))\rho')$$

on $(B_\rho)_{\rho'} \cap B(\Lambda, \nu)$. Because $\rho \geq 2\kappa$ after choosing the earlier fixed power $\kappa = K^{-C_2}$ sufficiently small, nesting gives

$$|B_{2\rho}| \leq |B_2| \leq e^{d_1} |X|, \quad |B_{\rho/2}| \geq |B_\kappa| \geq |X|,$$

so $h(B_\rho) \leq d_1$. Taking

$$\rho' \asymp (K^2 d_1)^{-1}, \quad \nu \asymp (K^2(s+1))^{-1}$$

and putting $C = (B_\rho)_{\rho'} \cap B(\Lambda, \nu)$ yields

$$|1 - \gamma(x)| \leq c/K^2 \quad (x \in C, \gamma \in \Delta). \quad (19.4)$$

For $0 < \eta \leq 1$, the Bohr-intersection estimate gives

$$|C_\eta| \geq (c\eta\nu)^s |B_{\rho\rho'\eta/2}|.$$

Choose

$$t = C_3 d_1 \log(2d_1 K) = O(\log^6(2K))$$

with C_3 large and take $\eta \in [1/(8(3t+1)), 1/(4(3t+1))]$. Because $\rho\rho'\eta/2 \geq \kappa$ once C_2 is fixed sufficiently large, the last Bohr set contains B_κ and hence has size at least $|X|$. Thus

$$|C_\eta| \geq (c\eta\nu)^s |X|.$$

On the other hand $C_{(3t+1)\eta} \subseteq B_2$, so

$$\frac{|C_{(3t+1)\eta}|}{|C_\eta|} \leq e^{d_1} (c\eta\nu)^{-s} < 2^t$$

by the definition of t , since $s = O(d_1)$, $\nu^{-1} \leq K^{O(1)} d_1^{O(1)}$, and $\eta^{-1} = O(t)$. Proposition 19.10 therefore supplies a proper coset progression $P \subseteq C_\eta$ of rank at most t and

$$\begin{aligned} |P| &\geq \exp(-Ct \log(2t)) |C_\eta| \\ &\geq \exp(-Ct \log(2t) - Cs \log(2t/\nu)) |X| \\ &\geq \exp(-C \log^6(2K) \log \log(4K)) |A|, \end{aligned} \quad (19.5)$$

where the final line also uses (19.1) and absorbs the lower-order polylogarithmic losses. Every point of P obeys the annihilation estimate (19.4).

Step 4: convert correlation into exact containment. Lemma 19.12, using (19.2), shows that every probability measure supported on P has

$$\|1_{A+S'} * \sigma\|_\infty \geq 15/16.$$

Translate and, if necessary, shrink P by an absolute factor so that it is symmetric; this costs $e^{O(t)}$ in size. Write its progression coordinates as

$$P_\theta = H + \{n_1 v_1 + \cdots + n_t v_t : |n_i| \leq \theta L_i\}.$$

Among the $O(t)$ nested scales between $1/2$ and 1 there are adjacent scales $\theta < \theta + \delta$, with $\delta \asymp 1/t$, satisfying

$$|P_{\theta+\delta}| < \sqrt{2} |P_\theta|.$$

Apply the correlation conclusion to the probability measure which is the average of the uniform measures on these two scales. For some translate $z + A + S'$, the average of the two occupation densities is at least $15/16$; since each density is at most 1 , each is therefore at least $7/8$. Because $|P_{\theta+\delta}| < \sqrt{2} |P_\theta|$, the two occupied portions have total cardinality strictly larger than $|P_{\theta+\delta}|$. Hence for every $u \in P_\delta$, the two occupied subsets

$$(z + A + S') \cap P_{\theta+\delta} \quad \text{and} \quad u + ((z + A + S') \cap P_\theta)$$

therefore intersect. Thus

$$u \in (A + S') - (A + S') \subseteq 2A - 2A.$$

Consequently $P_\delta \subseteq 2A - 2A$. Shrinking from P to P_δ loses at most $\exp(O(t \log(2t)))$, already present in (19.5). Taking $M = P_\delta$ proves the theorem. $\square$

Corollary 19.14 (Progression and subspace consequences). *Under the hypotheses of Theorem 19.13:*

1. $2A - 2A$ contains a proper coset progression of rank $O(\log^6(2K))$ and relative size $\exp(-O(\log^7(2K)))$;
2. if the ambient group has exponent at most m , then $4A - 4A$ contains a subgroup H with

$$|H| \geq \exp(-O_m(\log^7(2K)))|A|;$$

3. in a cyclic group, whenever the subgroup part is trivial, one of the one-dimensional coordinate progressions has length at least $|M|^{1/r}$, where $r = \text{rank } M$.

Proof. The first assertion is the theorem. Write the resulting coset progression as $M = H + P$, where P is proper of rank r . In exponent m , each coordinate side has at most m distinct values, so $|P| \leq m^r$. Since $H \subseteq M - M \subseteq 4A - 4A$,

$$|H| = |M|/|P| \geq m^{-r}|M|,$$

which gives the second claim. For the third, properness gives $|M| = \prod_{i=1}^r (2L_i + 1)$ when H is trivial, so at least one factor is at least $|M|^{1/r}$. $\square$

Chapter 20

The Kelley–Meka Regularization Mechanism

Warning 20.1 (Normalization used in this chapter). We use normalized Haar measure throughout. Thus, for a nonempty set X ,

$$\mu(X) = |X|/|G|, \quad \mu_X = 1_X/\mu(X),$$

so that $\mathbb{E}_x \mu_X(x) = 1$, and

$$f * g(x) = \mathbb{E}_y f(y)g(x - y), \quad f \circ g(x) = \mathbb{E}_y f(x + y)\overline{g(y)}.$$

Local L^p norms are taken against the displayed probability measure. With this normalization a random-looking convolution supported on a Bohr set B naturally has size $\mu(B)^{-1}$. The older Croot–Sisask formulas written with cardinalities are converted to this convention by dividing each raw indicator convolution by the corresponding power of $|G|$.

20.1 Why a separate potential is needed

The classical Fourier increment scheme asks whether the density of a set rises on a single structured region. The Kelley–Meka mechanism adds a second diagnostic: high moments of the difference convolution. A large high moment detects a small set of unusually popular differences; dependent random choice then converts those popular differences into a pair of moderately dense auxiliary sets. Almost periodicity turns the latter pair into a genuine structured density increment.

For $A \subseteq G$ of density α put

$$f_A = \mu_A \circ \mu_A - 1.$$

Its Fourier transform is nonnegative away from the trivial character: $\widehat{\mu_A \circ \mu_A} = |\widehat{\mu_A}|^2$. This positivity is the source of the “boosting” step below.

Lemma 20.2 (Spreadness stopping-time decomposition). *Let $G = \mathbb{F}_q^n$, let $A \subseteq G$ have density $\alpha > 0$, let $\gamma > 1$, and let $r \geq 1$. Then there are an affine subspace $x + V$ and $A' = (A - x) \cap V$ such that*

1. $\text{codim } V \leq r \lceil \log(1/\alpha) / \log \gamma \rceil$;
2. $\mu_V(A') \geq \alpha$;
3. A' is (γ, r) -spread in V : if $W \leq V$ has $\text{codim}_V W \leq r$, then every affine translate $y + W$ satisfies

$$\frac{|A' \cap (y + W)|}{|W|} < \gamma \mu_V(A').$$

Proof. Begin with $V_0 = G$, $A_0 = A$, $\alpha_0 = \alpha$. If A_j is not (γ, r) -spread in V_j , choose an affine subspace $y + W \subseteq V_j$ of codimension at most r on which the relative density is at least $\gamma\alpha_j$. Translate $y + W$ to W , put $V_{j+1} = W$, and let A_{j+1} be the translated restriction. Thus

$$\alpha_{j+1} \geq \gamma\alpha_j, \quad \text{codim } V_{j+1} \leq \text{codim } V_j + r.$$

After t steps we have $1 \geq \alpha_t \geq \gamma^t \alpha$, so $t \leq \log(1/\alpha)/\log \gamma$. Hence the process terminates within the asserted codimension, and termination is exactly spreadness. $\square$

Lemma 20.3 (Spectral unbalancing). *Let ν be a probability measure on a finite abelian group with $\widehat{\nu} \geq 0$, and let $f : G \rightarrow \mathbb{R}$ satisfy $\widehat{f} \geq 0$. For every fixed $0 < \varepsilon \leq 1$ there is C_ε such that*

$$\|f\|_{L^p(\nu)} \geq \varepsilon \implies \|1 + f\|_{L^{p'}(\nu)} \geq 1 + \varepsilon/2$$

for some integer $p' \leq C_\varepsilon \max(2, p)$.

Proof. Increase p by at most an absolute factor and assume that $p \geq 5$ is odd. For every integer $k \geq 1$, Parseval and Fourier positivity give

$$\int f^k d\nu = \langle \widehat{\nu}, \widehat{f}^{(*k)} \rangle \geq 0.$$

Let $P = \{f \geq 0\}$. Since $f^{p-1} = |f|^{p-1}$,

$$2 \int_P f^p d\nu = \int (f + |f|) f^{p-1} d\nu \geq \int |f|^p d\nu \geq \varepsilon^p.$$

Put $T = \{x \in P : f(x) \geq 3\varepsilon/4\}$. As $p \geq 5$,

$$\int_T f^p d\nu \geq \frac{1}{2} \varepsilon^p - (3\varepsilon/4)^p \geq \frac{1}{4} \varepsilon^p.$$

Cauchy–Schwarz therefore yields

$$\nu(T)^{1/2} \|f\|_{L^{2p}(\nu)}^p \geq \frac{1}{4} \varepsilon^p.$$

If $\|1 + f\|_{L^{2p}(\nu)} > 2$, the conclusion already holds with $p' = 2p$. Otherwise the triangle inequality gives $\|f\|_{2p} \leq 1 + \|1 + f\|_{2p} \leq 3$, whence

$$\nu(T) \geq (\varepsilon/12)^{2p}.$$

For every $r \geq 1$ we consequently have

$$\|1 + f\|_{L^r(\nu)} \geq (1 + 3\varepsilon/4)(\varepsilon/12)^{2p/r}.$$

Taking $r = C_\varepsilon p$, with C_ε sufficiently large, makes the right side at least $1 + \varepsilon/2$. Replacing r by the next integer proves the claim. $\square$

Lemma 20.4 (Sifting / dependent random choice). *Let $p \geq 2$ be even, let $B_1, B_2, A \subseteq G$ be nonempty, put $\alpha = |A|/|G|$ and $\nu = \mu_{B_1} \circ \mu_{B_2}$, and let $F : G \rightarrow \mathbb{R}_{\geq 0}$. There are $A_i \subseteq B_i$ such that*

$$\langle \mu_{A_1} \circ \mu_{A_2}, F \rangle \leq 2 \frac{\langle (\mu_A \circ \mu_A)^p, F \rangle_\nu}{\|\mu_A \circ \mu_A\|_{L^p(\nu)}^p}$$

and

$$\min_i \frac{|A_i|}{|B_i|} \geq \frac{1}{4} \alpha^{2p} \|\mu_A \circ \mu_A\|_{L^p(\nu)}^{2p}.$$

Consequently, for fixed $\varepsilon, \delta > 0$, if

$$S = \{x : \mu_A \circ \mu_A(x) > (1 - \varepsilon) \|\mu_A \circ \mu_A\|_{L^p(\nu)}\},$$

then, after increasing p by $O_{\varepsilon, \delta}(1)$ if necessary, there are $A_i \subseteq B_i$ with

$$\langle \mu_{A_1} \circ \mu_{A_2}, 1_S \rangle \geq 1 - \delta$$

and

$$\min_i \frac{|A_i|}{|B_i|} \geq (c_{\varepsilon, \delta} \alpha \|\mu_A \circ \mu_A\|_{L^p(\nu)})^{2p + O_{\varepsilon, \delta}(1)}.$$

Proof. For $\mathbf{s} = (s_1, \dots, s_p) \in G^p$ set

$$A_i(\mathbf{s}) = B_i \cap (A + s_1) \cap \dots \cap (A + s_p)$$

and let $a_i(\mathbf{s}) = |A_i(\mathbf{s})|/|G|$. Expanding the p -th power and averaging the shifts gives the exact identities

$$\langle (\mu_A \circ \mu_A)^p, F \rangle_\nu = \frac{\alpha^{-2p} |G|^2}{|B_1| |B_2|} \mathbb{E}_{\mathbf{s}} \langle 1_{A_1(\mathbf{s})} \circ 1_{A_2(\mathbf{s})}, F \rangle$$

and, on taking $F \equiv 1$,

$$\|\mu_A \circ \mu_A\|_{L^p(\nu)}^p = \frac{\alpha^{-2p} |G|^2}{|B_1| |B_2|} \mathbb{E}_{\mathbf{s}} a_1(\mathbf{s}) a_2(\mathbf{s}).$$

Put

$$M = \frac{1}{2} \alpha^p \left(\frac{|B_1| |B_2|}{|G|^2} \right)^{1/2} \|\mu_A \circ \mu_A\|_{L^p(\nu)}^p.$$

By Cauchy–Schwarz and $\mathbb{E}_{\mathbf{s}} a_i(\mathbf{s}) = \alpha^p |B_i|/|G|$, the portion of $\mathbb{E} a_1 a_2$ coming from $a_1 a_2 < M^2$ is less than one half of the total. Therefore at least half of the weighted mass lies on samples for which $a_1 a_2 \geq M^2$. Comparing the two displayed identities shows that one such sample also has

$$\frac{\langle 1_{A_1} \circ 1_{A_2}, F \rangle}{a_1 a_2} \leq 2 \frac{\langle (\mu_A \circ \mu_A)^p, F \rangle_\nu}{\|\mu_A \circ \mu_A\|_{L^p(\nu)}^p}.$$

Since $a_1 a_2 \geq M^2$ and $a_i \leq |B_i|/|G|$, each relative density $|A_i|/|B_i|$ is at least $\frac{1}{4} \alpha^{2p} \|\mu_A \circ \mu_A\|_{L^p(\nu)}^{2p}$. This proves the first assertion.

For the consequence use $F = 1_{G \setminus S}$. By definition of S ,

$$\frac{\langle (\mu_A \circ \mu_A)^p, 1_{G \setminus S} \rangle_\nu}{\|\mu_A \circ \mu_A\|_{L^p(\nu)}^p} \leq (1 - \varepsilon)^p.$$

Taking p larger by $O(\varepsilon^{-1} \log(1/\delta))$ makes twice this quantity at most δ . The density estimate changes only by the stated $O_{\varepsilon, \delta}(1)$ in the exponent. $\square$

Proposition 20.5 (Finite-field almost-periodicity). *Let $G = \mathbb{F}_q^n$ for a fixed prime q , and let $A_1, A_2, S \subseteq G$ with $\mu(A_i) \geq \beta > 0$. For every $0 < \varepsilon < 1$ there is a subspace $V \leq G$ with*

$$\text{codim } V \leq C_{\varepsilon, q} \log^4(2/\beta)$$

such that

$$|\langle \mu_V * \mu_{A_1} \circ \mu_{A_2}, 1_S \rangle - \langle \mu_{A_1} \circ \mu_{A_2}, 1_S \rangle| \leq \varepsilon.$$

Proof. Apply the Croot–Sisask sampling theorem proved earlier with the ambient shift set equal to G , with exponent $p = 2$, and with error $c\varepsilon$. Since $|A_1 + G|/|A_1| = \beta^{-1}$, it gives a set $T \subseteq G$ of density

$$\tau \geq \beta^{C_\varepsilon}$$

such that every $t \in T - T$ changes $\mu_{A_1} \circ \mu_{A_2}$ by at most $c\varepsilon$ in L^2 , and hence changes its pairing with 1_S by at most $c\varepsilon$ by Cauchy–Schwarz. The same is true for sums of four such almost-periods, with error at most $4c\varepsilon$.

Apply the finite-field consequence of the already-proved Bogolyubov–Ruzsa theorem to T . It gives a subspace $V \subseteq 4T - 4T$ of codimension

$$O(\log^4(2/\tau)) = O_\varepsilon(\log^4(2/\beta)).$$

Thus each $v \in V$ changes the pairing with 1_S by at most $O(c\varepsilon)$. Averaging this statement over $v \in V$ gives the claimed convolution by μ_V , once c is chosen sufficiently small. $\square$

Proposition 20.6 (Finite-field Kelley–Meka density increment). *Fix $0 < \varepsilon < 1/10$. Let $G = \mathbb{F}_q^n$ with q a fixed odd prime, let $A, C \subseteq G$ have densities $\alpha, \gamma > 0$. Then either*

$$|\langle \mu_A * \mu_A, \mu_C \rangle - 1| \leq \varepsilon,$$

or there is a subspace $V \leq G$ such that

$$\text{codim } V \leq C_{\varepsilon, q} \log^4(2/\gamma) \log^4(2/\alpha)$$

and

$$\|1_A * \mu_V\|_\infty \geq (1 + c_\varepsilon)\alpha.$$

Proof. If the first alternative fails, Hölder with $\|\mu_C\|_{p'} = \gamma^{-1/p}$ gives an even integer $p \ll_\varepsilon \log(2/\gamma)$ for which

$$\|\mu_A * \mu_A - 1\|_p \geq c\varepsilon.$$

By Fourier positivity,

$$\|\mu_A \circ \mu_A - 1\|_p \geq \|\mu_A * \mu_A - 1\|_p.$$

Lemma 20.3, applied to $f = \mu_A \circ \mu_A - 1$ and uniform ν , yields an exponent $p' \ll_\varepsilon p$ for which

$$\|\mu_A \circ \mu_A\|_{p'} \geq 1 + c\varepsilon.$$

Use Lemma 20.4 with $B_1 = B_2 = G$ and a fixed small δ . We obtain sets A_1, A_2 of densities at least

$$\beta \geq \alpha^{C_\varepsilon p'}$$

whose difference convolution gives mass at least $1 - \delta$ to

$$S = \{x : \mu_A \circ \mu_A(x) \geq 1 + c_1\varepsilon\}.$$

Proposition 20.5 supplies V of codimension

$$O_\varepsilon(\log^4(2/\beta)) = O_\varepsilon(\log^4(2/\gamma) \log^4(2/\alpha))$$

for which smoothing the A_1, A_2 correlation changes its mass on S by at most δ . Hence

$$1 + c_2\varepsilon \leq \langle \mu_V * \mu_{A_1} \circ \mu_{A_2}, \mu_A \circ \mu_A \rangle.$$

Using the adjoint identity for convolution and difference convolution, the right-hand side is at most

$$\alpha^{-1} \|1_A * \mu_V\|_\infty \|\mu_A * \mu_{A_2} \circ \mu_{A_1}\|_1 = \alpha^{-1} \|1_A * \mu_V\|_\infty.$$

This proves the density increment. $\square$

Theorem 20.7 (Finite-field spreadness/regularization theorem). *Let $G = \mathbb{F}_q^n$ for a fixed prime q , let $A \subseteq G$ have density $\alpha > 0$, let $p \geq 2$, and let $0 < \varepsilon < 1/10$. There are an affine subspace $x + V$ and $A' = (A - x) \cap V$ such that*

$$\text{codim } V \leq C_{\varepsilon, q} p^4 \log^5(2/\alpha), \quad \mu_V(A') \geq (1 - \varepsilon)\alpha,$$

and

$$\|\mu_{A'} * \mu_{A'} - \mu_V\|_{L^p(\mu_V)} \leq \varepsilon \frac{|G|}{|V|}.$$

Proof. Work successively inside affine subspaces, translating each chosen affine subspace to a vector subspace. Start with density $\alpha_0 = \alpha$. If the displayed L^p balance fails at stage j , the Hölder/unbalancing/sifting argument in the proof of Proposition 20.6, now with the witness to the L^p imbalance in place of C , gives a multiplicative density increment

$$\alpha_{j+1} \geq (1 + c_\varepsilon)\alpha_j$$

on a subspace whose additional codimension is at most

$$C_\varepsilon p^4 \log^4(2/\alpha_j) \leq C_\varepsilon p^4 \log^4(2/\alpha).$$

There can be at most $C_\varepsilon \log(2/\alpha)$ such increments, because the density never exceeds one. Consequently the total codimension is at most $C_\varepsilon p^4 \log^5(2/\alpha)$. At the terminal stage the L^p imbalance is at most ε by definition of termination. If desired, begin each increment with a $(1 - \varepsilon/2t)$ -dense restriction, where t is the preceding bound on the number of stages; the product of these retention factors is at least $1 - \varepsilon$. This gives the stated density retention. $\square$

20.2 Local L^∞ almost-periodicity on Bohr sets

The next three results isolate the analytic bridge that turns a large set of Croot–Sisask almost-periods into a *structured* set of almost-periods. This bridge is used later in the integer Kelley–Meka argument. We prove it here so that the phrase “local Chang converts the almost-periods into a Bohr set” never hides a theorem-sized input.

Lemma 20.8 (Three-fold L^∞ Croot–Sisask bootstrap). *Let A, M, L, S be finite subsets of a finite abelian group. Assume*

$$|A + S| \leq K|A|, \quad \eta := \min\{1, |M|/|L|\} > 0.$$

For $0 < \varepsilon < 1$ and an integer $k \geq 1$ there is $T \subseteq S$ with

$$|T| \geq \exp\left(-Ck^2\varepsilon^{-2} \log(2/\eta) \log(2K)\right) |S|$$

such that every $t \in kT - kT$ satisfies

$$\|1_A * 1_M * 1_L(\cdot + t) - 1_A * 1_M * 1_L\|_\infty \leq \varepsilon \mu(A) \mu(M).$$

Proof. Put

$$p = \lceil C_0 \log(2/\eta) \rceil, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

Apply the Croot–Sisask sampling proof given earlier to $1_A * 1_L$, with accuracy $\varepsilon/(4e)$ and with the collision parameter chosen so that the conclusion holds on $kT - kT$. The usual k -fold triangle argument in that proof gives

$$\|1_A * 1_L(\cdot + t) - 1_A * 1_L\|_p \leq \frac{\varepsilon}{2e} \mu(A) \mu(L)^{1/p} \quad (t \in kT - kT)$$

and

$$|T| \geq \exp\left(-Ck^2\varepsilon^{-2}p\log(2K)\right)|S|.$$

Convolving with 1_M and using Hölder gives

$$\begin{aligned} & \|1_A * 1_M * 1_L(\cdot + t) - 1_A * 1_M * 1_L\|_\infty \\ & \leq \mu(M)^{1/q} \|1_A * 1_L(\cdot + t) - 1_A * 1_L\|_p \\ & \leq \frac{\varepsilon}{2e} \mu(A) \mu(M) \left(\frac{|L|}{|M|}\right)^{1/p}. \end{aligned}$$

The choice of p makes $\eta^{-1/p} \leq e$; increasing C_0 absorbs the factor 2. Since $p \ll \log(2/\eta)$, the asserted size bound follows. $\square$

Proposition 20.9 (Local Chang–Sanders annihilation package). *Let $B = \text{Bohr}(\Gamma, \rho)$ be a regular Bohr set of rank d , and let $X \subseteq B$ have relative density $\xi > 0$. For $0 < \delta, \nu \leq 1$ there is a regular Bohr set*

$$B^\sharp = \text{Bohr}(\Gamma \cup \Lambda, \rho^\sharp) \leq B$$

with

$$|\Lambda| \leq C\delta^{-2}\log(2/\xi)$$

and, writing $d^\sharp = d + |\Lambda|$,

$$\rho^\sharp \geq \frac{c\rho\nu\delta^2}{d^2(d^\sharp + 1)\log(2/\xi)}$$

such that

$$|1 - \gamma(t)| \leq \nu \quad (\gamma \in \text{Spec}_\delta(\mu_X), t \in B^\sharp).$$

Proof. Apply the local Chang theorem to $f = 1_X$ with the uniform measure on B . Since

$$\|1_X\|_{L^2(\mu_B)} / \|1_X\|_{L^1(\mu_B)} = \xi^{-1/2},$$

the δ -large local spectrum has relative entropy at most $C\delta^{-2}\log(2/\xi)$. Choose a maximal relatively dissociated subset Λ of that spectrum. The proof of Lemma 19.5, applied with target error ν , gives an inner regular Bohr scale on which every character in the spectrum is ν -annihilated. Adjoining the frequencies in Λ and then decreasing the radius by at most an absolute factor to a regular value gives the displayed $B^\sharp$. The rank estimate is the entropy bound. Keeping the losses from the entropy-annihilator proof instead of suppressing them gives the conservative radius lower bound stated here. $\square$

Theorem 20.10 (L^∞ almost-periodicity relative to a Bohr set). *Let $B = \text{Bohr}(\Gamma, \rho)$ be regular of rank d . Let $S \subseteq B$ have relative density at least $\sigma > 0$, and let A, M, L satisfy*

$$|A + S| \leq K|A|, \quad 0 < \eta := |M|/|L| \leq 1.$$

For $0 < \varepsilon < 1$ there is a regular Bohr set $B' \leq B$ of rank $d + d'$ with

$$d' \leq C\varepsilon^{-2}\log^2\frac{2}{\varepsilon\eta}\log\frac{2}{\eta}\log(2K) + C\log\frac{2}{\sigma}$$

and radius

$$\rho' \geq \frac{c\rho\varepsilon\eta^{1/2}}{d^2(d + d' + 1)^2}$$

such that, for every $t \in B'$,

$$\|1_A * 1_M * 1_L(\cdot + t) - 1_A * 1_M * 1_L\|_\infty \leq C\varepsilon \mu(A) \mu(M).$$

Consequently

$$\|1_A * 1_M * 1_L * \mu_{B'} - 1_A * 1_M * 1_L\|_\infty \leq C\varepsilon \mu(A) \mu(M).$$

Proof. Choose

$$k = \left\lceil C_0 \log \frac{2}{\varepsilon \eta} \right\rceil.$$

Lemma 20.8 gives $T \subseteq S$ with

$$\frac{|T|}{|B|} \geq \sigma \exp\left(-Ck^2\varepsilon^{-2} \log(2/\eta) \log(2K)\right)$$

such that every $u \in kT - kT$ is an $O(\varepsilon\mu(A)\mu(M))$ almost-period of

$$F = 1_A * 1_M * 1_L.$$

Fix $z \in T$ and put $X = T - z$. Then $kX \subseteq kT - kT$, and averaging the almost-period estimate over $x_1, \dots, x_k \in X$ gives

$$\|F * \mu_X^{(*k)} - F\|_\infty \leq C\varepsilon \mu(A)\mu(M). \quad (1)$$

Apply Proposition 20.9 to X with $\delta = 1/2$ and target error

$$\nu = c\varepsilon\eta^{1/2}.$$

The logarithm of the reciprocal relative density of X is bounded by

$$Ck^2\varepsilon^{-2} \log(2/\eta) \log(2K) + \log(2/\sigma),$$

which gives the stated d' . After a harmless regularization of the radius, we obtain $B' \leq B$ of the asserted size on which

$$|1 - \gamma(t)| \leq c\varepsilon\eta^{1/2} \quad (\gamma \in \text{Spec}_{1/2}(\mu_X), t \in B'). \quad (2)$$

It remains to show that $F * \mu_X^{(*k)}$ is nearly invariant under translation by B' . By Fourier inversion, for $t \in B'$ the pointwise difference is at most

$$\mathbb{E}_{\gamma \in \widehat{G}} |\widehat{1_A}(\gamma)| |\widehat{1_M}(\gamma)| |\widehat{1_L}(\gamma)| |\widehat{\mu_X}(\gamma)|^k |1 - \gamma(t)|.$$

On $\text{Spec}_{1/2}(\mu_X)$ use (2). Off that spectrum, $|\widehat{\mu_X}(\gamma)|^k \leq 2^{-k} \leq c\varepsilon\eta^{1/2}$. Thus the last display is at most

$$C\varepsilon\eta^{1/2} \mathbb{E}_\gamma |\widehat{1_A}(\gamma)| |\widehat{1_M}(\gamma)| |\widehat{1_L}(\gamma)|.$$

Use $|\widehat{1_A}| \leq \mu(A)$ and Cauchy–Schwarz plus Parseval on the other two factors:

$$\mathbb{E}_\gamma |\widehat{1_M}(\gamma)| |\widehat{1_L}(\gamma)| \leq \mu(M)^{1/2} \mu(L)^{1/2}.$$

Since $\eta^{1/2} \mu(M)^{1/2} \mu(L)^{1/2} \leq \mu(M)$, this is $O(\varepsilon\mu(A)\mu(M))$. Combining with (1) and its translate proves the pointwise almost-periodicity. Averaging over $t \in B'$ proves the convolution statement. $\square$

20.3 The localized integer mechanism

For the integer theorem we use the cyclic model and regular Bohr sets. Two normalisations have to be kept separate. If $A \subseteq B$ has relative density α , then its ambient density is

$$a := \frac{|A|}{|G|} = \alpha \mu(B).$$

The sifting lemma is written in terms of a , while the local difference convolution has natural size $\mu(B)^{-1}$. Their product is therefore of size α ; this cancellation is the reason that no ambient factor $\mu(B)$ is lost in the local argument.

Proposition 20.11 (Local pairing almost-periodicity). For every $0 < \epsilon < 1/4$ there is C with the following property. Let B_0, B_1 be regular Bohr sets of the same rank d , with $B_1 \subset (B_0)_{C=d}$. Suppose

$$A_1 \subset B_0; \quad A_2 \subset B_1 \quad x$$

have relative densities ρ_1, ρ_2 , and let $S \subset G$ satisfy $|S| \geq 2|B_0|$. Then there is a regular Bohr set $D \subset B_1$ with

$$\text{rk } D \leq d + C \log^3 \frac{2}{\rho_1} \log \frac{2}{\rho_2} \quad (10)$$

and

$$|D| \geq \exp(-C) d \log \frac{2d}{\rho_1 \rho_2} + \log^3 \frac{2}{\rho_1} \log \frac{2}{\rho_2} \log \frac{2d}{\rho_1 \rho_2} |B_1| \quad (11)$$

such that

$$\int_D (f_{A_1} - f_{A_2}) \cdot \mathbf{1}_S = \int_{A_1} f_{A_1} - \int_{A_2} f_{A_2} \cdot \mathbf{1}_S \quad (12)$$

Proof. Choose a regular inner dilate $B_1^{[1]} = (B_1)_{C_0=d}$ with $|B_1^{[1]}| \geq 2|B_1|$. Apply Theorem 20.10 with the substitutions

$$A = A_2; \quad M = A_1; \quad L = S; \quad S_{\text{samp}} = B_1^{[1]}; \quad B = B_1^{[1]}.$$

The small-growth constant is

$$K = \frac{|B_1^{[1]}|}{|A_2|} \leq \frac{2}{\rho_2}.$$

The sampling density is one. Moreover, because $|S| \geq 2|B_0|$ and $|A_1| = \rho_1|B_0|$, the truncated ratio appearing in Theorem 20.10 satisfies

$$= \min \{1, |A_1|/|S|\} \geq 1/2.$$

Taking the theorem's L^1 error to be ϵ gives a regular $D \subset B_1$ whose rank increment is bounded by the right side of (10). Write $d_D = d$ for the rank increment of D . In the construction of Theorem 20.10, D is obtained by shrinking the old d frequencies of B_1 and adjoining $d_D = d$ new frequencies. Apply Lemma 18.4 with these two radius ratios. Substituting the theorem's radius lower bound and the rank-increment bound (10), and absorbing the harmless regularization constants, gives exactly the exponential lower bound (11).

It remains only to identify the smoothed quantity. At the origin the three-fold count

$$\mathbf{1}_{A_2} \cdot \mathbf{1}_{A_1} \cdot \mathbf{1}_S(0)$$

is exactly the number of pairs $(a_1, a_2) \in A_1 \times A_2$ for which $a_1 - a_2 \in S$. Dividing the normalized almost-periodicity estimate by $|A_1| |A_2|$ therefore turns its value into $\int_{A_1} f_{A_1} - \int_{A_2} f_{A_2} \cdot \mathbf{1}_S$; averaging the translates over D turns it into the first term in (12). The normalizing factors cancel, and (12) follows. $\square$

Lemma 20.12 (Localized Kelley Meka density increment). There are absolute $c, C > 0$ with the following property. Let $0 < \epsilon < 1/4$, let $p \geq 2$ be even, and let $k \geq 1$ be coprime to p . Let B, B^0, B^{00} be regular Bohr sets of rank d with $B^{00} \subset (B^0)_{C=d}$, and let $A \subset B$ have relative density ρ . If

$$k |A| \leq |A| k_{L^p(\cdot)} \leq (1 + \epsilon) |B|^{-1}; \quad \rho := \frac{|B^0|}{|B|} \leq \frac{|B^0|}{|B|} \leq \frac{|B^{00}|}{|B|} \leq \frac{|B^{00}|}{|B|}, \quad (13)$$

then there is a regular Bohr set $D \subset B^{00}$ satisfying

$$\text{rk } D \leq d + C \epsilon^p \log^4 \frac{2}{\rho}; \quad (14)$$

References

- [1] Thomas F. Bloom and Olof Sisask. “Breaking the logarithmic barrier in Roth’s theorem on arithmetic progressions”. In: *Ann. of Math.* 199 (2024), pp. 3–59.
- [2] Ernie Croot, Vsevolod Lev, and Peter Pach. “Progression-free sets in $\mathbb{Z}_4^n$ are exponentially small”. In: *Ann. of Math.* 185 (2017), pp. 331–337.
- [3] Jordan S. Ellenberg and Dion Gijswijt. “On large subsets of $\mathbb{F}_q^n$ with no three-term arithmetic progression”. In: *Annals of Mathematics* 185 (2017), pp. 339–343.
- [4] W. T. Gowers and Julia Wolf. “Linear forms and higher-degree uniformity for functions on $\mathbb{F}_p^n$ ”. In: *Geom. Funct. Anal.* 21 (2011), pp. 36–69.
- [5] Zander Kelley and Raghu Meka. *Strong bounds for 3-progressions*. Revised 2024. 2023. arXiv: [2302.05537](#).
- [6] James Leng, Ashwin Sah, and Mehtaab Sawhney. “Improved bounds for Szemerédi’s theorem”. In: *arXiv:2402.17995* (2024).
- [7] James Leng, Ashwin Sah, and Mehtaab Sawhney. “Quasipolynomial bounds on the inverse theorem for the Gowers $U^{s+1}[N]$ -norm”. In: *arXiv:2402.17994* (2024).
- [8] Rushil Raghavan. “Improved Bounds for 3-Progressions”. In: *arXiv:2603.27045* (2026). preprint.

Additive Combinatorics
Volume IV
Szemerédi's Theorem and Gowers Uniformity
Version 1.10

14 September 2026

Contents

1	The quantitative Szemerédi proof architecture	1
1.1	The three layers of the proof	1
1.2	Why we follow Gowers’s route rather than use the later inverse theorem . . .	1
2	Short Bridges in Gowers-Norm Arguments	3
2.1	Gowers-norm calculus	3
2.2	Quadratic calculus	4
2.3	Progression and transference bridges	4
2.4	Finite-field homothetic grids from partite hypergraph removal	5
3	Quantitative Gowers Localization and Density Increment	8
3.1	Uniformity degree: the convention used in this volume	8
3.2	The localization theorem	9
3.3	Structure of the proof	10
3.4	The $k = 4$ model is already internal	11
3.5	An auxiliary direct-correlation tail	11
3.6	A worked higher-order model: symmetry before integration	13
3.7	Global phase integration	14
3.8	Primary-source architecture of the Gowers localization core	16
3.9	Purification and multilinear extraction in the Gowers localization argument .	16
3.9.1	Arrangements and purification	17
3.9.2	From respected arrangements to a bilinear box	23
3.9.3	Higher-dimensional respected arrangements	27
3.9.4	The multiply-linear extraction induction	28
3.9.5	The local Section 17 twist	32
3.9.6	Degree lowering, polynomial untwisting, and the final partition bias .	33
4	From localization to quantitative Szemerédi	40
5	A proof notebook for the Gowers localization mechanism	43
5.1	From many additive quadruples to a Freiman map	43
5.2	The bilinear model	43
5.3	Why a multilinear phase can be integrated locally	44
5.4	The density-increment logic	44

Chapter 1

The quantitative Szemerédi proof architecture

Gowers’s quantitative proof of Szemerédi’s theorem combines several ideas that are individually familiar but difficult to coordinate: uniformity norms, generalized von Neumann inequalities, derivative structure, additive-combinatorial purification, multilinearization, and a density-increment iteration. The purpose of this opening chapter is to show how those pieces fit together before we enter the long localization argument.

1.1 The three layers of the proof

The proof is best understood as three interacting layers.

Uniformity and counting. The U^{k-1} norm measures whether a balanced function can influence the count of length- k progressions. Repeated Cauchy–Schwarz gives a generalized von Neumann inequality: if one input is sufficiently uniform, the whole progression average is small. Conversely, a progression deficit forces non-uniformity.

Structure and localization. Non-uniformity is not yet a density increment. For $k = 3$ Fourier analysis identifies a large frequency directly. For $k = 4$ the relevant object is quadratic. For general k , Gowers’s argument manufactures coherent derivative-frequency data, purifies it by Balog–Szemerédi–Gowers/Freiman methods, extracts multilinear structure, integrates that structure to a local polynomial phase, and finally localizes the phase on long progressions. Chapters 2–3 develop these interfaces carefully.

Iteration. Once one has a subprogression on which the density rises by a controlled amount, the rest is a stopping-time argument. Density cannot exceed one, so only finitely many increments occur; the quantitative difficulty is to ensure that the progression length does not collapse before the iteration stops. The final chapter carries out that bookkeeping and obtains the quantitative Szemerédi theorem.

1.2 Why we follow Gowers’s route rather than use the later inverse theorem

A later higher-order inverse theorem would shorten the proof substantially, but it would hide the mechanism that historically produced the first quantitative bounds for all progression lengths. This volume therefore contains the complete Gowers localization route itself. That

makes the logical relation between Volumes IV and V transparent: Volume IV proves Szemerédi quantitatively from the original uniformity/density-increment technology; Volume V then develops the stronger nilsequence inverse theory for its own sake.

Worked Example 1.1 (The $k = 3$ prototype). Let $A \subset \mathbb{Z}/N\mathbb{Z}$ have density α and put $f = 1_A - \alpha$. If the number of three-term progressions is smaller than the random main term, the Fourier expansion of the progression count forces a nonzero frequency γ with $|\widehat{f}(\gamma)| \gg_{\alpha} 1$. On a long progression on which $\gamma(x)$ varies little, the average of f cannot stay near zero; one of the cells therefore has density $> \alpha$. The general proof keeps this final logic but replaces the single frequency by a hierarchy of derivative-frequency data and the one-dimensional phase by a local polynomial object.

Chapter 2

Short Bridges in Gowers-Norm Arguments

This chapter records the short interfaces that connect the major proof blocks. They are proved here rather than hidden inside phrases such as “by standard Fourier analysis” or “by a routine regularity argument.”

2.1 Gowers-norm calculus

Theorem 2.1 (Positivity and seminorm property). *Let G be a finite abelian group with normalized counting measure. For $d \geq 1$ set*

$$\|f\|_{U^d}^{2^d} = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{\omega \in \{0,1\}^d} \mathcal{C}^{|\omega|} f(x + \omega \cdot h).$$

The displayed quantity is nonnegative and real. For $d \geq 2$, $\|\cdot\|_{U^d}$ is a seminorm, while $\|f\|_{U^1} = |\mathbb{E}f|$.

Proof. For $d = 1$, a change of variables gives $\|f\|_{U^1}^2 = |\mathbb{E}f|^2$. Pair the vertices of a d -cube according to the last coordinate to obtain

$$\|f\|_{U^d}^{2^d} = \mathbb{E}_h \|\Delta_h f\|_{U^{d-1}}^{2^{d-1}}, \quad \Delta_h f(x) = f(x + h) \overline{f(x)}.$$

Induction gives nonnegativity. Expanding the cube form for $f + g$ and applying Gowers–Cauchy–Schwarz to every mixed term bounds the term with j copies of g by $\|f\|_{U^d}^{2^{d-j}} \|g\|_{U^d}^j$. Summing the binomial expansion and taking the positive 2^d -th root gives the triangle inequality. $\square$

Lemma 2.2 (Derivative identity). *For $d \geq 2$,*

$$\|f\|_{U^d}^{2^d} = \mathbb{E}_h \|\Delta_h f\|_{U^{d-1}}^{2^{d-1}}.$$

Proof. Write a cube vertex as (ω', ε) . The two factors with the same ω' and last coordinate 0, 1 combine into the corresponding conjugate of $\Delta_{h_d} f(x + \omega' \cdot h')$. Averaging the resulting $(d-1)$ -cube and then h_d proves the formula. $\square$

Proposition 2.3 (Monotonicity). *For every finite abelian group and $d \geq 1$, $\|f\|_{U^d} \leq \|f\|_{U^{d+1}}$.*

Proof. In a $(d+1)$ -cube put f on the vertices with last coordinate 0 and 1 on those with last coordinate 1. The cube average is $\|f\|_{U^d}^{2^d}$. Gowers–Cauchy–Schwarz in dimension $d+1$ bounds it by $\|f\|_{U^{d+1}}^{2^d}$. $\square$

2.2 Quadratic calculus

Lemma 2.4 (Polarization). *Let V be a vector space over $\mathbb{F}_p$, p odd, and let $q : V \rightarrow \mathbb{F}_p$ have degree at most two. Then*

$$B_q(x, y) = q(x + y) - q(x) - q(y) + q(0)$$

is symmetric bilinear and

$$q(x) = q(0) + \ell(x) + \frac{1}{2}B_q(x, x)$$

for a unique linear form ℓ .

Proof. The constant and linear terms cancel under the second additive derivative, while every quadratic monomial polarizes bilinearly. Subtracting $q(0) + \frac{1}{2}B_q(x, x)$ leaves a polynomial with vanishing second additive derivative and zero constant term, hence a linear form. $\square$

Theorem 2.5 (Quadratic Gauss estimate by rank). *If B_q has rank r and $\ell \in V^*$, then*

$$|\mathbb{E}_x e_p(q(x) + \ell(x))| \leq p^{-r/2}.$$

The average vanishes unless the induced linear character is trivial on $\text{rad } B_q$; in that case equality holds in magnitude.

Proof. Let S denote the average. Squaring and writing $y = x + h$ gives

$$|S|^2 = \mathbb{E}_h e_p(q(h) - q(0) + \ell(h)) \mathbb{E}_x e_p(B_q(x, h)).$$

The inner average is 1 precisely for $h \in \text{rad } B_q$ and 0 otherwise. On the radical the remaining phase is linear, so its average is either 0 or $|\text{rad } B_q|/|V| = p^{-r}$. $\square$

Corollary 2.6 (Bias versus rank). *If $|\mathbb{E}_x e_p(q(x))| \geq \eta$, then $\text{rank}(B_q) \leq 2 \log_p(1/\eta)$.*

Proof. Combine the preceding theorem with $\eta \leq p^{-r/2}$. $\square$

2.3 Progression and transference bridges

Proposition 2.7 (Uniformity obstruction). *Fix $k \geq 3$. Assume the generalized von Neumann bound*

$$|\Lambda_k(f_0, \dots, f_{k-1})| \leq \min_i \|f_i\|_{U^{k-1}}$$

for 1-bounded functions in the ambient finite group. If A has density α and, after the explicitly separated degenerate contribution, its k -AP count differs from the constant-model main term by at least η , then for $f = 1_A - \alpha$,

$$\|f\|_{U^{k-1}} \geq \frac{\eta}{2^k - 1}.$$

Proof. Expand $1_A = \alpha + f$ in the multilinear progression average. The constant term is the main term; single- f terms vanish in the translation-invariant setting. At most $2^k - 1$ remaining terms carry the discrepancy, so one has magnitude at least $\eta/(2^k - 1)$ and is bounded by generalized von Neumann. $\square$

Lemma 2.8 (Finite-dimensional separation). *If $K \subset \mathbb{R}^m$ is nonempty compact convex and $x \notin K$, then a vector v strictly separates x from K .*

Proof. Choose $y_0 \in K$ minimizing $|x - y|$. Differentiating $|x - y_0 - t(y - y_0)|^2$ at $t = 0^+$ gives $(x - y_0) \cdot (y - y_0) \leq 0$ for every $y \in K$. Take $v = x - y_0$. $\square$

Proposition 2.9 (Counting stability). *Let Λ be a fixed multilinear configuration average with a generalized von Neumann estimate at U^{s+1} . If $f_i = f_{i,\text{str}} + f_{i,\text{unf}} + f_{i,\text{err}}$, the structured factors have fixed L^∞ bounds, $\|f_{i,\text{unf}}\|_{U^{s+1}} \leq \eta$, and $\|f_{i,\text{err}}\|_2 \leq \varepsilon$, then*

$$\Lambda(f_1, \dots, f_t) = \Lambda(f_{1,\text{str}}, \dots, f_{t,\text{str}}) + O_t(\eta + \varepsilon)$$

under the fixed norm hypotheses for the other slots.

Proof. Telescope one slot at a time. A uniform factor costs $O(\eta)$ by generalized von Neumann. An error factor costs $O(\varepsilon)$ by Cauchy–Schwarz after an affine change of variables. There are finitely many terms depending only on t . $\square$

Theorem 2.10 (Abstract relative Szemerédi theorem). *Fix $k \geq 3$ and $\delta > 0$. Let (ν_N) be a sequence of nonnegative majorants on finite ambient groups with $\mathbb{E}\nu_N \rightarrow 1$, and assume that every finite test-product error required by the dense-model theorem and the relative generalized von Neumann estimate tends to 0 as $N \rightarrow \infty$. If $0 \leq f_N \leq \nu_N$ and $\mathbb{E}f_N \geq \delta$ for every N , then there is $c = c(k, \delta) > 0$ such that*

$$\liminf_{N \rightarrow \infty} \Lambda_k^{\text{nd}}(f_N) \geq c,$$

where Λ_k^{nd} is the normalized nondegenerate k -term arithmetic-progression average on the ambient group.

Proof. Use the dense-model theorem to approximate f_N by $g_N \in [0, 1]$ with density at least $\delta/2$ against the full k -AP test family. Dense Szemerédi plus Varnavides gives $\Lambda_k(g_N) \geq c(k, \delta) > 0$. Telescope $\Lambda_k(f_N) - \Lambda_k(g_N)$; each difference slot is controlled by the relative generalized von Neumann/test-family estimate, so the total is $o(1) + O_k(\varepsilon)$. Choose $\varepsilon < c(k, \delta)/(4k)$. $\square$

2.4 Finite-field homothetic grids from partite hypergraph removal

The finite-field inverse theorem in Volume V needs a multidimensional recurrence input. Rather than importing the Furstenberg–Katznelson/Bergelson–Leibman–McCutcheon theorem, we derive exactly the required finitary supersaturation statement from the partite hypergraph-removal theorem proved in the hypergraph-removal appendix of Volume III.

Theorem 2.11 (Finite-field multidimensional Szemerédi, Varnavides form). *Fix a prime p , an integer $m \geq 1$, and $\alpha > 0$. There exists $c = c(p, m, \alpha) > 0$ such that for every finite-dimensional $\mathbb{F}_p$ -vector space V and every $A \subseteq V^m$ with $|A| \geq \alpha|V|^m$, the number of tuples $(a_1, \dots, a_m, h) \in V^{m+1}$ satisfying*

$$(a_1 + \omega_1 h, \dots, a_m + \omega_m h) \in A \quad \text{for every } \omega = (\omega_1, \dots, \omega_m) \in \mathbb{F}_p^m$$

is at least $c|V|^{m+1}$.

Proof. Put $\Omega = \mathbb{F}_p^m$ and $K = |\Omega| = p^m$. We apply the internally proved partite hypergraph-removal theorem to the boundary of the $(K - 1)$ -simplex, whose vertex set is Ω and whose top faces are

$$e_\omega = \Omega \setminus \{\omega\} \quad (\omega \in \Omega).$$

Every vertex class is a copy of V . Thus a labelled simplex is a tuple $x = (x_\nu)_{\nu \in \Omega} \in V^\Omega$.

For $\omega \in \Omega$ define the face map

$$\psi_\omega(x_{e_\omega}) := \sum_{\nu \neq \omega} (\omega - \nu)x_\nu \in V^m,$$

where a vector $\lambda = (\lambda_1, \dots, \lambda_m) \in \mathbb{F}_p^m$ acts on $x \in V$ by $\lambda x = (\lambda_1 x, \dots, \lambda_m x)$. The coefficient of x_ω is zero, so ψ_ω really is a function of the face variables x_{e_ω} . Declare x_{e_ω} to be an edge of type ω exactly when $\psi_\omega(x_{e_\omega}) \in A$.

Now set

$$h(x) := \sum_{\nu \in \Omega} x_\nu \in V, \quad a(x) := - \sum_{\nu \in \Omega} \nu x_\nu \in V^m.$$

Then for every $\omega \in \Omega$,

$$a(x) + \omega h(x) = \sum_{\nu \in \Omega} (\omega - \nu) x_\nu = \psi_\omega(x_{e_\omega}).$$

Consequently labelled simplices are exactly the preimages of the desired homothetic Ω -grids.

We next compute the multiplicity of this encoding. The linear map

$$\Phi : V^\Omega \longrightarrow V^m \times V, \quad x \longmapsto (a(x), h(x))$$

is surjective. Indeed, the coefficient vectors $(1, -\nu) \in \mathbb{F}_p^{m+1}$, $\nu \in \Omega$, span $\mathbb{F}_p^{m+1}$: the vectors corresponding to $\nu = 0, e_1, \dots, e_m$ already do so. Hence the scalar coefficient matrix has rank $m + 1$, and tensoring with V shows that every fibre of Φ has cardinality

$$|V|^{K-m-1}.$$

If N_A denotes the number of desired tuples (a, h) , the number of labelled simplices is therefore exactly

$$N_A |V|^{K-m-1}. \quad (*)$$

Choose

$$\varepsilon < \frac{\alpha}{2K}$$

and let $\delta = \delta(K, \varepsilon) > 0$ be the removal threshold supplied by the partite hypergraph-removal theorem. Suppose for contradiction that $N_A < \delta |V|^{m+1}$. By (*) the simplex density is then less than δ , so one can destroy all labelled simplices by deleting at most $\varepsilon |V|^{K-1}$ edges from each of the K edge types.

But the degenerate grids $h = 0$ alone create too many simplices. For every $a \in A$ the pair $(a, 0)$ is a valid grid, and its fibre under Φ has size $|V|^{K-m-1}$. Thus there are at least

$$|A| |V|^{K-m-1} \geq \alpha |V|^{K-1}$$

degenerate labelled simplices. A fixed edge of type ω belongs to at most one such degenerate simplex: after the $K - 1$ face variables have been fixed, the equation $h(x) = 0$ uniquely determines the missing variable x_ω . Hence all deleted edges together can destroy at most

$$K\varepsilon |V|^{K-1} < \frac{\alpha}{2} |V|^{K-1}$$

degenerate simplices, a contradiction. Therefore $N_A \geq \delta |V|^{m+1}$, and we may take $c = \delta$. $\square$

Idea Why the simplex encoding is the right one

The formula

$$\psi_\omega(x) = a(x) + \omega h(x)$$

is the multidimensional analogue of the simplex–progression dictionary used earlier for arithmetic progressions. The key design condition is that the coefficient of x_ω vanishes in ψ_ω , so membership of the ω th pattern point in A can be represented by an edge on

the opposite face. The removal lemma then turns “few complete grids” into a small deletion, while the constant grids $h = 0$ supply the large diagonal family that makes such a deletion impossible.

Chapter 3

Quantitative Gowers Localization and Density Increment

3.1 Uniformity degree: the convention used in this volume

The original Gowers paper indexes uniformity by the *degree of the pseudorandomness condition*, not by the index of the Gowers norm. That distinction is easy to lose because the two indices differ by one. We therefore fix the convention before stating the localization theorem.

Definition 3.1 (Cyclic degree- d uniformity mass). Let $G = \mathbb{Z}/N\mathbb{Z}$, let $h : G \rightarrow \mathbb{C}$ satisfy $|h| \leq 1$, and let $d \geq 0$. Define

$$U_d(h; G) := \|h\|_{U^{d+1}(G)}^{2^{d+1}}.$$

Equivalently,

$$U_d(h; G) = N^{-(d+2)} \sum_{a_1, \dots, a_d \in G} \left| \sum_{x \in G} \Delta_{a_1, \dots, a_d} h(x) \right|^2, \quad (\text{U.1})$$

where the empty derivative is h when $d = 0$. We say that h is *uniform of degree d at level ρ* if

$$U_d(h; G) \leq \rho,$$

and that it *fails degree- d uniformity at level ρ* otherwise.

Thus degree 0 uniformity is the mean condition $|\mathbb{E}_G h|^2 \leq \rho$, degree 1 uniformity is the U^2 condition, and in general

$$\boxed{\text{Gowers degree } d \iff U^{d+1}.} \quad (\text{U.2})$$

In particular, the uniformity condition controlling k -term arithmetic progressions has *degree $k - 2$* and is measured by U^{k-1} .

Definition 3.2 (Degree uniformity on a progression). Let

$$P = \{a + jd : 0 \leq j < L\}$$

be an arithmetic progression with distinct terms, either in the integers or in a prime cyclic group, and let $h : P \rightarrow \mathbb{C}$. Transport h to $[L] = \{0, \dots, L-1\}$ by $j \mapsto a + jd$. Choose a prime $M > 4(d_0 + 2)L$, where d_0 is the uniformity degree under discussion, extend the transported function by zero to $\mathbb{Z}/M\mathbb{Z}$, and put

$$U_{d_0}(h; P) := \frac{\|\tilde{h}\|_{U^{d_0+1}(\mathbb{Z}/M\mathbb{Z})}^{2^{d_0+1}}}{\|1_{[L]}\|_{U^{d_0+1}(\mathbb{Z}/M\mathbb{Z})}^{2^{d_0+1}}}. \quad (\text{U.3})$$

If $\mathcal{P}$ is a partition of P into progressions, define the relative uniformity mass

$$U_{d_0}(h \mid \mathcal{P}) := \sum_{Q \in \mathcal{P}} \frac{|Q|}{|P|} U_{d_0}(h; Q). \quad (\text{U.4})$$

Lemma 3.3 (The progression definition is intrinsic). *The ratio in (U.3) is independent of the auxiliary prime M once $M > 4(d_0 + 2)L$. Moreover*

$$0 \leq U_{d_0}(h; P) \leq 1 \quad (|h| \leq 1),$$

and for $d_0 = 0$ one has

$$U_0(h; P) = |\mathbb{E}_{x \in P} h(x)|^2.$$

Proof. After zero extension, a nonzero $(d_0 + 1)$ -cube has all of its vertices in $[L]$. Since $M > 4(d_0 + 2)L$, such a cube cannot wrap around the auxiliary cyclic group: the increments determined by the one-coordinate edges have unique integer lifts of modulus $< L$, and every remaining vertex has integer lift of modulus $< (d_0 + 2)L < M/4$. Hence the *unnormalized* cube sum in the numerator is an integer-interval cube sum independent of M . The same is true for the denominator. The normalized U^{d_0+1} powers in numerator and denominator both contain the same factor $M^{-(d_0+2)}$, so it cancels.

For $|h| \leq 1$, every numerator cube has modulus at most the corresponding indicator cube, and Gowers–Cauchy–Schwarz makes the cube sum nonnegative; hence the ratio lies in $[0, 1]$. When $d_0 = 0$, the numerator is $M^{-2} |\sum_P h|^2$ and the denominator is $M^{-2} |P|^2$, giving the last formula. $\square$

3.2 The localization theorem

For general k , the density-increment mechanism starts from failure of *degree $k - 2$* uniformity, equivalently from a large U^{k-1} statistic. The threshold is the same one used in the uniform counting alternative of Lemma 4.1; this makes the dichotomy in Chapter 4 literal rather than notational.

Theorem 3.4 (Quantitative Gowers localization theorem). *For every $k \geq 3$ there are computable positive functions*

$$\iota_k : (0, 1] \rightarrow (0, 1], \quad \sigma_k : (0, 1] \rightarrow (0, 1], \quad N_{0,k} : (0, 1] \rightarrow \mathbb{N}$$

with the following property. Let P be an arithmetic progression of length $N \geq N_{0,k}(\alpha)$ and let $A \subset P$ have relative density α . Put

$$f = 1_A - \alpha \quad \text{on } P.$$

If f fails degree- $(k - 2)$ uniformity at the level $\eta_k(\alpha)$ fixed in Lemma 4.1, that is,

$$U_{k-2}(f; P) > \eta_k(\alpha), \quad (\text{U.5})$$

then there is a subprogression $P' \subset P$ such that

$$|P'| \geq N^{\sigma_k(\alpha)}, \quad \text{dens}_{P'}(A) \geq \alpha + \iota_k(\alpha). \quad (\text{U.6})$$

The functions may be chosen nondecreasing in the density parameter. They are obtained by a finite recursion in the uniformity degree $d = k - 2$ from the Fourier base case and, from degree two onward, the quantitative constants in the multilinear-extraction theorem, the local polynomial-twist proposition, the polynomial-phase subdivision lemma, and the elementary bias-selection lemma below. In particular, no later nilsequence inverse theorem is used.

Proof location. The complete degree-lowering induction and the passage from partition bias to (U.6) are in Section 3.9.6. The statement above is the exact dependency interface used by the iteration chapter.

Idea Where the real difficulty lies

The cases $k = 3$ and $k = 4$ already illustrate the two basic phenomena: Fourier localization and quadratic localization. For arbitrary k , however, one must construct and integrate higher multilinear derivative data. The long argument below follows the mechanisms of Gowers’s Sections 12–17 rather than replacing them by a later nilsequence inverse theorem.

3.3 Structure of the proof

The proof is decomposed into the following steps.

- G1. Approximate frequency selection.** Repeated Gowers–Cauchy–Schwarz converts failure of degree- $(k - 2)$ uniformity—equivalently, a large U^{k-1} statistic—into many derivatives carrying coherent large Fourier data.
- G2. Additive-quadruple amplification.** The selected frequency map respects many additive quadruples; quantitative BSG extracts a large subset on which it behaves as a high-order Freiman homomorphism.
- G3. Bihomomorphism stage.** For $k \geq 5$, derivative parameters lead to a partially defined bihomomorphism; one must strengthen it on a dense structured domain.
- G4. Respected-arrangement calculus.** Cube identities supply many compatible additive arrangements and a density increment among arrangements.
- G5. Bilinear-piece extraction.** From the respected arrangements one extracts a large box on which the top derivative data is genuinely bilinear up to controlled lower-order errors.
- G6. Multilinearization.** Inductively pass from a bilinear piece to the $(k - 3)$ -multilinear frequency object supplied by failure of degree- $(k - 2)$ uniformity. This is the index appearing in Corollary 3.39.
- G7. Symmetrization.** Commutation of additive derivatives kills the antisymmetric defects or pushes them into lower-degree terms already controlled inductively.
- G8. Phase integration/local polynomial model.** Integrate the symmetric multilinear datum to the polynomial twist used in the main inductive step.
- G9. One-degree descent.** Proposition 3.42 converts that multilinear box into failure of degree- $(k - 3)$ uniformity for a polynomially twisted function, relative to a progression partition.
- G10. Induction on uniformity degree.** Apply the already-closed degree- $(k - 3)$ case on the nonuniform cells. This produces a partition on which the *twisted* function has large total cell bias.
- G11. Polynomial untwisting.** Subdivide the cells so that the polynomial phase is almost constant and transfer twisted cell bias back to the original balanced function.
- G12. Density increment and parameter closure.** Properize the partition, select a long positively biased cell, and verify that all losses give positive computable functions $\iota_k, \sigma_k, N_{0,k}$.

G1–G2 are supplied by the Gowers calculus, BSG, and Freiman theory already developed earlier in this book, while the $k = 4$ case supplies the quadratic integration model.

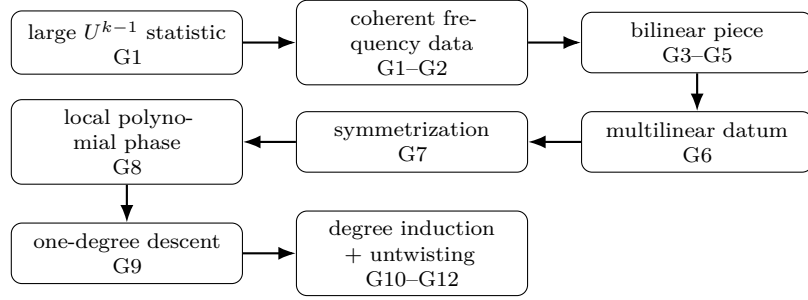


Figure 3.1: Proof architecture of quantitative Gowers localization. The middle passage G3–G9 is the genuinely higher-order part: additive derivative data must be purified, multilinearized, symmetrized, integrated, and finally localized before the elementary density-increment tail can begin.

The genuinely new general-order work is G3–G9: bilinear extraction, multilinearization, symmetrization, integration, and one-degree descent. G10–G12 then close the induction by reusing the lower-degree case and removing the polynomial twist. We prove those mechanisms explicitly rather than appeal to an unnamed “higher-order Fourier analysis” theorem.

3.4 The $k = 4$ model is already internal

For length four, the general picture collapses to the quadratic case. The recovered cyclic U^3 inverse theorem produces local quadratic phases, while the quadratic Gauss/rank calculus and simultaneous localization turn such a phase into a density increment on a long subprogression. Consequently the difficult novelty of IV.3.003 is not the quadratic base case but the induction producing and integrating higher multilinear data.

Idea Why this boundary matters

If one replaced G3–G9 by the later Green–Tao–Ziegler inverse theorem, the deduction of Szemerédi would become much shorter, but Volume IV would then depend forward on Volume V. We instead prove Gowers’s quantitative route here, preserving the logical order from uniformity to density increment before higher-order inverse theory is introduced.

3.5 An auxiliary direct-correlation tail

A polynomial correlation can sometimes be localized directly when major-arc coefficient control is already available. The next two results record that useful auxiliary mechanism. They are *not* the general G9-to-density-increment step: Proposition 3.42 produces lower-degree relative nonuniformity, and the general proof closes that output by the degree induction in Section 3.9.6.

Lemma 3.5 (Major-arc polynomial phases are constant on long progression blocks). *Fix $d, m \geq 1$ and $0 < \varepsilon < 1/10$. Let*

$$\phi_j(n) = e\left(\sum_{r=0}^d \theta_{j,r} n^r\right), \quad 1 \leq j \leq m,$$

on an interval $I \subset [-N, N]$. Suppose there are integers $q, H \geq 1$ and integers $a_{j,r}$ such that,

for every j and $1 \leq r \leq d$,

$$\theta_{j,r} = \frac{a_{j,r}}{q} + \frac{\delta_{j,r}}{q}, \quad |\delta_{j,r}| \leq \frac{\varepsilon}{C_d m d N^{r-1} H},$$

where, for definiteness, one may take

$$C_d := 4\pi(d+1).$$

Then every intersection of I with a block

$$\{b + q(t_0 + h) : 0 \leq h < H\}$$

on a fixed residue class modulo q has

$$\sup_{n, n' \text{ in the block}} |\phi_j(n) - \phi_j(n')| \leq \varepsilon \quad (1 \leq j \leq m).$$

Proof. Take two points n, n' in the same block. Since $n - n'$ is divisible by q , so is $n^r - (n')^r$ for every r . Hence the rational part contributes an integer:

$$\frac{a_{j,r}}{q} (n^r - (n')^r) \in \mathbb{Z}.$$

For the error part, factorization gives

$$|n^r - (n')^r| \leq r N^{r-1} |n - n'| \leq r N^{r-1} q H,$$

because $|n|, |n'| \leq N$ and $|n - n'| < qH$. Hence

$$\left| \sum_{r=1}^d \frac{\delta_{j,r}}{q} (n^r - (n')^r) \right| \leq \frac{\varepsilon}{C_d m d} \sum_{r=1}^d r = \frac{d+1}{2C_d m} \varepsilon.$$

Using $|e(x) - e(y)| \leq 2\pi|x - y|$ and $C_d = 4\pi(d+1)$ gives

$$|\phi_j(n) - \phi_j(n')| \leq \frac{\varepsilon}{4m} \leq \varepsilon.$$

This proves the asserted uniform oscillation bound. $\square$

Lemma 3.6 (Correlation with an almost constant phase gives a density increment). *Let P be a finite progression, $A \subset P$ have relative density α , and $f = 1_A - \alpha$. Suppose P is partitioned into cells Q together with an exceptional set E , with $|E| \leq \rho|P|$. Let $\psi : P \rightarrow \mathbb{C}$ satisfy $|\psi| \leq 1$, and assume that on every nonexceptional cell there is a constant z_Q , $|z_Q| \leq 1$, such that*

$$\sup_{n \in Q} |\psi(n) - z_Q| \leq \varepsilon.$$

If

$$|\mathbb{E}_{n \in P} f(n) \psi(n)| \geq \kappa,$$

then, provided $\kappa > 2(\varepsilon + \rho)$, one of the nonexceptional cells satisfies

$$\text{dens}_Q(A) \geq \alpha + \frac{\kappa - 2(\varepsilon + \rho)}{4}.$$

Proof. Rotate ψ by a scalar of modulus one so that the displayed correlation has real part at least κ . Write

$$d_Q = \mathbb{E}_{n \in Q} f(n) = \text{dens}_Q(A) - \alpha.$$

Replacing ψ by z_Q on each good cell changes the normalized correlation by at most ε ; the exceptional set contributes at most ρ , since $|f| \leq 1$. Hence

$$\sum_Q \frac{|Q|}{|P|} \Re(z_Q d_Q) \geq \kappa - (\varepsilon + \rho).$$

Therefore

$$\sum_Q \frac{|Q|}{|P|} |d_Q| \geq \kappa - (\varepsilon + \rho).$$

On the other hand $\mathbb{E}_P f = 0$. Removing the exceptional set changes the weighted mean of the d_Q by at most ρ , so the total positive weighted mass of the d_Q is at least

$$\frac{\kappa - 2(\varepsilon + \rho)}{2}.$$

If every positive d_Q were smaller than $(\kappa - 2(\varepsilon + \rho))/4$, their weighted total would be smaller than that quantity, a contradiction. Thus some good Q has the asserted increment. $\square$

Corollary 3.7 (Direct phase correlation gives a density increment). *Suppose a separate argument produces a correlation*

$$|\mathbb{E}_P f(n) \Psi(n)| \geq \kappa$$

and a partition of all but at most $\rho|P|$ points of P into subprogressions of length at least $|P|^c$ on which Ψ oscillates by at most ε , with $\varepsilon + \rho < \kappa/2$. Then one of those subprogressions has density at least

$$\alpha + \frac{\kappa - 2(\varepsilon + \rho)}{4}.$$

Thus a genuine phase correlation has an elementary density-increment tail. The general Gowers argument below does not assume that Proposition 3.42 already provides such a correlation.

Proof. Apply Lemma 3.6 to the cells supplied by G9. $\square$

3.6 A worked higher-order model: symmetry before integration

The G7–G8 interface is technically much harder over long progressions, but its algebraic reason can already be seen in the clean finite-field model below. This example is not used to certify G7 or G8; it is included so that the student can see why symmetrization is structurally necessary before one tries to integrate the top derivative.

Proposition 3.8 (Integrating a symmetric multilinear form in high characteristic). *Let $r \geq 2$, let $p > r$ be prime, and let W be an $\mathbb{F}_p$ -vector space. If*

$$T : W^r \rightarrow \mathbb{F}_p$$

is symmetric and r -linear, then

$$P(x) := \frac{1}{r!} T(x, \dots, x)$$

is a classical polynomial of additive degree at most r and

$$\Delta_{h_1} \cdots \Delta_{h_r} P(x) = T(h_1, \dots, h_r)$$

for every $x, h_1, \dots, h_r \in W$.

Proof. Expand

$$T(x + u, \dots, x + u)$$

by multilinearity. It is a sum of terms obtained by choosing, in each of the r slots, either x or u . Applying a difference Δ_h removes the term in which no slot uses h ; after r successive differences only those terms survive in which every one of $h_1, \dots, h_r$ occurs exactly once. There are $r!$ such assignments to the r slots. Symmetry of T makes all of them equal to $T(h_1, \dots, h_r)$. Hence

$$\Delta_{h_1} \cdots \Delta_{h_r} T(x, \dots, x) = r! T(h_1, \dots, h_r).$$

Because $p > r$, $r!$ is invertible in $\mathbb{F}_p$, proving the displayed identity for P . One additional difference vanishes because the right-hand side is independent of x , so P has additive degree at most r . $\square$

Worked Example 3.9 (Why the symmetry step cannot simply be skipped). Take $r = 2$. If a scalar phase P has a bilinear second derivative

$$B(h, k) = \Delta_h \Delta_k P(x),$$

then commutativity of additive differences forces $B(h, k) = B(k, h)$. Thus an arbitrary bilinear object extracted from additive energy cannot already be the Hessian of a phase: its antisymmetric defect must first be eliminated or pushed into a lower-order error. For $r > 2$ the same observation applies to every permutation of the derivative directions. Proposition 3.8 then shows what integration looks like once the symmetry problem has genuinely been solved.

3.7 Global phase integration

The preceding discussion isolated G7–G9 conceptually. We now reconstruct the *exact global algebraic step* used in Gowers’s Section 17. This is worth doing separately because it cleanly distinguishes a theorem that is genuinely elementary once multilinearity has been obtained from the much harder local-box argument that follows it. Throughout this section N is prime and $N > r$ for every multilinearity order r under discussion.

Write $e_N(t) = \exp(2\pi it/N)$. For a function $f : \mathbb{Z}_N \rightarrow \mathbb{C}$ define

$$\Delta_{h_1, \dots, h_r} f(x) := \prod_{\omega \in \{0, 1\}^r} \mathcal{C}^{|\omega|} f(x - \omega \cdot h),$$

where $\mathcal{C}$ denotes complex conjugation, and put

$$\widehat{F}(\xi) := \sum_{x \in \mathbb{Z}_N} F(x) e_N(-\xi x).$$

Thus

$$\|f\|_{U^r(\mathbb{Z}_N)}^{2r} = N^{-(r+1)} \sum_{x, h_1, \dots, h_r} \Delta_{h_1, \dots, h_r} f(x).$$

Lemma 3.10 (Polarisation of a multilinear frequency). *Let $\sigma : \mathbb{Z}_N^r \rightarrow \mathbb{Z}_N$ be r -linear. Then there is a polynomial $q : \mathbb{Z}_N \rightarrow \mathbb{Z}_N$ of degree at most r such that*

$$\Delta_{h_1}^{\text{add}} \cdots \Delta_{h_r}^{\text{add}} q(x) = \sigma(h_1, \dots, h_r) \quad (\text{GI.1})$$

for every $x, h_1, \dots, h_r \in \mathbb{Z}_N$, where $\Delta_h^{\text{add}} q(x) = q(x+h) - q(x)$. Equivalently,

$$\sigma(h_1, \dots, h_r) = \sum_{\omega \in \{0,1\}^r} (-1)^{r-|\omega|} q(x + \omega \cdot h). \quad (\text{GI.2})$$

Proof. Because each variable ranges in the one-dimensional vector space $\mathbb{Z}_N$ over itself, multilinearity forces

$$\sigma(h_1, \dots, h_r) = c h_1 \cdots h_r$$

for the scalar $c = \sigma(1, \dots, 1)$. Since $N > r$, the factorial $r!$ is invertible modulo N . Take

$$q(x) = \frac{c}{r!} x^r.$$

The r -fold additive difference of x^r is independent of the base point and equals $r! h_1 \cdots h_r$. One way to see this without suppressing a step is to expand $(x + \omega \cdot h)^r$ in the alternating cube sum. Every monomial missing one of the variables $h_1, \dots, h_r$ cancels after summing over the corresponding cube coordinate. Since the total degree is r , the only surviving term contains every h_i exactly once; it occurs with coefficient $r!$. Dividing by $r!$ gives (GI.1), and expanding the difference operators gives (GI.2). $\square$

Proposition 3.11 (Global multilinear frequency can be integrated by a polynomial twist). *Let $|f| \leq 1$ on $\mathbb{Z}_N$, let $\sigma : \mathbb{Z}_N^k \rightarrow \mathbb{Z}_N$ be k -linear, and suppose*

$$\sum_{h \in \mathbb{Z}_N^k} |\widehat{\Delta_h f}(\sigma(h))|^2 \geq \alpha N^{k+2}. \quad (4.22)$$

Then there is a polynomial $q : \mathbb{Z}_N \rightarrow \mathbb{Z}_N$ of degree at most $k+1$ such that, for $g(x) = f(x)e_N(-q(x))$,

$$\|g\|_{U^{k+1}(\mathbb{Z}_N)}^{2^{k+1}} \geq \alpha. \quad (4.23)$$

In particular, a coherent global k -linear choice of large derivative frequencies lowers the uniformity problem by one degree after a polynomial phase twist.

Proof. For a fixed $h = (h_1, \dots, h_k)$, expand the square of the Fourier coefficient:

$$\begin{aligned} |\widehat{\Delta_h f}(\sigma(h))|^2 &= \sum_{x,y} \Delta_h f(x) \overline{\Delta_h f(y)} e_N(-\sigma(h)(x-y)) \\ &= \sum_{x,u} \Delta_{h_1, \dots, h_k, u} f(x) e_N(-u\sigma(h)), \end{aligned}$$

where in the second line we put $u = x - y$; a translation of x if necessary only changes the harmless orientation convention in the cube derivative.

The map

$$\rho(h_1, \dots, h_k, u) = u\sigma(h_1, \dots, h_k)$$

is $(k+1)$ -linear. By Lemma 3.10, choose a polynomial q of degree at most $k+1$ with

$$\Delta_{h_1}^{\text{add}} \cdots \Delta_{h_k}^{\text{add}} \Delta_u^{\text{add}} q(x) = \rho(h, u).$$

Multiplying f by $e_N(-q)$ therefore inserts exactly the phase $e_N(-\rho(h, u))$ into its $(k + 1)$ -dimensional multiplicative cube derivative:

$$\Delta_{h_1, \dots, h_k, u} g(x) = \Delta_{h_1, \dots, h_k, u} f(x) e_N(-u \sigma(h)).$$

Summing the preceding Fourier-square identity over h and using (4.22) gives

$$\sum_{x, h_1, \dots, h_k, u} \Delta_{h_1, \dots, h_k, u} g(x) \geq \alpha N^{k+2}.$$

After division by N^{k+2} this is precisely (4.23). $\square$

Worked Example 3.12 (The quadratic case of the polarisation identity). For a bilinear frequency $\sigma(h_1, h_2) = ch_1 h_2$, take $q(x) = cx^2/2$. Then

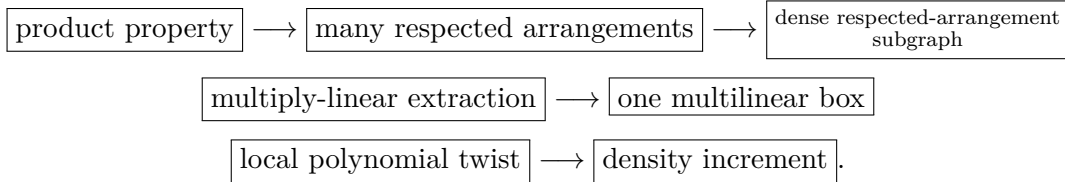
$$q(x + h_1 + h_2) - q(x + h_1) - q(x + h_2) + q(x) = ch_1 h_2.$$

Thus the phase attached to a second derivative really is the Hessian of a quadratic phase. The higher-degree lemma is the same finite-difference identity, not a new inverse theorem.

3.8 Primary-source architecture of the Gowers localization core

The proof follows the original article's order rather than compressing several distinct mechanisms into one synthetic step. The central theorem is the multiply-linear extraction theorem corresponding to Gowers's Theorem 16.2. Its proof must include the induction base and all induction lemmas needed to pass from the product property to a covering by boundedly many multilinear graphs on almost all base points. Only after that theorem is internal may Corollary 16.11 be used.

For the reader, the logical route is



The point of recording this route explicitly is that none of the middle three arrows may be replaced by a citation to a later higher-order inverse theorem: doing so would make Volume IV depend forward on Volume V.

3.9 Purification and multilinear extraction in the Gowers localization argument

The proof follows the order of Gowers's original §§12–17. We isolate the combinatorial objects first, then prove purification and multilinear extraction, and only at the end perform the local polynomial twist. This order matters: the local polynomial phase is a consequence of the multilinear box, not a substitute for the extraction theorem. This organization mirrors the logical order of the argument and keeps the dependence of each localization step explicit.

Throughout, N is prime and all additive variables lie in $\mathbb{Z}_N$. A r -box is a product $P = P_1 \times \dots \times P_r$ of arithmetic progressions having the same common difference. We write $\text{wd}(P)$ for the minimum length of its coordinate progressions. Constants depending only on r are denoted by $C_r, c_r > 0$; their exact numerical values are immaterial for the iteration, while their dependence on the density parameters is recorded explicitly.

3.9.1 Arrangements and purification

Definition 3.13 (Gowers d -arrangements in two variables). Let $B \subseteq \mathbb{Z}_N^2$ and $\phi : B \rightarrow \mathbb{Z}_N$. For a height $h \in \mathbb{Z}_N$ write

$$\phi_h(x, y) := \phi(x, y + h) - \phi(x, y),$$

whenever both points (x, y) and $(x, y + h)$ lie in B . A d -arrangement of height h is the sequence of $4d$ points

$$(x_i, y_i), (x_i, y_i + h) \quad (1 \leq i \leq 2d)$$

contained in B , with

$$x_1 + \cdots + x_d = x_{d+1} + \cdots + x_{2d}. \quad (4.1)$$

It is *respected* by ϕ if

$$\sum_{i=1}^d \phi_h(x_i, y_i) = \sum_{i=d+1}^{2d} \phi_h(x_i, y_i). \quad (4.2)$$

Thus an 8-arrangement has thirty-two vertices and, before membership in B is imposed, 32 free scalar parameters: the 16 variables x_i , the 16 variables y_i , and h , subject to the single relation (4.1). This is the arrangement used in Gowers's §12; it should not be confused with a 4×4 grid.

Definition 3.14 (Higher-dimensional arrangements). Let $k \geq 1$ and $B \subseteq \mathbb{Z}_N^{k+1}$. A d -arrangement in B is a sequence $C_1, \dots, C_{2d}$ of congruent k -dimensional cubes, with C_j lying in the cross-section whose last coordinate is r_j , and

$$r_1 + \cdots + r_d = r_{d+1} + \cdots + r_{2d}. \quad (4.3)$$

Equivalently it is a map

$$\rho : \{0, 1\}^k \times \{1, \dots, 2d\} \longrightarrow \mathbb{Z}_N^{k+1}, \quad \rho(\omega, j) = (y_1^j + \omega_1 h_1, \dots, y_k^j + \omega_k h_k, r_j). \quad (4.4)$$

Put $\pi(\omega) = (-1)^{|\omega|}$ and

$$\eta_0(\omega, j) = \begin{cases} \pi(\omega), & 1 \leq j \leq d, \\ -\pi(\omega), & d < j \leq 2d. \end{cases}$$

For $\phi : B \rightarrow \mathbb{Z}_N$ we say that ϕ *respects* ρ when

$$\sum_{\omega, j} \eta_0(\omega, j) \phi(\rho(\omega, j)) = 0. \quad (4.5)$$

For $k = 1$ this is exactly Definition 3.13. The total number of unrestricted d -arrangements is $N^{(2d+1)k+2d-1}$.

For the next amplification lemma only, we follow Gowers's original convention and use the *unnormalised* Fourier transform

$$\widehat{g}(r) := \sum_{x \in \mathbb{Z}_N} g(x) e_N(-rx), \quad e_N(t) := e^{2\pi i t / N}.$$

This convention is responsible for the explicit powers of N in the statement.

Lemma 3.15 (Weighted additive amplification). *Let $\lambda_x \geq 0$ and let $\psi_1, \dots, \psi_m$ be maps from a set $A \subseteq \mathbb{Z}_N$ to $\mathbb{Z}_N$. Suppose bounded functions F_i satisfy*

$$\sum_{x \in A} \lambda_x \prod_{i=1}^m |\widehat{\Delta_x F_i}(\psi_i(x))|^2 \geq \eta N^{2m+1}.$$

Then the weighted number of quadruples $a - b = c - d$ satisfying simultaneously

$$\psi_i(a) - \psi_i(b) = \psi_i(c) - \psi_i(d) \quad (1 \leq i \leq m)$$

is at least $\eta^4 N^3$ after the normalization $\sum_x \lambda_x \leq N$.

Proof. Extend λ_x by 0 off A and abbreviate $p = m$. Expanding the square of the unnormalised Fourier coefficient gives

$$\widehat{\Delta_x F_i}(\psi_i(x)) = \sum_s F_i(s) \overline{F_i(s-x)} e_N(-\psi_i(x)s),$$

up to the harmless choice of sign in the Fourier convention. Hence the hypothesis, after expanding all p squares and writing $u_i = s_i - t_i$, is

$$\sum_x \lambda_x \sum_{s_1, \dots, s_p} \sum_{u_1, \dots, u_p} \prod_{i=1}^p F_i(s_i) \overline{F_i(s_i-x)} \overline{F_i(s_i-u_i)} F_i(s_i-x-u_i) e_N(-u_i \psi_i(x)) \geq \eta N^{2p+1}. \quad (4.6a)$$

This is the point at which the proof begins to look less transparent if one simply says “apply Cauchy–Schwarz in the usual way.” We therefore keep the two eliminations visible.

Because $|F_i| \leq 1$, discard the factors $F_i(s_i)$ and $\overline{F_i(s_i-u_i)}$ outside the inner x -sum. Taking absolute values and then Cauchy–Schwarz in the (s, u) variables yields

$$\sum_{s, u} \left| \sum_x \lambda_x \prod_{i=1}^p \overline{F_i(s_i-x)} F_i(s_i-x-u_i) e_N(-u_i \psi_i(x)) \right|^2 \geq \eta^2 N^{2p+2}. \quad (4.6b)$$

Introduce a common translation variable z and put $v_i = z - s_i$. Summing over z multiplies both sides by N and rewrites the left side as

$$\sum_z \sum_{u, v} \left| \sum_x \lambda_x \prod_{i=1}^p \overline{F_i(z-v_i-x)} F_i(z-v_i-x-u_i) e_N(-u_i \psi_i(x)) \right|^2. \quad (4.6c)$$

For fixed (u, v) define functions of x

$$A_{u,v}(x) := \prod_{i=1}^p \overline{F_i(-v_i-x)} F_i(-v_i-x-u_i), \quad B_u(x) := \lambda_x e_N\left(-\sum_{i=1}^p u_i \psi_i(x)\right).$$

The inner expression in (4.6c) is a translate-correlation of $A_{u,v}$ and B_u . Parseval for the unnormalised transform therefore gives

$$\sum_{u,v} \sum_r |\widehat{A}_{u,v}(r)|^2 |\widehat{B}_u(r)|^2 \geq \eta^2 N^{2p+4}. \quad (4.6d)$$

Apply Cauchy–Schwarz to the sum over (u, v, r) :

$$\left(\sum_{u,v,r} |\widehat{A}_{u,v}(r)|^4 \right) \left(\sum_{u,v,r} |\widehat{B}_u(r)|^4 \right) \geq \eta^4 N^{4p+8}. \quad (4.6e)$$

For each fixed (u, v) , the fourth Fourier moment of $A_{u,v}$ is at most N^4 ; indeed it equals N times an additive energy and $|A_{u,v}| \leq 1$. There are N^{2p} choices of (u, v) , so

$$\sum_{u,v,r} |\hat{A}_{u,v}(r)|^4 \leq N^{2p+4}. \quad (4.6f)$$

Moreover B_u is independent of v , whence (4.6e)–(4.6f) imply

$$\sum_{u,r} \left| \sum_x \lambda_x e_N \left(-rx - \sum_{i=1}^p u_i \psi_i(x) \right) \right|^4 \geq \eta^4 N^{p+4}. \quad (4.6g)$$

Finally expand the fourth power. Orthogonality in r imposes $a - b = c - d$, while orthogonality in each u_i imposes

$$\psi_i(a) - \psi_i(b) = \psi_i(c) - \psi_i(d) \quad (1 \leq i \leq p).$$

Thus the left side of (4.6g) is exactly N^{p+1} times the weighted sum $\sum \lambda_a \lambda_b \lambda_c \lambda_d$ over simultaneously additive quadruples. Dividing by N^{p+1} gives the asserted lower bound $\eta^4 N^3$.

The proof is deliberately written in this expanded form because the two different uses of Cauchy–Schwarz do different jobs: the first removes the bounded functions, whereas the second converts a large mixed Fourier second moment into a fourth moment whose orthogonality relations are precisely simultaneous additivity. $\square$

Lemma 3.16 (Walsh rigidity of a cube coefficient). *Let $h_1, \dots, h_k \in \mathbb{Z}_N^\times$ and let $\eta : \{0, 1\}^k \rightarrow \mathbb{Z}_N$. Suppose that for every $A \subseteq \{1, \dots, k\}$ the polynomial*

$$\sum_{\omega \in \{0,1\}^k} \eta(\omega) \prod_{i \in A} (y_i + \omega_i h_i)$$

is independent of $y_1, \dots, y_k$. Then η is a scalar multiple of the parity function $\pi(\omega) = (-1)^{|\omega|}$.

Proof. The functions $\omega \mapsto \prod_{i \in A} (-1)^{\omega_i}$ form the Walsh basis of functions on $\{0, 1\}^k$. It is therefore enough to show that η is orthogonal to every Walsh character indexed by a proper subset $A \subsetneq [k]$.

We prove this by induction on $|A|$. Fix $j \notin A$ and apply the hypothesis to $A \cup \{j\}$. The coefficient of y_j in the resulting polynomial is

$$\sum_{\omega} \eta(\omega) \prod_{i \in A} (y_i + \omega_i h_i),$$

so this polynomial vanishes identically. Expanding it in the variables y_i gives

$$0 = \sum_{B \subseteq A} \left(\prod_{i \in A \setminus B} y_i \right) \left(\prod_{i \in B} h_i \right) \sum_{\omega} \eta(\omega) \prod_{i \in B} \omega_i. \quad (4.6)$$

Starting with $A = \emptyset$ and increasing $|A|$, equation (4.6) shows that all moments of η of degree $< k$ vanish. The monomials $\prod_{i \in B} \omega_i$ span the same subspace as the proper Walsh characters, so all those Walsh coefficients vanish. Only the top parity coefficient can remain; hence η is a multiple of π . $\square$

Lemma 3.17 (Zero set of a nonconstant multilinear polynomial). *If $\mu : \mathbb{Z}_N^k \rightarrow \mathbb{Z}_N$ is multilinear and nonconstant, then for every $a \in \mathbb{Z}_N$,*

$$|\{y : \mu(y) = a\}| \leq N^k - (N - 1)^k \leq kN^{k-1}. \quad (4.7)$$

Proof. Induct on k . Write

$$\mu(y_1, \dots, y_k) = y_k \mu_1(y_1, \dots, y_{k-1}) + \mu_2(y_1, \dots, y_{k-1}).$$

If μ_1 is nonzero, the induction hypothesis applied to μ_1 at level 0 shows that its zero set has size at most $E := N^{k-1} - (N-1)^{k-1}$. For each base point with $\mu_1 \neq 0$ there is at most one choice of y_k , while above the zero set there are at most N choices. Thus

$$|\{\mu = a\}| \leq N^{k-1} + (N-1)E = N^k - (N-1)^k.$$

If $\mu_1 = 0$, then μ_2 is nonconstant and the induction hypothesis, followed by free choice of y_k , gives the same (indeed stronger) bound. Finally $N^k - (N-1)^k \leq kN^{k-1}$ by the binomial theorem. $\square$

Lemma 3.18 (Degenerate arrangements are lower-dimensional). *Fix $k, d \geq 1$. Call an arrangement ρ degenerate if some coefficient map*

$$\eta : \{0, 1\}^k \times \{1, \dots, 2d\} \rightarrow \{-1, 0, 1\}$$

which is not a multiple of η_0 nevertheless satisfies

$$\sum_{\omega, j} \eta(\omega, j) \prod_{i \in A} \rho(\omega, j)_i = 0 \quad (4.8)$$

for every $A \subseteq \{1, \dots, k+1\}$. Then the number of degenerate d -arrangements is

$$O_{k,d}(N^{(2d+1)k+2d-2}). \quad (4.9)$$

Proof. We reproduce the two cases in Gowers's §15. First fix nonzero side lengths $h_1, \dots, h_k$ and cross-sections $r_1, \dots, r_{2d}$ satisfying (4.3). If for some j the function $\omega \mapsto \eta(\omega, j)$ is not a multiple of parity, Lemma 3.16 supplies a set $A \subseteq [k]$ for which the left side of (4.8), viewed as a function of the base point y^j , is a nonconstant multilinear polynomial. By Lemma 3.17, at most kN^{k-1} of the N^k choices of y^j can satisfy it. All remaining base points are free, so this loses one full power of N .

Otherwise write $\eta(\omega, j) = c_j \pi(\omega)$. Taking $A = [k+1]$ in (4.8) and alternating over the cube gives

$$(-1)^k h_1 \cdots h_k \sum_{j=1}^{2d} c_j r_j = 0. \quad (4.10)$$

Because η is not a multiple of η_0 , the coefficient vector $(c_1, \dots, c_{2d})$ is not proportional to $(1, \dots, 1, -1, \dots, -1)$. Hence (4.10) is independent of the defining relation (4.3) and again removes one power of N from the choices of the r_j . There are only 3^{2d^2} possible coefficient maps η , and arrangements with some $h_i = 0$ also form $O_k(N^{-1})$ of the total. A union bound proves (4.9). $\square$

Lemma 3.19 (Riesz-product purification). *Fix $k \geq 1$ and put $D_k = 17k + 15$. Let $B \subseteq \mathbb{Z}_N^{k+1}$ have size βN^{k+1} , and suppose $\phi : B \rightarrow \mathbb{Z}_N$ respects at least*

$$\alpha \beta^{15} N^{D_k}$$

8-arrangements. For every $0 < \varepsilon < 1/2$ there is $N_0 = N_0(k, \alpha, \beta, \varepsilon)$ such that, whenever the prime $N \geq N_0$, there is $B' \subseteq B$ containing at least

$$c_k(\alpha \varepsilon)^{C_k} \beta^{15} N^{D_k} \quad (4.11)$$

8-arrangements, at least a proportion $1 - \varepsilon$ of which are respected by ϕ . The constants $c_k, C_k > 0$ depend only on k .

Proof. This is the dependent random restriction in Gowers's Lemmas 12.5 and 15.5. We give the calculation because ordinary independent hashing does not distinguish a respected arrangement from a bad one.

For $A \subseteq [k+1]$ put $m_A(y) = \prod_{i \in A} y_i$. Choose an integer m later. For each $1 \leq \ell \leq m$, choose independently and uniformly $t_\ell \in \mathbb{Z}_N$ and coefficients $s_{A,\ell} \in \mathbb{Z}_N$. Conditional on these choices, retain the points $y \in B$ independently with probability

$$p(y) = 2^{-m} \prod_{\ell=1}^m \left(1 + \cos \frac{2\pi}{N} \left[t_\ell \phi(y) + \sum_{A \subseteq [k+1]} s_{A,\ell} m_A(y) \right] \right). \quad (4.12)$$

The crucial dependence is through the common random coefficients $t_\ell, s_{A,\ell}$.

Let $M = 16 \cdot 2^k$ be the number of vertices of an 8-arrangement. For one layer ℓ , expand every cosine in (4.12) into its two characters. Averaging over all $s_{A,\ell}$ kills a coefficient pattern $\eta : \{0, 1\}^k \times [16] \rightarrow \{-1, 0, 1\}$ unless it satisfies every monomial cancellation (4.8). On a nondegenerate arrangement, Lemma 3.18 says that the only surviving patterns are 0 and the two signs $\pm \eta_0$. Averaging over t_ℓ kills the last two unless the arrangement is respected. Keeping track of the two ways to produce the zero coefficient in each factor therefore gives, for a nondegenerate arrangement,

$$\Pr(\rho \subseteq B') = \begin{cases} q_m R_k^m, & \rho \text{ respected,} \\ q_m, & \rho \text{ not respected,} \end{cases} \quad (4.13)$$

where

$$q_m = 2^{-Mm}, \quad R_k = 1 + 2^{1-M} > 1. \quad (4.14)$$

This is the amplification: a respected arrangement receives the fixed factor R_k for every Riesz layer.

Let X be the number of retained respected nondegenerate arrangements and Y the number of retained bad nondegenerate arrangements. From (4.13),

$$\mathbb{E}X \geq q_m R_k^m (\alpha \beta^{15} N^{D_k} - O_k(N^{D_k-1})), \quad \mathbb{E}Y \leq q_m \beta^{15} N^{D_k}. \quad (4.15)$$

Choose the least m for which $\alpha \varepsilon R_k^m \geq 4$. Since $R_k > 1$ depends only on k ,

$$q_m \geq c_k (\alpha \varepsilon)^{C_k}. \quad (4.16)$$

For N above a threshold depending on the displayed parameters, the $O_k(N^{D_k-1})$ term and the total number of degenerate arrangements are each at most $q_m \beta^{15} N^{D_k} / 4$. Consequently

$$\mathbb{E}(\varepsilon X - Y - Z) \geq \frac{1}{2} q_m \beta^{15} N^{D_k}, \quad (4.17)$$

where Z is the number of retained degenerate arrangements. Some realization thus satisfies (4.17). In particular $Y + Z < \varepsilon X$, and also $X \gg_k q_m \varepsilon^{-1} \beta^{15} N^{D_k}$. Using (4.16) and enlarging C_k absorbs the harmless factor ε^{-1} and proves (4.11). $\square$

Lemma 3.20 (Purification by random restriction). *Suppose $B \subseteq \mathbb{Z}_N^2$ has size βN^2 and $\phi : B \rightarrow \mathbb{Z}_N$ respects at least $\alpha \beta^{15} N^{32}$ 8-arrangements. For every $0 < \varepsilon < 1/2$ there is a subset $B' \subseteq B$ containing at least $c(\alpha \varepsilon)^C \beta^{15} N^{32}$ 8-arrangements, of which at least a proportion $1 - \varepsilon$ are respected.*

Proof. Apply Lemma 3.19 with $k = 1$. The exponent is $D_1 = 32$, and Definition 3.14 reduces exactly to Definition 3.13. $\square$

Lemma 3.21 (Fourier graph $\Rightarrow$ many matching parallelogram pairs). *Let $f : \mathbb{Z}_N \rightarrow \mathbb{D}$, let $B \subseteq \mathbb{Z}_N^2$ have size βN^2 , and suppose*

$$|\widehat{\Delta_{x,y} f}(\phi(x, y))| \geq \gamma N \quad ((x, y) \in B).$$

Then there are at least $\beta^{16} \gamma^{48} N^8$ pairs of vertical parallelograms having the same width and height and the same ϕ -increment. Here a vertical parallelogram consists of two vertical edges $(x, y), (x, y + h)$ and $(x + w, z), (x + w, z + h)$; its ϕ -increment is $\phi_h(x, y) - \phi_h(x + w, z)$.

Proof. This is the counting step that is easy to hide inside the phrase “apply Cauchy–Schwarz twice.” We keep Gowers’s two stages visible.

First fix a vertical coordinate x . The Fourier hypothesis, Parseval, and Hölder give an additive-energy lower bound in the section $B_x = \{y : (x, y) \in B\}$. Averaging in x gives at least $(\beta \gamma^2)^4 N^3$ additive quadruples in a typical weighted section. Equivalently, if a vertical edge $e = ((x, y), (x, y + h))$ is assigned the weight $q(e)$ equal to the number of partner edges of the same height having the same ϕ -difference, then

$$\mathbb{E}_e q(e) \geq (\beta \gamma^2)^4 N. \quad (4.18a)$$

Write $\zeta = (\beta \gamma^2)^4$, and for a fixed height h denote the corresponding average by $\zeta(h)N$. Thus $\mathbb{E}_h \zeta(h) \geq \zeta$.

For fixed h choose a base ordinate y for which the sum of the edge weights is at least its average and put

$$\lambda_x = N^{-1} q((x, y), (x, y + h)).$$

Then $0 \leq \lambda_x \leq 1$ and $\sum_x \lambda_x \geq \zeta(h)N$. Apply Lemma 3.15 with the two bounded functions corresponding to the two vertical edges at heights y and $y + h$ and with the two frequency maps $x \mapsto \phi(x, y)$ and $x \mapsto \phi(x, y + h)$. The Fourier lower bound supplies the hypothesis with parameter $\zeta(h)\gamma^4$. Hence the weighted number of simultaneously additive quadruples is at least $(\zeta(h)\gamma^4)^4 N^3$. Every such quadruple, after restoring the choices counted by the four q -weights, produces $N^4 \lambda_a \lambda_b \lambda_c \lambda_d$ pairs of vertical parallelograms of the same width and height with equal ϕ -increment. Therefore the number for this height is at least $(\zeta(h)\gamma^4)^4 N^7$.

Summing over h and using Hölder once more,

$$\sum_h (\zeta(h)\gamma^4)^4 N^7 \geq \zeta^4 \gamma^{16} N^8 = \beta^{16} \gamma^{48} N^8.$$

This is the required bound. The deliberately extravagant exponents are useful: they keep every loss explicit and are still more than sufficient for the later density increment. $\square$

Lemma 3.22 (Matching parallelogram pairs $\Rightarrow$ respected arrangements). *Suppose there are at least θN^8 pairs of vertical parallelograms of the same width and height and the same ϕ -increment. Then ϕ respects at least $\theta^7 N^{32}$ 8-arrangements.*

Proof. For $u, h \in \mathbb{Z}_N$ define

$$g_u(x, y) = 1_B(x, y) e_N(u \phi(x, y)), \quad F_{u,h}(x) = \sum_y g_u(x, y + h) \overline{g_u(x, y)}.$$

Thus

$$F_{u,h}(x) = \sum_y 1_B(x, y) 1_B(x, y + h) e_N(u \phi_h(x, y)).$$

With the unnormalised Fourier transform one has

$$\sum_{u,r} |\widehat{F}_{u,h}(r)|^2 \leq N^5, \quad (4.18b)$$

by Parseval and $|F_{u,h}(x)| \leq N$. Expanding the fourth moment and using orthogonality in r and u shows that

$$\sum_{u,r} |\widehat{F}_{u,h}(r)|^4$$

is exactly N^2 times the number of matching parallelogram pairs of height h . After summing in h , the hypothesis therefore says

$$\sum_{u,r,h} |\widehat{F}_{u,h}(r)|^4 \geq \theta N^{10}, \quad \sum_{u,r,h} |\widehat{F}_{u,h}(r)|^2 \leq N^6. \quad (4.18c)$$

We now interpolate the fourth and sixteenth moments. For nonnegative a_j , log-convexity of L^p norms (or Hölder written as $\sum a_j^4 \leq (\sum a_j^2)^{6/7} (\sum a_j^{16})^{1/7}$) gives

$$\sum_{u,r,h} |\widehat{F}_{u,h}(r)|^{16} \geq \frac{(\theta N^{10})^7}{(N^6)^6} = \theta^7 N^{34}. \quad (4.18d)$$

Expanding the sixteenth power and summing over u and r imposes exactly the two relations

$$x_1 + \cdots + x_8 = x_9 + \cdots + x_{16},$$

and

$$\phi_h(x_1, y_1) + \cdots + \phi_h(x_8, y_8) = \phi_h(x_9, y_9) + \cdots + \phi_h(x_{16}, y_{16}).$$

Each respected 8-arrangement is counted with the factor N^2 coming from the two orthogonality sums. Division by N^2 in (4.18d) gives at least $\theta^7 N^{32}$ respected arrangements. $\square$

Proposition 3.23 (Strengthened bihomomorphism). *Let $f : \mathbb{Z}_N \rightarrow \mathbb{D}$ and let $B \subseteq \mathbb{Z}_N^2$ have density β . Assume that for every $(h_1, h_2) \in B$ one has chosen a frequency $\phi(h_1, h_2)$ satisfying*

$$|\widehat{\Delta_{h_1, h_2}} f(\phi(h_1, h_2))| \geq \gamma N.$$

Then for every $\varepsilon > 0$ there is $B' \subseteq B$ containing at least $c\beta^C \gamma^C \varepsilon^C N^{32}$ 8-arrangements, of which at least $1 - \varepsilon$ are respected by ϕ .

Proof. Lemma 3.21 gives $\theta = \beta^{16} \gamma^{48}$ in Lemma 3.22. Hence the original set B contains at least

$$\beta^{112} \gamma^{336} N^{32}$$

respected 8-arrangements. This is the source-level amplification step: the large Fourier graph has first produced matching vertical parallelograms and then a high moment has glued those pairs into the thirty-two vertices of an 8-arrangement. Finally apply Lemma 3.20 with $\alpha = \beta^{97} \gamma^{336}$ (since the total arrangement normalization there is $\beta^{15} N^{32}$). The resulting lower bound has the form $c\beta^C \gamma^C \varepsilon^C N^{32}$ and at least a proportion $1 - \varepsilon$ are respected. This is exactly the strengthening needed before extracting a bilinear piece. $\square$

3.9.2 From respected arrangements to a bilinear box

Lemma 3.24 (Small graph difference set gives an order- k Freiman restriction). *Let $A \subseteq \mathbb{Z}_N$, let $\psi : A \rightarrow \mathbb{Z}_N$, and let*

$$\Gamma = \{(x, \psi(x)) : x \in A\} \subseteq \mathbb{Z}_N^2.$$

If $|\Gamma - \Gamma| \leq K|\Gamma|$, then for every fixed $k \geq 2$ there is $A' \subseteq A$ with

$$|A'| \geq \frac{|A|}{8kK^{4k}}$$

such that $\psi|_{A'}$ is a Freiman homomorphism of order k .

Proof. This is Gowers's graph version of Ruzsa's small-difference-set argument. By the Plünnecke–Petridis estimate already proved in Volume II,

$$|4k\Gamma - 4k\Gamma| \leq K^{4k}|\Gamma|. \quad (4.18)$$

We first isolate one *common* exceptional set in the vertical coordinate. If for some first coordinate x the fibre of $2k\Gamma - 2k\Gamma$ above x contained more than K^{4k} distinct second coordinates, then translating that fibre by the at least $|\Gamma|$ distinct first coordinates occurring in $2k\Gamma - 2k\Gamma$ would create more than $K^{4k}|\Gamma|$ points of $4k\Gamma - 4k\Gamma$, contradicting (4.18). In particular the vertical fibre

$$D := \{y : (0, y) \in 2k\Gamma - 2k\Gamma\}$$

has size at most K^{4k} . Consequently, if $(x, y), (x, y') \in k\Gamma - k\Gamma$, then

$$(0, y - y') \in 2k\Gamma - 2k\Gamma, \quad \text{so } y - y' \in D. \quad (4.19)$$

This is the uniform statement needed below; merely knowing that the individual fibres are small would not be enough.

Let $M < N/2$ be even. For nonzero $w \in \mathbb{Z}_N$ there are exactly $2M$ nonzero d for which $w \in \{jd : -M \leq j \leq M\}$. Hence at most $2MK^{4k}$ values of d make this short progression meet $D \setminus \{0\}$. Choose $M \geq N/(4K^{4k})$ with $2MK^{4k} < N - 1$ and then choose a nonzero d such that

$$D \cap \{jd : -M \leq j \leq M\} = \{0\}. \quad (4.20)$$

Let $L \leq M/(2k)$ and put $Q = \{0, d, \dots, Ld\}$. For $a \in \mathbb{Z}_N$ define

$$\Gamma_a = \{(x, y) \in \Gamma : y \in a + Q\}.$$

We claim that every nonempty Γ_a is the graph of an order- k Freiman homomorphism. Indeed, suppose two k -term signed sums of points of Γ_a have the same first coordinate. Their second coordinates y, y' lie in the interval $\{jd : -kL \leq j \leq kL\}$, so $y - y'$ lies in the progression in (4.20). On the other hand both (x, y) and (x, y') lie in $k\Gamma - k\Gamma$, hence (4.19) gives $y - y' \in D$. Equation (4.20) forces $y = y'$. Thus equal k -term sums of inputs have equal k -term sums of outputs.

Finally average $|\Gamma_a|$ over a . Every point of Γ belongs to exactly $L + 1$ translates $a + Q$, so

$$\mathbb{E}_a |\Gamma_a| = \frac{L + 1}{N} |\Gamma|.$$

Taking $L \geq M/(2k) - 1$ gives some slice with $|\Gamma_a| \geq |\Gamma|/(8kK^{4k})$. Projecting that slice to its first coordinate produces A' . $\square$

Lemma 3.25 (Additive quadruples give an order-8 Freiman restriction). *Let $A \subseteq \mathbb{Z}_N$ have density δ , and suppose the graph of $\psi : A \rightarrow \mathbb{Z}_N$ contains at least $\eta|A|^3$ additive quadruples. Then there is $A' \subseteq A$, $|A'| \geq c(\delta, \eta)N$, on which ψ is a Freiman homomorphism of order 8.*

Proof. Apply the asymmetric Balog–Szemerédi–Gowers theorem, Theorem I.2.015 of Volume I, to the graph of ψ . It gives $\Gamma' \subseteq \Gamma$ of size at least $c_0(\delta, \eta)N$ and

$$|\Gamma' - \Gamma'| \leq K(\delta, \eta)|\Gamma'|.$$

Lemma 3.24 with $k = 8$ then gives a subset of size at least $|\Gamma'|/(64K^{32})$ on which the graph is order-8 Freiman. Projecting to the first coordinate yields A' . This is the precise replacement for the formerly vague phrase “pass to a subset preserving all order-8 relations.” $\square$

Lemma 3.26 (Bohr extension of an order-8 Freiman map). *Let $A \subseteq \mathbb{Z}_N$, $|A| = \delta N$, and let $\psi : A \rightarrow \mathbb{Z}_N$ be a Freiman homomorphism of order 8. Put*

$$K = \{r \in \mathbb{Z}_N : |\widehat{1_A}(r)| \geq \delta^{3/2} N/4\}.$$

Then $|K| \leq 16\delta^{-2}$ and there is a local homomorphism

$$\Psi : B(K, \delta/(32\pi)) \longrightarrow \mathbb{Z}_N,$$

meaning that $\Psi(u+v) = \Psi(u) + \Psi(v)$ whenever $u, v, u+v$ all lie in the displayed Bohr set such that

$$\psi(x) - \psi(y) = \Psi(x - y)$$

whenever $x, y \in A$ and $x - y$ belongs to that Bohr set.

Proof. Parseval gives $|K| \leq 16\delta^{-2}$. Consider the fourfold correlation

$$g(d) = \#\{(a_1, a_2, a_3, a_4) \in A^4 : a_1 + a_2 - a_3 - a_4 = d\}.$$

With the Fourier normalization of Chapter II,

$$g(d) = N^{-1} \sum_r |\widehat{1_A}(r)|^4 e_N(rd).$$

Split the sum into K and its complement. On the complement, the fourth moment is bounded by the threshold squared times Parseval, while on K a displacement $d \in B(K, \delta/(32\pi))$ changes each phase by at most the corresponding arc length. Consequently

$$|g(d) - g(0)| < g(0)$$

for every such d , and hence $g(d) > 0$. Thus

$$B(K, \delta/(32\pi)) \subseteq 2A - 2A. \tag{4.21}$$

Because ψ has Freiman order 8, the value

$$\psi(a_1) + \psi(a_2) - \psi(a_3) - \psi(a_4)$$

depends only on $a_1 + a_2 - a_3 - a_4$ throughout $2A - 2A$: two representations can be compared after adding the same four elements to both sides, producing an order-8 relation. This defines the induced Freiman homomorphism on $2A - 2A$; restricting it to the Bohr set gives Ψ . In particular the usual additivity identity holds whenever all three of $u, v, u+v$ remain in the Bohr set. Taking the representation $x + x - x - y = x - y$ gives $\Psi(x - y) = \psi(x) - \psi(y)$. $\square$

Lemma 3.27 (Simultaneous progression linearization). *Let R be an arithmetic progression of length L , and for $1 \leq i \leq q$ let $A_i \subseteq R$ satisfy $|A_i| \geq \delta L$. Suppose $\psi_i : A_i \rightarrow \mathbb{Z}_N$ is an order-8 Freiman homomorphism. For every integer m with*

$$1 \leq m \leq L^{c(\delta, q)}$$

there is a partition of R into arithmetic progressions of a common difference, each of length m or $m+1$, such that every restriction $\psi_i|_{A_i \cap S}$ is affine on every cell S .

Proof. Identify R Freiman-isomorphically of order 8 with an interval in a prime cyclic group $\mathbb{Z}_p$ with $L \leq p < 16L$. Apply Lemma 3.26 simultaneously to the q transported sets. The union of their large spectra together with the frequency 1 has cardinality at most $Cq\delta^{-2}$. By the elementary Bohr-size lemma of Volume II,

$$B\left(K, \frac{c\delta}{m^2}\right)$$

contains a non-zero d once $m \leq L^{c(\delta, q)}$. The inclusion of 1 in K also makes the least integer representative of d small enough that $\mathbb{Z}_p$ can be partitioned into genuine d -progressions of length at least m^2 . Subdivide them into progressions of length m or $m+1$.

If x, y lie in one such cell, then $x - y = jd$ with $|j| \leq m$. Since jd stays inside the larger Bohr set, Lemma 3.26 gives

$$\psi_i(x) - \psi_i(y) = \Psi_i(jd) = j\Psi_i(d).$$

Thus ψ_i is affine on $A_i \cap S$. Pull the partition back to R . □

Theorem 3.28 (Bilinear-piece extraction). *There are functions $c_1, c_2 > 0$ such that the following holds. If $f : \mathbb{Z}_N \rightarrow \mathbb{D}$ fails degree-3 uniformity at level α (equivalently, its U^4 mass exceeds α), then there are arithmetic progressions P, Q with a common difference and $|P|, |Q| \geq N^{c_1(\alpha)}$, a set $C \subseteq P \times Q$ of density at least $c_2(\alpha)$, and a bilinear-affine map*

$$L(x, y) = axy + bx + cy + d$$

on $P \times Q$ such that

$$|\widehat{\Delta_{x,y} f}(L(x, y))| \geq \alpha N/2 \quad ((x, y) \in C).$$

Proof. The frequency-selection argument in the preceding chapter gives a set B of density $\gg \alpha^C$ on which the displayed Fourier coefficient is at least $\alpha N/2$. Proposition 3.23, with a small fixed purification error, gives a large subgraph in which almost every relevant arrangement is respected.

Fix a horizontal fibre occurring with the typical density. The respected-arrangement count gives a positive proportion of additive quadruples for its frequency map, so Lemma 3.25 produces an order-8 Freiman restriction. Do this on all typical fibres and discard the exceptional ones. Repeat in the other coordinate. We therefore reach a positive-density subgraph whose surviving horizontal and vertical fibre maps are order-8 Freiman homomorphisms.

We now use the simultaneous form rather than linearizing each fibre independently. Lemma 3.27, applied to the finitely many fibre maps selected inside one long first-coordinate progression, partitions the second coordinate into common-difference progressions on which all surviving vertical maps are affine. Write the affine form as

$$\phi(x, y) = axy + b_x.$$

The respected parallelogram identities imply that the coefficient maps $x \mapsto a_x$ and $x \mapsto b_x$ themselves preserve order-8 additive relations on a positive-density set. Apply Lemma 3.27 a second time, now simultaneously to these coefficient maps. On one pair of long common-difference progressions P, Q we obtain

$$a_x = ax + c, \quad b_x = bx + d,$$

on a positive-density subset. Hence

$$\phi(x, y) = axy + bx + cy + d.$$

All restrictions were made inside the original frequency graph, so the Fourier lower bound survives. The Bohr-rank and cell-count losses are finite functions of α ; absorbing their reciprocal powers into $c_1(\alpha), c_2(\alpha)$ proves the theorem. □

3.9.3 Higher-dimensional respected arrangements

Definition 3.29 (Product property and quantitative multiple linearity). Let $\Gamma \subseteq \mathbb{Z}_N^r \times \mathbb{Z}_N$ and let $\pi(\Gamma) \subseteq \mathbb{Z}_N^r$ denote its base projection. A *coordinate cross-section* is obtained by fixing any subset of the r base coordinates. If D is the base projection of a nonempty cross-section and $\phi : D \rightarrow \mathbb{Z}_N$ is a single-valued selection from that cross-section, put

$$E_D(\phi) := \#\{(x_1, x_2, x_3, x_4) \in D^4 : x_1 - x_2 = x_3 - x_4, \phi(x_1) - \phi(x_2) = \phi(x_3) - \phi(x_4)\}$$

and

$$E(D) := \#\{(x_1, x_2, x_3, x_4) \in D^4 : x_1 - x_2 = x_3 - x_4\}.$$

We say that Γ has the *product property with parameter γ* if, for every nonempty coordinate cross-section, there is a single-valued selection ϕ for which

$$E_D(\phi) \geq \gamma E(D).$$

Let $P_0 \subseteq \mathbb{Z}_N^r$ be an r -box, let $0 < \kappa \leq 1$, and let $q \geq 1$. We say that Γ is (γ, q, κ) -*multiply r -linear on P_0* if there are pairwise disjoint r -boxes $P_a \subseteq P_0$ such that

$$\left| P_0 \setminus \bigcup_a P_a \right| \leq \gamma |P_0|, \quad \text{wd}(P_a) \geq \text{wd}(P_0)^\kappa,$$

and, for every a , there are maps $\lambda_{a,1}, \dots, \lambda_{a,q} : P_a \rightarrow \mathbb{Z}_N$ that are affine-linear in each box coordinate separately and satisfy

$$\Gamma \cap (P_a \times \mathbb{Z}_N) \subseteq \bigcup_{j=1}^q \{(h, \lambda_{a,j}(h)) : h \in P_a\}.$$

Equivalently, after parametrizing P_a by intervals in $\mathbb{Z}$, each $\lambda_{a,j}$ is a multi-affine polynomial $\sum_{S \subseteq [r]} c_S \prod_{i \in S} u_i$ modulo N . We call such a map r -linear on the box, following the localization terminology used in this chapter. If $P_0 = \mathbb{Z}_N^r$ we omit the words “on P_0 ”. Thus the scale exponent κ , the exceptional proportion γ , and the covering number q are all part of the definition.

Lemma 3.30 (Arrangement counting from the product property). *For every $r \geq 2$ there are $c_r, C_r > 0$ such that a graph $\Gamma \subseteq \mathbb{Z}_N^r \times \mathbb{Z}_N$ of density β with the product property at level γ contains at least $c_r(\beta\gamma)^{C_r} N^{d_r}$ respected 8-arrangements, where d_r is the number of free base variables in the arrangement.*

Proof. Induct on r . The case $r = 2$ is Proposition 3.23 before purification. For the induction step freeze the last coordinate. By the product property, all but a controlled set of fibres inherit the $(r - 1)$ -dimensional product property. The induction hypothesis gives many respected arrangements inside each such fibre. Regard their alternating defects as frequency choices in the remaining coordinate and apply Lemma 3.15. Four compatible $(r - 1)$ -arrangements glue to an r -arrangement. Hölder bounds the loss from the fibre densities by a fixed power, giving the stated constants. $\square$

Proposition 3.31 (Higher-dimensional purification). *For every $r \geq 1$ and $\varepsilon > 0$, the graph in Lemma 3.30 has a subgraph containing $c_r(\beta\gamma\varepsilon)^{C_r} N^{17r+15}$ 8-arrangements, at least $1 - \varepsilon$ of which are respected.*

Proof. Lemma 3.30 supplies at least $c_r(\beta\gamma)^{C_r} N^{17r+15}$ respected 8-arrangements. The total base set has density β , so after absorbing the first constant into the parameter α we may apply Lemma 3.19 with $k = r$. Its degeneracy estimate is exactly $O_r(N^{17r+14})$, one power below the natural arrangement scale, and its Riesz selector therefore produces the stated subgraph. Renaming the fixed powers of $\beta, \gamma, \varepsilon$ gives the displayed constants. $\square$

3.9.4 The multiply-linear extraction induction

Lemma 3.32 (Induction base). *For every $\gamma, \theta > 0$ there are $q_1 = q_1(\gamma, \theta)$ and $\kappa_1 = \kappa_1(\gamma, \theta) > 0$ such that the conclusion of Theorem 3.38 holds when $r = 1$.*

Proof. If the projection of the graph to the base has size below θN , delete that projection and there is nothing to cover. Otherwise choose one graph value above each of at least θN base points. The product property gives additive energy $\gg \gamma^4 N^3$ in the selected graph. Theorem I.2.015 of Volume I and the quantitative Freiman theorem from Volume II yield, after a deletion budget chosen below, a bounded family of progression cells on each of which the selected map is affine. The quantitative progression theorem gives a positive exponent $\kappa_1 = \kappa_1(\gamma, \theta)$ for the cell lengths, while the number of candidate affine maps is at most $q_1 \leq \gamma^{-C} \theta^{-C}$. Partition the gaps between these cells into progression boxes and charge those not carrying the selected graph to the base-projection deletion. Taking the internal deletion budget so that the total omitted base mass is at most the required tolerance gives precisely the (γ, q_1, κ_1) conclusion. $\square$

Lemma 3.33 (Cross-section inheritance). *Assume Theorem 3.38 in dimension r . For every $\gamma, \theta > 0$ there are $q_r = q_r(r, \gamma, \theta)$ and $\kappa_r = \kappa_r(r, \gamma, \theta) > 0$ with the following property. If $\Gamma \subseteq \mathbb{Z}_N^{r+1} \times \mathbb{Z}_N$ has the product property, then, after deleting at most $\theta N^{r+1}/4$ base points, every proper coordinate cross-section of the remaining graph is (γ, q_r, κ_r) -multiply r -linear.*

Proof. For each coordinate and each fixed value apply the induction hypothesis to the r -dimensional cross-section, with exceptional tolerance $\theta/(4(r+1))$ and multilinearity tolerance γ . Summing the exceptional base sets over all fixed values and all $r+1$ coordinate choices deletes at most $\theta N^{r+1}/4$ points. Take q_r to be the maximum of the finitely many covering-number bounds and κ_r the maximum of the corresponding scale exponents. These depend only on r, γ, θ . $\square$

Lemma 3.34 (One-coordinate linearization from structured large spectra). *Let $P = Q \times I \subseteq \mathbb{Z}_N^{r+1}$ be a box. Suppose that for every h in a set $H \subseteq Q$ we have a finite set of frequencies K_h and a partial function $x \mapsto \phi_h(x)$, and suppose:*

1. *the frequency graph*

$$\Delta := \{(h, \xi) : h \in H, \xi \in K_h\}$$

is (τ, q, κ) -multiply r -linear on Q in the sense of Definition 3.29;

2. *there is $\zeta > 0$ such that ϕ_h is a $B(K_h, \zeta)$ -homomorphism: whenever $x - y \in B(K_h, \zeta)$,*

$$\phi_h(x) - \phi_h(y) = \Psi_h(x - y)$$

for an additive map Ψ_h on that Bohr domain.

Then, for every $\sigma > \tau$, after deleting at most a σ -fraction of P , one can partition the surviving set into subboxes $T \times J$ of width at least

$$\text{wd}(P)^{c(r, q, \kappa, \sigma, \zeta)}$$

such that, for every surviving $h \in H \cap T$, the restriction of $x \mapsto \phi_h(x)$ to J is affine.

Proof. Take the box partition supplied by the (τ, q, κ) hypothesis on Q . Discard its exceptional part and, on each surviving base box Q_0 , write the large spectrum in the form

$$K_h \subseteq \{\mu_1(h), \dots, \mu_q(h)\} \quad (h \in H \cap Q_0), \quad (4.26)$$

where each μ_j is r -linear. The simultaneous Dirichlet/box lemma proved earlier in this volume partitions Q_0 into further boxes T with common difference d_T and power-sized width so that

$$\|\mu_j(h)d_T/N\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{\zeta}{2L} \quad (h \in T, 1 \leq j \leq q), \quad (4.27)$$

where L is the target last-coordinate length. By (4.26), (4.27) says exactly that

$$d_T \in B(K_h, \zeta/L) \quad (h \in H \cap T). \quad (4.28)$$

Partition I into progressions J of common difference d_T and lengths L or $L + 1$. If $x, y \in J$, then $x - y = nd_T$ with $|n| \leq L$; additivity and (4.28) keep $x - y$ inside $B(K_h, \zeta)$ and give

$$\phi_h(x) - \phi_h(y) = n\Psi_h(d_T).$$

Thus ϕ_h is affine on J . The power loss in the width is the product of the explicit scale loss κ and the simultaneous Dirichlet loss; choosing the intermediate widths successively leaves a positive exponent depending only on the listed parameters. Since $\tau < \sigma$, allocate the remaining deletion budget $\sigma - \tau$ to boundary cells. The total discarded mass is therefore at most $\sigma|P|$. $\square$

Lemma 3.35 (Cube gluing). *Let $H \subseteq \mathbb{Z}_N^r$, let $0 < \eta \leq 1$, and let ϕ be a partial function on $\mathbb{Z}_N^{r+1}$. Suppose that for every $h = (h_1, \dots, h_r) \in H$ there is a set $Y_h \subseteq \mathbb{Z}_N^{r+1}$ with*

$$|Y_h| \geq \eta N^{r+1}$$

such that every $(x_1, \dots, x_r, x) \in Y_h$ satisfies

$$\phi(h, x) = \sum_{\omega \in \{0,1\}^r} (-1)^{|\omega|} \phi(x_1 + \omega_1 h_1, \dots, x_r + \omega_r h_r, x),$$

whenever all displayed values of ϕ are defined. Then there are fixed base points $x_1, \dots, x_r$ for which at least $\eta|H|N$ pairs $(h, x) \in H \times \mathbb{Z}_N$ satisfy the same identity.

Proof. Count the triples

$$(h, (x_1, \dots, x_r), x)$$

for which $(x_1, \dots, x_r, x) \in Y_h$. Their number is

$$\sum_{h \in H} |Y_h| \geq \eta|H|N^{r+1}.$$

Averaging this count over the N^r possible choices of the base $(x_1, \dots, x_r)$ shows that one base occurs in at least $\eta|H|N$ such triples. For that base the corresponding (h, x) pairs give the claim. $\square$

Lemma 3.36 (Closure under finite unions and sums). *Let P_0 be an r -box and let $m \geq 1$. There is*

$$\kappa_* = \kappa_*(r, m, \kappa_1, \dots, \kappa_m) > 0$$

with the following properties. If Γ_i is $(\gamma_i, q_i, \kappa_i)$ -multiply r -linear on P_0 , then $\bigcup_i \Gamma_i$ is

$$\left(\sum_{i=1}^m \gamma_i, \sum_{i=1}^m q_i, \kappa_* \right) \text{-multiply } r \text{-linear on } P_0.$$

If f_i are functions whose graphs have those same parameters on P_0 , then the graph of $\sum_i f_i$ is

$$\left(\sum_{i=1}^m \gamma_i, \prod_{i=1}^m q_i, \kappa_* \right) \text{-multiply } r \text{-linear on } P_0.$$

Proof. Take the finitely many box partitions supplied by the hypotheses. Apply the simultaneous progression-refinement lemma used in Lemma 3.34 to obtain a common refinement of all surviving cells. Because m is fixed and every input cell has width at least a fixed positive power of $\text{wd}(P_0)$, the refined cells have width at least $\text{wd}(P_0)^{\kappa_*}$ for some positive $\kappa_* = \kappa_*(r, m, \kappa_1, \dots, \kappa_m)$. The union of the omitted sets has relative size at most $\sum_i \gamma_i$. On every refined box, concatenate the candidate lists for a union. For a sum, choose one candidate from each list, giving at most $\prod_i q_i$ sums; a sum of boxwise multi-affine maps is boxwise multi-affine. $\square$

Lemma 3.37 (Completion of the induction step). *Assume Theorem 3.38 in dimension r . For every $\gamma, \theta > 0$ there is $\kappa_{r+1} = \kappa_{r+1}(r, \gamma, \theta) > 0$ such that, if the base projection has density at least θ , one can find a subgraph whose base projection has density at least $c_r(\theta, \gamma) > 0$ and which is $(\gamma, 1, \kappa_{r+1})$ -multiply $(r+1)$ -linear. Repeating finitely many times leaves base projection below θ .*

Proof. Apply Lemma 3.33. If the remaining base projection has density below θ , stop. Otherwise select a single graph value on a θ -dense subset. The arrangement-counting consequence of the product property (Lemma 3.30), applied to the selected graph and its proper cross-sections, gives a constant $\eta = \eta(r, \gamma, \theta) > 0$ and sets Y_h satisfying the hypotheses of Lemma 3.35. That lemma therefore provides fixed base points $x_1, \dots, x_r$ and a domain of size at least $\eta\theta N^{r+1}$ on which the selected value can be written

$$\phi_1(h, x) = \phi_*(h, x) - \sum_{\omega \neq (1, \dots, 1)} (-1)^{|\omega|} \phi_\omega(h, x). \quad (4.29)$$

Every ϕ_ω in the displayed sum comes from a proper cross-section. By the induction hypothesis and Lemma 3.36, their alternating sum is multiply $(r+1)$ -linear. Apply the induction hypothesis to the finitely many summands with exceptional tolerances whose sum is at most $\gamma/4$, and then use Lemma 3.36. Subtract it from the selected graph. The residual ϕ_1 inherits two kinds of information: its proper cross-sections are multiply multilinear, and the large spectra controlling its last-coordinate fibres form a multiply r -linear frequency graph. Lemma 3.34 therefore partitions almost every long box $S = T \times J$ so that each fibre of ϕ_1 is one of at most q affine functions

$$x \mapsto \lambda_t(h)x + \mu_t(h). \quad (4.30)$$

It remains to prove that the coefficient functions in (4.30) inherit multilinearity; this is the point that cannot be replaced by the sentence “two points determine a line.” Discard affine classes containing fewer than $\sigma|J|/q$ points. Their total mass is at most $\sigma|S|$. Choose $R \asymp q\sigma^{-2}$ random points $z_1, \dots, z_R \in J$. For a surviving affine class, the probability that fewer than two distinct sample points lie in it is at most

$$(1 - \sigma/q)^R + R(\sigma/q)(1 - \sigma/q)^{R-1},$$

which is $O(\sigma)$ for the indicated choice of R . Averaging therefore yields a single sample for which, after another $O(\sigma)$ deletion, every surviving affine class contains two sample points $z_i \neq z_j$.

On such a class,

$$\lambda_t(h) = \frac{\phi_1(h, z_i) - \phi_1(h, z_j)}{z_i - z_j}, \quad \mu_t(h) = \phi_1(h, z_i) - z_i \lambda_t(h). \quad (4.31)$$

For fixed z_i, z_j , the functions $h \mapsto \phi_1(h, z_i)$ and $h \mapsto \phi_1(h, z_j)$ are restrictions to proper cross-sections; by the induction hypothesis they are multiply r -linear. Scalar multiplication,

subtraction and Lemma 3.36 show that every candidate λ_{ij} and μ_{ij} in (4.31) is multiply r -linear. Hence

$$(h, x) \mapsto \lambda_{ij}(h)x + \mu_{ij}(h)$$

is (γ', q', κ') -multiply $(r+1)$ -linear for parameters depending only on r, γ, θ once each component tolerance γ' is chosen so that all finite-union losses sum to at most $\gamma/4$. There are only $O(R^2)$ pairs (i, j) , so a common refinement of their box partitions still has power-sized width and covers all but $O(\sigma)|S|$ points. Pigeonholing one candidate gives a genuine $(\gamma, 1, \kappa_{r+1})$ -multiply $(r+1)$ -linear subgraph with positive-density base projection.

Delete that subgraph and repeat. Since the original graph has at most $\gamma^{-2}N^{r+1}$ points and every extracted piece has at least $c_r(\theta, \gamma)N^{r+1}$ points, at most $\gamma^{-2}/c_r(\theta, \gamma)$ iterations occur. The finite-union clause of Lemma 3.36 then gives the required multiply-linear cover of the complement of a base set of density at most θ , after choosing all component exceptional tolerances so that their finite sum is at most γ . The scale exponent κ_{r+1} is the maximum of the finitely many scale exponents produced in these refinements. $\square$

Theorem 3.38 (Multiply-linear extraction). *For every $r \geq 1$, $\gamma > 0$ and $\theta > 0$ there are*

$$q = q(r, \gamma, \theta) \geq 1, \quad \kappa = \kappa(r, \gamma, \theta) > 0$$

such that, if $\Gamma \subseteq \mathbb{Z}_N^r \times \mathbb{Z}_N$ has size at most $\gamma^{-2}N^r$ and satisfies the product property with parameter γ , then there is $J \subseteq \mathbb{Z}_N^r$, $|J| \geq (1-\theta)N^r$, for which $\Gamma \cap (J \times \mathbb{Z}_N)$ is (γ, q, κ) -multiply r -linear on the ambient box $\mathbb{Z}_N^r$.

Proof. Induct on r . Lemma 3.32 is the base and Lemma 3.37 is the induction step. The covering number obeys a finite recursion of the form

$$q_{r+1} \leq \gamma^{-2}\theta^{-1}q_r^{C_r}\gamma^{-C_r}\theta^{-C_r},$$

while the scale exponent is a finite positive composition of the localization and common-refinement exponents produced in the same step. Hence both q and a positive κ depend only on r, γ, θ , not on N . $\square$

Corollary 3.39 (One multilinear box). *Let $f : \mathbb{Z}_N \rightarrow \mathbb{D}$. If*

$$U_{r+1}(f; \mathbb{Z}_N) > \alpha,$$

that is, if f fails degree- $(r+1)$ uniformity at level α in the sense of Definition 3.1, then there are a box $P \subseteq \mathbb{Z}_N^r$ of width at least $N^{c_r(\alpha)}$, an r -linear map $\sigma : P \rightarrow \mathbb{Z}_N$, and a subset $P' \subseteq P$ of relative density at least $c_r(\alpha)$ such that

$$|\widehat{\Delta_h f}(\sigma(h))| \geq c_r(\alpha)N \quad (h \in P').$$

Proof. The frequency graph associated with failure of uniformity has size $O(\alpha^{-2}N^r)$ by Parseval and satisfies the product property by Proposition 3.31. Apply Theorem 3.38 with a small fixed θ and then use the box partition in Definition 3.29. Since the frequency graph has positive base density and only a fixed fraction of the ambient base is omitted, one surviving box and one of its finitely many candidate multilinear graphs contain a positive proportion of the frequency graph. Take that box for P , its candidate for σ , and the corresponding base points for P' . $\square$

3.9.5 The local Section 17 twist

Lemma 3.40 (Local finite-difference integration on a box). *Let N be prime and let $1 \leq r < N$. Fix $d \in \mathbb{Z}_N^\times$, residues $a_1, \dots, a_r \in \mathbb{Z}_N$, and integer intervals $I_1, \dots, I_r$ of length $< N$. Put*

$$P_i := \{a_i + dn_i : n_i \in I_i\} \subseteq \mathbb{Z}_N, \quad P := P_1 \times \dots \times P_r.$$

Suppose $\sigma : P \rightarrow \mathbb{Z}_N$ is multi-affine in these progression coordinates, in the explicit sense that there are coefficients $(c_S)_{S \subseteq [r]} \subseteq \mathbb{Z}_N$ for which

$$\sigma(a_1 + dn_1, \dots, a_r + dn_r) = \sum_{S \subseteq [r]} c_S \prod_{i \in S} n_i \quad (n_i \in I_i).$$

Then there are one-variable polynomials $\Phi_\omega : \mathbb{Z}_N \rightarrow \mathbb{Z}_N$, indexed by $\omega \in \{0, 1\}^{r+1}$, each of degree at most $r+1$, such that for every $h = (h_1, \dots, h_r) \in P$ and every $u, s \in \mathbb{Z}_N$,

$$u\sigma(h) = \sum_{\omega \in \{0, 1\}^{r+1}} (-1)^{|\omega|} \Phi_\omega \left(s - \sum_{i=1}^r \omega_i h_i - \omega_{r+1} u \right) \pmod{N}.$$

Proof. Because $d \in \mathbb{Z}_N^\times$, the coordinate $n_i = d^{-1}(h_i - a_i)$ is an affine function of h_i . Hence the given multi-affine formula extends $u\sigma(h)$ from $P \times \mathbb{Z}_N$ to a polynomial in $(h_1, \dots, h_r, u)$ of total degree at most $r+1$. It is therefore enough, by linearity, to treat a monomial $cz_1 \cdots z_j$ with $0 \leq j \leq r+1$.

For $j = 0$ take a constant one-variable polynomial. For $j \geq 1$, the polarisation identity of Lemma 3.10, applied in $\mathbb{Z}_N$, gives

$$cz_1 \cdots z_j = \sum_{\eta \in \{0, 1\}^j} (-1)^{|\eta|} \Psi_\eta \left(s - \sum_{m=1}^j \eta_m z_m \right)$$

for suitable one-variable polynomials Ψ_η of degree j ; the factor $j!$ occurring in the usual polarisation formula is invertible because $j \leq r+1 < N$. Regard each z_m as one of $h_1, \dots, h_r, u$, extend η by zeros in the unused directions, and add the resulting representations for all monomials. Collecting terms with the same $\omega \in \{0, 1\}^{r+1}$ gives the stated Φ_ω , each of degree at most $r+1$. All identities are identities in the field $\mathbb{Z}_N$; no choice of integer lift or no-wrap convention is required. $\square$

Lemma 3.41 (Partition complexity). *Let $I = [-\ell, \ell] \subset \mathbb{Z}$ and let $w \asymp \ell/r$. Partition $\mathbb{Z}_N$ into consecutive mod- N progressions Q_j of lengths between $w-1$ and $w+1$. For $a \in I^r$ and $s \in \mathbb{Z}_N$, the function*

$$\omega \longmapsto j_\omega, \quad s - \omega \cdot a \in Q_{j_\omega},$$

has, modulo common translation of all j_ω , at most 2^{Cr^4} possible forms.

Proof. Lift s/w and a_i/w to real numbers. Then $j_\omega = \lfloor \alpha_0 + \omega \cdot \alpha \rfloor$ with $|\alpha_i| \leq r+1$. The value changes only when α crosses one of the hyperplanes $\omega \cdot \alpha = m$. There are $O(r2^r)$ relevant hyperplanes. The number of cells cut out by M hyperplanes in $\mathbb{R}^r$ is at most $\sum_{j=0}^r \binom{M}{j} \leq M^r$. Multiplying by the bounded possibilities for the integer part of α_0 , then quotienting by common translation, gives 2^{Cr^4} possible patterns. $\square$

Proposition 3.42 (Local box to polynomial twist). *Let $f : \mathbb{Z}_N \rightarrow \mathbb{D}$. Suppose $P = P_1 \times \dots \times P_r$ is a box of common difference d and odd width $m \leq N^{1/2}$, and $\sigma : P \rightarrow \mathbb{Z}_N$ is r -linear with*

$$\sum_{h \in P} |\widehat{\Delta_h f}(\sigma(h))|^2 \geq \alpha N^2 |P|.$$

Then there is a polynomial q of degree at most $r + 1$ and a partition $\mathcal{Q}$ of $\mathbb{Z}_N$ into mod- N arithmetic progressions of common difference d whose lengths differ by at most one and are at least $m/(4r)$, such that, for $g = fe_N(-q)$,

$$U_r(g \mid \mathcal{Q}) \geq c_r \alpha, \quad (4.33)$$

where $c_r \geq 2^{-Cr^4}$. Thus the conclusion is failure of degree r uniformity relative to the progression partition, one degree lower than the input condition that produced the r -linear frequency box.

Proof. Multiplication by the inverse of d is an automorphism of $\mathbb{Z}_N$, so take $d = 1$. Center each P_i at r_i and write $h = r + a$, $a \in I^r$ with $I = [-\ell, \ell]$ and $2\ell + 1 = m$. Expanding the Fourier square gives

$$\sum_{a \in I^r} \sum_{s, u} \Delta_{a+r, u} f(s) e_N(-u\sigma(a+r)) \geq \alpha N^2 m^r. \quad (4.32)$$

Decompose every translate of f according to the progression partition $\mathcal{Q}$ of Lemma 3.41. After expansion, a term is nonzero only for a cell pattern $\omega \mapsto j_\omega$ allowed by that lemma. There are at most $L = 2^{Cr^4}$ translation classes of patterns. Two applications of Cauchy–Schwarz to (4.32) therefore give one pattern class whose contribution is at least $L^{-2} \alpha N^2 m^r$.

Choose a representative pattern. Lemma 3.40 expresses the local frequency $u\sigma(a+r)$ as an alternating cube derivative of polynomials of degree at most $r + 1$. Absorb all lower-face polynomials into the corresponding translated factors. Among the finitely many face polynomials, another Cauchy–Schwarz/pigeonhole step selects one polynomial q for which

$$\sum_{Q \in \mathcal{Q}} \sum_{h_1, \dots, h_{r+1}} \sum_s \Delta_{h_1, \dots, h_{r+1}} (Q(fe_N(-q)))(s) \geq L^{-2} \alpha m^{r+2} |\mathcal{Q}|.$$

After dividing by the corresponding indicator-cube normalizations in Definition 3.2, the preceding inequality says

$$U_r(g \mid \mathcal{Q}) \geq L^{-2} \alpha;$$

the cell lengths differ by at most one, so the fixed normalization comparison loses only a factor depending on r , which we absorb by decreasing c_r once. The choice $w \asymp \ell/r$ makes every progression length at least $m/(4r)$. Taking $c_r \geq 2^{-Cr^4}$ after this harmless adjustment proves the claim. $\square$

3.9.6 Degree lowering, polynomial untwisting, and the final partition bias

The output of Proposition 3.42 is *not* a correlation of the original balanced function with one phase. It is failure of lower-degree uniformity for a polynomially twisted function on a progression partition. The next four lemmas close exactly that interface. Their architecture is Gowers’s Sections 5 and 18 [1], but all deductions used here are written out because the degree shift is load-bearing.

Lemma 3.43 (Weyl differencing estimate). *Fix $s \geq 2$ and $\varepsilon > 0$. There is $C_{s, \varepsilon}$ such that the following holds. Let*

$$P(n) = \theta n^s + \theta_{s-1} n^{s-1} + \dots + \theta_0$$

be a real polynomial and suppose

$$\left| \theta - \frac{a}{q} \right| \leq q^{-2}, \quad (a, q) = 1.$$

Then, for every $L \geq 1$,

$$\left| \sum_{n=0}^{L-1} e(P(n)) \right| \leq C_{s, \varepsilon} L^{1+\varepsilon} \left(q^{-1} + L^{-1} + qL^{-s} \right)^{2^{1-s}}. \quad (4.34)$$

Proof. Apply the van der Corput inequality successively $s - 1$ times. At the j th step the j th finite difference of P has degree $s - j$; after the last step the phase is linear with leading coefficient

$$s! \theta h_1 \cdots h_{s-1}.$$

The geometric-series bound

$$\left| \sum_{n=0}^{L'-1} e(\xi n) \right| \leq \min\{L', (2\|\xi\|_{\mathbb{R}/\mathbb{Z}})^{-1}\}$$

therefore gives, after absorbing the boundary factors from the $s - 1$ differencings,

$$|S|^{2^{s-1}} \leq C_s L^{2^{s-1}-s} \sum_{|h_1|, \dots, |h_{s-1}| < L} \min\left\{L, \frac{1}{2\|s! \theta h_1 \cdots h_{s-1}\|}\right\}. \quad (4.35)$$

Group the tuples according to the product $m = h_1 \cdots h_{s-1}$. The divisor bound proved in Volume I gives $O_{s,\varepsilon}(L^\varepsilon)$ representations of each nonzero m after the usual dyadic decomposition. For $|\theta - a/q| \leq q^{-2}$, split the remaining m into residue classes modulo q ; within each complete block the numbers am/q are q^{-1} -spaced. Summing the geometric-series bound over one block and then the incomplete final block yields

$$\sum_{|m| \ll L^{s-1}} L^\varepsilon \min\left\{L, \frac{1}{2\|s! \theta m\|}\right\} \ll_{s,\varepsilon} L^{s+\varepsilon} (q^{-1} + L^{-1} + qL^{-s}).$$

Substitution in (4.35) and taking the 2^{s-1} th root gives (4.34). This is the standard Weyl-differencing proof, with the only number-theoretic input being the divisor estimate already owned by Volume I. $\square$

Lemma 3.44 (Quantitative polynomial recurrence). *For every $s \geq 1$ there are constants $c_s > 0$ and L_s such that, whenever N is prime, $L_s \leq L \leq N$, and $a \in \mathbb{Z}_N$, there is an integer $1 \leq p \leq L^{1/2}$ for which*

$$\left\| \frac{ap^s}{N} \right\|_{\mathbb{R}/\mathbb{Z}} \leq L^{-c_s}. \quad (4.36)$$

Proof. For $s = 1$ this is Dirichlet approximation. Let $s \geq 2$. We give the quantitative contradiction argument because it is the point at which a naive one-shot Dirichlet argument for the coefficient a would be insufficient.

Choose a small parameter $\tau > 0$, depending only on s , and put $T = \lfloor L^\tau \rfloor$. Suppose that (4.36) fails for every $p \leq L^{1/2}$ with an exponent c_s to be chosen below. In particular the set

$$\mathcal{A} = \{an^s : 1 \leq n \leq T\} \subset \mathbb{Z}_N$$

avoids the interval of residues of distance at most $L^{-c_s}N$ from 0. The Fejér-kernel argument used in the proof of the one-frequency Dirichlet lemma then produces a nonzero integer r with

$$|r| \ll L^{2c_s}$$

and

$$\left| \sum_{n=1}^T e\left(-\frac{ra}{N}n^s\right) \right| \gg L^{-c_s}T. \quad (4.37)$$

Indeed, convolving the indicator of the forbidden interval with itself gives a nonnegative kernel whose Fourier transform is supported effectively on $|r| \ll L^{2c_s}$; if every nonzero Fourier

coefficient of $\mathcal{A}$ in that range were smaller than a fixed multiple of $L^{-c_s}T$, the zero-frequency term would force a point of $\mathcal{A}$ into the forbidden interval.

Apply Dirichlet to $-ra/N$ with denominator bound T . Thus $|-ra/N - b/q| \leq (qT)^{-1}$ for coprime b, q and $1 \leq q \leq T$. If q were larger than $L^{C_s c_s}$, with C_s chosen sufficiently large, then Lemma 3.43, used with a fixed $\varepsilon < \tau/2^{s+2}$, would make the left side of (4.37) $o(L^{-c_s}T)$ once c_s is sufficiently small in terms of s and τ . Consequently

$$q \leq L^{C_s c_s}. \quad (4.38)$$

Now multiply the Dirichlet approximation by $r^{s-1}q^s$. Since $br^{s-1}q^{s-1}$ is integral,

$$\left\| \frac{a(rq)^s}{N} \right\| \leq \frac{|r|^{s-1}q^{s-1}}{T} \ll L^{(2+C_s)(s-1)c_s-\tau}.$$

Choose first $\tau = 1/(20s)$ and then choose $c_s > 0$ so small that $(2 + C_s)(s - 1)c_s \leq \tau/2$ and $(2 + C_s)c_s \leq 1/2$. Then $p = |rq|$ satisfies $p \leq L^{1/2}$ and the last display is at most L^{-c_s} after decreasing c_s once more. This contradicts the assumed failure of (4.36). $\square$

Lemma 3.45 (Polynomial phases subdivide into almost-constant progression cells). *Fix $s \geq 1$ and $0 < \varepsilon < 1/4$. There are $\kappa_s(\varepsilon) > 0$, $C_s(\varepsilon) \geq 1$, and $L_s(\varepsilon)$ with the following property. Let Q be a mod- N arithmetic progression of length $L \geq L_s(\varepsilon)$ in a prime cyclic group, and let $q : \mathbb{Z}_N \rightarrow \mathbb{Z}_N$ be a polynomial of degree at most s . Then Q can be partitioned into at most*

$$C_s(\varepsilon)L^{1-\kappa_s(\varepsilon)} \quad (4.39)$$

arithmetic progressions R such that for every R there is $z_R \in \mathbb{C}$, $|z_R| = 1$, with

$$\sup_{x \in R} |e_N(q(x)) - z_R| \leq \varepsilon. \quad (4.40)$$

All constants are obtained recursively from Lemma 3.44.

Proof. Affine transport reduces to $Q = [L]$. We induct on s . The case $s = 1$ follows from Dirichlet approximation: choose a step $p \leq L^{1/2}$ for which the linear increment has size $O(L^{-1/2})$ modulo one, split into the p residue-class progressions, and cut each into blocks on which the accumulated phase variation is at most ε .

For $s \geq 2$, write

$$q(n) = an^s + q_{s-1}(n), \quad \deg q_{s-1} \leq s-1.$$

By Lemma 3.44, choose $p \leq L^{1/2}$ with $\|ap^s/N\| \leq L^{-c_s}$. Partition $[L]$ into p progressions of common difference p and then into consecutive blocks of length

$$u = \left\lfloor L^{c_s/(4s)} \right\rfloor.$$

On one such block, writing $n = n_0 + pt$, the leading term is

$$a(n_0 + pt)^s = ap^s t^s + R_{n_0}(t),$$

where R_{n_0} has degree at most $s-1$. The phase variation contributed by $ap^s t^s$ is at most

$$2\pi u^s L^{-c_s} \leq 2\pi L^{-3c_s/4},$$

which is $< \varepsilon/2$ for large L . Apply the induction hypothesis, with tolerance $\varepsilon/2$, to the degree- $(s-1)$ polynomial $R_{n_0}(t) + q_{s-1}(n_0 + pt)$ on each block. The resulting cells satisfy

(4.40). Since the first subdivision uses $O(L/u)$ blocks and the inductive refinement of a block uses $O_{s,\varepsilon}(u^{1-\kappa_{s-1}(\varepsilon/2)})$ cells, the total is

$$O_{s,\varepsilon}(Lu^{-\kappa_{s-1}(\varepsilon/2)}) \leq C_s(\varepsilon)L^{1-\kappa_s(\varepsilon)}$$

with, for example,

$$\kappa_s(\varepsilon) = \frac{c_s}{8s} \kappa_{s-1}(\varepsilon/2) > 0.$$

Increasing $C_s(\varepsilon)$ absorbs the finitely many smaller values of L . $\square$

Lemma 3.46 (Removing a polynomial twist from partition bias). *Fix $s \geq 1$ and $\gamma > 0$. Let $h : \mathbb{Z}_N \rightarrow \mathbb{D}$, let $q : \mathbb{Z}_N \rightarrow \mathbb{Z}_N$ have degree at most s , and let $Q_1, \dots, Q_M$ be arithmetic progressions partitioning $\mathbb{Z}_N$. If*

$$\sum_{i=1}^M \left| \sum_{x \in Q_i} h(x) e_N(-q(x)) \right| \geq \gamma N, \quad (4.41)$$

then there is a refinement into arithmetic progressions $R_1, \dots, R_L$ such that

$$L \leq C_{s,\gamma} M^\kappa N^{1-\kappa}, \quad \sum_{j=1}^L \left| \sum_{x \in R_j} h(x) \right| \geq \frac{\gamma}{2} N, \quad (4.42)$$

where $\kappa = \kappa_s(\gamma/4) > 0$ is from Lemma 3.45.

Proof. Apply Lemma 3.45 to each Q_i with tolerance $\gamma/4$. Refining a partition cannot decrease the sum of the absolute values of the twisted cell sums. On a refined cell R , choose z_R as in (4.40). Then

$$\left| \sum_{x \in R} h(x) e_N(-q(x)) \right| \leq \left| \sum_{x \in R} h(x) \right| + \frac{\gamma}{4} |R|.$$

Summing and using (4.41) gives the second part of (4.42) (in fact $3\gamma N/4$). If $m_i = |Q_i|$, the number of refined cells is at most

$$C_{s,\gamma} \sum_i m_i^{1-\kappa} \leq C_{s,\gamma} M^\kappa N^{1-\kappa}$$

by Hölder, proving the first part. This is the precise untwisting mechanism of Gowers's Lemma 5.14. $\square$

Lemma 3.47 (Properization of a cyclic progression partition). *A mod- N arithmetic progression of length m can be partitioned into at most $4m^{1/2} + 4$ progressions that lift without wrap-around to ordinary integer progressions. Consequently a partition of $\mathbb{Z}_N$ into M mod- N progressions has a refinement into at most*

$$8\sqrt{MN} + 8M \quad (4.43)$$

proper progressions.

Proof. Write $Q = \{a, a+d, \dots, a+(m-1)d\}$. Among the $\lfloor \sqrt{m} \rfloor + 1$ residues $0, d, \dots, \lfloor \sqrt{m} \rfloor d$ choose two whose least integer representatives differ by at most $N/\sqrt{m}$. Their difference is ℓd with $1 \leq \ell \leq \sqrt{m}$. Splitting Q by the parameter index modulo ℓ gives at most $\sqrt{m}$ progressions of common difference ℓd and length $\gg \sqrt{m}$. Subdivide each whenever its lifted span first crosses a multiple of N ; because one step has lifted size at most $N/\sqrt{m}$, at most

two cuts per block are needed after a final subdivision into blocks of parameter length at most $\sqrt{m} + 1$. This gives at most $4\sqrt{m} + 4$ proper pieces.

If the original partition has cell sizes $m_1, \dots, m_M$, sum this bound and use Cauchy–Schwarz:

$$\sum_i \sqrt{m_i} \leq \sqrt{M \sum_i m_i} = \sqrt{MN}.$$

This gives (4.43). $\square$

Lemma 3.48 (Partition bias contains a long positive-density cell). *Let P be a finite set, let $f : P \rightarrow [-1, 1]$ satisfy $\sum_P f = 0$, and let $P = P_1 \sqcup \dots \sqcup P_M$. If*

$$\sum_{j=1}^M \left| \sum_{x \in P_j} f(x) \right| \geq \beta |P|, \quad (4.44)$$

then for some j ,

$$|P_j| \geq \frac{\beta |P|}{4M}, \quad \frac{1}{|P_j|} \sum_{x \in P_j} f(x) \geq \frac{\beta}{4}. \quad (4.45)$$

Proof. Put $S_j = \sum_{P_j} f$ and $S_j^+ = \max(S_j, 0)$. Since $\sum_j S_j = 0$,

$$\sum_j S_j^+ = \frac{1}{2} \sum_j |S_j| \geq \frac{\beta}{2} |P|.$$

Cells with $|P_j| < \beta |P| / (4M)$ contain in total fewer than $\beta |P| / 4$ points and hence contribute less than $\beta |P| / 4$ to the positive sum. The remaining cells therefore contribute at least $\beta |P| / 4$. If every remaining cell had $S_j < (\beta / 4) |P_j|$, their total positive contribution would be $< \beta |P| / 4$, a contradiction. One remaining cell satisfies both inequalities in (4.45). $\square$

Theorem 3.49 (Degree-lowering partition-bias theorem). *For every $d \geq 0$ and $0 < \rho \leq 1$ there are computable positive quantities*

$$b_d(\rho), \quad c_d(\rho), \quad C_d(\rho), \quad L_d(\rho)$$

with the following property. Let P be an arithmetic progression of length $L \geq L_d(\rho)$ and let $h : P \rightarrow \mathbb{D}$. If

$$U_d(h; P) > \rho, \quad (4.46)$$

then P has a partition into arithmetic progressions $R_1, \dots, R_H$ satisfying

$$H \leq C_d(\rho) L^{1-c_d(\rho)} \quad (4.47)$$

and

$$\sum_{j=1}^H \left| \sum_{x \in R_j} h(x) \right| \geq b_d(\rho) L. \quad (4.48)$$

The final cells can be chosen proper when P is viewed inside an auxiliary cyclic group.

Proof. We induct on d .

Base $d = 0$. By Lemma 3.3, (4.46) says $|\mathbb{E}_P h| > \rho^{1/2}$. Take the one-cell partition. Thus one may set $b_0(\rho) = \rho^{1/2}$ and $c_0(\rho) = 1$.

Fourier base $d = 1$. Transport P to $[L]$ and zero-extend it to a prime cyclic group $\mathbb{Z}_N$ with $24L < N < 48L$. As in the general transfer below, (4.46) implies $\|\tilde{h}\|_{U^2(\mathbb{Z}_N)}^4 > c\rho$ for an absolute $c > 0$. With the unnormalized Fourier transform,

$$\|\tilde{h}\|_{U^2}^4 = N^{-4} \sum_{\xi \in \mathbb{Z}_N} |\hat{h}(\xi)|^4.$$

Parseval and $|\tilde{h}| \leq 1$ give $\sum_{\xi} |\hat{h}(\xi)|^2 \leq N^2$, so some frequency ξ satisfies

$$|\hat{h}(\xi)| \geq (c\rho)^{1/2} N.$$

Thus $\tilde{h}$ has correlation of size $(c\rho)^{1/2} N$ with the linear phase $e_N(\xi x)$. Apply Lemma 3.45 with $s = 1$ and tolerance $(c\rho)^{1/2}/4$. Refinement and the triangle inequality give an untwisted partition bias of size at least $(c\rho)^{1/2} N/2$ using $O_{\rho}(N^{1-\kappa_1})$ cells. Intersect with the support interval as at the end of the general step. This proves (4.47)–(4.48) for $d = 1$.

Inductive step. Let $d \geq 2$ and assume the theorem through degree $d - 1$. Transport P to an interval $[L]$ and embed it by zero extension in a prime cyclic group $\mathbb{Z}_N$ with

$$8(d+2)L < N < 16(d+2)L;$$

the elementary prime-in-a-fixed-dilate interval proved in Volume I supplies such an N after increasing $L_d(\rho)$. The denominator in (4.3) is bounded below by a positive constant depending only on d : restrict the cube average to x and increments of size at most $L/(4d+8)$. Hence (4.46) implies

$$U_d(\tilde{h}; \mathbb{Z}_N) > c'_d \rho \tag{4.49}$$

for an explicit $c'_d > 0$.

Put $r = d - 1$. Corollary 3.39, with parameter $c'_d \rho$, gives an r -linear frequency box of power-sized width and a positive-density set of points carrying Fourier coefficients of size $\gg_{d,\rho} N$. Summing their squares gives the hypothesis of Proposition 3.42 with a new computable parameter $\alpha_0(d, \rho) > 0$. Restricting to an odd centered subbox if necessary changes only the parameter constants. Proposition 3.42 then produces a polynomial q of degree at most d and a partition $\mathcal{Q}$ of $\mathbb{Z}_N$ into nearly equal mod- N progressions of common length at least $N^{a_d(\rho)}$, for some computable $a_d(\rho) > 0$, such that, with $g = \tilde{h}e_N(-q)$,

$$U_{d-1}(g \mid \mathcal{Q}) > \rho_1, \tag{4.50}$$

where $\rho_1 = \rho_1(d, \rho) > 0$ is computable.

Because every local uniformity mass lies in $[0, 1]$, the union of cells Q with

$$U_{d-1}(g; Q) > \rho_1/2 \tag{4.51}$$

has total size at least $(\rho_1/2)N$. Apply the induction hypothesis on each cell satisfying (4.51), with parameter $\rho_1/2$, using its natural progression parametrization; leave the other cells unrefined. Summing (4.48) over the good cells gives a refinement $\mathcal{S}$ such that

$$\sum_{S \in \mathcal{S}} \left| \sum_{x \in S} g(x) \right| \geq \gamma_d(\rho) N \tag{4.52}$$

for a computable $\gamma_d(\rho) > 0$. Moreover, if the original Q -length is $W \geq N^{a_d(\rho)}$, then (4.47) in each good cell and one cell for each bad Q give

$$|\mathcal{S}| \leq C'_d(\rho) N W^{-c_{d-1}(\rho_1/2)} \leq C'_d(\rho) N^{1-\delta_d(\rho)} \tag{4.53}$$

with $\delta_d(\rho) = a_d(\rho)c_{d-1}(\rho_1/2) > 0$ after decreasing it if necessary.

Apply Lemma 3.46 to (4.52). It gives a further progression refinement $\mathcal{T}$ for which

$$\sum_{T \in \mathcal{T}} \left| \sum_{x \in T} \tilde{h}(x) \right| \geq \frac{\gamma_d(\rho)}{2} N \quad (4.54)$$

and

$$|\mathcal{T}| \leq C_d''(\rho) N^{1-\delta_d'(\rho)} \quad (4.55)$$

for some computable $\delta_d'(\rho) > 0$. Properize using Lemma 3.47; the square-root loss in the number of cells still leaves an exponent strictly below 1, while refinement cannot decrease the left side of (4.54).

Finally intersect every proper cell with the original interval $[L]$. Because $\tilde{h}$ vanishes outside $[L]$, the cell sums in (4.54) are unchanged; the intersection of a proper arithmetic progression with an interval is an arithmetic progression or empty. Since $N \asymp_d L$, discarding the empty intersections and absorbing the fixed comparison constants gives (4.47)–(4.48) for P . At every stage the quantitative bounds are finite compositions of positive powers, minima and fixed multiplicative constants. Replacing each lower bound by its monotone lower envelope on $[\rho, 1]$ if necessary makes $b_d(\rho)$ and $c_d(\rho)$ nondecreasing in ρ without changing positivity. This defines b_d, c_d, C_d, L_d recursively and completes the induction. $\square$

Proof of Theorem 3.4. Set $d = k - 2$ and $\rho = \eta_k(\alpha)$. By hypothesis,

$$\mathsf{U}_d(f; P) > \rho.$$

Apply Theorem 3.49. We obtain a partition $P = R_1 \sqcup \cdots \sqcup R_H$ such that

$$H \leq C_d(\rho) |P|^{1-c_d(\rho)}, \quad \sum_{j=1}^H \left| \sum_{x \in R_j} f(x) \right| \geq b_d(\rho) |P|.$$

Since $f = 1_A - \alpha$ on P , it is real-valued and has mean zero. Apply Lemma 3.48 with $\beta = b_d(\rho)$. For one cell $P' = R_j$,

$$\text{dens}_{P'}(A) - \alpha = \mathbb{E}_{P'} f \geq \frac{b_d(\rho)}{4}, \quad (4.56)$$

and

$$|P'| \geq \frac{b_d(\rho)}{4C_d(\rho)} |P|^{c_d(\rho)}. \quad (4.57)$$

Increase $N_{0,k}(\alpha)$ so that the fixed prefactor in (4.57) is at least $|P|^{-c_d(\rho)/2}$. Then (4.57) implies $|P'| \geq |P|^{c_d(\rho)/2}$. Thus one may take

$$\iota_k(\alpha) = \frac{1}{4} b_{k-2}(\eta_k(\alpha)), \quad \sigma_k(\alpha) = \frac{1}{2} c_{k-2}(\eta_k(\alpha)),$$

with $N_{0,k}$ enlarged as just described. These are finite recursive functions of k and α , and (4.56)–(4.57) are exactly the conclusions of the theorem. $\square$

Chapter 4

From localization to quantitative Szemerédi

This chapter derives quantitative Szemerédi from the localization theorem of Chapter 3 together with the purification and multilinear-extraction argument developed there. No further structural theorem is used.

Lemma 4.1 (Uniform case supplies progressions). *Fix $k \geq 3$ and $\alpha > 0$. There are computable thresholds $\eta_k(\alpha) > 0$ and $N_{u,k}(\alpha)$ such that the following holds. If $A \subset P$ has relative density α on a progression P of length at least $N_{u,k}(\alpha)$, put $f = 1_A - \alpha$ on P . If f is uniform of degree $k - 2$ at level $\eta_k(\alpha)$, namely*

$$U_{k-2}(f; P) \leq \eta_k(\alpha), \quad (4.58)$$

then A contains a nonconstant k -term arithmetic progression.

Proof. By Definition 3.1, degree $k - 2$ is the U^{k-1} level: (4.58) says that the normalized local U^{k-1} norm is at most $\eta_k(\alpha)^{1/2^{k-1}}$. Affinely identify P with an interval and use the zero-extension convention of Definition 3.2. The generalized von Neumann inequality of Chapter 2, applied after this embedding, bounds every mixed term in the expansion of the normalized k -term progression count by $C_k \eta_k(\alpha)^{1/2^{k-1}}$, where C_k is the fixed boundary-normalization constant coming from the indicator of the interval.

Choose $\eta_k(\alpha)$ so that

$$(2^k - 1)C_k \eta_k(\alpha)^{1/2^{k-1}} \leq \frac{\alpha^k}{4}.$$

This is a completely explicit positive choice once k and α are fixed. Then all nonconstant mixed terms together contribute at most $\alpha^k/4$. Finally choose $N_{u,k}(\alpha)$ so that progressions with zero common difference and the boundary loss in passing between the interval and cyclic models contribute at most another $\alpha^k/4$. The nondegenerate k -AP count is therefore at least $\alpha^k/2 > 0$. $\square$

Lemma 4.2 (Finite number of density increments). *Suppose a density sequence satisfies*

$$\alpha_{j+1} \geq \alpha_j + \iota_k(\alpha_j), \quad 0 < \alpha_0 \leq \alpha_j \leq 1,$$

where ι_k is positive and nondecreasing. Then there are fewer than

$$J_k(\alpha_0) := \left\lceil \frac{1 - \alpha_0}{\iota_k(\alpha_0)} \right\rceil + 1$$

successful increments.

Proof. Monotonicity gives $\iota_k(\alpha_j) \geq \iota_k(\alpha_0)$, hence $\alpha_j \geq \alpha_0 + j\iota_k(\alpha_0)$. This cannot remain at most 1 for $j \geq J_k(\alpha_0)$. $\square$

Definition 4.3 (Recursive scale threshold). Fix k and α . Let $J = J_k(\alpha)$. Define density lower bounds recursively by $a_0 = \alpha$ and $a_{j+1} = a_j + \iota_k(a_j)$. Starting from the last possible stage, define scale thresholds

$$T_J = \max\{N_{u,k}(a_J), N_{0,k}(a_J)\},$$

and, for $j < J$,

$$T_j = \max\{N_{u,k}(a_j), N_{0,k}(a_j), T_{j+1}^{1/\sigma_k(a_j)}\}.$$

This is finite because the recursion has only J stages.

Theorem 4.4 (Gowers quantitative proof of Szemerédi, conditional form). *Assume Theorem 3.4. Fix $k \geq 3$ and $\alpha > 0$. Every arithmetic progression P of length at least $T_0(k, \alpha)$ and every $A \subset P$ of density at least α contains a nonconstant k -term arithmetic progression. In particular this gives an explicit recursive upper bound for the Szemerédi number $N(k, \alpha)$.*

Proof. Set $P_0 = P$, $A_0 = A$, and $\alpha_0 = \text{dens}_{P_0}(A_0)$. At stage j , put $f_j = 1_{A_j} - \alpha_j$ on P_j . If $U_{k-2}(f_j; P_j) \leq \eta_k(\alpha_j)$, Lemma 4.1 gives the desired progression and we stop.

Otherwise f_j fails degree- $(k-2)$ uniformity at exactly the same level $\eta_k(\alpha_j)$, so Theorem 3.4 gives a subprogression $P_{j+1} \subset P_j$ with

$$|P_{j+1}| \geq |P_j|^{\sigma_k(\alpha_j)}, \quad \alpha_{j+1} \geq \alpha_j + \iota_k(\alpha_j).$$

The recursive definition of T_j ensures that if $|P_j| \geq T_j$, then $|P_{j+1}| \geq T_{j+1}$, so the argument may be repeated. Lemma 4.2 shows that fewer than $J_k(\alpha)$ nonuniform steps are possible. Therefore the iteration must enter the uniform case before it exhausts its scale budget, and a nonconstant k -AP is found. $\square$

Corollary 4.5 (Supersaturation). *Assume Theorem 3.4. For every fixed $k \geq 3$ and $\alpha > 0$ there are $c = c(k, \alpha) > 0$ and $N_0 = N_0(k, \alpha)$ such that, for every integer $N \geq N_0$, every $A \subseteq [N]$ with $|A| \geq \alpha N$ contains at least cN^2 ordered nonconstant k -term arithmetic progressions.*

Proof. We spell out the Varnavides double count because it is the step that upgrades one progression to quadratically many. Let $A \subset [N]$ with $|A| \geq \alpha N$. For N sufficiently large, the fixed-dilate prime corollary of the elementary Chebyshev bounds proved in Volume I gives a prime

$$kN < q \leq BkN,$$

where B is an absolute constant; regard A as a subset of the cyclic group $\mathbb{Z}/q\mathbb{Z}$. Its cyclic density is

$$\beta := \frac{|A|}{q} \geq \frac{\alpha}{Bk}.$$

Fix once and for all

$$M \geq T_0\left(k, \frac{\alpha}{2Bk}\right),$$

with $M < q$, and consider all affine embeddings

$$\phi_{a,d} : [M] \longrightarrow \mathbb{Z}/q\mathbb{Z}, \quad \phi_{a,d}(j) = a + jd, \quad d \neq 0.$$

For a uniformly chosen pair (a, d) , each individual value $a + jd$ is uniform in $\mathbb{Z}/q\mathbb{Z}$, so

$$\mathbb{E}_{a,d} \frac{|\phi_{a,d}^{-1}(A)|}{M} = \beta.$$

If p is the proportion of pairs with $|\phi_{a,d}^{-1}(A)|/M \geq \beta/2$, then, since every density is at most 1,

$$\beta \leq p + (1-p)\frac{\beta}{2}, \quad \text{hence} \quad p \geq \frac{\beta}{2-\beta} \geq \frac{\beta}{2}. \quad (\text{Iter.1})$$

Thus at least $(\beta/2)q(q-1)$ affine copies contain a subset of $[M]$ of density at least $\beta/2 \geq \alpha/(2Bk)$. The theorem applied inside $[M]$ gives a nonconstant index progression

$$j_0, j_0 + r, \dots, j_0 + (k-1)r$$

in the preimage of A . Its image is a nonconstant cyclic k -term progression in A .

Because all image points lie in the ordinary interval $[N]$ and $q > kN$, this cyclic progression is in fact an ordinary integer progression. Indeed, if its terms are $x_0, \dots, x_{k-1} \in [N]$, then $x_j - x_0 \equiv j(x_1 - x_0) \pmod{q}$; but

$$|x_j - x_0 - j(x_1 - x_0)| \leq (j+1)N < q,$$

so the congruence is an equality for every $j < k$.

It remains to control multiplicity. Fix one ordered integer k -AP in A . For every possible index progression (j_0, r) inside $[M]$, the equations

$$a + j_0 d = x_0, \quad r d = x_1 - x_0 \pmod{q}$$

determine d and then a uniquely, because $0 < r < M < q$ and q is prime. There are at most M^2 choices of (j_0, r) . Hence any output progression is witnessed by at most M^2 affine copies. Dividing the lower bound in (Iter.1) by this multiplicity gives at least

$$\frac{\beta q(q-1)}{2M^2} \geq c(k, \alpha) N^2$$

ordered nonconstant k -term progressions, with for instance any fixed positive $c(k, \alpha)$ below the displayed coefficient for all sufficiently large N . $\square$

Chapter 5

A proof notebook for the Gowers localization mechanism

The localization theorem is the densest part of Volume IV. The original proof is a long sequence of increasingly rigid descriptions of a frequency graph. The following model calculations are not substitutes for the preceding proof; they are intended to make the transitions reproducible by a student.

5.1 From many additive quadruples to a Freiman map

The key point is not that a function with many respected quadruples is globally additive. Balog–Szemerédi–Gowers first produces a large piece of its graph with small difference set. Gowers’s graph lemma then slices the *second coordinate* by a short progression chosen to avoid the finite vertical fibre

$$K = \{y : (0, y) \in 2k\Gamma - 2k\Gamma\}.$$

Inside one slice, two different k -term outputs for the same k -term input would have a nonzero difference in both K and the chosen progression, which is impossible. This is the exact mechanism behind the order- k Freiman restriction.

Worked Example 5.1 (Why one needs a common vertical fibre). It is not enough to know that every fibre of $k\Gamma - k\Gamma$ is small: the small sets could vary with the first coordinate. What is stable is their *difference*: if $(x, y), (x, y') \in k\Gamma - k\Gamma$, then $(0, y - y') \in 2k\Gamma - 2k\Gamma$. Thus all fibre differences lie in the single set K above. This is the point at which the small-doubling hypothesis becomes usable for a uniform slicing argument.

5.2 The bilinear model

Worked Example 5.2 (Recovering a bilinear coefficient from two rows). Suppose on a box the frequency map has the form $\phi(x, y) = a_x y + b_x$. Choose two y -values $y_1 \neq y_2$. Then

$$a_x = \frac{\phi(x, y_1) - \phi(x, y_2)}{y_1 - y_2}, \quad b_x = \phi(x, y_1) - a_x y_1.$$

If the two cross-sections $x \mapsto \phi(x, y_i)$ are already Freiman-linear on a large progression, so are a_x and b_x . This two-point recovery is the elementary linear algebra hidden inside the higher-dimensional random sampling step.

5.3 Why a multilinear phase can be integrated locally

Worked Example 5.3 (The quadratic case). For a bilinear form $\sigma(h) = 2ch$, the identity

$$2chu = (s^2 - (s-h)^2 - (s-u)^2 + (s-h-u)^2)c$$

expresses the Fourier phase as a second finite difference of the quadratic polynomial cs^2 . Twisting f by $e_N(-cs^2)$ therefore removes the structured derivative frequency. The general polarization lemma is precisely the higher-degree version of this identity.

5.4 The density-increment logic

The final iteration has three scales: a global density, a long progression on which a lower-degree obstruction survives, and the polynomial phase used to expose it. At each successful step the density increases by a quantity depending only on the current density and the target progression length remains a positive power of the previous scale. Since density cannot exceed one, only finitely many increments occur. The enormous quantitative losses come from repeatedly composing these power-scale functions, not from an unrecorded compactness argument.

Additive Combinatorics
Volume V
Higher-Order Fourier Analysis and Inverse Theory
Version 1.10

14 September 2026

Contents

1	Nilpotent groups and Mal’cev geometry	1
2	Polynomial sequences and nilsequences	6
3	Quantitative equidistribution	10
4	Factorization and polynomial nilsequence counting	12
5	The inverse-theorem layer	14
5.1	What is already owned	14
5.2	Further quantitative inverse theorems	15
5.3	A rigorous local-to-global model on progressions	16
5.4	Polynomial-path suspension and exact cyclic periodicization	17
6	The all-degree finite-field inverse theorem	21
6.1	Statement of the theorem	21
6.2	Intrinsic polynomial language	21
6.3	Depth normal form and exact roots	22
6.3.1	The cyclic differential-operator identity	23
6.3.2	A divisibility estimate for wrapped monomials	23
6.3.3	The classical layer	25
6.3.4	Full depth classification	25
6.3.5	Exact roots	26
6.4	Rank and analytic-rank calculus: the elementary half of F3	27
6.4.1	Analytic rank under addition and restriction	28
6.4.2	Bounded-codimension stability: the missing elementary part of the article’s rank package	30
6.5	TZ4: multiplication by p , reduced to one earlier recurrence input	32
6.6	Beginning TZ5: the classical symmetric-multilinear algebra	33
6.7	Beginning F4: a complete classical regularization model	37
6.8	Primary-source architecture of the finite-field inverse theorem	39
6.9	How this branch differs from the cyclic Manners branch	39
6.10	The Kaufman–Lovett bias criterion: from average structure to exact rank . .	40
6.10.1	Step 1: bias gives an approximate derivative factor	40
6.10.2	Step 2: strongly regular polynomial factors	41
6.10.3	Step 3: strong regularity gives almost independence on cubes	43
6.10.4	Step 4: approximate computation becomes exact computation	45
6.10.5	Step 5: the finite-field bias criterion	47
6.10.6	The CSM bias criterion used by Tao–Ziegler	48
6.11	Regular CSM systems are jointly equidistributed	48

6.12	From equidistribution to the support-partition algebra	50
6.12.1	Two quantitative preliminaries	51
6.12.2	The support-partition theorem	52
6.13	Symmetrization and integration of the biased CSM form	54
6.14	Weighted polynomial calculus and exact roots on integer lattices	56
6.15	Positive-depth factors, cube equidistribution, and factorwise exact roots	61
6.15.1	Factors and depth extensions	61
6.15.2	The full positive-depth regularity lemma	62
6.15.3	The explicit cube group of a filtered factor	66
6.15.4	Fourier proof of cube equidistribution	66
6.15.5	Polynomial functions on a regular factor	68
6.15.6	Technical exact roots on a regular factor	69
6.16	The simultaneous GIP induction	70
6.16.1	From factorwise exact roots to exact roots of bounded-rank polynomials	70
6.16.2	The induction step	70
6.17	The Bergelson–Tao–Ziegler ergodic inverse theorem	72
6.17.1	Seminorm recursion and factor restriction	74
6.17.2	Finite-characteristic algebra of ergodic phase polynomials	74
6.17.3	Cocycles, coboundaries, and a free vertical fibre	75
6.17.4	Structure of the Bergelson–Tao–Ziegler argument	76
6.18	The Bergelson–Tao–Ziegler characteristic-factor argument	76
6.18.1	Cube measures and the universal characteristic factor	77
6.18.2	The Host–Kra compact abelian extension step	78
6.18.3	Type, cube coboundaries, and descent	81
6.18.4	From order to type	82
6.18.5	Vertical differentiation and the Conze–Lesigne reduction	83
6.18.6	The finite vertical group: zero curvature and exact integration	84
6.18.7	The Abramov conclusion and the ergodic inverse theorem	86
6.19	The weak inverse theorem: the correspondence-principle front	87
6.19.1	Random linear sampling of global cube averages	87
6.19.2	The compact shift system	89
6.19.3	The local-testability theorem, proved internally	91
6.19.4	What remains in the correspondence route	94
6.20	Tao–Ziegler Section 11: from the weak inverse theorem to the sharp degree	94
6.20.1	Transfer of the ergodic phase back to finite vector spaces	94
6.20.2	A finite compactness version of the Section-11 structured/pseudorandom split	95
6.20.3	Rank separation and the minimal weak degree	96
7	Manners Quantitative Inverse Theory	99
7.1	Quantitative inverse theorem	99
7.2	The symmetrized derivative and its Fourier graph	100
7.3	The first genuine Manners bottleneck: respecting many cubes	101
7.4	Lifting a circle-valued approximate polynomial to the real line	106
7.5	Dependencies in the quantitative argument	106
8	Manners quantitative inverse theory II: from partial cubes to global polynomial hierarchies	108
8.1	Cubes, faces, and the meaning of a partial polynomial	108
8.2	Popular systems of cubes	109
8.3	Robust connectivity and non-interacting pieces	112

8.4	A sampled expander remains connected	113
8.5	Extending one robust component to almost all of the group	114
8.6	Polynomial hierarchies and recursive global structure	116
9	Manners strong hierarchies: normal forms, cocycles, redundancy, and boosting	120
9.1	Why the soft hierarchy is not yet enough	120
9.2	Normal form for derivative coefficients	120
9.3	Strong derivative conditions and generalized cocycles	121
9.4	Reductions and how a failed strong condition exposes redundancy	127
9.4.1	Discrete Cauchy–Schwarz complexity	128
9.4.2	Upper-compatibility failure	129
9.4.3	Failure of the generalized cocycle law	130
9.5	Boosting approximate polynomials to derivative conditions	131
9.5.1	The additive-growth input	133
9.6	The strong-hierarchy upgrade	138
10	From strong hierarchies to quantitative nil-polynomials	142
10.1	Quantitative polynomial-map groups	142
10.2	Nil-polynomials produce polynomial hierarchies	144
10.3	Realizing one generalized cocycle	144
10.4	M6: the structure theorem for strong hierarchies	145
10.5	M7: quantitative nilmanifold reconciliation	147
11	Manners quantitative inverse theory: spectral covering, nilsequence smoothing, and descent	148
11.1	M8a: covering almost all Fourier mass by nil-polynomial graphs	148
11.2	M8b: turning a nil-polynomial frequency into a Lipschitz nilsequence	149
11.3	M9: diagonal sampling and removal of derivative dilations	152
11.4	M10: quantitative induction and the selected inverse theorem	153
12	Worked models for higher-order structure	155
I	Detailed Proof Appendices	156
A	The U^3 Inverse Theorem over Finite Fields	157
A.1	Statement and proof map	157
A.2	A notation for harmless bounded factors	158
A.3	Step 1: many derivatives have Fourier structure	158
A.4	Step 2: the derivative frequencies have many additive quadruples	159
A.5	Step 3: BSG, Plünnecke, and graph rectification	160
A.6	Step 4: Bogolyubov produces a linear derivative map	161
A.7	Step 5: the symmetry argument in full	162
A.8	Step 6: integrating the symmetric derivative	164
A.9	Step 7: from local quadratic bias to global correlation	165
A.10	A check on the model example	166
A.11	What was genuinely difficult	166

B	The Cyclic U^3 Inverse Theorem: Local Quadratics	167
B.1	The theorem	167
B.2	Derivative frequencies and a small-doubling graph	168
B.3	Killing the vertical kernel	169
B.4	A locally linear frequency map	170
B.5	The symmetry calculation	171
B.6	Integrating the local derivative phase	172
B.7	From local quadratics to bracket quadratics	173
B.8	What the proof teaches	175
C	Equidistribution on Nilmanifolds	176
C.1	Weyl’s torus criterion	176
C.2	A quantitative torus obstruction	176
C.3	Smoothness norms	176
C.4	Factorization	177
C.5	Why inverse theory needs factorization	177
C.6	Weyl’s inequality as the abelian prototype	177
C.7	Horizontal characters in the Heisenberg example	177
C.8	Rational subgroups and dimension drop	178
C.9	Smooth, rational, and irrational components	178
C.10	Why total equidistribution is often needed	179
D	The Green–Tao–Ziegler Inverse Theorem	180
D.1	The theorem and the proof contract	180
D.2	Bias notation	181
D.3	Base case: degree one	181
D.4	Step 1: many derivatives have large lower Gowers norm	181
D.5	Steps 2–5: the inner degree–rank linearisation	182
D.5.1	2A. Cauchy–Schwarz eliminates the function and the part already built	183
D.5.2	2B. Represent and regularise the residual frequencies	183
D.5.3	2C. Additive combinatorics forces bracket-linearity	183
D.5.4	2D. Realise the current rank and lower the residual rank	184
D.6	Step 6: symmetry	184
D.7	Step 7: integrate the symmetric symbol	185
D.8	Step 8: close the induction	185
D.9	What the proof actually taught us	186
D.10	The corrected filtration in the Green–Tao–Ziegler argument	186
E	A Heisenberg Nilsequence Laboratory	188
E.1	The group	188
E.2	Iterating one element	188
E.3	The integer lattice	189
E.4	Reduction to a fundamental domain	189
E.5	A bracket-polynomial orbit	189
E.6	Vertical characters	190
E.7	Horizontal characters	190
E.8	A two-variable cube calculation	190
E.9	Polynomial sequences with independent central motion	191
E.10	What generalizes	191

F Nilcharacters, Symbols, and Bias	192
F.1 Vertical tori and vertical frequencies	192
F.2 Vertical Fourier decomposition	192
F.3 Equivalence modulo lower degree	193
F.4 Degree–rank symbols and the inner induction	194
F.5 Derivatives lower degree	195
F.6 Universal horizontal-frequency representations	197
F.7 Bias and the top symbol	199
F.8 The converse estimate: structured correlation forces Gowers uniformity	201
F.9 Multidegree and polarization	202
F.10 Splitting multidegree factors	203
G Coherence, Sunflowers, and Bracket Linearisation	205
G.1 A Gowers–Cauchy–Schwarz coherence lemma	205
G.2 Frequency regularisation: core, petals, and representation	208
G.3 Why independent petals cannot interact	209
G.4 From many symbol relations to a Freiman graph	211
G.5 Freiman homomorphisms become bracket-linear	212
G.6 The degree–rank coherence theorem	216
H Building, Symmetrising, and Integrating the Nilobject	218
H.1 Free filtered nilpotent groups	218
H.2 Realising bracket-linear symbols	219
H.3 Symmetry from commuting derivatives	221
H.4 Diagonal integration	224
H.5 The final Gowers closing argument	225
I Rational Nilmanifolds and Mal’cev Coordinates	227
I.1 The objects and the convention	227
I.2 Strong Mal’cev bases	227
I.3 The metric and Lipschitz control	228
J Polynomial Orbits, Horizontal Characters, and Equidistribution	229
J.1 Polynomial sequences	229
J.2 Horizontal characters	229
J.3 Quantitative obstruction theorem	230
K Smooth–Equidistributed–Rational Factorization	231
K.1 Three kinds of motion	231
K.2 The factorization theorem	231
L Mal’cev Theory with Rationality Bounds	233
L.1 The exponential map and rational Lie algebras	233
L.2 A rational flag and a strong basis	237
L.3 Triangular polynomial coordinate laws	238
L.4 Quantitative rational linear algebra	239
M Quantitative Leibman Theory and Factorization	241
M.1 Why filtered polynomial maps form a group	241
M.2 Polynomial phases on a torus	242
M.3 Vertical Fourier analysis and dimension descent	246

M.4	Lifting phase decompositions	253
M.5	The factorization descent	255
N	The Equal-Sides Multiparameter Counting Theorem	257
N.1	Multivariate smoothness norms	257
N.2	From bad cubes to one fixed horizontal character	258
N.3	Multiparameter factorization	259
N.4	Freezing and counting	260
	References	261

Chapter 1

Nilpotent groups and Mal'cev geometry

Proof architecture Why this chapter is here

Higher-order characters live on nilmanifolds. The algebraic content needed later is not a slogan: one needs coordinates in which multiplication is polynomial, lattices are rational, rational subgroups have controlled bases, and all of these statements carry quantitative height bounds. The detailed Mal'cev appendix proves these coordinate, lattice, rational-subgroup, and height statements in full.

Definition 1.1 (Lie-group vocabulary used in this volume). A *finite-dimensional real Lie group* is a finite-dimensional smooth real manifold G equipped with a group law for which multiplication and inversion are smooth. Its *Lie algebra* is the tangent space $\mathfrak{g} = T_e G$ at the identity, with bracket defined by identifying tangent vectors with left-invariant vector fields and taking their vector-field commutator. For $X \in \mathfrak{g}$, the *exponential* $\exp X$ is the value at time 1 of the one-parameter subgroup whose tangent vector at the identity is X .

For group elements we use $[x, y] = xyx^{-1}y^{-1}$ unless a proof explicitly announces the inverse convention. For subgroups $A, B \leq G$, the notation $[A, B]$ means the subgroup generated by all $[a, b]$ with $a \in A$ and $b \in B$. A subgroup $\Gamma < G$ is *discrete* when it is a discrete subset of the manifold topology, and *cocompact* when G/Γ is compact.

Definition 1.2 (Nilpotent and nilmanifold terminology). A group G is *s-step nilpotent* if its lower central series $G_1 = G$, $G_{i+1} = [G, G_i]$ satisfies $G_{s+1} = \{e\}$. Throughout this volume, a *lattice* in a Lie group means a discrete *cocompact* subgroup Γ . Thus G/Γ is compact by definition, and a *nilmanifold* is a quotient G/Γ with G connected, simply connected and nilpotent and Γ a lattice in this sense. Some texts define a lattice by finite invariant covolume and then prove cocompactness in the nilpotent setting; that equivalence is not needed here, so we do not hide it inside the terminology.

Definition 1.3 (Filtration). A *filtration* on a group G is a descending sequence

$$G = G_0 \geq G_1 \geq G_2 \geq \cdots$$

satisfying $[G_i, G_j] \subseteq G_{i+j}$ for all i, j . It has *degree at most s* when $G_{s+1} = \{e\}$. This is the group-theoretic notion used when we say that a Mal'cev basis is adapted to a filtration; the additional rationality condition needed for filtered nilmanifolds is defined after rational subgroups have been introduced.

Definition 1.4 (Strong lattice-adapted Mal'cev basis). Let G be connected, simply connected and nilpotent, with Lie algebra $\mathfrak{g}$, and let $\Gamma < G$ be a lattice. A basis $\mathcal{X} = (X_1, \dots, X_D)$ of $\mathfrak{g}$ is a *strong Mal'cev basis* for G/Γ when every tail

$$\text{span}(X_j, \dots, X_D)$$

is an ideal and its connected subgroup $G_j^* := \exp(\text{span}(X_j, \dots, X_D))$ satisfies that $G_j^* \cap \Gamma$ is a lattice in G_j^* , every $g \in G$ has a unique second-kind representation

$$g = \exp(t_1 X_1) \cdots \exp(t_D X_D),$$

and successive reduction of these coordinates modulo Γ uses integer translations in the coordinate directions. It is *adapted to a filtration* $G_\bullet$ when each Lie algebra $\text{Lie}(G_i)$ is spanned by a terminal block of the basis. These are the coordinates in which our lattice and quantitative rationality data are measured.

Definition 1.5 (Rational subgroups and horizontal characters). Fix a strong lattice-adapted Mal'cev basis $\mathcal{X}$ for G/Γ . A connected closed subgroup $H \leq G$ is Γ -*rational* when $\text{Lie}(H)$ admits a basis whose coordinates in $\mathcal{X}$ are rational. In the nilpotent setting this is equivalent to $H \cap \Gamma$ being a lattice in H ; the equivalence is proved in the detailed Mal'cev appendix. A *horizontal character* is a continuous homomorphism $\eta : G \rightarrow \mathbb{R}/\mathbb{Z}$ that annihilates both Γ and $[G, G]$. Relative to $\mathcal{X}$ it is represented on the abelianization by an integer coefficient vector, and its *height* or *complexity* is the size of that vector. Thus rationality and horizontal characters are not coordinate-free slogans here: the fixed Mal'cev basis is part of the quantitative data used throughout the volume.

Definition 1.6 (Rational filtrations and filtered nilmanifolds). For a nilmanifold G/Γ , a filtration $G_\bullet$ is Γ -*rational* when every connected filtration subgroup G_i is Γ -rational in the preceding sense. A *filtered nilmanifold* $G_\bullet/\Gamma$ means a nilmanifold equipped with such a filtration. These are the group-theoretic data fixed before polynomial maps are defined in the next chapter.

Definition 1.7 (Quantitative Mal'cev presentation, complexity, and metric). Fix a filtered nilmanifold $G_\bullet/\Gamma$ and a strong Mal'cev basis $\mathcal{X} = (X_1, \dots, X_D)$ adapted both to the filtration and to the lattice. For a rational number a/b in lowest terms, its *height* is $\max\{|a|, |b|\}$; the height of a rational vector or matrix is the maximum height of its entries. Write

$$[X_i, X_j] = \sum_{k=1}^D c_{ijk} X_k.$$

We call $(G_\bullet/\Gamma, \mathcal{X})$ an *explicit quantitative Mal'cev presentation of complexity at most M* when the filtration degree and dimension are at most M , the rational structure constants c_{ijk} have height at most M , and $\mathcal{X}$ is lattice-adapted in the strong sense of Definition 1.4: successive second-kind reduction modulo Γ uses integer coordinate translations. A filtered nilmanifold has *complexity at most M* when such a presentation has been fixed.

The same presentation fixes the metric conventions used throughout Volume V. On G we use the left-invariant Riemannian metric d_G for which $X_1, \dots, X_D$ is orthonormal at the identity. On G/Γ we use the quotient metric

$$d_{G/\Gamma}(g\Gamma, h\Gamma) := \inf_{\gamma \in \Gamma} d_G(g, h\gamma).$$

On bounded coordinate boxes these metrics are quantitatively equivalent to the coordinate metrics used in the detailed Mal'cev appendix, with polynomial losses when dimension and step are fixed. Thus phrases such as “bounded-complexity nilmanifold”, “ M -Lipschitz”, and “ $d_G(e, r(x)) \leq M$ ” have a fixed meaning before they enter the inverse-theorem statements.

Worked Example 1.8 (Complexity one on the circle). Take $G = \mathbb{R}$, $\Gamma = \mathbb{Z}$, $X_1 = 1$, and the degree-one filtration $G_0 = G_1 = \mathbb{R}$, $G_2 = \{0\}$. There are no non-zero bracket structure constants, so the presentation has complexity 1. The metric d_G is ordinary Euclidean distance and the quotient metric on $\mathbb{R}/\mathbb{Z}$ is

$$d_{\mathbb{R}/\mathbb{Z}}(x + \mathbb{Z}, y + \mathbb{Z}) = \min_{n \in \mathbb{Z}} |x - y - n|.$$

This is the quantitative model behind the first nil-polynomial example in the next chapter.

Proposition 1.9 (Lower central series calculus). *Let G be a group and $G_1 = G$, $G_{i+1} = [G, G_i]$. Then $G_{i+1} \triangleleft G_i \triangleleft G$, $[G_i, G_j] \subseteq G_{i+j}$, and each quotient G_i/G_{i+1} is abelian. In particular, if $G_{s+1} = \{e\}$, every iterated commutator of weight $> s$ vanishes.*

Proof. Normality follows inductively from $g[x, y]g^{-1} = [gxg^{-1}, gyg^{-1}]$. For the commutator estimate use the following consequence of the Hall–Witt identity: if A, B, C are normal subgroups, then

$$[[A, B], C] \subseteq [[B, C], A] [[C, A], B]. \quad (*)$$

For completeness, use the temporary convention $\langle x, y \rangle = x^{-1}y^{-1}xy$ and $w^g = g^{-1}wg$. Then the standard Hall–Witt identity is

$$\langle \langle x, y^{-1} \rangle, z \rangle^y \langle \langle y, z^{-1} \rangle, x \rangle^z \langle \langle z, x^{-1} \rangle, y \rangle^x = e.$$

This is a literal free-group identity, and we record the cancellation so it is not being used as a black box. The three displayed factors expand respectively to

$$\begin{aligned} x^{-1}y^{-1}xz^{-1}x^{-1}yx y^{-1}zy, \\ y^{-1}z^{-1}yx^{-1}y^{-1}zy z^{-1}xz, \\ z^{-1}x^{-1}zy^{-1}z^{-1}xz x^{-1}yx. \end{aligned}$$

Multiplying the first two and freely cancelling gives

$$x^{-1}y^{-1}xz^{-1}x^{-1}zyz^{-1}xz;$$

after multiplication by the third factor, adjacent inverse pairs cancel successively from the join outward until the empty word remains. Thus the Hall–Witt identity has been verified directly in the free group. Modulo the normal product $[[B, C], A][[C, A], B]$ the last two factors vanish, so the first does too; this is $(*)$. Normality also makes the subgroup commutators independent of which of the two inverse commutator conventions is used.

We now induct on i , for all j at once. The case $i = 1$ is the definition $[G, G_j] = G_{j+1}$. Assuming $[G_i, G_j] \subseteq G_{i+j}$ for every j , apply $(*)$ to $A = G$, $B = G_i$, $C = G_j$:

$$[G_{i+1}, G_j] \subseteq [[G_i, G_j], G] [[G_j, G], G_i] \subseteq G_{i+j+1};$$

the second factor is controlled by the same induction hypothesis with $j + 1$. Thus $[G_i, G_j] \subseteq G_{i+j}$ for all i, j . Taking $j = i$ gives $[G_i, G_i] \subseteq G_{2i} \subseteq G_{i+1}$, so G_i/G_{i+1} is abelian. An iterated commutator of weight r lies in G_r , and hence vanishes when $r > s$. $\square$

Theorem 1.10 (BCH truncation in the nilpotent case). *If $\mathfrak{g}$ is an s -step nilpotent real Lie algebra, then*

$$\log(\exp X \exp Y) = X + Y + \frac{1}{2}[X, Y] + \cdots$$

is a finite Lie polynomial containing only commutators of weight at most s . Consequently multiplication and inversion in exponential coordinates are polynomial maps whose degrees depend only on s .

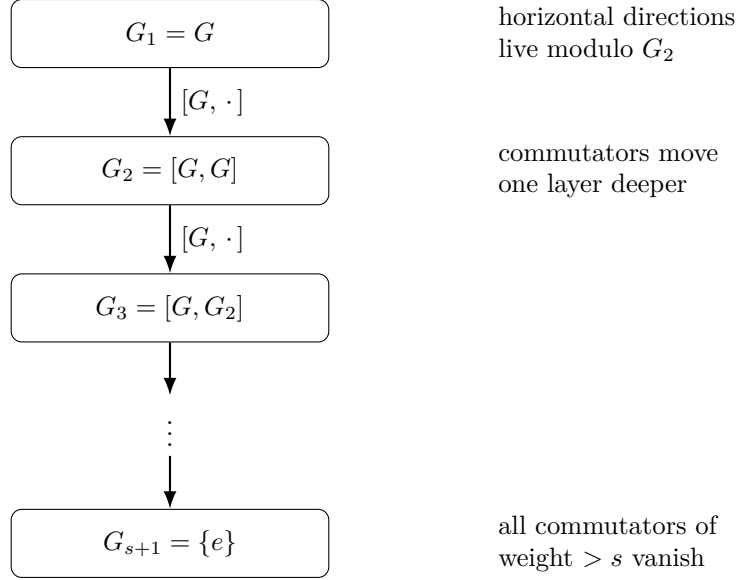


Figure 1.1: The lower central filtration. Each commutator with G pushes information one level down. In the Heisenberg example the diagram stops at $G_3 = \{e\}$: the x - and y -directions are horizontal, while their commutator is the central z -direction in G_2 .

Proof. The nontrivial point is that $\log(e^X e^Y)$ is a Lie series in the first place; mere nilpotence only explains why such a series would truncate. The detailed proof is Lemma L.1 in the Mal'cev appendix. There one works in the completed free associative algebra, differentiates $Z(t) = \log(e^{tX} e^{tY})$, and obtains the recursion

$$Z'(t) = \frac{\text{ad}_{Z(t)}}{e^{\text{ad}_{Z(t)}} - 1} (X + e^{t \text{ad}_X} Y).$$

Induction on homogeneous degree then proves that the coefficient of t^r is a Lie polynomial of bracket weight r . In an s -step nilpotent algebra every term of weight $> s$ vanishes, leaving a finite polynomial. The same appendix proves associativity from the formal exponential/logarithm identity and proves the local Lie-group BCH identity by deriving the differential of the exponential map. Thus multiplication is the finite BCH polynomial and inversion is $X \mapsto -X$. $\square$

Theorem 1.11 (Quantitative Mal'cev coordinate package). *Let G be a connected, simply connected, s -step nilpotent Lie group, let $\Gamma < G$ be a lattice, and let $\mathcal{X} = (X_1, \dots, X_D)$ be a strong lattice-adapted Mal'cev basis. Then:*

1. $\exp : \mathfrak{g} \rightarrow G$ is a global diffeomorphism;
2. Γ determines a rational structure on $\mathfrak{g}$;
3. the second-kind coordinate box $[0, 1)^D$ is a fundamental domain for G/Γ ;
4. multiplication, inversion, and first/second-kind coordinate conversion are triangular rational polynomial maps of degree $O_s(1)$;
5. quantitatively, if the structure constants in $\mathcal{X}$ have height at most $M \geq 2$ and a connected rational Lie subgroup $H = \exp \mathfrak{h}$ is cut out in $\mathcal{X}$ -coordinates by rational linear equations of height at most M , then $H/(H \cap \Gamma)$ admits a strong Mal'cev basis whose coordinate vectors have height at most $M^{C(D,s)}$. In particular, if η is a horizontal character with $|\eta| \leq M$, the connected kernel $(\ker \eta)^\circ$ has such a basis with the same bound. The same

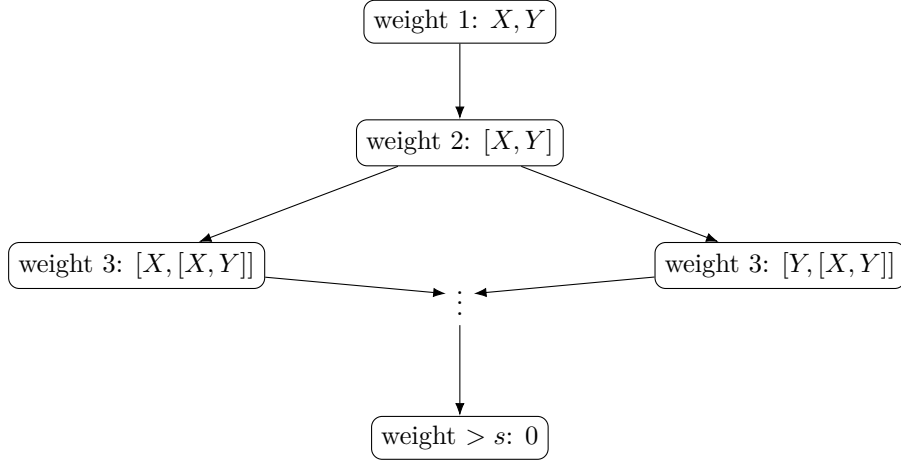


Figure 1.2: BCH is organized by commutator weight. The formal proof first shows that there are no non-Lie terms hiding in $\log(e^X e^Y)$; nilpotence then kills every branch above weight s , converting an infinite formal series into a finite group-law polynomial.

polynomial bound survives at most D successive kernel/intersection operations with the fixed rational filtration.

Proof. Appendix L proves the required exponential coordinates, lattice rationality, rational central flags, strong lattice-adapted Mal'cev bases, first/second-kind coordinate conversion, polynomial group laws, and height-controlled kernel statements in that order. Together these ingredients establish all assertions of the theorem. $\square$

Chapter 2

Polynomial sequences and nilsequences

Linear characters are enough for U^2 , but higher uniformity norms repeatedly differentiate structured objects. The correct nonabelian analogue of an ordinary polynomial is therefore not defined by coordinates first: it is defined by the rule that each discrete derivative falls one step deeper in a filtration. This makes the notion invariant under changes of Mal'cev coordinates and, crucially, compatible with the nilpotent commutator structure that appears in inverse theorems.

The model to keep in mind is ordinary finite-difference calculus: Δ_h lowers the degree of a polynomial by one. A filtered nilpotent group replaces “degree at most r ” by “lies in G_r ”. The first theorem below says that this analogy survives multiplication, but its proof requires genuine noncommutative cube calculus rather than formal commutation.

Definition 2.1 (Polynomial sequences). Recall from Definition 1.3 that a filtration $G_\bullet$ is descending and satisfies $[G_i, G_j] \subseteq G_{i+j}$. The degree- s filtrations used for s -step nilsequences satisfy $G_{s+1} = \{e\}$; when $G_0 = G_1 = G$, the terminal group G_s is then central because $[G, G_s] \subseteq G_{s+1} = \{e\}$. For a map $g : \mathbb{Z}^d \rightarrow G$ and $h \in \mathbb{Z}^d$ define the left discrete derivative

$$\partial_h g(n) := g(n+h)g(n)^{-1}.$$

We call g *polynomial adapted to $G_\bullet$* when, for every $r \geq 1$ and every $h_1, \dots, h_r$,

$$\partial_{h_r} \cdots \partial_{h_1} g(n) \in G_r \quad \text{for every } n.$$

For multidegree filtrations used later, the same definition is made with the corresponding multi-index layer; that refinement is introduced at the point where it is needed.

A filtration is called *proper* when $G_0 = G_1$. If A is an abelian group and $r \geq 0$, its *standard degree- r filtration* $A_{(r)}$ is

$$(A_{(r)})_0 = \cdots = (A_{(r)})_r = A, \quad (A_{(r)})_{r+1} = \{0\}.$$

In particular $\mathbb{R}_{(r)}$ denotes the additive group $\mathbb{R}$ with this filtration. These conventions are the ones used in the quantitative Manners structure arguments later in the volume.

Worked Example 2.2 (A degree-two polynomial sequence in the Heisenberg group). Write the real Heisenberg group in matrix coordinates

$$(x, y, z) := \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}, \quad (x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy').$$

Its commutator subgroup is the centre $G_2 = \{(0, 0, z)\}$; take the filtration $G_0 = G_1 = G$, $G_2 = Z(G)$, $G_3 = \{e\}$. Choose $a \in G$ and a central element $c \in G_2$, and define

$$g(n) = a^n c^{\binom{n}{2}}.$$

Because c is central,

$$\partial_1 g(n) = g(n+1)g(n)^{-1} = ac^n,$$

and therefore

$$\partial_1^2 g(n) = c \in G_2, \quad \partial_1^3 g(n) = e \in G_3.$$

The same binomial-difference calculation works for arbitrary increments. Thus g is polynomial of filtered degree two. Notice the point of the filtration: the first derivative need not be close to the identity, but the second derivative is forced into the centre, where noncommutativity has disappeared.

Definition 2.3 (Nil-polynomial; Manners convention). Let H be an abelian group. A function $f : H \rightarrow \mathbb{R}$ is a *nil-polynomial of degree s , dimension D , and complexity at most M* if there are

1. a degree- s filtered nilmanifold G/Γ of dimension D and complexity at most M ;
2. a lift $r : H \rightarrow G$ satisfying $d_G(e, r(x)) \leq M$ for every x , such that the quotient map $x \mapsto r(x)\Gamma$ is polynomial; and
3. a polynomial map $F : G \rightarrow \mathbb{R}_{(s)}$

such that $f = F \circ r$. Here $\mathbb{R}_{(s)}$ denotes the filtration $\mathbb{R}_0 = \dots = \mathbb{R}_s = \mathbb{R}$ and $\mathbb{R}_{s+1} = \{0\}$ from the preceding definition of standard degree filtrations.

The bounded lift is part of the definition: a nil-polynomial is not merely an arbitrary polynomial function on a nilmanifold. The lift records the controlled choice of representatives needed when one passes between a periodic quotient and real-valued bracket-type coordinates.

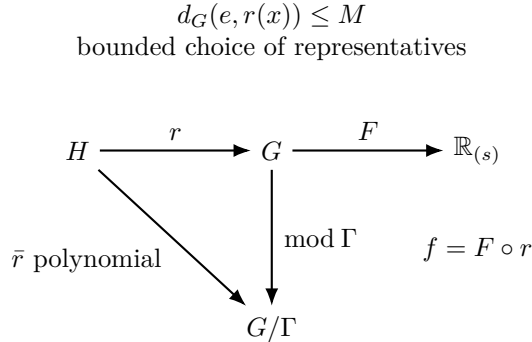


Figure 2.1: A nil-polynomial separates quotient polynomiality from the bounded lift. The map $\bar{r} = r \bmod \Gamma$ must be polynomial on the compact nilmanifold; the representative r need not itself be polynomial, but it must stay in a controlled region so that the real-valued polynomial F can be evaluated with quantitative bounds.

Worked Example 2.4 (The fractional-part function as a degree-one nil-polynomial). Let $H = \mathbb{Z}/N\mathbb{Z}$, $G = \mathbb{R}$, $\Gamma = \mathbb{Z}$, and use the degree-one filtration $G_0 = G_1 = \mathbb{R}$, $G_2 = \{0\}$. For the residue class $x \in H$, let $\tilde{x} \in \{0, 1, \dots, N-1\}$ be its standard integer representative and put

$$r(x) = \frac{\tilde{x}}{N} \in [0, 1), \quad F(t) = t.$$

Then $d_G(0, r(x)) \leq 1$, while

$$r(x) \bmod \mathbb{Z} = \frac{x}{N} \pmod{1}$$

is a homomorphism $H \rightarrow \mathbb{R}/\mathbb{Z}$, hence a polynomial map of degree one. The identity map $F : \mathbb{R} \rightarrow \mathbb{R}_{(1)}$ is polynomial, so

$$f(x) = F(r(x)) = \frac{\tilde{x}}{N} = \left\{ \frac{x}{N} \right\}$$

is a degree-one nil-polynomial. The example explains why the lift r is allowed to jump at the boundary: r itself is not a homomorphism into $\mathbb{R}$, but its class modulo the lattice is polynomial, and the bounded representative is exactly what turns that quotient-polynomial behaviour into a real-valued bracket coordinate.

Definition 2.5 (Nilsequences and vertical characters). Let G/Γ be a nilmanifold, let $g : \mathbb{Z}^d \rightarrow G$ be a polynomial sequence, and let $F : G/\Gamma \rightarrow \mathbb{C}$ be continuous. The sequence

$$n \mapsto F(g(n)\Gamma)$$

is a *polynomial nilsequence*. If the terminal layer G_s is central and projects to the compact torus $G_s/(G_s \cap \Gamma)$, a function F has *vertical frequency* ξ when

$$F(ux) = e(\xi(u))F(x) \quad (u \in G_s),$$

where ξ is a character of that torus. Such a vertically pure nilsequence is called a *nilcharacter*. Horizontal characters were defined in the previous chapter; vertical characters live on the opposite, top-central end of the filtration.

Theorem 2.6 (Filtered polynomial calculus). *Polynomial sequences adapted to a filtration are closed under pointwise multiplication and inversion. In Mal'cev coordinates each coordinate is an ordinary polynomial combination of binomial-coefficient monomials, with degree controlled by the filtration.*

Proof. With our left-derivative convention, direct multiplication gives

$$\partial_h(gg')(n) = (\partial_h g)(n) g(n) (\partial_h g')(n) g(n)^{-1}, \quad \partial_h(g^{-1})(n) = g(n+h)^{-1} (\partial_h g(n))^{-1} g(n+h). \quad (2.1)$$

These identities explain why closure is nontrivial: further derivatives create nested conjugations, and one must know that every commutator error lands in the correct deeper filtration layer.

The complete argument is Proposition M.2 in the Detailed Proof Appendices. There, a k -dimensional parallelepiped is sent to the Host–Kra cube group generated by codimension- i faces labelled by G_i . The face-collection lemma uses $[G_i, G_j] \subseteq G_{i+j}$ to prove that these cube sets are groups, and the cube characterization proves that membership for every parallelepiped is equivalent to the iterated-derivative definition above. Pointwise products and inverses then stay inside every cube group, giving closure with no hidden commutator step.

For the coordinate assertion, choose an adapted Mal'cev basis and descend through the central series. In an abelian quotient, iterated discrete differentiation is ordinary finite-difference calculus, so the coordinates are linear combinations of $\binom{n}{0}, \dots, \binom{n}{s}$. Lifting across one central layer, the polynomial multiplication and inversion laws from Theorem 1.11 add only polynomial expressions whose filtration weights are sums of weights already present. Thus the coordinate degrees remain controlled. The fully quantified coordinate conversion is part of the same detailed polynomial-orbit appendix. $\square$

Theorem 2.7 (Taylor expansion and rational periodicity). *For a filtered polynomial sequence $g : \mathbb{Z} \rightarrow G$ there are Taylor coefficients $g_i \in G_i$ such that, in ordered form,*

$$g(n) = g_0 g_1^{\binom{n}{1}} g_2^{\binom{n}{2}} \cdots g_s^{\binom{n}{s}}.$$

If the coefficients are rational of denominator dividing Q in lattice-adapted coordinates, then $g(n)\Gamma$ is periodic with period bounded by a power of Q depending only on the filtered complexity.

Proof. Successive discrete derivatives at the origin determine the Taylor coefficients; induction down the filtration reconstructs the ordered product because the error after removing the first r coefficients has vanishing derivatives through order r . For rational coefficients, choose a common multiple of the denominators after the triangular Mal'cev coordinate conversion. The binomial differences $\binom{n+q}{j} - \binom{n}{j}$ are divisible by the required denominators once q is a sufficiently large controlled multiple, so every coordinate of $g(n+q)g(n)^{-1}$ is integral and the quotient points agree. $\square$

Theorem 2.8 (Horizontal and vertical Fourier structure). *Horizontal characters are precisely continuous homomorphisms $G \rightarrow \mathbb{R}/\mathbb{Z}$ annihilating Γ and the commutator subgroup. Smooth/Lipschitz functions on a nilmanifold admit a vertical Fourier decomposition along the top central torus, with quantitative truncation sufficient for the inverse-theorem arguments.*

Proof. The horizontal statement follows by factoring through the maximal horizontal torus $G/[G, G]\Gamma$. For the vertical statement, restrict the function to fibers of the compact top-central torus and convolve in that torus with a multidimensional Fejer kernel. Fourier expansion gives finitely many vertical frequencies plus a uniform approximation error controlled by the Lipschitz norm and cutoff. The quantitative truncation lemma is proved in Appendix M; the detailed nilcharacter implementation is given in the proof appendices later in this volume. $\square$

Chapter 3

Quantitative equidistribution

The equidistribution results in this chapter are the quantitative nilsequence inputs used later in factorization and Möbius–nilsequence arguments.

Definition 3.1 (Quantitative equidistribution). Let (X, d) be a compact metric probability space with probability measure m_X . A finite sequence $x_1, \dots, x_N \in X$ is δ -*equidistributed* if for every Lipschitz $F : X \rightarrow \mathbb{C}$,

$$\left| \frac{1}{N} \sum_{n=1}^N F(x_n) - \int_X F dm_X \right| \leq \delta \|F\|_{\text{Lip}},$$

where $\|F\|_{\text{Lip}} = \|F\|_{\infty} + \sup_{x \neq y} |F(x) - F(y)|/d(x, y)$. It is *totally* δ -*equidistributed* when the analogous estimate, with normalized averaging over the progression, holds on every arithmetic progression $P \subset [N]$ with $|P| \geq \delta N$. On a torus or nilmanifold we use its fixed metric and Haar probability measure.

Theorem 3.2 (Quantitative Weyl criterion on a torus). *If a polynomial orbit in $\mathbb{T}^m$ fails δ -equidistribution on a box of sidelength N , then there is a nonzero character $k \in \mathbb{Z}^m$ of size $\delta^{-O_m(1)}$ such that every nonconstant coefficient of $k \cdot P$ is within $\delta^{-O_m(1)} N^{-j}$ of an integer at degree j .*

Proof. Fejer approximation converts failure of equidistribution into a bounded nonzero Fourier coefficient. Repeated differencing/Weyl’s inequality then gives many near-integral values of the corresponding polynomial phase. The dense coefficient-recovery lemmas recover each coefficient, first in one variable and then in several variables. Full constants and the multivariable recovery are proved in Appendices M and N. $\square$

Definition 3.3 (C^∞ norm of a polynomial phase). If

$$P(n) = \sum_{j=0}^s \alpha_j \binom{n}{j} \pmod{1},$$

define

$$\|P\|_{C^\infty[N]} := \max_{1 \leq j \leq s} N^j \|\alpha_j\|_{\mathbb{R}/\mathbb{Z}}.$$

For a multivariate polynomial phase

$$P(\mathbf{n}) = \sum_{0 < |\mathbf{i}| \leq s} \alpha_{\mathbf{i}} \binom{\mathbf{n}}{\mathbf{i}} \pmod{1},$$

define

$$\|P\|_{C^\infty[N]^d} := \max_{0 < |\mathbf{i}| \leq s} N^{|\mathbf{i}|} \|\alpha_{\mathbf{i}}\|_{\mathbb{R}/\mathbb{Z}}.$$

Theorem 3.4 (Quantitative Leibman theorem). *Fix integers $m, s \geq 1$. Let G/Γ be an m -dimensional filtered nilmanifold of degree at most s equipped with an M_0 -rational Mal'cev basis, where $M_0 \geq 2$. There is a constant $C = C(m, s) \geq 1$ such that the following holds for every $0 < \delta < 1/2$ and $N \geq 1$. If $g \in \text{poly}(\mathbb{Z}, G_\bullet)$ and the orbit $(g(n)\Gamma)_{1 \leq n \leq N}$ is not δ -equidistributed, then there is a nonzero horizontal character $\eta : G \rightarrow \mathbb{R}/\mathbb{Z}$ satisfying*

$$|\eta| \leq M_0^C \delta^{-C}, \quad \|\eta \circ g\|_{C^\infty[N]} \leq M_0^C \delta^{-C}.$$

Here $|\eta|$ is the Euclidean norm of the integer coefficient vector of η in the fixed Mal'cev coordinates. Equivalently, if

$$\eta(g(n)) = \alpha_0 + \sum_{j=1}^s \alpha_j \binom{n}{j} \pmod{1},$$

then

$$\|\alpha_j\|_{\mathbb{R}/\mathbb{Z}} \leq M_0^C \delta^{-C} N^{-j} \quad (1 \leq j \leq s).$$

Proof. This is exactly the detailed quantitative theorem proved in Appendix M, Theorem M.19. For orientation, its induction is lexicographic in filtration degree and nonlinearity rank: vertical Fourier truncation produces a controlled vertical frequency; van der Corput passes to the relative square; quotienting the diagonal top-central subgroup lowers degree; the resulting horizontal characters are synchronized by polynomial coefficient recovery; and the reducible case lowers the nonlinearity rank after a smooth/rational splitting. The displayed $M_0^C \delta^{-C}$ bounds are preserved at every induction step by the quantitative Mal'cev linear-algebra lemmas proved earlier in the appendix. $\square$

Corollary 3.5 (Total-equidistribution criterion). *Under the hypotheses of Theorem 3.4, there is $C_* = C_*(m, s)$ such that if every nonzero horizontal character η with $|\eta| \leq M_0^{C_*} \delta^{-C_*}$ satisfies*

$$\|\eta \circ g\|_{C^\infty[N]} > M_0^{C_*} \delta^{-C_*},$$

then $(g(n)\Gamma)_{n \leq N}$ is totally δ -equidistributed, after enlarging C_ by an absolute factor depending only on the fixed convention for total equidistribution.*

Proof. If total δ -equidistribution fails, there is an arithmetic progression $P = \{a + qr : 1 \leq r \leq L\} \subset [N]$ with $L \geq \delta N$ on which the orbit fails the corresponding quantitative equidistribution estimate. Apply Theorem 3.4 to $r \mapsto g(a + qr)$ and use the triangular binomial-change-of-variables lemma from Appendix M to transfer its $C^\infty[L]$ bound back to $\eta \circ g$. Since $q \leq \delta^{-1}$ and $L \geq \delta N$, only a further power of δ^{-1} is lost. $\square$

Theorem 3.6 (Multiparameter quantitative Leibman theorem, equal sides). *Fix $d, m, s \geq 1$ and $M_0 \geq 2$. Let G/Γ be as in Theorem 3.4 and let $g \in \text{poly}(\mathbb{Z}^d, G_\bullet)$. There is $C = C(d, m, s)$ such that, whenever $(g(\mathbf{n})\Gamma)_{\mathbf{n} \in [N]^d}$ is not δ -equidistributed, one can find a single nonzero horizontal character η for which*

$$|\eta| \leq M_0^C \delta^{-C}, \quad \|\eta \circ g\|_{C^\infty[N]^d} \leq M_0^C \delta^{-C}.$$

If the side lengths of the box lie in $[cN, C_0N]$ with fixed $0 < c \leq C_0$, the same conclusion holds with the exponent and implied constants also depending on c, C_0 .

Proof. This is Theorem N.4 in the equal-sides appendix. The proof first finds many one-dimensional restrictions that fail equidistribution. Pigeonholing is performed over the finite box of possible horizontal characters before the direction is changed, giving one fixed character η on a dense set of directions. Writing $\eta \circ g$ in multivariate Taylor coordinates, finite-difference coefficient recovery and the integer-box Schwartz–Zippel lemma then bound every mixed coefficient simultaneously. This is the point that justifies the phrase “one common horizontal obstruction.” $\square$

Chapter 4

Factorization and polynomial nilsequence counting

Definition 4.1 (Smooth and rational polynomial sequences). Let G/Γ be equipped with a fixed Mal'cev metric. A polynomial sequence $\varepsilon : \mathbb{Z} \rightarrow G$ is (M, N) -smooth if

$$d(\varepsilon(n), e) \leq M, \quad d(\varepsilon(n+1), \varepsilon(n)) \leq M/N \quad (1 \leq n < N).$$

A polynomial sequence γ is M -rational if each of its Taylor coefficients is rational of height at most M in the fixed rational Mal'cev basis. The periodicity lemma in the detailed appendix then gives a period at most $M^{O_{m,s}(1)}$ for $\gamma(n)\Gamma$.

Theorem 4.2 (Smooth–equidistributed–rational factorization). *Fix integers $m, s \geq 1$, an m -dimensional filtered nilmanifold G/Γ of degree at most s with rationality complexity at most $M_0 \geq 2$, and a real number $A > 0$. There is a constant $C = C(m, s, A)$ such that for every $g \in \text{poly}(\mathbb{Z}, G_\bullet)$ and $N \geq 1$ there is an integer M with*

$$M_0 \leq M \leq M_0^C$$

and a factorization

$$g(n) = \varepsilon(n)g'(n)\gamma(n)$$

for which:

- (i) ε is (M, N) -smooth;
- (ii) there is a connected M -rational subgroup $G' \leq G$ such that $g'(n) \in G'$ for every n and $(g'(n)(G' \cap \Gamma))_{n \leq N}$ is totally M^{-A} -equidistributed in $G'/(G' \cap \Gamma)$;
- (iii) γ is M -rational and $\gamma(n)\Gamma$ is periodic with period at most M^C after enlarging C if necessary.

Proof. This is the quantitative factorization theorem proved in Theorem K.2 and Appendix M, Theorem M.23. At a failed M^{-A} -equidistribution stage, Theorem 3.4 gives a nonzero horizontal character. Simultaneous Diophantine approximation splits its Taylor phase into a part of size $O(M^{O(1)}N^{-j})$ in degree j and a bounded-denominator rational part. The layerwise lifting lemmas produce the corresponding smooth factor ε_1 and rational periodic factor γ_1 without leaving the filtration layer; the residual polynomial lies in the connected rational kernel of the character. That kernel has strictly smaller dimension. Iterating at most m times proves termination, while the quantitative rational-linear-algebra lemmas show that all rationality parameters grow by bounded powers only. Choosing the successive thresholds recursively as fixed powers of M_0 gives the displayed $M \leq M_0^C$ bound. $\square$

Theorem 4.3 (Equal-scale multiparameter factorization). *Fix $d, m, s \geq 1$, $A > 0$, and $M_0 \geq 2$. Let $G_\bullet/\Gamma$ be an m -dimensional filtered nilmanifold of degree at most s with rationality complexity at most M_0 . For every polynomial map $g \in \text{poly}(\mathbb{Z}^d, G_\bullet)$ and every $N \geq 1$ there exists M with*

$$M_0 \leq M \leq M_0^{C(d,m,s,A)}$$

and

$$g(\mathbf{n}) = \varepsilon(\mathbf{n})g'(\mathbf{n})\gamma(\mathbf{n}) \quad (\mathbf{n} \in [N]^d)$$

such that:

- (i) *changing one coordinate of $\mathbf{n}$ by 1 changes ε by at most CM/N in the Mal'cev metric, and $\varepsilon([N]^d)$ lies in an M -bounded coordinate box;*
- (ii) *g' takes values in a connected M -rational subgroup G' and is totally M^{-A} -equidistributed on every comparable subbox and residue class of relative density at least M^{-A} ;*
- (iii) *$\gamma(\mathbf{n})\Gamma$ is periodic in every coordinate with a common period at most M^C .*

Proof. Apply the rational-kernel descent used above with the precise multiparameter obstruction Theorem 3.6. The layerwise lifts preserve multidegree, so a unit coordinate difference of the degree- $\mathbf{i}$ smooth Taylor term has size $O(M^{O(1)}/N)$; bounded-denominator rational Taylor generators give one common period. At each failure the connected middle subgroup is replaced by a proper rational kernel, so dimension again drops and the process stops after at most m stages. This is the argument of Theorem N.5. $\square$

Theorem 4.4 (Polynomial nilsequence counting on equal-scale boxes). *Fix $d, m, s \geq 1$, $M_0 \geq 2$, and constants $0 < c \leq C_0$. Let $G_\bullet/\Gamma$ be an m -dimensional filtered nilmanifold of degree at most s and rationality complexity at most M_0 , let $g \in \text{poly}(\mathbb{Z}^d, G_\bullet)$, and let $N \geq 1$. Let $B \subset [-N, N]^d$ be a box all of whose side lengths lie in $[cN, C_0N]$. For every $0 < \delta < 1/2$ and every Lipschitz function $F : G/\Gamma \rightarrow \mathbb{C}$, there is a partition of $B \cap \mathbb{Z}^d$ into $O_{\delta,c,C_0,m,s,M_0}(1)$ cells $\mathcal{C}$, each the intersection of a residue class of modulus $O_{\delta,c,C_0,m,s,M_0}(1)$ with a box, such that on each cell the factorization from Theorem 4.3, with an equidistribution exponent chosen as a function of δ , may be chosen and*

$$\mathbb{E}_{\mathbf{n} \in \mathcal{C}} F(g(\mathbf{n})\Gamma) = \int_{Y_{\mathcal{C}}} F_{\mathcal{C}} dm_{Y_{\mathcal{C}}} + O_{c,C_0,m,s,M_0}(\delta \|F\|_{\text{Lip}}).$$

Here $Y_{\mathcal{C}}$ is the rational orbit closure of the middle factor on that cell and $F_{\mathcal{C}}$ is the translate of F obtained by freezing the smooth and rational factors at the cell's reference point. The same conclusion holds for a convex region whose boundary contribution is $O(\delta)$ after the corresponding box partition.

Proof. Apply Theorem 4.3 at an equidistribution scale smaller than δ . Split first into residue classes on which the rational factor is constant modulo Γ , then into coordinate boxes of side length $\asymp \delta N/M$. Telescoping the coordinate increments moves the smooth factor by $O_d(\delta)$ on each cell, so it may be frozen at a cost $O(\delta \|F\|_{\text{Lip}})$. Total equidistribution of the middle factor replaces its cell average by Haar measure on $Y_{\mathcal{C}}$ with the same order of error. This is Theorem N.6; the convex-region form follows by separating the $O(\delta)$ boundary cells before applying the equal-scale argument. $\square$

Chapter 5

The inverse-theorem layer

5.1 What is already owned

Theorem 5.1 (U^3 inverse theorem in nilsequence language). *For every $0 < \delta \leq 1$ there exist constants $c_3(\delta) > 0$ and $M_3(\delta) \geq 1$ with the following property. In each of the cyclic and finite-field settings treated in the detailed U^3 appendices, every 1-bounded function f satisfying*

$$\|f\|_{U^3} \geq \delta$$

has a two-step polynomial nilsequence ψ of complexity at most $M_3(\delta)$ and $\|\psi\|_\infty \leq 1$ such that

$$\left| \mathbb{E}_x f(x) \overline{\psi(x)} \right| \geq c_3(\delta).$$

In the coordinate models constructed in those proofs, the structured phase is a quadratic or bracket-quadratic phase, and the Heisenberg/nilcharacter dictionary identifies it with the top-frequency coordinate of a two-step nilsequence.

Proof. The finite-field U^3 inverse theorem and the cyclic local-quadratic inverse theorem are proved in Detailed Proof Appendices A and B. Their conclusions give explicit positive correlation and finite complexity depending only on δ . The Heisenberg model in the nilsequence appendices sends each quadratic/bracket-quadratic phase occurring there to a two-step nilcharacter without changing the correlation. Taking $M_3(\delta)$ to be the maximum of the finitely many complexity envelopes in the two constructions and $c_3(\delta)$ to be the minimum of their positive correlation envelopes proves the stated uniform formulation. $\square$

Theorem 5.2 (Qualitative Green–Tao–Ziegler inverse theorem). *Fix an integer $s \geq 1$ and $0 < \delta \leq 1$. There exist*

$$N_0 = N_0(s, \delta) \in \mathbb{N}, \quad c = c(s, \delta) > 0, \quad M = M(s, \delta) \geq 1$$

such that for every $N \geq N_0$ and every function $f : [N] \rightarrow \mathbb{C}$ with $|f(n)| \leq 1$ and

$$\|f\|_{U^{s+1}[N]} \geq \delta,$$

there are an s -step filtered nilmanifold G/Γ of complexity at most M , a polynomial sequence $g : \mathbb{Z} \rightarrow G$, and an M -Lipschitz function $F : G/\Gamma \rightarrow \mathbb{C}$ with $\|F\|_\infty \leq 1$ such that

$$\left| \mathbb{E}_{1 \leq n \leq N} f(n) \overline{F(g(n)\Gamma)} \right| \geq c.$$

Proof. This is the theorem proved in Detailed Proof Appendix D, Theorem D.1. For $s = 1$ the proof is Fourier inversion, and for $s = 2$ it is the complete U^3 argument summarized in

Theorem 5.1. For $s \geq 3$, the proof applies the exact Gowers derivative identity to obtain a positive-density set of shifts h for which $\Delta_h f$ has large U^s norm, invokes the induction hypothesis, and then performs the coherence, bracket-linearization, symbol symmetrization, and integration steps of Appendix D and its supporting Appendices F–H. The geometric operations invoked in those steps use the precise Mal’cev rationality, quantitative Leibman, and factorization results proved in Chapters 1–4 and Appendices L–N. Thus every occurrence of “controlled” in the detailed proof is instantiated by a parameter depending only on (s, δ) , yielding the displayed N_0, c, M . $\square$

Proposition 5.3 (Recursive effectivity of the qualitative GTZ proof). *Fix $s \geq 1$. Suppose that for every quantitative lemma used in the induction of Theorem 5.2 we choose the explicit monotone parameter bound proved in its statement, and order these finitely many bounds by their first use in the induction. Let $\mathcal{L}_s$ denote that finite ordered list. Then there are recursively computable monotone functions*

$$M_s : (0, 1] \rightarrow [1, \infty), \quad c_s : (0, 1] \rightarrow (0, 1], \quad N_s : (0, 1] \rightarrow \mathbb{N}$$

obtained from $\mathcal{L}_1, \dots, \mathcal{L}_s$ by finite composition, maximum, minimum, integer ceiling, addition, multiplication, and substitution, such that the conclusion of Theorem 5.2 holds with $M = M_s(\delta)$, $c = c_s(\delta)$ and $N_0 = N_s(\delta)$. This proposition asserts recursive computability only; the sharper single/double-exponential Manners envelope is stated and proved later in the quantitative dependency estimates in Chapter 6.

Proof. Induct on s . For $s = 1$, Fourier inversion gives explicit algebraic choices for the correlation and complexity parameters. Assume the claim through degree $s - 1$. The degree- s proof contains only finitely many invocations of quantitative lemmas. Each such lemma has an explicit monotone output bound in terms of previously fixed inputs; after substituting the recursively computable degree- $(s - 1)$ bounds, its outputs remain recursively computable. A pigeonhole step over at most K alternatives replaces a correlation ρ by at least ρ/K ; an intersection of finitely many positive-density sets multiplies or minimizes their density bounds; finite coverings, vertical Fourier truncation, BSG, Pluennecke/Freiman steps, quantitative Leibman, and factorization each contribute one of the maps in $\mathcal{L}_s$. The degree-rank and rational-kernel descents have at most a number of iterations bounded by the relevant dimension parameter, hence are finite loops whose iteration counts are themselves bounded by the preceding maps. Therefore the final complexity is a finite maximum/composition of computable bounds, the final positive correlation is a finite minimum/product/quotient of positive computable bounds, and the threshold $N_s(\delta)$ is a finite maximum of the thresholds encountered. These operations preserve recursive computability and monotonicity after taking the monotone envelope. This constructs M_s, c_s, N_s and proves the proposition. $\square$

5.2 Further quantitative inverse theorems

Beyond the qualitative Green–Tao–Ziegler theorem, two quantitative refinements are needed in this volume:

1. V.5.002: the all-degree inverse theorem over fixed finite fields at the selected strength, including the low-characteristic/nonclassical-polynomial branch where needed;
2. V.5.005: Manners’s quantitative cyclic inverse theorem with explicit complexity bounds.

They are proved in the dedicated Tao–Ziegler/Bergelson–Tao–Ziegler and Manners chapters later in this volume. The cyclic local-to-global theorem is proved in this chapter by polynomial-path suspension and periodicization. Recursive effectivity from the qualitative proof is combined later with the explicit Manners complexity propagation.

Proposition 5.4 (Why nilpotent geometry is the higher-order character theory). *Ordinary characters are exactly the structured objects detected by U^2 ; quadratic phases and two-step nilcharacters are the objects detected by U^3 ; in general, iterated discrete differentiation lowers the degree of a polynomial nilsequence one step at a time, so s -step nilcharacters play the same structural role for U^{s+1} that linear characters play for U^2 .*

Proof. For a linear phase one derivative is constant. For a degree- s polynomial sequence adapted to a filtration, one derivative moves the top component into the next filtration level and reduces polynomial degree. After $s+1$ derivatives the top phase vanishes. Conversely the inverse-theorem induction starts from many derivatives with large U^s norm, models them by degree- $(s-1)$ nilcharacters, proves coherence, and integrates the coherent derivative data to one degree- s nilcharacter. Thus the same differentiation/integration architecture generalizes Fourier frequency. $\square$

5.3 A rigorous local-to-global model on progressions

The full cyclic local-to-global theorem must account for carry phenomena. The first nontrivial part, however, is completely elementary once polynomial orbits and product nilmanifolds have been developed: a structured phase defined on a long progression can be globalized without losing its polynomial degree by adjoining a one-dimensional torus which records the position inside the progression.

Lemma 5.5 (Smooth cutoff is a one-step nilsequence). *Let $N \geq 1$ and let $\chi : \mathbb{R}/\mathbb{Z} \rightarrow [0, 1]$ be Lipschitz. Then*

$$n \mapsto \chi(n/N \bmod 1)$$

is a one-step polynomial nilsequence on the torus $\mathbb{R}/\mathbb{Z}$, of Lipschitz complexity $O(\|\chi\|_{\text{Lip}})$.

Proof. Take $G = \mathbb{R}$, $\Gamma = \mathbb{Z}$, polynomial orbit $g(n) = n/N$, and observable $F = \chi$ on G/Γ . This is precisely the definition of a one-step polynomial nilsequence. $\square$

Lemma 5.6 (Product and restriction calculus). *If $\psi_1(n) = F_1(g_1(n)\Gamma_1)$ and $\psi_2(n) = F_2(g_2(n)\Gamma_2)$ are polynomial nilsequences of degrees at most s_1, s_2 , then $\psi_1\psi_2$ is a polynomial nilsequence on the product nilmanifold of degree at most $\max(s_1, s_2)$. The dimension, rationality and Lipschitz complexities are bounded by the corresponding sums/products of the two input complexities.*

Proof. Use the product filtered group $G_1 \times G_2$, lattice $\Gamma_1 \times \Gamma_2$, polynomial orbit (g_1, g_2) and observable $(x_1, x_2) \mapsto F_1(x_1)F_2(x_2)$. Polynomiality follows coordinatewise from the derivative definition, and the complexity estimates follow directly from product Mal'cev coordinates and the product metric. $\square$

Proposition 5.7 (Classical progression local-to-global theorem). *Fix $s \geq 1$ and $M \geq 2$. Let*

$$P = a + q\{0, 1, \dots, L-1\} \subset \mathbb{Z}, \quad q, L \geq 1,$$

and suppose

$$\psi(a + qt) = F(g(t)\Gamma), \quad 0 \leq t < L,$$

where $G_\bullet/\Gamma$ has complexity at most M , $g : \mathbb{Z} \rightarrow G$ is polynomial of degree at most s , and $F : G/\Gamma \rightarrow \mathbb{C}$ satisfies $\|F\|_\infty \leq 1$ and $\|F\|_{\text{Lip}} \leq M$. Then there is a polynomial nilsequence $\tilde{\psi} : \mathbb{Z} \rightarrow \mathbb{C}$ of degree at most s on

$$(G/\Gamma) \times (\mathbb{R}/\mathbb{Z})^2$$

whose nilmanifold complexity is $O_s(M + q)$ and whose defining observable has Lipschitz norm $O(Mq)$, such that

$$\tilde{\psi}(a + qt) = \psi(a + qt) \quad (\lceil 0.05L \rceil \leq t \leq \lfloor 0.95L \rfloor),$$

and

$$\tilde{\psi}(n) = 0 \quad \text{unless} \quad n \equiv a \pmod{q} \quad \text{and} \quad 0.01L < t < 0.99L, \quad t = (n - a)/q.$$

(The decimal constants may be replaced by any fixed nested cutoffs $0 < \alpha < \beta < 1/2$.)

Proof. Choose a Lipschitz function $\chi : \mathbb{R}/\mathbb{Z} \rightarrow [0, 1]$ which is 1 on $[0.05, 0.95]$, is supported in $[0.01, 0.99]$, and has absolute Lipschitz constant $O(1)$. On the residue class $n \equiv a \pmod{q}$, write $t = (n - a)/q$ and define

$$\tilde{\psi}(n) = F(g(t)\Gamma) \chi(t/L \bmod 1).$$

The first factor is an s -step polynomial nilsequence in t ; the second is a one-step nilsequence by Lemma 5.5; their product is an s -step nilsequence by Lemma 5.6.

To encode the residue class globally, use a second circle factor with orbit $n \mapsto n/q \pmod{1}$. Choose a Lipschitz bump β_q supported in the arc of radius $1/(3q)$ around a/q and equal to 1 at a/q . On the discrete orbit $\{n/q : n \in \mathbb{Z}\}$ this bump is exactly the indicator of $n \equiv a \pmod{q}$. Multiplying by this one-step torus nilsequence therefore extends the displayed formula to all integers without introducing a finite-group “nilmanifold.” On the central 90% of P the first cutoff is identically one, giving exact agreement. Outside the slightly enlarged support it vanishes. The product nilmanifold adds two one-dimensional tori. The first cutoff has Lipschitz norm $O(1)$ in the normalized progression coordinate and the residue-class bump has Lipschitz norm $O(q)$; hence the product observable has Lipschitz norm $O(Mq)$ and the presentation complexity is $O_s(M + q)$, as stated. $\square$

5.4 Polynomial-path suspension and exact cyclic periodicization

The genuine cyclic construction is easiest to write first with *left* quotients, exactly because integer-valued polynomial paths then act on the left at the evaluation point. This is only a notational device: inversion gives a filtered diffeomorphism

$$\widehat{\Gamma} \backslash \widehat{G} \longleftrightarrow \widehat{G} / \widehat{\Gamma}, \quad \widehat{\Gamma}x \longmapsto x^{-1}\widehat{\Gamma},$$

and filtered polynomiality is stable under inversion. Thus the construction produces structured objects in the right-quotient convention used elsewhere in this book.

Definition 5.8 (Polynomial path group). Fix an s -step rationally filtered nilpotent Lie group $G_\bullet$ with lattice Γ . Write

$$\mathcal{P}_s(G_\bullet) = \text{Poly}(\mathbb{R}, G_\bullet)$$

for the real polynomial paths whose restrictions to $\mathbb{Z}$ are polynomial in the filtered-difference sense. In Taylor coordinates an element is written

$$q(t) = q_0 q_1^{\binom{t}{1}} \cdots q_s^{\binom{t}{s}}, \quad q_i \in G_i.$$

Let

$$\Lambda_s(\Gamma) = \{q \in \mathcal{P}_s(G_\bullet) : q(n) \in \Gamma \text{ for every } n \in \mathbb{Z}\},$$

and let $T_u q(t) = q(t + u)$.

Lemma 5.9 (Polynomial paths, their lattice, and the suspension filtration). *The following statements hold.*

- (i) $\mathcal{P}_s(G_\bullet)$ is a finite-dimensional nilpotent Lie group under pointwise multiplication, and $\Lambda_s(\Gamma)$ is a discrete cocompact subgroup.
- (ii) $u \mapsto T_u$ is a one-parameter group of Lie-group automorphisms and $T_m \Lambda_s(\Gamma) = \Lambda_s(\Gamma)$ for $m \in \mathbb{Z}$.
- (iii) Consequently

$$\widehat{G} = \mathcal{P}_s(G_\bullet) \rtimes_T \mathbb{R}, \quad \widehat{\Gamma} = \Lambda_s(\Gamma) \rtimes_T \mathbb{Z}$$

is a Lie group with lattice, where

$$(q, u)(r, v) = (q T_u r, u + v).$$

- (iv) With

$$\widehat{G}_0 = \widehat{G}, \quad \widehat{G}_1 = \text{Poly}(\mathbb{R}, G_{\bullet+1}) \rtimes_T \mathbb{R},$$

and, for $i \geq 2$,

$$\widehat{G}_i = \text{Poly}(\mathbb{R}, G_{\bullet+i}) \rtimes_T \{0\},$$

the group $\widehat{G}$ is filtered of degree at most s . The underlying abstract group may have nilpotency class $s+1$ because $\widehat{G}_0$ need not equal $\widehat{G}_1$; this does not increase the degree of the filtered nilsequence.

If the original rational Mal'cev data have complexity at most M , then the dimension and rationality complexity of $\widehat{G}/\widehat{\Gamma}$, together with the coordinate-polynomial degrees occurring in (i)–(iv), are bounded by a function of s, M only.

Proof. The Taylor theorem for filtered polynomial sequences, Theorem 2.7, identifies a path with the finite tuple $(q_0, \dots, q_s)$, $q_i \in G_i$. The product and inverse of two such paths are again polynomial by Theorem 2.6; in Mal'cev coordinates their new Taylor coefficients are polynomial functions of the old ones. This gives a finite-dimensional nilpotent Lie-group structure on $\mathcal{P}_s(G_\bullet)$.

We next identify the lattice without hiding an infinite family of conditions. Suppose first that $q(j) \in \Gamma$ for $0 \leq j \leq s$. The Taylor formula is triangular: from $q(0)$ one recovers q_0 ; from $q(1)$ and q_0 one recovers q_1 ; continuing, $q(j)$ and the previously recovered coefficients determine q_j . Since all operations are group words and $q_j \in G_j$, induction gives

$$q_j \in \Gamma \cap G_j \quad (0 \leq j \leq s).$$

For every integer n , the binomial coefficients $\binom{n}{j}$ are integers, and therefore the Taylor product for $q(n)$ lies in Γ . Hence membership in $\Lambda_s(\Gamma)$ is equivalent to the finitely many conditions $q(0), \dots, q(s) \in \Gamma$.

Because the filtration is rational, $\Gamma \cap G_j$ is a lattice in G_j . In the triangular Taylor chart, $\Lambda_s(\Gamma)$ is therefore commensurable with the product of these coefficient lattices. It follows that it is discrete and cocompact in $\mathcal{P}_s(G_\bullet)$. This proves (i).

Translation preserves filtered degree because each real binomial polynomial $\binom{t+u}{j}$ is a linear combination of $\binom{t}{k}$, $k \leq j$, with coefficients depending polynomially on u . It also respects pointwise multiplication, so T_u is an automorphism. For integer m , an integer-valued path remains integer-valued after T_m , proving (ii). The lattice assertion in (iii) follows: a compact fundamental domain for $\mathcal{P}_s/\Lambda_s$ together with a unit interval in the suspension coordinate gives a compact fundamental domain for the semidirect quotient.

Finally, commutation with the suspension direction replaces a path by a translate-difference $T_u q q^{-1}$, which moves one step deeper in the shifted path filtration. The filtered polynomial commutator rule therefore gives

$$[\widehat{G}_i, \widehat{G}_j] \subseteq \widehat{G}_{i+j}.$$

Since $G_{s+1} = \{e\}$, the displayed filtration has $\widehat{G}_{s+1} = \{e\}$. This proves (iv). The quantitative assertion follows from the same triangular Mal'cev-coordinate descriptions: only finitely many polynomial coordinate maps of degree bounded in terms of s are introduced. $\square$

Lemma 5.10 (The suspension absorbs the carry). *Let $p : \mathbb{Z} \rightarrow G$ be polynomial of filtered degree at most s , extend it to its real Taylor path, and set*

$$q(t) = p(Nt) \in \mathcal{P}_s(G_\bullet).$$

In the left nilmanifold $\widehat{\Gamma} \backslash \widehat{G}$ define

$$a_n = (e, n/N)(q, 0) = (T_{n/N}q, n/N).$$

Then $n \mapsto \widehat{\Gamma}a_n$ is a filtered polynomial orbit of degree at most s and is exactly N -periodic. On the fundamental chart $0 \leq n/N < 1$, evaluation of the path coordinate at 0 gives the original value:

$$(T_{n/N}q)(0) = q(n/N) = p(n).$$

Proof. The map $n \mapsto (e, n/N)$ is a one-parameter polynomial sequence lying in $\widehat{G}_1$, and multiplication by the fixed element $(q, 0)$ preserves filtered polynomiality. For periodicity, write $t = n/N$. Using the semidirect-product law and the inverse formula gives

$$a_{n+N}a_n^{-1} = (T_{t+1}q, t+1)(T_{-t}(T_tq)^{-1}, -t) = (e, 1) \in \widehat{\Gamma}.$$

Thus $\widehat{\Gamma}a_{n+N} = \widehat{\Gamma}a_n$. The evaluation identity is immediate from the definition of T_t . $\square$

Theorem 5.11 (Periodicization of a polynomial nilsequence). *Fix s , a complexity bound M , and $0 < \eta < 1/10$. There is $M' = M'(s, M, \eta)$ with the following property. Let*

$$\psi(n) = F(p(n)\Gamma), \quad |F| \leq 1,$$

be a degree- s polynomial nilsequence of complexity at most M , and let $N \geq 1$. Then there is a filtered nilmanifold of degree at most s and complexity at most M' , an exactly N -periodic polynomial orbit $\tilde{p}$, and a one-bounded M' -Lipschitz observable $\tilde{F}$ such that

$$\tilde{F}(\tilde{p}(n)) = \psi(n) \quad (\eta N \leq n \leq (1 - \eta)N),$$

while the new nilsequence vanishes whenever the normalized suspension coordinate lies in an $\eta/2$ -neighborhood of the boundary of a period. The ambient nilmanifold and its complexity depend on s, M, η but not on N or on the Taylor coefficients of p .

Proof. Use the suspension of Lemma 5.9 and the left-periodic orbit a_n of Lemma 5.10. Choose a Lipschitz cutoff $\chi : [0, 1] \rightarrow [0, 1]$ that is 1 on $[\eta, 1 - \eta]$, vanishes on the $\eta/2$ -neighborhood of $\{0, 1\}$, and has Lipschitz norm $O(\eta^{-1})$.

A left coset in $\widehat{\Gamma} \backslash \widehat{G}$ has a representative (r, t) with $0 \leq t < 1$. On the interior of this fundamental suspension chart set

$$\widehat{F}(\widehat{\Gamma}(r, t)) = \chi(t)F(r(0)\Gamma).$$

This is well defined with respect to the path-lattice identification: if $\lambda \in \Lambda_s(\Gamma)$ then $(\lambda, 0)(r, t) = (\lambda r, t)$ and

$$(\lambda r)(0) = \lambda(0)r(0), \quad \lambda(0) \in \Gamma,$$

so the value of F is unchanged on the nilmanifold. The only additional identification occurs when t crosses an integer. Because χ vanishes in a neighborhood of 0 and 1, the displayed

formula extends across that seam without a compatibility condition. Hence it defines a global continuous function on the compact suspension quotient.

In fixed Mal'cev coordinates, evaluation $r \mapsto r(0)$ is a polynomial, hence Lipschitz, map on a compact fundamental domain. Thus $\widehat{F}$ is Lipschitz with norm bounded by a function of s, M, η , independent of N and of p .

For $\eta N \leq n \leq (1 - \eta)N$, the representative of the orbit is $(T_{n/N}q, n/N)$ and $\chi(n/N) = 1$, so

$$\widehat{F}(\widehat{\Gamma}a_n) = F((T_{n/N}q)(0)\Gamma) = F(q(n/N)\Gamma) = F(p(n)\Gamma) = \psi(n).$$

The vanishing statement is immediate from the cutoff.

Finally return to the right-quotient convention of the book by inversion. Put $\tilde{p}(n) = a_n^{-1}\widehat{\Gamma}$ and $\tilde{F}(a^{-1}\widehat{\Gamma}) = \widehat{F}(\widehat{\Gamma}a)$. Inversion preserves filtered polynomiality and has controlled Lipschitz complexity on the compact quotient. Exact left periodicity of a_n becomes exact right-coset periodicity of $\tilde{p}(n)$. This proves the theorem in the convention used elsewhere in Volume V. $\square$

Corollary 5.12 (Cyclic local-to-global nilsequence model). *Let $f : \mathbb{Z}/N\mathbb{Z} \rightarrow \mathbb{C}$ be one-bounded. Suppose that for some interval $J \subset \mathbb{Z}/N\mathbb{Z}$ with $N/20 \leq |J| \leq N/2$, the restriction of f to J has $U^{s+1}[J]$ norm at least δ . Then the qualitative interval inverse theorem together with Theorem 5.11 produces an exactly N -periodic degree- s nilsequence of complexity $O_{s,\delta}(1)$ whose correlation with f is $\gg_{s,\delta} 1$.*

More generally, if an interval restriction already correlates by ρ with a degree- s nilsequence of complexity M , then after choosing the cutoff width $\eta \ll \rho$ the resulting cyclic periodic nilsequence has correlation at least $c\rho - O(\eta)$, where $c > 0$ is the relative size of the chosen interval.

Proof. For the first statement, choose a smooth partition of unity on the circle into a bounded number of arcs of length between $N/20$ and $N/2$ after discretization. The Gowers triangle inequality and the elementary comparison between a cyclic cube lying wholly inside such an arc and the corresponding integer cube show that one localized piece retains U^{s+1} norm $\gg_s \delta$. Apply the internally owned qualitative interval inverse theorem, Theorem 5.2, to that piece. Its correlating nilsequence may be multiplied by the same smooth cutoff without changing the degree, by Lemmas 5.5 and 5.6. Theorem 5.11 then turns it into an exactly N -periodic nilsequence whose ambient complexity is independent of N .

The correlation loss is only the bounded partition-of-unity factor and the boundary region where the cutoff changes. Choosing η as a sufficiently small function of the pre-periodization correlation gives a positive lower bound depending only on s, δ . The second statement is the same estimate without the initial Gowers-norm localization step. $\square$

Chapter 6

The all-degree finite-field inverse theorem

6.1 Statement of the theorem

Low-characteristic inverse theory requires non-classical polynomial phases. We begin by stating the exact all-degree theorem and then develop the algebraic and analytic ingredients used in its proof.

Theorem 6.1 (Finite-field inverse theorem). *Fix a prime p , an integer $s \geq 1$, and $\delta > 0$. There is*

$$\varepsilon = \varepsilon(p, s, \delta) > 0$$

such that for every finite-dimensional $\mathbb{F}_p$ -vector space V and every 1-bounded function $f : V \rightarrow \mathbb{C}$ satisfying

$$\|f\|_{U^{s+1}(V)} \geq \delta,$$

there is a (not necessarily classical) polynomial

$$P : V \rightarrow \mathbb{R}/\mathbb{Z}, \quad \deg P \leq s,$$

for which

$$|\mathbb{E}_{x \in V} f(x) e(-P(x))| \geq \varepsilon.$$

The same conclusion for a vector space over a fixed prime-power field follows by restriction of scalars to its prime field.

Remark 6.2 (Proof organization). This is the Green–Tao/Tao–Ziegler inverse theorem in the low-characteristic category of non-classical polynomials. Its proof is distributed across this chapter and the subsequent factor, Bergelson–Tao–Ziegler, and Tao–Ziegler endpoint chapters; the final degree-sharpening step explicitly returns to Theorem 6.1.

6.2 Intrinsic polynomial language

Definition 6.3 (Non-classical polynomial). A map $P : V \rightarrow \mathbb{R}/\mathbb{Z}$ has degree at most d if

$$\Delta_{h_1} \cdots \Delta_{h_{d+1}} P(x) = 0$$

for every $x, h_1, \dots, h_{d+1} \in V$.

Lemma 6.4 (Derivative recursion). *For $d \geq 1$, a map P has degree at most d if and only if every first derivative $\Delta_h P$ has degree at most $d - 1$.*

Proof. This is immediate from commutativity of additive differences:

$$\Delta_{h_1} \cdots \Delta_{h_d}(\Delta_h P) = \Delta_h \Delta_{h_1} \cdots \Delta_{h_d} P.$$

The forward implication follows at once. Conversely, if the right side vanishes for every h and every $h_1, \dots, h_d$, then every $(d+1)$ -fold derivative of P vanishes. $\square$

Lemma 6.5 (Polynomial phases are genuine U^{s+1} obstructions). *If $P : V \rightarrow \mathbb{R}/\mathbb{Z}$ has degree at most s , then*

$$\|e(P)\|_{U^{s+1}(V)} = 1.$$

More generally, if $|f| \leq 1$ and

$$|\mathbb{E} f(x) e(-P(x))| \geq \rho,$$

then $\|f\|_{U^{s+1}(V)} \geq \rho$.

Proof. For the first assertion, the $(s+1)$ -cube product defining the Gowers norm is

$$e(\Delta_{h_1} \cdots \Delta_{h_{s+1}} P(x)) = 1$$

pointwise.

For the second assertion apply the Gowers–Cauchy–Schwarz inequality to the correlation with $e(P)$. Each successive Cauchy–Schwarz removes one copy of f while differentiating the phase. After $s+1$ steps the phase contribution is identically 1, so the original correlation is bounded by $\|f\|_{U^{s+1}}$. $\square$

Worked Example 6.6 (Why low characteristic forces non-classical phases). For $V = \mathbb{F}_2$, set $P(0) = 0$ and $P(1) = 1/4$. Then $\Delta_1 P$ alternates between $1/4$ and $-1/4$, its second derivative is the constant $1/2$, and its third derivative vanishes. Thus P has additive degree 2. But $e(P(1)) = i$, so $e(P)$ cannot be an ordinary $\mathbb{F}_2$ -valued quadratic phase, whose values are only $\{1, -1\}$. This is the smallest model of the depth phenomenon that the sharp inverse theorem must accommodate.

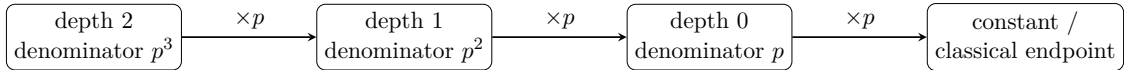


Figure 6.1: The depth ladder for a non-classical polynomial. Multiplication by p removes one denominator level and, by Lemma 6.7, costs exactly the characteristic-dependent amount of degree that drives the Tao–Ziegler induction. Reading the arrows backwards explains why taking a p th root may force a degree increase.

6.3 Depth normal form and exact roots

The first two finite-field proof obligations can in fact be closed without importing the later rank theorem. The key mechanism is elementary but genuinely characteristic- p : on the cyclic group $\mathbb{F}_p$, multiplication by p behaves like a differential operator of order $p-1$. We develop that point carefully because it is the source of both the depth parameter and the exact-roots phenomenon.

6.3.1 The cyclic differential-operator identity

For a direction $h \in V$ write $T_h f(x) = f(x + h)$ and $\Delta_h = T_h - 1$. When a formal power series in the commuting Δ_h acts on a polynomial of bounded degree, only finitely many terms survive, so formal Neumann-series inverses are legitimate.

Lemma 6.7 (Multiplication by p is a differential operator of order $p - 1$). *For every $h \in V$ there is an invertible formal differential operator I_h , with integer coefficients and constant term -1 , such that*

$$p\Delta_h = \Delta_h^p I_h.$$

Consequently, if $P \in \text{Poly}_{\leq d}(V \rightarrow G)$ for an additive group G , then

$$pP \in \text{Poly}_{\leq \max(d-p+1, 0)}(V \rightarrow G).$$

Proof. Because $ph = 0$ in V , one has $T_h^p = T_{ph} = 1$. Hence

$$0 = (1 + \Delta_h)^p - 1 = p\Delta_h + \binom{p}{2}\Delta_h^2 + \cdots + \Delta_h^p.$$

For $1 \leq r \leq p - 1$ the coefficient $\binom{p}{r}$ is divisible by p , so

$$\Delta_h^p = -p\Delta_h \left(1 + \frac{1}{p} \binom{p}{2} \Delta_h + \cdots + \frac{1}{p} \binom{p}{p-1} \Delta_h^{p-2} \right).$$

The operator in parentheses has constant term 1 and is therefore invertible by its formal Neumann series. Rearranging gives the first assertion, with I_h equal to minus that inverse.

Suppose first that $d \geq p - 1$. Apply $d - p + 2$ differences to pP . Using the identity on the first one gives

$$\Delta_{h_1} \cdots \Delta_{h_{d-p+2}}(pP) = \Delta_{h_1}^p I_{h_1} \Delta_{h_2} \cdots \Delta_{h_{d-p+2}} P.$$

The right side contains $d + 1$ additive differences; it vanishes because P has degree at most d . Thus pP has degree at most $d - p + 1$. If $d < p - 1$, the same calculation with a single difference gives

$$\Delta_h(pP) = \Delta_h^p I_h P = 0,$$

since $p > d$; hence pP is constant. This is exactly the displayed bound. $\square$

This lemma is the finitary exact mechanism behind the heuristic that division by p can cost $p - 1$ degrees; compare Tao–Ziegler’s differential-operator proof [6].

6.3.2 A divisibility estimate for wrapped monomials

We next prove directly that the candidate depth monomials really have the degrees claimed. For $0 \leq a < p$ put $g_a(x) = |x|^a$, regarded as an integer-valued function on $\mathbb{F}_p$.

Lemma 6.8 (Repeated differences gain powers of p). *Let $D = \Delta_1$ on $\mathbb{F}_p$. If $0 \leq a < p$ and $r \geq 0$, then $D^r g_a$ is pointwise divisible by*

$$p^{\nu_a(r)}, \quad \nu_a(r) := \begin{cases} 0, & r \leq a, \\ \left\lfloor \frac{r-a}{p-1} \right\rfloor, & r > a. \end{cases}$$

(When $a = 0$ and $r > 0$ the derivative is identically zero, so the assertion is understood in the strongest possible sense.)

Proof. Modulo p , the function g_a is the classical polynomial x^a on $\mathbb{F}_p$. Therefore $D^{a+1}g_a$ is divisible pointwise by p .

Return to the finite operator appearing before the Neumann-series inversion in the proof of Lemma 6.7:

$$D^p = -pDA(D), \quad A(D) = 1 + \frac{1}{p}\binom{p}{2}D + \cdots + \frac{1}{p}\binom{p}{p-1}D^{p-2},$$

where $A(D)$ has integer coefficients. Hence, for every $u \geq 1$,

$$D^{u+p-1} = -pD^u A(D).$$

This identity is an identity of finite difference operators, so it may be applied to the integer-valued function g_a without any convergence issue. Starting at $u = a + 1$ and iterating, each additional block of $p - 1$ differences contributes another factor of p . Extra differences preserve pointwise divisibility. This is precisely the exponent $\nu_a(r)$ displayed above. $\square$

Lemma 6.9 (Depth monomials have the expected weighted degree). *Let $i = (i_1, \dots, i_n)$ with $0 \leq i_t < p$ and $|i| = i_1 + \cdots + i_n > 0$. For $j \geq 0$ the function*

$$M_{i,j}(x) = \frac{|x_1|^{i_1} \cdots |x_n|^{i_n}}{p^{j+1}} \pmod{1}$$

is a non-classical polynomial of degree at most

$$|i| + j(p - 1).$$

Proof. It is enough to test differences in the coordinate directions: every Δ_h is a finite integer linear combination of monomials in the commuting coordinate differences $D_t := \Delta_{e_t}$ with no constant term, because $T_h = \prod_t (1 + D_t)^{|h_t|}$.

Consider therefore a mixed derivative $D_1^{r_1} \cdots D_n^{r_n}$ with

$$r_1 + \cdots + r_n > |i| + j(p - 1).$$

If some $i_t = 0 < r_t$, the derivative vanishes. Otherwise the numerator separates by coordinates:

$$D_1^{r_1} \cdots D_n^{r_n} \prod_t |x_t|^{i_t} = \prod_t D_t^{r_t} |x_t|^{i_t}.$$

By Lemma 6.8, this integer-valued function is divisible by p^L , where

$$L = \sum_t \nu_{i_t}(r_t) \geq \left\lceil \frac{\sum_t \max(r_t - i_t, 0)}{p - 1} \right\rceil.$$

Since

$$\sum_t \max(r_t - i_t, 0) \geq \sum_t r_t - |i| > j(p - 1),$$

we have $L \geq j + 1$. The mixed derivative of $M_{i,j}$ is therefore integer-valued, hence zero in $\mathbb{R}/\mathbb{Z}$. All derivatives of order $|i| + j(p - 1) + 1$ vanish, proving the claimed degree bound. $\square$

6.3.3 The classical layer

We also record the finite-field polynomial classification needed at depth zero.

Lemma 6.10 (Classical coordinate normal form). *A function $Q : \mathbb{F}_p^n \rightarrow \mathbb{F}_p$ has additive degree at most d if and only if it has a unique representation*

$$Q(x) = \sum_{\substack{0 \leq i_1, \dots, i_n < p \\ |i| \leq d}} a_i x_1^{i_1} \cdots x_n^{i_n}, \quad a_i \in \mathbb{F}_p.$$

Proof. For $n = 1$ and $0 \leq d \leq p - 1$, the monomials $1, x, \dots, x^d$ are linearly independent as functions on $\mathbb{F}_p$. On the other hand $\Delta_1 : \text{Poly}_{\leq d} \rightarrow \text{Poly}_{\leq d-1}$ has kernel exactly the constants, so the dimension can increase by at most one at each degree. Thus these monomials exhaust the degree- d polynomials. For $d \geq p - 1$ they already span all p functions on $\mathbb{F}_p$.

Induct on n . Write $x = (x', x_n)$. For each fixed x' , the one-dimensional result expands

$$Q(x', x_n) = \sum_{0 \leq r < p} Q_r(x') x_n^r$$

uniquely, with $Q_r = 0$ for $r > d$ when $d < p - 1$. We show by descending induction on r that Q_r has degree at most $d - r$. Suppose the assertion is already known for all larger exponents and subtract the corresponding terms. This subtraction does not increase the total degree, by the discrete Leibniz rule and the fact that x_n^t has degree at most t . The remaining polynomial has x_n -degree at most r . Applying $\Delta_{e_n}^r$ gives

$$r! Q_r(x'),$$

because the r th difference of x_n^r is the nonzero constant $r!$ and all lower powers disappear. Now apply any $d - r + 1$ further differences in the first $n - 1$ coordinates. The total number of differences is $d + 1$, so the result is zero. Since $r < p$, the scalar $r!$ is invertible in $\mathbb{F}_p$, and hence Q_r has degree at most $d - r$. The induction hypothesis in dimension $n - 1$ applied to each coefficient now gives the required total-degree expansion. Uniqueness follows from uniqueness in the last coordinate followed by the induction hypothesis. $\square$

6.3.4 Full depth classification

We can now close F1.

Theorem 6.11 (Coordinate/depth normal form for non-classical polynomials). *Let $P : \mathbb{F}_p^n \rightarrow \mathbb{R}/\mathbb{Z}$ and $d \geq 0$. Then $P \in \text{Poly}_{\leq d}$ if and only if it has a unique expansion*

$$P(x) = \alpha + \sum_{\substack{0 \leq i_1, \dots, i_n < p, \ j \geq 0 \\ 0 < |i| \leq d - j(p-1)}} \frac{c_{i,j} |x_1|^{i_1} \cdots |x_n|^{i_n}}{p^{j+1}} \pmod{1}, \quad (6.1)$$

where $c_{i,j} \in \{0, \dots, p - 1\}$ and $\alpha \in \mathbb{R}/\mathbb{Z}$. Consequently the depth is at most $\lfloor (d - 1)/(p - 1) \rfloor$, and if $d < p$ every nonconstant degree- d polynomial is classical up to an additive constant.

Proof. Every displayed summand has degree at most its weighted degree $|i| + j(p - 1)$ by Lemma 6.9, so every expression of form (6.1) has degree at most d .

For the converse and uniqueness, induct on d . The case $d = 0$ consists exactly of constant functions. Let $d \geq 1$ and put $d' = \max(d - p + 1, 0)$. By Lemma 6.7, pP has degree at most $d' < d$. The induction hypothesis gives a unique normal form

$$pP = \beta + \sum_{\substack{i, j \\ 0 < |i| \leq d' - j(p-1)}} \frac{c_{i,j} |x|^i}{p^{j+1}}.$$

Choose α with $p\alpha = \beta$ and define

$$R = \alpha + \sum_{i,j} \frac{c_{i,j}|x|^i}{p^{j+2}}.$$

If the sum defining R is nonempty, then $d' \geq 1$, hence $d \geq p$ and $d' = d - p + 1$. For every nonconstant term occurring in that case,

$$|i| + (j+1)(p-1) \leq d' + p - 1 = d.$$

If the sum is empty, R is constant. In either case Lemma 6.9 shows that R has degree at most d . Moreover $pR = pP$. Thus $S := P - R$ is a degree- d polynomial killed by p . There is therefore a unique function $Q : \mathbb{F}_p^n \rightarrow \mathbb{F}_p$ with $S = Q/p \pmod{1}$. Because the embedding $a \mapsto a/p \pmod{1}$ is injective and additive, all $(d+1)$ -fold derivatives of Q vanish. By Lemma 6.10, write uniquely

$$Q = q_0 + Q_0,$$

where $q_0 \in \mathbb{F}_p$ is the constant coefficient and Q_0 is a sum of nonconstant monomials with $0 \leq i_t < p$ and $|i| \leq d$. Replace α by $\alpha + |q_0|/p$; this does not change $p\alpha = \beta$ in $\mathbb{R}/\mathbb{Z}$. Then Q_0/p supplies exactly the missing nonconstant depth-zero terms of (6.1). This proves existence.

For uniqueness, first evaluate at 0 to identify α . Multiplying two putative normal forms by p kills all depth-zero terms and lowers every positive depth by one. The induction hypothesis for degree d' therefore identifies all coefficients of depth $j \geq 1$. After subtracting them, only a depth-zero identity remains. Such an identity is exactly the image under the injective embedding $\mathbb{F}_p \hookrightarrow \mathbb{R}/\mathbb{Z}$, $a \mapsto |a|/p$, of an identity between classical polynomial functions. Hence Lemma 6.10 gives uniqueness of the depth-zero coefficients as well.

The depth bound follows because a nonconstant term of depth j has weighted degree at least $1 + j(p-1)$. If $d < p$, no term of positive depth can occur, giving the final assertion. $\square$

Worked Example 6.12 (Depth really costs $p-1$ degrees). For $p = 2$, the function $x \mapsto |x|/4$ has depth 1 and degree 2; its second derivative is the nonzero constant $1/2$ and its third derivative vanishes. More generally, Theorem 6.11 says that every extra factor $1/p$ in the denominator costs exactly $p-1$ degrees. This is why low characteristic forces a larger phase category while high characteristic $d < p$ collapses back to ordinary polynomial phases.

6.3.5 Exact roots

With F1 in place, exact roots are no longer a separate black box.

Theorem 6.13 (Exact roots). *If $P \in \text{Poly}_{\leq d}(\mathbb{F}_p^n \rightarrow \mathbb{R}/\mathbb{Z})$, then there exists*

$$Q \in \text{Poly}_{\leq d+p-1}(\mathbb{F}_p^n \rightarrow \mathbb{R}/\mathbb{Z}) \quad \text{with} \quad pQ = P.$$

Multiplication by p lowers every positive depth by one and lowers its weighted degree by $p-1$; depth-zero nonconstant terms are killed.

Proof. Write P in the unique form of Theorem 6.11. Choose a p th root β of its constant term α , and increase every denominator by one:

$$Q(x) = \beta + \sum_{i,j} \frac{c_{i,j}|x|^i}{p^{j+2}} \pmod{1}.$$

Then $pQ = P$. A term of P of weighted degree $|i| + j(p-1)$ becomes in Q a term of weighted degree $|i| + (j+1)(p-1)$, at most $d + p - 1$; hence the normal-form theorem puts Q in $\text{Poly}_{\leq d+p-1}$. The final statement follows directly by multiplying a normal-form summand by p . $\square$

6.4 Rank and analytic-rank calculus: the elementary half of F3

The next layer is the calculus used to measure when a polynomial is genuinely of top degree. We separate the elementary inequalities from the deep implication “bounded analytic rank $\Rightarrow$ bounded algebraic rank.” The latter is F6, not a basic property of the definitions.

Fix $s \geq 0$ and let $P \in \text{Poly}_{\leq s+1}(V \rightarrow \mathbb{R}/\mathbb{Z})$.

Definition 6.14 (Algebraic rank). The rank $\text{rank}_s(P)$ is the least m for which there are $Q_1, \dots, Q_m \in \text{Poly}_{\leq s}(V \rightarrow \mathbb{R}/\mathbb{Z})$ and a function $F : (\mathbb{R}/\mathbb{Z})^m \rightarrow \mathbb{R}/\mathbb{Z}$ with

$$P = F(Q_1, \dots, Q_m).$$

For $s = 0$ a nonconstant linear polynomial is assigned infinite rank.

Definition 6.15 (Analytic rank). The top derivative

$$d^{s+1}P(h_1, \dots, h_{s+1}) := \Delta_{h_1} \cdots \Delta_{h_{s+1}} P(x)$$

is independent of x . Its bias is

$$\text{bias}(P) := \mathbb{E}_{h_1, \dots, h_{s+1}} e\left(d^{s+1}P(h_1, \dots, h_{s+1})\right).$$

Repeated averaging in the last variable shows that this quantity is a real number in $[0, 1]$. We define

$$\text{arank}_s(P) := -\log_p \text{bias}(P),$$

with value $+\infty$ when the bias is zero. Equivalently,

$$\text{bias}(P) = \|e(P)\|_{U^{s+1}}^{2^{s+1}}.$$

Lemma 6.16 (Finite range at bounded degree). *For each s there is $C = C(p, s)$ such that every polynomial $Q \in \text{Poly}_{\leq s}(V \rightarrow \mathbb{R}/\mathbb{Z})$ takes values in a coset of $\frac{1}{p^C}\mathbb{Z}/\mathbb{Z}$. In particular Q takes at most p^C values, uniformly in $\dim V$.*

Proof. This is immediate from Theorem 6.11: a nonconstant term of depth j can occur only when $1 + j(p-1) \leq s$. Thus all nonconstant denominators divide $p^{1+\lfloor (s-1)/(p-1) \rfloor}$; the remaining constant term merely translates the finite subgroup. $\square$

Lemma 6.17 (Elementary algebraic-rank calculus). *For $P, Q \in \text{Poly}_{\leq s+1}$,*

$$\text{rank}_s(P) = \text{rank}_s(-P), \quad \text{rank}_s(P + Q) \leq \text{rank}_s(P) + \text{rank}_s(Q).$$

Also $\text{arank}_s(P) = \text{arank}_s(-P)$.

Proof. If $P = F(Q_1, \dots, Q_m)$ then $-P = (-F)(Q_1, \dots, Q_m)$, proving equality of the algebraic ranks. If $P = F(Q_1, \dots, Q_m)$ and $Q = G(R_1, \dots, R_r)$, then

$$P + Q = H(Q_1, \dots, Q_m, R_1, \dots, R_r), \quad H(u, v) = F(u) + G(v),$$

which gives subadditivity. Finally $d^{s+1}(-P) = -d^{s+1}P$; complex conjugation does not change the (real nonnegative) bias. $\square$

6.4.1 Analytic rank under addition and restriction

The remaining elementary closure property of analytic rank is best understood at the level of the top multilinear derivative. We first isolate the small tensor lemma that makes the argument transparent. The proof is the finite-field “lower-order terms cannot increase top bias” argument; in this form it also yields the exact subadditivity theorem for analytic rank (compare Lovett’s tensor formulation [4]).

Lemma 6.18 (Lower-order perturbations cannot increase multilinear bias). *Let $T : V^r \rightarrow \mathbb{R}/\mathbb{Z}$ be additive in each variable. For every proper subset $I \subsetneq [r]$, let $R_I : V^I \rightarrow \mathbb{R}/\mathbb{Z}$ be additive in each of its variables (with $R_\emptyset$ a constant). Then*

$$\left| \mathbb{E}_{x_1, \dots, x_r} e \left(T(x_1, \dots, x_r) + \sum_{I \subsetneq [r]} R_I(x_I) \right) \right| \leq \mathbb{E}_{x_1, \dots, x_r} e(T(x_1, \dots, x_r)).$$

The quantity on the right is real and nonnegative.

Proof. Write $R_{[r]} = T$ and

$$\Phi(x_1, \dots, x_r) = \sum_{I \subseteq [r]} R_I(x_I).$$

Fix a variable, say x_i , and all the other variables. The summands with $i \in I$ combine to an additive character phase $L_i(x_i)$, while the summands with $i \notin I$ combine to a constant C_i with respect to x_i . Character orthogonality gives

$$\mathbb{E}_{x_i} e(\Phi) = e(C_i) 1_{L_i \equiv 0}.$$

Consequently

$$|\mathbb{E}_x e(\Phi(x))| \leq \mathbb{P}_{x_i}(L_i \equiv 0) = \mathbb{E}_x e \left(\sum_{I \ni i} R_I(x_I) \right).$$

The last average is nonnegative: after averaging in x_i it is exactly the probability that the corresponding additive functional vanishes.

Apply this elimination successively for $i = 1, 2, \dots, r$. After the first step all summands missing variable 1 have disappeared; after the second, all summands missing one of 1, 2 have disappeared; and so on. After the r th step only $R_{[r]} = T$ remains. This proves the inequality. The same final character-orthogonality calculation shows directly that the bias of T is a probability, hence is real and nonnegative. $\square$

Theorem 6.19 (Exact subadditivity of analytic rank). *For $P, Q \in \text{Poly}_{\leq s+1}(V \rightarrow \mathbb{R}/\mathbb{Z})$,*

$$\text{bias}(P + Q) \geq \text{bias}(P) \text{bias}(Q), \quad \text{arank}_s(P + Q) \leq \text{arank}_s(P) + \text{arank}_s(Q).$$

In particular this is stronger than the constant-loss sum inequality needed in the finite-field regularization argument.

Proof. Put $r = s + 1$, $T = d^r P$ and $S = d^r Q$. For $x = (x_1, \dots, x_r)$ and $y = (y_1, \dots, y_r)$, the change of variables $(x, y) \mapsto (x, x + y)$ is a bijection of $V^r \times V^r$. Hence

$$\text{bias}(P) \text{bias}(Q) = \mathbb{E}_{x, y} e(T(x) + S(x + y)).$$

By multilinearity,

$$S(x + y) = \sum_{I \subseteq [r]} S_I(x_I, y_{I^c}),$$

where $S_{[r]}(x) = S(x)$ and, for fixed y , every $S_I(\cdot, y_{I^c})$ with $I \subsetneq [r]$ is a multilinear form in only the variables indexed by I . Define

$$A(y) := \mathbb{E}_x e(T(x) + S(x + y)).$$

The average $\mathbb{E}_y A(y)$ is the nonnegative real number $\text{bias}(P) \text{bias}(Q)$, so for some $y = b$,

$$|A(b)| \geq \text{Re } A(b) \geq \text{bias}(P) \text{bias}(Q).$$

For this fixed b , Lemma 6.18, with top form $T + S$, gives $|A(b)| \leq \text{bias}(P + Q)$. Combining the two inequalities proves the bias estimate. Taking $-\log_p$ gives analytic-rank subadditivity, with the usual extended-real convention when a bias vanishes. $\square$

Corollary 6.20 (Restriction monotonicity). *If $U \leq V$ is a subspace, then*

$$\text{arank}_s(P|_U) \leq \text{arank}_s(P).$$

Equivalently, restricting all variables of the top derivative to a subspace cannot decrease its bias.

Proof. Choose a complement W with $V = U \oplus W$, and put $T = d^{s+1}P$. Write each $x_i = u_i + w_i$. For fixed $w = (w_1, \dots, w_r)$, multilinearity expands $T(u + w)$ as the top term $T(u_1, \dots, u_r)$ plus terms depending on proper subsets of the u -variables. Lemma 6.18 therefore yields

$$|\mathbb{E}_{u \in U^r} e(T(u + w))| \leq \text{bias}(T|_{U^r}).$$

Averaging over w and using that $\text{bias}(T)$ is nonnegative gives

$$\text{bias}(T) \leq \mathbb{E}_w |\mathbb{E}_u e(T(u + w))| \leq \text{bias}(T|_{U^r}).$$

Taking $-\log_p$ proves the claim. $\square$

Lemma 6.21 (Generalized Cauchy–Schwarz against the top derivative). *Put $r = s + 1$ and $T = d^r P$. If $f_j : V^r \rightarrow \mathbb{C}$ are 1-bounded and f_j is independent of h_j , then*

$$\left| \mathbb{E}_{h_1, \dots, h_r} e(T(h_1, \dots, h_r)) \prod_{j=1}^r f_j(h_1, \dots, h_r) \right| \leq \text{bias}(P)^{1/2^r}.$$

Proof. We prove the more general assertion for every multilinear $T : V^r \rightarrow \mathbb{R}/\mathbb{Z}$ by induction on r . For $r = 1$ the only factor f_1 is constant, and

$$|\mathbb{E}_h e(T(h)) f_1| \leq \text{bias}(T) \leq \text{bias}(T)^{1/2}.$$

Assume $r \geq 2$. Apply Cauchy–Schwarz in all variables except the last one (equivalently, view the average as an inner product over $(h_1, \dots, h_{r-1})$). The factor f_r , which is independent of h_r , is harmless. Expanding the square introduces two last variables u, v . After the change of variables $k = u - v$, multilinearity gives

$$T(h_1, \dots, h_{r-1}, u) - T(h_1, \dots, h_{r-1}, v) = T(h_1, \dots, h_{r-1}, k).$$

For fixed v and k , the factor

$$f_j(h_1, \dots, h_{r-1}, v + k) \overline{f_j(h_1, \dots, h_{r-1}, v)}$$

is still 1-bounded and independent of h_j for every $j < r$. Hence the induction hypothesis applies for each fixed pair (v, k) and bounds the inner $(r - 1)$ -variable average by

$$\text{bias}(T_k)^{1/2^{r-1}}, \quad T_k(h_1, \dots, h_{r-1}) = T(h_1, \dots, h_{r-1}, k).$$

Averaging this bound over the redundant variable v and then over k gives

$$|A|^2 \leq \mathbb{E}_k \text{bias}(T_k)^{1/2^{r-1}} \leq (\mathbb{E}_k \text{bias}(T_k))^{1/2^{r-1}},$$

where the last step is concavity of $t \mapsto t^{1/2^{r-1}}$. Finally

$$\mathbb{E}_k \text{bias}(T_k) = \text{bias}(T),$$

so $|A| \leq \text{bias}(T)^{1/2^r}$, as required. $\square$

Proposition 6.22 (Bounded algebraic rank implies bounded analytic rank). *There is $C_{p,s} > 0$ such that*

$$\text{arank}_s(P) \leq C_{p,s} \text{rank}_s(P)$$

for every $P \in \text{Poly}_{\leq s+1}$ of finite algebraic rank.

Proof. Let $m = \text{rank}_s(P)$ and write $P = F(Q_1, \dots, Q_m)$ with $\deg Q_i \leq s$. By Lemma 6.16, after translating constants each Q_i takes values in a cyclic group of order at most p^C , where $C = C(p, s)$. Hence $H := e \circ F$ restricted to the product of these value groups has a finite Fourier expansion

$$H(y) = \sum_{\xi \in \widehat{G}} c_\xi \xi(y), \quad |G| \leq p^{Cm}.$$

Parseval gives $\sum_\xi |c_\xi|^2 = 1$, and therefore $\sum_\xi |c_\xi| \leq |G|^{1/2}$. Substituting $y = (Q_1(x), \dots, Q_m(x))$ and taking the inner product with $e(P) = H(Q_1, \dots, Q_m)$ yields

$$1 \leq \sum_\xi |c_\xi| \left| \mathbb{E}_x e(P(x)) \overline{\xi(Q_1(x), \dots, Q_m(x))} \right|.$$

Thus one character ξ has correlation at least $|G|^{-1/2}$. The phase $\xi(Q_1, \dots, Q_m)$ is $e(R)$ for a polynomial R of degree at most s (integer multiples and sums preserve degree). Lemma 6.5 then gives

$$\|e(P)\|_{U^{s+1}} \geq |G|^{-1/2} \geq p^{-Cm/2}.$$

Raising to the power 2^{s+1} and taking $-\log_p$ proves the result, for example with $C_{p,s} = 2^s C$. $\square$

6.4.2 Bounded-codimension stability: the missing elementary part of the article's rank package

The primary Tao–Ziegler induction repeatedly passes to a subspace of bounded index. For that reason one needs more than the one-sided restriction monotonicity above: the *property of having bounded rank* must be insensitive to bounded-codimension restriction. We record the finitary form needed in this volume.

Lemma 6.23 (Algebraic rank is stable under bounded-codimension restriction). *Fix p, s, c . Let $U \leq V$ have codimension at most c , and let $P \in \text{Poly}_{\leq s+1}(V \rightarrow \mathbb{R}/\mathbb{Z})$. If $\text{rank}_s(P|_U) \leq m$, then*

$$\text{rank}_s(P) \leq C_{p,s,c}(m + 1)$$

for a constant depending only on p, s, c .

Proof. Choose a complement W of U , so $V = U \oplus W$ and $|W| \leq p^c$, and let $\pi : V \rightarrow U$ be the projection. By hypothesis

$$P|_U = F(Q_1, \dots, Q_m), \quad \deg Q_i \leq s,$$

where no polynomiality is assumed of the outer function F . This last point matters: one must not form $P - F(Q_1 \circ \pi, \dots, Q_m \circ \pi)$ and silently call the result a polynomial.

For each $w \in W$ put

$$D_w(u) := \Delta_w P(u) = P(u + w) - P(u), \quad u \in U.$$

Every D_w has degree at most s on U . Extend it to V by $\tilde{D}_w = D_w \circ \pi$, and similarly put $\tilde{Q}_i = Q_i \circ \pi$. Choose linear coordinates $\ell_1, \dots, \ell_c : V \rightarrow \mathbb{F}_p$ cutting out U (padding with zero coordinates if necessary). Their tuple $L = (\ell_1, \dots, \ell_c)$ determines the unique $w \in W$ for which $x = \pi x + w$.

Now the exact identity

$$P(x) = P(\pi x) + \Delta_w P(\pi x) = F(\tilde{Q}_1(x), \dots, \tilde{Q}_m(x)) + \tilde{D}_w(x), \quad w = w(L(x)),$$

shows directly that P is a function of

$$\tilde{Q}_1, \dots, \tilde{Q}_m, \ell_1, \dots, \ell_c, (\tilde{D}_w)_{0 \neq w \in W}.$$

Every displayed coordinate is a genuine polynomial of degree at most s on V . There are at most $m + c + p^c - 1$ of them, so $\text{rank}_s(P) \leq m + c + p^c - 1$, proving the claim. $\square$

Lemma 6.24 (Analytic rank changes by at most the codimension loss). *Let $r = s + 1 \geq 2$ and let $U \leq V$ have codimension c . Then*

$$\text{arank}_s(P|_U) \leq \text{arank}_s(P) \leq \text{arank}_s(P|_U) + (r + 1)c.$$

In particular bounded analytic rank is stable under bounded-codimension restriction in both directions, with no appeal to the deep analytic-rank-to-algebraic-rank theorem.

Proof. The first inequality is Corollary 6.20. For the converse, write $f = e(P)$ and $\mu = |U|/|V| = p^{-c}$. Since U is a subgroup, a cube $(x + \omega \cdot h)_{\omega \in \{0,1\}^r}$ lies entirely in U exactly when $x, h_1, \dots, h_r \in U$. Therefore

$$\|f 1_U\|_{U^r(V)}^{2^r} = \mu^{r+1} \|f|_U\|_{U^r(U)}^{2^r}. \quad (1)$$

Let $U^\perp$ be the annihilator of U in the dual group of V . Fourier inversion for the subgroup indicator gives

$$1_U(x) = \mu \sum_{\gamma \in U^\perp} \gamma(x).$$

For $r \geq 2$ the U^r seminorm is a norm, and multiplication by a linear character does not change it. Hence the triangle inequality yields

$$\|f 1_U\|_{U^r(V)} \leq \mu \sum_{\gamma \in U^\perp} \|f\gamma\|_{U^r(V)} = \mu |U^\perp| \|f\|_{U^r(V)} = \|f\|_{U^r(V)}. \quad (2)$$

Combining (1) and (2),

$$\|f\|_{U^r(V)}^{2^r} \geq \mu^{r+1} \|f|_U\|_{U^r(U)}^{2^r}.$$

Our bias convention is precisely $\text{bias}_s(P) = \|e(P)\|_{U^r}^{2^r}$, so

$$\text{bias}_s(P) \geq p^{-(r+1)c} \text{bias}_s(P|_U).$$

Taking $-\log_p$ proves the second inequality. $\square$

6.5 TZ4: multiplication by p , reduced to one earlier recurrence input

The original article proves this step before the symmetric-multilinear inverse theorem. The proof is short only after one has isolated the correct multidimensional recurrence statement. We therefore separate the recurrence input from the algebraic argument.

Proposition 6.25 (Finite-field multidimensional Szemerédi input). *Fix $p, m \geq 1$ and $\alpha > 0$. There is $c = c(p, m, \alpha) > 0$ such that, for every finite-dimensional $\mathbb{F}_p$ -vector space V and every $A \subseteq V^m$ with $|A| \geq \alpha|V|^m$, there are at least*

$$c|V|^{m+1}$$

tuples $(a_1, \dots, a_m, h) \in V^{m+1}$ for which

$$(a_1 + \omega_1 h, \dots, a_m + \omega_m h) \in A \quad \text{for every } (\omega_1, \dots, \omega_m) \in \mathbb{F}_p^m.$$

Proof. This is the finite-field homothetic-grid theorem proved in Volume IV from the internally owned partite hypergraph-removal lemma. We repeat the short encoding because it is the precise interface used here.

Put $\Omega = \mathbb{F}_p^m$ and $K = p^m$. Introduce one vertex variable $x_\nu \in V$ for every $\nu \in \Omega$ and, for each $\omega \in \Omega$, define

$$\psi_\omega(x_{\Omega \setminus \{\omega\}}) = \sum_{\nu \neq \omega} (\omega - \nu) x_\nu \in V^m.$$

Writing

$$h = \sum_{\nu \in \Omega} x_\nu, \quad a = - \sum_{\nu \in \Omega} \nu x_\nu,$$

one has $\psi_\omega = a + \omega h$. Thus the condition $\psi_\omega \in A$ is an edge condition on the face opposite ω , and a labelled boundary simplex is exactly a preimage of a desired homothetic grid.

The map $x \mapsto (a, h)$ is surjective because the coefficient vectors $(1, -\nu)$ span $\mathbb{F}_p^{m+1}$; hence every grid has exactly $|V|^{K-m-1}$ preimages. If fewer than $\delta|V|^{m+1}$ grids existed, the partite hypergraph-removal theorem, with deletion parameter $\varepsilon < \alpha/(2K)$ and its corresponding threshold δ , would allow us to destroy every simplex by deleting at most $\varepsilon|V|^{K-1}$ edges of each type. On the other hand, the constant grids $h = 0$ with $a \in A$ already yield at least $\alpha|V|^{K-1}$ labelled simplices. A fixed deleted edge lies in at most one such constant simplex, since $h = 0$ uniquely determines the missing vertex variable. Thus all K deletion sets can destroy fewer than $K\varepsilon|V|^{K-1} < \alpha|V|^{K-1}/2$ constant simplices, a contradiction. Therefore the number of grids is at least $\delta|V|^{m+1}$. $\square$

Theorem 6.26 (Multiplication by p preserves positive top bias above the endpoint). *Fix $k > p$. For every $\eta > 0$ there is $c = c(p, k, \eta) > 0$ such that, if $P : V \rightarrow \mathbb{R}/\mathbb{Z}$ has degree at most k and*

$$\mathbb{E}_{h_1, \dots, h_k} e(d^k P(h_1, \dots, h_k)) \geq \eta,$$

then pP has degree at most $k - p + 1$ and

$$\mathbb{E}_{h, h_{p+1}, \dots, h_k} e(d^{k-p+1} (pP)(h, h_{p+1}, \dots, h_k)) \geq c.$$

In particular, bounded analytic rank of P implies bounded analytic rank of pP in the lowered degree.

Proof. The degree assertion is Lemma 6.7. Put

$$T = d^k P : V^k \rightarrow \mathbb{R}/\mathbb{Z}.$$

For fixed $(h_1, \dots, h_{k-1})$, the map $u \mapsto T(h_1, \dots, h_{k-1}, u)$ is an additive character. Averaging that character in u therefore gives either 0 or 1. Since the total bias of T is at least η , the set

$$A := \{(h_1, \dots, h_{k-1}) \in V^{k-1} : T(h_1, \dots, h_{k-1}, \cdot) \equiv 0\}$$

has density at least η .

Write $z = (h_{p+1}, \dots, h_{k-1}) \in V^{k-p-1}$ and let

$$A_z := \{(h_1, \dots, h_p) \in V^p : (h_1, \dots, h_p, z) \in A\}.$$

Since the average density of the fibres A_z is at least η , the set of z for which A_z has density at least $\eta/2$ has density at least $\eta/2$. For every such z , apply Proposition 6.25 with $m = p$ and density threshold $\eta/2$. Summing over the good fibres gives a positive proportion, depending only on p, k, η , of tuples

$$(h, h_1, \dots, h_{k-1})$$

for which

$$(h_1 + \omega_1 h, \dots, h_p + \omega_p h, h_{p+1}, \dots, h_{k-1}) \in A \text{ for every } \omega \in \mathbb{F}_p^p. \quad (5.1)$$

For each such tuple the additive functional in the final variable vanishes for every choice in (5.1). Take one alternating difference in each of the first p base variables, using the increment h in every coordinate (only the 0 and 1 choices from the full $\mathbb{F}_p^p$ grid are needed). Multilinearity of T collapses the result to

$$T(\underbrace{h, \dots, h}_{p \text{ times}}, h_{p+1}, \dots, h_{k-1}, \cdot) \equiv 0. \quad (5.2)$$

Thus (5.2) holds for a positive proportion of the $(k-p)$ -tuples $(h, h_{p+1}, \dots, h_{k-1})$. Character orthogonality in the remaining last variable gives

$$\mathbb{E}_{h, h_{p+1}, \dots, h_k} e(T(h, \dots, h, h_{p+1}, \dots, h_k)) \geq c. \quad (5.3)$$

It remains to identify this repeated-argument derivative with the top derivative of pP . From the finite difference identity $p\Delta_h = \Delta_h^p I_h$ used in Lemma 6.7, taking exactly the top homogeneous part gives

$$d^k P(\underbrace{h, \dots, h}_p, h_{p+1}, \dots, h_k) = -d^{k-p+1}(pP)(h, h_{p+1}, \dots, h_k),$$

with the sign depending only on the difference convention. Complex conjugation does not change bias, so (5.3) is precisely the desired lower bound for pP . $\square$

6.6 Beginning TZ5: the classical symmetric-multilinear algebra

The original Tao–Ziegler proof does not jump directly from bias to a polynomial phase. Its Sections 6–8 first identify the exact algebra generated by top derivatives of classical polynomials. We reconstruct that algebra before introducing the genuinely deep bias-to-rank input. This separation is useful pedagogically: everything in this section is algebraic and explicit, whereas the later equidistribution theorem contains a separate structural theorem.

Definition 6.27 (Classical symmetric multilinear form). Let $r \geq 1$. A map $T : V^r \rightarrow \mathbb{F}_p$ is a *classical symmetric r -linear form* if it is multilinear, invariant under every permutation of its arguments, and

$$T(h_1, \dots, h_r) = 0$$

whenever at least p of the arguments are equal. We write $\text{CSM}^r(V)$ for the vector space of these forms. When $r < p$ the final condition is vacuous.

Proposition 6.28 (Top derivatives and CSM forms). *For every $r \geq 1$, the top derivative maps classical polynomials onto classical symmetric multilinear forms:*

$$0 \longrightarrow \text{Poly}_{\leq r-1}(V \rightarrow \mathbb{F}_p) \longrightarrow \text{Poly}_{\leq r}(V \rightarrow \mathbb{F}_p) \xrightarrow{d^r} \text{CSM}^r(V) \longrightarrow 0.$$

Proof. If P is classical of degree at most r , commutativity of differences and the identity

$$\Delta_{a+b} = \Delta_a + \Delta_b + \Delta_a \Delta_b$$

show that $d^r P$ is symmetric and multilinear: in the top derivative the extra product term contributes an $(r+1)$ -fold derivative and therefore vanishes. Because $pP = 0$, Lemma 6.7 gives

$$d^r P(\underbrace{h, \dots, h}_p, h_{p+1}, \dots, h_r) = 0,$$

so $d^r P$ is classical. Its kernel is exactly the polynomials of degree at most $r-1$.

For surjectivity choose coordinates $V = \mathbb{F}_p^n$. Write

$$T(h_1, \dots, h_r) = \sum_{i_1, \dots, i_r} c_{i_1, \dots, i_r} h_{1, i_1} \cdots h_{r, i_r}.$$

Symmetry makes the coefficient tensor symmetric. The classical cancellation forces a coefficient to vanish whenever some coordinate index occurs at least p times: test the form on standard basis vectors with p identical arguments and polarize the remaining ones. It is therefore enough to integrate one orbit of a multi-index at a time.

Let a_j be the multiplicity of coordinate j in such a multi-index. Then $0 \leq a_j < p$ and $\sum_j a_j = r$. Since each $a_j!$ is invertible in $\mathbb{F}_p$, the polynomial

$$Q_a(x) = \prod_{j=1}^n \frac{x_j^{a_j}}{a_j!}$$

has top derivative equal to the symmetrized coordinate monomial associated with that multi-index. A linear combination of these Q_a therefore has top derivative T . $\square$

Definition 6.29 (Concatenation). If $S \in \text{CSM}^k(V)$ and $T \in \text{CSM}^\ell(V)$, define $S * T \in \text{CSM}^{k+\ell}(V)$ by

$$(S * T)(h_1, \dots, h_{k+\ell}) = \sum_{\substack{A \sqcup B = [k+\ell] \\ |A|=k, |B|=\ell}} S(h_A) T(h_B).$$

The notation $S(h_A)$ is unambiguous because S is symmetric.

Lemma 6.30 (Discrete product rule for top derivatives). *If $P : V \rightarrow \mathbb{F}_p$ has degree at most k and $Q : V \rightarrow \mathbb{F}_p$ has degree at most ℓ , then*

$$d^{k+\ell}(PQ) = d^k P * d^\ell Q.$$

Proof. Apply successive finite differences to PQ . At every differentiation the discrete Leibniz rule assigns the new difference to the P factor, the Q factor, or to both. After $k + \ell$ derivatives, every term in which P receives more than k derivatives or Q receives more than ℓ derivatives vanishes. A term in which one derivative is assigned to both factors forces the total number received by the two factors to exceed $k + \ell$, and hence also vanishes. Thus the surviving terms are exactly those partitions of the $k + \ell$ directions into k directions falling on P and ℓ falling on Q . Summing them is precisely the displayed concatenation. $\square$

Definition 6.31 (Symmetric power). Let $S \in \text{CSM}^k(V)$ with $k \geq 2$ and let $m \geq 1$. Define $\text{Sym}^m(S) \in \text{CSM}^{mk}(V)$ by summing

$$\prod_{A \in \mathcal{A}} S(h_A)$$

over all set partitions $\mathcal{A}$ of $[mk]$ into m unordered blocks of size k .

Lemma 6.32 (Elementary symmetric-power identities). *The form $\text{Sym}^m(S)$ is classical, symmetric and multilinear, and*

$$m! \text{Sym}^m(S) = \underbrace{S * \cdots * S}_{m \text{ factors}}.$$

Proof. Multilinearity and symmetry follow directly from the definition. In the iterated concatenation every unordered block partition occurs once for every ordering of its m blocks, hence exactly $m!$ times.

For classicality, suppose p input variables are all equal to h . Fix the blocks formed by the remaining $mk - p$ labelled variables and fix how many of the p equal arguments are inserted into each block. If some block receives all p , its factor S vanishes. Otherwise the multiplicities are integers $c_1, \dots, c_m$ with $0 \leq c_j < p$ and $\sum_j c_j = p$. All labelled allocations with these multiplicities give the same product, while their number is

$$\frac{p!}{c_1! \cdots c_m!}.$$

The numerator is divisible by p and none of the denominator factorials is, so this multiplicity is 0 in $\mathbb{F}_p$. Every grouped contribution therefore vanishes, and hence so does the whole symmetric power. $\square$

Lemma 6.33 (Divided-power lifting of a symmetric power). *Let $k \geq 2$, $m \geq 1$, and $S \in \text{CSM}^k(V)$. There exists a classical polynomial $Q : V \rightarrow \mathbb{F}_p$ of degree at most mk such that*

$$d^{mk}Q = \text{Sym}^m(S).$$

If $m \geq 2$, Q may be chosen with $\text{rank}_{mk-1}(Q) \leq 1$.

Proof. By Proposition 6.28, choose a classical polynomial $P : V \rightarrow \mathbb{F}_p$ of degree at most k with $d^k P = S$. The formal expression $P^m/m!$ would integrate the symmetric power when $m!$ is invertible. To avoid division by $m!$ we use divided powers.

Choose $M \geq 0$ so that

$$p^M \leq m < p^{M+1}$$

(with $M = 0$ allowed). Apply Theorem 6.13 M times to $\iota(P)$, choosing the constants in the root construction inside the corresponding root-of-unity groups. We obtain a polynomial

$$\tilde{P}_M : V \rightarrow \frac{1}{p^{M+1}}\mathbb{Z}/\mathbb{Z}, \quad \deg \tilde{P}_M \leq k + M(p-1), \quad p^M \tilde{P}_M = \iota(P).$$

After identifying $\frac{1}{p^{M+1}}\mathbb{Z}/\mathbb{Z}$ with $\mathbb{Z}/p^{M+1}\mathbb{Z}$, write the corresponding polynomial as P_M ; then $P_M \equiv P \pmod{p}$.

Because $m < p^{M+1}$, the integer-valued function $n \mapsto \binom{n}{m} \pmod{p}$ is periodic modulo p^{M+1} . Indeed, Vandermonde's identity gives

$$\binom{n+p^{M+1}}{m} - \binom{n}{m} = \sum_{j=1}^m \binom{p^{M+1}}{j} \binom{n}{m-j},$$

and every coefficient $\binom{p^{M+1}}{j}$ with $0 < j < p^{M+1}$ is divisible by p . Thus the classical function

$$Q := \binom{P_M}{m} \pmod{p}$$

is well defined.

We first prove $\deg Q \leq mk$. More generally, if $F_j = \Delta_{h_1} \cdots \Delta_{h_j} P_M$, we claim that for $0 \leq m' < p^{M+1}$,

$$\deg \binom{F_j}{m'} \leq k - j + (m' - 1) \max(k - j, 1), \quad (6.4^*)$$

whenever the right-hand side is nonnegative; if it is negative, the expression is zero modulo p . The zero case follows from the depth normal form: sufficiently many finite differences make F_j divisible by a high power of p , while $\binom{n}{m'}$ is divisible by p whenever $p^a \mid n$ and $m' < p^a$.

For the nonzero cases use downward induction on j , and for fixed j upward induction on m' . The identity

$$\Delta_h \binom{F}{m'} = \sum_{i=1}^{m'} \binom{\Delta_h F}{i} \binom{F}{m' - i} \quad (6.5^*)$$

follows from Vandermonde's formula $\binom{u+v}{m'} = \sum_i \binom{u}{i} \binom{v}{m'-i}$. The two induction hypotheses bound the degree of the i th summand by

$$[k - j - 1 + (i - 1) \max(k - j - 1, 1)] + [k - j + (m' - i - 1) \max(k - j, 1)],$$

which is at most $k - j + (m' - 1) \max(k - j, 1) - 1$. The derivative recursion Lemma 6.4 proves (6.4*). Taking $j = 0, m' = m$ gives $\deg Q \leq mk$.

Now compute the top derivative. In (6.5*) with $F = P_M$ and $m' = m$, the summand with $i > 1$ has degree at most $mk - i < mk - 1$ and hence does not contribute to $d^{mk}Q$. Only $i = 1$ survives at top degree, giving

$$d^{mk}Q(h_1, \dots, h_{mk}) = d^{mk-1} \left((\Delta_{h_{mk}} P) \binom{P_M}{m-1} \right) (h_1, \dots, h_{mk-1}),$$

because $\Delta_h P_M \equiv \Delta_h P \pmod{p}$ in the top classical component. Apply Lemma 6.30. Induction on m then gives exactly one term for each partition of the mk directions into m blocks of size k , hence

$$d^{mk}Q = \text{Sym}^m(d^k P) = \text{Sym}^m(S).$$

Finally suppose $m \geq 2$. The polynomial Q is a function of the single polynomial P_M . Moreover

$$\deg P_M \leq k + M(p - 1) < mk.$$

For the last inequality, $M = 0$ is immediate; for $M \geq 1$, use $p^M \leq m$ and $k \geq 2$ to obtain $M(p - 1) \leq m - 1 \leq (m - 1)k - 1$. Hence P_M has degree at most $mk - 1$, so Q has rank at most one at level $mk - 1$. $\square$

Definition 6.34 (Bounded-rank CSM form). A form $T \in \text{CSM}^r(V)$ has *CSM rank at most* R if there are at most R lower-order classical symmetric multilinear forms $S_1, \dots, S_R$, with $\text{ord}(S_i) < r$, such that $T(h_1, \dots, h_r)$ is a function of the values obtained by applying the S_i to subuples of $(h_1, \dots, h_r)$. A finite CSM system is *regular* if, at every order r , no nonzero linear combination of its order- r generators has bounded CSM rank in the quantitative range under consideration.

Lemma 6.35 (Regularization of a finite CSM system). *Fix an order bound r_0 and a rank threshold R . Every finite family of CSM forms of orders $< r_0$ is measurable with respect to an R -regular finite CSM system of orders $< r_0$. Its size is bounded in terms of the original size, r_0 , R , and p only.*

Proof. Induct on r_0 . Work first at the top order $r_0 - 1$. If some nonzero linear combination

$$L = \sum_i c_i T_i$$

of the current top-order generators has CSM rank at most R , choose i_0 with $c_{i_0} \neq 0$. By the definition of rank, L is a function of at most R CSM forms of strictly smaller order, evaluated on subuples. Solve for T_{i_0} :

$$T_{i_0} = c_{i_0}^{-1} \left(L - \sum_{i \neq i_0} c_i T_i \right).$$

Delete T_{i_0} and adjoin the lower-order witness forms occurring in a rank representation of L . This preserves measurability of the old system, creates no new top-order generator, and lowers the number of top-order generators by one.

Repeat. After at most the original number of top-order generators, the process stops, and then every nonzero linear combination of the remaining top-order generators has rank $> R$. The total number of newly adjoined lower-order forms is bounded in terms of R and the starting top-order complexity.

Now apply the induction hypothesis to the enlarged collection of forms of order $< r_0 - 1$. This does not alter the surviving top-order forms, so their R -regularity is preserved. The lower-order refined system determines every lower-order witness, hence also every deleted top-order generator and therefore every form in the original system. Termination and a bound for the final complexity follow recursively because every repair moves strictly downward in order. $\square$

6.7 Beginning F4: a complete classical regularization model

Before confronting depth extensions, it is useful to isolate the depth-zero argument in a form that is completely quantitative. This is not merely motivational: it identifies the well-founded replacement mechanism that the non-classical proof must preserve while explaining exactly where positive depth creates a new difficulty.

Definition 6.36 (Classical polynomial factor and fixed-threshold regularity). Let $\mathcal{B} = (P_1, \dots, P_m)$ be a finite list of classical polynomials $P_i : V \rightarrow \mathbb{F}_p$, each assigned its actual additive degree $d_i \leq d$. For $R \geq 1$, call $\mathcal{B}$ *R -regular above the linear level* if for every $2 \leq k \leq d$ and every nonzero coefficient vector $(c_i)_{d_i=k} \in \mathbb{F}_p^{\{i:d_i=k\}}$,

$$\text{rank}_{k-1} \left(\sum_{d_i=k} c_i P_i \right) > R.$$

Constants and linear polynomials are carried by the factor but are not included in this regularity test.

Theorem 6.37 (Quantitative classical polynomial-factor regularization). *Fix $d \geq 2$ and $R \geq 1$. Every list $\mathcal{B} = (P_1, \dots, P_m)$ of classical polynomials of degree at most d has a semantic refinement $\mathcal{B}'$ with the following properties:*

1. *every P_i is a function of the polynomials in $\mathcal{B}'$;*
2. *every member of $\mathcal{B}'$ has degree at most d ;*
3. *$\mathcal{B}'$ is R -regular above the linear level;*
4. *one may take*

$$|\mathcal{B}'| \leq m(1 + R)^{d-1}.$$

Proof. We regularize degree levels in descending order $d, d-1, \dots, 2$. Suppose we are at level k and the current factor is not R -regular there. Then there is a nonzero linear combination

$$L = \sum_{d_i=k} c_i P_i$$

with $\text{rank}_{k-1}(L) = r \leq R$. By the definition of algebraic rank there are polynomials $Q_1, \dots, Q_r$ of degree at most $k-1$ and a function F such that

$$L = F(Q_1, \dots, Q_r).$$

Choose i_0 with $c_{i_0} \neq 0$. Since c_{i_0} is invertible in $\mathbb{F}_p$,

$$P_{i_0} = c_{i_0}^{-1} \left(F(Q_1, \dots, Q_r) - \sum_{\substack{d_i=k \\ i \neq i_0}} c_i P_i \right).$$

Hence deleting P_{i_0} and adjoining $Q_1, \dots, Q_r$ produces a semantic refinement: the new factor still determines the deleted coordinate and therefore every coordinate of the old factor. Crucially, no polynomial of degree $\geq k$ is added, while the number of degree- k coordinates drops by one.

Repeat until no bad combination remains at degree k , then move to $k-1$. The process terminates because at a fixed level each repair deletes one level- k coordinate and never creates another, and later repairs occur strictly below that level. Regularity already achieved at a higher degree can therefore never be destroyed.

It remains only to record the quantitative complexity. If the factor has M coordinates when a degree level is entered, there are at most M repairs at that level. Each repair deletes one coordinate and adds at most R , so after that level the complexity is at most

$$M + MR \leq M(1 + R).$$

There are at most $d-1$ levels $d, d-1, \dots, 2$. Starting from m gives the displayed bound $m(1 + R)^{d-1}$. The construction never raises degree, proving all four claims. $\square$

Worked Example 6.38 (A cubic factor whose top coordinates are redundant). Suppose P_1, P_2 are classical cubics and

$$P_2 - 2P_1 = F(Q_1, Q_2)$$

for quadratic Q_1, Q_2 . If $p \neq 2$, then the factor generated by P_1, P_2 is semantically refined by P_1, Q_1, Q_2 , because $P_2 = 2P_1 + F(Q_1, Q_2)$. The number of cubic coordinates falls from two to one, while all new coordinates lie one degree lower. This is the elementary mechanism behind the well-founded regularization algorithm.

6.8 Primary-source architecture of the finite-field inverse theorem

Tao–Ziegler’s proof is easiest to understand when its ingredients are separated by mathematical role. In particular, the implication “bounded analytic rank $\Rightarrow$ bounded algebraic rank” is the substance of the GIP theorem itself, not one detachable elementary lemma. The following proof roadmap records the order in which the ingredients are proved in this volume.

- TZ1. Coordinate/depth normal form.** Unique non-classical coordinate expansion, weighted degree and depth. This is already proved in Theorem 6.11.
- TZ2. Elementary exact roots.** Multiplication by p lowers degree by $p - 1$ and every polynomial has a p th root with exactly that degree loss. This is already proved in Theorem 6.13.
- TZ3. Elementary rank package.** Rank/analytic-rank definitions, finite range, subadditivity, restriction monotonicity, the easy implication rank $\Rightarrow$ analytic rank, and bounded-codimension algebraic-rank stability. These are proved in this chapter.
- TZ4. Multiplication by p above the endpoint.** Reconstruct the article’s Theorem 4.1: bounded analytic rank of a degree- k polynomial with $k > p$ forces bounded analytic rank after multiplication by p , now viewed in degree $k - p + 1$. The proof uses the finite-field multidimensional Szemerédi theorem and is part of the simultaneous induction, not a formal consequence of TZ2.
- TZ5. Classical symmetric-multilinear inverse theorem.** Reconstruct the article’s Theorem 4.6 and Sections 6–8: multilinear concatenation, equidistribution of multilinear maps, and the inverse theorem for biased classical symmetric multilinear forms.
- TZ6. Regular factors and equidistribution.** Reconstruct the non-classical polynomial-factor regularity and cube-equidistribution machinery of Section 9. The classical depth-zero theorem above is only its base model.
- TZ7. Technical exact roots on regular factors.** Reconstruct Theorem 9.7 and the integer-lattice lifting argument of Section 10. This is the step that makes exact roots compatible with a regular factor, not merely the coordinate-level existence of a root from TZ2.
- TZ8. GIP induction.** Combine TZ4, TZ5 and TZ7, together with lower-degree GIP, to prove that bounded analytic rank is equivalent to bounded algebraic rank at the next degree.
- TZ9. Weak inverse theorem of bounded degree.** Obtain from large U^{s+1} norm correlation with a polynomial phase of some bounded degree $d(s, p)$. The Tao–Ziegler paper imports this as its Theorem 11.3 from the earlier Bergelson–Tao–Ziegler characteristic-factor theorem and the correspondence principle. Because the present book forbids load-bearing named black boxes, this child needs an internal owner rather than the citation used in the article.
- TZ10. Degree sharpening and final inverse theorem.** Reconstruct Section 11: choose minimal weak degree, decompose into structured and pseudorandom parts relative to bounded-degree polynomial factors, use GIP to rule out the top degree, and descend until the correlating phase has the sharp degree s .

6.9 How this branch differs from the cyclic Manners branch

The two inverse-theorem branches solve different problems. V.5.002 is algebraic and finite-field: low characteristic enlarges the polynomial category and the proof must understand

depth, rank, and exact roots. V.5.005 is quantitative and cyclic: the structured category is nilpotent geometry and the main difficulty is to recover it with usable dimension and complexity bounds. Neither theorem is a legitimate substitute for the other, so it is useful to keep their proof architectures separate.

6.10 The Kaufman–Lovett bias criterion: from average structure to exact rank

The preceding CSM argument has reached a point at which one genuinely needs a theorem about ordinary finite-field polynomials. This theorem is often compressed to the slogan “bias implies low rank.” That slogan hides the difficult direction. A biased polynomial is first shown to be *approximately* determined by a bounded collection of derivatives; the main work is then to upgrade approximate determination to exact determination. Kaufman–Lovett accomplish this by replacing the derivative factor by a strongly regular refinement, proving an almost-independence theorem on additive cubes, and then using the vanishing of a $(d+1)$ -fold derivative to repair first almost-good factor cells and then the remaining bad cells.

We reconstruct that mechanism here because it is the load-bearing input in the converse direction of Tao–Ziegler’s CSM bias criterion. Throughout this section the prime p and the degree bound are fixed. Write

$$e_p(t) := \exp(2\pi it/p), \quad t \in \mathbb{F}_p.$$

For a polynomial $P : \mathbb{F}_p^n \rightarrow \mathbb{F}_p$ put

$$\text{bias}(P) := |\mathbb{E}_x e_p(P(x))|.$$

6.10.1 Step 1: bias gives an approximate derivative factor

Lemma 6.39 (Derivative sampling: the Bogdanov–Viola approximation mechanism). *Let $P : V \rightarrow \mathbb{F}_p$ be any function on a finite vector space V and suppose*

$$|\mathbb{E}_x e_p(P(x))| \geq \delta > 0.$$

For every $\varepsilon > 0$ there are

$$s = s(p, \delta, \varepsilon)$$

directions $a_1, \dots, a_s \in V$ and a function $F : \mathbb{F}_p^s \rightarrow \mathbb{F}_p$ such that

$$\mathbb{P}_x[P(x) \neq F(\Delta_{a_1} P(x), \dots, \Delta_{a_s} P(x))] < \varepsilon.$$

If P is a polynomial of degree at most d , each input polynomial $\Delta_{a_j} P$ has degree at most $d-1$.

Proof. Set

$$c := \mathbb{E}_y e_p(P(y)), \quad |c| \geq \delta.$$

For a fixed x and a uniformly random $a \in V$, translation invariance gives the exact identity

$$\mathbb{E}_a e_p(\Delta_a P(x)) = \mathbb{E}_a e_p(P(x+a) - P(x)) = c e_p(-P(x)). \quad (\text{KL.1})$$

Thus $P(x)$ is encoded in the mean of a bounded complex random variable. The p possible means

$$c, c e_p(-1), \dots, c e_p(-(p-1))$$

are separated. More precisely, with

$$\sigma_p := \min_{1 \leq t < p} |1 - e_p(t)| > 0,$$

any two are at distance at least $\delta\sigma_p$.

Choose $a_1, \dots, a_s$ independently and uniformly and form

$$Z_x := \frac{1}{s} \sum_{j=1}^s e_p(\Delta_{a_j} P(x)).$$

Both real and imaginary parts are averages of independent variables in $[-1, 1]$. Hoeffding's inequality therefore gives, for an absolute constant $c_0 > 0$,

$$\mathbb{P}_{a_1, \dots, a_s} \left[|Z_x - ce_p(-P(x))| > \frac{\delta\sigma_p}{3} \right] \leq 4 \exp(-c_0 s \delta^2 \sigma_p^2). \quad (\text{KL.2})$$

Choose s so that the right-hand side is $< \varepsilon$. Define the decoder on a vector $(u_1, \dots, u_s) \in \mathbb{F}_p^s$ by taking the unique $t \in \mathbb{F}_p$ whose centre $ce_p(-t)$ is closest to

$$s^{-1} \sum_j e_p(u_j),$$

breaking ties arbitrarily. The separation just noted shows that this decoder is correct whenever the event in (KL.2) fails.

Averaging first over x and then over the random directions shows that the expected fraction of incorrectly decoded points is $< \varepsilon$. Hence some deterministic choice of $a_1, \dots, a_s$ has error $< \varepsilon$, proving the claim. The degree statement is the elementary derivative recursion. $\square$

Remark 6.40 (Why this lemma is stronger than the slogan “take a majority”). The values of P need not themselves have a unique majority value. What the bias gives is a non-zero Fourier coefficient of the value distribution. Formula (KL.1) turns the unknown value $P(x)$ into a rotation of that fixed Fourier coefficient; a small sample of derivatives estimates the rotation. This is the exact information used later, and it works in every fixed prime characteristic.

6.10.2 Step 2: strongly regular polynomial factors

Definition 6.41 (Polynomial factors and semantic refinement). A polynomial factor is a finite ordered family

$$\mathcal{G} = (G_1, \dots, G_m), \quad G_i : V \rightarrow \mathbb{F}_p.$$

Its cells are

$$R_c := \{x \in V : (G_1(x), \dots, G_m(x)) = c\}, \quad c \in \mathbb{F}_p^m.$$

A factor $\mathcal{H}$ *semantically refines* $\mathcal{G}$ if every G_i is a function of the coordinates of $\mathcal{H}$. For a family $\mathcal{G}$ write $\text{Der}(\mathcal{G})$ for all first derivatives $\Delta_a G_i$, and $\text{Der}^{\leq C}(\mathcal{G})$ for the polynomials obtainable by at most C successive derivatives.

Definition 6.42 (Strong regularity). Let $\Phi : \mathbb{N} \rightarrow \mathbb{N}$ be increasing. A pair $(\mathcal{G}, \Delta)$, where

$$\mathcal{G} = (G_1, \dots, G_m), \quad 1 \leq \Delta(G_i) \leq \deg G_i,$$

is *strongly Φ -regular* if the following two properties hold.

(SR1) Higher supports are determined by lower supports. For every i and every $r > \Delta(G_i)$, the value

$$G_i(X + Y_1 + \dots + Y_r)$$

is a function of the collection

$$\{G_j(X + Y_J) : 1 \leq j \leq m, J \subseteq [r], |J| \leq \Delta(G_j)\}, \quad (\text{KL.3})$$

where $Y_J := \sum_{j \in J} Y_j$.

(SR2) No supported linear relation has low derivative rank. For every $r \geq 0$ and every non-zero coefficient family $\alpha_{i,J}$ with $|J| \leq \Delta(G_i)$, put

$$A(X, Y_1, \dots, Y_r) := \sum_{i,J} \alpha_{i,J} G_i(X + Y_J), \quad (\text{KL.4})$$

and let $\mathcal{G}' \subseteq \mathcal{G}$ be the coordinates that actually occur. Then A is not a function of at most $\Phi(m)$ polynomials of the form

$$H(X + Y_I), \quad H \in \text{Der}(\mathcal{G}'), \quad I \subseteq [r]. \quad (\text{KL.5})$$

If only (SR1) is imposed, we call the pair *pre-strongly regular*.

The purpose of $\Delta(G_i)$ is worth emphasizing. In small characteristic the full degree of G_i can be larger than the number of genuinely independent cube variables carried by that coordinate. The strong factor records the smaller number and asks that all larger supports be algebraically determined by lower-support data.

Lemma 6.43 (Strong regularization). *Fix an increasing growth function Φ . If $\mathcal{F} = (F_1, \dots, F_s)$ is a bounded family of polynomials of degree at most k , then it admits a semantic refinement $\mathcal{G} = (G_1, \dots, G_m)$ and degree bounds $\Delta(G_i)$ such that*

1. $(\mathcal{G}, \Delta)$ is strongly Φ -regular;
2. every G_i has degree at most k ;
3. m is bounded in terms of p, k, s, Φ only, independently of $\dim V$;
4. for some bounded $C = C(p, k, s, \Phi)$,

$$\mathcal{G} \subseteq \text{Der}^{\leq C}(\mathcal{F}).$$

Proof. Start with $\mathcal{G} = \mathcal{F}$ and $\Delta(F_i) = \deg F_i$. This is pre-strongly regular: if $r > \deg F_i$, then the vanishing of the r -fold derivative gives

$$F_i(X + Y_{[r]}) = \sum_{J \subseteq [r]} (-1)^{r-|J|+1} F_i(X + Y_J), \quad (\text{KL.6})$$

and repeated use of the same identity reduces the right-hand side to supports of size at most $\deg F_i$.

Suppose now that (SR2) fails. Thus a non-zero combination A as in (KL.4) can be written as a function of

$$H_1(X + Y_{I_1}), \dots, H_\ell(X + Y_{I_\ell}), \quad \ell \leq \Phi(m), \quad (\text{KL.7})$$

with $H_j \in \text{Der}(\mathcal{G}')$. Choose a participating coordinate G_{i_0} of maximal degree, and among the supports carrying a non-zero coefficient of that coordinate choose one, J_0 , maximal by inclusion. Set all variables outside J_0 equal to zero. After collecting terms, the coefficient of $G_{i_0}(X + Y_{J_0})$ remains non-zero: a second term of the same coordinate could collapse onto this support only from a strict superset of J_0 , excluded by maximality. We may therefore solve for $G_{i_0}(X + Y_{J_0})$ as a function of

- values $G_{i_0}(X + Y_J)$ with $J \subsetneq J_0$;
- supported values of the other current coordinates;

- supported values of the derivatives $H_1, \dots, H_\ell$.

Adjoin $H_1, \dots, H_\ell$ to the factor and give each the initial bound $\Delta(H_j) = \deg H_j$. Replace

$$\Delta(G_{i_0}) \quad \text{by} \quad |J_0| - 1,$$

deleting G_{i_0} if this number is zero. The displayed solved relation handles support size $|J_0|$; substituting it into the old pre-strong relations recursively handles every larger support. Hence (SR1) survives, and the new factor still determines the old one.

Every newly adjoined polynomial is a derivative of a participating coordinate and hence has degree strictly smaller than the maximal degree chosen at that repair. This makes the iteration well founded. More quantitatively, at the current maximal degree q the sum of the integers $\Delta(G_i)$ over degree- q coordinates decreases by at least one at every repair, while no new degree- q coordinate is introduced. Hence there are only boundedly many degree- q repairs. During them the total number of lower-degree coordinates grows by repeated applications of

$$M \mapsto M + \Phi(M),$$

so remains bounded by a recursive function of the old complexity. After the degree- q repairs stop, descend to $q - 1$. Induction on q gives a bound for the total number of repairs and the final complexity depending only on p, k, s, Φ . One additional derivative is taken at each repair, so the same argument bounds the derivative depth C . Termination means precisely that (SR2) now holds. $\square$

6.10.3 Step 3: strong regularity gives almost independence on cubes

For an integer $D \geq 1$ define the number of independent low-support coordinates

$$N_D(\mathcal{G}, \Delta) := \sum_{i=1}^m \sum_{j=1}^{\Delta(G_i)} \binom{D}{j}. \quad (\text{KL.8})$$

The next result is the technical heart of the Kaufman–Lovett argument.

Lemma 6.44 (Differentiation preserves bias). *If H is a polynomial in variables $Y_1, \dots, Y_k$ and $\text{bias}(H) \geq \rho$, then after taking additive derivatives in any r of the variable blocks, with new independent directions, the resulting polynomial H' satisfies*

$$\text{bias}(H') \geq \rho^{2^r}.$$

Proof. For one derivative, writing all undifferentiated variables collectively as Z ,

$$\begin{aligned} \mathbb{E}_{Y,Z,Ae_p}(H(Y+A,Z) - H(Y,Z))^2 &= \mathbb{E}_Z |\mathbb{E}_Y e_p(H(Y,Z))|^2 \\ &\geq |\mathbb{E}_{Y,Z} e_p(H(Y,Z))|^2 \end{aligned}$$

by Cauchy–Schwarz. Iterate r times. $\square$

Lemma 6.45 (Almost independence for a strongly regular factor). *Fix integers $d_0, D \geq 1$ and an error $\gamma > 0$. Assume inductively that the Kaufman–Lovett bias-implies-low-rank theorem is already known for all polynomial degrees $< d_0$. There exists a growth function Φ , depending only on p, d_0, D, γ , with the following property.*

Let $(\mathcal{G}, \Delta)$ be strongly Φ -regular, with every $\deg G_i < d_0$. For each non-empty $I \subseteq [D]$ choose a base point $x_I \in V$ and coefficients $a_j^{(I)} \in \mathbb{F}_p$ that are non-zero precisely when $j \in I$. For independent uniform $Y_1, \dots, Y_D \in V$, the joint distribution of

$$G_i \left(x_I + \sum_{j \in I} a_j^{(I)} Y_j \right), \quad 1 \leq |I| \leq \Delta(G_i), \quad (\text{KL.9})$$

is within total-variation distance γ of the uniform distribution on $\mathbb{F}_p^{N_D(\mathcal{G}, \Delta)}$.

Proof. Let $M = N_D(\mathcal{G}, \Delta)$. If the distribution in (KL.9) is not γ -close to uniform, Fourier inversion on $\mathbb{F}_p^M$ supplies a non-trivial linear character whose expectation has magnitude at least

$$\rho = \rho(p, M, \gamma) > 0. \quad (\text{KL.10})$$

Thus there are coefficients $\alpha_{i,I}$, not all zero, for which

$$H(Y_1, \dots, Y_D) := \sum_{i,I} \alpha_{i,I} G_i \left(x_I + \sum_{j \in I} a_j^{(I)} Y_j \right)$$

has bias at least ρ .

Choose a support I_0 maximal by inclusion among supports with a non-zero coefficient; put $r = |I_0|$. By rescaling the variables in I_0 we may suppose its coefficients are all 1, and after relabeling $I_0 = [r]$. Differentiate once in each of $Y_1, \dots, Y_r$, introducing directions $Z_1, \dots, Z_r$. Terms whose supports omit one of these variables vanish. We obtain

$$H'(W, Z_1, \dots, Z_r) = \sum_i \alpha_{i,[r]} \sum_{\omega \in \{0,1\}^r} (-1)^{|\omega|} G_i(W + Z_\omega), \quad (\text{KL.11})$$

where W is a uniform translate of $Y_1 + \dots + Y_r$ and $Z_\omega = \sum_j \omega_j Z_j$. By Lemma 6.44,

$$\text{bias}(H') \geq \rho^{2^r}. \quad (\text{KL.12})$$

Also $\deg H' < d_0$. The inductive bias-implies-low-rank theorem therefore expresses H' as a function of at most

$$C = C(p, d_0, \rho, D)$$

first derivatives of H' .

A first derivative of (KL.11), under an arbitrary simultaneous translation of $(W, Z_1, \dots, Z_r)$, is a sum of at most 2^r terms of the form

$$\Delta_a G_i(W + Z_\omega), \quad G_i \text{ participating in } H'. \quad (\text{KL.13})$$

Hence H' is a function of at most $2^r C$ supported evaluations of derivatives of the participating G_i . On the other hand, (KL.11) is precisely a non-zero supported linear combination of the kind forbidden by (SR2), because maximality of I_0 ensures that at least one top supported coefficient survives. Choose the growth function so that

$$\Phi(m) > 2^D C(p, d_0, \rho(p, N_D, \gamma), D).$$

after replacing the right side by a monotone majorant in m . This contradicts strong regularity and proves the lemma. $\square$

Corollary 6.46 (Matching all cube values). *Under the hypotheses of Lemma 6.45, let $x, x' \in V$ lie in the same factor cell and fix $y'_1, \dots, y'_D \in V$. Then, for independent uniform $Y_1, \dots, Y_D$,*

$$\mathbb{P}\left[G_i(x + Y_I) = G_i(x' + y'_I) \text{ for every } i \text{ and } I \subseteq [D]\right] = p^{-N_D(\mathcal{G}, \Delta)}(1 + O(\gamma)), \quad (\text{KL.14})$$

where $y'_I := \sum_{j \in I} y'_j$.

Proof. By (SR1), it is enough to impose the equalities for $1 \leq |I| \leq \Delta(G_i)$. Those low-support values have, by Lemma 6.45, a distribution within γ of uniform, so the specified pattern has probability $p^{-N_D}(1 + O(\gamma))$ after decreasing the chosen error to a sufficiently small function of the bounded number N_D . The support-zero equality is the assumption that x and x' are in the same cell, and (SR1) then propagates all remaining equalities. $\square$

6.10.4 Step 4: approximate computation becomes exact computation

Lemma 6.47 (Uniformity of factor cells). *Fix $m \geq 1$. There is $\gamma_0 = \gamma_0(p, m) > 0$ with the following property. Let $(\mathcal{G}, \Delta)$ be a strongly regular factor with m generators for which Lemma 6.45 is available with $D = 1$ and total-variation error $0 < \gamma \leq \gamma_0$. For*

$$R_c := \{x \in V : (G_1(x), \dots, G_m(x)) = c\}, \quad c \in \mathbb{F}_p^m,$$

every cell R_c is nonempty and

$$||R_c| - p^{-m}|V|| \leq C_{p,m}\gamma|V|. \quad (\text{KL.15})$$

Proof. If R_c is non-empty, choose $x \in R_c$. Apply Corollary 6.46 with $D = 1$, $x' = x$, and $y'_1 = 0$. The event is exactly that $x + Y_1$ lies in R_c , so it has probability $p^{-m}(1 + O(\gamma))$; here $N_1 = m$ because every $\Delta(G_i) \geq 1$.

If one cell were empty, at most $p^m - 1$ cells would remain, each of size at most $p^{-m}|V|(1 + C\gamma)$. Choosing $\gamma < c_m p^{-m}$ makes their total size strictly less than $|V|$, a contradiction. $\square$

For the cube-repair step put

$$L := N_{d+1}(\mathcal{G}, \Delta).$$

The next lemma records both one-cube and paired-cube counts. The paired version is what prevents an exceptional subset of a cell from capturing too many cube vertices.

Lemma 6.48 (One-cube and paired-cube counts in a cell). *Let R be a factor cell and $x \in R$. For independent uniform $Y_1, \dots, Y_{d+1}$,*

$$p_R := \mathbb{P}[x + Y_I \in R \text{ for every } I \subseteq [d+1]] = p^{-L}(1 + O(\gamma)). \quad (\text{KL.16})$$

Moreover, fix a non-empty $I_0 \subseteq [d+1]$, and let $Y_1, \dots, Y_{d+1}, Z_1, \dots, Z_{d+1}$ be uniform subject to $Y_{I_0} = Z_{I_0}$. Then

$$\begin{aligned} & \mathbb{P}[x + Y_I \in R, x + Z_I \in R \text{ for all } I \subseteq [d+1] \mid Y_{I_0} = Z_{I_0}] \\ & \leq p^m p^{-2L}(1 + O(\gamma)). \end{aligned} \quad (\text{KL.17})$$

Proof. For (KL.16), apply Corollary 6.46 with $x' = x$ and all comparison directions equal to zero.

For (KL.17), after relabeling write $I_0 = [s]$. Make the invertible linear coordinate change

$$Y'_s = Y_1 + \dots + Y_s,$$

keeping the other Y'_j as suitable free coordinates, and do the same for Z . The condition $Y_{I_0} = Z_{I_0}$ says precisely that the two systems share the one coordinate $Y'_s = Z'_s$. Every sum Y_I and Z_I becomes a linear combination whose non-zero coefficients involve exactly the corresponding support set; coefficients may be ± 1 but are non-zero on their support. Applying the almost-independence lemma to the union of the two systems counts two copies of the L low-support constraints, except that the m one-variable constraints attached to the shared coordinate occur only once. Thus the probability is

$$p^{-(2L-m)}(1 + O(\gamma)) = p^m p^{-2L}(1 + O(\gamma)),$$

which proves the claim. $\square$

Lemma 6.49 (An almost-good cell is completely good). *Let $P : V \rightarrow \mathbb{F}_p$ have degree at most d , and let R be a factor cell. Suppose there is $b \in \mathbb{F}_p$ such that*

$$\mathbb{P}_{x \in R}[P(x) \neq b] < 2^{-2(d+1)}. \quad (\text{KL.18})$$

There is $\gamma_0 = \gamma_0(p, d) > 0$ such that, if the strong-regularity error satisfies $0 < \gamma \leq \gamma_0$, then $P(x) = b$ for every $x \in R$.

Proof. Let $B = \{x \in R : P(x) \neq b\}$ and suppose $B \neq \emptyset$. Fix $x \in B$. By (KL.16), with probability p_R a random $(d+1)$ -cube based at x stays entirely inside R .

Fix a non-empty vertex I_0 . Let $p_B(I_0)$ be the probability that the cube stays in R and $x + Y_{I_0} \in B$. Writing the last event as a sum over the possible point of B and applying Cauchy–Schwarz gives

$$p_B(I_0)^2 \leq |B| |V|^{-1} \mathbb{P}[\text{two } R\text{-cubes based at } x \mid Y_{I_0} = Z_{I_0}]. \quad (\text{KL.19})$$

Use (KL.17), (KL.16), and $|R| = p^{-m} |V| (1 + O(\gamma))$ to obtain

$$p_B(I_0) \leq p_R \sqrt{|B|/|R|} (1 + O(\gamma)) < 2^{-(d+1)} p_R (1 + O(\gamma)). \quad (\text{KL.20})$$

There are $2^{d+1} - 1$ non-zero vertices. For sufficiently small γ , the union bound shows that the event

$$x + Y_I \in R \setminus B \quad \text{for every non-empty } I$$

has positive probability. Choose such a cube. Every non-zero vertex has P -value b . Since P has degree at most d ,

$$0 = \Delta_{Y_1} \cdots \Delta_{Y_{d+1}} P(x) = \sum_{I \subseteq [d+1]} (-1)^{d+1-|I|} P(x + Y_I).$$

All non-zero terms are b , and the alternating sum of their coefficients is the negative of the coefficient at the empty set. Hence $P(x) = b$, contradicting $x \in B$. Thus B is empty. $\square$

Lemma 6.50 (If almost every cell is good, every cell is good). *Let P have degree at most d . Suppose P is constant on all but a proportion at most $2^{-(d+2)}$ of the cells of a strongly regular factor. Then P is constant on every cell.*

Proof. Fix one cell R and points $x, x' \in R$. We show $P(x) = P(x')$. Choose $y'_1, \dots, y'_{d+1}$ independently. Cell uniformity and the union bound show that, with positive probability, every non-zero vertex $x' + y'_I$ lies in a good cell: there are fewer than 2^{d+1} such vertices and the bad-cell proportion is at most $2^{-(d+2)}$, up to the arbitrarily small cell-uniformity error. Fix one such $y'_1, \dots, y'_{d+1}$.

By Corollary 6.46, there exist directions $y_1, \dots, y_{d+1}$ such that for every I and every factor coordinate,

$$G_i(x + y_I) = G_i(x' + y'_I). \quad (\text{KL.21})$$

For non-empty I the point on the right lies in a good cell, so the two points in (KL.21) lie in that same cell and have equal P -values. Apply the vanishing $(d+1)$ -cube identity to the two cubes. All non-empty vertex values agree, hence the empty vertex values agree as well: $P(x) = P(x')$. $\square$

Proposition 6.51 (Approximation by a bounded derivative factor implies exact computation). *Fix a prime p , integers $d \geq 2$ and $s \geq 1$, and assume the bias-implies-low-rank theorem in every degree $< d$. There exist*

$$\varepsilon_d = \varepsilon_d(p, d) > 0, \quad R = R(p, d, s) \in \mathbb{N},$$

with the following property. Let V be a finite-dimensional $\mathbb{F}_p$ -vector space, let $P : V \rightarrow \mathbb{F}_p$ have degree at most d , and suppose there are directions $a_1, \dots, a_s \in V$ and a map $\Phi : \mathbb{F}_p^s \rightarrow \mathbb{F}_p$ such that, writing $F_i = \Delta_{a_i} P$,

$$|\{x \in V : P(x) \neq \Phi(F_1(x), \dots, F_s(x))\}| < \varepsilon_d |V|.$$

Then there exist directions $h_1, \dots, h_r \in V$ with $r \leq R$ and a map $\Psi : \mathbb{F}_p^r \rightarrow \mathbb{F}_p$ such that

$$P(x) = \Psi(\Delta_{h_1} P(x), \dots, \Delta_{h_r} P(x)) \quad \text{for every } x \in V.$$

Proof. Choose a growth function Φ so large, pointwise in the possible output complexity, that Lemma 6.45 is available for $D = d + 1$ with every error needed in Lemmas 6.47–6.50. Apply Lemma 6.43 to obtain a strongly regular refinement $\mathcal{G} = (G_1, \dots, G_m)$. The factor $\mathcal{G}$ still determines $\mathcal{F}$, so the original decoder becomes a decoder on the cells of $\mathcal{G}$.

For a cell R_c , let η_c be its conditional error rate. Cell uniformity gives comparable weights to all cells, and the global average of the η_c is $O(\varepsilon_d)$. Set

$$\theta := 2^{-(d+1)}.$$

Mark a cell bad when $\eta_c > \theta$. Markov's inequality and cell uniformity show that the proportion of bad cells is $O(\varepsilon_d/\theta)$. Choose ε_d so small that this proportion is at most $2^{-(d+2)}$. Every non-bad cell satisfies the hypothesis of Lemma 6.49, and is therefore a cell on which P is exactly constant. Lemma 6.50 then shows that the remaining cells are also constant. Hence P is a function of the coordinates $G_1, \dots, G_m$.

Finally, the regularization lemma gives $G_i \in \text{Der}^{\leq C}(\mathcal{F})$ for bounded C . Since every F_j is a first derivative $\Delta_a P$, any iterated derivative of F_j is a bounded linear combination of translates of P with coefficients summing to zero. Rewriting each translate $P(X + b) - P(X)$ as the first derivative $\Delta_b P(X)$ expresses every G_i as a linear combination of at most 2^{C+1} first derivatives of P . Collect the finitely many derivative directions appearing in all the G_i . Their number is bounded in terms of p, d, s only, and their values determine every G_i , hence determine P . $\square$

6.10.5 Step 5: the finite-field bias criterion

Theorem 6.52 (Kaufman–Lovett: bias implies low algebraic rank). *For every prime p , integer $d \geq 1$, and $\delta > 0$, there is a constant $C = C(p, d, \delta)$ such that whenever $P : \mathbb{F}_p^n \rightarrow \mathbb{F}_p$ has degree at most d and*

$$\text{bias}(P) \geq \delta,$$

one has

$$\text{rank}_{d-1}(P) \leq C.$$

Moreover, the lower-degree generators may be chosen to be first derivatives $\Delta_a P$.

Proof. Induct on d . If $d = 1$, every non-constant linear polynomial has zero bias, so a biased degree-one polynomial is constant and has rank zero.

Assume $d \geq 2$ and the theorem holds below degree d . Let ε_d be the threshold from Proposition 6.51. Apply Lemma 6.39 to P with this error. We obtain a bounded family of first derivatives of P whose values approximate P . The proposition, whose proof used only the inductive bias theorem in degrees $< d$, upgrades this to exact computation of P from a bounded family of first derivatives. This is precisely the asserted bound on $\text{rank}_{d-1}(P)$. $\square$

Worked Example 6.53 (What strong regularity repairs). Let $p = 2$ and suppose a cubic factor contains two coordinates G_1, G_2 for which a cube expression such as

$$G_2(X + Y_1 + Y_2) - G_2(X)$$

can already be recovered from a bounded family of derivatives of G_1 and G_2 . Ordinary factor regularity only sees linear combinations $c_1 G_1 + c_2 G_2$ at one point. Strong regularity sees this two-variable dependence. The repair lowers $\Delta(G_2)$ and adjoins the relevant derivatives. After finitely many such repairs, all genuinely independent cube coordinates are visible, which is exactly what the paired-cube argument needs.

6.10.6 The CSM bias criterion used by Tao–Ziegler

Proposition 6.54 (Finite CSM bias criterion). *Fix $k \geq 2$ and $\delta > 0$. There exists $C = C(p, k, \delta)$ such that if*

$$T : V^k \rightarrow \mathbb{F}_p$$

is a classical symmetric multilinear form and

$$|\mathbb{E}_{h_1, \dots, h_k} e_p(T(h_1, \dots, h_k))| \geq \delta,$$

then T is a function of at most C lower-order classical symmetric multilinear forms on proper subsets of the k variables. Conversely, bounded such rank forces the bias to be bounded below by a positive constant depending only on the rank data.

Proof. Regard T as a degree- k polynomial on the product vector space V^k . Theorem 6.52 expresses it as a function of boundedly many first derivatives. A derivative in a direction $a = (a_1, \dots, a_k)$ expands, by multilinearity, as

$$\Delta_a T(h_1, \dots, h_k) = \sum_{\emptyset \neq J \subseteq [k]} T(z_1^{(J)}, \dots, z_k^{(J)}), \quad (\text{KL.22})$$

where $z_j^{(J)} = a_j$ for $j \in J$ and $z_j^{(J)} = h_j$ otherwise. Every summand depends on only the variables outside the non-empty set J , hence is a multilinear form of order $< k$. Polarizing/symmetrizing the finitely many coefficient forms if necessary expresses these summands through a bounded family of CSM forms of orders $< k$, evaluated on proper subsets of $(h_1, \dots, h_k)$. Thus T has bounded CSM rank.

For the converse, suppose T is a function of boundedly many lower-order forms. Fourier expand the resulting finite function on their joint value space. At least one Fourier term has non-negligible correlation with $e_p(T)$. Each such term factorizes into bounded functions, one missing each variable represented in the corresponding proper support. Repeated Cauchy–Schwarz in the missing variables bounds that correlation by a fixed power of the top multilinear bias. Since the correlation is bounded below and all complexities are fixed, the bias of T is bounded below as well. $\square$

6.11 Regular CSM systems are jointly equidistributed

The regularization lemma is useful only if high CSM rank has a concrete probabilistic consequence. This section proves that consequence in the precise finite configuration form needed by the symmetrization argument. The architecture is the one used in Tao–Ziegler’s counting lemma: Fourier-expand the desired joint distribution, isolate a maximal-order supported form, and remove every other term by Cauchy–Schwarz in a variable that it misses.

Lemma 6.55 (Missing-variable Cauchy–Schwarz). *Let $r \geq 2$, let $T : V^r \rightarrow \mathbb{F}_p$ be multilinear, and let Y be any finite probability space. For each $j \in [r]$ let*

$$F_j : V^{r-1} \times Y \rightarrow \mathbb{C}$$

be 1-bounded, where F_j is evaluated on all x_i with $i \neq j$ and on $y \in Y$. Then

$$\left| \mathbb{E}_{x_1, \dots, x_r, y} e_p(T(x_1, \dots, x_r)) \prod_{j=1}^r F_j(x_{[r] \setminus \{j\}}, y) \right|^{2^r} \leq |\mathbb{E}_{h_1, \dots, h_r} e_p(T(h_1, \dots, h_r))|.$$

Proof. This is the multilinear form of the Cauchy–Schwarz–Gowers argument, but we record the mechanism because it is the load-bearing step of the counting lemma. Apply Cauchy–Schwarz first in every variable except x_1 and then in x_1 , introducing two copies of each variable. At the step involving x_j , the factor F_j disappears because it is independent of x_j ; every remaining 1-bounded factor may simply be discarded once its modulus has appeared. After all r steps the phase is the alternating sum over an r -dimensional cube,

$$\sum_{\omega \in \{0,1\}^r} (-1)^{|\omega|} T(x_1^{\omega_1}, \dots, x_r^{\omega_r}).$$

By multilinearity this alternating sum equals, up to the harmless global sign $(-1)^r$,

$$T(x_1^1 - x_1^0, \dots, x_r^1 - x_r^0).$$

The differences $x_j^1 - x_j^0$ are independent and uniform on V . The auxiliary variable y has disappeared during the same Cauchy–Schwarz chain. Taking absolute values gives the displayed inequality. $\square$

Definition 6.56 (Finite evaluation pattern of a CSM system). Let

$$\mathcal{T} = \{T_{r,a} : V^r \rightarrow \mathbb{F}_p : 2 \leq r \leq r_0, a \in A_r\}$$

be a finite family of classical symmetric multilinear forms. For an integer $D \geq r_0$ write

$$\mathcal{I}_D(\mathcal{T}) := \{(r, a, I) : 2 \leq r \leq r_0, a \in A_r, I \subseteq [D], |I| = r\}.$$

For $h = (h_1, \dots, h_D) \in V^D$ define

$$\Xi_{\mathcal{T}, D}(h) := (T_{r,a}(h_I))_{(r,a,I) \in \mathcal{I}_D(\mathcal{T})} \in \mathbb{F}_p^M, \quad M := |\mathcal{I}_D(\mathcal{T})|,$$

where the entries of h_I are listed in increasing order. Symmetry makes this ordering irrelevant.

Lemma 6.57 (Tao–Ziegler counting lemma for a regular CSM system). *Fix a prime p , integers $r_0 \geq 2$ and $D \geq r_0$, a complexity bound $m \geq 1$, and $\gamma > 0$. There is a rank threshold*

$$R = R(p, r_0, D, m, \gamma)$$

with the following property. Let

$$\mathcal{T} = \{T_{r,a} : V^r \rightarrow \mathbb{F}_p : 2 \leq r \leq r_0, a \in A_r\}$$

be a finite family of classical symmetric multilinear forms as in Definition 6.56, with $|A_r| \leq m$ for every r . Assume that, at each order r , every non-zero linear combination of the $T_{r,a}$ has CSM rank greater than R . Put $M := |\mathcal{I}_D(\mathcal{T})|$. Then, for independent uniform $h_1, \dots, h_D \in V$ and every $b \in \mathbb{F}_p^M$,

$$\left| \mathbb{P}[\Xi_{\mathcal{T}, D}(h) = b] - p^{-M} \right| \leq \gamma.$$

Proof. Let $\lambda = (\lambda_{r,a,I}) \in \mathbb{F}_p^M$ be non-zero. The corresponding Fourier coefficient is

$$\hat{\mu}(\lambda) = \mathbb{E}_h e_p \left(\sum_{r,a,I} \lambda_{r,a,I} T_{r,a}(h_I) \right). \quad (\text{TZ.7.8.1})$$

Choose the largest order r for which some coefficient is non-zero, and among those choose a support $I_0 \subseteq [D]$, $|I_0| = r$, for which the coefficient vector $(\lambda_{r,a,I_0})_{a \in A_r}$ is non-zero. Combine the terms on that one support into

$$T^* := \sum_{a \in A_r} \lambda_{r,a,I_0} T_{r,a}. \quad (\text{TZ.7.8.2})$$

By regularity, T^* has CSM rank greater than R .

Every other term in the phase of (TZ.7.8.1) misses at least one variable h_j with $j \in I_0$. Indeed, a term of smaller order uses fewer than r variables, while another order- r term has a support $I \neq I_0$ of the same cardinality and therefore omits some element of I_0 . Assign each such term to one omitted $j \in I_0$. After exponentiating, all terms assigned to j form a 1-bounded factor F_j independent of h_j . Variables outside I_0 are absorbed into the auxiliary probability space Y in Lemma 6.55. Consequently

$$|\hat{\mu}(\lambda)|^{2^r} \leq |\mathbb{E}_{x_1, \dots, x_r} e_p(T^*(x_1, \dots, x_r))|. \quad (\text{TZ.7.8.3})$$

Now use the contrapositive of the internal CSM bias criterion, Proposition 6.54. For every $\eta > 0$ there is $C = C(p, r, \eta)$ such that bias at least η forces CSM rank at most C . Choose R larger than all the finitely many values

$$C(p, r, \eta^{2^r}), \quad 2 \leq r \leq r_0,$$

with $\eta := \gamma$ (or any smaller number). Then $|\hat{\mu}(\lambda)| < \gamma$ for every non-zero λ .

Finally Fourier inversion on the finite group $\mathbb{F}_p^M$ gives

$$\mathbb{P}[\Xi_{\mathcal{T}, D} = b] - p^{-M} = p^{-M} \sum_{\lambda \neq 0} \hat{\mu}(\lambda) e_p(-\lambda \cdot b).$$

Hence the absolute value is at most $p^{-M}(p^M - 1)\gamma < \gamma$, as required. $\square$

Worked Example 6.58 (Why maximal support is the right choice). Suppose the top order is 3 and the Fourier phase contains

$$T_1(h_1, h_2, h_3) + T_2(h_1, h_2, h_4) + S(h_2, h_4).$$

Choose $I_0 = \{1, 2, 3\}$. The second cubic term misses h_3 and the quadratic term misses h_1 and h_3 . Thus, after exponentiation, everything except $e_p(T_1(h_1, h_2, h_3))$ can be assigned to factors missing one of the three variables. Repeated Cauchy–Schwarz reduces the entire Fourier coefficient to the bias of T_1 . The general proof differs only in bookkeeping.

6.12 From equidistribution to the support-partition algebra

The counting lemma tells us that a sufficiently regular CSM system has no accidental algebraic relations on a fixed finite family of evaluations. Tao–Ziegler use this in a particularly efficient way: a biased symmetric multilinear form is first made measurable with respect to a regular lower-order CSM system, and the multilinearity of the original form is then transferred to the finite coordinate function describing that measurability. The resulting finite function has a very rigid support algebra. We now reconstruct that argument quantitatively.

6.12.1 Two quantitative preliminaries

Lemma 6.59 (CSM rank is stable under bounded-codimension restriction). *Fix $k \geq 2$. Let $U \leq V$ have codimension c , and let $T \in \text{CSM}^k(V)$. If $T|_{U^k}$ has CSM rank at most R , then T has CSM rank at most*

$$C_{k,c}(R+1)$$

for an explicit constant $C_{k,c}$ depending only on k and c . Consequently, for every target rank R_0 there is $R_1 = R_1(k, c, R_0)$ such that

$$\text{rank}_{\text{CSM}}(T) > R_1 \implies \text{rank}_{\text{CSM}}(T|_{U^k}) > R_0.$$

Proof. Choose a complement W with $V = U \oplus W$ and a basis $e_1, \dots, e_c$ of W . Write

$$h = u(h) + \sum_{a=1}^c \lambda_a(h) e_a, \quad u(h) \in U,$$

where the $\lambda_a : V \rightarrow \mathbb{F}_p$ are linear forms. By multilinearity,

$$T(h_1, \dots, h_k) = \sum_{J \subseteq [k]} \sum_{\mathbf{a} \in [c]^J} \left(\prod_{j \in J} \lambda_{a_j}(h_j) \right) T((e_{a_j})_{j \in J}, (u(h_i))_{i \notin J}). \quad (8.R1)$$

The $J = \emptyset$ term is $T|_{U^k}$ evaluated at $u(h_1), \dots, u(h_k)$ and therefore is a function of at most R lower-order CSM forms on proper subtuples, after precomposing those forms with the projection $u : V \rightarrow U$. Such a precomposition is again a classical symmetric multilinear form of the same order.

If $J \neq \emptyset$, the final factor in (8.R1) is a multilinear form in only $k - |J| < k$ variable slots. It is symmetric in those slots and classical: if p of the remaining arguments are equal, then the original form T already has p equal arguments and vanishes. The scalar coefficients $\lambda_a(h_j)$ are order-one CSM forms. Hence every summand with $J \neq \emptyset$ is a function of lower-order CSM forms on proper subtuples.

There are at most

$$\sum_{j=1}^k \binom{k}{j} c^j = (1+c)^k - 1$$

such complement types, up to the bounded collection of coefficient and frozen-slot forms. Together with the R witnesses for the U^k term, this gives a bound of the form $C_{k,c}(R+1)$. The contrapositive gives the second assertion. $\square$

Lemma 6.60 (Growth-function regularization of finite CSM systems). *Fix $r_0 \geq 2$ and an increasing function*

$$\mathcal{R} : \mathbb{N} \rightarrow \mathbb{N}.$$

Every finite CSM system of orders $< r_0$ is measurable with respect to a finite CSM system

$$\mathcal{S} = \{S_{r,a} : 1 \leq r < r_0, a \in A_r\}$$

of complexity $M = \sum_r |A_r|$, bounded in terms of the initial complexity, r_0 , p and $\mathcal{R}$, such that every nonzero linear combination of the order- r generators has CSM rank greater than $\mathcal{R}(M)$.

Proof. We use the same descending-order replacement mechanism as Lemma 6.35, but we test regularity against the *current* complexity. Whenever a nonzero combination L of generators of some order r has CSM rank at most $\mathcal{R}(M_{\text{cur}})$, delete one generator occurring with nonzero coefficient and adjoin a rank representation of L using forms of orders strictly below r . Measurability of the previous system is preserved exactly as in the proof of Lemma 6.35.

The only point requiring care is termination, because the threshold is allowed to grow when lower-order witnesses are adjoined. Order the vector of generator counts

$$(m_{r_0-1}, m_{r_0-2}, \dots, m_1)$$

lexicographically from the highest order down. A repair at order r decreases m_r by one and changes only lower coordinates. Hence every repair strictly decreases this vector in the well-founded lexicographic order on $\mathbb{N}^{r_0-1}$. In particular, there can be only finitely many repairs. A completely explicit (though enormous) complexity bound is obtained recursively by replacing each possible deletion at order r by at most $\mathcal{R}(M_{\text{cur}})$ new lower-order witnesses; only the existence of such a bound will be needed below.

At termination, if a nonzero order- r combination had rank at most $\mathcal{R}(M)$ for the final complexity M , the algorithm would still permit another repair. Therefore no such combination exists, which is the required regularity statement. $\square$

Remark 6.61 (Why the growth-function formulation matters). A fixed-threshold regularization lemma by itself is not sufficient for the next argument: the rank threshold required by the counting lemma depends on the final number of coordinates, while that final number can depend on the threshold used during regularization. Lemma 6.60 removes this apparent fixed-point problem. It is the finite quantitative counterpart of the “choose the regularity function sufficiently rapidly” convention in the original ultralimit proof.

6.12.2 The support-partition theorem

Fix an integer $D \geq 2$. Let

$$\mathcal{S} = \{S_{r,a} : V^r \rightarrow \mathbb{F}_p : 2 \leq r < D, a \in A_r\}$$

be a finite regular CSM system. Introduce the index set

$$\mathcal{A} := \{\alpha = (r, a, J) : 2 \leq r < D, a \in A_r, J \subseteq [D], |J| = r\}.$$

For $\alpha = (r, a, J)$ put $J_\alpha := J$ and

$$S_\alpha(h_1, \dots, h_D) := S_{r,a}(h_J).$$

The symmetric group $\text{Sym}(D)$ acts on $\mathcal{A}$ by permuting the support J .

Lemma 6.62 (Block additivity transferred from multilinearity). *Assume that the joint evaluation maps of $\mathcal{S}$ on D and $D+1$ variables are surjective. Suppose*

$$T(h_1, \dots, h_D) = F((S_\alpha(h))_{\alpha \in \mathcal{A}})$$

for a function $F : \mathbb{F}_p^{\mathcal{A}} \rightarrow \mathbb{F}_p$, and suppose T is multilinear. For $j \in [D]$ set

$$\mathcal{A}_{\ni j} := \{\alpha : j \in J_\alpha\}, \quad \mathcal{A}_{\not\ni j} := \mathcal{A} \setminus \mathcal{A}_{\ni j}.$$

Then for all $x, y \in \mathbb{F}_p^{\mathcal{A}_{\ni j}}$ and $z \in \mathbb{F}_p^{\mathcal{A}_{\not\ni j}}$,

$$F(x + y, z) = F(x, z) + F(y, z). \tag{8.2}$$

In particular $F(0, z) = 0$, and for fixed z the map $x \mapsto F(x, z)$ is an $\mathbb{F}_p$ -linear functional.

Proof. By symmetry of the notation it is enough to treat $j = D$. We must realize prescribed values x, y, z by vectors

$$h_1, \dots, h_{D-1}, h_D, h'_D \in V.$$

Apply surjectivity of the complete $(D+1)$ -variable evaluation pattern. Prescribe the coordinates supported entirely in $[D-1]$ to be z ; prescribe every coordinate whose support contains D but not $D+1$ to be x ; and prescribe the corresponding coordinate with D replaced by $D+1$ to be y . All remaining evaluation coordinates involving both D and $D+1$ may be assigned arbitrarily. Surjectivity provides the desired vectors.

For a support containing D , multilinearity of the generator gives

$$S_{r,a}(h_{J \setminus \{D\}}, h_D + h'_D) = S_{r,a}(h_{J \setminus \{D\}}, h_D) + S_{r,a}(h_{J \setminus \{D\}}, h'_D).$$

Supports not containing D are unchanged. Multilinearity of T in its last variable therefore yields

$$F(x+y, z) = T(h_1, \dots, h_{D-1}, h_D + h'_D) = T(h_1, \dots, h_D) + T(h_1, \dots, h'_D) = F(x, z) + F(y, z).$$

An additive map between vector spaces over the prime field is automatically $\mathbb{F}_p$ -linear. $\square$

Proposition 6.63 (Support-partition multilinearity; Tao–Ziegler Proposition 8.1). *Under the hypotheses of Lemma 6.62, every function $F : \mathbb{F}_p^{\mathcal{A}} \rightarrow \mathbb{F}_p$ satisfying (8.2) for every $j \in [D]$ is a linear combination of square-free monomials*

$$x_{\alpha_1} \cdots x_{\alpha_q} \tag{8.3}$$

for which the supports

$$J_{\alpha_1}, \dots, J_{\alpha_q}$$

are pairwise disjoint and have union $[D]$.

Proof. We induct on D . The assertion is vacuous for $D = 1$ and immediate for $D = 2$. Assume $D \geq 2$ and the result known for smaller ambient support sets.

For $\alpha \in \mathcal{A}_{\ni D}$ let e_α be the corresponding coordinate basis vector and put

$$F_\alpha := \Delta_{e_\alpha} F.$$

If $J_\beta \cap J_\alpha \neq \emptyset$, choose $j \in J_\beta \cap J_\alpha$. For fixed coordinates outside $\mathcal{A}_{\ni j}$, equation (8.2) says that F is a linear functional of the entire block of coordinates in $\mathcal{A}_{\ni j}$. Its coefficient in the e_α direction is therefore independent of every coordinate in that same block, in particular of x_β . Thus F_α depends only on coordinates x_β whose support is disjoint from J_α .

Let

$$\mathcal{A}_{J_\alpha^\perp} := \{\beta \in \mathcal{A} : J_\beta \subseteq [D] \setminus J_\alpha\}.$$

The block-additivity relations for indices $j \notin J_\alpha$ descend to the resulting function

$$F_\alpha : \mathbb{F}_p^{\mathcal{A}_{J_\alpha^\perp}} \rightarrow \mathbb{F}_p.$$

After relabeling the complement $[D] \setminus J_\alpha$, the induction hypothesis therefore expresses F_α as a linear combination of monomials whose supports partition that complement. Consequently

$$x_\alpha F_\alpha$$

is a linear combination of monomials of the desired form, with J_α supplying one block of the partition of $[D]$.

It remains to reconstruct F from these derivatives. Since (8.2) makes F linear in the full block $\mathcal{A}_{\supset D}$, we have exactly

$$F(x) = \sum_{\alpha \in \mathcal{A}_{\supset D}} x_{\alpha} F_{\alpha}(x). \quad (8.4)$$

Indeed the right side is the coordinate expansion of that linear functional; equivalently, the difference between the two sides is invariant in every e_{α} direction with $\alpha \in \mathcal{A}_{\supset D}$ and vanishes when all those coordinates are zero, because $F(0, z) = 0$. Substituting the inductive descriptions of the F_{α} into (8.4) proves the proposition. $\square$

Worked Example 6.64 (Why overlapping supports are forbidden). Let $D = 4$ and suppose B, B' are bilinear generators. The monomial

$$B(h_1, h_2)B'(h_3, h_4)$$

has supports $\{1, 2\}$ and $\{3, 4\}$, which partition $[4]$, so it is permitted. By contrast

$$B(h_1, h_2)B'(h_1, h_3)$$

uses h_1 twice and misses h_4 . It cannot be linear in the entire block of CSM coordinates involving h_1 in the sense of (8.2); Proposition 6.63 therefore excludes it. This is the finite-coordinate form of the multilinearity of the original four-linear form.

6.13 Symmetrization and integration of the biased CSM form

We now close the classical multilinear block. The argument follows Tao–Ziegler’s Theorem 6.6 and its deduction of Theorem 4.6, but we keep the quantitative choices visible so that no circular regularity threshold is hidden.

Theorem 6.65 (Explicit inverse theorem for biased classical symmetric multilinear forms). *Fix a prime p , an integer $s \geq 2$, and $\delta > 0$. Assume the lower-degree GIP/rank implications required by the internal CSM bias criterion up to degree $s - 1$. Then there exist constants $c, M < \infty$, depending only on p, s, δ , with the following property.*

If V is a finite-dimensional $\mathbb{F}_p$ -vector space and $T \in \text{CSM}^{s+1}(V)$ satisfies

$$|\mathbb{E}_{h_1, \dots, h_{s+1}} e_p(T(h_1, \dots, h_{s+1}))| \geq \delta,$$

then there is a subspace $U \leq V$ of codimension at most c such that on U^{s+1} , T is a linear combination of at most M expressions

$$\text{Sym}^{m_1}(S_1) * \dots * \text{Sym}^{m_{\ell}}(S_{\ell}), \quad (8.5)$$

where

$$2 \leq k_i < s + 1, \quad S_i \in \text{CSM}^{k_i}(U), \quad \sum_{i=1}^{\ell} m_i k_i = s + 1.$$

Proof. Step 1: make T measurable with respect to lower-order forms. By Proposition 6.54, the bias lower bound implies that T is a function of a bounded family of lower-order CSM forms evaluated on proper subtuples of its $s + 1$ variables. Enlarge this family by its finite orbit under $\text{Sym}(s + 1)$; this keeps the complexity bounded and makes the family permutation-stable.

Step 2: regularize with a sufficiently rapid growth function. For a system of complexity m , let $R_{\text{count}}(m)$ be a rank threshold from Lemma 6.57 strong enough that both

the $(s+1)$ - and $(s+2)$ -variable evaluation patterns put positive mass on every point of their finite value spaces. Let

$$R_{\text{res}}(m) := \max_{2 \leq k \leq s} R_1(k, m, R_{\text{count}}(m)),$$

where R_1 is supplied by Lemma 6.59. Apply Lemma 6.60 with a growth function $\mathcal{R}$ chosen larger than $R_{\text{res}}(m)$ at every m . We obtain a regular system

$$\mathcal{S} = \{S_{k,i} : 1 \leq k \leq s, 1 \leq i \leq m_k\}$$

of bounded final complexity m , with respect to which T is measurable.

Step 3: eliminate the linear forms. Let

$$U := \bigcap_{i=1}^{m_1} \ker S_{1,i}.$$

Then $\text{codim } U \leq m_1 \leq m$. On U all linear coordinates vanish. By our choice of $\mathcal{R}$ and Lemma 6.59, every nonzero combination of the remaining order- k generators, $2 \leq k \leq s$, still has rank greater than the counting threshold $R_{\text{count}}(m)$. Hence the restricted system on U is sufficiently regular for Lemma 6.57.

Step 4: obtain an exact finite coordinate function. Put $D = s+1$ and let $\mathcal{A}$ consist of all triples (k, i, J) with $2 \leq k \leq s$ and $J \subseteq [D]$, $|J| = k$. Since the restricted system is regular, choose the error in Lemma 6.57 smaller than the uniform mass of one point in the relevant finite value space. The D -variable evaluation map is then surjective. Thus there is a uniquely defined function

$$F : \mathbb{F}_p^{\mathcal{A}} \rightarrow \mathbb{F}_p$$

such that

$$T(h_1, \dots, h_D) = F((S_\alpha(h))_{\alpha \in \mathcal{A}}).$$

The same choice for $D+1$ variables supplies the augmented surjectivity needed in Lemma 6.62.

Because T is symmetric and the evaluation map is surjective, F is invariant under the natural action of $\text{Sym}(D)$ on supports. Because T is multilinear, Lemma 6.62 applies. Proposition 6.63 therefore gives a unique expansion

$$F(x) = \sum_{\mathcal{P}} c_{\mathcal{P}} \prod_{\alpha \in \mathcal{P}} x_{\alpha}, \quad (8.6)$$

where every $\mathcal{P}$ is a collection of indices whose supports partition $[D]$. The square-free monomials in (8.6) are linearly independent as functions on $\mathbb{F}_p^{\mathcal{A}}$ (induct on the number of coordinates, or evaluate successively at 0 and 1), so the coefficients $c_{\mathcal{P}}$ are uniquely determined. Symmetry of F consequently makes them constant on permutation orbits.

Step 5: identify each permutation orbit with a basic symmetric form. Fix one orbit. Group its blocks according to the generator type (k, i) . Suppose the distinct types are $(k_1, i_1), \dots, (k_{\ell}, i_{\ell})$ and type q occurs m_q times. The sum over all distinct support partitions in this orbit is exactly

$$\text{Sym}^{m_1}(S_{k_1, i_1}) * \dots * \text{Sym}^{m_{\ell}}(S_{k_{\ell}, i_{\ell}}).$$

Indeed, Sym^{m_q} forgets the order of the m_q identical blocks, while concatenation distributes the mutually distinguished generator types over disjoint blocks; hence every support partition in the orbit occurs exactly once. Since all linear generators have been removed, every $k_q \geq 2$. The blocks partition $[D]$, so $\sum_q m_q k_q = D = s+1$. Splitting (8.6) into permutation orbits proves (8.5), with all complexities bounded in terms of p, s, δ only. $\square$

Theorem 6.66 (Inverse theorem for biased classical top derivatives; Tao–Ziegler Theorem 4.6). *Under the hypotheses of Theorem 6.65, there is a bounded codimension subspace $U \leq V$ and a classical polynomial $Q : U \rightarrow \mathbb{F}_p$ of degree at most $s + 1$ and bounded algebraic rank at level s such that*

$$d^{s+1}Q = T|_{U^{s+1}}.$$

All bounds depend only on p, s, δ .

Proof. It suffices by Theorem 6.65 and linearity to integrate one basic expression

$$B = \text{Sym}^{m_1}(S_1) * \cdots * \text{Sym}^{m_\ell}(S_\ell), \quad \sum_i m_i k_i = s + 1,$$

with $2 \leq k_i < s + 1$ and $S_i \in \text{CSM}^{k_i}(U)$. By Proposition 6.28, choose classical polynomials P_i of degree at most k_i with $d^{k_i}P_i = S_i$. By Lemma 6.33, there are classical polynomials Q_i of degree at most $m_i k_i$ satisfying

$$d^{m_i k_i} Q_i = \text{Sym}^{m_i}(S_i),$$

and Q_i has rank at most one at level $m_i k_i - 1$ whenever $m_i \geq 2$.

Set

$$Q_B := \prod_{i=1}^{\ell} Q_i.$$

Repeated use of Lemma 6.30 yields

$$d^{s+1}Q_B = B. \tag{8.7}$$

We claim that Q_B has bounded rank at level s . If $\ell \geq 2$, each factor has positive degree strictly less than $s + 1$ because the positive degrees $m_i k_i$ sum to $s + 1$; hence Q_B is a function of the bounded family $Q_1, \dots, Q_\ell$ of degree at most s . If $\ell = 1$, then $m_1 k_1 = s + 1$ while $k_1 < s + 1$, so $m_1 \geq 2$ and the rank-one conclusion of Lemma 6.33 applies directly. Thus every basic expression has a bounded-rank polynomial primitive.

Finally sum the finitely many primitives with the coefficients supplied by Theorem 6.65. Algebraic-rank subadditivity, proved earlier in this chapter, keeps the resulting rank bounded, and (8.7) gives the desired top derivative. $\square$

Worked Example 6.67 (The quartic low-characteristic pattern). When $s + 1 = 4$ and $B \in \text{CSM}^2(U)$, one allowed orbit is

$$\text{Sym}^2(B)(a, b, c, d) = B(a, b)B(c, d) + B(a, c)B(b, d) + B(a, d)B(b, c).$$

In characteristic two this cannot be obtained by formally dividing $B * B$ by $2!$. Lemma 6.33 instead constructs a quartic Q with $d^4 Q = \text{Sym}^2(B)$ using divided powers and exact roots. The support-partition theorem explains why this is not an isolated trick: every orbit allowed by multilinearity is built from the same symmetric-power and concatenation operations.

6.14 Weighted polynomial calculus and exact roots on integer lattices

The technical exact-root theorem for a regular factor is eventually reduced to a purely integer-valued problem. The relevant notion of degree is not the ordinary total degree: translation by $p^j e_i$ is assigned the weight $D_i + j(p - 1)$. Tao–Ziegler’s Section 10 classifies the resulting

functions explicitly in the binomial basis. We reconstruct that classification here because it is the algebraic heart of the positive-depth root argument.

Fix integers $m \geq 1$ and initial degrees $D_1, \dots, D_m \geq 2$. For $i \in [m]$ and $j \geq 0$ call

$$p^j e_i \in \mathbb{Z}^m$$

a *basic generator* and assign it weight

$$w(p^j e_i) := D_i + j(p-1).$$

For a tuple $\mathbf{v} = (v_1, \dots, v_r)$ of basic generators define $w(\mathbf{v}) := \sum_t w(v_t)$ and $\Delta_{\mathbf{v}} := \Delta_{v_1} \cdots \Delta_{v_r}$.

Definition 6.68 (Weighted degree). A function $f : \mathbb{Z}^m \rightarrow \mathbb{R}/\mathbb{Z}$ has *weighted degree at most d* if

$$\Delta_{\mathbf{v}} f = 0 \quad \text{whenever } w(\mathbf{v}) > d.$$

Lemma 6.69 (Elementary weighted-degree calculus). *If f has weighted degree at most d , then:*

- (i) $\Delta_{p^j e_i} f$ has weighted degree at most $d - D_i - j(p-1)$;
- (ii) if $D_i + j(p-1) > d$, then f is periodic with period $p^j e_i$;
- (iii) sums of functions of weighted degrees at most d_1, d_2 have weighted degree at most $\max(d_1, d_2)$.

Proof. For (i), append $p^j e_i$ to any multigenerator whose total weight is larger than $d - D_i - j(p-1)$. The resulting total weight exceeds d , so the corresponding derivative of f vanishes. Part (ii) is the special case in which the first derivative itself is zero, and (iii) follows from linearity of finite differences. $\square$

For $n \in \mathbb{Z}$ and $r \geq 0$ write

$$\binom{n}{r} := \frac{n(n-1) \cdots (n-r+1)}{r!},$$

with $\binom{n}{0} = 1$. These are integer-valued for all integers n , including negative n .

Lemma 6.70 (p -adic divisibility of a prime-power binomial coefficient). *Let $k \geq 0$ and $1 \leq \ell \leq p^k$. If $t = v_p(\ell)$, then*

$$v_p \binom{p^k}{\ell} \geq k - t.$$

Proof. Use

$$\binom{p^k}{\ell} = \frac{p^k}{\ell} \binom{p^k - 1}{\ell - 1}.$$

The second factor is an integer, while $v_p(p^k/\ell) = k - t$. $\square$

Lemma 6.71 (Weighted degree of a binomial monomial). *Let $a_1, \dots, a_m \geq 0$, not all zero, and let $r \in \mathbb{Z}$. The function*

$$M(x_1, \dots, x_m) := \frac{c}{p^{r+1}} \binom{x_1}{a_1} \cdots \binom{x_m}{a_m} \pmod{1}, \quad (10.1)$$

where $c \in \mathbb{Z}$, has weighted degree at most

$$\sum_{i=1}^m D_i a_i + r(p-1). \quad (10.2)$$

If $r < 0$, the assertion is interpreted after reducing the rational coefficient modulo 1; terms that are integral vanish.

Proof. By linearity it is enough to take $c = 1$. We induct on the number $d_M := \sum_i D_i a_i + r(p-1)$. The assertion is trivial when the displayed function vanishes modulo 1, and hence for sufficiently negative d_M .

By Lemma 6.69, it is enough to prove that for every basic generator $p^k e_j$ the derivative $\Delta_{p^k e_j} M$ has weighted degree at most $d_M - D_j - k(p-1)$. Fix $j = m$ for notation. Vandermonde's identity gives

$$\Delta_{p^k e_m} \binom{x_m}{a_m} = \sum_{\ell=1}^{a_m} \binom{p^k}{\ell} \binom{x_m}{a_m - \ell}. \quad (10.3)$$

A term with $\ell > p^k$ vanishes. For $1 \leq \ell \leq p^k$, put $t = v_p(\ell)$. Lemma 6.70 shows that $\binom{p^k}{\ell}$ contributes at least p^{k-t} , which can be cancelled against the denominator p^{r+1} . By the induction hypothesis, the resulting term has weighted degree at most

$$\sum_{i < m} D_i a_i + D_m(a_m - \ell) + (r - k + t)(p-1).$$

We compare this with the desired bound. Since $\ell \geq p^t$ and $D_m \geq 2$,

$$D_m(\ell - 1) \geq \ell - 1 \geq p^t - 1 \geq t(p-1).$$

Therefore

$$\sum_{i < m} D_i a_i + D_m(a_m - \ell) + (r - k + t)(p-1) \leq d_M - D_m - k(p-1).$$

Every term in (10.3) has the required degree, so the same is true of the derivative. This closes the induction. $\square$

Theorem 6.72 (Classification of weighted polynomials on $\mathbb{Z}^m$). *A function $f : \mathbb{Z}^m \rightarrow \mathbb{R}/\mathbb{Z}$ has weighted degree at most d if and only if it can be written*

$$f(x) = \alpha + \sum_{\substack{a_1, \dots, a_m \geq 0, \ r \geq 0 \\ a_1 + \dots + a_m > 0 \\ \sum_i D_i a_i + r(p-1) \leq d}} \frac{c_{\mathbf{a}, r}}{p^{r+1}} \prod_{i=1}^m \binom{x_i}{a_i} \pmod{1}, \quad (10.4)$$

for some $\alpha \in \mathbb{R}/\mathbb{Z}$ and integers $c_{\mathbf{a}, r}$. The sum is finite modulo 1.

Proof. The implication from the displayed representation to weighted degree at most d follows from Lemma 6.71 and linearity.

For the converse we use a double induction: first on d , and for fixed d on the dimension m . The statement is trivial for $d < \min_i D_i$ after subtracting a constant, and the case $m = 0$ is vacuous. Let $m \geq 1$ and suppose f has weighted degree at most d .

By Lemma 6.69, $\Delta_{e_m} f$ has weighted degree at most $d - D_m$. The induction hypothesis in weighted degree gives an expansion

$$\Delta_{e_m} f(x) = \alpha + \sum_{\mathbf{a}, r} \frac{c_{\mathbf{a}, r}}{p^{r+1}} \prod_{i=1}^m \binom{x_i}{a_i}, \quad (10.5)$$

where every occurring term satisfies $\sum_i D_i a_i + r(p-1) \leq d - D_m$.

Integrate the nonconstant terms in the x_m direction by replacing $\binom{x_m}{a_m}$ with $\binom{x_m}{a_m+1}$. Pascal's identity

$$\binom{x_m + 1}{a_m + 1} - \binom{x_m}{a_m + 1} = \binom{x_m}{a_m}$$

shows that the resulting function g satisfies

$$\Delta_{e_m} g = \Delta_{e_m} f - \alpha.$$

Each integrated term has weighted degree at most d , because its weight has increased by exactly D_m . Hence

$$f(x) = \alpha x_m + f(x_1, \dots, x_{m-1}, 0) + g(x). \quad (10.6)$$

The middle term has weighted degree at most d on $\mathbb{Z}^{m-1}$ and is therefore of the required form by the induction hypothesis in dimension. The function g is of the required form by construction. It remains only to classify the linear term αx_m .

Let $j \geq 0$ be the least integer for which

$$D_m + j(p-1) > d.$$

Because αx_m is the difference of functions of weighted degree at most d , Lemma 6.69(ii) gives

$$0 = \Delta_{p^j e_m}(\alpha x_m) = p^j \alpha.$$

Thus $\alpha = c/p^j \pmod{1}$ for some integer c . The function

$$\frac{c}{p^j} \binom{x_m}{1}$$

is one of the terms allowed in (10.4), because minimality of j gives

$$D_m + (j-1)(p-1) \leq d$$

when $j \geq 1$; if $j = 0$, the relation $\alpha = 0$ makes the term disappear. This proves the classification. $\square$

Corollary 6.73 (Periodicity and the top weighted layer). *If f has weighted degree at most d , then:*

- (i) f is periodic with period $p^j e_i$ whenever $D_i + j(p-1) > d$;
- (ii) modulo a function of the residue coordinates $x_i \pmod{p^{j+1}}$ with $D_i + j(p-1) < d$, the degree- d part of f is an integer linear combination of the coordinate functions

$$x \mapsto \frac{x_i}{p^{j+1}} \pmod{1} \quad \text{with } D_i + j(p-1) = d.$$

Proof. Part (i) is already Lemma 6.69(ii). For (ii), inspect (10.4). A term lying on the top weighted layer and linear in a coordinate is exactly of the displayed form. Every remaining term either has strictly smaller weighted degree or is determined by residue classes at a strictly lower weighted level, by the periodicity just proved. Grouping those terms gives the claimed lower-layer function. $\square$

Theorem 6.74 (Exact roots for weighted polynomials on $\mathbb{Z}^m$). *If $f : \mathbb{Z}^m \rightarrow \mathbb{R}/\mathbb{Z}$ has weighted degree at most d , then there exists $g : \mathbb{Z}^m \rightarrow \mathbb{R}/\mathbb{Z}$ of weighted degree at most $d + p - 1$ such that*

$$pg = f. \quad (10.7)$$

Moreover g can be chosen so that, modulo a function of lower weighted residue coordinates, its top layer is an integer linear combination of

$$x \mapsto \frac{x_i}{p^{j+1}} \pmod{1} \quad \text{with } D_i + j(p-1) = d + p - 1.$$

Consequently g is periodic with period $p^{j+1} e_i$ whenever $D_i + j(p-1) > d$.

Proof. Use the classification theorem. Choose a p th root of the constant α . For each nonconstant term

$$\frac{c_{\mathbf{a},r}}{p^{r+1}} \prod_i \binom{x_i}{a_i}$$

replace the coefficient by $c_{\mathbf{a},r}/p^{r+2}$. Multiplication by p recovers the original term, while the weighted-degree bound increases from $\sum_i D_i a_i + r(p-1)$ to

$$\sum_i D_i a_i + (r+1)(p-1),$$

which is at most $d + p - 1$. Summing these roots proves (10.7). The top-layer and periodicity assertions follow from Corollary 6.73 applied to g . $\square$

Proposition 6.75 (Periodic exact-root form used by regular factors). *Suppose f has weighted degree at most d and is periodic with period $p^{J_i+1}e_i$ for every i . Then the root g in Theorem 6.74 may be chosen so that:*

$$(i) \quad g \text{ is periodic with period } p^{j+1}e_i \quad \text{whenever } D_i + j(p-1) > d; \quad (10.8)$$

$$(ii) \quad g \text{ is periodic with period } p^{J_i+2}e_i \text{ for every } i.$$

Consequently, if $J'_i > J_i$ for every i , then g is automatically periodic with period $p^{J'_i+1}e_i$ and therefore descends to the depth-extended quotient.

Proof. Part (i) is the generic periodicity statement in the explicit binomial root construction. Indeed, a term of the root with denominator one p -power deeper than a term of f can survive translation by $p^{j+1}e_i$ only when the corresponding weighted level of the original term is at most d .

For part (ii), use the same explicit binomial construction but choose the binomial expansion of f on the quotient $\mathbb{Z}^m / (p^{J_1+1}\mathbb{Z} \times \dots \times p^{J_m+1}\mathbb{Z})$. Every summand of that expansion is therefore determined by the corresponding residue coordinate modulo p^{J_i+1} . The root construction changes only the coefficient by one additional factor of p ; it does not increase any binomial exponent. Hence the i th coordinate period of each rooted summand is enlarged by at most one p -adic level, from p^{J_i+1} to p^{J_i+2} . Summing the rooted terms gives the asserted periodicity of g .

This is the extra piece of information used tersely in Tao–Ziegler’s proof of their technical exact-root theorem: the generic weighted-degree periodicity in (i) is not the whole story; one also carries forward the periodicity already present in the function being rooted. Finally $J'_i > J_i$ implies $J'_i + 1 \geq J_i + 2$, so $p^{J'_i+1}e_i$ is an integer multiple of the period just obtained. $\square$

Worked Example 6.76 (One variable and a positive-depth root). Let $m = 1$, $D_1 = 2$, and $p = 2$. The function

$$f(x) = \frac{1}{4} \binom{x}{2} \pmod{1}$$

has weighted degree at most $2 \cdot 2 + 1 = 5$. Dividing the coefficient by 2 gives

$$g(x) = \frac{1}{8} \binom{x}{2} \pmod{1}, \quad 2g = f,$$

and the weighted degree increases by exactly $p - 1 = 1$. The point is not the numerical example but the mechanism: taking a root changes only the depth denominator, while the binomial basis makes its degree cost transparent.

6.15 Positive-depth factors, cube equidistribution, and factor-wise exact roots

The weighted integer-lattice theorem from the preceding section is only useful for the finite-field inverse theorem after one proves that a polynomial which is measurable with respect to a regular positive-depth factor is represented by a weighted polynomial on that lattice. This is the purpose of Tao–Ziegler’s Section 9. The original paper packages the argument in filtered-group language. We spell out the abelian cube group and its Fourier analysis because this is where the positive-depth bookkeeping actually enters the proof.

Throughout this section V is a finite-dimensional vector space over $\mathbb{F}_p$. All quantitative thresholds are allowed to depend on the displayed degree, depth and complexity parameters, but not on $\dim V$.

6.15.1 Factors and depth extensions

Definition 6.77 (Positive-depth polynomial factor). A factor of dimension m , initial degrees $D_1, \dots, D_m \geq 2$, and depths $J_1, \dots, J_m \geq 0$ is a family

$$\mathcal{P} = (P_{i,j})_{1 \leq i \leq m, 0 \leq j \leq J_i}$$

with

$$P_{i,j} : V \longrightarrow \frac{1}{p^{j+1}} \mathbb{Z}/\mathbb{Z}, \quad \deg P_{i,j} \leq D_i + j(p-1), \quad pP_{i,j} = P_{i,j-1}, \quad (9.1)$$

where $P_{i,-1} := 0$. Its degree is

$$\deg \mathcal{P} := \max_i (D_i + J_i(p-1)).$$

The associated top-coordinate map is

$$\tilde{P} : V \rightarrow G_{\mathcal{P}} := \prod_{i=1}^m \frac{1}{p^{J_i+1}} \mathbb{Z}/\mathbb{Z}, \quad \tilde{P}(x) = (P_{1,J_1}(x), \dots, P_{m,J_m}(x)).$$

A function is $\mathcal{P}$ -measurable if it is a function of $\tilde{P}$.

Definition 6.78 (Quantitative regularity). Let $R \geq 1$. The factor $\mathcal{P}$ is R -regular if, for every weighted level $k \geq 2$, every integer combination

$$Q = \sum_{D_i + j(p-1) = k} c_{i,j} P_{i,j}$$

in which the coefficients are not all divisible by p satisfies

$$\text{rank}_{k-1}(Q) > R.$$

Each theorem that invokes regularity states the required lower bound on R as an explicit threshold function of its fixed parameters.

Definition 6.79 (Depth extension). A factor $\mathcal{P}' = (P_{i,j})_{j \leq J'_i}$ is a *depth extension* of $\mathcal{P}$ if $J'_i > J_i$ for every i and the two chains agree up to depth J_i . Such extensions exist by the elementary exact-root theorem (Theorem 6.13).

The exclusion of initial degree one is structural, not cosmetic. It prevents the endpoint $k = p$ in the multiplication-by- p rank transfer, exactly as in the original argument.

Lemma 6.80 (Depth extension preserves regularity quantitatively). *Fix s_0 and a target rank R . Assume the lower-degree GIP implications needed in Corollary 5.2 of the Tao–Ziegler induction, equivalently the rank-transfer consequence already isolated in TZ4. Then there is $R' = R'(p, s_0, R)$ with the following property: if $\mathcal{P}$ has degree at most s_0 and is R' -regular, then every bounded-depth extension $\mathcal{P}'$ of $\mathcal{P}$ is R -regular.*

Proof. It is enough to add one new depth at a time. Suppose a one-step extension fails R -regularity at weighted level k . Then

$$L = \sum_{D_i + j(p-1) = k} c_{i,j} P'_{i,j}$$

has rank at most R , with some coefficient of a newly adjoined coordinate P'_{i,J_i+1} nonzero modulo p ; otherwise the same bad relation already existed in $\mathcal{P}$.

Before doing so, reduce every coefficient modulo p . Replacing a coefficient c by $c - pq$ changes $cP'_{i,j}$ by $qP'_{i,j-1}$, a polynomial of weighted degree $k - p + 1 < k$ and hence of rank at most one at level $k - 1$. Thus a bad rank- R relation can be normalized in this way at the cost of a bounded change in the rank threshold. Since the old factor was regular, after this normalization at least one newly adjoined coordinate has coefficient nonzero modulo p .

Because $D_i \geq 2$, the presence of a new depth forces $k > p$. Multiply the relation by p . The chain relation (9.1) sends every new coordinate down one depth, and the rank-transfer theorem from TZ4, together with the lower-degree GIP induction, bounds the rank of pL at weighted level $k - p + 1$ by a function $C_{p,s_0}(R)$. After dividing all integer coefficients by their common power of p , we obtain a nontrivial relation among coordinates already present in $\mathcal{P}$. Thus $\mathcal{P}$ fails $C_{p,s_0}(R)$ -regularity. Choosing $R' > C_{p,s_0}(R)$ is a contradiction.

Iterating this one-step argument through the bounded number of added depths proves the claim, after composing the threshold function finitely many times. $\square$

6.15.2 The full positive-depth regularity lemma

The preceding depth-extension lemma is only one half of the regular-factor machinery. We also need to manufacture a regular factor from an arbitrary bounded list of polynomials. Tao–Ziegler do this with the asymptotic language “bounded rank”. In a finite book one must make the hierarchy of rank thresholds explicit. The next elementary observation is the bookkeeping device that prevents a circular choice of constants.

Lemma 6.81 (Finite hierarchy of loss parameters). *Fix an integer $N \geq 0$, a starting threshold $R \geq 1$, and finitely many nondecreasing functions $L_1, \dots, L_q : \mathbb{N} \rightarrow \mathbb{N}$. There are integers*

$$R = H_0 < H_1 < \dots < H_N$$

such that, whenever $j < N$ and $1 \leq a \leq q$,

$$L_a(H_j) < H_{j+1}.$$

Proof. Having chosen H_j , take H_{j+1} larger than H_j and all the finitely many numbers $L_a(H_j)$. Repeat N times. $\square$

The point is that every time the top-degree presentation has to be changed, at least one top generator disappears. Hence only a bounded number of losses can occur, and a finite hierarchy suffices; no fixed point of a complexity function is being assumed.

Theorem 6.82 (Positive-depth polynomial-factor regularization). *Fix p, s_0, m and a target regularity threshold R . Assume, for every $t < s_0$,*

- (a) the finite GIP implication at degree t ;
- (b) the factorwise exact-root conclusion $ER(t)$: a polynomial of degree at most $t + 1$ which is measurable with respect to a regular factor has a p th root of degree at most $t + p$ measurable with respect to every depth extension in which each generator (D_i, J_i) is extended to (D_i, J'_i) with $D_i + J'_i(p - 1) > t$, such that, modulo a function of lower weighted levels, the root's top weighted part is an integer linear combination of factor coordinates of weighted level $t + p$;
- (c) the present regularization theorem has already been proved with s_0 replaced by t .

Then there is a constant

$$C = C(p, s_0, m, R)$$

with the following property. For every finite-dimensional $\mathbb{F}_p$ -vector space V and every list

$$P_1, \dots, P_m \in \text{Poly}_{\leq s_0}(V \rightarrow \mathbb{R}/\mathbb{Z})$$

there are a subspace $U \leq V$ of codimension at most C and an R -regular positive-depth factor $\mathcal{Q}$ on U , of degree at most s_0 and complexity at most C , such that every $P_a|_U$ is $\mathcal{Q}$ -measurable.

More precisely, if $d_a = \deg(P_a|_U)$, then

$$P_a|_U = \sum_{D_i + j(p-1) = d_a} n_{a,i,j} Q_{i,j} + F_a(\mathcal{Q}_{\leq d_a-1}) \quad (9.R)$$

for integers $n_{a,i,j}$ and some function F_a of the depth retraction below d_a .

Proof. We argue by induction on s_0 . When $s_0 \leq 1$, subtract constants and intersect the kernels of the finitely many resulting linear forms. The restrictions are constant, so the empty factor works. Assume now $s_0 \geq 2$ and that the theorem is known below s_0 .

Step 1: a finite top-degree decomposition. At weighted degree s_0 , multiplication by p lowers degree to at most $s_0 - p + 1 < s_0$. Consequently coefficients of top classes may be reduced modulo p once lower-degree errors are allowed. We perform the following finite elimination procedure, with rank thresholds chosen from Lemma 6.81. The hierarchy is taken long enough for at most $2m + 2$ top-generator eliminations and for all codimension, rank-transfer and lower-factor complexity losses occurring below.

Choose a maximal list

$$A_1, \dots, A_r \in \text{Poly}_{\leq s_0}(V \rightarrow \mathbb{R}/\mathbb{Z}), \quad r \leq m,$$

in the $\mathbb{F}_p$ -span of the degree- s_0 parts of the P_a with the property that no nonzero combination of the A_l is the sum of a classical degree- s_0 polynomial and a polynomial of rank below the current hierarchy threshold. Maximality implies that every top part of every P_a can be written as

$$\sum_l c_{a,l} A_l + C_a + S_a, \quad (9.R1)$$

where C_a is classical of degree at most s_0 and S_a has bounded rank $s_0 - 1$. Indeed, if adjoining a missing top part destroys independence, the resulting relation has a nonzero coefficient on that top part and can be solved for it.

Apply the same maximality argument inside the finite span of the classical remainders C_a . We obtain classical polynomials

$$B_1, \dots, B_q, \quad q \leq m,$$

such that every C_a is a linear combination of the B_b plus another bounded-rank polynomial, while every nonzero $\mathbb{F}_p$ -combination of the B_b has rank above the current threshold. Absorbing these new bounded-rank pieces into S_a , we have reduced all the original top-degree data to

$$A_1, \dots, A_r, \quad B_1, \dots, B_q,$$

plus bounded-rank errors. By definition of rank, each such error is a function of a bounded list of polynomials of degree at most $s_0 - 1$.

If a later bounded-codimension restriction creates a new low-rank relation among the surviving A_l modulo classical polynomials, eliminate one A_l using that relation and add its lower-degree witnesses to the lower-degree data. Likewise, a new low-rank relation among the B_b eliminates one B_b . Each repair strictly decreases $r + q$, so there are at most $2m$ repairs. The finite hierarchy guarantees that after the last repair the surviving top generators are still R_* -independent on the final subspace, where R_* is chosen larger than every top-level loss used below. This is the finitary substitute for the paper's quotient by the vector space of bounded-rank polynomials. Lemma 6.23 is exactly what converts a low-rank relation after restriction into the corresponding controlled loss one level up the hierarchy.

Step 2: regularize all lower-degree data simultaneously. For each surviving nonclassical top generator A_l , the polynomial pA_l has degree at most $\max(s_0 - p + 1, 0)$. Collect

- every lower-degree polynomial occurring in the original P_a ;
- all lower-degree witnesses for the bounded-rank errors in (9.R1);
- the polynomials $pA_1, \dots, pA_r$.

The number of collected polynomials is bounded in terms of the fixed parameters and the finite hierarchy. Apply the induction hypothesis at degree $s_0 - 1$ (padding lower levels if necessary). After passing to a bounded-codimension subspace U , all these polynomials are measurable with respect to one highly regular factor

$$\mathcal{Q} = (Q_{i,j})$$

of degree at most $s_0 - 1$. We choose its regularity threshold large enough, through the finite hierarchy, to survive the finitely many depth extensions and rank-transfer losses below.

Use elementary exact roots to extend every chain until

$$s_0 - p < D_i + J_i(p - 1) \leq s_0 - 1. \quad (9.R2)$$

Then extend each chain one further step, writing the resulting factor as $\mathcal{Q}^+$. Some of these extra roots can have weighted degree larger than s_0 ; they are temporary coordinates used only to invoke the lower exact-root property. The depth-extension lemma (Lemma 6.80) keeps $\mathcal{Q}^+$ highly regular, and its depth retraction $\mathcal{Q}_{\leq s_0}^+$ is therefore highly regular as well. For every newly adjoined coordinate Q_{i,J_i+1} we retain the freedom to replace it by

$$Q_{i,J_i+1} + H_i,$$

where H_i is an arbitrary classical polynomial of the same degree, because $pH_i = 0$ and hence the chain relation is unchanged.

Step 3: lift the nonclassical top generators. Each pA_l is measurable with respect to $\mathcal{Q}$ and has degree at most $s_0 - p + 1$. Apply the lower-stage exact-root property $\text{ER}(s_0 - p)$ when $s_0 \geq p$ (the assertion is elementary when $s_0 < p$). We obtain a polynomial U_l of degree at most s_0 , measurable with respect to $\mathcal{Q}^+$, satisfying

$$pU_l = pA_l. \quad (9.R3)$$

By the top-layer clause in $\text{ER}(s_0 - p)$,

$$U_l = \sum_{i \in I} c_{l,i} Q_{i,J_i+1} + E_l, \quad (9.R4)$$

where I consists of those indices with $D_i + (J_i + 1)(p - 1) = s_0$, and E_l is a function of weighted levels below s_0 . In particular U_l is already measurable with respect to the retraction $\mathcal{Q}_{\leq s_0}^+$; the temporary roots above level s_0 can now be discarded. Equation (9.R3) says that $A_l - U_l$ is classical.

The row vectors

$$\bar{c}_l = (c_{l,i} \bmod p)_{i \in I}$$

are linearly independent over $\mathbb{F}_p$. If $\sum_l a_l \bar{c}_l = 0$ with not all a_l zero, then (9.R4) makes $\sum_l a_l U_l$ a function of the lower factor. Since $A_l - U_l$ is classical, the same combination of the A_l is classical plus a polynomial of rank bounded by the complexity of the lower factor, contradicting the stable top independence from Step 1. The hierarchy was chosen so that this complexity is below the relevant independence threshold.

Because the matrix $(\bar{c}_{l,i})$ has full row rank, the map

$$(H_i)_{i \in I} \mapsto \left(\sum_{i \in I} c_{l,i} H_i \right)_{1 \leq l \leq r}$$

from families of classical degree- s_0 polynomials to r -tuples of such polynomials is surjective. Use the freedom in the new roots Q_{i,J_i+1} to choose H_i so that the classical discrepancy $A_l - U_l$ is cancelled for every l . After this harmless modification,

$$A_l = U_l \quad (1 \leq l \leq r). \quad (9.R5)$$

Thus every nonclassical top generator is now measurable with respect to the extended factor.

Step 4: adjoin the independent classical top generators and verify regularity.

Start from the retraction $\mathcal{Q}_{\leq s_0}^+$ and adjoin $B_1, \dots, B_q$ as depth-zero coordinates of initial degree s_0 ; call the resulting factor $\mathcal{Q}^{\text{fin}}$. Every original $P_a|_U$ is measurable with respect to it by Steps 1–3.

Levels below s_0 remain R -regular because they come from the highly regular lower factor and its controlled depth extensions. Consider a nontrivial level- s_0 combination of the new coordinates of rank at most R . First reduce the coefficients of all positive-depth coordinates modulo p . As in the proof of Lemma 6.80, subtracting a multiple of p replaces a new root by its predecessor and therefore changes the relation only by a polynomial of weighted degree at most $s_0 - p + 1 < s_0$; this costs only a controlled amount of rank_{s_0-1} .

If, after this normalization, a depth-extended coordinate Q_{i,J_i+1} occurs with coefficient nonzero modulo p , multiply the relation by p . The classical B_b disappear, the new roots descend to Q_{i,J_i} , and the rank-transfer consequence of TZ4 turns the assumed rank bound into a controlled low-rank relation in the old factor. This contradicts the chosen lower-factor regularity. If no new root remains, the normalized relation is a nontrivial combination of the B_b plus a lower-degree polynomial. Subtracting that one lower-degree coordinate changes rank by at most one, so the stable independence of the B_b from Step 1 gives the contradiction. Hence the final factor is R -regular.

Finally, every operation in Steps 1–4 respects weighted degree. Tracing the top-degree terms in (9.R1), (9.R4), and (9.R5) gives exactly (9.R). The number of eliminations, witnesses, depth extensions and codimension losses is bounded by the finite hierarchy, so all of them are absorbed into one constant $C(p, s_0, m, R)$. This completes the induction. $\square$

6.15.3 The explicit cube group of a filtered factor

For $r \geq 0$ define a descending filtration on $G_{\mathcal{P}}$ by letting $(G_{\mathcal{P}})_r$ be the subgroup generated by

$$p^{-(J_i-j+1)}e_i \quad (1 \leq i \leq m, \ 0 \leq j \leq J_i, \ r \leq D_i + j(p-1)). \quad (9.2)$$

Thus the deeper a coordinate is, the longer it survives under repeated differences.

For $k \geq 0$ and a configuration $t = (t_{\omega})_{\omega \in \{0,1\}^k} \in G_{\mathcal{P}}^{\{0,1\}^k}$ define its *Taylor coefficient* on $S \subseteq [k]$ by

$$a_S(t) := \sum_{T \subseteq S} (-1)^{|S|-|T|} t_{1_T}, \quad (9.3)$$

where 1_T is the vertex with support T .

Definition 6.83 (Filtered cube group). Set

$$C^k(G_{\mathcal{P},\bullet}) := \left\{ t : a_S(t) \in (G_{\mathcal{P}})_{|S|} \text{ for every } S \subseteq [k] \right\}.$$

Lemma 6.84 (Taylor coordinates parametrize the cube group). *The map*

$$t \mapsto (a_S(t))_{S \subseteq [k]}$$

is an isomorphism of finite abelian groups from $C^k(G_{\mathcal{P},\bullet})$ onto $\prod_{S \subseteq [k]} (G_{\mathcal{P}})_{|S|}$. Its inverse is

$$t_{\omega} = \sum_{S \subseteq \text{supp}(\omega)} a_S. \quad (9.4)$$

Proof. Equations (9.3) and (9.4) are Möbius inversion on the Boolean lattice. They are mutually inverse homomorphisms. The membership conditions defining the cube group become exactly $a_S \in (G_{\mathcal{P}})_{|S|}$. $\square$

Lemma 6.85 (The factor map sends affine cubes to filtered cubes). *For every $x, h_1, \dots, h_k \in V$ the configuration*

$$t_{\omega} = \tilde{P}(x + \omega_1 h_1 + \dots + \omega_k h_k)$$

belongs to $C^k(G_{\mathcal{P},\bullet})$.

Proof. The i th coordinate of $a_S(t)$ is the $|S|$ -fold additive derivative of P_{i,J_i} . If

$$D_i + j(p-1) < |S| \leq D_i + (j+1)(p-1),$$

then $P_{i,j}$ has vanishing $|S|$ -fold derivative. Since $p^{J_i-j}P_{i,J_i} = P_{i,j}$, the derivative of P_{i,J_i} lies in the subgroup generated by $p^{-(J_i-j+1)}$. This is exactly the i th coordinate condition in (9.2). $\square$

6.15.4 Fourier proof of cube equidistribution

We now reconstruct the load-bearing part of Tao–Ziegler Lemma 9.8. The finite form below is what the later factor-descent argument actually uses.

Lemma 6.86 (Dual coordinate reduction at a cube level). *Let χ be a nontrivial character of $(G_{\mathcal{P}})_K$. Then there are integers a_i , not all divisible by p , and depths $0 \leq j_i \leq J_i$ such that*

$$\chi(\tilde{P}(x)) = e \left(\sum_i a_i P_{i,j_i}(x) \right)$$

whenever the character is evaluated on a K th Taylor coefficient, and every occurring weighted level satisfies

$$D_i + j_i(p-1) \geq K. \quad (9.5)$$

At least one occurring coefficient lies on the maximal weighted level $D := \max_i (D_i + j_i(p-1))$.

Proof. The i th coordinate of $(G_{\mathcal{P}})_K$ is cyclic. Let j_i be the least depth for which $K \leq D_i + j_i(p-1)$ and for which the i th component of χ is nontrivial after restriction to that cyclic subgroup. Writing the character coefficient in p -adic valuation form and using $p^{J_i - j_i} P_{i, J_i} = P_{i, j_i}$ gives the displayed expression with coefficient prime to p . Coordinates on which the character is trivial are omitted. Nontriviality of χ guarantees that at least one coordinate remains. $\square$

Theorem 6.87 (Cube equidistribution of a sufficiently regular factor). *Fix $s_0, k, m, (D_i), (J_i)$ and $\varepsilon > 0$. Assume $\text{GIP}(r)$ for all $r < s_0$. There exists*

$$R_0 = R_0(p, s_0, k, m, (D_i), (J_i), \varepsilon)$$

such that, if $\mathcal{P}$ has degree at most s_0 and is R_0 -regular, then the random filtered cube

$$(\tilde{P}(x + \omega \cdot h))_{\omega \in \{0,1\}^k}$$

with $x, h_1, \dots, h_k$ uniform in V is ε -close to the uniform distribution on $C^k(G_{\mathcal{P}, \bullet})$ in total variation.

Proof. Because the target cube group is finite with size bounded in terms of the displayed parameters, Fourier inversion reduces the claim to the following: every nontrivial character η of $C^k(G_{\mathcal{P}, \bullet})$ has expectation at most a prescribed $\delta = o_\varepsilon(1)$ on the factor cubes.

Use Lemma 6.84 to write

$$\eta(t) = \prod_{S \subseteq [k]} \chi_S(a_S(t)),$$

where each χ_S is a character of $(G_{\mathcal{P}})_{|S|}$ and at least one is nontrivial. Choose S_* of maximal cardinality K for which χ_{S_*} is nontrivial, and relabel the cube variables so that $S_* = [K]$. By Lemma 6.86, the contribution of this top Taylor coefficient is

$$e(\Delta_{h_1} \cdots \Delta_{h_K} Q(x)), \quad Q = \sum_{i \in I} a_i P_{i, j_i}, \quad (9.6)$$

where every occurring level is at least K . Let

$$D = \max_{i \in I} (D_i + j_i(p-1)).$$

The degree- D portion of Q is a nontrivial combination of factor coordinates on weighted level D , hence R_0 -regularity gives

$$\text{rank}_{D-1}(Q) > R_0 - O_{p, s_0, m}(1). \quad (9.7)$$

The lower-weight summands of Q contribute only bounded rank when Q is regarded as a degree- D polynomial, so they are absorbed into the harmless error in (9.7).

By the contrapositive quantitative form of $\text{GIP}(D-1)$, choosing R_0 sufficiently large forces

$$\|e(Q)\|_{U^D(V)} < \delta. \quad (9.8)$$

If $K = 0$, the chosen character occurs only on the base Taylor coefficient. All other nontrivial Taylor characters are absent by maximality, so the character average is $|\mathbb{E}_x e(Q(x))|$. Since $D \geq 1$, repeated Cauchy–Schwarz (equivalently the $U^1 \leq U^D$ monotonicity inequality) bounds this by $\|e(Q)\|_{U^D} < \delta$.

Suppose now that $K \geq 1$. Since $K \leq D$, monotonicity gives $\|e(Q)\|_{U^K} < \delta$. It remains to see that every other character factor is harmless under the K Cauchy–Schwarz operations. If $S \neq S_*$ and χ_S is nontrivial, maximality of $|S_*|$ implies that S omits at least one variable h_r

with $r \in S_*$. Thus its contribution is independent of that variable after the Taylor-coordinate expansion. Applying the generalized Cauchy–Schwarz inequality successively in $h_1, \dots, h_K$ removes all such factors and bounds the original character average by $\|e(Q)\|_{U^K}$. This is at most δ by (9.8).

Finally choose δ smaller than ε divided by the number of characters of the finite cube group. Fourier inversion then gives the asserted total-variation bound. $\square$

Worked Example 6.88 (What the cube group remembers). Take one chain with $D = 2$, depth $J = 1$, and $p = 2$. The top coordinate takes values in $\frac{1}{4}\mathbb{Z}/\mathbb{Z}$ and has degree at most 3. On a two-dimensional cube, the second Taylor coefficient may still be any element of $\frac{1}{4}\mathbb{Z}/\mathbb{Z}$; on a three-dimensional cube, multiplication by 2 kills the third derivative, so the third Taylor coefficient is constrained to $\frac{1}{2}\mathbb{Z}/\mathbb{Z}$; on a four-dimensional cube it vanishes. This is precisely the filtration (9.2). Positive depth is therefore visible as extra p -power torsion in the low-order Taylor coefficients, not as an extra formal label.

6.15.5 Polynomial functions on a regular factor

Lemma 6.89 (Cube criterion for a function on the filtered factor). *Let $F : G_{\mathcal{P}} \rightarrow \mathbb{R}/\mathbb{Z}$. The following are equivalent:*

(i) *for every $(d+1)$ -cube $t \in C^{d+1}(G_{\mathcal{P}}, \bullet)$,*

$$\sum_{\omega \in \{0,1\}^{d+1}} (-1)^{|\omega|} F(t_{\omega}) = 0; \quad (9.9)$$

(ii) *F is a polynomial map of filtered degree at most d .*

Proof. The implication (ii) $\Rightarrow$ (i) is the definition of filtered degree expressed on an affine cube. Conversely, fix a filtered increment $u \in (G_{\mathcal{P}})_r$. Insert u as one edge of a formal cube in (9.9). The resulting identity says that $\Delta_u F$ satisfies the same cube criterion with degree lowered by r . Induction on d , beginning with constants, therefore gives the filtered finite-difference definition of degree at most d . $\square$

Theorem 6.90 (Polynomial descent to a regular factor). *Fix $p, s_0, d \geq 0$ and a factor-complexity bound $m \geq 1$. Assume $GIP(r)$ for all $r < s_0$. There is a rank threshold $R_0 = R_0(p, s_0, d, m)$ such that, if $\mathcal{P}$ is an R_0 -regular polynomial factor of degree at most s_0 and complexity at most m , and $F : G_{\mathcal{P}} \rightarrow \mathbb{R}/\mathbb{Z}$, then*

$$Q := F(\tilde{P}) : V \rightarrow \mathbb{R}/\mathbb{Z}$$

is a polynomial of degree at most d if and only if F is a filtered polynomial of degree at most d on $G_{\mathcal{P}}$.

Proof. If F is filtered polynomial, Lemma 6.85 and the finite-difference chain rule give that $F \circ \tilde{P}$ has ordinary degree at most d .

Conversely suppose $Q = F(\tilde{P})$ has degree at most d . Let $t \in C^{d+1}(G_{\mathcal{P}}, \bullet)$ be arbitrary. Apply Theorem 6.87 with error smaller than the reciprocal of twice the size of the finite cube group. The point mass at t is then positive, so there exist $x, h_1, \dots, h_{d+1}$ whose factor cube is exactly t . Since Q has degree at most d ,

$$0 = \Delta_{h_1} \cdots \Delta_{h_{d+1}} Q(x) = \sum_{\omega} (-1)^{|\omega|} F(t_{\omega}).$$

Thus (9.9) holds for every formal cube, and Lemma 6.89 completes the proof. $\square$

Define the filtered homomorphism

$$\phi : \mathbb{Z}^m \rightarrow G_{\mathcal{P}}, \quad \phi(a_1, \dots, a_m) = \left(\frac{a_1}{p^{J_1+1}}, \dots, \frac{a_m}{p^{J_m+1}} \right) \pmod{1}. \quad (9.10)$$

The filtration on $\mathbb{Z}^m$ is exactly the weighted filtration of the preceding section.

Corollary 6.91 (Weighted-lattice representation of a factor-measurable polynomial). *Under the hypotheses of Theorem 6.90, a function $Q : V \rightarrow \mathbb{R}/\mathbb{Z}$ is $\mathcal{P}$ -measurable and of degree at most d if and only if there exists a weighted polynomial $f : \mathbb{Z}^m \rightarrow \mathbb{R}/\mathbb{Z}$ of weighted degree at most d , periodic with period $p^{J_i+1}e_i$ in the i th coordinate, such that*

$$Q(x) = f(a_1, \dots, a_m) \quad (9.11)$$

whenever

$$P_{i,J_i}(x) = \frac{a_i}{p^{J_i+1}} \pmod{1}.$$

Proof. If $Q = F(\tilde{P})$, Theorem 6.90 makes F a filtered polynomial. Set $f = F \circ \phi$. Because ϕ is a filtered homomorphism, f has the required weighted degree, and its periods are immediate from (9.10). This proves one direction.

Conversely, periodicity lets f descend to a function F on $G_{\mathcal{P}}$. The weighted finite-difference condition says exactly that F satisfies the filtered cube criterion, hence F is filtered polynomial and $F(\tilde{P})$ has degree at most d . $\square$

6.15.6 Technical exact roots on a regular factor

Theorem 6.92 (Factorwise exact roots). *Fix $s_0 \geq 1$ and an integer $m \geq 1$. Assume $GIP(r)$ for every $r \leq s_0$ and the lower-stage exact-root induction needed to construct the regular factor. There is a threshold $R_{\text{root}} = R_{\text{root}}(p, s_0, m)$ with the following property. Let $\mathcal{P}$ be an R_{root} -regular factor of degree at most s_0 with at most m generators, and let*

$$Q \in \text{Poly}_{\leq s_0+1}(V \rightarrow \mathbb{R}/\mathbb{Z})$$

be $\mathcal{P}$ -measurable. Let $\mathcal{P}'$ be a depth extension of $\mathcal{P}$ satisfying

$$D_i + J'_i(p-1) > s_0 \quad \text{for every } i.$$

Then there exists

$$R \in \text{Poly}_{\leq s_0+p}(V \rightarrow \mathbb{R}/\mathbb{Z})$$

which is $\mathcal{P}'$ -measurable and satisfies $pR = Q$. Moreover its top weighted part is an integer linear combination of factor coordinates of weighted level $s_0 + p$, plus a function of lower weighted levels.

Proof. By Corollary 6.91, write Q as a weighted polynomial $f : \mathbb{Z}^m \rightarrow \mathbb{R}/\mathbb{Z}$ of weighted degree at most $d = s_0 + 1$, periodic with the original periods $p^{J_i+1}e_i$. Apply Theorem 6.74 to obtain a root g of weighted degree at most $s_0 + p$. Proposition 6.75 carries the old periods one p -adic level deeper. Because a depth extension satisfies $J'_i > J_i$, the function g is therefore periodic with every period $p^{J'_i+1}e_i$.

Hence g descends through the extended map

$$\phi' : \mathbb{Z}^m \rightarrow \prod_i \frac{1}{p^{J'_i+1}} \mathbb{Z}/\mathbb{Z}.$$

Using the reverse direction of Corollary 6.91 for the regular depth extension (the latter is regular by Lemma 6.80), the descended function gives a polynomial R of degree at most $s_0 + p$ which is measurable with respect to $\mathcal{P}'$. Multiplication by p recovers f and hence Q . The top-layer assertion is exactly the top-layer assertion in Theorem 6.74 translated through the factor coordinates. $\square$

6.16 The simultaneous GIP induction

The preceding sections have now proved the three mechanisms that Tao–Ziegler isolate before returning to the polynomial inverse theorem: multiplication by p , integration of a biased classical symmetric multilinear form, and exact roots compatible with regular positive-depth factors. We can therefore run the induction which the original paper places in Section 4. We keep the bounded-codimension passages visible, because hiding them is one of the easiest ways to make this argument circular.

6.16.1 From factorwise exact roots to exact roots of bounded-rank polynomials

Theorem 6.93 (Bounded-rank exact roots). *Fix $t \geq 0$ and a rank bound M . Assume $\text{GIP}(r)$ and the factorwise exact-root property $\text{ER}(r)$ for every $r < t$, and assume $\text{GIP}(t)$. Then there is $C = C(p, t, M)$ such that the following holds.*

If

$$S \in \text{Poly}_{\leq t+1}(V \rightarrow \mathbb{R}/\mathbb{Z}), \quad \text{rank}_t(S) \leq M,$$

then there is a subspace $U \leq V$ of codimension at most C and a polynomial

$$Q \in \text{Poly}_{\leq t+p}(U \rightarrow \mathbb{R}/\mathbb{Z})$$

with $pQ = S|_U$ and $\text{rank}_{t+p-1}(Q) \leq C$.

Proof. Choose polynomials $R_1, \dots, R_M$ of degree at most t which determine S . Apply Theorem 6.82 at degree t to the tuple $(R_1, \dots, R_M)$. After restricting to a subspace U of bounded codimension, they are all measurable with respect to a regular factor $\mathcal{R}$ of bounded complexity and degree at most t . Hence $S|_U$ is $\mathcal{R}$ -measurable.

Extend every depth of $\mathcal{R}$ until

$$t < D_i + J'_i(p-1) \leq t + p - 1.$$

The extension remains regular by Lemma 6.80. The factorwise exact-root theorem (Theorem 6.92, with $s_0 = t$) now gives a root Q of degree at most $t + p$ which is measurable with respect to the extended factor. Since the extended factor has bounded complexity and degree at most $t + p - 1$, its top coordinates themselves form a bounded list of polynomials of degree at most $t + p - 1$; therefore Q has bounded algebraic rank at level $t + p - 1$. All bounds depend only on the displayed parameters. $\square$

6.16.2 The induction step

Lemma 6.94 (The quadratic base of GIP). *GIP(0) and GIP(1) hold quantitatively over every fixed $\mathbb{F}_p$.*

Proof. Degree at most one is immediate. For degree at most two, put $B(h, k) = \Delta_h \Delta_k P$. This is a bilinear form with values in $\frac{1}{p}\mathbb{Z}/\mathbb{Z}$ after identifying that group with $\mathbb{F}_p$. For fixed h , the average over k of $e(B(h, k))$ equals 1 precisely when $B(h, \cdot) = 0$ and is 0 otherwise. Hence

$$\text{bias}(B) = \frac{|\ker(h \mapsto B(h, \cdot))|}{|V|} = p^{-r},$$

where r is the matrix rank of B . Thus bounded analytic rank gives bounded r . Choose linear coordinates $\ell_1, \dots, \ell_r$ whose joint kernel is the radical of B . On that radical the quadratic part of P vanishes, so P restricts there to a linear polynomial. Adding one further linear coordinate for this residual linear part, we see that P is a function of at most $r + 1$ linear polynomials. This is a quantitative bound for $\text{rank}_1(P)$. $\square$

Theorem 6.95 (GIP induction step; finite Tao–Ziegler form). *Fix $s \geq 2$ and $A \geq 0$. Assume GIP(r) for every $r < s$, the multiplication-by- p theorem of TZ4, the exact-root machinery of Sections 6.15 and Theorem 6.93, and the classical symmetric-multilinear inverse theorem Theorem 6.66. Then there is $C = C(p, s, A)$ such that*

$$P \in \text{Poly}_{\leq s+1}(V \rightarrow \mathbb{R}/\mathbb{Z}), \quad \text{arank}_s(P) \leq A \implies \text{rank}_s(P) \leq C.$$

Thus GIP(s) follows from the lower-degree instances.

Proof. Write $k = s + 1$. We separate the low-characteristic case $k > p$ from the endpoint/high characteristic case $k \leq p$.

Case 1: $s + 1 > p$. By Theorem 6.26, bounded analytic rank of P implies a bound, depending only on p, s, A , for the analytic rank of

$$pP \in \text{Poly}_{\leq s-p+2}(V \rightarrow \mathbb{R}/\mathbb{Z}).$$

Apply GIP($s - p + 1$) (which is a lower-degree instance) to obtain

$$\text{rank}_{s-p+1}(pP) \leq M(p, s, A). \quad (\text{G1})$$

The bounded-rank exact-root theorem with $t = s - p + 1$ then supplies a bounded-codimension subspace $U \leq V$ and

$$Q \in \text{Poly}_{\leq s+1}(U \rightarrow \mathbb{R}/\mathbb{Z})$$

of bounded rank_s such that

$$pQ = pP|_U. \quad (\text{G2})$$

Consequently $P|_U - Q$ takes values in $\frac{1}{p}\mathbb{Z}/\mathbb{Z}$. After the fixed identification $\iota : \mathbb{F}_p \rightarrow \frac{1}{p}\mathbb{Z}/\mathbb{Z}$, write

$$P|_U - Q = \iota(R) \quad (\text{G3})$$

for a classical polynomial $R : U \rightarrow \mathbb{F}_p$ of degree at most $s + 1$.

The bounded-codimension analytic-rank lemma (Lemma 6.24) keeps $P|_U$ of bounded analytic rank; Q has bounded analytic rank by Proposition 6.22; and analytic-rank subadditivity therefore gives bounded analytic rank for the residual $\iota(R)$. Equivalently, the symmetric multilinear form

$$T = d^{s+1}R : U^{s+1} \rightarrow \mathbb{F}_p$$

has bias bounded below by a constant depending only on p, s, A .

Apply Theorem 6.66. There is a further bounded-codimension subspace $U' \leq U$ and a bounded-rank classical polynomial $W : U' \rightarrow \mathbb{F}_p$ of degree at most $s + 1$ such that

$$d^{s+1}W = T|_{(U')^{s+1}}. \quad (\text{G4})$$

Hence

$$P|_{U'} - Q|_{U'} - \iota(W)$$

has vanishing $(s + 1)$ st derivative and therefore degree at most s . It contributes at most one additional coordinate to $\text{rank}_s(P|_{U'})$. Since both Q and W have bounded rank, $P|_{U'}$ has bounded rank. Finally Lemma 6.23 lifts that bound from U' back to V .

Case 2: $s + 1 \leq p$. Lemma 1.7(v), proved in TZ1–TZ2, gives

$$\deg(pP) \leq \max(s - p + 2, 0) \leq 1.$$

After passing to the kernel of its linear part, a subspace of codimension at most one, pP is constant. Choose a constant Q with $pQ = pP$ there. The residual $P - Q$ is classical, and the preceding classical-top-derivative argument applies verbatim: bounded analytic rank passes to the subspace, Theorem 6.66 integrates the biased top derivative, and the remaining error has degree at most s . Bounded-codimension rank stability returns the conclusion to V . $\square$

Lemma 6.96 (The ER(0) base). *The exact-root property ER(0) holds.*

Proof. A degree-one polynomial has the form

$$Q(x) = \alpha + \iota(\ell(x))$$

for a linear form $\ell : V \rightarrow \mathbb{F}_p$. Choose $\beta \in \mathbb{R}/\mathbb{Z}$ with $p\beta = \alpha$. The coordinate normal form from TZ1 shows directly that

$$R(x) = \beta + \frac{|\ell(x)|}{p^2} \pmod{1}$$

is a polynomial of degree at most p and satisfies $pR = Q$. It is a function of Q itself (choose the representative $|\ell(x)| \in \{0, \dots, p-1\}$ from the value of $Q - \alpha$), hence it is measurable with respect to every factor with respect to which Q is measurable, and therefore with respect to every depth extension. This is ER(0). $\square$

Theorem 6.97 (Simultaneous GIP–ER induction). *For every fixed $s \geq 0$, the finite quantitative properties GIP(s) and ER(s) hold. More precisely, all rank, codimension, factor-complexity and root bounds depend only on p, s and the relevant input rank/analytic-rank bounds, never on $\dim V$.*

Proof. We use strong induction on s . The GIP bases $s = 0, 1$ are Lemma 6.94, and ER(0) is Lemma 6.96. Suppose now that all GIP(r) and ER(r) with $r < s$ have been proved.

First apply Theorem 6.95. Every exact-root input used there occurs at a strictly smaller degree (in the low-characteristic case it is applied to pP , at level $s - p + 1$), so the induction hypothesis supplies it. Hence GIP(s) holds.

Now prove ER(s). Let $\mathcal{P}$ be a regular factor, let $Q \in \text{Poly}_{\leq s+1}$ be $\mathcal{P}$ -measurable, and let $\mathcal{P}'$ be a depth extension satisfying $D_i + J'_i(p-1) > s$ for every i . The positive-depth regularity and cube machinery in Section 6.15 require only GIP(r) and ER(r) below s , except for the cube-equidistribution step, which may use the just-proved GIP(s). Thus Theorem 6.92 applies and gives a $\mathcal{P}'$ -measurable root of degree at most $s + p$ with the required top-layer form. This is exactly ER(s).

The induction closes, proving both families simultaneously. $\square$

Corollary 6.98 (Finite polynomial GIP). *For every fixed s and A there is $C = C(p, s, A)$ such that*

$$\text{arank}_s(P) \leq A \implies \text{rank}_s(P) \leq C$$

for every finite-dimensional $\mathbb{F}_p$ -vector space and every $P \in \text{Poly}_{\leq s+1}(V \rightarrow \mathbb{R}/\mathbb{Z})$.

6.17 The Bergelson–Tao–Ziegler ergodic inverse theorem

The remaining input in the weak inverse theorem is ergodic rather than finitary. The primary source is Bergelson–Tao–Ziegler [1]. Its proof is not a single compactness lemma: it passes through the Host–Kra characteristic factor, abelian cohomology, abelian extensions, functions and cocycles of finite type, Conze–Lesigne equations, reduction of the vertical group to a finite group, and a finite-group integration argument. We keep those joints visible.

Definitions needed for the headline theorem

Definition 6.99 (*G*-systems and ergodicity). Let G be a countable abelian group. A *measure-preserving G-system* is a probability space (X, μ) together with measurable invertible maps $(T_g)_{g \in G}$ such that $T_{g+h} = T_g T_h$, $T_0 = \text{id}$, and $\mu(T_g^{-1}E) = \mu(E)$ for every measurable E . The system is *ergodic* if every measurable set E satisfying $T_g^{-1}E = E$ for all g has measure 0 or 1.

Lemma 6.100 (Mean ergodic projection for the finite-subgroup exhaustion). *Let $H_1 \leq H_2 \leq \dots$ be finite subgroups with union G , and let (T_g) be a measure-preserving action on (X, μ) . For $F \in L^2(X)$ put*

$$A_M F := \mathbb{E}_{h \in H_M} T_h F.$$

Then A_M is the orthogonal projection onto the closed subspace V_M of H_M -invariant functions, and $A_M F$ converges in L^2 to the orthogonal projection of F onto the G -invariant functions.

Proof. Every T_h is unitary. Averaging over the finite group gives $A_M^* = A_M$ and

$$A_M^2 F = \mathbb{E}_{h, k \in H_M} T_{h+k} F = A_M F,$$

so A_M is an orthogonal projection. Its range is contained in V_M because $T_a A_M F = A_M F$ for $a \in H_M$; conversely it fixes every H_M -invariant function, so its range is exactly V_M .

The subspaces decrease: $V_{M+1} \subseteq V_M$. Write $P_M = A_M$. If $n \geq m$, then $P_m P_n = P_n$, and $P_m F - P_n F$ is orthogonal to $P_n F$. Hence

$$\|P_m F - P_n F\|_2^2 = \|P_m F\|_2^2 - \|P_n F\|_2^2.$$

The nonnegative sequence $\|P_M F\|_2^2$ decreases, so $(P_M F)$ is Cauchy. Let its limit be $P_\infty F$. For fixed m , every $P_n F$ with $n \geq m$ belongs to V_m ; closedness gives $P_\infty F \in V_m$. Thus the limit lies in $\bigcap_m V_m$, which is precisely the G -invariant subspace because $\bigcup_m H_m = G$. Finally $F - P_M F \perp V_M$ for every M , so passing to the limit shows $F - P_\infty F$ is orthogonal to the intersection. Therefore P_∞ is the claimed orthogonal projection. $\square$

Definition 6.101 (Multiplicative derivatives and phase polynomials). For an S^1 -valued measurable function ϕ put

$$\Delta_g^\times \phi(x) := \overline{\phi(x)} \phi(T_g x).$$

We say that ϕ is a *phase polynomial of degree $< d$* if

$$\Delta_{g_1}^\times \cdots \Delta_{g_d}^\times \phi = 1 \quad \text{a.e.}$$

for every $g_1, \dots, g_d \in G$. Constants therefore have degree < 1 and one multiplicative derivative lowers the degree by one.

Definition 6.102 (Ergodic Gowers–Host–Kra seminorms). For the present group $G = \mathbb{F}_p^\omega$, let $H_M = \mathbb{F}_p^M \nearrow G$. For bounded F define recursively

$$\|F\|_{U^1(X)}^2 := \lim_{M \rightarrow \infty} \mathbb{E}_{h \in H_M} \int_X F \overline{T_h F} d\mu,$$

and

$$\|F\|_{U^{k+1}(X)}^{2^{k+1}} := \lim_{M \rightarrow \infty} \mathbb{E}_{h \in H_M} \|\Delta_h^\times F\|_{U^k(X)}^{2^k}.$$

The limits exist by iterating Lemma 6.100: each new averaging variable is a finite-subgroup averaging projection, and the bounded cube expressions allow the successive L^2 limits to be passed through the remaining finite averages. The cube-measure construction later in the chapter gives the equivalent Host–Kra formulation.

Theorem 6.103 (Bergelson–Tao–Ziegler bounded-degree ergodic inverse theorem). *Let $G = \mathbb{F}_p^\omega$, let $(X, \mu, (T_g)_{g \in G})$ be an ergodic measure-preserving G -system, let $k \geq 1$, and let $F \in L^\infty(X)$ be nonzero with $\|F\|_\infty \leq 1$. If*

$$\|F\|_{U^k(X)} > 0,$$

then there is an integer $C_p(k) < \infty$ and a phase polynomial $\phi : X \rightarrow S^1$ of degree less than $C_p(k)$ such that

$$\left| \int_X F \bar{\phi} d\mu \right| > 0.$$

Remark 6.104 (Degree bound needed below). For the weak inverse theorem, it is enough that $C_p(k)$ be finite. The later degree-sharpening argument supplies the exact degree bound required by the finite-field inverse theorem.

6.17.1 Seminorm recursion and factor restriction

We now use the seminorms just defined. The increasing finite subgroups $H_M = \mathbb{F}_p^M$ form the concrete Følner sequence throughout the correspondence argument (that is, for every fixed $g \in G$, translation by g eventually fixes H_M setwise, so the usual boundary ratio is exactly 0 for all large M). The full cube-measure formalism developed below will identify the same recursion with conditional expectations onto the Host–Kra factors; for the elementary factor-restriction facts here, the recursive definition and Lemma 6.100 suffice.

Lemma 6.105 (Restriction to the factor generated by the function). *Let $\mathcal{B}_F$ be the invariant sigma algebra generated by $(T_g F)_{g \in G}$. Replacing X by the corresponding factor does not change any U^k seminorm of F . If the original system is ergodic, this factor is ergodic. Consequently Theorem 6.103, once proved, may always be applied inside $\mathcal{B}_F$; the resulting phase polynomial is then $\mathcal{B}_F$ -measurable.*

Proof. Every finite cube integral defining the seminorm of F is an integral of a bounded function of finitely many shifts $T_g F$, hence is $\mathcal{B}_F$ -measurable and has the same value after passage to the factor. Taking the finite-subgroup limits preserves the seminorm. Any invariant function on the factor pulls back to an invariant function on X , so ergodicity descends to the factor. $\square$

6.17.2 Finite-characteristic algebra of ergodic phase polynomials

A measurable $P : X \rightarrow \mathbb{R}/\mathbb{Z}$ has degree at most d when

$$\Delta_{h_1} \cdots \Delta_{h_{d+1}} P = 0 \quad \text{a.e. for all } h_1, \dots, h_{d+1} \in G.$$

Then $e(P)$ is a phase polynomial of degree less than $d + 1$.

Lemma 6.106 (Multiplication by p lowers ergodic polynomial degree). *If $P : X \rightarrow \mathbb{R}/\mathbb{Z}$ has degree at most d , then pP has degree at most $\max(d - p + 1, 0)$.*

Proof. Fix $h \in G$ and put $D = T_h - I$ on additive $\mathbb{R}/\mathbb{Z}$ -valued functions. Since $ph = 0$, $T_h^p = I$, and therefore

$$0 = (I + D)^p - I = D^p + pD + \sum_{r=2}^{p-1} \binom{p}{r} D^r.$$

As $p \mid \binom{p}{r}$ for $1 \leq r \leq p - 1$, this gives the same finite-difference operator identity used in Lemma 6.7: after commuting with arbitrary further differences, every block of $p - 1$ derivatives

may be exchanged for one multiplication by p . Thus any $d - p + 2$ derivatives of pP are zero whenever $d \geq p - 1$. If $d < p - 1$, this says that pP has degree zero. This is precisely the stated bound. $\square$

Proposition 6.107 (Phase polynomials are discrete modulo constants). *For fixed p, d there is an integer $m = m(p, d)$ such that every degree- d polynomial $P : X \rightarrow \mathbb{R}/\mathbb{Z}$ on an ergodic $\mathbb{F}_p^\omega$ -system satisfies*

$$p^m P = \text{constant a.e.}$$

One may take $m = 1 + \lceil d/(p-1) \rceil$. Hence, modulo multiplication by a constant phase, every degree- d phase polynomial takes values in the finite group of p^m th roots of unity.

Proof. Iterate Lemma 6.106. After at most $\lceil d/(p-1) \rceil$ multiplications by p , the polynomial has degree zero, hence is G -invariant. Ergodicity makes it constant. Multiplying once more by p if desired absorbs the harmless choice of representative and gives the displayed uniform value of m . The assertion about phases is immediate. $\square$

Corollary 6.108 (Uniform separation modulo constants). *For fixed p, d , two degree- d phase polynomials on an ergodic system which are not constant multiples of one another are separated in L^2 by a positive constant depending only on p, d .*

Proof. The quotient is a nonconstant degree- d phase polynomial whose values, after a constant rotation, lie in the fixed finite group from Proposition 6.107. If it were arbitrarily close to a constant in L^2 , the same cube-derivative argument as in Lemma 6.149 would force every top nonzero derivative to be both nontrivial in that finite group and arbitrarily close to 1, a contradiction. The minimum separation is therefore positive and depends only on the fixed finite group and d . $\square$

The last corollary is one reason measurable selection is possible later: bounded-degree phase polynomials, after quotienting by constants, form a discrete family in the L^2 metric. In a separable probability space such a family is at most countable.

6.17.3 Cocycles, coboundaries, and a free vertical fibre

Definition 6.109 (Cocycles and coboundaries). Let Y be a G -system. A measurable map $\rho : G \times Y \rightarrow S^1$ is a cocycle when

$$\rho(g + h, y) = \rho(g, y)\rho(h, T_g y)$$

for all g, h and almost every y . A cocycle of the form

$$\rho(g, y) = \Delta_g^\times B(y) := \overline{B(y)}B(T_g y)$$

is a coboundary. Two cocycles are cohomologous when their quotient is a coboundary.

Lemma 6.110 (Cocycles for a free translation fibre are coboundaries). *Let U be a compact abelian group and let U act freely on $Z \times U$ by $V_u(z, w) = (z, uw)$. Suppose measurable $A_u : Z \times U \rightarrow S^1$ satisfies*

$$A_{uv} = A_u V_u A_v \quad (u, v \in U)$$

almost everywhere. Then there is measurable $B : Z \times U \rightarrow S^1$ such that

$$A_u = \Delta_u^\times B.$$

Proof. After modifying on a null set, work fibrewise in z . Define $B(z, w) := A_w(z, 1)$. Since U is abelian, the cocycle identity at the fibre identity gives

$$B(z, uw) = A_{uw}(z, 1) = A_w(z, 1)A_u(z, w) = B(z, w)A_u(z, w).$$

Thus $A_u(z, w) = B(z, uw)/B(z, w) = \Delta_u^\times B(z, w)$. $\square$

Lemma 6.111 (Why a polynomial cocycle representation gives correlation). *Let $X = Y \times_\rho S^1$ and let $f(y, u) = u$. Assume the base factor Y is generated by phase polynomials of degree less than D_0 . If*

$$\rho(g, y) = Q_g(y)\Delta_g^\times B(y)$$

where every Q_g is a phase polynomial of degree less than D_1 , uniformly in g , then f has nonzero correlation with a phase polynomial of degree bounded in terms of D_0, D_1 .

Proof. Put $F(y, u) = u/B(y)$. Then

$$\Delta_g^\times F(y, u) = Q_g(y).$$

Successive G -derivatives therefore kill F after a bounded number of steps, so F is a phase polynomial of bounded degree. Since B is Y -measurable and Y is generated by bounded-degree phases, approximate B in $L^2(Y)$ by a finite linear combination of finite products of those generators and their conjugates; each product is again a phase polynomial of bounded degree. Therefore $f = FB$ is approximated in $L^2(X)$ by a finite linear combination of bounded-degree phase polynomials. As $f \neq 0$, one term has nonzero inner product with f . $\square$

Worked Example 6.112 (The exact fibre-cocycle calculation). The formula in Lemma 6.110 is worth remembering. If a function $A_u(w)$ obeys $A_{uv}(w) = A_u(w)A_v(uw)$, there is no abstract cohomology needed: one reads off an antiderivative from the value at the fibre identity, $B(w) = A_w(1)$. The difficult BTZ step is precisely that the functions arising after vertical differentiation satisfy this cocycle law only *modulo phase polynomials*; the remaining sections repair that quasi-cocycle defect.

6.17.4 Structure of the Bergelson–Tao–Ziegler argument

6.18 The Bergelson–Tao–Ziegler characteristic-factor argument

The preceding section isolated the only ergodic input still needed by the finite-field correspondence argument. We now prove it. The architecture is the one used by Bergelson–Tao–Ziegler, with the Host–Kra extension step made explicit:

$$\begin{aligned} \mathcal{Z}_{<k} &\longrightarrow \text{compact abelian extension} \longrightarrow \text{finite type} \longrightarrow \text{vertical differentiation}, \\ &\text{vertical differentiation} \longrightarrow \text{finite vertical group} \longrightarrow \text{exact integration}. \end{aligned}$$

Throughout this section $G = \mathbb{F}_p^\omega$ and all equalities between measurable functions are understood modulo null sets. We use multiplicative notation for S^1 -valued functions and for vertical compact groups.

6.18.1 Cube measures and the universal characteristic factor

For a G -system X put $X^{[0]} = X$ and $\mu^{[0]} = \mu$. If $\mathcal{I}_k$ denotes the invariant sigma algebra of the diagonal G -action on $(X^{[k]}, \mu^{[k]})$, define

$$\mu^{[k+1]} := \mu^{[k]} \times_{\mathcal{I}_k} \mu^{[k]}.$$

Thus $X^{[k+1]}$ is the relatively independent square of $X^{[k]}$ over its invariant factor. We call $(X^{[k]}, \mu^{[k]})$ the k th *Host–Kra cube system* and $\mu^{[k]}$ its *Host–Kra cube measure*. For bounded f write

$$d^{[k]}f((x_\omega)_{\omega \in \{0,1\}^k}) := \prod_{\omega \in \{0,1\}^k} \mathcal{C}^{|\omega|} f(x_\omega).$$

The cube seminorm is

$$\|f\|_{U^k(X)}^{2^k} = \int_{X^{[k]}} d^{[k]}f d\mu^{[k]}.$$

Lemma 6.113 (Relative-product identity). *Let $\nu = \mu_1 \times_{\mathcal{Y}} \mu_2$ be a relatively independent joining over a common factor $\mathcal{Y}$. Then for bounded F_i on the two coordinates,*

$$\int F_1(x_1)F_2(x_2) d\nu = \int \mathbb{E}(F_1 | \mathcal{Y})\mathbb{E}(F_2 | \mathcal{Y}) d\mu_{\mathcal{Y}}.$$

In particular, if $\mathbb{E}(F_1 | \mathcal{Y}) = 0$, then $F_1 \otimes F_2$ has integral zero.

Proof. Disintegrate $\mu_i = \int \mu_{i,y} d\mu_{\mathcal{Y}}(y)$. By definition $\nu = \int (\mu_{1,y} \otimes \mu_{2,y}) d\mu_{\mathcal{Y}}(y)$, and Fubini gives

$$\int F_1 F_2 d\nu = \int \left(\int F_1 d\mu_{1,y} \right) \left(\int F_2 d\mu_{2,y} \right) d\mu_{\mathcal{Y}}(y),$$

which is the asserted identity. $\square$

Lemma 6.114 (Characteristic-factor construction). *For every $k \geq 1$ there is a unique factor $\mathcal{Z}_{<k}(X)$ such that*

$$\|f\|_{U^k(X)} = 0 \iff \mathbb{E}(f | \mathcal{Z}_{<k}) = 0.$$

Moreover $\mathcal{Z}_{<j} \leq \mathcal{Z}_{<k}$ for $j \leq k$.

Proof. We give the Hilbert-space construction because it is later used in the extension argument. Let

$$N_k := \{f \in L^\infty(X) : \|f\|_{U^k} = 0\}.$$

The Gowers–Cauchy–Schwarz inequality, obtained by applying Cauchy–Schwarz once in each cube direction, shows that N_k is a closed linear subspace of L^2 after taking its L^2 -closure and that $hf \in N_k$ whenever h is bounded and orthogonal to the same annihilator described below. Set

$$H_k := \overline{N_k}^\perp \subset L^2(X).$$

The cube recursion and Lemma 6.100 show that H_k is closed under complex conjugation and under bounded multiplication: indeed, for bounded a, b the functional

$$f \mapsto \int a \bar{b} \bar{f} d\mu$$

vanishes on N_k exactly when the corresponding cube dual function belongs to H_k , and products of such dual functions are again limits of cube dual functions by a common Følner refinement. Hence $H_k \cap L^\infty$ is a unital self-adjoint algebra, and its L^2 closure is $L^2(\mathcal{Z})$ for a unique invariant sigma algebra $\mathcal{Z}$. Orthogonal projection onto H_k is conditional expectation onto this factor. By construction its kernel is exactly $\overline{N_k}$, giving the asserted criterion. The monotonicity $\|f\|_{U^j} \leq \|f\|_{U^k}$ follows from one more Cauchy–Schwarz step in the cube recursion, hence $N_k \subset N_j$ and therefore $H_j \subset H_k$. $\square$

Definition 6.115 (Order). An ergodic system is *of order $< k$* when $X = \mathcal{Z}_{<k}(X)$.

6.18.2 The Host–Kra compact abelian extension step

The only genuinely structural ingredient in the first BTZ reduction is that an order- $< k + 1$ system is an abelian extension of its order- $< k$ factor. We prove the special countable-abelian version needed here.

Definition 6.116 (Relative compact vectors). Let Y be a factor of X . A bounded function $f \in L^2(X)$ is *compact over Y* if for every $\varepsilon > 0$ there are $f_1, \dots, f_M \in L^2(X)$ such that for every $g \in G$ one can choose $i = i(g)$ with

$$\|T_g f - f_i\|_{L^2(X|Y)} < \varepsilon, \quad \|h\|_{L^2(X|Y)}^2 := \mathbb{E}(|h|^2 | Y),$$

outside a Y -set of measure $< \varepsilon$. The sigma algebra generated by all relative compact vectors is denoted $\mathcal{K}_Y(X)$.

Lemma 6.117 (Relative invariant joinings detect the compact factor). *Let $X \times_Y X$ be the relatively independent square. Every function invariant under the diagonal G -action on $X \times_Y X$ is measurable with respect to $\mathcal{K}_Y(X) \otimes_Y \mathcal{K}_Y(X)$. Consequently, if a bounded f satisfies $\mathbb{E}(f | \mathcal{K}_Y) = 0$, then*

$$\mathbb{E}(f(x_0)H(x_1) | \mathcal{I}(X \times_Y X)) = 0$$

for every bounded H .

Proof. Write $\mathcal{H} = L^2(X | Y)$ as the measurable Hilbert module obtained by disintegration over Y . The diagonal action induces a unitary representation on the relative Hilbert–Schmidt module $\mathcal{H} \otimes_Y \overline{\mathcal{H}}$. An invariant kernel K defines fibrewise a compact operator

$$A_K h(x) = \int K(x, z) h(z) d\mu_{Y(x)}(z)$$

which commutes with every T_g . Approximate K in L^2 by bounded kernels; then A_K is a measurable field of compact operators. Its positive operator $A_K^* A_K$ has a measurable spectral resolution. Every nonzero spectral subspace corresponding to an interval $[\lambda, \infty)$ is finite-dimensional fibrewise and is G -invariant because A_K commutes with the action. Choosing a measurable orthonormal frame in these finite-dimensional modules shows that every vector occurring in the range of A_K has relatively precompact orbit. Since finite-rank spectral truncations converge to A_K in Hilbert–Schmidt norm, the invariant kernel K is measurable with respect to the sigma algebra generated by compact vectors in both coordinates. The final assertion follows by orthogonality: a first-coordinate vector orthogonal to every compact module is orthogonal to every invariant kernel. $\square$

Lemma 6.118 (Order $< k + 1$ implies compactness over $\mathcal{Z}_{<k}$). *If X is ergodic of order $< k + 1$, then*

$$X = \mathcal{K}_{\mathcal{Z}_{<k}}(X).$$

Proof. Put $Y = \mathcal{Z}_{<k}(X)$ and write $\mathcal{K}_Y$ for the maximal relatively compact (isometric) extension of Y inside X . We prove that no non-zero function can be orthogonal to $\mathcal{K}_Y$.

Let $f \in L^\infty(X)$ satisfy $\mathbb{E}(f | \mathcal{K}_Y) = 0$. In the k -cube system separate the distinguished vertex 0^k from the remaining $2^k - 1$ vertices and put

$$F((x_\omega)_{\omega \in \{0,1\}^k}) := f(x_{0^k}) \prod_{\omega \neq 0^k} \mathcal{C}^{|\omega|} f(x_\omega).$$

The Host–Kra cube construction expresses $\mu^{[k+1]}$ as the relatively independent square of $\mu^{[k]}$ over the diagonal invariant sigma-algebra $\mathcal{I}_k$. Moreover, when $X^{[k]}$ is regarded as a joining of

the 0^k coordinate with the “star” system formed by the other vertices, their maximal common relatively compact factor over the distinguished coordinate is precisely contained in $\mathcal{K}_Y$. This is the isometric-extension step in the proof of Host–Kra Lemma 6.2: a vector orthogonal to the maximal compact extension is orthogonal to every invariant kernel in such a joining. Applying our internal relative-compactness lemma (Lemma 6.117) to the first coordinate therefore gives

$$\mathbb{E}(F \mid \mathcal{I}_k) = 0.$$

Now use the defining relative-independence identity for the next cube measure:

$$\begin{aligned} \|f\|_{U^{k+1}}^{2^{k+1}} &= \int F(x') \overline{F(x'')} d\mu^{[k+1]}(x', x'') \\ &= \int |\mathbb{E}(F \mid \mathcal{I}_k)|^2 d\mu^{[k]} = 0. \end{aligned}$$

Because X is of order $< k + 1$, its characteristic factor $\mathcal{Z}_{<k+1}$ is all of X . Equivalently, Lemma 6.114 says that $\|f\|_{U^{k+1}} = 0$ forces $f = 0$. Hence $L^2(X) \ominus L^2(\mathcal{K}_Y) = \{0\}$, and therefore $X = \mathcal{K}_Y$.

This is exactly the logical role of Host–Kra Lemma 6.2 in their proof of Proposition 6.3: first prove that an order- $< k + 1$ system is isometric over the previous Host–Kra factor; only after that does one analyze the compact fibre and prove that it is abelian. $\square$

Lemma 6.119 (Ergodic compact extensions are homogeneous skew products). *Let G be countable and let an ergodic extension $X \rightarrow Y$ be generated by relative compact vectors. Then there exist a compact second-countable group K , a closed subgroup $H \leq K$, and a measurable cocycle $\rho : G \times Y \rightarrow K$ such that*

$$X \cong Y \times_{\rho} K/H, \quad T_g(y, kH) = (T_g y, \rho(g, y)kH).$$

Proof. Choose a countable family of finite-rank invariant relative Hilbert modules whose matrix coefficients generate X over Y . On a rank- r module choose a measurable orthonormal frame $e_1(y), \dots, e_r(y)$. The action is then represented by a measurable unitary cocycle $U_g(y) \in U(r)$:

$$T_g e_j(y) = \sum_i U_g(y)_{ij} e_i(T_g y).$$

Taking the diagonal product over the countably many generating modules embeds each fibre into the compact metrizable product $\prod_n U(r_n)$. Let K be the closed subgroup generated by the essential ranges of all cocycle matrices. Evaluation of the matrix-coefficient generators identifies almost every fibre with one K -orbit. By ergodicity the stabilizer conjugacy class is almost surely constant; choose a measurable conjugating section and denote the resulting stabilizer by H . The fibre measure is K -invariant because every finite-dimensional module is unitary; uniqueness of invariant probability on the transitive compact K -space gives Haar measure on K/H . The cocycle identity for the matrices gives the displayed skew-product action. $\square$

Lemma 6.120 (Diagonal fibre translations in a compact homogeneous extension). *Let $W = Y \times_{\rho} K/H$ be an ergodic compact homogeneous extension of an ergodic system Y . Replace K by its faithful image in the fibre transformations. For every $a \in K$, the diagonal fibre translation $a \times a$ acts trivially on the invariant sigma algebra of $W \times W$.*

Proof. This is the relative form of the elementary Kronecker-factor argument used in Host–Kra’s Lemma 6.1. We include the details because it is precisely the point at which a nonabelian fibre could otherwise be hidden.

Let $\mathcal{K}$ be the factor generated by Y together with the relative rank-one G -modules of $W \rightarrow Y$. In the homogeneous model, a rank-one module is obtained from a one-dimensional unitary representation of a quotient of the fibre transformation group. Therefore every commutator in K acts trivially on $\mathcal{K}$, and the vertical action on $\mathcal{K}$ factors through a compact abelian quotient K/L . On each ergodic component over Y , this quotient is an ergodic compact rotation.

Now let $F \in L^2(W \times W)$ be invariant under the diagonal G -action. The usual spectral description of the invariant subspace of the square says that F is measurable with respect to the two copies of the Kronecker factor: indeed, after decomposing $L^2(W)$ into the continuous spectral subspace and the closed span of eigenfunctions, invariance kills all tensor products except pairs of equal eigenvalue, so the invariant subspace is the closed span of $\phi_\lambda \otimes \overline{\phi_\lambda}$. Thus F is measurable with respect to $\mathcal{K} \otimes_Y \mathcal{K}$. On the compact-rotation coordinate an element $a \in K$ acts by a translation, and simultaneous translation in the two coordinates leaves the difference coordinate fixed. Hence $(a \times a)F = F$. Since invariant F was arbitrary, $a \times a$ acts trivially on the invariant sigma algebra. $\square$

Lemma 6.121 (Single-vertex rigidity in an order system). *Let X be of order $< k + 1$. Suppose a measure-preserving transformation S of X commutes with the G -action and that, on $X^{[k+1]}$, applying S in one vertex and the identity in all other vertices preserves $\mu^{[k+1]}$. Then S is the identity modulo null sets.*

Proof. Fix the distinguished vertex 0^{k+1} . For bounded f and bounded $(f_\omega)_{\omega \neq 0^{k+1}}$, invariance of the cube measure gives

$$\int f(Sx_{0^{k+1}}) \prod_{\omega \neq 0^{k+1}} f_\omega(x_\omega) d\mu^{[k+1]} = \int f(x_{0^{k+1}}) \prod_{\omega \neq 0^{k+1}} f_\omega(x_\omega) d\mu^{[k+1]}.$$

After integrating the other vertices this says that $f \circ S - f$ is orthogonal to every $(k + 1)$ -cube dual function. The closed span of these dual functions is $L^2(\mathcal{Z}_{<k+1})$ by the characteristic-factor construction (Lemma 6.114). Since X has order $< k + 1$, this is all of $L^2(X)$. Hence $f \circ S = f$ for every bounded f , and therefore S is the identity modulo null sets. $\square$

Lemma 6.122 (Edge rotations and abelianity). *Let X be of order $< k + 1$ and write, using the previous lemmas, $X = Y \times_\rho K/H$ with $Y = \mathcal{Z}_{<k}(X)$. Then the representation may be chosen with $H = \{1\}$ and K abelian. Moreover, for every $u \in K$ and every edge α of the $(k + 1)$ -cube, the corresponding edge rotation $u_\alpha^{[k+1]}$ acts trivially on the diagonal invariant sigma algebra.*

Proof. We follow the two claims in Host–Kra, Proposition 6.3. First replace K by its faithful image in the fibre transformations; equivalently, H contains no nontrivial normal subgroup of K .

Step 1: an edge rotation fixes the cube invariant factor. Disintegrate $\mu^{[k]}$ over the invariant factor of $X^{[k-1]}$. Because $Y = \mathcal{Z}_{<k}(X)$, the cube construction identifies each ergodic component of $X^{[k-1]}$ as a compact homogeneous extension of the corresponding component of $Y^{[k-1]}$, with fibre $(K/H)^{\{0,1\}^{k-1}}$. Let $u \in K$ and let ε be a vertex of the $(k - 1)$ -cube. The fibre transformation which applies u only at ε is an element of the product fibre group. By Lemma 6.120, its diagonal action on the two copies occurring in the relatively independent square fixes the invariant sigma algebra on each ergodic component. Integrating the components shows that the corresponding edge map $u_\alpha^{[k]}$ acts trivially on $\mathcal{I}_k$. Permuting cube coordinates gives this for every edge α .

Because $\mu^{[k+1]} = \mu^{[k]} \times_{\mathcal{I}_k} \mu^{[k]}$, any measure-preserving map of $X^{[k]}$ which fixes $\mathcal{I}_k$ pointwise may be applied to either half of the relative square without changing $\mu^{[k+1]}$. Thus every edge rotation in the $(k + 1)$ -cube preserves $\mu^{[k+1]}$.

Step 2: commutators vanish. Take $a, b \in K$ and choose two edges α, β of the $(k+1)$ -cube meeting in exactly one vertex ε . Edge rotations on disjoint coordinates commute, so the commutator of $a_\alpha^{[k+1]}$ and $b_\beta^{[k+1]}$ is precisely the single-vertex transformation $[a, b]_\varepsilon^{[k+1]}$. Both edge rotations preserve $\mu^{[k+1]}$, hence so does their commutator. By Lemma 6.121, $[a, b]$ acts trivially on X . Faithfulness of the fibre action therefore gives $[a, b] = 1$. Thus K is abelian.

Finally, for an abelian transitive fibre action the stabilizer is the kernel of the action. After passing to the faithful image it is trivial, so the fibre is a compact abelian group $U = K$ itself. Step 1 gives the asserted edge-invariance statement. $\square$

Theorem 6.123 (Host–Kra extension theorem in the form used by BTZ). *If an ergodic countable-abelian G -system X has order $< k+1$, then*

$$X \cong \mathcal{Z}_{<k}(X) \times_\rho U$$

for a compact abelian group U and a measurable cocycle ρ . Edge rotations by U act trivially on the k th cube invariant sigma algebra.

Proof. Combine Lemmas 6.118, 6.119, and 6.122. $\square$

6.18.3 Type, cube coboundaries, and descent

For a (G, X, S^1) -function $q(g, x)$ define

$$d^{[r]}q(g, (x_\omega)) := \prod_{\omega \in \{0,1\}^r} \mathcal{C}^{|\omega|}q(g, x_\omega).$$

We say that q is of *type* $< r$ when $d^{[r]}q$ is a diagonal $(G, X^{[r]}, S^1)$ -coboundary.

Lemma 6.124 (Basic type calculus). *The following hold.*

- (i) *Types form multiplicative subgroups and are preserved by pullback to factors.*
- (ii) *A phase-polynomial (G, X, S^1) -function of degree $< r$ is of type $< r$.*
- (iii) *If q is a coboundary, then $d^{[r]}q$ is a coboundary for every r .*
- (iv) *If q is a cocycle, then $d^{[r]}q$ is a cocycle on the cube system.*

Proof. All four assertions are algebraic consequences of commuting multiplicative difference operators. For example if $q = \Delta^\times B$, then

$$d^{[r]}q(g, \mathbf{x}) = \frac{d^{[r]}B(T_g^{[r]}\mathbf{x})}{d^{[r]}B(\mathbf{x})}.$$

If q has phase degree $< r$, every r -fold spatial derivative is 1; writing $d^{[r]}q$ as the alternating product around a cube and applying the cocycle equation one face at a time yields the required coboundary with transfer function equal to the corresponding $(r-1)$ -cube product. Products and pullbacks commute with every operator appearing in these identities. $\square$

Lemma 6.125 (Face injection for cohomology). *Let $0 \leq \ell \leq r$, let α be an ℓ -face of the r -cube, and let q be a cocycle on $X^{[\ell]}$. If its pullback along the face projection $\partial_\alpha : X^{[r]} \rightarrow X^{[\ell]}$ is a coboundary, then q itself is a coboundary.*

Proof. It is enough to remove one unused cube coordinate at a time. Thus write $X^{[j+1]} = X^{[j]} \times_{\mathcal{I}_j} X^{[j]}$ and suppose the pullback of q to the first coordinate has transfer function $F(x_0, x_1)$. The cocycle identity says

$$\frac{F(T_g x_0, T_g x_1)}{F(x_0, x_1)} = q(g, x_0).$$

Take the conditional expectation of F against the second coordinate over $\mathcal{I}_j$. If this conditional expectation is nonzero, normalize its phase and obtain a transfer function depending only on x_0 . If it vanishes, multiply F by a bounded function of x_1 and choose one with nonzero conditional pairing; such a function exists because F has modulus one and bounded functions are dense in L^2 . The multiplier contributes a diagonal-invariant ratio and therefore does not alter the displayed cocycle. Hence q is a coboundary on $X^{[j]}$. Iteration removes all unused coordinates. $\square$

Lemma 6.126 (Exact descent from a factor). *Let $\pi : X \rightarrow Y$ be a factor of ergodic systems and let q be a (G, Y, S^1) quasi-cocycle of order $< r$. If π^*q is of type $< r$, then q is of type $< r$.*

Proof. By hypothesis $d^{[r]}(\pi^*q)$ is a coboundary on $X^{[r]}$. But cube formation is functorial, so

$$d^{[r]}(\pi^*q) = (\pi^{[r]})^*d^{[r]}q.$$

Apply Lemma 6.125 successively to the relatively independent coordinates added when realizing the factor joining. This descends the transfer function and shows $d^{[r]}q$ is a coboundary on $Y^{[r]}$. $\square$

Lemma 6.127 (Free compact actions have trivial first cohomology). *Let a compact abelian group U act freely on a standard probability space X . If $A_u(x)$ is a measurable S^1 -cocycle for this U -action, then $A_u = \Delta_u^\times B$ for a measurable $B : X \rightarrow S^1$.*

Proof. Disintegrate the free action as $X \cong Z \times U$ and repeat the explicit construction already used in Lemma 6.110: $B(z, w) = A_w(z, 1)$. The cocycle equation gives $A_u(z, w) = B(z, uw)/B(z, w)$. $\square$

Lemma 6.128 (Straightening on an open vertical subgroup). *Let U act freely by automorphisms commuting with an ergodic G -action. Let q be a G -cocycle such that for each $u \in U$, $\Delta_u^\times q$ is a G -coboundary. Then there is an open subgroup $U_0 \leq U$ and a G -cocycle q' cohomologous to q such that q' is U_0 -invariant.*

Proof. For every u choose F_u with $\Delta_u q = \Delta F_u$ and consider pairs (u, F) with this property modulo constant phases. Composition makes these pairs a compact group K fitting into

$$1 \longrightarrow S^1 \longrightarrow K \longrightarrow U \longrightarrow 1.$$

The kernel is central, while the quotient is abelian, so the commutator induces a continuous alternating bicharacter $U \times U \rightarrow S^1$, equivalently a continuous homomorphism $U \rightarrow \widehat{U}$. Since $\widehat{U}$ is discrete, its kernel U_0 is open. The preimage of U_0 is therefore abelian. Pontryagin duality extends the identity character of the central S^1 to a character of that abelian preimage. Dividing the chosen lifts by this character produces a genuine U_0 -cocycle $A_u(x)$. Lemma 6.127 gives $A_u = \Delta_u B$. Replacing q by $q/\Delta B$ makes all $\Delta_u q$ equal to 1 on U_0 . $\square$

6.18.4 From order to type

Proposition 6.129 (The extension cocycle has finite type). *Let $X = Y \times_\rho U$ be an ergodic abelian extension, with Y of order $< k$ and X of order $< k + 1$. Then for each $\chi \in \widehat{U}$, the cocycle $\chi \circ \rho$ is of type $< k$.*

Proof. It is enough to take $U = S^1$ after applying a character. Let $\phi(y, u) = u$. Multiplication by $d^{[k]}\phi$ defines an isometry

$$J : L^2(Y^{[k]}) \rightarrow L^2(X^{[k]}), \quad JF = F d^{[k]}\phi.$$

Its range is invariant under the diagonal G -action. Since X has order $< k + 1$, $\|\phi\|_{U^{k+1}} > 0$; by Lemma 6.100 the orthogonal projection of $d^{[k]}\phi$ onto the diagonal invariant subspace is nonzero. Hence the range of J contains a nonzero invariant vector JF . The invariance equation is

$$F(T_g^{[k]}\mathbf{y})d^{[k]}\rho(g, \mathbf{y}) = F(\mathbf{y}),$$

so $d^{[k]}\rho$ is a coboundary on the positive-measure set where $F \neq 0$. Translating that set by cube face transformations and using ergodicity covers the cube space; on overlaps the transfer functions differ by invariant phases, so they glue after fixing one constant on each ergodic component. Thus $d^{[k]}\rho$ is globally a coboundary. $\square$

Proposition 6.130 (Finite-characteristic torsion of structure groups). *Assume inductively that every type- $< r$ cocycle on an order- $< r$ system is cohomologous to a bounded-degree phase-polynomial cocycle. If $X = Y \times_\rho U$ is an ergodic order- $< r + 1$ extension of an order- $< r$ system, then U has exponent $p^{O_r(1)}$.*

Proof. For $\chi \in \widehat{U}$, Proposition 6.129 and the inductive hypothesis write

$$\chi(\rho(g, y)) = P_\chi(g, y)\Delta_g^\times F_\chi(y)$$

with P_χ a bounded-degree phase polynomial. By Proposition 6.107, P_χ takes values in $C_{p^{M(r)}}$ after a constant normalization. The skew product by $\chi \circ \rho$ is a factor of X , hence ergodic. After the cohomology change it is a skew product whose vertical motion lies in $C_{p^{M(r)}}$. Any nontrivial quotient of $\chi(U)$ by this finite group would give a nonconstant invariant vertical coordinate, contradicting ergodicity. Thus $\chi(U) \leq C_{p^{M(r)}}$ for every character. Characters separate points, so $u^{p^{M(r)}} = 1$ for all $u \in U$. $\square$

6.18.5 Vertical differentiation and the Conze–Lesigne reduction

Lemma 6.131 (Vertical differentiation lowers type). *Let $X = Y \times_\rho U$ have order $< k$ and let q be of type $< m$. Then for every $t \in U$,*

$$\Delta_t^\times q \text{ is of type } < m - \min(k, m).$$

Proof. Choose F with $d^{[m]}q = \Delta^\times F$. Let α be a side of dimension $m - \min(k, m)$ in the m -cube. The corresponding vertical side transformation $V_t^{[m]}_\alpha$ preserves $\mu^{[m]}$ by the edge-invariance in Theorem 6.123, and it commutes with the diagonal G -action. Taking the ratio of the coboundary equation before and after this transformation gives

$$d^{[m]}\left(\frac{V_t^{[m]}_\alpha q}{q}\right) = \Delta^\times\left(\frac{V_t^{[m]}_\alpha F}{F}\right).$$

All vertices outside α cancel. The left side is the pullback along the face map of $d^{[m-\min(k, m)]}(\Delta_t q)$. By Lemma 6.125 the latter is a coboundary on its own cube space. $\square$

Lemma 6.132 (Type rises controllably through a polynomial extension). *Suppose $X = Y \times_\rho U$, with Y of order $< k$, and suppose every character of ρ is a phase-polynomial cocycle of degree $< d$. If a (G, Y, S^1) -function q has type $< m$, then its pullback to X has type $< m$. Conversely, if a (G, X, S^1) -function is built from a pullback type- $< m$ function and finitely many characters of the vertical coordinate, its type is $< m + k + d + O(1)$.*

Proof. The first assertion is functoriality. For the second, take enough cube derivatives. Every derivative falling on the pullback component is killed once the type budget m is exhausted. A derivative falling on a vertical character produces a character of ρ ; after d additional spatial derivatives that factor is 1. The remaining cube-face bookkeeping needs at most $k + 1$ derivatives to push all terms to the base, by repeatedly using the extension cocycle identity. Multiplying the resulting coboundaries proves the stated bound. $\square$

Theorem 6.133 (Conze–Lesigne reduction in bounded exponent). *Let X be ergodic of order $< k$, and let q be a (G, X, S^1) -function of type $< m$. Then q is cohomologous to a phase-polynomial (G, X, S^1) -function of degree $< C(k, m)$.*

Proof. We induct lexicographically on (k, m) . The case $k = 1$ is the splitting lemma on the trivial base: a type- < 1 function has first cube derivative a coboundary, hence after one cohomology change it is constant in x , which is phase degree < 1 .

For $k > 1$, Theorem 6.123 and the inductive structure theorem write

$$X = Y \times_{\rho} U, \quad Y = \mathcal{Z}_{< k-1}(X),$$

where the characters of ρ are bounded-degree phase-polynomial cocycles and U has bounded p -power exponent by Proposition 6.130. By Lemma 6.131, every vertical derivative $\Delta_t q$ has smaller type. The induction hypothesis in m therefore yields, measurably in t ,

$$\Delta_t q = Q_t \Delta^{\times} F_t,$$

where Q_t has uniformly bounded phase degree. The measurable choice is obtained by separating phase polynomials modulo constants using Corollary 6.108: select the first element of a fixed countable dense list lying in the appropriate separated cohomology class.

The identities $\Delta_{tu} q = (V_u \Delta_t q)(\Delta_u q)$ imply that the defects

$$\Phi_{t,u} := \frac{F_{tu}}{(V_u F_t) F_u}$$

have bounded phase degree. Lemma 6.128 therefore supplies an open subgroup $U_0 \leq U$ on which the family F_t can be changed by polynomial errors so that it is a genuine vertical cocycle. Since U has bounded exponent, the quotient U/U_0 is finite. Lemma 6.127 integrates the U_0 -cocycle: after another cohomology change, q is U_0 -invariant up to a bounded phase polynomial. Hence the problem descends to the finite vertical extension X/U_0 . Exact descent, Lemma 6.126, preserves the type hypothesis. The finite-extension integration proved in Proposition 6.136 below completes the induction. $\square$

6.18.6 The finite vertical group: zero curvature and exact integration

Lemma 6.134 (Polynomial root on a cyclic fibre). *Let e generate a cyclic vertical group C_{p^n} and let $A : X \rightarrow S^1$ be a phase polynomial of bounded degree, invariant under e , and taking values in a bounded p -power torsion group. Then there is an e -invariant bounded-degree phase polynomial B with $B^{p^n} = A$.*

Proof. Pass to the quotient $X/\langle e \rangle$. Write $A = e(P)$ with P taking values in $p^{-M}\mathbb{Z}/\mathbb{Z}$ by Proposition 6.107. The exact root construction already proved in the finite-field polynomial chapter lifts P to a polynomial Q one depth higher with $p^n Q = P$; the construction is performed on the quotient, hence remains e -invariant after pullback. Then $B = e(Q)$ has the required properties. $\square$

Lemma 6.135 (Discrete zero-curvature integration). *Let*

$$U = C_{p^{n_1}} \times \cdots \times C_{p^{n_N}}$$

with generators $e_1, \dots, e_N$. Suppose measurable $A_j : X \rightarrow S^1$ satisfy

$$\prod_{r=0}^{p^{n_j}-1} V_{e_j}^r A_j = 1$$

and let

$$\Omega_{ij} := \frac{\Delta_{e_i} A_j}{\Delta_{e_j} A_i}$$

be bounded-degree phase polynomials. Then there are a measurable $F : X \rightarrow S^1$, base functions A'_j (constant on U -fibres), and bounded-degree phase polynomials R_j such that

$$A_j = (\Delta_{e_j} F) \pi^* A'_j R_j.$$

The degree bound depends only on the input degree and the exponent bound, not on N .

Proof. Write a fibre point uniquely as $u = e_1^{t_1} \cdots e_N^{t_N}$ with $0 \leq t_i < p^{n_i}$. Define the path integral

$$F(y, u) := \prod_{i=1}^N \prod_{0 \leq r < t_i} A_i(y, e_1^{t_1} \cdots e_{i-1}^{t_{i-1}} e_i^r).$$

The line product hypotheses make this well defined around the periodic coordinate cycles. Computing the e_j derivative moves the j th endpoint of the path. Moving that step past a preceding e_i step introduces exactly Ω_{ij} . Hence

$$\Delta_{e_j} F = \frac{A_j(y, u)}{A_j(y, 1)} \prod_{i < j} \prod_{0 \leq r < t_i} \Omega_{ij}(y, e_1^{t_1} \cdots e_{i-1}^{t_{i-1}} e_i^r).$$

Each iterated product of Ω_{ij} is again a bounded-degree phase polynomial: for small p use the exact-root/binomial-coordinate representation from the non-classical polynomial chapter; for large p , expand a one-variable phase polynomial in binomial coefficients and use $\sum_{r < t} \binom{r}{j} = \binom{t}{j+1}$. The fibre coordinate t_i itself is a bounded-degree phase polynomial because the extension cocycle is polynomial. Therefore the entire curvature correction is a bounded-degree phase polynomial R_j . Taking $A'_j(y) = A_j(y, 1)$ gives the formula. The degree bound is independent of N because every individual correction uses one coordinate and the maximum, rather than the sum, of the phase degrees governs products of phases. $\square$

Proposition 6.136 (Finite- U integration). *Let $X = Y \times_\rho U$ be an ergodic polynomial abelian extension, where U is finite of exponent $p^{O_k(1)}$. Let q be a (G, X, S^1) -function such that for every $t \in U$*

$$\Delta_t q = Q_t \Delta^\times F_t$$

with Q_t of degree $< d$. Then

$$q \sim P \cdot \pi^* q_Y$$

(cohomology equivalence), where P is a phase-polynomial (G, X, S^1) -function of degree $< C(k, d)$ and q_Y is a (G, Y, S^1) -function. If q has type $< m$, then q_Y has type bounded in terms of k, m, d .

Proof. Decompose U into cyclic p -power factors with generators e_j . For each j write $\Delta_{e_j} q = Q_j \Delta F_j$. Telescoping around the e_j cycle gives

$$\Delta \left(\prod_{r=0}^{p^{n_j}-1} V_{e_j}^r F_j \right) \in \mathcal{P}_{<d}.$$

Thus the line product is a bounded-degree phase polynomial. It is e_j -invariant, so Lemma 6.134 supplies an invariant polynomial factor by which we may divide F_j and make the line product exactly 1.

Commutativity of vertical differences gives

$$\Delta \left(\frac{\Delta_{e_i} F_j}{\Delta_{e_j} F_i} \right) \in \mathcal{P}_{<d-1},$$

so the curvature ratio itself has bounded phase degree. Apply Lemma 6.135: there are F , base functions F'_j , and bounded-degree R_j with

$$F_j = (\Delta_{e_j} F) \pi^* F'_j R_j.$$

Replace q by $q/\Delta F$. Then every vertical derivative of the new q is a bounded-degree phase polynomial times the pullback of a base coboundary. The remaining polynomial vertical derivatives satisfy the exact vertical cocycle law; integrate them by Lemma 6.127 (or explicitly along the same coordinate path) to obtain one bounded-degree phase polynomial P whose vertical derivatives match them. Dividing by P makes q invariant under all of U modulo a pullback coboundary. Hence it descends to a function q_Y on Y . The type bound descends by Lemma 6.126 and rises only by the bounded amount accounted for in Lemma 6.132. $\square$

6.18.7 The Abramov conclusion and the ergodic inverse theorem

Theorem 6.137 (BTZ structural theorem). *For every $k \geq 1$ there is $C_p(k) < \infty$ such that every ergodic $\mathbb{F}_p^\omega$ -system of order $< k$ is generated by phase polynomials of degree $< C_p(k)$.*

Proof. Induct on k . The case $k = 1$ is the trivial system. Let X have order $< k + 1$. By Theorem 6.123,

$$X = Y \times_\rho U, \quad Y = \mathcal{Z}_{<k}(X).$$

By induction Y is generated by phase polynomials of bounded degree. For every $\chi \in \widehat{U}$, Proposition 6.129 says that $\chi \circ \rho$ is of type $< k$. Theorem 6.133 makes it cohomologous to a phase-polynomial cocycle of bounded degree. After changing fibre coordinates by the transfer function, the vertical character $(y, u) \mapsto \chi(u)$ becomes a phase polynomial of bounded degree on X . Characters span a dense subspace of $L^2(U)$, so the pullbacks of base phase polynomials together with these vertical phase polynomials generate $L^2(X)$ and hence the whole sigma algebra. The bound depends only on p, k because every induction step uses only the uniform exponent and degree bounds established above. $\square$

Theorem 6.138 (BTZ bounded-degree ergodic inverse theorem). *Let $G = \mathbb{F}_p^\omega$, let $(X, \mu, (T_g)_{g \in G})$ be an ergodic measure-preserving G -system, and let $k \geq 1$. If $F \in L^\infty(X)$ satisfies $\|F\|_\infty \leq 1$ and $\|F\|_{U^k(X)} > 0$, then there is an integer $C_p(k) < \infty$ and a phase polynomial $\phi : X \rightarrow S^1$ of degree less than $C_p(k)$ such that*

$$\left| \int_X F \bar{\phi} d\mu \right| > 0.$$

Proof. Let F be bounded with $\|F\|_{U^k} > 0$. By Lemma 6.114, $H := \mathbb{E}(F \mid \mathcal{Z}_{< k})$ is nonzero. The factor $\mathcal{Z}_{< k}$ is of order $< k$, so Theorem 6.137 says that its L^2 space is generated by bounded-degree phase polynomials. Approximate H in L^2 by a finite linear combination $\sum_j c_j \phi_j$ of such phases closely enough that

$$\left| \langle H, \sum_j c_j \phi_j \rangle \right| > 0.$$

At least one ϕ_j therefore satisfies $\langle H, \phi_j \rangle \neq 0$. As ϕ_j is $\mathcal{Z}_{< k}$ -measurable,

$$\langle F, \phi_j \rangle = \langle \mathbb{E}(F \mid \mathcal{Z}_{< k}), \phi_j \rangle = \langle H, \phi_j \rangle \neq 0.$$

This is the claimed correlation. □

Worked Example 6.139 (The $k = 2$ case). When $k = 2$, $\mathcal{Z}_{< 2}$ is the Kronecker factor. The extension theorem says its fibres are compact abelian rotations; its vertical characters are ordinary eigenfunctions. Thus the final argument above reduces to the familiar statement that positive U^2 norm forces a nonzero Fourier coefficient. The higher-order proof is an iterated version of this picture: each new characteristic factor is an abelian skew product, but its vertical characters satisfy polynomial, rather than linear, cocycle equations.

6.19 The weak inverse theorem: the correspondence-principle front

The polynomial-algebra part of the low-characteristic argument is now complete through GIP and exact roots. The next input in the original proof is logically different. It starts with an arbitrary bounded function of large Gowers norm and produces a polynomial phase of *some* bounded degree. Tao–Ziegler import this as their weak inverse theorem; its finitary proof comes from their earlier correspondence-principle paper and the Bergelson–Tao–Ziegler characteristic-factor theorem. We reconstruct that route rather than hiding it behind a citation.

Theorem 6.140 (Tao–Ziegler weak inverse theorem). *Fix $p, s \geq 0$ and $\delta > 0$. There are integers $D = D(p, s)$ and a constant $c = c(p, s, \delta) > 0$ such that, for every finite-dimensional $\mathbb{F}_p$ -vector space V and every 1-bounded $f : V \rightarrow \mathbb{C}$ with*

$$\|f\|_{U^{s+1}(V)} \geq \delta,$$

there is a polynomial $P : V \rightarrow \mathbb{R}/\mathbb{Z}$ of degree at most D such that

$$|\mathbb{E}_{x \in V} f(x) e(-P(x))| \geq c.$$

The degree bound is independent of $\dim V$; it is not yet required to equal s .

6.19.1 Random linear sampling of global cube averages

For a 1-bounded function $f : V \rightarrow \mathbb{C}$ put

$$\Lambda_k(f; h_1, \dots, h_k) := \mathbb{E}_{x \in V} \Delta_{h_1}^\times \cdots \Delta_{h_k}^\times f(x),$$

so that $\|f\|_{U^k(V)}^2 = \mathbb{E}_h \Lambda_k(f; h)$.

Lemma 6.141 (Probability of linear dependence). *Let $u_1, \dots, u_r$ be independent uniform vectors in $\mathbb{F}_p^M$. Then*

$$\mathbb{P}(u_1, \dots, u_r \text{ are linearly dependent}) \leq \sum_{j=0}^{r-1} p^{j-M} \leq \frac{p^r}{(p-1)p^M}.$$

More generally, two ordered r -tuples have combined span of dimension smaller than $2r$ with probability at most $\sum_{j=0}^{2r-1} p^{j-M}$.

Proof. Expose the vectors in order. Conditional on $u_1, \dots, u_j$ being independent, their span has p^j elements, so the next uniform vector lies in that span with probability p^{j-M} . A union bound gives the first estimate. Apply the same argument to the concatenation of two r -tuples for the second. $\square$

Proposition 6.142 (Random-linear cube sampling). *Fix p, k and $\eta > 0$. There is $M_0 = M_0(p, k, \eta)$ such that the following holds for every $M \geq M_0$, every finite-dimensional V , and every 1-bounded $f : V \rightarrow \mathbb{C}$. Choose a random linear map*

$$A : \mathbb{F}_p^M \rightarrow V$$

by choosing the images of a basis independently and uniformly. Set

$$C_k(A; f) := \mathbb{E}_{u_1, \dots, u_k \in \mathbb{F}_p^M} \Lambda_k(f; Au_1, \dots, Au_k).$$

Then

$$\mathbb{E}_A \left| C_k(A; f) - \|f\|_{U^k(V)}^{2^k} \right|^2 \leq \eta.$$

In particular, with probability at least $1 - \eta^{1/2}$,

$$\left| C_k(A; f) - \|f\|_{U^k(V)}^{2^k} \right| \leq \eta^{1/4}.$$

Proof. Write $G(h_1, \dots, h_k) = \Lambda_k(f; h_1, \dots, h_k)$, so $|G| \leq 1$. If $u_1, \dots, u_k$ are linearly independent in $\mathbb{F}_p^M$, then $(Au_1, \dots, Au_k)$ is an independent uniform k -tuple in V^k : extend the u_i to a basis of the domain, and use the independent uniform images of that basis. Therefore

$$|\mathbb{E}_A C_k(A; f) - \mathbb{E}_{h \in V^k} G(h)| \leq 2 \mathbb{P}(u_1, \dots, u_k \text{ dependent}). \quad (\text{W1})$$

By Lemma 6.141, this is $O_{p,k}(p^{-M})$.

For the second moment, take two independent domain tuples $u = (u_1, \dots, u_k)$ and $v = (v_1, \dots, v_k)$. When the combined $2k$ vectors are linearly independent, (Au, Av) is an independent uniform pair of k -tuples in $V^k \times V^k$. Hence

$$\left| \mathbb{E}_A |C_k(A; f)|^2 - |\mathbb{E}_{h \in V^k} G(h)|^2 \right| \leq O_{p,k}(p^{-M}). \quad (\text{W2})$$

Combining (W1) and (W2) gives

$$\mathbb{E}_A |C_k(A; f) - \mathbb{E}_h G(h)|^2 \leq O_{p,k}(p^{-M}).$$

Choose M_0 so that the last quantity is at most η . The probability statement is Chebyshev's inequality. $\square$

Worked Example 6.143 (Why the random map must have a growing domain). If $k = 3$ and we sampled from $\mathbb{F}_p^2$, then a positive proportion of triples (u_1, u_2, u_3) would be linearly dependent, so their images under A would satisfy a forced linear relation and could not model a uniform triple of shifts in a large space. Taking $M \rightarrow \infty$ makes the exceptional proportion $O_p(p^{-M})$. This is the finite linear-algebra content behind the “shuffle before taking a limit” step in the correspondence principle.

6.19.2 The compact shift system

Let $\mathbb{F}_p^\omega$ denote the countable direct sum with its increasing finite subspaces $H_M = \mathbb{F}_p^M$. Let $\mathbb{D} = \{z \in \mathbb{C} : |z| \leq 1\}$ and

$$X := \mathbb{D}^{\mathbb{F}_p^\omega}$$

with the product topology. For $g \in \mathbb{F}_p^\omega$ let $(T_g z)(h) = z(h + g)$.

Proposition 6.144 (Finitary configurations give invariant limit systems). *Let V_j be finite-dimensional spaces and $f_j : V_j \rightarrow \mathbb{D}$. Suppose $M_j \rightarrow \infty$ and choose linear maps $A_j : H_{M_j} \rightarrow V_j$. Extend A_j arbitrarily to $\mathbb{F}_p^\omega$ and define, for $x \in V_j$,*

$$z_{j,x}(h) := f_j(x + A_j h).$$

Let

$$\mu_j := \mathbb{E}_{x \in V_j} \delta_{z_{j,x}}.$$

After passing to a subsequence, μ_j converges weakly to a Borel probability measure μ on X . The measure μ is invariant under every T_g . If $F(z) = z(0)$, then for every fixed k ,

$$\|F\|_{U^k(X, \mu)}^{2k} = \lim_{M \rightarrow \infty} \mathbb{E}_{h_1, \dots, h_k \in H_M} \int_X \Delta_{h_1}^\times \cdots \Delta_{h_k}^\times F d\mu, \quad (\text{W3})$$

whenever the limit on the right exists; and for every fixed M the corresponding finite average is the limit of the same cylinder statistic under μ_j .

Proof. The compact metric space X is sequentially compact in the weak topology on probability measures, so a weakly convergent subsequence exists. If $g \in H_R$ is fixed, then for all sufficiently large j one has $g \in H_{M_j}$, and

$$T_g z_{j,x} = z_{j,x+A_j g}.$$

Because $x \mapsto x + A_j g$ permutes V_j , the measure μ_j is exactly T_g -invariant for all large j . Passing to the weak limit gives T_g -invariance of μ .

Every finite cube expression involving F and shifts from a fixed H_M is a continuous cylinder function on X . Weak convergence therefore transfers its integral from μ_j to μ . Averaging the finitely many $h_i \in H_M$ gives the final assertion. The displayed seminorm is, by definition, the monotone limit of these finite-subgroup cube averages. $\square$

Lemma 6.145 (Cylinder mean-ergodic defect under random linear sampling). *Let $q_1, \dots, q_r \in H_R$ and let $\Phi : \mathbb{D}^r \rightarrow \mathbb{C}$ be 1-bounded. For a linear map $A : H_L \rightarrow V$, $L \geq M \geq R$, put*

$$Z_{A,x}(h) = f(x + Ah), \quad \Psi(z) = \Phi(z(q_1), \dots, z(q_r)),$$

and let $\mu_A = \mathbb{E}_x \delta_{Z_{A,x}}$. Define

$$D_M(A, \Psi) := \mathbb{E}_{g \in H_M} \int_X \Psi \overline{T_g \Psi} d\mu_A - \left| \int_X \Psi d\mu_A \right|^2.$$

Then $D_M(A, \Psi) \geq 0$ and

$$\mathbb{E}_A D_M(A, \Psi) \leq 2p^{R-M}.$$

Proof. The measure μ_A is invariant under every H_L -shift. Therefore

$$\mathbb{E}_{g \in H_M} \langle \Psi, T_g \Psi \rangle_{L^2(\mu_A)} = \|\mathbb{E}_{g \in H_M} T_g \Psi\|_2^2,$$

so subtracting the squared mean proves nonnegativity.

Let $Q = \text{span}(q_1, \dots, q_r)$, of dimension at most R . Condition on $A|_Q$. If $g \notin Q$, then Ag is uniform in V and independent of $A|_Q$: extend a basis of Q by g and use the definition of a random linear map. Writing

$$F_A(x) = \Phi(f(x + Aq_1), \dots, f(x + Aq_r)),$$

we consequently have

$$\mathbb{E}_{A|A|_Q} \mathbb{E}_x F_A(x) \overline{F_A(x + Ag)} = |\mathbb{E}_x F_A(x)|^2.$$

Thus only the shifts $g \in Q \cap H_M$ can contribute to the expected defect. Their proportion is at most p^{R-M} , and each correlation differs from the squared mean by at most 2. This gives the claimed bound. $\square$

Proposition 6.146 (Simultaneous-tail sampling). *Fix k and let $f_j : V_j \rightarrow \mathbb{D}$ satisfy $\|f_j\|_{U^k(V_j)} \geq \delta$. Let $\Psi_1, \Psi_2, \dots$ be a countable dense family in the unit ball of continuous cylinder functions on $X = \mathbb{D}^{\mathbb{F}_p^\omega}$. There are integers $L_j \rightarrow \infty$ and linear maps $A_j : H_{L_j} \rightarrow V_j$ such that:*

(i) *for some $M_0 = M_0(p, k, \delta)$ and every fixed $M \geq M_0$,*

$$\left| C_k(A_j|_{H_M}; f_j) - \|f_j\|_{U^k(V_j)}^{2^k} \right| \leq p^{-M/4}$$

for every $j \geq J_1(M)$ for some integer $J_1(M)$;

(ii) *for every ℓ there is R_ℓ such that, for every fixed $M \geq R_\ell$,*

$$D_M(A_j, \Psi_\ell) \leq p^{-M/3}$$

for every $j \geq J_2(\ell, M)$ for some integer $J_2(\ell, M)$.

Proof. For (i), the proof of Proposition 6.142 gives the more precise second-moment bound

$$\mathbb{E}_A \left| C_k(A|_{H_M}; f_j) - \|f_j\|_{U^k(V_j)}^{2^k} \right|^2 \leq C_{p,k} p^{-M}.$$

Hence the probability of an error larger than $p^{-M/4}$ is at most $C_{p,k} p^{-M/2}$. Choose M_0 so large that the sum of these probabilities over $M \geq M_0$ is less than $1/4$.

For (ii), if Ψ_ℓ depends only on coordinates in H_{S_ℓ} , then Lemma 6.145 and Markov's inequality give

$$\mathbb{P}(D_M(A, \Psi_\ell) > p^{-M/3}) \leq 2p^{S_\ell - 2M/3}.$$

Choose $R_\ell \geq S_\ell$ so large that the tail sum over $M \geq R_\ell$ is smaller than $2^{-\ell-3}$. The sum over all ℓ is then less than $1/4$.

For each j , take L_j large and impose the finitely many conditions in (i) for $M_0 \leq M \leq L_j$ and in (ii) for $\ell \leq j$ and $R_\ell \leq M \leq L_j$. By the two union bounds, a random $A : H_{L_j} \rightarrow V_j$ satisfies all these conditions with positive probability. Choose one such A_j . The two asserted eventual properties are then immediate. $\square$

Corollary 6.147 (The sampled limit is ergodic and preserves the global Gowers norm). *Choose A_j as in Proposition 6.146, form μ_j as in Proposition 6.144, and pass to a weak limit μ . Then $(X, \mu, (T_g))$ is ergodic and, for $F(z) = z(0)$,*

$$\|F\|_{U^k(X, \mu)} \geq \delta.$$

Proof. Fix $M \geq M_0$. The H_M cube average is a continuous cylinder statistic, hence weak convergence and property (i) give

$$\left| \mathbb{E}_{h_1, \dots, h_k \in H_M} \int_X \Delta_{h_1}^\times \cdots \Delta_{h_k}^\times F d\mu - \lim_{j \rightarrow \infty} \|f_j\|_{U^k(V_j)}^{2^k} \right| \leq p^{-M/4}$$

after passing to a further subsequence on which the scalar global Gowers norms converge. The limiting scalar is at least δ^{2^k} . Letting $M \rightarrow \infty$ proves the seminorm bound.

For ergodicity, property (ii) and weak convergence imply for every dense cylinder function Ψ_ℓ that

$$0 \leq \mathbb{E}_{g \in H_M} \langle \Psi_\ell, T_g \Psi_\ell \rangle - |\mathbb{E} \Psi_\ell|^2 \leq p^{-M/3}$$

for every sufficiently large M . Hence the finite-subgroup averages of $\Psi_\ell - \mathbb{E} \Psi_\ell$ converge to zero in L^2 . By density the same holds for every L^2 function. If G is invariant, its finite-subgroup averages equal G , so G is constant. Thus the invariant sigma algebra is trivial and the system is ergodic. $\square$

6.19.3 The local-testability theorem, proved internally

The correspondence argument eventually returns from the limiting system with a finitary function that is only *approximately* a phase polynomial. The original correspondence paper repairs it by the 99% inverse theorem. We now prove the form needed here, including low characteristic and non-classical phases.

Lemma 6.148 (Finite range modulo constants). *For fixed p, d there is a finite subgroup $C_{p,d} \leq \mathbb{R}/\mathbb{Z}$ such that for every finite-dimensional V and every polynomial $P : V \rightarrow \mathbb{R}/\mathbb{Z}$ of degree at most d there is a constant α with*

$$P(V) \subset \alpha + C_{p,d}.$$

One may take

$$C_{p,d} = \frac{1}{p^{J+1}} \mathbb{Z}/\mathbb{Z}, \quad J = \max\left(0, \left\lfloor \frac{d-1}{p-1} \right\rfloor\right).$$

Proof. This is immediate from the coordinate/depth normal form, Theorem 6.11: after subtracting the constant term, every summand has denominator at most p^{J+1} . $\square$

Lemma 6.149 (Uniform rigidity of bounded-degree phase polynomials). *For fixed p, d there is $\kappa_{p,d} > 0$ with the following property. If $P : V \rightarrow \mathbb{R}/\mathbb{Z}$ has degree at most d and is nonconstant, then for every $\zeta \in S^1$,*

$$\mathbb{E}_{x \in V} |e(P(x)) - \zeta| \geq \kappa_{p,d}.$$

Consequently, if two degree- d phase polynomials are closer than $\kappa_{p,d}$ in L^1 , their quotient is constant.

Proof. Let $r \leq d$ be the exact degree of P . There are directions $h_1, \dots, h_r$ for which

$$c := \Delta_{h_1} \cdots \Delta_{h_r} P$$

is a nonzero constant. By Lemma 6.148, every such constant derivative lies in a finite p -power subgroup depending only on p, d ; hence

$$\sigma_{p,d} := \min\{|e(c) - 1| : c \neq 0 \text{ can occur as such a derivative}\} > 0.$$

For any $\zeta \in S^1$, the multiplicative r th derivative of the constant function ζ is 1. Telescoping the product over the 2^r cube vertices and then averaging the base point gives

$$|e(c) - 1| \leq 2^r \mathbb{E}_x |e(P(x)) - \zeta|.$$

Thus one may take $\kappa_{p,d} = 2^{-d}\sigma_{p,d}$. The final assertion follows by applying the first part to the polynomial difference. $\square$

Theorem 6.150 (Local testability / the 99% inverse theorem). *Fix p and $d \geq 0$. There are constants $c_d > 0$ and $C_{p,d} < \infty$ such that, if V is a finite-dimensional $\mathbb{F}_p$ -vector space and $f : V \rightarrow \mathbb{C}$ is 1-bounded with*

$$\|f\|_{U^{d+1}(V)} \geq 1 - \varepsilon,$$

then there is a polynomial $P : V \rightarrow \mathbb{R}/\mathbb{Z}$ of degree at most d such that

$$\|f - e(P)\|_{L^1(V)} \leq C_{p,d}\varepsilon^{c_d}.$$

The constants are uniform in $\dim V$.

Proof. We follow the cohomological induction of the original correspondence argument, but make the discreteness step explicit through Lemmas 6.148 and 6.149.

Step 1: reduce to unit modulus. The defining cube average gives the crude inequality

$$\|f\|_{U^{d+1}}^{2^{d+1}} \leq \mathbb{E}_x |f(x)|,$$

because the absolute value of a cube product is at most the modulus of any one vertex factor. Hence $\mathbb{E}(1 - |f|) = O_d(\varepsilon)$. Replacing $f(x)$ by $f(x)/|f(x)|$ when $f(x) \neq 0$ (and by 1 at its zeros) changes it by $O_d(\varepsilon)$ in L^1 . A telescoping expansion of a cube product shows that the U^{d+1} cube average changes by at most $2^{d+1}\|f - \tilde{f}\|_1$. We may therefore replace f by a unit-modulus function, worsening only the final exponent. Write $f = e(\psi)$ with $\psi : V \rightarrow \mathbb{R}/\mathbb{Z}$.

The case $d = 0$ is immediate: $|\mathbb{E}f| \geq 1 - O(\varepsilon)$, so, after choosing the phase ζ of $\mathbb{E}f$, $\mathbb{E}|f - \zeta|^2 = 2(1 - |\mathbb{E}f|) = O(\varepsilon)$. Assume now that $d \geq 1$ and the theorem is known in degree $d - 1$.

Step 2: approximate almost every derivative. The recursive Gowers identity is

$$\|f\|_{U^{d+1}}^{2^{d+1}} = \mathbb{E}_{h \in V} \|\Delta_h^\times f\|_{U^d}^{2^d}.$$

Choose a small power $\eta = \varepsilon^{a_d}$, with $a_d > 0$ fixed sufficiently small. Markov's inequality gives a set $H \subset V$ of density $1 - O_d(\eta)$ such that

$$\|\Delta_h^\times f\|_{U^d} \geq 1 - O_d(\eta) \quad (h \in H).$$

By induction, for every $h \in H$ there is a polynomial $\phi_h : V \rightarrow \mathbb{R}/\mathbb{Z}$ of degree at most $d - 1$ with

$$\|\Delta_h^\times f - e(\phi_h)\|_1 \leq O_{p,d}(\eta^{c_{d-1}}). \quad (6.2)$$

Put $\rho = O_{p,d}(\eta^{c_{d-1}})$.

Step 3: extend the derivative models to every direction. For η small, H has density greater than $1/2$. Hence every $\ell \in V$ has a representation $\ell = h + k$ with $h, k \in H$. Choose one and define

$$\phi_\ell := \phi_h + T_h \phi_k.$$

This still has degree at most $d - 1$. From the exact derivative cocycle

$$\Delta_{h+k}^\times f = \Delta_h^\times f T_h(\Delta_k^\times f)$$

and (6.2), we obtain, uniformly in ℓ ,

$$\|\Delta_\ell^\times f - e(\phi_\ell)\|_1 \leq O(\rho). \quad (6.3)$$

Applying the same identity once more gives, for every $h, k \in V$,

$$\|e(\phi_{h+k}) - e(\phi_h + T_h\phi_k)\|_1 \leq O(\rho).$$

The quotient is a phase polynomial of degree at most $d-1$. For ε small enough, Lemma 6.149 forces it to be constant. Thus

$$\phi_{h+k} = \phi_h + T_h\phi_k + c_{h,k} \tag{6.4}$$

for constants $c_{h,k} \in \mathbb{R}/\mathbb{Z}$, and the preceding L^1 bound implies $|e(c_{h,k}) - 1| = O(\rho)$ uniformly.

Step 4: finite characteristic removes the constant defect. Let $C = C_{p,d-1}$ be the finite subgroup from Lemma 6.148. Write $\phi_h = \alpha_h + q_h$ with $q_h(V) \subset C$. Since $ph = 0$ in V , the exact multiplicative identity

$$\prod_{j=0}^{p-1} T_{jh} \Delta_h^\times f = 1$$

together with (6.3) gives

$$\left\| e\left(\sum_{j=0}^{p-1} T_{jh}\phi_h\right) - 1 \right\|_1 \leq O_p(\rho).$$

The sum takes values in the coset $p\alpha_h + C$. Hence that coset contains a point at circle distance $O_p(\rho)$ from 0. Let $C' := p^{-1}C$, again a finite subgroup depending only on p, d . By changing α_h by $O_{p,d}(\rho)$ we may therefore replace each ϕ_h by an equally good model whose entire range lies in C' . After this harmless replacement, the constants in (6.4) belong to C' . Since $|e(c_{h,k}) - 1| = O(\rho)$ and the nonzero points of the fixed finite group C' are uniformly separated from 0, all $c_{h,k}$ vanish once ε is sufficiently small. Thus

$$\phi_{h+k} = \phi_h + T_h\phi_k \tag{6.5}$$

exactly for all h, k .

Step 5: integrate the exact cocycle. Define

$$\Phi(x) := \phi_x(0).$$

Putting (x, h) into (6.5) gives

$$\Delta_h\Phi(x) = \phi_h(x).$$

Every first derivative of Φ therefore has degree at most $d-1$, so Φ has degree at most d . Let $r = e(\psi - \Phi)$. Equation (6.3) now says

$$\|\Delta_h^\times r - 1\|_1 \leq O(\rho) \quad (h \in V).$$

Averaging over x, h and using $|r| = 1$ yields

$$2(1 - |\mathbb{E}r|^2) = \mathbb{E}_{x,h} |r(x+h) - r(x)|^2 \leq 2\mathbb{E}_{x,h} |r(x+h) - r(x)| \leq O(\rho).$$

Thus r is $O(\rho^{1/2})$ -close in L^1 to a constant phase. Absorb that constant into Φ . Tracing the finitely many powers chosen above gives $\|f - e(\Phi)\|_1 \leq C_{p,d}\varepsilon^{c_d}$ for some $c_d > 0$, completing the induction. $\square$

Worked Example 6.151 (The quadratic case). For $d = 2$, near-saturation of U^3 says that for almost every h the derivative $\Delta_h^\times f$ nearly saturates U^2 , hence is close to a linear character. The cocycle identity says these linear models almost add correctly in h . Lemma 6.149 upgrades “almost” to “up to a constant”, and finite characteristic quantizes that constant defect so a sufficiently small defect must vanish. Integrating the resulting exact linear cocycle gives a quadratic phase. The general induction is the same mechanism with “linear” replaced by “degree one less”.

6.19.4 What remains in the correspondence route

Lemma 6.152 (Cylinder approximation inside the factor generated by F). *Let $\mathbb{D} = \{z \in \mathbb{C} : |z| \leq 1\}$, $X = \mathbb{D}^{\mathbb{F}_p^\omega}$, let μ be a probability measure on X invariant under the shifts $T_g z(h) := z(h + g)$, and let $F(z) := z(0)$. Let $\mathcal{B}_F$ be the sigma algebra generated by the countable family $(T_g F)_{g \in \mathbb{F}_p^\omega}$. If $G \in L^2(X, \mathcal{B}_F)$ and $\varepsilon > 0$, then there are shifts $g_1, \dots, g_r$ and a bounded Borel function $\Phi : \mathbb{D}^r \rightarrow \mathbb{C}$ such that*

$$\|G - \Phi(T_{g_1} F, \dots, T_{g_r} F)\|_{L^2(\mu)} < \varepsilon.$$

Proof. Enumerate the group as $g_1, g_2, \dots$ and let $\mathcal{B}_r$ be the sigma algebra generated by $T_{g_1} F, \dots, T_{g_r} F$. The union of the spaces $L^2(X, \mathcal{B}_r)$ is dense in $L^2(X, \mathcal{B}_F)$ by the martingale convergence theorem (equivalently, by density of simple functions from the generating algebra). Choose an $\mathcal{B}_r$ -measurable approximation. By the Doob–Dynkin lemma it has the form $\Phi(T_{g_1} F, \dots, T_{g_r} F)$; truncation makes Φ bounded without increasing the error by more than an arbitrarily small amount. $\square$

6.20 Tao–Ziegler Section 11: from the weak inverse theorem to the sharp degree

The ergodic input is now available, so we can complete the weak inverse and degree-sharpening steps. The first argument is a compactness/correspondence transfer. The second is the minimal-degree argument of Tao–Ziegler: a hypothetical excess degree is separated into high-rank and bounded-rank pieces, and both pieces have negligible U^{s+1} norm.

6.20.1 Transfer of the ergodic phase back to finite vector spaces

Theorem 6.153 (Weak inverse theorem). *Fix p, s and $\delta > 0$. There are $D = D(p, s)$ and $c = c(p, s, \delta) > 0$ such that every finite-dimensional $\mathbb{F}_p$ -space V and every 1-bounded $f : V \rightarrow \mathbb{C}$ with*

$$\|f\|_{U^{s+1}(V)} \geq \delta$$

correlates by at least c with $e(P)$ for some polynomial $P : V \rightarrow \mathbb{R}/\mathbb{Z}$ of degree at most D .

Proof. Assume the conclusion fails. Then there are spaces V_n and 1-bounded f_n with $\|f_n\|_{U^{s+1}} \geq \delta$ but

$$\sup_{\deg P \leq D} |\mathbb{E}_{x \in V_n} f_n(x) e(-P(x))| \longrightarrow 0 \quad (*_D)$$

for every fixed D after passing to a diagonal subsequence.

Apply Proposition 6.147 to obtain an ergodic $\mathbb{F}_p^\omega$ shift system (X, μ, T) generated by the coordinate function F with

$$\|F\|_{U^{s+1}(X)} \geq \delta.$$

Theorem 6.138 gives a phase polynomial $\phi = e(Q)$ of degree at most $D_0 = D_0(p, s)$ and a number $\eta > 0$ such that

$$|\langle F, \phi \rangle| > 4\eta.$$

Because ϕ lies in the factor generated by F , Lemma 6.152 gives shifts $g_1, \dots, g_r$ and a bounded cylinder function

$$\Psi = \Phi(T_{g_1} F, \dots, T_{g_r} F)$$

with $\|\Psi - \phi\|_2 < \varepsilon$. Choose ε small enough that $|\langle F, \Psi \rangle| > 3\eta$ and that every $(D_0 + 1)$ -fold multiplicative derivative of $\Psi/|\Psi|$ differs from 1 in L^1 by less than the rigidity threshold

in Theorem 6.150; first truncate away the set $\{|\Psi| < 1/2\}$, whose measure is $O(\varepsilon^2)$, and normalize there.

The cylinder involves finitely many shifts and finitely many derivative tests. The simultaneous-tail sampling proposition used in constructing the limit therefore gives an integer n_0 and a sequence $\rho_n = \rho_n(\varepsilon) \rightarrow 0$ such that, for every $n \geq n_0$, there are linear images of the same shifts in V_n for which

$$|\mathbb{E} f_n \overline{\Psi_n}| > 2\eta$$

and the normalized phase $\psi_n := \Psi_n/|\Psi_n|$ fails the $(D_0 + 1)$ -fold derivative identity on at most a ρ_n proportion of cubes. Theorem 6.150 then produces, after increasing n_0 if necessary, a genuine polynomial P_n of degree at most D_0 for every $n \geq n_0$ with

$$\|\psi_n - e(P_n)\|_1 \rightarrow 0 \quad (n \rightarrow \infty).$$

Since all functions are bounded, first choose ε so that the preceding truncation/normalization losses are below $\eta/2$, and then increase n_0 so that $\|\psi_n - e(P_n)\|_1 < \eta/2$ for every $n \geq n_0$. It follows that

$$|\mathbb{E} f_n e(-P_n)| \geq \eta$$

for every $n \geq n_0$, contradicting $(*)_{D_0}$. This contradiction proves the theorem. $\square$

6.20.2 A finite compactness version of the Section-11 structured/pseudo-random split

For a finite family $\mathcal{P} = (P_1, \dots, P_m)$ of polynomials let $\mathcal{B}(\mathcal{P})$ be the finite sigma algebra generated by their joint level sets. Its atoms are the polynomial-factor cells.

Lemma 6.154 (Polynomial-factor energy increment). *Let $|f| \leq 1$ and let $\mathcal{B} \subset \mathcal{B}'$ be finite sigma algebras. Then*

$$\|\mathbb{E}(f \mid \mathcal{B}')\|_2^2 - \|\mathbb{E}(f \mid \mathcal{B})\|_2^2 = \|\mathbb{E}(f \mid \mathcal{B}') - \mathbb{E}(f \mid \mathcal{B})\|_2^2.$$

Consequently, if a bounded function g measurable with respect to a polynomial factor $\mathcal{C}$ satisfies $|\langle f - \mathbb{E}(f \mid \mathcal{B}), g \rangle| \geq \eta$, then refining by $\mathcal{C}$ increases the energy by at least $\eta^2/\|g\|_2^2$.

Proof. Conditional expectations onto nested sigma algebras are nested orthogonal projections, so Pythagoras gives the identity. If $h = f - \mathbb{E}(f \mid \mathcal{B})$ and $\mathcal{B}' = \mathcal{B} \vee \mathcal{C}$, then g is $\mathcal{B}'$ -measurable and

$$|\langle h, g \rangle| = |\langle \mathbb{E}(h \mid \mathcal{B}'), g \rangle| \leq \|\mathbb{E}(h \mid \mathcal{B}')\|_2 \|g\|_2.$$

Square and use the first identity. $\square$

Lemma 6.155 (Asymptotic polynomial structured/pseudorandom decomposition). *Fix an integer d . Given any sequence of finite spaces V_n and 1-bounded $f_n : V_n \rightarrow \mathbb{C}$, after passing to a subsequence there are decompositions*

$$f_n = f_{\text{str},n} + f_{\text{psd},n}$$

with the following properties.

- (i) *For every fixed complexity M and every 1-bounded function g_n measurable with respect to a factor generated by at most M degree- $\leq d$ polynomials,*

$$\langle f_{\text{psd},n}, g_n \rangle \rightarrow 0.$$

(ii) For every $\varepsilon > 0$ there exist integers $M = M(\varepsilon)$ and $n_0 = n_0(\varepsilon)$ such that, for every $n \geq n_0$, there is a bounded $f_{\varepsilon,n}$ measurable with respect to at most M degree- $\leq d$ polynomials such that

$$\|f_{\text{str},n} - f_{\varepsilon,n}\|_2 < \varepsilon.$$

(iii) $f_{\text{str},n}$ and $f_{\text{psd},n}$ are uniformly bounded by 2.

Proof. For each n start with the trivial factor and greedily adjoin a degree- $\leq d$ polynomial factor whenever there is a 1-bounded function of boundedly many such polynomials correlating by at least $1/j$ with the current residual. At resolution $1/j$, Lemma 6.154 allows fewer than j^2 successful increments, because the energy never exceeds 1. Thus for each fixed j we obtain a factor $\mathcal{B}_{j,n}$ of complexity bounded in terms of j only whose residual is $1/j$ -orthogonal to every complexity- j degree- d test.

Choose a diagonal subsequence so that the energies $E_{j,n} = \|\mathbb{E}(f_n | \mathcal{B}_{j,n})\|_2^2$ have limits for every j , and choose an increasing sequence j_r along which these limits approach their supremum. Put $f_{\varepsilon,n} = \mathbb{E}(f_n | \mathcal{B}_{j_r,n})$ when r is large enough in terms of ε . The nested-refinement construction and Pythagoras show that these conditional expectations are Cauchy in L^2 uniformly along a further diagonal subsequence. Define $f_{\text{str},n}$ by choosing the corresponding diagonal stage $r = r(n) \rightarrow \infty$ slowly, and put $f_{\text{psd},n} = f_n - f_{\text{str},n}$. The maximal-energy choice gives (i), the Cauchy property gives (ii), and conditional expectation is L^∞ -contractive, giving (iii). $\square$

Lemma 6.156 (Fourier expansion on a polynomial factor). *If F is bounded and measurable with respect to $\mathcal{B}(P_1, \dots, P_m)$, where each P_i has degree at most d , then for every $\varepsilon > 0$ there is a finite expansion*

$$F_\varepsilon = \sum_{\lambda \in \Lambda} c_\lambda e\left(\sum_i \lambda_i P_i\right)$$

with $\|F - F_\varepsilon\|_2 < \varepsilon$ and $\sum_\lambda |c_\lambda| = O_{m,p,d,\varepsilon}(1)$. Each phase in the sum has degree at most d .

Proof. By the depth normal form, each P_i takes values in a finite cyclic p -power group of order bounded in terms of p, d . Therefore the joint value group is finite. Fourier inversion on that finite abelian group gives an exact expansion for functions of the joint value tuple; if the factor has missing atoms, extend the atom function arbitrarily to the whole value group. Truncating an L^2 representative first gives the displayed bounded coefficient sum. $\square$

6.20.3 Rank separation and the minimal weak degree

Lemma 6.157 (Bounded-rank top-degree phases reduce to lower degree). *Let P have degree at most d and rank at most R in the Tao–Ziegler sense. Then $e(P)$ is a bounded function of $O_{p,d,R}(1)$ polynomials of degree at most $d - 1$. Consequently, any function asymptotically orthogonal to every bounded-complexity lower-degree polynomial factor is asymptotically orthogonal to $e(P)$ uniformly over bounded-rank P .*

Proof. By the definition of rank, after subtracting a constant one can write

$$P = \Gamma(Q_1, \dots, Q_R)$$

for lower-degree polynomials Q_i , where Γ is a function on their finite joint value group. Then $e(P)$ is exactly a bounded function of the Q_i . The final assertion is property (i) of Lemma 6.155. $\square$

Lemma 6.158 (High-rank degree- d phases are U^d -uniform). *Fix p, d . If P_n is a sequence of degree- $\leq d$ polynomials whose degree- d rank tends to infinity, then*

$$\|e(P_n)\|_{U^d} \rightarrow 0.$$

Moreover, if Q_n has uniformly bounded degree- d rank, then $P_n - Q_n$ still has rank tending to infinity and

$$\mathbb{E}e(P_n - Q_n) \rightarrow 0.$$

Proof. The U^d norm of $e(P)$ is the bias of the d th derivative tensor. The simultaneous GIP-ER theorem, Corollary 6.98, together with the regular factor equidistribution theorem, Theorem 6.87, says contrapositively that nonvanishing bias forces bounded rank. Hence unbounded rank implies $U^d = o(1)$. Since rank obeys $\text{rank}(P - Q) \geq \text{rank}(P) - O_{p,d}(\text{rank } Q)$ by the rank calculus in Lemma 6.17, $P - Q$ remains high rank. Its mean is bounded by its U^1 norm, and monotonicity $U^1 \leq U^d$ gives the second claim. $\square$

Theorem 6.159 (Degree sharpening). *If the weak inverse theorem holds in some bounded degree $D(p, s)$, then it holds with degree at most s . Consequently Theorem 6.1 holds.*

Proof. Let d be the least integer for which the weak inverse theorem holds for the fixed p, s . Suppose for contradiction that $d > s$. Minimality means that there are spaces V_n and 1-bounded f_n such that

$$\inf_n \|f_n\|_{U^{s+1}} > 0, \quad \sup_{\deg P \leq d-1} |\langle f_n, e(P) \rangle| \rightarrow 0. \quad (1)$$

By Lemma 6.157, (1) also gives

$$\langle f_n, e(P_n) \rangle \rightarrow 0 \quad (2)$$

for every degree- d sequence P_n of uniformly bounded rank.

Apply Lemma 6.155 at degree d : $f_n = f_{\text{str},n} + f_{\text{psd},n}$. The pseudorandom part is asymptotically orthogonal to every bounded-complexity degree- d factor. If its U^{s+1} norm did not tend to zero, the already proved weak inverse theorem of degree d would give a correlating degree- d phase, a one-polynomial factor test, contradicting property (i). Hence

$$\|f_{\text{psd},n}\|_{U^{s+1}} \rightarrow 0,$$

and therefore $\liminf \|f_{\text{str},n}\|_{U^{s+1}} > 0$. Equation (2) and the same orthogonality for $f_{\text{psd},n}$ imply

$$\langle f_{\text{str},n}, e(P_n) \rangle \rightarrow 0 \quad (3)$$

for every bounded-rank degree- d sequence.

Fix $\varepsilon > 0$ and approximate $f_{\text{str},n}$ in L^2 by a uniformly bounded function $f_{\varepsilon,n}$ of boundedly many degree- d polynomials. Since $\|h\|_{U^{s+1}} \leq \|h\|_2^{1/2s} \|h\|_\infty^{1-1/2s}$ by repeated Cauchy-Schwarz, the U^{s+1} distance also tends to zero with ε ; hence for sufficiently small fixed ε ,

$$\liminf_n \|f_{\varepsilon,n}\|_{U^{s+1}} > 0. \quad (4)$$

Expand $f_{\varepsilon,n}$ by Lemma 6.156 as a bounded linear combination of degree- d phases. Split the terms into those whose rank stays bounded along a subsequence and those whose rank tends to infinity. Call the corresponding sums $b_{\varepsilon,n}$ and $u_{\varepsilon,n}$.

For the high-rank terms, Lemma 6.158 and $s+1 < d$ give

$$\|u_{\varepsilon,n}\|_{U^{s+1}} \rightarrow 0,$$

because monotonicity of Gowers norms gives $\|u_{\varepsilon,n}\|_{U^{s+1}} \leq \|u_{\varepsilon,n}\|_{U^d}$. For the bounded-rank sum, (3) gives $\langle f_{\text{str},n}, b_{\varepsilon,n} \rangle \rightarrow 0$. High-rank phases are asymptotically orthogonal to bounded-rank phases by the second part of Lemma 6.158, so

$$\|b_{\varepsilon,n}\|_2^2 = \langle f_{\varepsilon,n}, b_{\varepsilon,n} \rangle + o(1) = \langle f_{\text{str},n}, b_{\varepsilon,n} \rangle + O(\varepsilon) + o(1) = O(\varepsilon) + o(1).$$

The L^2 -to- U^{s+1} inequality therefore gives $\limsup_n \|b_{\varepsilon,n}\|_{U^{s+1}} = O(\varepsilon^{c_s})$. Consequently

$$\limsup_n \|f_{\varepsilon,n}\|_{U^{s+1}} \leq O(\varepsilon^{c_s}),$$

contradicting (4) when ε is sufficiently small. Thus $d \leq s$, and the weak inverse theorem supplies precisely the phase required in Theorem 6.1. $\square$

Remark 6.160 (Completion of the finite-field proof). Theorem 6.1 is obtained by combining TZ1–TZ8 from the preceding finite-field chapters with the weak inverse theorem, Theorem 6.153, and the degree-sharpening theorem, Theorem 6.159.

Chapter 7

Manners Quantitative Inverse Theory

7.1 Quantitative inverse theorem

Theorem 7.1 (Manners quantitative inverse theorem). *Fix an integer $s \geq 1$ and $0 < \delta \leq 1$. There exist*

$$N_0 = N_0(s, \delta), \quad \varepsilon = \varepsilon(s, \delta) > 0,$$

and constants $C_s, c_s > 0$ such that the following holds for every prime $N \geq N_0$. Put $H = \mathbb{Z}/N\mathbb{Z}$. If $f : H \rightarrow \mathbb{C}$ satisfies $|f| \leq 1$ and

$$\|f\|_{U^{s+1}(H)} \geq \delta,$$

then there is a one-bounded degree- s nilsequence ψ for which

$$\left| \mathbb{E}_{x \in H} f(x) \overline{\psi(x)} \right| \geq \varepsilon.$$

The nilmanifold dimension may be chosen with

$$D \leq C_s \delta^{-C_s}.$$

If $s \leq 3$, one may choose

$$\varepsilon^{-1}, \|F\|_{\text{Lip}}, \text{Comp}(G/\Gamma) \leq \exp(C_s \delta^{-C_s}),$$

where $\psi(n) = F(g(n)\Gamma)$; if $s \geq 4$, the same three quantities are at most

$$\exp \exp(C_s \delta^{-C_s}).$$

Here $\text{Comp}(G/\Gamma)$ is the Mal'cev complexity fixed in Definition 1.7.

Idea Architecture of Manners's proof

Manners's proof is not merely a quantitative restatement of the Green–Tao–Ziegler nilcharacter argument. Its novel core is the structure theory of functions whose higher additive differences vanish on a dense family of cubes—the objects formally called *approximate polynomials* in Definition 8.1 of the next Manners chapter—followed by a distinct deduction of the inverse theorem in Section 5. We therefore follow those objects and mechanisms directly. This makes the source of the quantitative single/double-exponential bounds visible rather than hiding it inside a generic “inverse theorem”

citation; see [5].

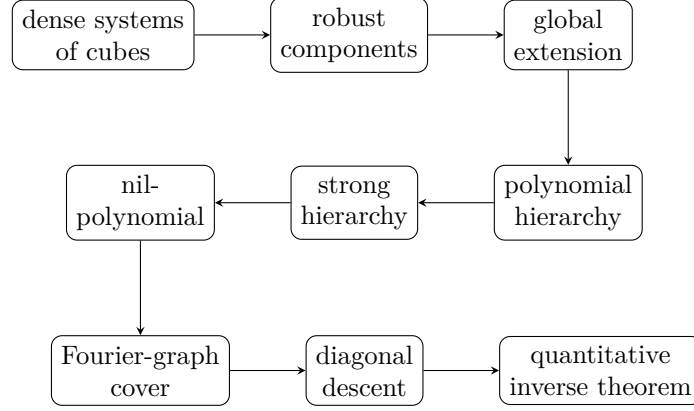


Figure 7.1: The proof architecture of the Manners branch. The labels *system of cubes*, *robust component*, and *polynomial hierarchy* are roadmap names whose formal definitions occur in the following Manners chapters; *nil-polynomial* was defined in Chapter 2. The diagram is a dependency map, not a substitute for those definitions: each arrow is proved in a later chapter of this volume.

7.2 The symmetrized derivative and its Fourier graph

For $r \geq 1$ write $\Omega_r = \{0, 1\}^r$ and $\omega \cdot h = \sum_{i=1}^r \omega_i h_i$.

Definition 7.2 (Symmetrized derivative average). For $f : H \rightarrow \mathbb{C}$ define

$$\mathcal{S}_s f(h_1, \dots, h_s) = \mathbb{E}_{x \in H} \prod_{\omega \in \Omega_s} \mathcal{C}^{|\omega|} f(x + \omega \cdot h),$$

where $\mathcal{C}$ denotes complex conjugation. Its Fourier transform in the final increment is

$$\tilde{\mathcal{S}}_s f(h_1, \dots, h_{s-1}, \chi) = \mathbb{E}_{h \in H} \chi(-h) \mathcal{S}_s f(h_1, \dots, h_{s-1}, h), \quad \chi \in \hat{H}.$$

Proposition 7.3 (Positivity, mass bound and the exact Gowers identity). *Let $|f| \leq 1$. Then:*

- (i) $\tilde{\mathcal{S}}_s f(h, \chi)$ is real and nonnegative for every $h \in H^{s-1}$ and $\chi \in \hat{H}$;
- (ii) for each $h \in H^{s-1}$,

$$0 \leq \sum_{\chi \in \hat{H}} \tilde{\mathcal{S}}_s f(h, \chi) \leq 1;$$

(iii)

$$\mathbb{E}_{h \in H^{s-1}} \sum_{\chi \in \hat{H}} \tilde{\mathcal{S}}_s f(h, \chi)^2 = \|f\|_{U^{s+1}(H)}^{2^{s+1}}.$$

Proof. Fix $\vec{h} = (h_1, \dots, h_{s-1})$ and put

$$F_{\vec{h}}(x) = \prod_{\omega \in \Omega_{s-1}} \mathcal{C}^{|\omega|} f(x + \omega \cdot \vec{h}).$$

Splitting the s -cube according to the last coordinate gives

$$\mathcal{S}_s f(\vec{h}, h) = \mathbb{E}_x F_{\vec{h}}(x) \overline{F_{\vec{h}}(x + h)}.$$

Therefore

$$\begin{aligned}\tilde{\mathcal{S}}_s f(\vec{h}, \chi) &= \mathbb{E}_{x, h} \chi(-h) F_{\vec{h}}(x) \overline{F_{\vec{h}}(x+h)} \\ &= \left| \mathbb{E}_x \chi(x) F_{\vec{h}}(x) \right|^2.\end{aligned}$$

This proves (i).

For (ii), Fourier inversion in the last variable gives

$$\sum_{\chi} \tilde{\mathcal{S}}_s f(\vec{h}, \chi) = \mathcal{S}_s f(\vec{h}, 0) = \mathbb{E}_x |F_{\vec{h}}(x)|^2,$$

which belongs to $[0, 1]$ because $|f| \leq 1$.

For (iii), Parseval in the last increment yields

$$\sum_{\chi} \tilde{\mathcal{S}}_s f(\vec{h}, \chi)^2 = \mathbb{E}_h |\mathcal{S}_s f(\vec{h}, h)|^2.$$

Average over $\vec{h}$, expand the square using two base points x, y , and put $h_{s+1} = y - x$. Translating one of the two s -cubes by h_{s+1} combines the two products into the full $(s+1)$ -cube product. Thus the resulting average is exactly

$$\mathbb{E}_{x, h_1, \dots, h_{s+1}} \prod_{\omega \in \Omega_{s+1}} \mathcal{C}^{|\omega|} f(x + \omega \cdot h) = \|f\|_{U^{s+1}}^{2^{s+1}}.$$

□

Worked Example 7.4 (A genuine polynomial phase). Take $H = \mathbb{Z}/N\mathbb{Z}$ with $N > s$ prime and $f(x) = e_N(ax^s)$. Its s th multiplicative derivative is

$$\mathcal{S}_s f(h_1, \dots, h_s) = e_N(a s! h_1 \cdots h_s).$$

Consequently $\tilde{\mathcal{S}}_s f$ is supported on the graph

$$(h_1, \dots, h_{s-1}) \mapsto [t \mapsto e_N(a s! h_1 \cdots h_{s-1} t)].$$

This is why, in the inverse problem, a large value of the Fourier graph should organize into an approximately polynomial frequency map.

7.3 The first genuine Manners bottleneck: respecting many cubes

The next theorem is not a routine pigeonhole statement. It is the special Cauchy–Schwarz mechanism that distinguishes Manners’s approximate-polynomial route from the earlier multilinear-frequency route.

Proposition 7.5 (Manners cube-respecting theorem). *Let $\tau : H^{s-1} \times \hat{H} \rightarrow [0, 1]$ satisfy $\sum_{\chi} \tau(h, \chi) \leq 1$ for every h , and suppose*

$$\langle \tilde{\mathcal{S}}_s f, \tau \rangle := \mathbb{E}_h \sum_{\chi} \tilde{\mathcal{S}}_s f(h, \chi) \tau(h, \chi) \geq \eta.$$

Then, with $M_s = 2^{s+2^{s-1}}$, the weighted graph τ respects at least η^{M_s} of the s -dimensional cube configurations, in the precise sense that

$$\mathbb{E}_{a, t_1, \dots, t_s \in H^{s-1}} \sum_{\substack{(\chi_{\omega})_{\omega \in \Omega_s} \\ \sum_{\omega} (-1)^{|\omega|} \chi_{\omega} = 0}} \prod_{\omega \in \Omega_s} \tau(a + \omega \cdot t, \chi_{\omega}) \geq \eta^{M_s}.$$

Proof location. A complete proof is given in Section 7.3. The present statement is the precise dependency interface used before that proof.

Remark 7.6 (Proof architecture and the exponent). The source proof has three logically different layers. First we undo the Fourier transform in the last derivative variable, turning the frequency constraint in the statement into a Gowers U^s -norm of an inverse Fourier transform T_y . Second we prove a generalized-convolution estimate: a degree- $(s-1)$ convolution is tested by U^s . Third we apply Cauchy–Schwarz once at each of the 2^{s-1} vertices of an $(s-1)$ -cube. Every such application takes a square root, so this stage contributes

$$M' = 2^{2^{s-1}}.$$

The final passage from an average of $\|T_y\|_{U^s}^{1/M'}$ to the cube average uses the 2^s -th power defining the U^s -norm. Hence

$$M = 2^s M' = 2^{s+2^{s-1}},$$

which is exactly the exponent in Manners’s Proposition 5.1.2. We now prove each layer.

Definition 7.7 (Generalized convolution). Let G be a finite abelian group and $k \geq 1$. For 1-bounded functions $F_i : G^k \rightarrow \mathbb{C}$ ($1 \leq i \leq k+1$) define

$$\star(F_1, \dots, F_{k+1})(x) := \mathbb{E}_{\substack{y_1, \dots, y_{k+1} \in G \\ y_1 + \dots + y_{k+1} = x}} \prod_{i=1}^{k+1} F_i(y_1, \dots, \widehat{y}_i, \dots, y_{k+1}).$$

The constrained average is normalized: for each x it is the average over the $|G|^k$ solutions of the displayed equation.

Lemma 7.8 (Generalized convolution is tested by the Gowers norm). *Let $r : G \rightarrow \mathbb{C}$ and let $F_1, \dots, F_{k+1}$ be 1-bounded. Then*

$$|\mathbb{E}_{x \in G} r(x) \star (F_1, \dots, F_{k+1})(x)| \leq \|r\|_{U^{k+1}(G)}.$$

Proof. Eliminate x by writing $x = y_1 + \dots + y_{k+1}$. The left-hand side is

$$\Lambda = \mathbb{E}_{y_1, \dots, y_{k+1}} r(y_1 + \dots + y_{k+1}) \prod_{i=1}^{k+1} F_i(y_1, \dots, \widehat{y}_i, \dots, y_{k+1}).$$

We apply Cauchy–Schwarz successively in $y_1, \dots, y_{k+1}$. At the i th application the factor F_i is independent of the variable being doubled, and hence is discarded using $|F_i| \leq 1$. To make the bookkeeping explicit, after the first j applications the remaining copies of r are indexed by $\epsilon \in \{0, 1\}^j$ and have arguments

$$z + \epsilon_1 h_1 + \dots + \epsilon_j h_j + y_{j+1} + \dots + y_{k+1},$$

with conjugation $\mathcal{C}^{|\epsilon|}$; all other surviving factors are still 1-bounded. This statement follows by induction, since squaring an average introduces a second copy of the current variable and their difference becomes the new increment h_j .

After $k+1$ applications no F_i remains. Translating the base variable gives

$$|\Lambda|^{2^{k+1}} \leq \mathbb{E}_{z, h_1, \dots, h_{k+1}} \prod_{\epsilon \in \{0, 1\}^{k+1}} \mathcal{C}^{|\epsilon|} r(z + \epsilon \cdot h) = \|r\|_{U^{k+1}}^{2^{k+1}}.$$

Taking the 2^{k+1} -st root proves the claim. □

For a family $\mathbf{k} = (k_\omega)_{\omega \in \Omega_{s-1}}$ put

$$\mathcal{A}(\mathbf{k}; R) := \mathbb{E}_{h \in H^{s-1}} R(h) \mathbb{E}_{x \in H} \prod_{\omega \in \Omega_{s-1}} \mathcal{C}^{|\omega|} k_\omega(x + \omega \cdot h).$$

Thus, if every $k_\omega = k$, then $\mathcal{A}(\mathbf{k}; R) = \langle \mathcal{S}_{s-1} k, R \rangle$ with the bilinear pairing convention used in this chapter.

Lemma 7.9 (Terminal generalized-convolution estimate). *Suppose that every $k_\omega : H \rightarrow \mathbb{C}$ is a generalized convolution of degree $s - 1$ of 1-bounded functions. If $\|R\|_\infty \leq 1$, then*

$$|\mathcal{A}(\mathbf{k}; R)| \leq \|R\|_{U^s(H^{s-1})}.$$

Proof. Write

$$k_\omega(u) = \mathbb{E}_{\substack{y_{\omega,1}, \dots, y_{\omega,s} \in H \\ y_{\omega,1} + \dots + y_{\omega,s} = u}} \prod_{j=1}^s F_{\omega,j}(y_{\omega,1}, \dots, \widehat{y}_{\omega,j}, \dots, y_{\omega,s}),$$

where every $F_{\omega,j}$ is 1-bounded. Substitute these formulae into $\mathcal{A}$. For each ω split

$$y_{\omega,j} = z_{\omega,j} + \omega \cdot t_j, \quad \sum_{j=1}^s z_{\omega,j} = x, \quad \sum_{j=1}^s t_j = h,$$

with $t_j \in H^{s-1}$. This is a constant-fibre change of variables, so normalized averages are unchanged.

Fix the outer variables x and $z_{\omega,j}$. The remaining average, now over $t_1, \dots, t_s \in H^{s-1}$ with sum h , has the form

$$\mathbb{E}_h R(h) \mathbb{E}_{t_1 + \dots + t_s = h} \prod_{j=1}^s \Phi_j(t_1, \dots, \widehat{t}_j, \dots, t_s),$$

where

$$\Phi_j(t_1, \dots, \widehat{t}_j, \dots, t_s) = \prod_{\omega \in \Omega_{s-1}} \mathcal{C}^{|\omega|} F_{\omega,j}(z_{\omega,1} + \omega \cdot t_1, \dots, \widehat{z_{\omega,j} + \omega \cdot t_j}, \dots, z_{\omega,s} + \omega \cdot t_s).$$

Each Φ_j is 1-bounded. The inner constrained average is therefore a generalized convolution of degree $s - 1$ on the group $G = H^{s-1}$. Lemma 7.8, with $k = s - 1$, bounds its pairing with R by $\|R\|_{U^s(G)}$. This bound is uniform in the fixed outer variables, so averaging them and using the triangle inequality proves the lemma. $\square$

Lemma 7.10 (One vertex Cauchy-Schwarz step). *Assume $\|R\|_\infty \leq 1$ and all k_ω are 1-bounded. Fix $\eta \in \Omega_{s-1}$. There is a new 1-bounded function $k'_\eta : H \rightarrow \mathbb{C}$ which is a generalized convolution of degree $s - 1$ of 1-bounded functions, while $k'_\omega = k_\omega$ for $\omega \neq \eta$, such that*

$$|\mathcal{A}(\mathbf{k}; R)| \leq \|k_\eta\|_2 |\mathcal{A}(\mathbf{k}'; R)|^{1/2}.$$

Proof. For $y \in H$ write

$$B_\eta(y) := \mathbb{E}_{\substack{x \in H, h \in H^{s-1} \\ x + \eta \cdot h = y}} R(h) \prod_{\omega \neq \eta} \mathcal{C}^{|\omega|} k_\omega(x + \omega \cdot h).$$

Then

$$\mathcal{A}(\mathbf{k}; R) = \mathbb{E}_y \mathcal{C}^{|\eta|} k_\eta(y) B_\eta(y),$$

so Cauchy-Schwarz gives

$$|\mathcal{A}(\mathbf{k}; R)|^2 \leq \|k_\eta\|_2^2 \mathbb{E}_y |B_\eta(y)|^2.$$

Define

$$k'_\eta(y) := \mathbb{E}_{x+\eta \cdot h=y} \mathcal{C}^{|\eta|+1} R(h) \prod_{\omega \neq \eta} \mathcal{C}^{|\omega|+|\eta|+1} k_\omega(x + \omega \cdot h).$$

Thus $\mathcal{C}^{|\eta|} k'_\eta(y) = \overline{B_\eta(y)}$, and $|k'_\eta| \leq 1$. Substituting this definition into $\mathcal{A}(\mathbf{k}'; R)$ shows directly that

$$\mathcal{A}(\mathbf{k}'; R) = \mathbb{E}_y |B_\eta(y)|^2.$$

Indeed, the outer cube in $\mathcal{A}(\mathbf{k}'; R)$ and the cube inside the definition of k'_η share the vertex $y = x + \eta \cdot h$; expanding both normalized averages parametrizes exactly an ordered pair of cubes sharing that vertex, which is the expansion of $\mathbb{E}_y B_\eta(y) \overline{B_\eta(y)}$.

It remains to verify the structural assertion about k'_η . Reflect the cube in those coordinates for which $\eta_i = 1$. Equivalently, put

$$u_i = (1 - 2\eta_i)h_i, \quad y = x + \eta \cdot h.$$

Then every other vertex can be written

$$x + \omega \cdot h = y + \sum_{i: \omega_i \neq \eta_i} u_i.$$

Introduce s variables

$$z_i = -u_i \quad (1 \leq i \leq s-1), \quad z_s = y + u_1 + \cdots + u_{s-1}; \quad z_1 + \cdots + z_s = y.$$

The factor coming from R depends only on $z_1, \dots, z_{s-1}$ and hence omits z_s . For a vertex $\omega \neq \eta$, let $j = \min\{i : \omega_i \neq \eta_i\}$. Its argument is

$$y + \sum_{i: \omega_i \neq \eta_i} u_i = z_s + \sum_{i: \omega_i = \eta_i} z_i,$$

which is independent of z_j . Group together all vertex factors having the same first differing coordinate j . We obtain $s-1$ bounded factors, the j th omitting z_j , and the R -factor omitting z_s . This is exactly a generalized convolution of degree $s-1$ in the variables $z_1 + \cdots + z_s = y$. $\square$

Lemma 7.11 (Manners's discrete Cauchy-Schwarz estimate). *For 1-bounded $k : H \rightarrow \mathbb{C}$ and $R : H^{s-1} \rightarrow \mathbb{C}$,*

$$|\langle \mathcal{S}_{s-1} k, R \rangle| \leq \|R\|_{U^s(H^{s-1})}^{1/M'}, \quad M' = 2^{2^{s-1}}.$$

Proof. Initially set $k_\omega = k$ for all $\omega \in \Omega_{s-1}$. Enumerate the $m = 2^{s-1}$ vertices as $\eta_1, \dots, \eta_m$. Apply Lemma 7.10 successively at these vertices. Every function that appears remains 1-bounded, so all L^2 factors are at most 1. Once a vertex has been processed, its replacement is a generalized convolution, and later steps leave that vertex function unchanged. Consequently

$$|\langle \mathcal{S}_{s-1} k, R \rangle| \leq |\mathcal{A}(\mathbf{k}^{(m)}; R)|^{2^{-m}},$$

where every component of $\mathbf{k}^{(m)}$ is a degree- $(s-1)$ generalized convolution of 1-bounded functions. Lemma 7.9 gives

$$|\mathcal{A}(\mathbf{k}^{(m)}; R)| \leq \|R\|_{U^s}.$$

Since $2^m = 2^{2^{s-1}} = M'$, the claimed estimate follows. $\square$

Proof of Proposition 7.5. Define the inverse Fourier transform in the frequency coordinate by

$$T(h_1, \dots, h_{s-1}, y) := \sum_{\chi \in \hat{H}} \chi(-y) \tau(h_1, \dots, h_{s-1}, \chi).$$

Because $\tau \geq 0$ and every fibre has mass at most 1, $\|T\|_\infty \leq 1$. Fourier inversion and the definition of $\tilde{\mathcal{S}}_s f$ give

$$\eta \leq \mathbb{E}_h \sum_{\chi} \tilde{\mathcal{S}}_s f(h, \chi) \tau(h, \chi) = \mathbb{E}_{h,y} \mathcal{S}_s f(h, y) T(h, y).$$

For fixed y put

$$k_y(x) := f(x) \overline{f(x+y)}, \quad T_y(h) := T(h, y).$$

Splitting the s -cube according to its last coordinate yields

$$\mathcal{S}_s f(h_1, \dots, h_{s-1}, y) = \mathcal{S}_{s-1} k_y(h_1, \dots, h_{s-1}).$$

Hence, by the triangle inequality and Lemma 7.11,

$$\eta \leq \mathbb{E}_y |\langle \mathcal{S}_{s-1} k_y, T_y \rangle| \leq \mathbb{E}_y \|T_y\|_{U^s}^{1/M'}.$$

The U^s -norm is at most 1. Monotonicity of L^q norms on a probability space, with exponents $1/M' < 2^s$, therefore gives

$$\eta \leq \left(\mathbb{E}_y \|T_y\|_{U^s}^{2^s} \right)^{1/(2^s M')}.$$

Thus

$$\mathbb{E}_y \|T_y\|_{U^s}^{2^s} \geq \eta^{2^s M'} = \eta^{M_s}.$$

Finally expand the left-hand side. Fourier expansion at each of the 2^s vertices and averaging in y kills every tuple of frequencies except those satisfying

$$\sum_{\omega \in \Omega_s} (-1)^{|\omega|} \chi_\omega = 0.$$

What remains is exactly

$$\mathbb{E}_{a, t_1, \dots, t_s} \sum_{\substack{(\chi_\omega)_{\omega \in \Omega_s} \\ \sum_{\omega} (-1)^{|\omega|} \chi_\omega = 0}} \prod_{\omega \in \Omega_s} \tau(a + \omega \cdot t, \chi_\omega),$$

which proves the proposition. $\square$

Worked Example 7.12 (Where the exponent comes from when $s = 2$). Now $\Omega_{s-1} = \{0, 1\}$ has two vertices. Two vertexwise Cauchy–Schwarz steps give $M' = 2^2 = 4$. The final U^2 -cube average is a fourth moment, so $M = 2^2 M' = 16$. The apparently ungainly exponent is therefore not mysterious: it is the cost of two different operations, vertex dualization and the final Gowers moment.

Corollary 7.13 (Approximate polynomial frequency map). *Assume Proposition 7.5. If $\|f\|_{U^{s+1}} \geq \delta$, then there are a set $S \subset H^{s-1}$ of density $\delta^{O_s(1)}$ and a map $\kappa : S \rightarrow \hat{H}$ such that*

$$\tilde{\mathcal{S}}_s f(x, \kappa(x)) \geq \delta^{O_s(1)} \quad (x \in S),$$

and κ is an approximate polynomial of degree $s - 1$ with parameter $\delta^{O_s(1)}$.

Proof. By Proposition 7.3(iii), the squared mass of $\tilde{\mathcal{S}}_s f$ has average at least $\delta^{2^{s+1}}$. Since its values are in $[0, 1]$ and every fibre has total mass at most 1, a dyadic pigeonhole selects a threshold $\varepsilon = \delta^{O_s(1)}$ and a set of base points S of density $\delta^{O_s(1)}$ for which some frequency has weight at least ε . Choose one such frequency $\kappa(x)$ for each $x \in S$ and let τ be the indicator of its graph. Then

$$\langle \tilde{\mathcal{S}}_s f, \tau \rangle \geq \varepsilon \mu(S).$$

Apply Proposition 7.5. Because τ is the graph of a function, a surviving cube contributes exactly when the alternating sum of the vertex frequencies is zero. That is precisely the vanishing sth derivative condition for a degree- $(s-1)$ approximate polynomial. The parameter is a fixed power of $\varepsilon \mu(S)$. $\square$

7.4 Lifting a circle-valued approximate polynomial to the real line

The second article-level step in Section 5 is elementary enough to internalize now.

Lemma 7.14 (Random integer lift). *Let G be a finite abelian group, $X \subseteq G$, and let*

$$\kappa : X \rightarrow \frac{1}{N}\mathbb{Z}/\mathbb{Z}$$

be an approximate polynomial of degree r with parameter η . Then there exists a lift $\tilde{\kappa} : X \rightarrow \frac{1}{N}\mathbb{Z}$ with $\tilde{\kappa} \bmod 1 = \kappa$ and approximate-polynomial parameter at least

$$\eta' \geq \frac{\eta}{3^{2^{r+1}}} - \frac{2^{2^{r+2}}}{|G|}.$$

Proof. Write $\{y\} \in [0, 1)$ for the standard representative of $y \in \mathbb{R}/\mathbb{Z}$. Choose independently for each $x \in X$ a random correction $R(x) \in \{-1, 0, 1\}$, uniformly, and put

$$\widetilde{K}(x) = \{\kappa(x)\} + R(x).$$

Consider an $(r+1)$ -cube c all of whose 2^{r+1} vertices are distinct and on which $\partial^{r+1}\kappa(c) = 0$ in $\mathbb{R}/\mathbb{Z}$. Then $\partial^{r+1}\{\kappa\}(c)$ is an integer. Its absolute value is less than 2^{r+1} , so among the $3^{2^{r+1}}$ possible choices of the corrections at these vertices there is at least one for which $\partial^{r+1}\widetilde{K}(c) = 0$ exactly in $\mathbb{R}$. Hence

$$\mathbb{P}(\partial^{r+1}\widetilde{K}(c) = 0) \geq 3^{-2^{r+1}}.$$

It remains to bound degenerate cubes. For two distinct vertices $\omega \neq \omega'$ the condition $c(\omega) = c(\omega')$ is one nontrivial linear condition on the $r+2$ cube parameters, hence has at most $|G|^{r+1}$ solutions. By a union bound, the number of cubes with any repeated pair of vertices is at most $2^{2^{r+2}}|G|^{r+1}$. Since at least $\eta|G|^{r+2}$ cubes are good for κ , the expected number good for the random lift is at least

$$3^{-2^{r+1}}\eta|G|^{r+2} - 2^{2^{r+2}}|G|^{r+1}.$$

Some realization of R attains at least this expectation. As κ takes values in $\frac{1}{N}\mathbb{Z}/\mathbb{Z}$ and the corrections are integral, the resulting lift lies in $\frac{1}{N}\mathbb{Z}$, proving the claim. $\square$

7.5 Dependencies in the quantitative argument

The remaining proof follows the architecture of [5]. The main stages are as follows.

- M1. Section 5.1: locate an approximate polynomial.** Proposition 7.3 is proved; Proposition 7.5 is proved by the generalized-convolution and vertexwise Cauchy–Schwarz proof above.
- M2. Section 5.2: real-valued lift.** Lemma 7.14 is proved.
- M3. Section 2.2: systems of cubes.** proved in Chapter 8: pruning, near-full cleanup, and recovery are proved.
- M4. Section 2.3: robust connectivity and weak interaction.** proved in Chapter 8, including the quantitative component split.
- M5. Sections 2.4–2.5: global extension and polynomial hierarchies.** proved in Chapter 8, together with the Appendix-A random induced-connectivity lemma needed by the sampling argument.

- M6. Sections 3.2–3.6: improve the hierarchy.** Put the auxiliary b -functions in normal form, extract cocycles, impose the strong derivatives condition, remove degeneracies/redundancies, locate polynomial pieces by discrete Cauchy–Schwarz, and boost to the full derivative conditions.
- M7. Section 4.1: structure theorem for polynomial hierarchies.** Convert the strengthened hierarchy to a nil-polynomial with the required polynomial dimension bound and single/double-exponential complexity bounds.
- M8. Section 4.2 and Appendix C: quantitative nilmanifold realization.** Construct the filtered nilpotent groups, lattices, polynomial maps and metrics and verify all complexity estimates. Earlier Mal’cev material in this volume supplies part, but not automatically all, of these quantitative owners.
- M9. Sections 5.3–5.4: cover all large Fourier mass and correlate $S_s f$.** Replace the single partial graph by finitely many structured graphs, apply the approximate-polynomial structure theorem, use a partition of unity to avoid an extra exponential at each induction step, and obtain correlation with a finite sum of nilsequences.
- M10. Section 5.5: descend the correlation from $S_s f$ to f .** Reconstruct the sampling/Cauchy–Schwarz argument and the lower-degree induction without importing the older symmetry/anti-differentiation route.
- M11. Global quantitative propagation.** Propagate dimension, nilmanifold complexity, Lipschitz parameter and correlation losses to the exact theorem envelope: polynomial dimension; single exponential for $s \leq 3$ and double exponential for $s \geq 4$ in the remaining parameters.

Chapter 8

Manners quantitative inverse theory II: from partial cubes to global polynomial hierarchies

8.1 Cubes, faces, and the meaning of a partial polynomial

For $k \geq 0$ write $\Omega_k = \{0, 1\}^k$. An affine k -cube in a finite abelian group H is a map

$$c = \angle(x; h_1, \dots, h_k) : \Omega_k \rightarrow H, \quad c(\omega) = x + \sum_{i=1}^k \omega_i h_i.$$

We denote the set of all such cubes by $\mathcal{C}^k(H)$. Thus $|\mathcal{C}^k(H)| = |H|^{k+1}$. If $a \in H$, the translated cube is $c + a$, obtained by adding a to every vertex. If $c \in \mathcal{C}^k(H)$ and $a \in H$, then

$$[c, c + a] \in \mathcal{C}^{k+1}(H)$$

means the $(k + 1)$ -cube whose lower face is c and upper face is $c + a$. A face is obtained by fixing some coordinates of ω to 0 or 1. Coordinate permutations and the reflections $\omega_i \mapsto 1 - \omega_i$ are called cube symmetries.

For a real-valued function F defined on all vertices of c , set

$$\partial^k F(c) = \sum_{\omega \in \Omega_k} (-1)^{|\omega|} F(c(\omega)).$$

If $X \subseteq H$, write $\mathcal{C}^k(X)$ for cubes all of whose vertices lie in X . Normalized counting measure on whichever cube space is under discussion is written μ .

Definition 8.1 (Approximate polynomial in the Manners sense). Let $F : X \rightarrow \mathbb{R}$, let $s \geq 0$, and let $0 < \delta \leq 1$. We call F an *approximate polynomial of degree s and parameter δ* if

$$\mu\{c \in \mathcal{C}^{s+1}(X) : \partial^{s+1} F(c) = 0\} \geq \delta,$$

where the density is measured in the full cube space $\mathcal{C}^{s+1}(H)$. Thus the hypothesis remembers both that the cube lies in the domain and that the $(s + 1)$ st additive derivative vanishes.

Idea Why a system of cubes is needed

A dense set of good $(s + 1)$ -cubes need not use every vertex equally often. If one only keeps the top-dimensional cubes, later restrictions to faces may leave the domain. Manners therefore prunes simultaneously in every dimension. The resulting object says not only that a cube is good, but that it has many good extensions one dimension up. This is the exact robustness needed by the later random-translation argument.

8.2 Popular systems of cubes

Definition 8.2 (System of cubes). A *system of cubes of degree s and parameter $\delta > 0$* is a collection

$$\mathcal{S}_0, \mathcal{S}_1, \dots, \mathcal{S}_{s+1}, \quad \mathcal{S}_k \subseteq \mathcal{C}^k(H),$$

satisfying:

- (i) **symmetry:** every $\mathcal{S}_k$ is invariant under all cube symmetries;
- (ii) **faces:** every face of a cube in $\mathcal{S}_k$ belongs to the corresponding lower-dimensional $\mathcal{S}_j$;
- (iii) **popularity:** for $0 \leq k \leq s$ and $c \in \mathcal{S}_k$, at least $\delta|H|$ elements $a \in H$ satisfy $[c, c + a] \in \mathcal{S}_{k+1}$.

The system is *nonempty* if one, hence every, level forced by the face condition is nonempty.

Lemma 8.3 (Automatic density at every level). *If $\mathcal{S}_0, \dots, \mathcal{S}_{s+1}$ is a nonempty system with parameter δ , then*

$$\mu(\mathcal{S}_k) \geq \delta^{k+1} \quad (0 \leq k \leq s + 1).$$

In particular $\mu(\mathcal{S}_0) \geq \delta$.

Proof. Nonemptiness gives some $x_0 \in \mathcal{S}_0$. Popularity at level 0 says that at least $\delta|H|$ translates $x_0 + a$ lie in $\mathcal{S}_0$ (they are the upper vertices of members of $\mathcal{S}_1$). Thus $|\mathcal{S}_0| \geq \delta|H|$. Now suppose $|\mathcal{S}_j| \geq \delta^{j+1}|H|^{j+1}$. Every $c \in \mathcal{S}_j$ has at least $\delta|H|$ increments a for which $[c, c + a] \in \mathcal{S}_{j+1}$. In the standard parameterization of affine cubes, the pair (c, a) determines the extended cube uniquely. Hence

$$|\mathcal{S}_{j+1}| \geq \delta|H| |\mathcal{S}_j| \geq \delta^{j+2}|H|^{j+2}.$$

Induction gives $|\mathcal{S}_k| \geq \delta^{k+1}|H|^{k+1}$ for every k , which is the asserted normalized density bound. $\square$

Lemma 8.4 (Pruning dense top-dimensional cubes). *Fix $s \geq 0$. Suppose sets $\mathcal{A}_k \subseteq \mathcal{C}^k(H)$, $0 \leq k \leq s + 1$, satisfy the symmetry and face conditions of Definition 8.2, and suppose $\mu(\mathcal{A}_{s+1}) \geq \eta$. Then there is a nonempty cube system $\mathcal{S}_k \subseteq \mathcal{A}_k$ with parameter*

$$\delta \geq c_s \eta$$

for a constant $c_s > 0$ depending only on s .

Proof. Set

$$\delta = 2^{-2s-4}(s+1)^{-2}\eta.$$

Starting with the sets $\mathcal{A}_k$, repeatedly perform the following deletion. If $c \in \mathcal{A}_k$ has fewer than $\delta|H|$ extensions $[c, c + a] \in \mathcal{A}_{k+1}$, delete c . To preserve the face axiom, also delete every cube in dimension $k + 1$ having c as a codimension-one face, and then every higher cube

having one of those deleted cubes as a face. After each complete deletion the symmetry and face axioms remain true.

We only have to show that the top level cannot be exhausted. A fixed deleted k -cube creates at most $2(k+1)\delta|H|$ deleted $(k+1)$ -cubes: there are $2(k+1)$ choices of codimension-one face, and the failure of popularity bounds the possible new increments. Crude face counting then shows that at most

$$4^{s+1}2(s+1)\delta|H|^{s+1-k}$$

top-dimensional cubes are ultimately deleted because of this k -cube. Since there are at most $|H|^{k+1}$ possible k -cubes and each is deleted at most once, the total number removed from the top level is at most

$$\sum_{k=0}^s 4^{s+1}2(s+1)\delta|H|^{s+2} \leq \frac{\eta}{2}|H|^{s+2}$$

by the chosen value of δ . At least half of the original top-dimensional good cubes survive. The process terminates on a nonempty family satisfying popularity, and hence gives the required system. $\square$

Corollary 8.5 (Good derivative cubes produce a cube system). *If $F : X \rightarrow \mathbb{R}$ is an approximate polynomial of degree s and parameter η , then there is a nonempty system $\mathcal{S}_0, \dots, \mathcal{S}_{s+1}$ of parameter $\gg_s \eta$ such that $\mathcal{S}_0 \subseteq X$ and*

$$\partial^{s+1}F(c) = 0 \quad (c \in \mathcal{S}_{s+1}).$$

Proof. Take $\mathcal{A}_k = \mathcal{C}^k(X)$ for $0 \leq k \leq s$ and take $\mathcal{A}_{s+1}$ to be the good $(s+1)$ -cubes for F . These sets are symmetric and closed under faces, so Lemma 8.4 applies. $\square$

The next result is the near-full-density counterpart. It is essential later because random constructions lose a small exceptional set at several dimensions at once.

Lemma 8.6 (Near-full sets contain an almost-full cube system). *For every s there are constants $C_s, c_s > 0$ such that the following holds. If $\mathcal{A}_k \subseteq \mathcal{C}^k(H)$ and $\mu(\mathcal{A}_k) \geq 1 - \eta$ for $0 \leq k \leq s+1$, then there is a nonempty cube system $\mathcal{S}_k \subseteq \mathcal{A}_k$ with parameter at least*

$$1 - C_s \eta^{c_s},$$

provided this lower bound is positive.

Proof. We give the cleanup argument because its two-stage form is used repeatedly later. First replace each $\mathcal{A}_k$ by the subset on which every coordinate permutation and reflection remains in $\mathcal{A}_k$. There are only $O_s(1)$ such symmetries, so this changes the error from η to at most $C_s \eta$. Next retain only cubes all of whose faces lie in the already symmetrized lower-dimensional sets. A random face of a uniform random affine cube is uniform in its own cube space, and there are at most 4^k faces. A union bound therefore leaves sets $\mathcal{B}_k$ of density $1 - O_s(\eta)$ which satisfy symmetry and the face axiom.

We now prune failures of popularity without losing a macroscopic set. Put $\mathcal{S}_{s+1}^{(2)} = \mathcal{B}_{s+1}$ and work downwards. For $0 \leq k \leq s$ let T_k be the set of k -cubes c for which more than a proportion θ of increments a fail to satisfy $[c, c+a] \in \mathcal{S}_{k+1}^{(2)}$. Let U_k be the set of c for which more than a proportion θ of the translates $c+a$ belong to T_k , and put

$$\mathcal{S}_k^{(2)} = \mathcal{C}^k(H) \setminus (T_k \cup U_k).$$

This downward definition really stays inside the face-closed family $\mathcal{B}_k$. Indeed the assertion is clear at the top level. Inductively, if $c \in \mathcal{S}_k^{(2)}$, then $c \notin T_k$, so at least one increment a has

$[c, c + a] \in \mathcal{S}_{k+1}^{(2)} \subseteq \mathcal{B}_{k+1}$. The face axiom for $\mathcal{B}$ then gives $c \in \mathcal{B}_k$. Thus $\mathcal{S}_k^{(2)} \subseteq \mathcal{B}_k \subseteq \mathcal{A}_k$ at every level. Fubini gives

$$\mu(T_k) \leq \frac{1 - \mu(\mathcal{S}_{k+1}^{(2)})}{\theta}, \quad \mu(U_k) \leq \frac{1 - \mu(\mathcal{S}_{k+1}^{(2)})}{\theta^2}.$$

Consequently, if $e_k = 1 - \mu(\mathcal{S}_k^{(2)})$, then

$$e_k \leq (\theta^{-1} + \theta^{-2})e_{k+1}.$$

Thus

$$e_0 \leq C_s \eta (\theta^{-1} + \theta^{-2})^{s+1}.$$

The point here is only to keep the zero level nonempty; the intermediate $\mathcal{S}_k^{(2)}$ need not yet have density $1 - O(\eta^{c_s})$. Choose

$$\theta = A_s \eta^{1/(2(s+1))}$$

with A_s large enough (and then restrict to the range in which $1 - C_s \eta^{1/(2(s+1))} > 0$). Since $\theta^{-1} + \theta^{-2} \leq 2\theta^{-2}$ for $\theta \leq 1$, this makes $e_0 < 1$, so $\mathcal{S}_0^{(2)}$ is nonempty.

Finally retain from $\mathcal{S}_k^{(2)}$ only those cubes every one of whose faces lies in the corresponding $\mathcal{S}_j^{(2)}$. This restores the face axiom exactly, and at level zero it deletes nothing, so nonemptiness is preserved. For a retained k -cube c , consider a random increment a . Every face of $[c, c + a]$ is either a face of c , a translate of a face of c , or an extension of a lower-dimensional face of c . In the first case there is no failure; in the second, $c|_F \notin U_j$ bounds the bad proportion by θ ; in the third, $c|_F \notin T_j$ does the same. Since there are $O_s(1)$ faces, a union bound shows that at least $(1 - C_s \theta)|H|$ increments are good. Hence the final sets form a cube system with parameter

$$1 - C_s \theta \geq 1 - C'_s \eta^{1/(2(s+1))}.$$

This proves the lemma (with, for example, $c_s = 1/(2(s+1))$ after changing the constant). Notice that near-full density of the final levels now follows from Lemma 8.3, not from the intermediate recurrence above. $\square$

Corollary 8.7 (Recovery after deleting a few cubes). *Let $\mathcal{S}_0, \dots, \mathcal{S}_{s+1}$ be a cube system of parameter δ . Suppose $\mathcal{A}_k \subseteq \mathcal{S}_k$ and $\mu(\mathcal{S}_k \setminus \mathcal{A}_k) \leq \eta$ for every k . Then, for constants $C_s, c_s > 0$, there is a nonempty subsystem $\mathcal{S}'_k \subseteq \mathcal{A}_k$ with parameter*

$$\delta' \geq \delta - C_s \eta^{c_s},$$

provided the right side is positive; moreover $\mu(\mathcal{S}_k \setminus \mathcal{S}'_k) \leq C_s \eta^{c_s}$.

Proof. Apply Lemma 8.6 to $\mathcal{C}^k(H) \setminus (\mathcal{S}_k \setminus \mathcal{A}_k)$, obtaining an almost-full system $\mathcal{T}_k$ with parameter $1 - \rho$, where $\rho \leq C_s \eta^{c_s}$. Then $\mathcal{S}'_k = \mathcal{S}_k \cap \mathcal{T}_k$ is a system of parameter at least $\delta - \rho$: an extension is good for both systems except for at most the sum of their bad proportions. By Lemma 8.3, $\mu(\mathcal{T}_k) \geq (1 - \rho)^{k+1} \geq 1 - (s+2)\rho$, so $\mu(\mathcal{S}_k \setminus \mathcal{S}'_k) = O_s(\rho)$ as claimed. $\square$

Worked Example 8.8 (Why face closure is not cosmetic). Suppose many parallelograms $(x, x + a, x + b, x + a + b)$ satisfy a quadratic identity, but a small exceptional set of vertices appears disproportionately often. Keeping only the parallelograms does not ensure that a chosen good edge can be extended again. The cube system removes such exceptional vertices, then the edges that depend on them, and so on, until every surviving face has many surviving extensions. The bookkeeping in Lemma 8.4 is precisely what prevents this cascading deletion from destroying all top-dimensional cubes.

8.3 Robust connectivity and non-interacting pieces

For a finite graph $G = (V, E)$ and $A \subseteq V$ with $0 < |A| \leq |V|/2$, write $E(A, V \setminus A)$ for crossing oriented pairs and define

$$H(G) = \min_A \frac{|E(A, V \setminus A)|}{|A|}, \quad h(G) = \frac{H(G)}{|V|}.$$

Thus $h(G)$ is the normalized Cheeger constant.

Lemma 8.9 (Robust component decomposition). *Fix $s \geq 0$. Let $\mathcal{S}_0, \dots, \mathcal{S}_{s+1}$ be a nonempty system with parameter δ . There is $\varepsilon_0 = \varepsilon_0(\delta, s) > 0$ such that, whenever $0 < \varepsilon \leq \varepsilon_0$, there is a subsystem $\mathcal{S}'_0, \dots, \mathcal{S}'_{s+1}$ with parameter*

$$\delta' \geq \delta - C_s \frac{\varepsilon^{c_s}}{\delta}$$

and with $\mu(\mathcal{S}_k \setminus \mathcal{S}'_k) \leq C_s \varepsilon^{c_s} / \delta$, such that the graph

$$V = \mathcal{S}'_0, \quad xy \in E \iff x \neq y \text{ and } [x, y] \in \mathcal{S}'_1$$

has at most $1/\delta'$ connected components and every component V_i satisfies

$$H(G[V_i]) \geq \varepsilon |H|.$$

Every cube in $\mathcal{S}'_k$ has all of its vertices in a single component, so each component supports its own non-interacting subsystem with the same parameter δ' .

Proof. Let $\rho = C_s \varepsilon^{c_s}$ be the loss supplied by Corollary 8.7, with constants chosen so that $\rho \geq \varepsilon$ and $\rho \leq \delta^2/4$. Begin with the given system. At stage i let its parameter be $\delta_i = \delta - i\rho$ and let G_i be the graph defined by its first two levels. Popularity gives minimum degree at least $\delta_i |H| - 1$, hence every connected component contains at least $\delta_i |H|$ vertices and there are at most $1/\delta_i$ components.

If every component has Cheeger constant at least $\varepsilon |H|$, stop. Otherwise one component $A \sqcup B$ has $0 < |A| \leq |B|$ and fewer than $\varepsilon |A| |H|$ crossing edges. The minimum-degree bound gives

$$|E(A, B)| \geq |A|(\delta_i |H| - |A|),$$

so $|A| > (\delta_i - \varepsilon) |H|$. Delete from the first cube level all oriented edges between A and B . Their normalized number is at most $O(\varepsilon)$. Apply Corollary 8.7 to repair all higher face and popularity conditions, losing at most ρ in normalized density and parameter. No new crossing edge between A and B can reappear because the repaired first level is a subset of the cut graph.

Hence each unsuccessful stage increases the number of connected components by at least one, unless an entire one of the already separated pieces disappears. But every such piece has size at least $(\delta_i - \varepsilon) |H|$, whereas at most $\rho |H|$ vertices are removed in one repair. For $i \leq T = \lfloor 2/\delta \rfloor$,

$$\delta_i - \varepsilon - \rho \geq \delta - (T + 1)\rho > 0,$$

so no piece disappears. If the procedure survived all the way to time T , the graph would have at least $T + 1 > 2/\delta$ components; but it has at most $1/(\delta - T\rho) \leq 2/\delta$, a contradiction. Thus it stops earlier. The total loss is at most $T\rho = O_s(\varepsilon^{c_s}/\delta)$.

Finally, the one-dimensional faces of an affine k -cube form a connected cube graph. By the face axiom every such edge belongs to $\mathcal{S}'_1$, so all vertices of a surviving cube lie in one graph component. Intersecting each level with cubes whose vertices lie in a fixed component gives the claimed non-interacting subsystems. $\square$

8.4 A sampled expander remains connected

The global extension proof samples finitely many translates. The next lemma is the Appendix-A input used by Manners; we include its proof so M4 has no hidden graph theorem.

Lemma 8.10 (Random induced connectivity). *Let $G = (V, E)$ have normalized Cheeger constant $0 < h \leq 1$. Choose $x_1, \dots, x_m$ independently and uniformly from V and form the graph G' on $[m]$ by joining i and j whenever $x_i x_j \in E$. Then*

$$\mathbb{P}(G' \text{ is connected}) \geq 1 - (m+1) \exp\left(-\frac{(m-1)h^4}{64 \log(2/h^2)}\right).$$

Repeated samples are allowed.

Proof. We expose the samples in two blocks. Put $\ell = \lceil m/2 \rceil$ and first expose $x_1, \dots, x_\ell$. Write $\Gamma(v)$ for the neighbors of v in G . Set $S_1 = \Gamma(x_1)$ and, for $2 \leq i \leq \ell$, let

$$S_i = \begin{cases} S_{i-1} \cup \Gamma(x_i), & x_i \in S_{i-1}, \\ S_{i-1}, & x_i \notin S_{i-1}. \end{cases}$$

The indices at which the first alternative occurs, together with 1, select a connected subgraph of the sampled graph, and the union of their neighborhoods contains S_ℓ .

For every $S \subseteq V$, the normalized Cheeger bound gives

$$\mathbb{E}_{y \in V} \mathbf{1}_{y \in S} |\Gamma(y) \setminus S| = \frac{|E(S, V \setminus S)|}{|V|} \geq h \min(|S|, |V \setminus S|).$$

Singletons show $|S_1| \geq h|V|$; therefore the last quantity is at least $h^2(|V| - |S|)$ whenever $S = S_i$. If $T_i = 1 - |S_i|/|V|$, then

$$\mathbb{E}(T_i \mid T_1, \dots, T_{i-1}) \leq (1 - h^2)T_{i-1}.$$

Let $R_i = T_i/T_{i-1}$ (with an arbitrary value if $T_{i-1} = 0$) and cap it below by $R'_i = \max(h^2/2, R_i)$. Then $h^2/2 \leq R'_i \leq 1$ and

$$\mathbb{E}(R'_i \mid R'_1, \dots, R'_{i-1}) \leq 1 - h^2/2.$$

Consequently

$$L_i = i \log(1 - h^2/2) - \sum_{j=1}^i \log R'_j$$

is a submartingale with increments bounded in absolute value by $\log(2/h^2)$. Azuma–Hoeffding implies

$$\mathbb{P}(T_\ell \geq (1 - h^2/2)^{\ell/2}) \leq \exp\left(-\frac{\ell h^4}{32 \log(2/h^2)}\right).$$

Thus, except with this probability, S_ℓ misses at most $\rho|V|$ vertices, where $\rho \leq e^{-\ell h^2/4}$.

Now expose $x_{\ell+1}, \dots, x_m$. With probability at least $1 - (m - \ell)\rho$, every new sample lies in S_ℓ and hence has a neighbor in the connected selected first-block subgraph. Also every first-block sample has degree at least $h|V|$, so the chance it has no neighbor in the second block is at most $(1 - h)^{m-\ell}$. A union bound therefore gives failure probability at most

$$p + (m - \ell)\rho + \ell(1 - h)^{m-\ell},$$

where p is the first-stage Azuma failure probability. With $\ell = \lceil m/2 \rceil$, each term is bounded by

$$\exp\left(-\frac{(m-1)h^4}{64 \log(2/h^2)}\right),$$

and the asserted estimate follows. $\square$

8.5 Extending one robust component to almost all of the group

Theorem 8.11 (Manners global extension lemma). *Let $s \geq 1$. Suppose $\mathcal{S}_0, \dots, \mathcal{S}_{s+1}$ is a nonempty system of parameter $\delta > 0$, $F : \mathcal{S}_0 \rightarrow \mathbb{R}$ satisfies*

$$\partial^{s+1} F(c) = 0 \quad (c \in \mathcal{S}_{s+1}),$$

and the graph on $\mathcal{S}_0$ with edge xy when $[x, y] \in \mathcal{S}_1$ has normalized Cheeger constant $h > 0$. For every $\varepsilon > 0$ there exist:

- (i) $X \subseteq H$, $\mu(X) \geq 1 - \varepsilon$, and $\tilde{F} : X \rightarrow \mathbb{R}$ agreeing with F on $X \cap \mathcal{S}_0$;
- (ii) a set $\mathcal{Q} \subseteq \mathcal{C}^{s+1}(H)$ of density at least $1 - \varepsilon$, all of whose vertices lie in X ;
- (iii) integers K, M satisfying

$$K, M \ll_s \left(\delta^{-1} h^{-4} \log \frac{2}{\varepsilon} \log \frac{2}{\delta} \log^2 \frac{2}{h} \right)^{O_s(1)};$$

- (iv) for $1 \leq r \leq K$, a nonempty degree- $(s-1)$ cube system $\mathcal{S}_0^{(r)}, \dots, \mathcal{S}_s^{(r)}$ of parameter δ and a function $G_r : \mathcal{S}_0^{(r)} \rightarrow \mathbb{R}$ with $\partial^s G_r = 0$ on $\mathcal{S}_s^{(r)}$;

such that for every $c \in \mathcal{Q}$ there are vectors $b(c, \omega) \in \mathbb{Z}^K$, $\|b(c, \omega)\|_1 \leq M$, with

$$\partial^{s+1} \tilde{F}(c) = \sum_{\omega \in \Omega_{s+1}} (-1)^{|\omega|} \sum_{r=1}^K b(c, \omega)_r G_r(c(\omega)),$$

and every displayed value $G_r(c(\omega))$ is defined whenever its coefficient is nonzero.

Proof. Step 1: choose a sample size. Choose L so that

$$L \ll_s \delta^{-1} h^{-4} \log \frac{2}{\varepsilon} \log \frac{2}{\delta} \log^2 \frac{2}{h}$$

and, simultaneously,

$$\begin{aligned} e^{-\delta L/8} &\leq 2^{-s-1}(\varepsilon/40)(L+1)^{-s-1}, \\ (\delta L/2 + 1)e^{-(\delta L/2 - 1)h^4/(64 \log(2/h^2))} &\leq 2^{-s-1}(\varepsilon/40)(L+1)^{-s-1}, \\ 2e^{-\delta(L-1)/2} &\leq 2^{-s-1}(\varepsilon/20)(L+1)^{-s-1}, \\ (s+1)e^{-\delta L} &\leq \varepsilon/10. \end{aligned}$$

Such an L exists because an exponential eventually dominates any fixed power; the stated logarithmic envelope is obtained by applying this observation to the four inequalities.

Choose $a_1, \dots, a_L$ independently and uniformly in H . For $a, b \in H$ define

$$\mathcal{S}_k^{(a,b)} = \{c \in \mathcal{C}^k(H) : [c+a, c+b] \in \mathcal{S}_{k+1}\}, \quad 0 \leq k \leq s.$$

Whenever nonempty these sets form a degree- $(s-1)$ system with parameter δ . Define

$$G_{a,b}(x) = F(x+a) - F(x+b) \quad (x \in \mathcal{S}_0^{(a,b)}).$$

For $c \in \mathcal{S}_s^{(a,b)}$,

$$\partial^s G_{a,b}(c) = \partial^{s+1} F([c+a, c+b]) = 0.$$

Step 2: glue sampled translates at a typical point. Fix $x \in H$ and let $I_x = \{i : x + a_i \in \mathcal{S}_0\}$. Chernoff gives $|I_x| \geq \delta L/2$ except with probability at most $e^{-\delta L/8}$. Conditional on the selected indices, the points $x + a_i$ with $i \in I_x$ are independent uniform samples from $\mathcal{S}_0$. Lemma 8.10 and the second choice of L therefore imply that, except with probability $O_s(\varepsilon L^{-s-1})$, the sampled edge graph on I_x is connected. Hence for any $k, \ell \in I_x$ a path $k = i_0, i_1, \dots, i_m = \ell$ gives the exact telescoping identity

$$F(x + a_k) - F(x + a_\ell) = \sum_{j=0}^{m-1} G_{a_{i_j}, a_{i_{j+1}}}(x).$$

Orient each pair increasingly; the coefficients are then in $\{-1, 0, 1\}$ and each unordered pair is used at most once. Let X_0 be the set of x for which this gluing conclusion holds. Averaging in x gives

$$\mathbb{E}\mu(X_0) \geq 1 - O_s(\varepsilon L^{-s-1}).$$

For $x \in X_0$, define $F'(x) = F(x + a_i)$ for the least i with $x + a_i \in \mathcal{S}_0$.

Step 3: bridge one arbitrary displacement. Fix $x, u \in H$. Split the indices into two halves. With probability at least $1 - e^{-\delta(L-1)/2}$, one first-half sample $x + a_k$ lies in $\mathcal{S}_0$. Given such a point, popularity in the cube system says that at least $\delta|H|$ points y satisfy $[x + a_k, y] \in \mathcal{S}_1$. Therefore a second-half sample $x + u + a_\ell$ hits this good set except with probability $e^{-\delta L/2}$. Thus, except with the third prescribed failure bound, some k, ℓ satisfy

$$[x + a_k, x + u + a_\ell] \in \mathcal{S}_1.$$

If also $x, x + u \in X_0$, compare the representatives defining $F'(x)$ and $F'(x + u)$ to a_k and a_ℓ . The path identities from Step 2 show

$$F'(x) - F'(x + u) = G_{a_k, a_\ell + u}(x) + \sum_{i < j} (\beta_{ij} G_{a_i, a_j}(x) + \beta'_{ij} G_{a_i, a_j}(x + u))$$

with coefficients in $\{-1, 0, 1\}$, all terms being defined.

Step 4: make the bridges work simultaneously for a high derivative. Choose fresh independent uniform shifts $u_1, \dots, u_L$ and put $T = \{0, u_1, \dots, u_L\}$. Let $(s+1)T$ be the $(s+1)$ -fold sumset. Define X_1 to be the set of points for which the bridge assertion in Step 3 succeeds for every $u \in (s+1)T$, and define

$$X_2 = \{x \in X_1 : x + t \in X_0 \text{ for every } t \in (s+1)T\}.$$

The preceding failure estimates and $|(s+1)T| \leq (L+1)^{s+1}$ give

$$\mathbb{E}\mu(X_2) \geq 1 - 2^{-s-1}\varepsilon/10.$$

List as $G'_1, \dots, G'_K$ all nonempty functions of the forms

$$G_{a_i, a_j}, \quad G_{a_i + t, a_j + t}, \quad G_{a_i, a_j + t} \quad (i < j, \ t \in (s+1)T).$$

Then $K \leq 3L^2(L+1)^{s+1} \leq 3(L+1)^{s+3}$, and each function still has a parameter- δ degree- $(s-1)$ cube system on which its s th derivative vanishes.

Now fix $c = \angle(x; v_1, \dots, v_{s+1})$ with every vertex in X_2 , and choose k with $x + a_k \in \mathcal{S}_0$. Starting with that point, select $t_1, \dots, t_{s+1} \in T$ recursively. If a current m -face belongs to $\mathcal{S}_m$, popularity supplies at least $\delta|H|$ valid next increments; one of the independent choices $v_{m+1} + u_j$ hits them except with probability $e^{-\delta L}$. By the fourth choice of L , with probability at least $1 - \varepsilon/10$ we obtain

$$c' = \angle(x + a_k; v_1 + t_1, \dots, v_{s+1} + t_{s+1}) \in \mathcal{S}_{s+1}.$$

Since $\partial^{s+1}F(c') = 0$,

$$\partial^{s+1}F'(c) = \sum_{\omega} (-1)^{|\omega|} (F'(c(\omega)) - F'(c'(\omega))).$$

For a fixed vertex, write $t = \omega \cdot (t_1, \dots, t_{s+1}) \in (s+1)T$. First compare $F'(c(\omega) + t)$ to $F'(c'(\omega))$ using the path identity of Step 2; this costs at most L^2 terms among the G'_r . Then compare $F'(c(\omega))$ with $F'(c(\omega) + t)$ using Step 3; this costs at most $2L^2 + 1$ further terms. Hence

$$F'(c(\omega)) - F'(c'(\omega)) = \sum_{r=1}^K b(c, \omega)_r G'_r(c(\omega)), \quad \|b(c, \omega)\|_1 \leq 3L^2 + 1.$$

This is exactly the required derivative representation.

Step 5: freeze one successful random choice and translate back. Let $\mathcal{Q}'$ be the set of cubes for which the Step 4 construction succeeds. Averaging over the random choices gives $\mathbb{E}\mu(\mathcal{Q}') \geq 1 - \varepsilon/5$. Markov's inequality therefore allows us to fix all random translates so that simultaneously $\mu(X_2) \geq 1 - \varepsilon$ and $\mu(\mathcal{Q}') \geq 1 - \varepsilon$. Translate everything by a_1 : put $X = X_2 + a_1$, $\tilde{F}(y) = F'(y - a_1)$, and translate every auxiliary function and cube system similarly. Because the representative rule uses the least index, for $y \in X \cap \mathcal{S}_0$ the first translate is available and $\tilde{F}(y) = F(y)$. The derivative identity is translation invariant, and $M = 3L^2 + 1$ together with the bound for K has the asserted quantitative envelope. $\square$

Worked Example 8.12 (The fractional-part product). For $H = \mathbb{Z}/N\mathbb{Z}$ and

$$F(x) = \left\{ \frac{\alpha x}{N} \right\} \left\{ \frac{\beta x}{N} \right\},$$

F is not a global quadratic polynomial as a real-valued function because each fractional-part coordinate jumps at an integer boundary. Nevertheless its third additive derivative is a bounded integer linear combination of the lower-degree coordinates $\{\alpha x/N\}$, $\{\beta x/N\}$ and 1 evaluated at cube vertices. Theorem 8.11 abstracts exactly this phenomenon: a partial degree- s object acquires a global representative whose $(s+1)$ st derivatives are controlled by finitely many degree- $(s-1)$ objects.

8.6 Polynomial hierarchies and recursive global structure

The global extension theorem does not finish the problem: its auxiliary functions are only lower-degree approximate polynomials. We therefore iterate the construction and record the dependency in a graded hierarchy.

Definition 8.13 (Derivative condition and polynomial hierarchy). Let $X \subseteq H$ and let

$$\mathbf{F} = (F_{i,j} : 0 \leq i \leq s, 1 \leq j \leq d_i)$$

be real-valued functions on X . Write $\mathbf{F}_{\leq t}$ for all levels at most t . Let $G : X \rightarrow \mathbb{R}$, let $\mathcal{Q} \subseteq \mathcal{C}^k(H)$ have all vertices in X , and let $0 \leq t \leq s+1$. We say that G satisfies the $(\mathbf{F}, k, t, M)$ *derivative condition* on $\mathcal{Q}$ if there are integer coefficient vectors

$$b(c, \omega) \in \bigoplus_{r=0}^{t-1} \mathbb{Z}^{d_r}$$

such that

$$\partial^k G(c) = \sum_{\omega \in \Omega_k} (-1)^{|\omega|} b(c, \omega) \cdot \mathbf{F}_{\leq t-1}(c(\omega))$$

and the level- r part obeys

$$\|b_{=r}(c, \omega)\|_1 \leq M^{t-r} \quad (0 \leq r < t).$$

For $t = 0$ this means simply $\partial^k G(c) = 0$ on $\mathcal{Q}$.

Now let $\mathcal{S}_0, \dots, \mathcal{S}_{s+1}$ be a nonempty cube system with $\mathcal{S}_0 \subseteq X$. We call $\mathbf{F}$ an $(s, \mathbf{d}, M)$ *polynomial hierarchy* on this system if every coordinate $F_{i,j}$ satisfies the $(\mathbf{F}_{\leq i-1}, i+1, i, M)$ derivative condition on $\mathcal{S}_{i+1}$. The total dimension is $D = \sum_i d_i$.

Remark 8.14 (Why the graded bound is M^{t-r}). One could replace all coefficient bounds by a larger common power of M when s is fixed. The graded form is preferable because later substitution of a derivative condition of degree r into one of degree t adds exponents in exactly the direction encoded by M^{t-r} .

Theorem 8.15 (Recursive global structure; Manners Corollary 2.5.4). *Let $s \geq 1$, let $X \subseteq H$, and let $G : X \rightarrow \mathbb{R}$ be an approximate polynomial of degree s and parameter $\delta > 0$. For every $\varepsilon > 0$ there exist*

- a system $\mathcal{S}_0, \dots, \mathcal{S}_{s+1}$ of parameter $1 - \varepsilon$;
- an $(s-1, \mathbf{d}, M)$ polynomial hierarchy $\mathbf{F}$ on the first $s+1$ levels;
- a set $Y \subseteq X \cap \mathcal{S}_0$ with $\mu(Y) \gg_s \delta^{O_s(1)}$;
- a function $\tilde{G} : \mathcal{S}_0 \rightarrow \mathbb{R}$ agreeing with G on Y and satisfying the $(\mathbf{F}, s+1, s, M)$ derivative condition on $\mathcal{S}_{s+1}$;

with

$$D = \sum_{i=0}^{s-1} d_i \ll_s (\varepsilon \delta)^{-O_s(1)}, \quad M \ll_s (\varepsilon \delta)^{-O_s(1)}.$$

Proof. Apply Corollary 8.5 to obtain a system $\mathcal{T}_0^{(0)}, \dots, \mathcal{T}_{s+1}^{(0)}$ of parameter $\delta_0 \gg_s \delta$ on which $\partial^{s+1} G$ vanishes. Apply Lemma 8.9 with a threshold $h_0 = \delta^{O_s(1)}$ small enough that the surviving parameter remains at least $\delta_0/2$. Pass to one connected component. It still carries a cube system of parameter $\delta_1 \geq \delta_0/2$, its normalized Cheeger constant is at least h_0 , and its zero-level set has density at least δ_1 .

We prove the remaining statement by induction on the degree t . The inductive claim is as follows. Suppose $\mathcal{T}_0, \dots, \mathcal{T}_{t+1}$ has parameter $\rho > 0$, its first-level graph has normalized Cheeger constant at least $h > 0$, and $g : \mathcal{T}_0 \rightarrow \mathbb{R}$ has $\partial^{t+1} g = 0$ on $\mathcal{T}_{t+1}$. For any $\eta > 0$ there is a parameter- $(1 - \eta)$ system $\mathcal{R}_0, \dots, \mathcal{R}_{t+1}$, a degree- $(t-1)$ hierarchy $\mathbf{P}$, and $\tilde{g} : \mathcal{R}_0 \rightarrow \mathbb{R}$ such that

$$g = \tilde{g} \text{ on } \mathcal{R}_0 \cap \mathcal{T}_0,$$

$\tilde{g}$ obeys the $(\mathbf{P}, t+1, t, M)$ derivative condition, and

$$D, M \ll_t (\eta \rho h)^{-O_t(1)}. \quad (*)$$

For $t = 0$, connectedness makes g constant on $\mathcal{T}_0$, so this is immediate.

Assume $t \geq 1$. Apply Theorem 8.11 to g with a small error η_0 . It gives an almost-global extension $\tilde{g}$, a set $X_0 \subseteq H$ with $\mu(X_0) \geq 1 - \eta_0$, a set of good $(t+1)$ -cubes $\mathcal{Q}_0 \subseteq \mathcal{C}^{t+1}(X_0)$ with $\mu(\mathcal{Q}_0) \geq 1 - \eta_0$, and

$$K, M_0 \ll_t (\rho h \eta_0)^{-O_t(1)}$$

lower-degree functions $g_1, \dots, g_K$. Each g_i is supported on a parameter- ρ degree- $(t-1)$ cube system and has vanishing t th derivative on its top level; on every cube of $\mathcal{Q}_0$, the $(t+1)$ st derivative of $\tilde{g}$ is an integer combination, with coefficient size at most M_0 , of the values of these g_i at the cube vertices.

Before recursing on the g_i , turn the two near-full sets just obtained into a genuine near-full cube system. Apply Lemma 8.6 to

$$\mathcal{A}_0 = X_0, \quad \mathcal{A}_k = \mathcal{C}^k(H) \ (1 \leq k \leq t), \quad \mathcal{A}_{t+1} = \mathcal{Q}_0.$$

We obtain a system $\mathcal{R}'_0, \dots, \mathcal{R}'_{t+1}$ of parameter $1 - O_t(\eta_0^{c_t})$ with $\mathcal{R}'_0 \subseteq X_0$ and $\mathcal{R}'_{t+1} \subseteq \mathcal{Q}_0$. Thus the derivative formula from the global extension theorem already holds on the top level of a genuine near-full system.

Now apply Lemma 8.9 separately to the degree- $(t-1)$ system attached to each g_i , with threshold $h_1 = \rho^{O_t(1)}$ chosen so that the surviving parameter ρ_1 is at least $\rho/2$. Each g_i splits into at most $1/\rho_1$ robust components. Write $g_{i,j}$ for the restriction to the j th component. The inductive hypothesis at degree $t-1$, with error η_1 , supplies on each such component a hierarchy $\mathbf{P}^{(i,j)}$, a near-full system $\mathcal{U}_0^{(i,j)}, \dots, \mathcal{U}_t^{(i,j)}$ of parameter $1 - \eta_1$, and an extension $\tilde{g}_{i,j}$ satisfying

$$D_{i,j}, M_{i,j} \ll_t (\eta_1 \rho_1 h_1)^{-O_t(1)}.$$

The system $\mathcal{U}^{(i,j)}$ has only degree $t-1$, whereas the final common system must have levels through $t+1$. By Lemma 8.3, every level of $\mathcal{U}^{(i,j)}$ has density $1 - O_t(\eta_1)$. Apply Lemma 8.6 once more, now taking these levels as the prescribed sets for $0 \leq k \leq t$ and taking the full cube space in level $t+1$. This produces an extended degree- t system $\hat{\mathcal{U}}_0^{(i,j)}, \dots, \hat{\mathcal{U}}_{t+1}^{(i,j)}$ of parameter $1 - O_t(\eta_1^{c_t})$ whose first $t+1$ levels lie inside the inductive system.

Intersect $\mathcal{R}'$ with all the $\hat{\mathcal{U}}^{(i,j)}$. The intersection of systems whose parameters are $1 - \alpha_1, \dots, 1 - \alpha_m$ has parameter at least $1 - (\alpha_1 + \dots + \alpha_m)$, because an increment can fail in the intersection only if it fails in one of the factors. Since there are at most K/ρ_1 component systems, the common system has parameter $1 - \eta_2$ with

$$\eta_2 \ll_t \eta_0^{c_t} + \frac{K}{\rho_1} \eta_1^{c_t}. \quad (**)$$

On this common system, concatenate all lower levels of the $\mathbf{P}^{(i,j)}$ and use the extensions $\tilde{g}_{i,j}$ themselves as the new level $t-1$. This is a polynomial hierarchy: every lower-level derivative condition is inherited from the corresponding $\mathbf{P}^{(i,j)}$, and the derivative condition for each new top-level function $\tilde{g}_{i,j}$ is exactly the one supplied by the inductive hypothesis. Whenever the original global-extension formula calls for $g_i(c(\omega))$, that vertex belongs to a unique robust component j of the domain of g_i ; on the common zero level we have $\tilde{g}_{i,j} = g_{i,j} = g_i$. Replacing that value by the corresponding top-level hierarchy coordinate therefore preserves the formula and gives the required $(\mathbf{P}, t+1, t, M)$ derivative condition for $\tilde{g}$.

There are at most K/ρ_1 new top-level coordinates, and the remaining coordinates come from their lower-level hierarchies. Consequently

$$D \leq \frac{K}{\rho_1} + \sum_{i,j} D_{i,j} \ll_t \frac{K}{\rho_1} (\eta_1 \rho_1 h_1)^{-O_t(1)},$$

while M can be taken to be the maximum of M_0 and all $M_{i,j}$ (after enlarging it by a fixed t -dependent power to respect the graded convention M^{t-r}). Choose first $h_1 = \rho^{O_t(1)}$ so that $\rho_1 \geq \rho/2$, then $\eta_0 = \eta^{O_t(1)}$ so that the first term of $(**)$ is at most $\eta/2$, and finally $\eta_1 = (\eta\rho/K)^{O_t(1)}$ so that the second term is at most $\eta/2$. These choices keep every reciprocal parameter polynomially bounded in $(\eta\rho h)^{-1}$, and hence give exactly $(*)$.

Apply the inductive claim with $t = s$, $\rho = \delta_1$, $h = h_0$, and $\eta' = \min(\varepsilon, \delta_1/2)$. The common zero-level set removes at most η' from the robust component, so

$$\mu(Y) \geq \delta_1 - \eta' \geq \delta_1/2 \gg_s \delta^{O_s(1)}.$$

Since δ_1, h_0 are fixed powers of δ , estimate $(*)$ gives $D, M \ll_s (\varepsilon\delta)^{-O_s(1)}$, completing the proof. $\square$

Worked Example 8.16 (A two-level hierarchy). Suppose a cubic-looking partial function has fourth derivative controlled by a finite family of quadratic approximate polynomials Q_j . The global-extension theorem says that, on almost all 4-cubes,

$$\partial^4 \tilde{G}(c) = \sum_{\omega, j} (-1)^{|\omega|} b_{\omega, j} Q_j(c(\omega)).$$

Applying the same theorem to each Q_j expresses its third derivatives in terms of linear approximate polynomials $L_{j, r}$. A degree-2 polynomial hierarchy records these two facts simultaneously rather than repeatedly expanding a nested formula. The hierarchy is therefore a bookkeeping device for a genuine recursive construction, not an abstract regularity factor introduced after the fact.

Chapter 9

Manners strong hierarchies: normal forms, cocycles, redundancy, and boosting

9.1 Why the soft hierarchy is not yet enough

The hierarchy produced in Chapter 7 and Theorem 8.15 records all derivatives of the auxiliary functions by bounded integer combinations of lower levels. It does *not* yet say that those coefficients vary coherently from cube to cube. Manners's Sections 3.2–3.6 repair exactly this problem. The argument has four layers:

normal form $\longrightarrow$ cocycle rigidity $\longrightarrow$ redundancy detection $\longrightarrow$ boosting.

The result is a *strong hierarchy*. This is the last combinatorial input before the nil-polynomial structure theorem of Section 4 of the source paper.

9.2 Normal form for derivative coefficients

Write $\Omega_k = \{0, 1\}^k$, identified with the subsets of $[k]$. If $\omega', \eta, \omega \in \Omega_k$, the notation $\omega \supseteq \eta \supseteq \omega'$ has its obvious coordinatewise meaning. For $0 \leq r \leq k$ define

$$Z_r(\omega, \omega') := \sum_{\substack{\eta: \omega \supseteq \eta \supseteq \omega' \\ |\eta| \leq r}} (-1)^{|\omega| - |\eta|}. \quad (9.1)$$

Definition 9.1 (Normal form). Let

$$b : \Omega_k \longrightarrow \bigoplus_{r=0}^{t-1} \mathbb{Z}^{d_r}.$$

We say that b is in *normal form* if

$$b_{=r}(\omega) = 0 \quad \text{whenever } |\omega| \geq r + 1.$$

For a family $b(c, \omega)$ indexed by cubes, normality is required for every fixed cube c .

Lemma 9.2 (Boolean discrete Leibniz rule). *For functions $\alpha, \beta : \Omega_k \rightarrow R$ with values in a commutative ring,*

$$\sum_{\omega} (-1)^{|\omega|} \alpha(\omega) \beta(\omega) = \sum_{\eta} (-1)^{|\eta|} \left(\sum_{\omega \subseteq \eta} (-1)^{|\omega|} \alpha(\omega) \right) \left(\sum_{\omega' \supseteq \eta} (-1)^{|\omega'|} \beta(\omega') \right).$$

Proof. After expanding the right side, the coefficient of $\alpha(\omega)\beta(\omega')$ is

$$(-1)^{|\omega|+|\omega'|} \sum_{\eta: \omega \subseteq \eta \subseteq \omega'} (-1)^{|\eta|}.$$

If $\omega = \omega'$, this equals $(-1)^{|\omega|}$. Otherwise it vanishes by the binomial identity $(1-1)^{|\omega'|+|\omega|-|\omega|} = 0$. Thus only the diagonal terms remain. $\square$

Lemma 9.3 (Normal-form reduction; Manners Lemma 3.2.2). *Let $\mathbf{F}$ be an $(s, \mathbf{d}, M)$ polynomial hierarchy on a system of cubes. Suppose G obeys an $(\mathbf{F}, k, t, M)$ derivative condition on a family $\mathcal{Q}$, and every face of every $c \in \mathcal{Q}$ of dimension at most $s+1$ lies in the corresponding level of the system. Then the coefficient function may be replaced by one in normal form, at the cost of replacing M by $O_{k,t}(M)$. On the top level the normalized coefficient is explicitly*

$$b'_{=t-1}(c, \omega') = \sum_{\omega \in \Omega_k} Z_{t-1}(\omega, \omega') b_{=t-1}(c, \omega). \quad (9.2)$$

Proof. Fix one cube c ; all operations are cube-local. We normalize successively from level $t-1$ down to level 0. Suppose the levels $> r$ have already been normalized and consider

$$\sum_{\omega} (-1)^{|\omega|} b_{=r}(\omega) \cdot \mathbf{F}_{=r}(c(\omega)).$$

Apply Lemma 9.2 componentwise. The terms are indexed by $\eta \in \Omega_k$ and contain

$$\partial^{|\eta|} \mathbf{F}_{=r}(c|_{F_\eta}) \quad \text{and} \quad B_r(\eta) := \sum_{\omega \supseteq \eta} (-1)^{|\omega|} b_{=r}(\omega).$$

Whenever $|\eta| \geq r+1$, the defining hierarchy condition expresses the derivative of each coordinate of $\mathbf{F}_{=r}$ as a bounded integer combination of the lower levels $\mathbf{F}_{\leq r-1}$. Hence the contribution of all such η can be transferred to lower levels without changing the total derivative identity. We may therefore force $B_r(\eta) = 0$ for every $|\eta| \geq r+1$.

Möbius inversion on the Boolean lattice gives the unique coefficient function with these high-face sums zero:

$$b'_{=r}(\omega') = \sum_{\omega} Z_r(\omega, \omega') b_{=r}(\omega).$$

The inner alternating sum defining Z_r vanishes whenever $|\omega'| \geq r+1$, so $b'_{=r}$ is normal. Since $|Z_r(\omega, \omega')| \leq 2^k$ and the hierarchy substitutions use coefficients bounded by fixed powers of M , the new lower-level coefficients remain bounded by $O_{k,t}(M)^{t-i}$ at level i . Iterating over $r = t-1, t-2, \dots, 0$ proves the claim, and the first step gives (9.2). $\square$

Worked Example 9.4 (Quadratic normal form). Take $t = 2$. The level-1 coefficients are permitted only at vertices of Hamming weight at most 1. A coefficient placed at a vertex of weight 2 can be moved to the adjacent weight-1 and weight-0 vertices. The error introduced is a second derivative of a level-1 hierarchy function, hence is already controlled by level 0. Normal form is therefore the precise algebraic version of “push high-face coefficients downward.”

9.3 Strong derivative conditions and generalized cocycles

Let $T_{k,i}$ denote the gluing triples (c_0, c_1, c) obtained by joining two k -cubes along a common face in the i th coordinate, with c the glued cube.

Definition 9.5 (Almost upper compatibility). Let $S \subseteq \mathcal{C}^k(H)$ and $b : S \times \Omega_k \rightarrow \mathbb{Z}^d$. We call b δ -almost upper compatible if for every $i \in [k]$ and every $\omega \in \Omega_k$ with $\omega_i = 1$, all but at most $\delta|H|^{k+2}$ gluing triples $(c_0, c_1, c) \in T_{k,i}$ with $c_0, c_1 \in S$ satisfy

$$b(c_0, \omega) = b(c_1, \omega).$$

When $\delta = 0$ this says that $b(c, \omega)$ depends only on the upper face visible from ω .

Definition 9.6 (Generalized cocycle). Let $b : S \times \Omega_k \rightarrow \mathbb{Z}^d$ be normal and let $0 \leq r \leq k-1$. We call b a *generalized k -cocycle of type r and loss δ* if it is δ -almost upper compatible and, for every direction i , all but $\delta|H|^{k+2}$ admissible gluing triples satisfy the following normal-form gluing law. Form the raw coefficient function

$$q(\omega) = \begin{cases} b(c_0, \omega), & \omega_i = 0, \\ b(c_1, \omega \setminus \{i\}), & \omega_i = 1, \end{cases}$$

and normalize it at level r by

$$q^{\text{nf}}(\omega') = \sum_{\omega} Z_r(\omega, \omega') q(\omega).$$

Then $b(c, \omega') = q^{\text{nf}}(\omega')$ for every ω' .

Definition 9.7 (Strong derivative condition). Suppose G and b obey the $(\mathbf{F}, k, t, M)$ derivative condition on S . We say they obey the $(\mathbf{F}, k, t, M, \delta)$ *strong derivative condition* if

- (i) b is in normal form;
- (ii) $b_{=r}$ is δ -almost upper compatible for every $1 \leq r \leq t-1$;
- (iii) $b_{=t-1}$ is a generalized k -cocycle of type $t-1$ and loss δ .

The key rigidity statement says that generalized cocycles arise, up to a small exceptional set, by normalizing vertex values of a single polynomial map.

Theorem 9.8 (Robust generalized-cocycle classification; Manners Lemma 3.3.5). *Let $\mathcal{S}_0, \dots, \mathcal{S}_k$ be a system of parameter $1 - \varepsilon$, with $\varepsilon \leq \varepsilon_0(k)$, and let $\rho : \mathcal{S}_k \times \Omega_k \rightarrow \mathbb{Z}^d$ be a generalized k -cocycle of type r and loss δ . Then there exists $\lambda : H \rightarrow \mathbb{R}^d$ such that, writing*

$$\Lambda(c, \omega') = \sum_{\omega \in \Omega_k} Z_r(\omega, \omega') \lambda(c(\omega)), \tag{9.3}$$

we have:

- (i) $\rho(c, \omega) = \Lambda(c, \omega)$ for every ω on all but $O_k(\delta^{1/O_k(1)})|\mathcal{C}^k(H)|$ cubes $c \in \mathcal{S}_k$;
- (ii) $\partial^{k-r} \lambda(c) \in \mathbb{Z}^d$ for every $(k-r)$ -cube c , equivalently $\lambda \bmod \mathbb{Z}^d : H \rightarrow \mathbb{R}^d / \mathbb{Z}^d$ is polynomial of degree at most $k-r-1$;
- (iii) $\sup_x \|\lambda(x)\|_1 \ll_k \sup_{c, \omega} \|\rho(c, \omega)\|_1 + d$;
- (iv) $\Lambda(c, \omega) \in \mathbb{Z}^d$ for every cube c and every vertex ω .

Proof location. A complete proof is given later under the heading “Strong derivative conditions and generalized cocycles.” The present statement is the precise dependency interface used before that proof.

We prove this theorem in three steps. The first integrates ordinary cocycles, the second repairs an almost polynomial to an exact polynomial modulo the integer lattice, and the third reduces a generalized cocycle to ordinary cocycles on faces.

Definition 9.9 (Ordinary cube cocycle). Let A be an abelian group. A map $\rho : S \rightarrow A$, $S \subseteq \mathcal{C}^k(H)$, is a k -cocycle of loss δ if, for every gluing direction, all but $\delta|H|^{k+2}$ admissible triples obey

$$\rho(c) = \rho(c_0) - \rho(c_1).$$

Lemma 9.10 (Approximate integration of a cocycle). Let $\mathcal{S}_0, \dots, \mathcal{S}_k$ be a system of parameter $1 - \varepsilon$ with $0 \leq \varepsilon \leq \varepsilon_0(k)$. If $\rho : \mathcal{S}_k \rightarrow \mathbb{Z}^d$ is an ordinary k -cocycle of loss δ in the sense of the preceding definition, then there is $\lambda_0 : H \rightarrow \mathbb{R}^d$ such that

$$\partial^k \lambda_0(c) = \rho(c)$$

for all but $O_k(\delta^{1/O_k(1)})|\mathcal{C}^k(H)|$ cubes $c \in \mathcal{S}_k$, and

$$\sup_x \|\lambda_0(x)\|_1 \ll_k \sup_c \|\rho(c)\|_1.$$

Proof. We give the linear-algebra correction explicitly. It is enough to work one coordinate at a time, so assume first that $d = 1$.

Pruning the error set. For each direction i , let $I_i^\circ \subseteq T_{k,i}$ consist of those gluing triples (c_0, c_1, c) for which all three cubes lie in $\mathcal{S}_k$ and $\rho(c) = \rho(c_0) - \rho(c_1)$. The cocycle hypothesis says that only $O_k(\delta)|H|^{k+2}$ eligible triples are omitted. Markov's inequality therefore gives, for each i , a set of top cubes on which at least $(1 - 2\varepsilon - \delta^{1/2})|H|$ gluing triples ending at that cube belong to I_i° . Intersect these good-cube sets over the k directions and apply the near-full-system lemma 8.6 to remove the exceptional cubes hereditarily. We obtain a subsystem with top level $S \subseteq \mathcal{S}_k$ such that

$$\mu(\mathcal{S}_k \setminus S) = O_k(\delta^{1/O_k(1)}),$$

and sets $I_i \subseteq I_i^\circ$ satisfying

$$\#\{(c_0, c_1, c) \in I_i : c = c'\} \geq (1 - \eta)|H| \quad (c' \in S), \quad (9.4)$$

where $\eta \leq 2\varepsilon + O_k(\delta^{1/O_k(1)})$. In addition every triple in I_i has all three cubes in S . These are the only combinatorial properties used below.

The exact complete-complex identity. Let U be the real vector space of functions $H \rightarrow \mathbb{R}$, V the space of functions $\mathcal{C}^k(H) \rightarrow \mathbb{R}$, and W_i the space of functions $T_{k,i} \rightarrow \mathbb{R}$. Abbreviate the k th derivative map by $\partial : U \rightarrow V$. Define

$$(\Delta_i q)(c_0, c_1, c) = q(c_0) - q(c_1) - q(c),$$

$$(\sigma q)(x) = \mathbb{E}_{h_1, \dots, h_k} q(\angle(x; h_1, \dots, h_k)),$$

and

$$(\tau_i w)(c') = \mathbb{E}_{(c_0, c_1, c) \in T_{k,i} : c=c'} w(c_0, c_1, c).$$

A direct expansion gives

$$\partial \sigma = \prod_{i=1}^k (I + \tau_i \Delta_i). \quad (9.5)$$

Indeed, if $c' = \angle(x; h_1, \dots, h_k)$, then

$$((I + \tau_i \Delta_i)q)(c') = \mathbb{E}_{h'_i} [q(\angle(x; h_1, \dots, h'_i, \dots, h_k)) - q(\angle(x + h_i; h_1, \dots, h'_i, \dots, h_k))].$$

Composing these identities for $i = 1, \dots, k$ produces

$$\mathbb{E}_{h'_1, \dots, h'_k} \sum_{\omega \in \Omega_k} (-1)^{|\omega|} q(\angle(x + \omega \cdot h; h'_1, \dots, h'_k)),$$

which is exactly $\partial(\sigma q)(c')$.

Restricted operators. Let $\pi_S : V \rightarrow V$ and $\pi_{I_i} : W_i \rightarrow W_i$ be multiplication by the corresponding indicator functions. Set

$$Z' = \pi_S \partial \sigma - \prod_{i=1}^k \pi_S (I + \tau_i \pi_{I_i} \Delta_i)$$

and

$$V' = \text{im } \pi_S \cap \bigcap_{i=1}^k \ker(\pi_{I_i} \Delta_i).$$

Because every triple of I_i lies in S^3 , projecting a function to S does not change its defect on I_i ; hence $Z'(V') \subseteq V'$.

Equip all spaces with the sup norm. We have $\|\partial\| \leq 2^k$, $\|\Delta_i\| \leq 3$, and $\|\sigma\|, \|\tau_i\|, \|\pi_S\|, \|\pi_{I_i}\| \leq 1$. Put $\Phi_i = I + \tau_i \Delta_i$ and $\Phi'_i = I + \tau_i \pi_{I_i} \Delta_i$; both have norm at most 4. From (9.4), if a function is supported outside S , then for $c' \in S$ at most an η -fraction of the gluing average can see it. Thus

$$\|\pi_S \Phi_i (I - \pi_S)\| \leq 2\eta.$$

Likewise, at most an η -fraction of the triples ending at $c' \in S$ lie outside I_i , so

$$\|\pi_S (\Phi_i - \Phi'_i)\| \leq 3\eta.$$

Telescoping the k factors first to insert π_S between successive complete operators and then to replace each Φ_i by Φ'_i gives

$$\|Z'\| \leq 5k4^{k-1}\eta.$$

Choose the original smallness thresholds so that this is at most $1/2$.

Now let $q = \pi_S \rho$. Since ρ obeys the cocycle identity on every I_i , we have $q \in V'$. On V' , all restricted defect terms vanish, and therefore

$$q = (I + Z')g = (\pi_S \partial \sigma)g$$

for a unique $g \in V'$; here $I + Z'$ is invertible because $\|Z'\| \leq 1/2$. Set $\lambda_0 = \sigma g$. Then

$$\partial^k \lambda_0(c) = \rho(c) \quad (c \in S).$$

Moreover $\|g\|_\infty \leq 2\|q\|_\infty$ and hence $\|\lambda_0\|_\infty \leq 2\|\rho\|_\infty$. Restoring the d coordinates separately and summing their bounds gives the stated ℓ^1 estimate. The only exceptional top cubes are those removed in the pruning step, whose density is $O_k(\delta^{1/O_k(1)})$. $\square$

Lemma 9.11 (Property testing for polynomial maps; Manners Lemma 3.3.9). *Let A be an abelian group and $f : H \rightarrow A$. Suppose*

$$\mu\{c \in \mathbb{C}^{m+1}(H) : \partial^{m+1} f(c) = 0\} \geq 1 - \varepsilon, \quad \varepsilon < 2^{-2m-2}.$$

Let X consist of those x for which at least half of all $(m+1)$ -cubes based at x have zero $(m+1)$ st derivative. Then $\mu(X) \geq 1 - 2\varepsilon$, and there exists $\tilde{f} : H \rightarrow A$ with $\partial^{m+1} \tilde{f} \equiv 0$ and $\tilde{f} = f$ on X .

Proof. For $x \in H$ and $h = (h_1, \dots, h_{m+1})$ define the candidate base value

$$g_h(x) = - \sum_{\omega \in \Omega_{m+1} \setminus \{0\}} (-1)^{|\omega|} f(x + \omega \cdot h).$$

Thus $g_h(x) = f(x)$ precisely when the $(m+1)$ -cube based at x with increments h is good. We first show that, for every fixed x , the value $g_h(x)$ has a strict majority as h varies.

Take two increment tuples h, h' . There is a tricube identity

$$\partial^{m+1} f(\angle(x; h')) = \sum_{\eta \in \Omega_{m+1}} (-1)^{|\eta|} \partial^{m+1} f(\angle(x_\eta; h_\eta)),$$

where $x_\eta = x + \eta \cdot h'$ and the i th component of h_η is h_i if $\eta_i = 0$ and $h_i - h'_i$ if $\eta_i = 1$. The internal vertices cancel pairwise; this is simply the $(m+1)$ -dimensional discrete inclusion–exclusion identity. The term $\eta = 0$ is the derivative on $\angle(x; h)$.

Fix x and choose (h, h') uniformly. For every $\eta \neq 0$, the auxiliary cube $\angle(x_\eta; h_\eta)$ is uniform in $\mathcal{C}^{m+1}(H)$, so it is good with probability at least $1 - \varepsilon$. A union bound shows that

$$\partial^{m+1} f(\angle(x; h)) = \partial^{m+1} f(\angle(x; h'))$$

for at least a $1 - (2^{m+1} - 1)\varepsilon$ proportion of pairs (h, h') . Equivalently, $g_h(x) = g_{h'}(x)$ for that proportion of pairs. Since this exceeds $1/2$, the collision probability forces one value to occur on at least a $1 - 2^{m+1}\varepsilon$ proportion of the h 's. Define $\tilde{f}(x)$ to be this majority value. This definition is made for *every* $x \in H$.

If $x \in X$, at least half of the cubes based at x are good, so $g_h(x) = f(x)$ for at least half of h . Since the majority value is unique, $\tilde{f}(x) = f(x)$.

It remains to prove exact polynomiality. Fix an arbitrary cube $\angle(x; h')$. The same tricube cancellation gives, for every auxiliary h ,

$$\sum_{\eta \in \Omega_{m+1}} (-1)^{|\eta|} g_{h_\eta}(x_\eta) = 0.$$

With x, h' fixed and h uniform, each h_η is uniform. The probability that $g_{h_\eta}(x_\eta) = \tilde{f}(x_\eta)$ simultaneously for all 2^{m+1} vertices is at least $1 - 2^{2m+2}\varepsilon > 0$. Choose such an h and substitute into the last display. We obtain

$$\partial^{m+1} \tilde{f}(\angle(x; h')) = 0.$$

Because the original cube was arbitrary, the derivative vanishes everywhere. Finally, averaging the number of bad cubes based at x shows that points with more than half bad based cubes have density at most 2ε , so $\mu(X) \geq 1 - 2\varepsilon$. $\square$

Corollary 9.12 (Robust ordinary cocycle classification). *Under the hypotheses of Lemma 9.10, there exists $\lambda : H \rightarrow \mathbb{R}^d$ such that $\partial^k \lambda(c) = \rho(c)$ on all but $O_k(\delta^{1/O_k(1)})$ of the top cubes, while $\partial^k \lambda(c) \in \mathbb{Z}^d$ for every k -cube c .*

Proof. Apply Lemma 9.10 to obtain λ_0 . Reduce λ_0 modulo $\mathbb{Z}^d$. Its k th derivative is zero on almost every cube. Apply Lemma 9.11 coordinatewise in the torus $\mathbb{R}^d/\mathbb{Z}^d$ to replace it by an exact degree- $(k-1)$ polynomial map on all of H , without changing it at the popular points. Lift the correction back to $\mathbb{R}^d$ by choosing representatives in a fixed fundamental cube. The lift changes λ_0 by a uniformly bounded vector plus an integer-valued function, so the norm bound is preserved and every k th derivative becomes integral. On cubes whose vertices lie in the unchanged popular set, the original identity with ρ remains valid; the exceptional density is still a fixed power of δ . $\square$

Proof of Theorem 9.8. We follow the face-by-face reduction in the source.

Step 1: prune to hereditary good gluing triples. For each direction i , let I_i° be the set of gluing triples for which both upper compatibility and the generalized cocycle identity hold at every vertex. The total excluded proportion is $O_k(\delta)$. As in the pruning argument

of Lemma 9.10, remove cubes that participate in too few good triples, and then pass to a hereditary subsystem $\mathcal{S}'_0, \dots, \mathcal{S}'_k$ with

$$\mu(\mathcal{S}_j \setminus \mathcal{S}'_j) = O_k(\delta^{1/O_k(1)}).$$

For every surviving top cube and every direction, at least $(1 - \eta)|H|$ gluing triples are good in all three slots, where $\eta = 2\varepsilon + O_k(\delta^{1/O_k(1)})$.

Step 2: face sums descend to ordinary cocycles. For a face $F \subseteq \Omega_k$ of codimension r , define on top cubes

$$\rho_F(c) = \sum_{\omega \in F} (-1)^{|\omega|} \rho(c, \omega).$$

Consider a good gluing triple in direction i . If F lies in the common upper face, upper compatibility gives $\rho_F(c_0) = \rho_F(c_1)$. If F lies in the lower face, the generalized normal-form gluing identity and the Boolean cancellation $\sum_{\omega \in E} (-1)^{|\omega|} = 0$ on every positive-dimensional face give $\rho_F(c_0) = \rho_F(c)$. If F crosses the i th direction, the same calculation gives

$$\rho_F(c) = \rho_F(c_0) - \rho_F(c_1).$$

Finally, opposite parallel faces have face sums differing by the expected sign.

These identities imply that $\rho_F(c)$ depends only on the restricted cube $c|_F$. To see this without making an arbitrary-choice assumption, take two extensions of the same $(k - r)$ -cube. Change the r inactive coordinates one at a time. In each coordinate, at least $(1 - \eta)|H|$ good intermediate extensions are available, and for η sufficiently small two such large sets intersect. The first two identities above show that ρ_F is unchanged at each step. Thus there is a well-defined function

$$\rho'_F : \mathcal{S}'_{k-r} \longrightarrow \mathbb{Z}^d, \quad \rho_F(c) = \rho'_F(c|_F).$$

Moreover ρ'_F is an ordinary $(k - r)$ -cocycle with loss $O_k(\delta^{1/O_k(1)})$. Indeed, if a lower-dimensional gluing triple violated its cocycle law, then by the popularity of the subsystem it would lift in at least $|H|^r/2$ distinct ways to top-dimensional gluing triples violating the crossing-face identity above. Since only $O_k(\delta^{1/O_k(1)})|H|^{k+2}$ top triples are bad, only the same fixed-power proportion of lower triples can fail. If F, F' are parallel codimension- r faces, the opposite-face identity similarly gives

$$\rho'_{F'} = (-1)^{\sigma(F, F')} \rho'_F$$

for the parity $\sigma(F, F')$ of the translation from one face to the other.

Step 3: integrate every face cocycle. Apply Corollary 9.12 to every ρ'_F . We get $\lambda_F : H \rightarrow \mathbb{R}^d$ with

$$\partial^{k-r} \lambda_F = \rho'_F$$

on all but a fixed-power exceptional set, with all $(k - r)$ th derivatives integral everywhere and with uniform norm $O_k(\|\rho\|_\infty + d)$. For parallel faces choose the primitives with the same sign relation as the ρ'_F ; additive constants remain free.

We now show that all these primitives differ only by global constants and signs. It is enough first to compare two lower codimension- r faces. For every top cube in the pruned system, differentiating $\partial^{k-r} \lambda_F = \rho'_F$ in the remaining r coordinates gives

$$\partial^k \lambda_F(c) = \sum_{\omega \in \Omega_k} (-1)^{|\omega|} \rho(c, \omega),$$

independently of the chosen lower face F . Hence the difference $\tau = \lambda_F - \lambda_{F'}$ has $\partial^k \tau = 0$ on the near-full top system. Apply Lemma 9.11 to extend τ from the popular vertex set to a global function $\tilde{\tau}$ with $\partial^k \tilde{\tau} \equiv 0$. Averaging this identity over all k increments based at x gives

$$0 = \mathbb{E}_{h_1, \dots, h_k} \partial^k \tilde{\tau}(\angle(x; h_1, \dots, h_k)) = \tilde{\tau}(x) - \mathbb{E}_y \tilde{\tau}(y),$$

so $\tilde{\tau}$ is constant. Thus the original primitives agree up to constants on the entire surviving vertex set. Combining this with the parallel-face sign relation, choose one primitive λ so that for every codimension- r face $F = \{\omega : \eta_0 \subseteq \omega \subseteq \eta_1\}$,

$$\lambda_F(x) = (-1)^{|\eta_0|} \lambda(x) + C_F.$$

Step 4: recover the original generalized cocycle. Constants disappear under the $(k-r)$ th derivative. Therefore, for every surviving top cube c and every face F of dimension $k-r$,

$$\sum_{\omega \in F} (-1)^{|\omega|} \lambda(c(\omega)) = \sum_{\omega \in F} (-1)^{|\omega|} \rho(c, \omega). \quad (9.6)$$

By partitioning a larger face into faces of dimension $k-r$, the same identity holds for every face of dimension at least $k-r$.

Now expand (9.3) using (9.1) and reverse the order of summation:

$$\Lambda(c, \omega') = \sum_{\substack{\eta \supseteq \omega' \\ |\eta| \leq r}} (-1)^{|\eta|} \sum_{\omega \supseteq \eta} (-1)^{|\omega|} \lambda(c(\omega)).$$

Use (9.6) on the upper faces $\{\omega : \omega \supseteq \eta\}$ and replace λ by ρ . Since normal form gives $\rho(c, \omega) = 0$ for $|\omega| > r$, the restriction $|\eta| \leq r$ becomes harmless. The inner Boolean sum

$$\sum_{\eta: \omega \supseteq \eta \supseteq \omega'} (-1)^{|\eta|}$$

vanishes unless $\omega = \omega'$, in which case it equals $(-1)^{|\omega|}$. Thus $\Lambda(c, \omega') = \rho(c, \omega')$. This proves (i).

Property (ii) is inherited from the ordinary cocycle classification of each face primitive. The same face-sum formula, now on an arbitrary cube c , shows that every expression entering $\Lambda(c, \omega')$ is integral, proving (iv). The sup norm of λ is controlled by one face primitive plus a bounded constant adjustment, proving (iii). All discarded sets came from finitely many k -dependent pruning and integration steps, so their total density remains $O_k(\delta^{1/O_k(1)})$. $\square$

9.4 Reductions and how a failed strong condition exposes redundancy

Definition 9.13 (Reduction of a hierarchy). Let $\mathbf{F}$ and $\mathbf{F}'$ be tuples of real-valued functions on the same set, graded by levels $0, \dots, s$. We say that $\mathbf{F}'$ *reduces* $\mathbf{F}$ with parameter K if for every coordinate $F_{i,j}$ and every point x there is an integer vector $v(x)$ supported on levels $\leq i$ with $\|v(x)\|_1 \leq K$ and

$$F_{i,j}(x) = v(x) \cdot \mathbf{F}'_{\leq i}(x).$$

Lemma 9.14 (Reductions preserve derivative conditions). *If G obeys an $(\mathbf{F}, k, t, M)$ derivative condition and $\mathbf{F}'$ reduces $\mathbf{F}$ with parameter K , then G obeys an $(\mathbf{F}', k, t, KM)$ derivative condition.*

Proof. Substitute the bounded integer expression for each occurrence of $F_{r,j}(c(\omega))$ into the original derivative formula. The new coefficient at level i is the sum of at most the old ℓ^1 mass times K , so the graded bound is dominated by $(KM)^{t-i}$ after enlarging the harmless constants. $\square$

The proof that failed compatibility creates a lower-degree approximate polynomial is a discrete Cauchy–Schwarz argument. We therefore internalize its abstract form first.

9.4.1 Discrete Cauchy–Schwarz complexity

Definition 9.15 (Cauchy–Schwarz complexity with denominator). Let $\phi_1, \dots, \phi_m : \mathbb{Z}^d \rightarrow \mathbb{Z}$ be linear forms. They have *Cauchy–Schwarz complexity at most q with denominator Q at position j* if there are sets $\Sigma_1, \dots, \Sigma_{q+1} \subseteq [m] \setminus \{j\}$ covering all other indices and vectors $\sigma_1, \dots, \sigma_{q+1} \in \mathbb{Z}^d$ such that

$$\phi_i(\sigma_r) = 0 \quad (i \in \Sigma_r), \quad 0 \neq \phi_j(\sigma_r) \mid Q.$$

Lemma 9.16 (Fiber Cauchy–Schwarz). *Let $\pi : X \rightarrow Y$ be a homomorphism of finite abelian groups. If functions $F : X \rightarrow A$ and $G : Y \rightarrow A$ into an abelian group agree on at least a δ fraction of X , then*

$$\mu\{(x, x') \in X^2 : \pi x = \pi x', F(x) = G(x')\} \geq \delta^2$$

when the fiber-pair space is given its natural normalized measure.

Proof. For $y \in Y$ let α_y be the fraction of the fiber $\pi^{-1}(y)$ on which $F = G(y)$. The desired pair density is at least $\mathbb{E}_y \alpha_y^2$, whereas the hypothesis is $\mathbb{E}_y \alpha_y \geq \delta$. Jensen or Cauchy–Schwarz gives $\mathbb{E} \alpha_y^2 \geq \delta^2$. $\square$

Lemma 9.17 (Iterated fiber Cauchy–Schwarz). *Let X be a finite abelian group and let $K_1, \dots, K_r \leq X$. Suppose an identity*

$$F_0(x) + \sum_{i \in I} F_i(x) = 0$$

holds on a set of density δ . Assume the nuisance indices are covered by sets $I_1, \dots, I_r$, and F_i is constant on cosets of K_a whenever $i \in I_a$. Then, after taking successive same-coset differences in the directions $K_1, \dots, K_r$, the r -fold difference of F_0 vanishes on a set of the resulting fiber-cube space of density at least δ^{2^r} .

Proof. At stage a , apply Lemma 9.16 to the current identity and the quotient by K_a . Pair two surviving configurations in the same K_a -coset and subtract their identities. Every term whose index belongs to I_a cancels. Terms already cancelled never reappear; a term assigned to a later class becomes a finite difference of the original term, and because translations by K_a commute with translations by K_b , it remains constant on K_b -cosets when its assigned step b arrives. The density is squared at each application. After r steps every nuisance term has been cancelled at least once, while F_0 has become its r -fold finite difference. The surviving density is at least δ^{2^r} . $\square$

Theorem 9.18 (Discrete Cauchy–Schwarz complexity; Manners Lemma 3.5.3). *Suppose $\phi_1, \dots, \phi_m : \mathbb{Z}^d \rightarrow \mathbb{Z}$ have Cauchy–Schwarz complexity at most q with denominator Q at j , and let $F_1, \dots, F_m : H \rightarrow \mathbb{R}$. If*

$$\mu \left\{ x \in H^d : \sum_{i=1}^m F_i(\phi_i(x)) = 0 \right\} \geq \delta$$

and $\gcd(|H|, Q) = 1$, then

$$\mu\{c \in \mathbb{C}^{q+1}(H) : \partial^{q+1} F_j(c) = 0\} \geq \delta^{2^{q+1}}.$$

Proof. For the vector σ_r in the complexity certificate, let $K_r = \{h\sigma_r : h \in H\} \leq H^d$. If $i \in \Sigma_r$, then $\phi_i(x + h\sigma_r) = \phi_i(x)$, so $F_i \circ \phi_i$ factors through H^d/K_r . Apply Lemma 9.17 to the zero identity, using $K_1, \dots, K_{q+1}$ and selecting the j th term. We obtain, on a set of density at least $\delta^{2^{q+1}}$,

$$\Delta_{q_1 h_1} \cdots \Delta_{q_{q+1} h_{q+1}} F_j(\phi_j(x)) = 0, \quad q_r = \phi_j(\sigma_r).$$

Each q_r divides Q and is nonzero. Since $\gcd(|H|, Q) = 1$, multiplication by q_r is an automorphism of H . Thus the variables $q_r h_r$ are uniformly distributed independent increments. The base point $\phi_j(x)$ is also uniform on the image relevant to the identity; translating by one of the increments removes any harmless coset offset. Hence the surviving configurations are exactly a set of $(q+1)$ -cubes of density at least $\delta^{2^{q+1}}$, and the displayed difference is $\partial^{q+1} F_j$. $\square$

9.4.2 Upper-compatibility failure

Lemma 9.19 (Failure of upper compatibility gives a lower-degree approximate polynomial). *Assume $t \geq 2$ and $\gcd(|H|, (t-2)!) = 1$. Let G, b obey an $(\mathbf{F}, k, t, M)$ derivative condition on $S \subseteq \mathbb{C}^k(H)$, with b normal. Suppose that for some $1 \leq R \leq t-1$, some direction i , a vertex ω with $\omega_i = 1$, and a nonzero integer vector v supported on levels $\leq R$, there are at least $\varepsilon|H|^{k+2}$ gluing triples (c_0, c_1, c) such that*

- (a) $c_0, c_1 \in S$;
- (b) $b_{\leq R}(c_0, \omega) - b_{\leq R}(c_1, \omega) = v$;
- (c) all coefficients above level R agree on the common upper face.

Then $v \cdot \mathbf{F}_{\leq R}$ is an approximate polynomial of degree $R-1$ on a proportion

$$O_{k,t}(M)^{-O_{k,t}(D)} \varepsilon^{O_{k,t}(1)}$$

of all R -cubes.

Proof. Pigeonhole all coefficient values $b(c_0, \omega')$ and $b(c_1, \omega')$. Their ℓ^1 norms are $M^{O_{k,t}(1)}$, so at least $\varepsilon O(M)^{-O_{k,t}(D)}$ of the original triples have fixed coefficient patterns $B_0(\omega')$, $B_1(\omega')$. On those triples, subtract the two derivative formulas. All terms away from the common upper face can be absorbed into one-variable functions of the corresponding cube vertices. On the common face, normality kills every level- $\leq R$ coefficient at vertices of weight $> R$, while hypothesis (c) cancels the levels $> R$. Thus we obtain an identity

$$\sum_{\eta \in \Omega_k} (-1)^{|\eta|} G_\eta(c(\eta)) + \sum_{\substack{\nu \in F_i \\ |\nu| \leq R}} (-1)^{|\nu|} (B_0(\nu) - B_1(\nu)) \cdot \mathbf{F}_{\leq R}(c_0(\nu)) = 0,$$

with the summand at $\nu = \omega$ equal to $v \cdot \mathbf{F}_{\leq R}(c_0(\omega))$.

Parametrize the gluing triple by $(x, h_1, \dots, h_k, h'_i)$. The first family of arguments is

$$\phi_\eta = x + \eta \cdot h,$$

and the common-face family is

$$\psi_\nu = x + h'_i + \nu \cdot h.$$

At the distinguished form ψ_ω , these forms have Cauchy–Schwarz complexity at most $R - 1$ with denominator $(t - 2)!$. An explicit certificate is: one class containing all ϕ_η , killed by the vector varying only h'_i ; one class for each coordinate $j \neq i$ with $\omega_j = 1$, containing forms ψ_ν with $\nu_j = 0$ and killed by varying only h_j ; and, for each remaining weight $r > |\omega|$ up to R , one class of forms of weight r , killed by the vector with all $h_j = 1$ and base coordinate $-(r + 1)$. The distinguished form takes values among $\pm 1, \dots, \pm(t - 2)$ on these certificate vectors, hence each divides $(t - 2)!$. Theorem 9.18 now gives an R th derivative zero on a fixed power of the pigeonholed density, exactly as claimed. $\square$

Corollary 9.20 (Generic upper-compatibility dichotomy). *If some level $b_{=r}$ of a normal derivative condition fails δ -almost upper compatibility, then, for some $1 \leq R \leq t - 1$, there is a bounded integer vector v with nonzero top component $v_{=R}$ for which $v \cdot \mathbf{F}_{\leq R}$ is an approximate polynomial of degree $R - 1$ with parameter*

$$O_{k,t}(M)^{-O_{k,t}(D)} \delta^{O_{k,t}(1)}.$$

Proof. Choose the largest level R where upper compatibility fails. Pigeonhole the direction, vertex, and nonzero difference at level R . At every higher level the corresponding upper-face coefficients agree outside a smaller exceptional set by maximality. A union bound leaves the hypotheses of Lemma 9.19, and the difference vector has bounded norm because every coefficient does. $\square$

9.4.3 Failure of the generalized cocycle law

Lemma 9.21 (Cocycle failure gives a lower-level redundancy). *Let $t \geq 1$, assume $\gcd(|H|, (t - 1)!) = 1$, and let $\mathbf{F}$ be a $(t - 1, \mathbf{d}, M)$ hierarchy. Suppose G, b obey an $(\mathbf{F}, k, t, M)$ derivative condition and b is normal and upper compatible on the relevant gluing triples. If the top coefficient $b_{=t-1}$ fails the generalized-cocycle law with loss δ , then there is a bounded integer vector v supported on levels $\leq t - 1$, with $v_{=t-1} \neq 0$, and a parameter*

$$\nu \geq O(M)^{-O_{k,t}(D)} \delta^{O_{k,t}(1)},$$

with the following conclusion.

- (i) If $t \geq 2$, then $v \cdot \mathbf{F}_{\leq t-1}$ is an approximate polynomial of degree $t - 2$ and parameter at least ν .
- (ii) If $t = 1$, then $v \cdot \mathbf{F}_{=0}$ vanishes on a set of vertices of density at least ν ; since every level-zero hierarchy coordinate is constant, this combination vanishes identically on the hierarchy's vertex system.

Proof. For a bad gluing triple, glue the two coefficient configurations and normalize the result using Lemma 9.3. Subtract this normalized configuration from the coefficient configuration already attached to the glued cube. By hypothesis there is a vertex ω at which the top-level difference z is nonzero. Pigeonhole the entire pair of bounded coefficient patterns; on a positive-power fraction of bad triples they become fixed patterns B, B' , so the derivative identity reduces to

$$\sum_{\substack{\eta \in \Omega_k \\ |\eta| \leq t-1}} (-1)^{|\eta|} (B(\eta) - B'(\eta)) \cdot \mathbf{F}(c(\eta)) = 0.$$

The coefficient at the chosen ω has nonzero level- $(t - 1)$ part.

If $t = 1$, the displayed identity contains only level-zero hierarchy coordinates. Pigeonholing has already made its coefficient vector v fixed and nonzero, so it shows $v \cdot \mathbf{F}_{=0} = 0$ on a

positive-density set of vertices. Every level-zero hierarchy coordinate is constant by definition, hence the same combination is zero on the entire vertex system.

Assume now that $t \geq 2$. The vertex forms $x + \eta \cdot h$, $|\eta| \leq t - 1$, have Cauchy–Schwarz complexity at most $t - 2$ at the chosen vertex, with denominator $(t - 1)!$: separate first according to coordinates present in ω , and then according to the remaining Hamming weights. On the corresponding certificate vectors the distinguished form takes a nonzero integer in $\{\pm 1, \dots, \pm(t - 1)\}$. Theorem 9.18 therefore forces the selected linear combination of hierarchy coordinates to have zero $(t - 1)$ st derivative on a fixed power of the pigeonholed set. This is exactly the asserted degree- $(t - 2)$ approximate polynomial. $\square$

Corollary 9.22 (Failure of strongness detects redundancy). *Suppose G obeys a normal $(\mathbf{F}, k, t, M)$ derivative condition and $\gcd(|H|, (t - 1)!) = 1$. If it does not satisfy the $(\mathbf{F}, k, t, M, \delta)$ strong condition, then one of the following holds.*

- (i) *There is a level $1 \leq R \leq t - 1$ and a bounded integer combination $v \cdot \mathbf{F}_{\leq R}$ with $v_{=R} \neq 0$ which is an approximate polynomial of degree $R - 1$, with parameter at least $O(M)^{-O_{k,t}(D)} \delta^{O_{k,t}(1)}$.*
- (ii) *$t = 1$ and there is a nonzero bounded $v \in \mathbb{Z}^{d_0}$ such that $v \cdot \mathbf{F}_{=0} \equiv 0$ on the vertex system.*

Proof. For $t = 1$ there is no upper-compatibility condition, so apply Lemma 9.21; its second alternative gives (ii). For $t \geq 2$, a failure of strongness is either an upper-compatibility failure, handled by Corollary 9.20, or a failure of the top generalized-cocycle identity, handled by Lemma 9.21. In either case the coprimality assumption implies the hypotheses of the relevant discrete Cauchy–Schwarz step, and alternative (i) follows. $\square$

9.5 Boosting approximate polynomials to derivative conditions

We now prove the mechanism in Manners Section 3.6. A lower-degree approximate-polynomial relation is initially known only on a positive-density set of cubes. The aim is to enlarge that set until the derivative relation holds on almost every cube of the ambient near-full cube system.

Definition 9.23 (Gluing closure). If $S \subseteq \mathcal{C}^k(H)$, $i \in [k]$, and $\gamma > 0$, let $S +_{i,\gamma} S$ be the set of cubes c admitting at least $\gamma|H|$ gluing representations $(c_0, c_1, c) \in T_{k,i}$ with $c_0, c_1 \in S$.

Lemma 9.24 (Stability of gluing closure). *If $S' \subseteq S$ and $\gamma > \gamma'$, then*

$$|(S +_{i,\gamma} S) \setminus (S' +_{i,\gamma'} S')| \leq \frac{2|S \setminus S'|}{\gamma - \gamma'}.$$

Proof. Every cube on the left has at least $(\gamma - \gamma')|H|$ representations for which at least one of c_0, c_1 lies in $S \setminus S'$. Count such incidences. A removed cube can occur in at most $|H|$ gluing triples in either slot, giving at most $2|S \setminus S'| |H|$ incidences. $\square$

Lemma 9.25 (Derivative conditions propagate through gluing). *Suppose ϕ obeys an $(\mathbf{F}, k, t, m)$ derivative condition on S . Given $\nu, \gamma > 0$, one of the following holds:*

- (a) *on a subset $S' \subseteq S +_{i,\gamma} S$ whose complement in the gluing closure has density at most ν , ϕ obeys an $(\mathbf{F}, k, t, O_{k,t}(m))$ derivative condition;*
- (b) *for some $1 \leq R \leq t - 1$ there is a bounded integer combination with nonzero level- R component that is an approximate polynomial of degree $R - 1$ with parameter $O(m)^{-O_{k,t}(D)} (\nu\gamma)^{O_{k,t}(1)}$.*

Proof. Normalize the coefficient function by Lemma 9.3. If it is not $(\nu\gamma)$ -almost upper compatible, Corollary 9.20 gives (b). Otherwise call a gluing representation compatible when the two coefficient patterns agree on the common upper face. A cube in the gluing closure but not admitting any compatible representation accounts for at least $\gamma|H|$ incompatible triples. There are at most $\nu\gamma|H|^{k+2}$ such triples, so at most $\nu|H|^{k+1}$ cubes are lost. On a compatible triple, subtracting the two derivative formulas and cancelling the common-face coefficients gives the required formula on the glued cube; normalization changes the coefficient bound only by $O_{k,t}(1)$ powers. $\square$

Lemma 9.26 (Near-full systems propagate a derivative condition to almost every supported cube). *For every $k, t \geq 1$ there is $\varepsilon_0 = \varepsilon_0(k, t) > 0$ with the following property. Let $\mathcal{S}'_0, \dots, \mathcal{S}'_k$ be a system of parameter $1 - \varepsilon'$ with $0 < \varepsilon' \leq \varepsilon_0$. If ϕ obeys an $(\mathbf{F}, k, t, m)$ derivative condition on $\mathcal{S}'_k$, then for every $\nu > 0$ either*

- (a) *the same function obeys an $(\mathbf{F}, k, t, O_{k,t}(m))$ derivative condition on all but a ν -fraction of the k -cubes whose vertices lie in $\mathcal{S}'_0$;*
- (b) *the lower-degree redundancy alternative of Lemma 9.25 occurs with parameter $O(m)^{-O_{k,t}(D)}\nu^{O_{k,t}(1)}$.*

Proof. For $0 \leq a \leq k$, let $Z^{(a)}$ be the set of k -cubes all of whose 2^a codimension- a faces in the first a directions lie in $\mathcal{S}'_{k-a}$. Thus $Z^{(0)} = \mathcal{S}'_k$, while $Z^{(k)}$ consists exactly of the k -cubes supported on $\mathcal{S}'_0$. Popularity in the cube system and a union bound over the 2^{a-1} pairs of opposite faces give

$$Z^{(a)} = Z^{(a-1)} +_{a, 1-2^a\varepsilon'} Z^{(a-1)}.$$

Starting from the top level, apply Lemma 9.25 successively in directions $1, \dots, k$, always with gluing threshold $1/4$ and error budget $4^{-k}\nu$. If the redundancy alternative never occurs, Lemma 9.24 gives the recurrence

$$|Z^{(a)} \setminus S^{(a)}| \leq 4|Z^{(a-1)} \setminus S^{(a-1)}| + 4^{-k}\nu|H|^{k+1}.$$

Starting from zero error, after k steps this is at most $\nu|H|^{k+1}$. The coefficient parameter changes only $O_{k,t}(1)$ times. $\square$

The next operation is translation rather than gluing. We use the globally available $(t+1)$ -derivative condition to compare the t th derivative on two parallel t -cubes.

Lemma 9.27 (Translation propagation). *Assume ϕ has an $(\mathbf{F}, t+1, t, M)$ derivative condition on all but a ν_0 -fraction of the $(t+1)$ -cubes supported on the ambient vertex set. Suppose it also has an $(\mathbf{F}, t, t, m)$ derivative condition on $S \subseteq \mathcal{C}^t(H)$. For $\alpha > 0$ put*

$$W = \{c : \#\{h : c + \square_t(h) \in S\} \geq \alpha|H|\}.$$

Then either, after deleting at most $3\nu_0\alpha^{-1}|\mathcal{C}^t(H)|$ cubes from W , the $(\mathbf{F}, t, t, O_t(M+m))$ derivative condition holds there, or a lower-degree redundancy occurs with parameter

$$O(M+m)^{-O_t(D)}\nu_0^{O_t(1)}.$$

Proof. Discard cubes $c_0 \in W$ for which fewer than $\alpha|H|/2$ translations $c_1 = c_0 + \square_t(h)$ lie in S and form a good $(t+1)$ -cube $[c_0, c_1]$. Double counting shows that the discarded set has size at most $2\nu_0\alpha^{-1}|\mathcal{C}^t(H)|$.

For a surviving pair put $c_1 = c_0 + \square_t(h)$ and let $c = [c_0, c_1]_{t+1}$. Choose normalized coefficient functions b and b' for the known $(t+1)$ - and t -derivative formulae. Since

$$\partial^t \phi(c_0) = \partial^{t+1} \phi(c) + \partial^t \phi(c_1),$$

the coefficient pattern for c_0 is obtained by placing $b(c, \omega)$ on the lower face and $b(c, \omega) - b'(c_1, \omega|_{[t]})$ on the upper face, and then applying the normal-form lemma. Call the resulting pattern $\bar{b}_c$. If

$$\bar{b}_c(\omega) = 0 \quad (\omega \in \mathcal{F}^{t+1}),$$

then all surviving terms lie on the lower face and immediately give an $(\mathbf{F}, t, t, O_t(M + m))$ derivative formula at c_0 .

Let W' be the set of c_0 for which at least one good translation has this vanishing property. If $|W'' \setminus W'| \leq \nu_0 \alpha^{-1} |\mathcal{C}^t(H)|$, where W'' is the set surviving the first pruning, then

$$|W \setminus W'| \leq (2\nu_0/\alpha + \nu_0/\alpha) |\mathcal{C}^t(H)|$$

after the preceding double count, and we are done.

Assume instead that $|W'' \setminus W'| > \nu_0 \alpha^{-1} |\mathcal{C}^t(H)|$. Each such c_0 has at least $(\alpha/2)|H|$ good translations, so there are at least $(\nu_0/2)|\mathcal{C}^{t+1}(H)|$ corresponding $(t+1)$ -cubes on which some upper-face coefficient of $\bar{b}_c$ is nonzero. Pigeonhole first the highest level R , then an upper-face vertex ω with $\omega(t+1) = 1$, and finally the complete bounded coefficient pattern. We obtain

$$\gg_t (M + m)^{-O_t(D)} \nu_0 |\mathcal{C}^{t+1}(H)|$$

cubes with one fixed pattern B , with $B_{=R}(\omega) \neq 0$, and with every coefficient on $\mathcal{F}^{t+1}$ above level R equal to zero. On those cubes the identity for $\partial^t \phi(c_0)$ becomes a fixed bounded linear combination of hierarchy values indexed only by vertices satisfying either $\eta(t+1) = 0$ or $|\eta| \leq R$.

We now verify, rather than merely invoke, the Cauchy–Schwarz certificate needed to apply Theorem 9.18 at the distinguished form $\varphi_\omega(x, h) = x + \omega \cdot h$. For each coordinate $j \leq t+1$ with $\omega(j) = 1$, take

$$\Sigma_j = \{\varphi_\eta : \eta(j) = 0\}, \quad \sigma_j = e_j.$$

For the remaining stages $q = |\omega| + 1, \dots, R$, take

$$\Sigma_q = \{\varphi_\eta : |\eta| = q\}, \quad \sigma_q = (-q, 1, \dots, 1).$$

These R classes cover every nuisance form. For a coordinate class one has $\varphi_\omega(e_j) = 1$. For a weight- q class,

$$\varphi_\omega(-q, 1, \dots, 1) = |\omega| - q \in \{-1, -2, \dots, -(R - |\omega|)\}.$$

Thus every distinguished value is nonzero and divides $(t-1)!$ (because $R - |\omega| \leq t-1$). The system therefore has Cauchy–Schwarz complexity at most $R-1$ with denominator $(t-1)!$. The discrete Cauchy–Schwarz theorem gives a nonzero integer combination $v \cdot \mathbf{F}_{\leq R}$, with top component $v_{=R} \neq 0$ and $\|v\|_1 = O_t(M + m)^{O_t(1)}$, satisfying

$$\mu\{d \in \mathcal{C}^R(H) : \partial^R(v \cdot \mathbf{F}_{\leq R})(d) = 0\} \geq O(M + m)^{-O_t(D)} \nu_0^{O_t(1)}.$$

This is precisely the lower-degree redundancy alternative. Thus the only remaining case is the good one, and the total deletion is the stated $3\nu_0/\alpha$ fraction. $\square$

9.5.1 The additive-growth input

We now internalize the near-maximal-energy statement used in the source’s growth lemma. The proof uses Kneser’s theorem, already proved in Volume II, rather than leaving Fournier’s stability theorem as a black box.

Lemma 9.28 (Near-maximal energy forces a near coset). *There are absolute constants $\beta_0, C > 0$ such that the following holds. Let A be a nonempty finite subset of an abelian group, let $0 < \beta \leq \beta_0$, and suppose*

$$E(A) = \#\{a_1 - a_2 = a_3 - a_4 : a_i \in A\} \geq (1 - \beta)|A|^3.$$

Then there are a subgroup K and a coset $x + K$ such that

$$|A \cap (x + K)| \geq (1 - C\sqrt{\beta}) \max(|A|, |K|).$$

Proof. For d write $r(d) = |A \cap (A + d)|$. Then $\sum_d r(d) = |A|^2$ and $\sum_d r(d)^2 = E(A)$. Averaging $r(a - b)$ over $(a, b) \in A^2$ therefore gives at least $(1 - \beta)|A|$. By Markov, there is $a_0 \in A$ such that for a set $A' \subseteq A$ of size at least $(1 - \sqrt{\beta})|A|$ one has

$$r(a - a_0) \geq (1 - \sqrt{\beta})|A| \quad (a \in A').$$

Equivalently, A and $A + (a - a_0)$ have symmetric difference at most $2\sqrt{\beta}|A|$. For $a, b \in A'$, the triangle inequality for symmetric difference gives

$$|A \Delta (A + a - b)| \leq 4\sqrt{\beta}|A|,$$

so $r(a - b) \geq (1 - 2\sqrt{\beta})|A|$. Since $\sum_d r(d) = |A|^2$,

$$|A' - A'| \leq \frac{|A|}{1 - 2\sqrt{\beta}} \leq (1 + C_1\sqrt{\beta})|A'|.$$

For sufficiently small β the last ratio is $< 3/2$.

Apply Kneser's addition theorem (proved in Volume II) to A' and $-A'$, and let K be the stabilizer of $A' - A'$. If $A' + K$ occupied at least two K -cosets, Kneser would give

$$|A' - A'| \geq 2|A' + K| - |K| \geq \frac{3}{2}|A' + K| \geq \frac{3}{2}|A'|,$$

a contradiction. Hence A' lies in one coset $x + K$, and $|K| = |A' - A'| \leq (1 + C_1\sqrt{\beta})|A|$. Since $A' \subseteq A \cap (x + K)$ and $|A'| \geq (1 - \sqrt{\beta})|A|$, the displayed conclusion follows after enlarging C . $\square$

Lemma 9.29 (Cube-set growth trichotomy; Manners Claim 3.6.5). *Fix $t \geq 1$ and $c_1 > 0$. There are constants $c_2, c_3 > 0$ depending only on t, c_1 such that the following holds. Let S be a set of t -cubes, supported on the vertex set of a cube system of parameter $1 - \varepsilon$, with density δ' . Let*

$$W = \{c : \#\{h : c + \square_t(h) \in S\} \geq (\delta'/2)|H|\}.$$

Then for some $i \in [t]$ and $\gamma \gg_{t, c_1} (\delta')^2$, at least one of the following occurs:

- (i) $|W +_{i, \gamma} W| \geq (1 + c_2 - O_t(\varepsilon))|S|$;
- (ii) H has a proper subgroup of order at least $c_3\delta'|H|$;
- (iii) $\delta' \geq 1 - c_1$.

Proof. Parameterize a t -cube as $\angle(x; h_1, \dots, h_t)$ and let

$$V = \{h \in H^t : \#\{x : \angle(x; h) \in S\} \geq (\delta'/2)|H|\}.$$

Counting S by increment vector gives

$$|V| \geq (\delta'/2)|H|^t. \tag{9.7}$$

Fix a direction i and all increments except h_i , writing the resulting fiber as $V_{h,i} \subseteq H$. On the almost-full set of allowed base points, form the directed graph whose edges are differences in this fiber. If $r(x, y)$ is the number of common out-neighbors, then Cauchy–Schwarz and double counting give, for any small $\gamma > 0$,

$$\#\{(x, y) : r(x, y) \geq \gamma|H|\} \geq |H||V_{h,i}| \frac{|V_{h,i}|^3}{E(V_{h,i})} (1 - O_t(\varepsilon) - O(\gamma|H|^2/|V_{h,i}|^2)). \quad (9.8)$$

Every pair counted here contributes a cube to $W +_{i,\gamma} W$.

Choose small constants ρ, β, c_3 successively. For each fiber, classify it as *small* if $|V_{h,i}| \leq 2c_3\delta'|H|$, *large* if it has size at least $(1 - \rho)|H|$, and *medium* otherwise with $E(V_{h,i}) \leq (1 - \beta)|V_{h,i}|^3$. Lemma 9.28 shows that these cases exhaust all fibers unless alternative (ii) already holds: a high-energy non-small fiber is close to a coset of a subgroup of size $\gg \delta'|H|$; if that subgroup is proper we are in (ii), and if it is all of H the fiber is large.

Write

$$\sigma_i = |V|^{-1} \sum_{\tilde{h} \in L_{\text{med},i}} |V_{\tilde{h},i}|.$$

The small fibers contain at most $4c_3|V|$ elements of V . Taking $\gamma = 2c_3^3(\delta')^2$, the common-neighbor estimate can therefore be summed over the medium and large fibers to give the explicit lower bound

$$|W +_{i,\gamma} W| \geq |H||V| \left(1 - 2^{t+1}\varepsilon - 5c_3\right) (1 + \beta\sigma_i). \quad (9.9)$$

Here we used $(1 - \beta)^{-1} \geq 1 + \beta$ on medium fibers and the baseline factor 1 on large fibers. This is the source’s quantitative form of the assertion that medium additive energy creates growth.

Suppose now that the growth alternative fails in every direction. Put

$$B = \{v \in V : \pi_i(v) \in L_{\text{big},i} \text{ for every } i\}.$$

Since the medium and small fibers are the only ways to leave B ,

$$|B|/|V| \geq 1 - \sum_{i=1}^t \sigma_i - 4tc_3.$$

Every $\tilde{h} \in L_{\text{big},i}$ contributes at least $(1 - \rho)|H|$ elements to V , so

$$|\pi_i(B)| \leq |L_{\text{big},i}| \leq (1 - \rho)^{-1}|V|/|H|.$$

Loomis–Whitney consequently yields

$$|V| \geq |H|^t \left(1 - t \sum_{i=1}^t \sigma_i - t\rho - 4t^2c_3\right). \quad (9.10)$$

This is the exact quantitative mechanism that turns failure of growth into near-full density.

We now choose constants in the source order. Assume $c_1 \leq 1$ and set

$$\rho = \frac{c_1}{10t}, \quad \beta = \frac{c_1^2}{10^4t^2}, \quad c_2 = c_3 = \frac{c_1^3}{10^6t^2}, \quad \gamma = 2c_3^3(\delta')^2.$$

After decreasing c_3 if necessary these choices satisfy the preceding high-energy classification. If simultaneously

$$|V| \leq \delta'(1 + c_1/10)|H|^t, \quad \sigma_i \leq c_1/(10t^2) \quad (1 \leq i \leq t),$$

then (9.10) gives $|V| \geq (1 - 3c_1/10)|H|^t$, and comparison with the displayed upper bound forces $\delta' \geq 1 - c_1$, which is alternative (iii).

Thus, if (iii) fails, either $|V| \geq \delta'(1 + c_1/10)|H|^t$ or $\sigma_i \geq c_1/(10t^2)$ for some i . In the first case (9.9), together with $|S| = \delta'|H|^{t+1}$, gives

$$|W +_{i,\gamma} W| \geq (1 + c_1/20 - O_t(\varepsilon))|S|$$

for any i . In the second case the factor $1 + \beta\sigma_i$ in (9.9) gives

$$|W +_{i,\gamma} W| \geq (1 + c_2 - O_t(\varepsilon))|S|.$$

Both are alternative (i), and the proof is complete. $\square$

Worked Example 9.30 (Why a subgroup obstruction appears). If H has a large proper subgroup K and S consists of cubes whose increments all lie in K , translation and gluing never escape K^t . The density of S can be positive while every closure operation remains trapped. Claim 9.29 isolates precisely this obstruction. For the final application $H = \mathbb{Z}/N\mathbb{Z}$ with N prime, no such intermediate subgroup exists.

Theorem 9.31 (Boosting lemma; Manners Lemma 3.4.9). *Let $t \geq 1$, let $\mathcal{S}_0, \dots, \mathcal{S}_{t+1}$ be a cube system of parameter $1 - \varepsilon$ with $\varepsilon \leq \varepsilon_0(t)$, and let $\mathbf{F}$ be a $(t - 1, \mathbf{d}, M)$ polynomial hierarchy. Suppose $\phi : \mathcal{S}_0 \rightarrow \mathbb{R}$ obeys the $(\mathbf{F}, t + 1, t, M)$ derivative condition on $\mathcal{S}_{t+1}$ and*

$$\mu\{c \in \mathcal{C}^t(H) : \partial^t \phi(c) = 0\} \geq \delta.$$

Let $\eta > 0$, assume $\gcd(|H|, (t - 1)!) = 1$, There is $J_0 = J_0(t) \geq 1$ such that, if H has no proper subgroup of order at least $\delta|H|/J_0$, then one of the following holds:

- (A) *ϕ obeys an $(\mathbf{F}, t, t, M')$ derivative condition on a set $S \subseteq \mathcal{S}_t$ with $\mu(\mathcal{S}_t \setminus S) \leq \eta$, where $M' \ll_t M\delta^{-O_t(1)}$;*
- (B) *for some $1 \leq R \leq t - 1$ there is an integer vector v supported on levels $\leq R$, with $v_{=R} \neq 0$ and $\|v\|_1 \ll_t (M/\delta)^{O_t(1)}$, such that*

$$\mu\{c \in \mathcal{C}^R(H) : \partial^R(v \cdot \mathbf{F}_{\leq R})(c) = 0\} \geq O(M/\delta)^{-O_t(D)}\eta^{O_t(1)}.$$

When $t = 1$, alternative (B) is empty, so (A) always holds.

Proof. We keep the parameter bookkeeping visible, because this proof is where the source's revised version corrected a genuine dependence on the current derivative coefficient bound.

Step 1: make the ambient $(t + 1)$ -derivative condition essentially global on the vertex set. Apply Lemma 9.26 to the top-level system with an error parameter $\nu_0 > 0$ to be fixed at the end. Unless alternative (B) already holds with acceptable parameters, we obtain a set Y containing all but a ν_0 -fraction of the $(t + 1)$ -cubes supported on $\mathcal{S}_0$ on which ϕ obeys an $(\mathbf{F}, t + 1, t, O_t(M))$ derivative condition.

Step 2: start from the zero-derivative cubes and grow. Put

$$U_0 = \{c \in \mathcal{C}^t(H) : c(\omega) \in \mathcal{S}_0 \ \forall \omega, \ \partial^t \phi(c) = 0\}.$$

After intersecting with the supported cubes, $\mu(U_0) \gg_t \delta$; changing $J(t)$ and the harmless powers of δ absorbs this fixed loss. On U_0 there is the derivative condition with coefficient bound $m_0 = 0$.

Assume inductively that $\mu(U_j) \geq \delta$ and that ϕ obeys an $(\mathbf{F}, t, t, m_j)$ derivative condition there. Apply Lemma 9.29 with a small constant $c_1 = c_1(t)$. The subgroup alternative cannot

occur once $J(t) > 1/c_3$. If the density alternative occurs, stop. Otherwise there are a direction i , a set of translation-popular cubes W , and $\gamma \gg_t \mu(U_j)^2 \geq c_t \delta^2$ such that

$$\mu(W +_{i,\gamma} W) \geq (1 + c_2 - O_t(\varepsilon))\mu(U_j).$$

Apply Lemma 9.27 to W , using the ambient good set Y . Outside $O_t(\nu_0/\delta)$ of the cubes we get a derivative condition of parameter $O_t(M + m_j)$. Then use Lemma 9.25 at threshold $\gamma/2$. Unless (B) occurs, we obtain U_{j+1} with

$$\mu(U_{j+1}) \geq (1 + c_2/2)\mu(U_j), \quad m_{j+1} \leq C_t(M + m_j), \quad (*)$$

provided $\nu_0 \ll_t \delta^2$ and $\varepsilon_0(t)$ is sufficiently small.

The dependence on m_j in $(*)$ is essential: gluing combines the newly obtained coefficient pattern with the one already known on U_j . Iterating for at most $K = O_t(\log(1/\delta))$ steps gives

$$\mu(U_K) \geq 1 - c_1, \quad m_K \ll_t M\delta^{-O_t(1)}. \quad (**)$$

The second estimate follows by solving the affine recurrence $(*)$; the exponential in K becomes a fixed power of $1/\delta$.

Step 3: convert weak almost-everywhere control into a near-full cube system. Let

$$W_K = \{c : \#\{h : c + \square_t(h) \in U_K\} \geq c_1|H|\}.$$

By Lemma 9.27, after removing $O_t(\nu_0/c_1)$ of the cubes, the derivative condition with parameter $O_t(M + m_K)$ holds on a subset $W'_K \subseteq W_K$.

We now build a subsystem whose top level lies in W'_K . Let $V \subseteq H^t$ be the increment vectors for which at least $c_1|H|$ base points give a cube in U_K . Since $\mu(U_K) \geq 1 - c_1$, the translation-invariant cube family

$$\tilde{V} = \{\angle(x; h) : h \in V\}$$

has density at least $1 - 2c_1$. Apply the near-full-system lemma 8.6 to $\tilde{V}$, and separately to the complement of the small error set $W_K \setminus W'_K$. Intersect these two systems with the original $\mathcal{S}_0, \dots, \mathcal{S}_t$. By choosing $c_1 = c_1(t)$ first and then $\nu_0 \leq (\eta c_1)^{C_t}/C_t$, we obtain a system $\mathcal{S}'_0, \dots, \mathcal{S}'_t$ of parameter $1 - \varepsilon'$ with $\varepsilon' \leq \varepsilon'_0(t)$,

$$\mathcal{S}'_r \subseteq \mathcal{S}_r, \quad \mu(\mathcal{S}_0 \setminus \mathcal{S}'_0) \leq 2^{-t-1}\eta, \quad \mathcal{S}'_t \subseteq W'_K.$$

On its top level the derivative condition has parameter $O_t(M + m_K) \ll_t M\delta^{-O_t(1)}$ by $(**)$.

Step 4: fill almost all supported t -cubes. Apply Lemma 9.26 to this new system with error $\eta/2$. Unless (B) occurs, the derivative condition holds on all but $\eta/2$ of the t -cubes supported on $\mathcal{S}'_0$. At most $2^t \mu(\mathcal{S}_0 \setminus \mathcal{S}'_0) \leq \eta/2$ further cubes of $\mathcal{S}_t$ use a removed vertex. Hence the total loss is at most η .

Finally choose ν_0 to satisfy simultaneously the constraints above. Every occurrence of the redundancy alternative then has coefficient norm at most $(M/\delta)^{O_t(1)}$ and density at least $O(M/\delta)^{-O_t(D)}\eta^{O_t(1)}$, exactly the parameters in (B). If no such occurrence takes place, (A) follows with $M' \ll_t M\delta^{-O_t(1)}$. $\square$

Lemma 9.32 (Splitting a degree- t derivative condition; Manners Lemma 3.4.10). *For every $t \geq 1$, $M \geq 1$, $\mathbf{d} = (d_0, \dots, d_{t-1})$, and $\eta \in (0, 1)$ there is*

$$\varepsilon_0 = \varepsilon_0(t, M, \mathbf{d}, \eta) > 0$$

with the following property. Let $\mathcal{S}_0, \dots, \mathcal{S}_t$ be a cube system of parameter $1 - \varepsilon$ with $0 \leq \varepsilon \leq \varepsilon_0$, let $\mathbf{F}$ be a $(t-1, \mathbf{d}, M)$ polynomial hierarchy on this system, and suppose ϕ obeys an $(\mathbf{F}, t, t, M)$ derivative condition on $\mathcal{S}_t$. Then one of the following holds:

(A) after removing at most an η -fraction of top cubes, one can write $\phi = \phi_0 + \phi_1$ where ϕ_0 obeys an $(\mathbf{F}, t, t-1, O_t(M))$ derivative condition and

$$\phi_1(x) = v(x) \cdot \mathbf{F}_{=t-1}(x), \quad \|v(x)\|_1 \ll_t M + d_{t-1};$$

(B) a nontrivial bounded combination $v \cdot \mathbf{F}_{\leq R}$ at some $1 \leq R \leq t-1$ is an approximate polynomial of degree $R-1$ with parameter $O_t(M)^{-O_t(D)} \eta^{O_t(1)}$.

For $t = 1$, (A) holds unconditionally.

Proof. When $t = 1$, the level-zero hierarchy functions are constant. Fix $x_0 \in \mathcal{S}_0$ and put $\phi_0 \equiv \phi(x_0)$. For each y , choose a point z such that both edges $[x_0, z]$ and $[y, z]$ lie in the near-full first cube level. Subtracting the two first-derivative formulas expresses $\phi(y) - \phi(x_0)$ as a bounded integer combination of $\mathbf{F}_{=0}$, giving ϕ_1 .

Assume $t > 1$. Normalize the coefficient function. By Corollary 9.22, alternative (ii) of that corollary is impossible, so either (B) already holds or, after choosing a loss $\delta = \eta^{O_t(1)}/O_t(1)$, the derivative condition is strong. In particular the top coefficient $b_{=t-1}$ is a generalized t -cocycle of type $t-1$ and loss δ . Theorem 9.8 gives a function $\lambda : H \rightarrow \mathbb{R}^{d_{t-1}}$ and a set S with $\mu(\mathcal{S}_t \setminus S) \leq \eta$ such that the normal form of the vertex configuration $\lambda(c(\omega))$ equals $b_{=t-1}(c, \omega)$ for every $c \in S$. Since here $k - r = t - (t-1) = 1$, every first difference of λ is integral. Subtract a global constant so that $\lambda(0) = 0$; then $\lambda(x) \in \mathbb{Z}^{d_{t-1}}$ for every x and $\|\lambda(x)\|_1 \ll_t M + d_{t-1}$.

Set

$$\phi_1(x) = \lambda(x) \cdot \mathbf{F}_{=t-1}(x), \quad \phi_0 = \phi - \phi_1.$$

The pointwise description of ϕ_1 is immediate. On $c \in S$, subtract the t th derivative of ϕ_1 from that of ϕ . Normalize the resulting coefficient pattern. The explicit top-level normalization formula (9.2) gives

$$b'_{=t-1}(c, \omega') = b_{=t-1}(c, \omega') - \sum_{\omega} Z_{t-1}(\omega, \omega') \lambda(c(\omega)) = 0.$$

Thus all level- $(t-1)$ coefficients vanish, leaving an $(\mathbf{F}, t, t-1, O_t(M))$ derivative condition for ϕ_0 . $\square$

9.6 The strong-hierarchy upgrade

We can now assemble the whole M5 chain.

Theorem 9.33 (Strong-hierarchy upgrade; Manners Theorem 3.4.3). *Let $s \geq t \geq 1$. Let $\mathcal{S}_0, \dots, \mathcal{S}_{s+1}$ be a cube system of parameter $1 - \varepsilon$, let $\mathbf{F}$ be an $(s-1, \mathbf{d}, M)$ polynomial hierarchy on its first $s+1$ levels, and suppose G obeys an $(\mathbf{F}, s+1, t, M)$ derivative condition on $\mathcal{S}_{s+1}$. Put $D = \sum_i d_i$ and let $\delta > 0$. For $\varepsilon \leq \varepsilon_0(s)$ there are a subsystem $\mathcal{S}'_0, \dots, \mathcal{S}'_{s+1}$ of parameter at least $1 - \varepsilon - \delta$, with $\mu(\mathcal{S}_k \setminus \mathcal{S}'_k) \leq \delta$, and a parameter M' satisfying the source bounds*

$$M' \leq O_s(M/\delta)^{O_s(D)^{O(1)}} \quad (t \leq 2),$$

and

$$M' \leq O_s(M/\delta)^{O_s(D)^{O_s(D)}} \quad (t \geq 3),$$

such that either H has a subgroup of size at least $|H|/M'$, or there is an $(s, \mathbf{d}', M')$ hierarchy $\mathbf{F}'$ reducing $\mathbf{F}$ with parameter M' for which G obeys the $(\mathbf{F}', s+1, t, M', \delta)$ strong derivative condition on $\mathcal{S}'_{s+1}$.

Proof. We separate the general dimension-induction argument from the simultaneous low-degree argument.

The case $t \geq 3$. Put

$$I = \sum_{r=0}^{t-1} (r+1) d_r.$$

We induct on I . Normalize the coefficient function b for the derivative condition of G . If b is already strong there is nothing to prove. Otherwise Corollary 9.22 supplies a level $1 \leq R \leq t-1$ and an integer vector v with $v_{=R} \neq 0$, $\|v\|_1 \leq M^{O_s(1)}$, such that

$$\mu\{c \in \mathbb{C}^R(H) : \partial^R(v \cdot \mathbf{F}_{\leq R})(c) = 0\} \geq \nu, \quad \nu \geq O_s(M)^{-O_s(D)} \delta^{O_s(1)}. \quad (9.11)$$

In particular both $\|v\|_1$ and ν^{-1} are bounded by an envelope of the form $O_s(M/\delta)^{O_s(D)^{O(1)}}$.

Let $\psi = v \cdot \mathbf{F}_{\leq R}$. Because the hierarchy already gives the derivative conditions for each coordinate of $\mathbf{F}$, the function ψ obeys an $(\mathbf{F}, R+1, R, M\|v\|_1)$ derivative condition. Apply Theorem 9.31 to (9.11), with an error budget $\eta = (\delta/I)^{C_s}/C_s$. If its lower-level alternative occurs, we obtain another relation of the same form at a strictly smaller level $R' < R$, with the same type of parameter envelope. Replace (R, v, ν) and repeat. There can be at most $t-2$ such descents. Thus either the subgroup alternative in the theorem has already occurred, or eventually ψ obeys an $(\mathbf{F}, R, R, M_1)$ derivative condition on all but an η -fraction of the relevant R -cubes.

Apply Lemma 9.32, with the same error scale. Another occurrence of its lower-level alternative is handled by the same descending procedure. Otherwise, after passing to a subsystem with losses $O_s(\eta^{1/O_s(1)})$, we have

$$v \cdot \mathbf{F}_{\leq R} = \phi_0 + \phi_1, \quad (9.12)$$

where ϕ_0 obeys an $(\mathbf{F}, R, R-1, M_2)$ derivative condition and

$$\phi_1(x) = u(x) \cdot \mathbf{F}_{\leq R-1}(x), \quad \|u(x)\|_1 \leq M_2,$$

with M_2 still inside the same $O_s(M/\delta)^{O_s(D)^{O_s(1)}}$ envelope. Choose j with $q := v_{R,j} \neq 0$.

We now perform the actual hierarchy reduction. Define $\mathbf{F}^{(1)}$ by

- keeping levels $r > R$ unchanged;
- replacing the d_R functions at level R by $q^{-1}F_{R,\ell}$ for $\ell \neq j$;
- replacing level $R-1$ by $q^{-1}F_{R-1,1}, \dots, q^{-1}F_{R-1,d_{R-1}}, q^{-1}\phi_0$;
- replacing every level $r < R-1$ by $q^{-1}\mathbf{F}_{=r}$.

Scaling all lower levels together preserves the derivative-condition coefficients. The new coordinate $q^{-1}\phi_0$ has the required hierarchy condition by the property of ϕ_0 . Finally, (9.12) solves for the deleted coordinate:

$$F_{R,j} = q^{-1}\phi_0 + q^{-1}\phi_1 - \sum_{\ell \neq j} q^{-1}v_{R,\ell}F_{R,\ell} - q^{-1}v_{<R} \cdot \mathbf{F}_{<R}.$$

Every term on the right is an integer combination of the new scaled coordinates: for instance $q^{-1}F_{R,\ell}$ is itself a new coordinate, while $q^{-1}\phi_1$ is an integer combination of the scaled lower levels. Thus $\mathbf{F}^{(1)}$ reduces $\mathbf{F}$ with parameter $(M_2|q|)^{O_s(1)}$. Its weighted dimension is

$$I(\mathbf{F}^{(1)}) = I(\mathbf{F}) - (R+1) + R = I - 1.$$

By Lemma 9.14, the derivative condition for G transfers to $\mathbf{F}^{(1)}$ with the enlarged parameter.

Choose C_s so that the subsystem loss in this reduction is at most $\delta/(2I)$. Apply the inductive hypothesis to $\mathbf{F}^{(1)}$ with remaining error $\delta(1 - 1/(2I))$. Reductions compose

multiplicatively, so after at most I stages we obtain one final reducing hierarchy. At each stage the logarithm of the current parameter is raised to a power $O_s(D)$; iterating at most $O_s(D)$ times gives the source envelope

$$M' \leq O_s(M/\delta)^{O_s(D)^{O_s(D)}}.$$

The cube losses form a summable budget bounded by δ . This proves the $t \geq 3$ case.

The simultaneous argument for $t = 1, 2$. Here every use of Theorem 9.31 or Lemma 9.32 that is needed below has internal degree 1; hence its lower-level redundancy alternative is empty. This permits all top-level redundancies to be removed in one basis change.

Fix a small threshold $\sigma > 0$, to be chosen as a fixed power of δ times a reciprocal power of M . When $t = 2$, let $V \subseteq \mathbb{Z}^{d_1}$ consist of every top-level difference that occurs on at least $\sigma|H|^{s+3}$ upper-incompatibility gluing triples. For $t = 1$ or 2 , let $W \subseteq \mathbb{Z}^{d_{t-1}}$ consist of the top-level discrepancies that occur on at least that many violations of the generalized-cocycle gluing law. Put $U = V \cup W$ for $t = 2$ and $U = W$ for $t = 1$. All $u \in U$ satisfy $\|u\|_1 \leq M^{O_s(1)}$. The low-degree conclusions must be read separately. If $t = 1$, Lemma 9.21 says that every $u \in U = W$ gives an exact relation $u \cdot \mathbf{F}_{=0} \equiv 0$. If $t = 2$, then Corollary 9.20 and Lemma 9.21 give a degree-0 approximate polynomial of the form $u \cdot \mathbf{F}_{=1} + v_0 \cdot \mathbf{F}_{=0}$, with parameter at least $O_s(M)^{-O_s(D)}\sigma^{O_s(1)}$. Since the level-zero hierarchy coordinates are constant, $u \cdot \mathbf{F}_{=1}$ itself is a degree-0 approximate polynomial with the same parameter. Thus no fictitious degree-(-1) object is used when $t = 1$, and the $t = 2$ input to the degree-one boosting step has the required degree 0. Choose $U' \subseteq U$ with $|U'| \leq d_{t-1}$ and $\text{span}_{\mathbb{Q}} U' = \text{span}_{\mathbb{Q}} U$.

If $t = 1$, every level-zero hierarchy coordinate is constant on the near-full vertex system. Thus an integer combination $u \cdot \mathbf{F}_{=0}$ which vanishes on a positive fraction is identically zero. Choose a set J of standard basis vectors such that

$$\mathcal{B} = \{e_j : j \in J\} \cup U'$$

is a basis of $\mathbb{Q}^{d_0}$, and let A be the matrix with these basis vectors as rows. Define the new level zero by

$$\mathbf{F}'_{=0} = \frac{1}{\det A} (F_{0,j})_{j \in J}.$$

If $\text{adj}(A)$ denotes the adjugate, the identity $A^{-1} = \text{adj}(A)/\det A$ expresses every old coordinate as an integer combination of the new ones; rows belonging to U' contribute zero because $u \cdot \mathbf{F}_{=0} \equiv 0$. Hadamard's inequality bounds the adjugate entries by $O_s(M)^{O_s(D)}$, so $\mathbf{F}'$ reduces $\mathbf{F}$ with that parameter.

For $t = 2$, apply Theorem 9.31 and Lemma 9.32 at degree 1 to every $u \cdot \mathbf{F}_{=1}$ with $u \in U'$. On one common subsystem obtained by intersecting the individual systems, write

$$u \cdot \mathbf{F}_{=1} = \phi_{0,u} + \phi_{1,u},$$

where $\phi_{0,u}$ has zero first derivative and is therefore constant on the surviving vertex set, and $\phi_{1,u}(x)$ is a bounded integer combination of $\mathbf{F}_{=0}(x)$. Choose $J \subseteq [d_1]$ so that $\{e_j : j \in J\} \cup U'$ is a basis of $\mathbb{Q}^{d_1}$, again with row matrix A . Define

$$\mathbf{F}'_{=1} = \frac{1}{\det A} (F_{1,j})_{j \in J}, \quad \mathbf{F}'_{=0} = \frac{1}{\det A} (\mathbf{F}_{=0}, (\phi_{0,u})_{u \in U'}).$$

For a row $u \in U'$, the relation $u \cdot \mathbf{F}_{=1} = \phi_{0,u} + \phi_{1,u}$ expresses the corresponding basis coordinate using $\mathbf{F}'_{=0}$; for a standard row it is one of the retained top coordinates. Multiplying by the adjugate matrix therefore expresses every old $F_{1,j}$ as an integer combination of the new levels. Again Hadamard's bound and the degree-one boosting bounds show that the reduction parameter is at most $O_s(M/\sigma)^{O_s(D)^{O_s(1)}}$.

It remains to verify strongness rather than merely reduction. Let b' be the coefficient function obtained from b after the fixed basis change. On the new top level it is a fixed integral linear image of $b_{=t-1}$ whose kernel is precisely the rational span of U' . If b' failed δ -upper compatibility, then pigeonholing the old coefficient values would produce a difference $v \notin \text{span}_{\mathbb{Q}} U'$ occurring on $\gg_s \delta M^{-O_s(D)} |H|^{s+3}$ triples. Taking $\sigma \leq c_s \delta M^{-C_s D}$ would put v in $V \subseteq U$, contradicting the definition of U' . If instead the generalized cocycle identity for b' failed with loss δ , the same pigeonholing, now using the cocycle-discrepancy calculation of Lemma 9.21, produces a vector outside $\text{span}_{\mathbb{Q}} U'$ belonging to W , the same contradiction. Thus b' is strong.

Choose σ as the stated reciprocal-polynomial multiple of δ and distribute the subsystem error as $\delta/d_{t-1}^{C_s}$ across the at most d_{t-1} degree-one boosting operations. The Hadamard/adjugate bounds and the polynomial degree-one losses combine to

$$M' \leq O_s(M/\delta)^{O_s(D)^{O(1)}}.$$

If at any use of the boosting theorem the no-large-subgroup hypothesis fails, the subgroup has order at least $|H|/M'$ after enlarging the common parameter, giving the first alternative of the theorem. Otherwise the constructions above give the required reducing strong hierarchy. This completes both low-degree cases and the proof. $\square$

Worked Example 9.34 (Redundancy in a quadratic hierarchy). Suppose the top level consists of Q_1, Q_2 and repeated compatibility tests show that $3Q_1 - 2Q_2$ has vanishing second derivative on many parallelograms. The boosting theorem first promotes this to a first-degree derivative condition on almost every allowed parallelogram. The splitting lemma then writes

$$3Q_1 - 2Q_2 = L + \kappa(x) \cdot \mathbf{F}_{\leq 0}(x),$$

with L genuinely linear in the hierarchy sense. After the basis change $(Q_1, Q_2) \mapsto (3Q_1 - 2Q_2, Q_2)$, the first coordinate can be moved down one level; the pointwise lower-level term is redundant. The weighted dimension has decreased, so the procedure cannot loop.

Chapter 10

From strong hierarchies to quantitative nil-polynomials

Proof architecture What changes in Section 4

Stages M4–M5 produced a global polynomial hierarchy and then removed the degeneracies which make its highest derivative representation ambiguous. The remaining task is algebraic rather than combinatorial: one has to manufacture a filtered nilpotent group in which the top derivative coefficients are honest polynomial coordinates. The construction is recursive in the hierarchy level. At each stage we build a nil-polynomial which realizes the top generalized cocycle, subtract it, and drop one hierarchy level. The quantitative point is that every auxiliary group is given by an explicit Mal'cev presentation, so that its dimension and rational complexity can be propagated instead of hidden in an existence theorem.

10.1 Quantitative polynomial-map groups

Proposition 10.1 (Polynomial-map group with an integral lattice). *Let $G_\bullet, G'_\bullet$ be explicitly presented filtered connected simply connected nilpotent groups of degrees at most s, s' , dimensions D, D' , and rational complexities at most M, M' , with lattices Γ, Γ' . Then $\text{poly}(G, G')$, under pointwise multiplication, has the shifted filtration*

$$\text{poly}(G, G')_i = \text{poly}(G, (G')^{+i}),$$

and the subgroup $\text{poly}(G, G')_1$ is an explicitly presented filtered nilpotent group. It admits Mal'cev one-parameter generators $p_{i,j}^\alpha$ such that

$$p = \prod_{i,j} p_{i,j}^{\alpha_{i,j}}$$

uniquely, with $p(\Gamma) \subseteq \Gamma'$ exactly when all $\alpha_{i,j}$ are integers. Moreover

$$\dim \text{poly}(G, G')_1 \ll_{s,s'} (DD')^{O_{s,s'}(1)}, \quad \text{comp} \ll_{s,s'} (DMD'M')^{O_{s,s'}(1)}.$$

Left and right translation on polynomial maps are triangular in this basis, and their coefficients have polynomial height in the translating element.

Proof. We give the construction because it is the quantitative owner needed later. Choose strong Mal'cev bases $X_1, \dots, X_D$ and $Y_1, \dots, Y_{D'}$ adapted to the filtrations and lattices by

Theorem 1.11. By Theorem 2.6, a polynomial map $p : G \rightarrow G'$ is determined recursively along the central series by finitely many binomial-coordinate coefficients. More precisely, after quotienting G' by its last non-trivial central layer, induction gives polynomial coordinates for the quotient map. The remaining central coordinate is an ordinary polynomial in the Mal'cev coordinates of G with weighted degree bounded by s' . Repeating down the filtration gives a finite coordinate list indexed by a layer i and a monomial index j ; the number of such coordinates is $\ll_{s,s'} (DD')^{O_{s,s'}(1)}$.

For each coordinate choose the one-parameter polynomial map $p_{i,j}^\alpha$ which changes that coordinate by α and leaves all earlier coordinates zero. Triangularity of the BCH law (Theorem 1.10) gives existence and uniqueness of the ordered factorization. Since the chosen Mal'cev bases are lattice adapted, evaluating on the lattice points with bounded coordinate size shows that the coefficient is integral iff the corresponding map sends Γ into Γ' . This is the usual integer-valued binomial-polynomial criterion, applied successively to the central coordinates.

The group law and inversion are polynomial in the coordinate coefficients. Their rational coefficients come from the BCH polynomials and from the two Mal'cev bases, so the common denominator and coefficient height are bounded by $(DMD'M')^{O_{s,s'}(1)}$. Hence the displayed one-parameter subgroups form an explicit Mal'cev presentation of $\text{poly}(G, G')_1$ with the claimed complexity.

Finally let $\mathfrak{s}(g, p)(x) = p(gx)$ and $\tau(p, g)(x) = p(xg)$. Discrete differentiation and the filtered product rule show that if p starts in filtration level i and $g \in G_{i'}$, then $p\mathfrak{s}(g, p)^{-1}$ and $p\tau(p, g)^{-1}$ start in level $i + i'$. Thus the coordinate matrices of $\mathfrak{s}(g, \cdot)$ and $\tau(\cdot, g)$ are triangular. Evaluating the translated basis polynomials on the bounded lattice test set which determines their coefficients and using the polynomial Mal'cev metric bounds gives

$$\sum_{a,b} |c_{a,b}| \ll_{s,s'} (DMD'M')^{O_{s,s'}(1)} (1 + d_G(e, g))^{O_{s,s'}(1)}.$$

This is the height estimate required later. No qualitative polynomial-map-group theorem is being used outside the internal Mal'cev and filtered-polynomial calculus. $\square$

Lemma 10.2 (Polynomial lifting from the lattice). *Every polynomial map $q : \Gamma \rightarrow G'$ extends uniquely to a polynomial map $G \rightarrow G'$. If the lattice coordinate coefficients of q have height at most L , the extension has coordinate height at most $(DMD'M'L)^{O_{s,s'}(1)}$.*

Proof. In the binomial-coordinate basis of Proposition 10.1, restriction to Γ records exactly the integer lattice values of the coordinate coefficients. Ordinary integer-valued polynomial interpolation in each successive central quotient recovers the unique real coefficient, and the triangular coordinate change has height bounded polynomially in the presentation complexity. Induction down the central series gives both uniqueness and the displayed height bound. $\square$

Lemma 10.3 (Translation matrix coefficients are polynomial). *Let $E = \text{poly}(G, \mathbb{R}_{(r)})$, with the lattice of integer-valued polynomial maps, and choose the basis from Proposition 10.1. If $e_{i,j} \in E_i$ and $e'_{i',j'}$ is the dual coordinate which vanishes on $E_{i'-1}$, then*

$$z \mapsto e'_{i',j'}(\mathfrak{s}(z, e_{i,j}))$$

is a real polynomial map of degree at most $i - i'$. Its expansion in the integral basis has integer coefficients of total size $(DM)^{O_s(1)}$.

Proof. Take derivatives in directions $h_a \in G_{r_a}$. The filtered product rule places $\partial_{h_1} \cdots \partial_{h_m} e_{i,j}$ in level $i - r_1 - \cdots - r_m$. Once $r_1 + \cdots + r_m > i - i'$, the dual coordinate $e'_{i',j'}$ annihilates that level. Hence the displayed matrix coefficient has weighted degree at most $i - i'$. It is integer

valued on Γ because both the translation action and the chosen polynomial lattice preserve integer-valued maps. Its coordinates therefore are integers. Applying the bounded-test-set coefficient estimate from the previous proposition to lattice elements of bounded Mal'cev size gives the asserted ℓ^1 bound. $\square$

10.2 Nil-polynomials produce polynomial hierarchies

Proposition 10.4 (Derivative hierarchy attached to a nil-polynomial). *Let $r : H \rightarrow G$ be such that $r \bmod \Gamma$ is a polynomial map and $d_G(e, r(x)) \leq M$. Let $F : G \rightarrow \mathbb{R}_{(t)}$ be polynomial and put $f = F \circ r$. Then there is a polynomial hierarchy $\mathbf{F} = (F_{i,j} \circ r)$ of degree t , dimension $\ll_t D^{O_t(1)}$ and complexity $(DM)^{O_t(1)}$, on which f satisfies the natural derivatives condition. Every right lattice translation defect*

$$P - \tau(P, \gamma)$$

of a level- i' basis polynomial is an integral combination of strictly higher-level basis polynomials, with coefficient norm

$$\ll_t (DM)^{O_t(1)} (1 + d_G(e, \gamma))^{O_t(1)}.$$

Proof. Apply Proposition 10.1 to $E = \text{poly}(G, \mathbb{R}_{(t)})$. Choose its integral filtered basis $P_{i,j}$ and set $f_{i,j} = P_{i,j} \circ r$. If $\gamma \in \Gamma$, triangularity of right translation gives

$$P_{i',j'} - \tau(P_{i',j'}, \gamma) = \sum_{i > i'} \sum_j b_{i,j} P_{i,j}$$

with integral $b_{i,j}$ and the stated bound. Now take a Host–Kra cube c in H . Polynomiality of $r \bmod \Gamma$ and the cube lifting statement contained in the filtered polynomial calculus allow us to choose lattice corrections $\gamma(c, \omega)$ of bounded Mal'cev size so that $r(c(\omega))\gamma(c, \omega)$ is an honest Host–Kra cube in G . Since a degree- t polynomial on G has alternating sum zero on a $(t+1)$ -cube, the alternating sum of $P_{i,j}(r(c(\omega)))$ equals a sum of the lattice-translation defects above. Substituting their triangular expansions gives exactly a hierarchy derivatives condition, with bounded integer coefficient vectors. The basis cardinality and all coefficient bounds were already quantified in Proposition 10.1. $\square$

10.3 Realizing one generalized cocycle

Lemma 10.5 (Cocycle realization by a nil-polynomial). *Let $H = (\mathbb{Z}/N^\kappa \mathbb{Z})^n$ with $N > s$. Let $f : H \rightarrow \mathbb{R}$ be a nil-polynomial of degree $t-1$, dimension D and complexity M , and let $\lambda : H \rightarrow \mathbb{R}$ be bounded by M such that $\lambda \bmod \mathbb{Z}$ is a polynomial map of degree $s+1-t$. Then, on $H' = (\mathbb{Z}/N^{2\kappa} \mathbb{Z})^n$, there is a degree- s nil-polynomial g and a hierarchy $\mathbf{f}'$ with $f'_{t-1,1}(x) = f(x \bmod N^\kappa)$ for which the $(s+1)$ -st derivative of g has top coefficient equal to the generalized cocycle built from λ . Moreover*

$$D_g \ll_{n,s} D^{O_{n,s}(1)}, \quad M_g \ll_{n,s} (DM)^{O_{n,s}(1)}.$$

Proof. Set $E = \text{poly}(G, \mathbb{R}_{(t-1)})$ and let $V = E^*$ with the dual filtration. The left translation action of G on E induces a triangular dual action on V : if $u \in V_i$ and $z \in G_j$, then $u - \mathfrak{s}(z, u) \in V_{i+j}$. Let $\Upsilon \subset V$ be the dual lattice to the integer-valued polynomial lattice. Define

$$G^* = G \ltimes (\mathbb{R} \times V \times V)$$

with multiplication chosen so that the first V -coordinate records the translation evaluation defect and the second records the bilinear product of that defect with the scalar λ . Concretely,

if we write elements as (z, a, u, v) , the multiplication is the triangular semidirect product obtained from the left action and the additional cross term $a\mathfrak{s}(z, u')$ in the last coordinate. The commutator identities are therefore

$$\begin{aligned} [(z, 0, 0, 0), (e, 0, u, 0)] &= (e, 0, u - \mathfrak{s}(z^{-1}, u), 0), \\ [(e, a, 0, 0), (e, 0, u, 0)] &= (e, 0, 0, -au), \end{aligned}$$

with all other single-coordinate commutators governed by the filtered action. Giving the second V copy the additional shift $s + 1 - t$ makes these commutators land in the sum of the filtration degrees. Thus G_1^* is a degree- s filtered nilpotent group. Proposition 10.1 provides an explicit Mal'cev presentation of dimension $D^{O_s(1)}$, complexity $(DM)^{O_s(1)}$, and lattice $\Gamma \times \mathbb{Z} \times \Upsilon \times \Upsilon$.

Lift $r \bmod \Gamma$ and $\lambda \bmod \mathbb{Z}$ to polynomial maps on $\mathbb{Z}^n$ using Lemma 10.2. The map

$$x \mapsto (r(x), \lambda(x), \text{ev}_{r(x)} - \text{ev}_e, 0)$$

is polynomial before reduction modulo the period. When x is shifted by $N^\kappa y$, the first three coordinates change by lattice elements. The cross term in the fourth coordinate may fail to be integral at period N^κ , but the binomial expansion of the two polynomial lifts shows that after the doubled period $N^{2\kappa}$ it lies in Υ . Consequently the map descends modulo the lattice to a polynomial map $H' \rightarrow G_1^*/\Gamma^*$.

The final coordinate evaluation functional $\Phi(z, a, u, v) = v(F)$ is a filtered polynomial of degree s . Set $g = \Phi \circ r^*$. On a cube, the Host–Kra alternating sum of Φ vanishes after the lattice corrections. Expanding the correction terms, the only top contribution is precisely the product of the scalar increments of λ with the right-translation defect of F ; this is the generalized cocycle from Theorem 9.8. All remaining terms are matrix coefficients of the translation action on E , and Lemma 10.3 expresses them as bounded integral combinations of the hierarchy in Proposition 10.4. Hence g has the required derivatives condition. Every construction used only direct products, semidirect products, the polynomial-map group, and a period doubling, so the displayed dimension and complexity bounds follow from the previous quantitative owners. $\square$

Lemma 10.6 (Period descent for nil-polynomials). *Let $H' = (\mathbb{Z}/N^{2\kappa}\mathbb{Z})^n$ and $H = (\mathbb{Z}/N^\kappa\mathbb{Z})^n$. If $q : H' \rightarrow \mathbb{R}$ is a degree- s nil-polynomial of dimension D and complexity M , then for every translation $a \in \ker(H' \rightarrow H)$ the pullback of $x \mapsto q(\iota(x) + a)$ to H is a degree- s nil-polynomial of dimension $D^{O_{n,s}(1)}$ and complexity $(DM)^{O_{n,s}(1)}$.*

Proof. Lift $q = F \circ r$ to $\mathbb{Z}^n$. Adjoin to the nilpotent group the additive translation coordinate $\mathbb{R}^n$ and the polynomial-map correction group which records the discrepancy between translating the lifted polynomial map and reducing to a fixed Mal'cev fundamental domain. This is the semidirect product construction of Proposition 10.1. The subgroup generated by the translation directions and the polynomial corrections is still degree s ; its dimension is polynomial in D and n , and its Mal'cev structure constants are polynomially bounded in DM . The graph map $x \mapsto (x/N^\kappa, r(\iota(x) + a))$ modulo the enlarged lattice is polynomial and F extends by ignoring the translation coordinate. Thus the descended function is a nil-polynomial with the claimed bounds. $\square$

10.4 M6: the structure theorem for strong hierarchies

Theorem 10.7 (Strong hierarchy to quantitative nil-polynomial). *Fix $n, s \geq 1$, $1 \leq t \leq s$, a hierarchy dimension vector d , a complexity bound $M \geq 1$, and $\eta > 0$. Put $D = \sum_i d_i$. There exist*

$$\varepsilon_0 = \varepsilon_0(n, s, t, d, M, \eta) > 0, \quad N_0 = N_0(n, s, t, d, M, \eta) \geq 1$$

with the following property. Let $H = (\mathbb{Z}/N^\kappa\mathbb{Z})^n$ with $N \geq N_0$, let $S_0, \dots, S_{s+1}$ be a cube system of parameter at least $1 - \varepsilon_0$, and let $\mathbf{f}$ be an $(s-1, d, M)$ polynomial hierarchy on $S_0, \dots, S_s$. Suppose $g : S_0 \rightarrow \mathbb{R}$ obeys a strong $(\mathbf{f}, s+1, t, M)$ derivatives condition on S_{s+1} . Then there is a degree- s nil-polynomial $q : H \rightarrow \mathbb{R}$ agreeing with g on all but $\eta|H|$ points of S_0 , with

$$\dim q \leq O_{n,s}(D)^{O_{n,s}(1)},$$

and, for $s \geq 3$,

$$\text{comp}(q) \leq (M/\eta)^{O_{n,s}(D)^{O_s(D)}}.$$

For $s \leq 2$ the exponent in the last bound may be chosen polynomial in D .

Proof. We induct on the hierarchy level t . The case $t = 0$ is the polynomial-property-testing endpoint: the $(s+1)$ -st derivative of g vanishes on the near-full cube system, so Lemma 9.11 extends g to a global degree- s additive polynomial; on $H = (\mathbb{Z}/N^\kappa\mathbb{Z})^n$ with $N > s$ the corresponding bounded real lift is constant, hence a zero-dimensional nil-polynomial.

Assume $t \geq 1$. By strongness and Theorem 9.8, every top-level coefficient is, after discarding a prescribed small cube error, represented by a generalized cocycle λ_j of degree $s+1-t$. By the induction hypothesis applied to each top-level hierarchy function $f_{t-1,j}$, each agrees outside a small exceptional set with a nil-polynomial of degree $t-1$. Enlarge the error budget so that all these replacements hold simultaneously; the dimension cost is additive and the complexity cost is the maximum of the individual bounds.

For each top-level coordinate apply Lemma 10.5. It constructs a degree- s nil-polynomial q_j whose top derivative coefficient is the same generalized cocycle as that coordinate. Summing the q_j is legitimate by direct product of the nilmanifolds. Therefore $g - \sum_j q_j$ has a derivatives condition in which the entire level $t-1$ cancels. The remaining coefficient functions involve only hierarchy levels $< t-1$ and their complexity has increased by at most a polynomial power of the current dimension and complexity. We may thus invoke the induction hypothesis at level $t-1$.

The cocycle-realization lemma temporarily works on the doubled group H' . Averaging over the fibers of $H' \rightarrow H$ selects one translate on which the agreement error is no larger than its mean. The period descent lemma then returns that translate to a nil-polynomial on H with polynomial losses in dimension and complexity.

It remains to verify the parameter envelope. At one level, the number of top coordinates is at most D . Choose the per-coordinate error $\eta_1 = (\eta/(CD))^{C_s}$ and the cocycle-strongness loss $\delta = \eta^{C_s}/C_s$. Lemmas 10.5 and 10.6 raise the current complexity to a power $O_{n,s}(D)^{O_s(1)}$. Iterating through at most s hierarchy levels, while the hierarchy itself may have passed through at most $O_s(D)$ rational reductions in Theorem 9.33, gives

$$M' \leq (M/\eta)^{O_{n,s}(D)^{O_s(D)}}.$$

When $s \leq 2$ the simultaneous redundancy removal in that theorem avoids the iterated exponent and yields $O_n(D)^{O(1)}$ in the exponent. The exceptional sets form a geometric error budget bounded by η . This completes the induction. $\square$

Corollary 10.8 (Approximate-polynomial structure theorem). *Let $X \subseteq (\mathbb{Z}/N\mathbb{Z})^n$ and let $f : X \rightarrow \mathbb{R}$ be an approximate polynomial of degree s with parameter ε . Then on a subset of size $\gg_s \varepsilon^{O_s(1)} N^n$, f agrees with a degree- s nil-polynomial of dimension $\varepsilon^{-O_{n,s}(1)}$ and complexity single exponential for $s \leq 2$ and double exponential for $s \geq 3$ in a polynomial power of $1/\varepsilon$.*

Proof. Corollary 8.5 and Theorem 8.15 give a near-full hierarchy on a positive-density subsystem. Theorem 9.33 makes it strong, and Theorem 10.7 converts it to a nil-polynomial.

Multiplying the density losses and substituting the polynomial hierarchy dimension and the single/double exponential complexity envelopes gives the stated bounds. $\square$

10.5 M7: quantitative nilmanifold reconciliation

Theorem 10.9 (Manners nilmanifold construction package). *Every nilmanifold used in Theorem 10.7 and in Theorem 11.8 can be chosen with: (i) degree equal to the required polynomial degree; (ii) dimension polynomial in the hierarchy dimension; (iii) a strong Mal'cev basis whose structure constants and lattice denominators are bounded by the complexity parameter; (iv) polynomial maps and fundamental-domain representatives with Lipschitz constants polynomial in dimension and complexity; and (v) all direct-product, polynomial-map, semidirect-product, period-descent, and Host–Kra cube constructions preserving these bounds by a fixed power depending only on the degree.*

Proof. Items (i)–(iii) follow from the explicit construction in Proposition 10.1, the semidirect-product commutator calculation in Lemma 10.5, and the lattice-adapted Mal'cev basis theorem 1.11. Item (iv) follows because multiplication, inversion and change between first- and second-kind Mal'cev coordinates are polynomial maps of bounded degree with rational coefficients of bounded height; on a bounded fundamental domain their derivatives are bounded by a polynomial in the coefficient height. Polynomial representatives of quotient maps are furnished by Lemma 10.2 and the Taylor theorem 2.7.

For item (v), direct products concatenate Mal'cev bases. In a semidirect product the new structure constants are the old structure constants together with the triangular action matrices; these were bounded in Proposition 10.1. The polynomial-map group has the same quantitative presentation by that proposition. Period descent was proved in Lemma 10.6. Finally the Host–Kra cube group is generated by face elements from the original filtration, so its Mal'cev coordinates and lattice are obtained by a bounded number (depending only on s) of products and commutators. BCH therefore bounds its structure constants by a fixed power of the original complexity. This reproduces the quantitative content of Manners Section 4.2 and Appendix C inside the present nilmanifold conventions. $\square$

Worked Example 10.10 (Why a semidirect product is unavoidable). For a quadratic nil-polynomial f and a linear scalar phase λ , the top derivative term contains $\lambda(x)(f(x+h) - f(x))$. A direct product can record f and λ separately but cannot turn their product into a polynomial coordinate. The last V -coordinate in the cocycle-realization group is designed so that the commutator of the scalar direction with the evaluation-defect direction is exactly that product. This is the nilpotent analogue of adjoining an antiderivative variable.

Chapter 11

Manners quantitative inverse theory: spectral covering, nilsequence smoothing, and descent

11.1 M8a: covering almost all Fourier mass by nil-polynomial graphs

Throughout this chapter $H = \mathbb{Z}/N\mathbb{Z}$ with N prime and, when diagonal descent is used, $N > s$. We retain the notation $\tilde{\mathcal{S}}_s f$ from Chapter 7.

Lemma 11.1 (Almost-all-mass graph covering). *Let $|f| \leq 1$ and let $\varepsilon > 0$. There exist*

$$L \leq \varepsilon^{-O_s(1)},$$

sets $A_1, \dots, A_L \subset H^{s-1}$, and degree- $(s-1)$ nil-polynomials $g_i : H^{s-1} \rightarrow \mathbb{R}$ such that $g_i(x) \in N^{-1}\mathbb{Z}$ for $x \in A_i$, with dimension $\varepsilon^{-O_s(1)}$ and with the Manners single/double-exponential complexity envelope, for which, if

$$S_i = \{(x, \chi_{g_i(x)}) : x \in A_i\}, \quad \chi_a(y) = e(ay), \quad S = \bigcup_i S_i,$$

then

$$\mathbb{E}_{x \in H^{s-1}} \sum_{\chi \in \widehat{H}} (1 - 1_S(x, \chi)) \tilde{\mathcal{S}}_s f(x, \chi)^2 \leq \varepsilon.$$

The graphs S_i may be chosen pairwise disjoint.

Proof. Put

$$R = \{(x, \chi) : \tilde{\mathcal{S}}_s f(x, \chi) \geq \varepsilon/2\}.$$

By Proposition 7.3(ii),

$$\begin{aligned} \mathbb{E}_x \sum_{\chi} 1_{R^c}(x, \chi) \tilde{\mathcal{S}}_s f(x, \chi)^2 &\leq \frac{\varepsilon}{2} \mathbb{E}_x \sum_{\chi} \tilde{\mathcal{S}}_s f(x, \chi)^2 \\ &\leq \varepsilon/2. \end{aligned} \tag{1}$$

We now select disjoint graphs iteratively. Suppose $S_1, \dots, S_k$ have been selected. If the desired estimate already holds, stop. Otherwise let

$$T = R \setminus \bigcup_{i \leq k} S_i, \quad U = \{x : \exists \chi (x, \chi) \in T\}.$$

Using (1) and the failure of the target inequality gives

$$\mu(U) \geq \mathbb{E}_x \sum_{\chi} 1_T(x, \chi) \tilde{\mathcal{S}}_s f(x, \chi)^2 \geq \varepsilon/2, \quad (2)$$

since the total Fourier mass above each x is at most one. Choose for every $x \in U$ one $\kappa(x)$ with $(x, \kappa(x)) \in T$. The already-proved cube-respecting theorem and the graph-selection corollary in Chapter 7 show that κ is an approximate polynomial on a subset of density $\varepsilon^{O_s(1)}$. The lift lemma 7.14 chooses representatives $\kappa'(x) \in N^{-1}\mathbb{Z}$. Applying the approximate-polynomial structure theorem 10.8 then yields a set $A_{k+1} \subset U$ of density

$$\eta \geq \varepsilon^{O_s(1)} \quad (3)$$

and a nil-polynomial g_{k+1} with $g_{k+1} = \kappa'$ on A_{k+1} and with the claimed dimension and complexity. Its graph lies inside T , so disjointness is preserved.

Each selected graph captures at least

$$\mu(A_i)(\varepsilon/2)^2 \geq \eta \varepsilon^2/4 \quad (4)$$

of the squared Fourier mass. By Proposition 7.3(iii), the total such mass is at most one. Consequently the procedure stops after at most $4\eta^{-1}\varepsilon^{-2} = \varepsilon^{-O_s(1)}$ steps. At termination (1) and the definition of the stopping condition give the required estimate. The parameter claims follow from (3), Theorem 10.7, and the quantitative package 10.9. $\square$

11.2 M8b: turning a nil-polynomial frequency into a Lipschitz nilsequence

Lemma 11.2 (Frequency graph to a degree- s nilsequence family). *Let $\phi : H^{s-1} \rightarrow \mathbb{R}$ be a degree- $(s-1)$ nil-polynomial of dimension D and complexity M , and let $A \subset H^{s-1}$ satisfy $\phi(x) \in N^{-1}\mathbb{Z}$ for $x \in A$. There are a degree- s nilmanifold $\tilde{G}/\tilde{\Gamma}$ of dimension $D^{O_s(1)}$ and complexity $(DM)^{O_s(1)}$, a polynomial map $p : H^{s-1} \times H \rightarrow \tilde{G}/\tilde{\Gamma}$, and a family of $(DM)^{O_s(1)}$ -Lipschitz functions F_z such that, for each $x \in A$, one can choose a bounded parameter $z = z(x)$ for which*

$$F_z(p(x, y)) = e(\phi(x)y) \quad (y \in H).$$

Proof. Write $\phi = F \circ r$ on an explicitly presented filtered group G as in the definition of a nil-polynomial. Represent $y \in H$ by the integer $-N/2 < y \leq N/2$ and enlarge the group to $G' = \mathbb{R} \times G$, with the real coordinate in filtration degree one. Set

$$r'(x, y) = (y/N, r(x)), \quad F'(t, g) = NtF(g). \quad (5)$$

Discrete derivatives and the product rule show inductively that multiplication by the degree-one coordinate raises polynomial degree by exactly one; hence F' is degree s . The cube definition of polynomial maps shows that $r' \bmod (\mathbb{Z} \times \Gamma)$ is polynomial.

The obstruction is that $e(F')$ need not be Lipschitz in a controlled way. We therefore put F' into the group rather than into the test function. Let

$$W = \text{poly}(G', \mathbb{R}_{(s)})$$

with the finite-dimensional polynomial basis and integral lattice supplied by Proposition 10.1. Decompose $F' = F_0 + F_1$, where F_0 lies in a fixed fundamental parallelepiped and F_1 belongs to the integral polynomial lattice Ψ . Form the semidirect product

$$\tilde{G} = G' \ltimes W, \quad (g, w)(g', w') = (gg', \tau(w, g') + w'), \quad (6)$$

with the translation action τ from Proposition 10.1; its filtration, Mal'cev basis, lattice $\tilde{\Gamma}$, and complexity bounds are those already verified in Theorem 10.9. Conjugating $(r'(x, y), 0)$ by the integral element $(1, F_1)$ gives a polynomial map

$$\tilde{r}(x, y) = (r'(x, y), \tau(F_1, r'(x, y)) - F_1) \in \tilde{G}_1. \quad (7)$$

Define on $\tilde{G}_1$

$$\mathcal{F}(g, w) = e(F_0(g) + w(1)). \quad (8)$$

By the coefficient estimates for W , $\mathcal{F}$ is quantitatively Lipschitz on bounded coordinate boxes. It is invariant under the W -part of the lattice but not necessarily under the original Γ -part. To descend it, take tent functions in the real coordinate and in a Mal'cev coordinate box centered at z . Summing their right translates over $\mathbb{Z} \times \Gamma$ gives

$$\mathcal{F}_z(\tilde{g}\tilde{\Gamma}) = \sum_{m \in \mathbb{Z}, \gamma \in \Gamma} T^{(1)}(t + m) T_z(g\gamma) \mathcal{F}((t + m, g\gamma), w^{m, \gamma}). \quad (9)$$

The coordinate box is supported in a fundamental domain, so at most one γ and at most two m contribute. Thus (9) is well defined and $(DM)^{O_s(1)}$ -Lipschitz.

For $x \in A$ choose z to be the Mal'cev coordinate vector of $r(x)$. Then the Γ -sum in (9) collapses to the identity translate, and (5)–(8) give

$$\mathcal{F}_z(\tilde{r}(x, y)) = \sum_{t \in y/N + \mathbb{Z}} T^{(1)}(t) e(N\phi(x)t).$$

Since $N\phi(x) \in \mathbb{Z}$ and the integer translates of the tent function sum to one, this equals $e(\phi(x)y)$. This is the required family. $\square$

Lemma 11.3 (Coordinate-fiber pruning). *If $X \subset H^s$ and $c > 0$, there is $X' \subset X$ such that for every $x \in X'$, every coordinate i , and the coordinate fiber through x ,*

$$|\{z \in X' : z_j = x_j \ (j \neq i)\}| \geq c|H|,$$

while $|X \setminus X'| \leq cs|H|^s$.

Proof. Repeatedly delete any coordinate fiber having fewer than $c|H|$ surviving points. For each of the $s|H|^{s-1}$ possible fibers, deletion occurs at most once and removes fewer than $c|H|$ points. The claimed loss follows, and the terminal set has the asserted fiber property. $\square$

Proposition 11.4 (Nilsequence partition of unity for $\mathcal{S}_s f$). *Suppose $|f| \leq 1$ and $\|f\|_{U^{s+1}} = \delta$. There is a one-bounded function $B : H^s \rightarrow \mathbb{C}$ satisfying*

$$|\langle \mathcal{S}_s f, B \rangle| \geq \frac{1}{2} \delta^{2^{s+1}}, \quad (10)$$

and a representation

$$B = \sum_{j=1}^T \alpha_j \rho_j, \quad |\alpha_j| \leq 1, \quad (11)$$

where each $\rho_j : H^s \rightarrow [0, 1]$ is a degree- s nilsequence. Its dimension is $\delta^{-O_s(1)}$, while T , complexity and Lipschitz parameter obey the Manners single-exponential bound for $s \leq 3$ and double-exponential bound for $s \geq 4$.

Proof. Apply Lemma 11.1 with a small parameter ε and let S be the union of the resulting graphs. Let Λ be the inverse Fourier transform of $1_S \tilde{\mathcal{S}}_s f$. Parseval gives

$$\mathbb{E}_h |\mathcal{S}_s f(h) - \Lambda(h)|^2 \leq \varepsilon. \quad (12)$$

For each graph nil-polynomial, Lemma 11.2 puts its characters into a common product nilmanifold. Write $p : H^s \rightarrow G/\Gamma$ for the resulting polynomial map and $d(u, v) = d_{G/\Gamma}(p(u), p(v))$. For a permutation π of the coordinates put $d_\pi(u, v) = d(\pi u, \pi v)$. Symmetry of $\mathcal{S}_s f$ and (12) imply, outside a set Ω_0 of measure $O_s(\varepsilon^{1/2})$, that changing only coordinate i obeys

$$|\mathcal{S}_s f(u) - \mathcal{S}_s f(v)| \leq O_s(\varepsilon^{1/4}) + K_1 d_{\pi_i}(u, v). \quad (13)$$

For $x, y \in H^s$ define the directed coordinate-walk distance

$$\Delta(x, y) = \sum_{i=1}^s d_{\pi_i}((y_{<i}, x_i, x_{>i}), (y_{\leq i}, x_{>i})). \quad (14)$$

This is itself a degree- s nilsequence in (x, y) . If every intermediate coordinate point avoids Ω_0 , telescoping (13) yields

$$|\mathcal{S}_s f(x) - \mathcal{S}_s f(y)| \ll_s K_1 \Delta(x, y) + \varepsilon^{1/4}. \quad (15)$$

We next establish enough nearby centers. Cover G/Γ by

$$Q \leq (C_s M D / \eta)^{O_s(D)} \quad (16)$$

balls of radius $\eta/(2s)$ and color points of H^s by the cells of the images appearing in the successive coordinate path. Apply Lemma 11.3 to every one of the Q^s color classes with $c = \sigma/(2sQ^s)$. Outside another set of measure at most $\sigma/2$, each point x has at least

$$\nu |H|^s, \quad \nu \geq (C_s M D / \eta)^{-O_s(D)} \sigma^s, \quad (17)$$

admissible y with $\Delta(x, y) \leq \eta$. Sampling $T \geq \nu^{-1} \log(2/\sigma)$ centers $y_1, \dots, y_T$, some choice covers all but another $\sigma/2$ fraction.

Define tent weights

$$\beta_j(x) = \min\{1, 2 \max(0, 1 - \Delta(x, y_j)/(2\eta))\}$$

and normalize by

$$\rho_j(x) = \frac{\beta_j(x)}{\max\{1, \sum_k \beta_k(x)\}}. \quad (18)$$

(The denominator in (18) is the equivalent normalization that ensures $\sum_j \rho_j \leq 1$; on the good set a center has $\beta_j = 1$, hence the sum is exactly one.) Functional calculus on the one-dimensional tent function and products/direct products of nilsequences show that every ρ_j is a degree- s nilsequence with Lipschitz parameter $O(1/\eta)$. Whenever $\rho_j(x) > 0$, (15) gives

$$|\mathcal{S}_s f(x) - \mathcal{S}_s f(y_j)| \ll_s K_1 \eta + \varepsilon^{1/4}. \quad (19)$$

Set

$$\alpha_j = \frac{\mathbb{E}_x \rho_j(x) \mathcal{S}_s f(x)}{\mathbb{E}_x \rho_j(x)}$$

when the denominator is nonzero (discard zero terms) and define B by (11). Then $|B| \leq 1$ and (19), together with the exceptional-set bounds, gives

$$\|B - \mathcal{S}_s f\|_2^2 \ll_s (K_1 \eta + \varepsilon^{1/4})^2 + \varepsilon^{1/2} + \sigma. \quad (20)$$

Choose ε, σ as fixed powers of δ and then $\eta \asymp_s \delta^{O_s(1)}/K_1$ so that the right side of (20) is at most $\delta^{2^{s+1}}/4$. Since $\|\mathcal{S}_s f\|_2^2 = \delta^{2^{s+1}}$ by Proposition 7.3(iii), Cauchy–Schwarz gives (10). Substituting the choices into (16)–(18) and the bounds from M6–M7 gives the stated parameter envelopes. $\square$

11.3 M9: diagonal sampling and removal of derivative dilations

For $h \in H$ write $\Delta_h u(x) = u(x) \overline{u(x+h)}$.

Lemma 11.5 (Diagonal sampling). *Assume N is coprime to s !. If $|f| \leq 1$ and*

$$|\langle \mathcal{S}_s f, B \rangle| \geq \eta,$$

then some $h_0 \in H^s$ and $g_1(x) = B(h_0 + (x, \dots, x))$ satisfy

$$\left| \mathbb{E}_{u_1, \dots, u_s} \mathcal{S}_s f(u_1, \dots, u_s) \overline{\mathcal{S}_s g_1(-u_1, -u_2/2, \dots, -u_s/s)} \right| \geq \eta^{2^s}. \quad (21)$$

Proof. Write $h = r + (t, \dots, t)$ and average first over r . Some r preserves the original correlation. After collecting the 2^s translates of f according to the Hamming weight of the vertex, this fixed r gives bounded functions $F_1, \dots, F_s$ for which

$$\left| \mathbb{E}_{x,t} \overline{g_1(t)} f(x) \prod_{j=1}^s F_j(x + jt) \right| \geq \eta. \quad (22)$$

We now eliminate $F_s, F_{s-1}, \dots, F_1$. For bounded $F_1, \dots, F_k$, changing variables $y = x + kt$ and applying Cauchy–Schwarz in y gives one derivative of f with increment u and one derivative of g_1 with increment $-u/k$. The remaining forms become the same configuration with $k-1$ auxiliary functions. Iterating, and using Holder at the averaging steps, yields

$$|(22)|^{2^s} \leq \mathbb{E}_{u_1, \dots, u_s, x, t} \Delta_{u_1} \cdots \Delta_{u_s} f(x) \overline{\Delta_{u_1} \Delta_{u_2/2} \cdots \Delta_{u_s/s} g_1(t)}.$$

By the derivative identity for $\mathcal{S}_s$, the right side is exactly the correlation in (21). $\square$

Lemma 11.6 (One-coordinate dilation removal). *Let $a_1, \dots, a_s$ and $k \geq 1$ be integers invertible modulo N . If $|f| \leq 1$, then for each $j \in [s]$ and each $g : H \rightarrow \mathbb{C}$ there are shifts $r_1, \dots, r_k$ and*

$$g'(x) = \prod_{i=1}^k g(x + r_i)$$

such that

$$\begin{aligned} & \left| \mathbb{E}_h \mathcal{S}_s f(h) \overline{\mathcal{S}_s g(h_1/a_1, \dots, h_s/a_s)} \right| \\ & \leq \left| \mathbb{E}_h \mathcal{S}_s f(h) \overline{\mathcal{S}_s g'(h_1/a_1, \dots, h_j/(ka_j), \dots, h_s/a_s)} \right|^{1/k}. \end{aligned}$$

Proof. Because the multiplicative derivatives commute, take $j = s$. Fix $h_1, \dots, h_{s-1}$ and set

$$F = \Delta_{h_1} \cdots \Delta_{h_{s-1}} f, \quad G = \Delta_{h_1/a_1} \cdots \Delta_{h_{s-1}/a_{s-1}} g.$$

Fourier expansion gives the nonnegative identity

$$\mathbb{E}_h (\mathbb{E}_x \Delta_h F(x)) \overline{(\mathbb{E}_y \Delta_{h/a_s} G(y))} = \sum_{\chi} |\widehat{F}(\chi)|^2 |\widehat{G}(a_s \chi)|^2. \quad (23)$$

Convexity bounds (23) by the k th root of the same weighted sum with $|\widehat{G}(a_s \chi)|^{2k}$. Expanding the $2k$ th power and retaining the diagonal condition gives

$$\sum_{\chi} |\widehat{F}(\chi)|^2 |\widehat{G}(a_s \chi)|^{2k} \leq \mathbb{E}_{r \in H^k} \mathbb{E}_h (\mathbb{E}_x \Delta_h F(x)) \overline{(\mathbb{E}_y \Delta_{h/(ka_s)} G_r(y))}, \quad (24)$$

where G_r is obtained from G by taking the product of k translates. Since derivatives commute with translation and products, $G_r = \Delta_{h_1/a_1} \cdots \Delta_{h_{s-1}/a_{s-1}} g_r$ with $g_r(x) = \prod_i g(x + r_i)$. Average (24) over $h_1, \dots, h_{s-1}$ and use convexity once more to move the outer average inside the k th root. Some particular r is at least the average, giving the claim. $\square$

Proposition 11.7 (Manners diagonal descent). *Assume $N > s$. If $|f| \leq 1$ and $|\langle \mathcal{S}_s f, B \rangle| \geq \eta$, then there are $L = s!^{s-1}$ shifts $h_1, \dots, h_L \in H^s$ and*

$$g(x) = \prod_{i=1}^L B(h_i - (x/s!, \dots, x/s!)) \quad (25)$$

for which

$$\|f\bar{g}\|_{U^s} \geq \eta^L. \quad (26)$$

Proof. Lemma 11.5 starts with a diagonal function g_1 and derivative scalings $(-1, -1/2, \dots, -1/s)$. Apply Lemma 11.6 successively to coordinate j with $k = s!/j$. After all coordinates the denominators have the common magnitude $s!$, at a total product loss

$$\prod_{j=1}^s (s!/j) = (s!)^{s-1} = L. \quad (27)$$

The resulting function g_3 is a product of L translates of g_1 , and therefore has the form (25) after absorbing the fixed diagonal base point into the shifts. The correlation obtained is

$$\left| \mathbb{E}_h \mathcal{S}_s f(h) \overline{\mathcal{S}_s g_3(h/s!)} \right| \geq \eta^{2^s L}. \quad (28)$$

The change of variables and the standard cube identity give

$$\mathbb{E}_h \mathcal{S}_s f(h) \overline{\mathcal{S}_s g_3(h/s!)} = \mathbb{E}_t \|x \mapsto f(x) \overline{g_3(x+t)}\|_{U^s}^{2^s}. \quad (29)$$

Choose t attaining at least the average and replace $g_3(x)$ by $g_3(x+t)$. Taking the 2^s th root of (28)–(29) gives (26). $\square$

11.4 M10: quantitative induction and the selected inverse theorem

Theorem 11.8 (Manners inverse theorem with propagated complexity). *Fix $s \geq 1$. Let $N > s$ be prime, $H = \mathbb{Z}/N\mathbb{Z}$, and $|f| \leq 1$ with $\|f\|_{U^{s+1}} \geq \delta$. Then there is a degree- s one-bounded nilsequence ψ for which*

$$|\mathbb{E}_x f(x) \overline{\psi(x)}| \geq \varepsilon_s(\delta), \quad (30)$$

whose dimension is $\delta^{-O_s(1)}$. The reciprocal correlation, Lipschitz parameter, and nilmanifold complexity are bounded by a single exponential in a power of $1/\delta$ for $s \leq 3$, and by a double exponential in a power of $1/\delta$ for $s \geq 4$.

Proof. We induct on s . For $s = 1$, Fourier inversion and Parseval give a character with squared Fourier coefficient at least $\|f\|_{U^2}^4$, which is a degree-one nilsequence.

Assume the theorem through degree $s-1$. Proposition 11.4 gives $B = \sum_{j \leq T} \alpha_j \rho_j$ with

$$|\langle \mathcal{S}_s f, B \rangle| \geq \eta := \delta^{2^{s+1}}/2. \quad (31)$$

Proposition 11.7 then produces g of the form (25), with $L = s!^{s-1}$, such that

$$\|f\bar{g}\|_{U^s} \geq \eta^L. \quad (32)$$

Apply the induction hypothesis to $f\bar{g}$ at parameter η^L . We obtain a degree- $(s-1)$ nilsequence ψ' satisfying

$$|\mathbb{E}_x f(x) \overline{g(x) \psi'(x)}| \geq \varepsilon'. \quad (33)$$

Expand every occurrence of B in (25) using (11). There are T^L terms; by the triangle inequality, some indices $j_1, \dots, j_L$ give

$$\left| \mathbb{E}_x f(x) \overline{\psi'(x)} \prod_{i=1}^L \overline{\rho_{j_i}(h_i - (x/s!, \dots, x/s!))} \right| \geq \varepsilon' / T^L. \quad (34)$$

Affine composition preserves polynomial sequences and degree. By direct products, the product in (34) is a degree- s nilsequence of dimension at most LD , and multiplying by ψ' remains a degree- s nilsequence. This proves (30).

It remains to check that the quantitative recursion has the advertised height. The fixed number $L = s!^{s-1}$ depends only on s , while η is a fixed power of δ . M8 produces T, D polynomial in the inverse small parameters before the M6 structure complexity enters. M6 has single-exponential complexity for the hierarchy degrees needed when the final inverse degree is at most three and double-exponential complexity from degree four onward; M7 and the frequency-to-nilsequence construction raise these quantities only to powers depending on s and polynomially enlarge dimension. The partition-of-unity net contributes at most an exponential in the dimension times logarithmic small-parameter factors. Finally the induction step evaluates the degree- $(s-1)$ bounds at $\eta^L = \delta^{O_s(1)}$ and multiplies a fixed number L of degree- s factors. Thus neither the M9 descent nor the induction introduces a new exponential level. The dimension recursion is polynomial in $1/\delta$, while the complexity, Lipschitz and reciprocal-correlation recursions have exactly the stated single/double-exponential envelope. $\square$

Remark 11.9 (Quantitative dependencies). The bounds in Theorem 11.8 are obtained, in order, from the graph-cover iteration Lemma 11.1, the M6 structure theorem 10.7, the M7 nilmanifold package 10.9, the frequency-conversion Lemma 11.2, the partition-of-unity Proposition 11.4, and the fixed-length descent Proposition 11.7. The final paragraph of the proof above composes these bounds without introducing a new exponential level.

Worked Example 11.10 (Why the factor $s!$ is structural). If $f(x) = e(P(x))$ for a genuine degree- s polynomial, then the diagonal specialization of its s -fold derivative contains $s!P(x)$ at the top degree. Thus the scalings $-1, -1/2, \dots, -1/s$ in Lemma 11.5 are not a proof artifact: their product records the polarization coefficient. Proposition 11.7 removes them by taking products of translates rather than by making a false symmetry assertion.

Chapter 12

Worked models for higher-order structure

Worked Example 12.1 (The Heisenberg nilmanifold and a bracket-quadratic phase). Let

$$G = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, z \in \mathbb{R} \right\}$$

and let Γ be the integer subgroup. The commutator is central: $[(x, y, z), (x', y', z')] = (0, 0, xy' - x'y)$. A polynomial orbit with linearly moving x - and y -coordinates therefore accumulates a quadratic central coordinate. A vertical character of the centre reads that coordinate as a quadratic/bracket- quadratic phase. This concrete model explains why the U^3 inverse theorem is the first place where ordinary linear characters are no longer sufficient.

Worked Example 12.2 (A non-classical polynomial in low characteristic). Over $\mathbb{F}_2^n$, the phase $x \mapsto |x|/4 \pmod{1}$ (with $|x|$ an integer lift of Hamming weight) is not an ordinary $\mathbb{F}_2$ -valued polynomial phase, but its iterated additive derivatives drop degree in the non-classical sense. Multiplication by 2 lowers the depth. This is the basic phenomenon behind the depth chains in the Tao–Ziegler regularity argument: in low characteristic one must track both degree and depth, not degree alone.

Worked Example 12.3 (Why quantitative and qualitative inverse theorems look different). The qualitative Green–Tao–Ziegler theorem says that large $U^{s+1}[N]$ norm forces correlation with some bounded-complexity s -step nilsequence, with no useful explicit bound required. Manners instead constructs a sequence of approximate-polynomial objects and repeatedly repairs them until the nilpotent structure is quantitative. The output is again a nilsequence, but now the dimension, rationality and Lipschitz parameters can be followed through the proof. The two theorems have the same structural endpoint and very different proof architectures.

Part I

Detailed Proof Appendices

Appendix A

The U^3 Inverse Theorem over Finite Fields

A.1 Statement and proof map

Let $V = \mathbb{F}_p^n$, where p is a fixed odd prime, and write

$$e_p(t) = e^{2\pi it/p}.$$

We identify V^* with the additive character group by $\xi \mapsto (x \mapsto e_p(\xi \cdot x))$.

Theorem A.1 (Finite-field U^3 inverse theorem). *For every $\delta > 0$ there is $c_p(\delta) > 0$ such that, if $f : V \rightarrow \mathbb{C}$ is 1-bounded and*

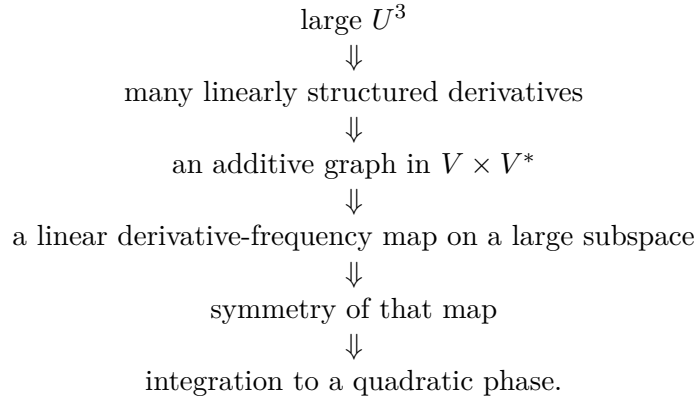
$$\|f\|_{U^3(V)} \geq \delta,$$

then there is a quadratic polynomial $Q : V \rightarrow \mathbb{F}_p$ for which

$$|\mathbb{E}_x f(x) e_p(-Q(x))| \geq c_p(\delta).$$

All losses in the proof below are bounded functions of (p, δ) only. In fact the argument gives explicit power-type and exponential-in-a-power bounds, but optimizing them would obscure the mechanism.

The proof has five genuinely different stages:



The first and fourth arrows are repeated Cauchy–Schwarz; the middle arrow is additive combinatorics; the last is linear algebra.

A.2 A notation for harmless bounded factors

Long Cauchy–Schwarz calculations become unreadable if every bounded factor is renamed after every change of variables. In this chapter, an expression

$$b(x, h, k)$$

means an unspecified complex function of the displayed variables bounded pointwise by 1; the letter b may denote a different such function on the next line. No estimate is hidden in this notation: only factors of modulus at most one are being suppressed. Whenever a phase is retained explicitly, it is because its algebraic dependence matters.

We also write

$$\nabla_a \xi_h := \xi_{h+a} - \xi_h$$

for a finite difference of a frequency-valued map.

A.3 Step 1: many derivatives have Fourier structure

Put

$$F_h(x) = \Delta_h f(x) = f(x+h) \overline{f(x)}.$$

The recursive Gowers identity and the U^2 Fourier formula give

$$\delta^8 \leq \|f\|_{U^3}^8 = \mathbb{E}_h \|F_h\|_{U^2}^4. \quad (\text{A.1})$$

For a 1-bounded function g , define its linear bias

$$\|g\|_{u^2} := \max_{\xi \in V^*} |\widehat{g}(\xi)|.$$

By Parseval,

$$\|g\|_{U^2}^4 = \sum_{\xi} |\widehat{g}(\xi)|^4 \leq \|g\|_{u^2}^2 \sum_{\xi} |\widehat{g}(\xi)|^2 \leq \|g\|_{u^2}^2.$$

Hence (A.1) implies

$$\mathbb{E}_h \|F_h\|_{u^2}^2 \geq \delta^8.$$

Let

$$H = \{h : \|F_h\|_{u^2}^2 \geq \delta^8/2\}.$$

Then

$$\frac{|H|}{|V|} \geq \frac{\delta^8}{2}. \quad (\text{A.2})$$

For each $h \in H$, choose $\xi_h \in V^*$ with

$$\left| \mathbb{E}_x f(x+h) \overline{f(x)} e_p(-\xi_h \cdot x) \right| \geq \frac{\delta^4}{\sqrt{2}}. \quad (\text{A.3})$$

The difficult issue is not finding ξ_h ; it is proving that these choices cohere additively.

A.4 Step 2: the derivative frequencies have many additive quadruples

Proposition A.2 (Coherent frequency graph). *Let*

$$\Gamma = \{(h, \xi_h) : h \in H\} \subseteq V \times V^*.$$

Then

$$E(\Gamma) \geq c(\delta)|V|^3,$$

where $E(\Gamma)$ counts ordered quadruples

$$z_1 + z_4 = z_2 + z_3, \quad z_i \in \Gamma.$$

Equivalently, for at least $c(\delta)|V|^3$ triples (h, a, b) with all four points in H ,

$$\xi_h - \xi_{h+a} - \xi_{h+b} + \xi_{h+a+b} = 0. \quad (\text{A.4})$$

Proof. Square (A.3), average over $h \in H$, and use (A.2). After introducing a second base point $x + k$ and changing variables, we obtain

$$|\mathbb{E}_{h,x,k} b(x+h, k) b(x, k) 1_H(h) e_p(\xi_h \cdot k)| \geq c_1(\delta). \quad (\text{A.5})$$

Here and below the constants are positive functions of δ only.

Apply van der Corput/Cauchy–Schwarz in the variable pair (x, k) to remove the bounded factor that does not contain h in a stable way. Introducing an increment a gives

$$|\mathbb{E}_{h,a,x,k} b(x+h+a, k) b(x+h, k) 1_H(h+a) 1_H(h) e_p((\nabla_a \xi_h) \cdot k)| \geq c_2(\delta). \quad (\text{A.6})$$

This is the first important transformation: one Cauchy–Schwarz step has differentiated the selected frequency map.

Apply van der Corput a second time, now introducing an increment b . The bounded functions again change but remain 1-bounded, while the phase becomes the second difference of ξ :

$$\left| \mathbb{E}_{h,a,b,x,k} b(h, a, b, x) \prod_{\omega \in \{0,1\}^2} 1_H(h + \omega_1 a + \omega_2 b) e_p((\nabla_b \nabla_a \xi_h) \cdot k) \right| \geq c_3(\delta). \quad (\text{A.7})$$

The variable k now occurs only in the displayed character. Taking absolute values in all other bounded factors and averaging first in k , character orthogonality gives

$$\mathbb{E}_k e_p(\eta \cdot k) = 1_{\eta=0}.$$

Therefore

$$\mathbb{E}_{h,a,b} \prod_{\omega \in \{0,1\}^2} 1_H(h + \omega_1 a + \omega_2 b) 1_{\nabla_b \nabla_a \xi_h = 0} \geq c_3(\delta).$$

Multiplying by $|V|^3$ yields the claimed number of additive parallelograms, and (A.4) is exactly the frequency relation saying that the corresponding four graph points form an additive quadruple. $\square$

Idea What the two Cauchy–Schwarz steps accomplished

The phase ξ_h was chosen separately for each derivative. Cauchy–Schwarz does not magically make those choices linear. It differentiates the map $h \mapsto \xi_h$ twice. Orthogo-

nality then says that the second derivative vanishes on many parallelograms. This is the precise source of additive structure in the frequency graph.

A.5 Step 3: BSG, Plünnecke, and graph rectification

Proposition A.2 and the asymmetric Balog–Szemerédi–Gowers theorem (Volume I, Section 2.2, Theorem 2.19 (project tag I.2.015)) give a subset

$$\Gamma' \subseteq \Gamma$$

such that

$$|\Gamma'| \geq K^{-1}|V|, \quad (\text{A.8})$$

and

$$|\Gamma' - \Gamma'| \leq K|\Gamma'|, \quad (\text{A.9})$$

where from now on $K = K(\delta) \geq 1$ denotes a quantity bounded in terms of δ only and is allowed to increase from line to line. By Plünnecke–Ruzsa, also proved in Volume I,

$$|m\Gamma' - \ell\Gamma'| \leq K^{m+\ell}|\Gamma'| \quad (\text{A.10})$$

after replacing K by a fixed power, for every fixed m, ℓ used below.

The set Γ' is still a graph, but its larger sumsets need not be graphs. We now remove this vertical ambiguity completely.

Lemma A.3 (Graph rectification). *There is a subset $\Gamma'' \subseteq \Gamma'$ with*

$$|\Gamma''| \geq K^{-1}|V|$$

such that $4\Gamma'' - 4\Gamma''$ is a graph over its first-coordinate projection. Equivalently,

$$(0, \eta) \in 8\Gamma'' - 8\Gamma'' \implies \eta = 0.$$

Proof. Let

$$A_{\text{vert}} = \{\eta \in V^* : (0, \eta) \in 8\Gamma' - 8\Gamma'\}.$$

Because Γ' itself is a graph, the translates

$$\Gamma' + (0, \eta), \quad \eta \in A_{\text{vert}},$$

are pairwise disjoint. Indeed, an equality

$$(h, \xi_h + \eta) = (h', \xi_{h'} + \eta')$$

forces $h = h'$, hence $\xi_h = \xi_{h'}$, and therefore $\eta = \eta'$. Moreover every such translate is contained in

$$9\Gamma' - 8\Gamma'.$$

Consequently, by (A.10),

$$|A_{\text{vert}}| |\Gamma'| \leq |9\Gamma' - 8\Gamma'| \leq K|\Gamma'|,$$

so

$$|A_{\text{vert}}| \leq K. \quad (\text{A.11})$$

Choose an integer m with

$$p^m > 2|A_{\text{vert}}|.$$

A uniformly random linear map

$$\Psi : V^* \rightarrow \mathbb{F}_p^m$$

sends any fixed nonzero η to 0 with probability p^{-m} . By the union bound and (A.11), with positive probability

$$\ker \Psi \cap (A_{\text{vert}} \setminus \{0\}) = \emptyset.$$

Fix such a Ψ .

Partition Γ' according to the value of $\Psi(\xi)$. One fiber, say

$$\Gamma'' = \{(h, \xi_h) \in \Gamma' : \Psi(\xi_h) = c\},$$

has size at least

$$|\Gamma'|/p^m \geq K^{-1}|V|.$$

If

$$(0, \eta) \in 8\Gamma'' - 8\Gamma'',$$

then $\eta \in A_{\text{vert}}$, while the eight positive and eight negative second coordinates all have the same Ψ -value c , so

$$\Psi(\eta) = 0.$$

The choice of Ψ forces $\eta = 0$.

Finally, if two points of $4\Gamma'' - 4\Gamma''$ have the same first coordinate, their difference has the form $(0, \eta) \in 8\Gamma'' - 8\Gamma''$, hence $\eta = 0$. Thus $4\Gamma'' - 4\Gamma''$ is a graph. $\square$

Idea Why the random map is the right repair

BSG supplies small doubling, but small doubling does not preserve the graph property. The only possible failure is vertical: two additive expressions may have the same h -coordinate and different frequencies. The set of possible vertical differences is small by Plünnecke. A low-dimensional random linear quotient separates those finitely many errors, and a single fiber then kills all of them simultaneously.

A.6 Step 4: Bogolyubov produces a linear derivative map

Let

$$H'' = \pi_1(\Gamma'') \subseteq V.$$

Since Γ'' is a graph,

$$|H''| = |\Gamma''| \geq K^{-1}|V|.$$

The finite-field Bogolyubov lemma proved in Volume I gives a subspace

$$W_0 \leq V$$

of codimension at most K such that

$$W_0 \subseteq 2H'' - 2H''. \tag{A.12}$$

Because $4\Gamma'' - 4\Gamma''$ is a graph, so is $2\Gamma'' - 2\Gamma''$. Thus for every $h \in W_0$ there is a unique frequency $\zeta(h)$ such that

$$(h, \zeta(h)) \in 2\Gamma'' - 2\Gamma''.$$

The uniqueness on the larger graph immediately makes ζ additive. Indeed, for $h, k \in W_0$,

$$(h, \zeta(h)) + (k, \zeta(k)) \in 4\Gamma'' - 4\Gamma'',$$

while

$$(h + k, \zeta(h + k)) \in 2\Gamma'' - 2\Gamma'' \subseteq 4\Gamma'' - 4\Gamma''.$$

These two points have the same first coordinate, so the graph property gives

$$\zeta(h + k) = \zeta(h) + \zeta(k).$$

Hence $\zeta : W_0 \rightarrow V^*$ is linear. Since p is odd, write

$$\zeta(h) = 2Mh$$

for a linear map $M : W_0 \rightarrow V^*$.

Define the graph subspace

$$Z = \{(h, 2Mh) : h \in W_0\} \subseteq 2\Gamma'' - 2\Gamma''.$$

The sum $Z + \Gamma''$ is contained in $3\Gamma'' - 2\Gamma''$, hence has size at most $K|V|$. It is a disjoint union of cosets of the subspace Z . Since

$$|Z| = |W_0| \geq K^{-1}|V|,$$

there are at most K such cosets. One coset

$$(h_0, \xi_0) + Z$$

therefore contains at least $K^{-1}|V|$ points of Γ'' .

Translating the first coordinate by h_0 , we obtain a set $H_0 \subseteq W_0$ of density at least K^{-1} such that

$$\xi_{h_0+h} = 2Mh + \xi_0 \quad (h \in H_0). \quad (\text{A.13})$$

Return to the derivative correlations (A.3). For every $h \in H_0$ choose a scalar $a(h)$ of modulus one which rotates that correlation to the positive real axis, and extend a by zero on $W_0 \setminus H_0$. Put

$$F(x) = f(x + h_0), \quad G(x) = \overline{f(x)} e_p(-\xi_0 \cdot x).$$

Then F, G are fixed 1-bounded functions and averaging the aligned correlations gives

$$|\mathbb{E}_{h \in W_0} \mathbb{E}_{x \in V} a(h) F(x + h) G(x) e_p(-2Mh \cdot x)| \geq \kappa, \quad (\text{A.14})$$

where $\kappa = \kappa(\delta) > 0$. Notice that no copy of f has been discarded: the two x -functions are fixed once and for all. This is the exact bilinear correlation to which the symmetry argument applies.

Nothing has yet forced M to be self-adjoint. That is the symmetry argument.

A.7 Step 5: the symmetry argument in full

For $x, h \in W_0$, define the antisymmetric bilinear form

$$\{x, h\} := Mx \cdot h - Mh \cdot x.$$

Lemma A.4 (Symmetry from a derivative correlation). *Suppose $M : W_0 \rightarrow V^*$ is linear and the fixed-function correlation (A.14) holds. Then the radical*

$$W = \{h \in W_0 : \{x, h\} = 0 \text{ for every } x \in W_0\}$$

has relative density at least κ^4 in W_0 . In particular,

$$\text{codim}_V W \leq \text{codim}_V W_0 + 4 \log_p(1/\kappa),$$

and M is self-adjoint on W :

$$Mh \cdot k = Mk \cdot h \quad (h, k \in W).$$

Proof. First decompose the x -average in (A.14) into cosets of W_0 . By the triangle inequality, one coset $x_0 + W_0$ has an average of magnitude at least κ . Write $x = x_0 + w$. The factor $e_p(-2Mh \cdot x_0)$ depends only on h and is absorbed into $a(h)$; define fixed functions

$$F_0(w) = F(x_0 + w), \quad G_0(w) = G(x_0 + w).$$

Thus

$$|\mathbb{E}_{w,h \in W_0} a_0(h) F_0(w+h) G_0(w) e_p(-2Mh \cdot w)| \geq \kappa$$

with $|a_0|, |F_0|, |G_0| \leq 1$.

We now carry out the symmetry calculation, renaming w as x . Cauchy–Schwarz in h removes $a_0(h)$. After expanding the square and writing the two copies of x as x and y , we obtain

$$\left| \mathbb{E}_{x,y,h \in W_0} F_0(x+h) \overline{F_0(y+h)} G_0(x) \overline{G_0(y)} e_p(-2Mh \cdot (x-y)) \right| \geq \kappa^2. \quad (\text{A.15})$$

Put

$$z = x + y + h.$$

Because W_0 is a vector space, this is a bijective change of variables. Expanding

$$Mh \cdot (x-y) = M(z-x-y) \cdot (x-y)$$

and using linearity of M , every term depending on only one of the pairs (z,y) , (z,x) , x , or y may be absorbed into the corresponding bounded factor. The only genuinely two-variable obstruction that survives is

$$\{x, y\} = Mx \cdot y - My \cdot x.$$

Thus (A.15) becomes

$$|\mathbb{E}_{x,y,z \in W_0} b(z,y) b(z,x) e_p(-2\{x,y\})| \geq \kappa^2. \quad (\text{A.16})$$

Pigeonhole in z gives

$$|\mathbb{E}_{x,y \in W_0} b(y) b(x) e_p(-2\{x,y\})| \geq \kappa^2. \quad (\text{A.17})$$

Cauchy–Schwarz in x now removes $b(x)$:

$$\kappa^4 \leq \mathbb{E}_{x,y,y' \in W_0} b(y) \overline{b(y')} e_p(-2\{x, y-y'\}).$$

Taking absolute values in the remaining bounded factors and setting $h = y - y'$,

$$\kappa^4 \leq \mathbb{E}_{h \in W_0} |\mathbb{E}_{x \in W_0} e_p(-2\{x, h\})|. \quad (\text{A.18})$$

For fixed h , the map $x \mapsto 2\{x, h\}$ is a linear functional on W_0 . Since p is odd, character orthogonality says that the inner average in (A.18) equals 1 precisely when $\{x, h\} = 0$ for every x , and equals 0 otherwise. Hence

$$\frac{|W|}{|W_0|} \geq \kappa^4.$$

Because W is the kernel of the linear map $h \mapsto (x \mapsto \{x, h\})$, it is a subspace, and the codimension bound follows from

$$p^{-\text{codim}_{W_0} W} = \frac{|W|}{|W_0|} \geq \kappa^4.$$

By definition of W , the restriction of M to W is self-adjoint. □

Proof architecture Why symmetry is not optional

Linearization reconstructs a bilinear candidate $B(h, x) = Mh \cdot x$. A bilinear form is the derivative of a scalar quadratic only when its antisymmetric part vanishes. The identity

$$\Delta_h \Delta_k f = \Delta_k \Delta_h f$$

is hidden inside the Cauchy–Schwarz calculation above; it is exactly what annihilates the antisymmetric part after passage to the large radical W .

A.8 Step 6: integrating the symmetric derivative

Return to (A.14). Decompose W_0 into cosets of the radical W found in Lemma A.4. One direction coset $h_1 + W$ retains a correlation $\gg_\delta 1$. Write $h = h_1 + k$. Absorb $a(h_1 + k)$ and the phase $e_p(-2Mh_1 \cdot x)$ into new fixed bounded coefficients, and replace $F(x + h_1 + k)$ by the fixed translate $F_1(x + k)$. We obtain

$$|\mathbb{E}_{k \in W} \mathbb{E}_{x \in V} a_1(k) F_1(x + k) G_1(x) e_p(-2Mk \cdot x)| \geq \kappa_1. \quad (\text{A.19})$$

Next decompose x into cosets of W and choose one coset $x_0 + W$ retaining the same lower bound. Write $x = x_0 + w$ and absorb $e_p(-2Mk \cdot x_0)$ into $a_1(k)$. Thus

$$|\mathbb{E}_{k, w \in W} a_2(k) F_2(w + k) G_2(w) e_p(-2Mk \cdot w)| \geq \kappa_1, \quad (\text{A.20})$$

where $F_2(w) = F_1(x_0 + w)$ and $G_2(w) = G_1(x_0 + w)$ are fixed 1-bounded functions.

Because M is self-adjoint on W , the scalar function

$$q(w) := Mw \cdot w$$

is quadratic on W and satisfies

$$q(w + k) - q(w) = 2Mk \cdot w + Mk \cdot k. \quad (\text{A.21})$$

Hence the bilinear phase in (A.20) may be integrated exactly:

$$e_p(-2Mk \cdot w) = e_p(-q(w + k)) e_p(q(w)) e_p(q(k)).$$

Absorb the last factor into $a_2(k)$ and put

$$U(w) = F_2(w) e_p(-q(w)), \quad V_1(w) = G_2(w) e_p(q(w)).$$

Then

$$|\mathbb{E}_{w, k \in W} V_1(w) a_3(k) U(w + k)| \geq \kappa_1. \quad (\text{A.22})$$

The following elementary Fourier lemma now extracts a linear phase; it is the final analytic step.

Lemma A.5 (A large local trilinear form has linear bias). *Let W be a finite vector space and let $F, u, v : W \rightarrow \mathbb{C}$ be 1-bounded. Then*

$$|\mathbb{E}_{x, h \in W} F(x) u(h) v(x + h)| \leq \max_{\lambda \in W^*} |\mathbb{E}_{x \in W} F(x) e_p(-\lambda \cdot x)|.$$

Proof. With normalized Fourier transform on W ,

$$\mathbb{E}_{x,h} F(x)u(h)v(x+h) = \sum_{\lambda \in W^*} \widehat{F}(\lambda) \widehat{u}(\lambda) \widehat{v}(-\lambda),$$

up to a harmless sign convention. Therefore

$$|\text{left-hand side}| \leq \|\widehat{F}\|_\infty \sum_{\lambda} |\widehat{u}(\lambda) \widehat{v}(-\lambda)|.$$

Cauchy–Schwarz and Parseval bound the final sum by $\|u\|_2 \|v\|_2 \leq 1$. $\square$

Apply the lemma to (A.22) with $F = V_1$. Some $\lambda \in W^*$ satisfies

$$|\mathbb{E}_{w \in W} G_2(w) e_p(q(w) - \lambda \cdot w)| \geq \kappa_1.$$

Unwinding the definitions of G_2, G_1, G shows that $G_2(w)$ is $\overline{f(x_0 + w)}$ multiplied by a fixed linear character and a constant phase. Taking complex conjugates, we obtain a quadratic polynomial Q_{x_0} on the affine coset $x_0 + W$ such that

$$|\mathbb{E}_{x \in x_0 + W} f(x) e_p(-Q_{x_0}(x))| \geq \kappa_1. \quad (\text{A.23})$$

Thus the symmetry calculation has been integrated without any further Cauchy–Schwarz loss.

A.9 Step 7: from local quadratic bias to global correlation

A quadratic polynomial on an affine subspace of a vector space over an odd field extends to a quadratic polynomial on the whole space.

Lemma A.6 (Extension of a quadratic from a coset). *Let $W \leq V$, $y \in V$, and let $Q_y : y + W \rightarrow \mathbb{F}_p$ have vanishing third additive derivative on every cube in $y + W$. Then there is a quadratic polynomial $Q : V \rightarrow \mathbb{F}_p$ whose restriction to $y + W$ is Q_y .*

Proof. Translate the coset to W . Over an odd field, every quadratic polynomial on W has the form

$$q(w) = \frac{1}{2} B(w, w) + \ell(w) + c$$

with B symmetric bilinear. Choose a complement U with $V = W \oplus U$. Extend B arbitrarily to a symmetric bilinear form on V and extend ℓ arbitrarily to a linear form on V . The same formula defines the required extension. Translating back by y completes the proof. $\square$

Extend the phase Q_{x_0} in (A.23) to a global quadratic Q . Let $d = \text{codim } W$. The indicator of the coset $x_0 + W$ has the Fourier expansion

$$1_{x_0 + W}(x) = p^{-d} \sum_{\eta \in W^\perp} e_p(\eta \cdot (x - x_0)).$$

Equation (A.23) implies

$$|\mathbb{E}_x 1_{x_0 + W}(x) f(x) e_p(-Q(x))| \geq \kappa_1 p^{-d}.$$

Substituting the Fourier expansion and using the triangle inequality, some $\eta \in W^\perp$ satisfies

$$|\mathbb{E}_x f(x) e_p(-Q(x) + \eta \cdot x)| \geq \kappa_1 p^{-d}.$$

The phase $Q(x) - \eta \cdot x$ is still quadratic, and $d = O_{p,\delta}(1)$. This proves Theorem A.1.

A.10 A check on the model example

If

$$f(x) = e_p\left(\frac{1}{2}B(x, x) + \ell(x)\right)$$

with B symmetric, then

$$\Delta_h f(x) = e_p\left(B(x, h) + \frac{1}{2}B(h, h) + \ell(h)\right).$$

Thus the selected derivative frequency is exactly

$$\xi_h = B(\cdot, h),$$

a linear self-adjoint map. Every step of the inverse proof is therefore reconstructing a property possessed exactly by a genuine quadratic phase.

A.11 What was genuinely difficult

No step can be replaced by the slogan “large U^3 means quadratic.”

1. The U^2 inverse theorem gives a frequency separately for each derivative.
2. Two Cauchy–Schwarz steps force those frequencies to satisfy many parallelogram relations.
3. BSG and Plünnecke turn partial additivity into small-doubling structure.
4. The random quotient removes the vertical kernel that would otherwise invalidate the graph argument.
5. Bogolyubov converts dense additive structure into a genuine subspace and hence a linear map.
6. A second Cauchy–Schwarz argument kills the antisymmetric part.
7. Only then can the derivative data be integrated to a scalar quadratic.

This chain is the first complete example of the general higher-order inverse philosophy: structured derivatives must first be made coherent and then integrated.

Appendix B

The Cyclic U^3 Inverse Theorem: Local Quadratics

The finite-vector-space argument in Chapter A has a luxury that disappears over $\mathbb{Z}/p\mathbb{Z}$: subspaces are global domains for addition. In a cyclic group the derivative-frequency argument still begins in exactly the same way, but the frequency map becomes linear only *locally*, on a Bohr neighbourhood. The purpose of this chapter is to prove that local theorem rather than merely use it as motivation.

Throughout $G = \mathbb{Z}/p\mathbb{Z}$ with p an odd prime, and $\widehat{G}$ is identified with G through the pairing

$$\xi \cdot x := \frac{\xi x}{p} \pmod{1}.$$

For $S \subseteq \widehat{G}$ put

$$\|x\|_S := \sup_{\gamma \in S} \|\gamma \cdot x\|_{\mathbb{R}/\mathbb{Z}}, \quad B(S, \rho) := \{x \in G : \|x\|_S \leq \rho\}.$$

Regular Bohr sets, their translation estimates, Bogolyubov's lemma, and the Ruzsa–Balog–Szemerédi machinery were proved in Volumes I–II and will be used without reproving them here.

B.1 The theorem

A phase $\phi : B \rightarrow \mathbb{R}/\mathbb{Z}$ is *locally quadratic* if

$$\partial_{h_1} \partial_{h_2} \partial_{h_3} \phi(x) = 0$$

whenever all eight vertices of the corresponding cube lie in B .

Theorem B.1 (Cyclic U^3 inverse theorem). *For every $\delta > 0$ there are constants $r = r(\delta)$, $\rho = \rho(\delta) > 0$ and $c = c(\delta) > 0$ with the following property. If $f : G \rightarrow \mathbb{C}$ is 1-bounded and*

$$\|f\|_{U^3(G)} \geq \delta,$$

then there is a regular Bohr set $B = B(S, \rho)$ with $|S| \leq r$ such that

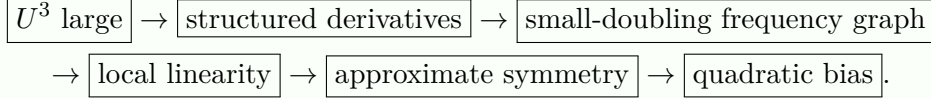
$$\mathbb{E}_{y \in G} \|f\|_{u^3(y+B)} \geq c,$$

where

$$\|f\|_{u^3(y+B)} := \sup_{\phi \text{ locally quadratic on } y+B} |\mathbb{E}_{x \in y+B} f(x) e(-\phi(x))|.$$

The proper-progression and bracket-coordinate lemmas proved later in this chapter turn the local quadratic furnished by the theorem into a bracket-quadratic model, with all rank, progression-length, and nilsequence-complexity losses quantified in those lemmas.

Proof architecture The five mechanisms



The difficult point is coherence: a Fourier frequency chosen separately for each derivative has no reason a priori to vary linearly with the increment.

B.2 Derivative frequencies and a small-doubling graph

The recursive identity gives

$$\|f\|_{U^3}^8 = \mathbb{E}_h \|\Delta_h f\|_{U^2}^4.$$

Hence there is $H \subseteq G$, $|H| \gg_\delta p$, for which

$$\|\Delta_h f\|_{U^2} \gg_\delta 1.$$

The U^2 inverse theorem supplies, for each $h \in H$, a frequency $\xi_h \in \widehat{G}$ such that

$$|\mathbb{E}_x \Delta_h f(x) e(-\xi_h \cdot x)| \gg_\delta 1. \quad (\text{B.1})$$

After multiplying by scalar phases we may take the real parts positive.

Proposition B.2 (Frequency parallelograms). *For $\gg_\delta p^3$ additive quadruples*

$$h_1 + h_2 = h_3 + h_4, \quad h_i \in H,$$

one has

$$\xi_{h_1} + \xi_{h_2} = \xi_{h_3} + \xi_{h_4}.$$

Proof. Average (B.1) in h , apply Cauchy–Schwarz once to remove a factor depending only on the base point, change variables so that the two copies of h form a parallelogram, and apply Cauchy–Schwarz again. The phase left after the four copies of f cancel is

$$e(-(\xi_{h_1} + \xi_{h_2} - \xi_{h_3} - \xi_{h_4}) \cdot x).$$

Averaging in x annihilates this term unless the displayed frequency is zero. The starting average is bounded below in terms of δ and all summands are bounded by 1, so a positive proportion of the $O(p^3)$ additive quadruples survive. $\square$

Thus

$$\Gamma := \{(h, \xi_h) : h \in H\} \subseteq G \times \widehat{G}$$

has energy $\gg_\delta |\Gamma|^3$. By Balog–Szemerédi–Gowers followed by Plünnecke–Ruzsa, after replacing Γ by a subset of comparable size we may assume that for some $K = K(\delta)$

$$K^{-1}p \leq |\Gamma| \leq |9\Gamma - 8\Gamma| \leq Kp. \quad (\text{B.2})$$

B.3 Killing the vertical kernel

The graph may cease to be a graph after taking differences: one might have $(0, \eta) \in 8\Gamma - 8\Gamma$ with $\eta \neq 0$. The following elementary separation lemma is the device that removes this ambiguity.

Lemma B.3 (Separating a finite set of frequencies). *Let $A \subseteq \widehat{G}$ contain 0 and have $|A| \leq L$. Then there is a set $S \subseteq G$, $|S| \leq 1 + 2 \log_2 L$, such that*

$$A \cap \{\eta : \|\eta \cdot s\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{1}{4} \text{ for every } s \in S\} = \{0\}.$$

Proof. For a fixed nonzero η , as s ranges uniformly over G the value $\eta \cdot s$ is uniform in $\mathbb{R}/\mathbb{Z}$ on the p -torsion grid. For $p > 4$,

$$\Pr_s(\|\eta \cdot s\| \leq \frac{1}{4}) \leq \frac{1}{2} + O(p^{-1}).$$

Choose $m = \lceil 2 \log_2 L \rceil + 1$ independent s_j . The expected number of nonzero $\eta \in A$ satisfying all m constraints is < 1 for sufficiently large p . Hence some choice has none. The finitely many smaller primes can be absorbed into the constants of the theorem. $\square$

Lemma B.4 (Graph refinement). *Under (B.2), there is a subset*

$$\Gamma'' = \{(h, \xi_h) : h \in H''\} \subseteq \Gamma, \quad |\Gamma''| \gg_K p,$$

such that $4\Gamma'' - 4\Gamma''$ is again a graph over its first coordinate.

Proof. Let

$$A := \{\eta \in \widehat{G} : (0, \eta) \in 8\Gamma - 8\Gamma\}.$$

For each $\eta \in A$, the translates $(0, \eta) + \Gamma$ lie in $9\Gamma - 8\Gamma$. They are pairwise disjoint because Γ itself is a graph. Hence

$$|A||\Gamma| \leq |9\Gamma - 8\Gamma| \leq Kp,$$

and (B.2) gives $|A| \leq K^2$.

Apply Lemma B.3 to obtain $S \subseteq G$ with $|S| = O(\log K)$ and

$$A \cap B_{\widehat{G}}(S, \frac{1}{4}) = \{0\}.$$

Map $\widehat{G}$ to $(\mathbb{R}/\mathbb{Z})^S$ by

$$\Psi(\eta) = (\eta \cdot s)_{s \in S}.$$

Partition the torus into cubes of side $1/64$. By pigeonhole, a cube contains the second coordinates of $\gg_K p$ elements of Γ ; let this subset be Γ'' .

Take two elements of $4\Gamma'' - 4\Gamma''$ with the same first coordinate. Their difference is $(0, \eta)$ with

$$(0, \eta) \in 8\Gamma'' - 8\Gamma'' \subseteq 8\Gamma - 8\Gamma,$$

so $\eta \in A$. The second coordinate η is a signed sum of sixteen frequencies whose images under Ψ all lie in the same cube. Hence, in every S -coordinate,

$$\|\eta \cdot s\|_{\mathbb{R}/\mathbb{Z}} \leq 16/64 = 1/4.$$

Thus $\eta \in A \cap B_{\widehat{G}}(S, 1/4) = \{0\}$. The two elements therefore have the same second coordinate. This is exactly the assertion that $4\Gamma'' - 4\Gamma''$ is a graph. $\square$

B.4 A locally linear frequency map

Put

$$\Gamma'' = \{(h, \xi_h) : h \in H''\}.$$

Bogolyubov's theorem from Volume II gives a regular Bohr set

$$B_0 = B(S_0, \tfrac{1}{4}) \subseteq 2H'' - 2H'', \quad |S_0| = O_K(1).$$

Because $2\Gamma'' - 2\Gamma''$ is a graph, there is a unique map $M : B_0 \rightarrow \widehat{G}$ such that

$$\{(h, 2M(h)) : h \in B_0\} \subseteq 2\Gamma'' - 2\Gamma''. \quad (\text{B.3})$$

The factor 2 is harmless because p is odd.

Lemma B.5 (Local Freiman linearity). *If $h, h' \in B_0$ and all the sums/differences under consideration lie in B_0 , then*

$$M(h \pm h') = M(h) \pm M(h').$$

In particular, on a smaller Bohr set $B(S_0, 1/8)$, M is additive on every local parallelogram.

Proof. Suppose $h_1 + h_2 = h_3 + h_4$ with all $h_i \in B_0$. By (B.3), the point

$$(0, 2M(h_1) + 2M(h_2) - 2M(h_3) - 2M(h_4))$$

lies in $4\Gamma'' - 4\Gamma''$. This set is a graph by Lemma B.4; its only point above 0 is $(0, 0)$. Hence the second coordinate vanishes. Taking $h_3 = h + h'$, $h_4 = 0$ gives additivity; subtraction is similar. $\square$

Choose a regular $B_1 = B(S_0, \rho_1)$ with $\rho_1 \in [1/16, 1/8]$. Averaging translates of H'' across B_1 gives $x_0 \in G$ for which

$$A := \{h \in B_1 : x_0 + h \in H''\}$$

has relative density $\gg_K 1$ in B_1 . For $h, h' \in A$,

$$(h - h', \xi_{x_0+h} - \xi_{x_0+h'}) \in \Gamma'' - \Gamma'',$$

and comparison with (B.3) gives

$$\xi_{x_0+h} - \xi_{x_0+h'} = 2M(h - h').$$

Using local additivity of M , there is a fixed ξ_0 such that

$$\xi_{x_0+h} = \xi_0 + 2M(h) \quad (h \in A). \quad (\text{B.4})$$

Return to the derivative correlations. For $h \in A$ choose a scalar $a(h)$ of modulus one which aligns the correlation in (B.1), and extend a by zero on $B_1 \setminus A$. Put

$$F(x) = f(x + x_0), \quad G(x) = \overline{f(x)} e(-\xi_0 \cdot x).$$

Then F, G are fixed 1-bounded functions, and (B.4) gives the coherent correlation

$$|\mathbb{E}_{h \in B_1} \mathbb{E}_{x \in G} a(h) F(x + h) G(x) e(-2M(h) \cdot x)| \gg_\delta 1. \quad (\text{B.5})$$

No copy of f has been discarded: the only auxiliary factor depending on h is the scalar alignment $a(h)$. This is the promised coherent derivative phase.

B.5 The symmetry calculation

Define

$$\{x, h\} := M(x) \cdot h - M(h) \cdot x.$$

If M were globally linear and came from a quadratic, this antisymmetric form would vanish. We now prove that it is small on a smaller Bohr set.

Lemma B.6 (Approximate symmetry). *There are constants $R = R(\delta) \in \mathbb{N}$ and $C = C(\delta) > 0$ and a Bohr set $B_3 = B(S_3, \rho_3) \subseteq B_1$ with $|S_3| \leq R$ such that*

$$\|\{x, z\}\|_{\mathbb{R}/\mathbb{Z}} \leq C\|x\|_{S_3} \quad (x, z \in B_3).$$

Consequently, for every $\varepsilon > 0$, if

$$\rho_4 := \min\{\rho_3, \varepsilon/C\}, \quad B_4 := B(S_3, \rho_4),$$

then

$$\|\{x, z\}\|_{\mathbb{R}/\mathbb{Z}} \leq \varepsilon \quad (x, z \in B_4).$$

Proof. Choose nested regular Bohr sets

$$B_2 \subseteq B_2^+ \subseteq B_1$$

so that adding an element of B_2 changes the normalized indicator of B_2^+ in L^1 by $o_\delta(1)$. Average the x -variable in (B.5) over translates $y + B_2^+$; since y is uniform on G , this decomposition is exact before restricting to interior points. Pigeonhole a translate on which the correlation remains $\gg_\delta 1$, and write $x = y + u$. The phase $M(h) \cdot y$ depends only on h and is absorbed into $a(h)$. After shrinking the h -set by a regularity error, we are therefore reduced to

$$\left| \mathbb{E}_{h \in B_2^+} \mathbb{E}_{u \in B_2^+} a_0(h) F_0(u + h) G_0(u) e(-2M(h) \cdot u) \right| \gg_\delta 1, \quad (\text{B.6})$$

where all displayed additions lie in B_1 and F_0, G_0 are fixed 1-bounded functions.

The algebra is now exactly the symmetry calculation proved in Lemma A.4; the only extra bookkeeping is that the Bohr sets are not groups. We spell out the point where regularity is used. Cauchy–Schwarz in h expands (B.6) into an average over u, v, h . Restrict u, v, h to the smaller set B_2 so that all translated arguments remain in B_2^+ ; the discarded boundary has relative measure $o_\delta(1)$. With

$$z = u + v + h$$

the phase is

$$-2M(z - u - v) \cdot (u - v).$$

Local additivity of M on B_1 expands this into terms depending on (z, u) , (z, v) , u , or v , together with the single mixed term

$$-2\{u, v\} = -2(M(u) \cdot v - M(v) \cdot u).$$

All one-pair terms are absorbed into fixed bounded factors. Pigeonholing z therefore gives

$$|\mathbb{E}_{u, v \in B_2} c(u) c'(v) e(-2\{u, v\})| \gg_\delta 1. \quad (\text{B.7})$$

Apply Cauchy–Schwarz in u . Expanding the square and then pigeonholing one value v' shows that there is $A_0 \subseteq B_2$ of relative density $\gg_\delta 1$ such that, for every $v \in A_0$,

$$|\mathbb{E}_{u \in B_2} e(-2\{u, v - v'\})| \gg_\delta 1. \quad (\text{B.8})$$

For fixed z in the local difference domain, the map $u \mapsto e(2\{u, z\})$ is an ordinary character on every permitted local sum, because M is locally additive. The large-character lemma for a regular Bohr set, proved in Volume II, therefore turns (B.8) into

$$\|2\{u, v - v'\}\|_{\mathbb{R}/\mathbb{Z}} \leq C_\delta \|u\|_{S_0} \quad (u \in B_2).$$

By local additivity in the second variable the same estimate holds for $z \in 2A_0 - 2A_0$ whenever the required sums stay in B_1 .

The local Bogolyubov lemma from Volume II supplies a Bohr set

$$B_3 = B(S_3, \rho_3) \subseteq (2A_0 - 2A_0) \cap B_2,$$

with $|S_3| = O_\delta(1)$ and $\rho_3 \gg_\delta 1$. Enlarge S_3 by S_0 and shrink the radius once. Then

$$\|2\{u, z\}\| \leq C_\delta \|u\|_{S_3} \quad (u, z \in B_3).$$

Since multiplication by 2 is invertible on the p -torsion character group, the same estimate, with a changed constant, holds for $\{u, z\}$. A final radius shrink gives the stated $\varepsilon(\delta)$ bound. $\square$

Idea Why exact symmetry is unnecessary

A tempting but incorrect shortcut is to try to make $M(x) \cdot h = M(h) \cdot x$ exactly on a Bohr set by refining cells. The proof does not need this. Approximate symmetry on a sufficiently small Bohr set is enough, because the error occurs inside an exponential and can be made smaller than the correlation already obtained.

B.6 Integrating the local derivative phase

On B_0 define

$$q(u) := M(u) \cdot u.$$

Local additivity of M makes q locally quadratic. More precisely, whenever $u, h, u + h \in B_0$,

$$2M(h) \cdot u = q(u + h) - q(u) - q(h) - \{u, h\}. \quad (\text{B.9})$$

Indeed, expanding $q(u + h)$ gives

$$q(u + h) = q(u) + q(h) + M(u) \cdot h + M(h) \cdot u,$$

and subtracting $\{u, h\} = M(u) \cdot h - M(h) \cdot u$ proves the identity.

We now localize the original coherent correlation rather than merely assert that this can be done. Choose a very small regular Bohr set $C \subseteq B_4$ such that $C + C \subseteq B_4$. Regularity of B_1 gives an approximate partition of its normalized measure by translates of C : for every 1-bounded H ,

$$\mathbb{E}_{h \in B_1} H(h) = \mathbb{E}_{t \in B_1} \mathbb{E}_{k \in C} H(t + k) + o_\delta(1)$$

after replacing B_1 by a harmless inner regular shrink. Apply this to the h -average in (B.5). Pigeonhole a translate $t + C$ retaining a controlled correlation. Write $h = t + k$. Local additivity of M gives $M(t + k) = M(t) + M(k)$; absorb the $M(t) \cdot x$ character into the fixed function G and translate F accordingly. Thus

$$|\mathbb{E}_{k \in C} \mathbb{E}_{x \in G} a_1(k) F_1(x + k) G_1(x) e(-2M(k) \cdot x)| \gg_\delta 1. \quad (\text{B.10})$$

Average the x -variable exactly over translates $y + C$:

$$\mathbb{E}_{x \in G} H(x) = \mathbb{E}_{y \in G} \mathbb{E}_{u \in C} H(y + u).$$

Some y preserves the lower bound in (B.10). Put $x = y + u$. The factor $e(-2M(k) \cdot y)$ depends only on k and is absorbed into $a_1(k)$. We obtain

$$|\mathbb{E}_{u,k \in C} a_2(k) F_2(u+k) G_2(u) e(-2M(k) \cdot u)| \gg_\delta 1. \quad (\text{B.11})$$

All points $u, k, u+k$ lie in B_4 , so Lemma B.6 and (B.9) give, uniformly on the averaging domain,

$$e(-2M(k) \cdot u) = e(-q(u+k)) e(q(u)) e(q(k)) + O(\varepsilon).$$

Choose the radius in the symmetry lemma so that the accumulated error is smaller than half the correlation. Absorb $e(q(k))$ into $a_2(k)$ and set

$$U(w) = F_2(w) e(-q(w)), \quad V(w) = G_2(w) e(q(w)).$$

Then

$$|\mathbb{E}_{u,k \in C} V(u) a_3(k) U(u+k)| \gg_\delta 1. \quad (\text{B.12})$$

We extract a character by a completely explicit Fourier argument. Let $\beta = |C|/|G|$ and $C_2 = C + C$. Extending the three factors by zero, (B.12) implies

$$|\mathbb{E}_{u,k \in G} (V1_C)(u) (a_31_C)(k) (U1_{C_2})(u+k)| \gg_\delta \beta^2.$$

Fourier inversion gives

$$\sum_{\xi \in \widehat{G}} \widehat{V1_C}(\xi) \widehat{a_31_C}(\xi) \widehat{U1_{C_2}}(-\xi).$$

By Cauchy–Schwarz and Parseval, its modulus is at most

$$\|\widehat{V1_C}\|_\infty \|a_31_C\|_2 \|U1_{C_2}\|_2 \leq \|\widehat{V1_C}\|_\infty \sqrt{\beta \mu(C_2)}.$$

Regularity and bounded rank give $\mu(C_2) \ll_\delta \beta$. Hence some $\xi \in \widehat{G}$ satisfies

$$|\mathbb{E}_{u \in C} V(u) e(-\xi \cdot u)| \gg_\delta 1.$$

Unwinding V and taking complex conjugates gives a locally quadratic phase ϕ_y on the translate $y + C$ such that

$$|\mathbb{E}_{x \in y+C} f(x) e(-\phi_y(x))| \gg_\delta 1.$$

Averaging over the preliminary choices of y (or, equivalently, retaining the average before the final pigeonhole) yields

$$\mathbb{E}_{y \in G} \|f\|_{u^3(y+C)} \gg_\delta 1.$$

This is the local quadratic conclusion of Theorem B.1.

B.7 From local quadratics to bracket quadratics

Remark B.7 (Scope of the local-to-global extension step). The December-2023 correction to Green–Tao’s original U^3 paper points out that its Proposition 3.2 is false for an arbitrary ambient abelian group. Nothing in the present proof uses that false generality. The finite-field argument occurs in a vector space over a finite field, where the required extension is ordinary linear algebra. In the cyclic argument below we first pass to a *proper* generalized progression inside the Bohr set; the coordinate map is therefore injective on the box and the subgroup part is trivial. The locally linear phase is extended only in this proper-box setting. We make this restriction explicit in the theorem and its proof, so no arbitrary-group extension principle is being imported.

The inverse theorem is already proved, but the form of the structured phase matters later. Volume II proved that a bounded-rank Bohr set contains a large proper generalized progression

$$P = \{a + n_1 v_1 + \cdots + n_r v_r : 0 \leq n_j < L_j\}.$$

On a proper progression, a locally linear map has coordinate form

$$M\left(\sum_j n_j v_j\right) = M(0) + \sum_j n_j \omega_j.$$

The coordinates n_j are recovered from the ambient cyclic variable by choosing a fundamental domain. Those choices create integer carries, and after replacing floors by $t - \{t\}$, one obtains expressions built from

$$n, \quad \{\alpha n + \theta\}.$$

Integrating once gives bracket quadratics generated by

$$\alpha n^2, \quad \beta n\{\gamma n + \theta\}, \quad \lambda\{\gamma n + \theta\}\{\gamma' n + \theta'\},$$

plus linear and constant terms. The coordinate calculation is the degree-two specialization of the bracket-linearisation proof in Appendix G.

Appendix E constructs these bracket phases explicitly as vertical characters on products of Heisenberg nilmanifolds and tori. A smooth partition of unity around the finitely many carry boundaries turns the cutoff progression phase into a bounded-complexity two-step nilsequence. This proves the last assertion of Theorem B.1.

Corollary B.8 (Global degree-two inverse theorem on intervals). *For every $\delta > 0$ there are $c, M > 0$ such that every 1-bounded $f : [N] \rightarrow \mathbb{C}$ with*

$$\|f\|_{U^3[N]} \geq \delta$$

correlates with a two-step polynomial nilsequence of complexity at most M :

$$\left| \mathbb{E}_{n \leq N} f(n) \overline{F(g(n)\Gamma)} \right| \geq c.$$

Proof. Use the zero-extension definition of $U^3[N]$ from Volume III. Fix a sufficiently large absolute constant C . The fixed-dilate prime corollary of the elementary Chebyshev bounds proved in Volume I gives, for all sufficiently large N , a prime

$$CN < p \leq BCN,$$

where B is absolute. Regard the zero extension of f as a function on $\mathbb{Z}/p\mathbb{Z}$. Since no relevant cube wraps around, the normalizing factor in the definition of $U^3[N]$ shows that its ambient $U^3(\mathbb{Z}/p\mathbb{Z})$ norm is bounded below by a constant depending only on δ and C .

Apply Theorem B.1. For some translate $y + B$ there is a locally quadratic phase ϕ with

$$|\mathbb{E}_{x \in y+B} f(x) e(-\phi(x))| \gg_\delta 1.$$

The rank and radius of B are bounded in terms of δ , hence $|B| \gg_\delta p$. We first record the localization step carefully. Choose a much smaller concentric regular Bohr set $B' \subseteq B$. Regularity gives

$$\|1_B(\cdot - b) - 1_B\|_{L^1(G)} \ll_\delta \varepsilon |B| \quad (b \in B')$$

when the radius of B' is an ε -fraction of that of B . Consequently, for every 1-bounded $g : B + B' \rightarrow \mathbb{C}$,

$$\mathbb{E}_{t \in B} \mathbb{E}_{b \in B'} g(t + b) = \mathbb{E}_{x \in B} g(x) + O_\delta(\varepsilon).$$

Translate the original Bohr neighbourhood to the origin and take $g(b) = f(y+b)e(-\phi(y+b))$ on the required small enlargement of B . Choose ε below the correlation constant. Some translate $y+t+B'$ therefore retains a controlled correlation.

Now choose an even smaller regular Bohr set $B'' \subseteq B'$ and, by the Bohr-set progression theorem of Volume II, a proper generalized progression $P \subseteq B''$ of bounded rank and size $|P| \gg_\delta |B''|$. Make B'' sufficiently small compared with B' that every $p_0 \in P$ is a small translation for B' . Repeating the preceding convolution argument, with P in place of B' , gives a translate $z+P \subseteq y+t+B'+B''$ on which

$$|\mathbb{E}_{x \in z+P} f(x)e(-\phi(x))| \gg_\delta 1.$$

All points involved remain in the original domain of local quadraticity after the initial harmless shrink of B .

The coordinate calculation preceding this corollary expresses $e(\phi(x))$ on $z+P$ as a bounded-complexity bracket quadratic. Appendix E realizes that bracket quadratic as a two-step nilsequence. It remains only to globalize the cutoff 1_{z+P} . The progression coordinates are given by boundedly many torus coordinates and carry conditions. Choose Lipschitz functions on those tori which agree with the indicator away from boundary strips of total relative measure at most ε . Their product is a bounded-complexity nilsequence κ of step at most two (after refining to pure Bohr coordinates one may take it to be one-step) satisfying

$$\|\kappa - 1_{z+P}\|_{L^1(\mathbb{Z}/p\mathbb{Z})} \ll_\delta \varepsilon.$$

Multiplying the bracket-quadratic nilsequence by κ remains a two-step nilsequence. Taking ε smaller than the local correlation loss turns the local average into a global correlation of size $\gg_\delta |P|/p \gg_\delta 1$ on $\mathbb{Z}/p\mathbb{Z}$.

Finally restrict this nilsequence to $[N]$. Since f vanishes outside $[N]$, the cyclic correlation is exactly a constant multiple of the normalized interval correlation. Complexity depends only on δ . $\square$

Worked Example B.9 (A carry creates a genuinely two-step phase). Let

$$\phi(n) = \alpha n \{\beta n\}.$$

Writing

$$\{\beta(n+h)\} = \{\beta n\} + \{\beta h\} - c(n, h), \quad c(n, h) \in \mathbb{Z},$$

gives

$$\Delta_h \phi(n) = \alpha h \{\beta n\} + \alpha(n+h) \{\beta h\} - \alpha(n+h) c(n, h).$$

On a Bohr cell where the carry $c(n, h)$ is frozen, this is locally linear in n ; globally the carry jumps. The Heisenberg central coordinate records exactly this jump. This is why “quadratic” over a cyclic group is naturally expressed by a two-step nilsequence rather than by an ordinary polynomial alone.

B.8 What the proof teaches

The proof has now established every additive-combinatorial step of the cyclic U^3 inverse theorem. The only previously proved inputs were the foundational results of Volumes I–II: BSG/Plünnecke, Bohr regularity, Bogolyubov, and elementary large-spectrum control. There was no appeal to the general higher-order inverse theorem.

The higher-degree argument will repeat the same five mechanisms with one replacement:

$$\text{single frequency } \xi_h \rightsquigarrow \text{top nilcharacter symbol } [\chi_h]_{s-1}.$$

The rest of Volume IV develops the invariant language required to make that replacement rigorous.

Appendix C

Equidistribution on Nilmanifolds

C.1 Weyl's torus criterion

A sequence $x_n \in \mathbb{T}^d$ is equidistributed exactly when every nontrivial character has vanishing average.

Proposition C.1 (Weyl criterion). *The sequence (x_n) is equidistributed in $\mathbb{T}^d$ if and only if for every nonzero $k \in \mathbb{Z}^d$,*

$$\frac{1}{N} \sum_{n \leq N} e(k \cdot x_n) \rightarrow 0.$$

Proof. Necessity follows by testing against characters. For sufficiency, approximate any continuous function uniformly by a trigonometric polynomial; all nonconstant Fourier modes vanish in the average and the constant mode gives the integral. $\square$

C.2 A quantitative torus obstruction

If $n\alpha$ is not δ -equidistributed on $[N]$, approximate a witnessing Lipschitz function by Fourier modes $|k| \leq M$. Some nonzero k with $|k| \leq \delta^{-O_d(1)}$ has

$$\left| \frac{1}{N} \sum_{n \leq N} e(nk \cdot \alpha) \right| \gg \delta^{O_d(1)}.$$

The geometric-sum formula then forces

$$\|k \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \ll \delta^{-O_d(1)} N^{-1}.$$

Thus failure of equidistribution is witnessed by a low-complexity rational resonance.

C.3 Smoothness norms

For

$$P(n) = \alpha_0 + \alpha_1 \binom{n}{1} + \cdots + \alpha_s \binom{n}{s} \in \mathbb{R}/\mathbb{Z},$$

define

$$\|P\|_{C^\infty[N]} = \max_{1 \leq j \leq s} N^j \|\alpha_j\|_{\mathbb{R}/\mathbb{Z}}.$$

Small smoothness norm means the phase is nearly rational/slow at the natural scale of each degree.

C.4 Factorization

Quantitative equidistribution leads to a factorization

$$g(n) = \varepsilon(n)g'(n)\gamma(n),$$

where ε is smooth, γ is rational and periodic, and g' is equidistributed in a rational subnilmanifold. If equidistribution fails, a horizontal character cuts the orbit down to a proper rational subgroup; dimension therefore decreases and the procedure terminates.

Worked Example C.2 (Quadratic major/minor arcs). For $P(n) = \alpha n^2 + \beta n$, a good approximation $\alpha = a/q + O(N^{-2}\delta^{-C})$ with small q creates structure on residue classes modulo q . Without such an approximation, Weyl differencing gives cancellation. The nilmanifold factorization theorem is the nonabelian, multi-degree analogue of this classical major-arc/minor-arc dichotomy.

C.5 Why inverse theory needs factorization

A nilsequence produced by induction may contain rational or slowly varying redundancies. Factorization isolates the genuinely irrational top-degree component, allowing one either to count it by equidistribution or to lower dimension/degree. Quantitative improvements here propagate directly into the inverse theorem and then into quantitative Szemerédi bounds.

C.6 Weyl’s inequality as the abelian prototype

The quantitative obstruction theorem should be compared with the classical statement for a polynomial phase

$$P(n) = \alpha_d n^d + \cdots + \alpha_0.$$

Repeated van der Corput differencing transforms

$$\sum_{n \leq N} e(P(n))$$

into averages of linear exponential sums. If the original sum is large, the final linear phases cannot all oscillate rapidly. One concludes that the leading coefficient α_d admits a rational approximation with controlled denominator.

Thus the classical conclusion is

large polynomial exponential sum $\Rightarrow$ Diophantine resonance of a coefficient.

The quantitative Leibman theorem replaces “coefficient” by “horizontal character applied to the polynomial orbit.”

C.7 Horizontal characters in the Heisenberg example

For the real Heisenberg group, the commutator subgroup is the central z -axis. A horizontal character must vanish on this commutator and therefore has the form

$$\eta(x, y, z) = kx + \ell y \pmod{1}, \quad k, \ell \in \mathbb{Z}.$$

It cannot directly see the vertical coordinate z . Nevertheless a failure of equidistribution caused by vertical structure eventually creates a horizontal obstruction after differencing or

factorization. This is a key conceptual point: the theorem detects nonabelian obstruction through an abelian quotient, but the quotient may only become visible after the orbit has been reorganized.

Worked Example C.3 (A visibly nonequidistributed orbit). Let

$$g(n) = (n\alpha, 0, 0)\Gamma$$

in a Heisenberg nilmanifold. If $\alpha = a/q$ is rational, then $g(n)$ is periodic with period q in the horizontal x coordinate. The horizontal character $\eta(x, y, z) = qx$ satisfies

$$\eta(g(n)) = qn\alpha = an = 0 \pmod{1}.$$

Its $C^\infty[N]$ norm is zero, giving the strongest possible obstruction. If α is merely within $O((qN)^{-1})$ of a/q , then the same character varies only slowly, producing quantitative nonequidistribution.

C.8 Rational subgroups and dimension drop

Given a horizontal character η , its kernel

$$G' = \ker \eta$$

is a closed subgroup containing the commutator subgroup. If η has rational coefficients relative to the lattice, then $G' \cap \Gamma$ is a lattice in G' , so

$$G'/(G' \cap \Gamma)$$

is a lower-dimensional nilmanifold. This is the geometric reason a resonance is useful: it does not merely identify a bad frequency; it gives a smaller ambient space in which the orbit can be reanalyzed.

C.9 Smooth, rational, and irrational components

The factorization

$$g = \varepsilon g' \gamma$$

separates three types of behavior.

Smooth component. The increment

$$\varepsilon(n+1)\varepsilon(n)^{-1}$$

is $O(N^{-1})$ in a fixed metric. On intervals of length $o(N)$, ε is almost constant and can be frozen.

Rational component. The orbit $\gamma(n)\Gamma$ is periodic with controlled period. One passes to residue classes and freezes it exactly.

Irrational component. The remaining orbit $g'(n)\Gamma'$ is quantitatively equidistributed and can be replaced, in Lipschitz averages, by Haar measure on its orbit closure.

This decomposition is the nilpotent analogue of writing a frequency as

$$\alpha = \frac{a}{q} + \theta$$

with a rational major-arc part and a small error, except that the irrational component may live on a proper nonabelian subnilmanifold.

C.10 Why total equidistribution is often needed

Ordinary equidistribution on $[N]$ does not guarantee equidistribution on long subprogressions. Density-increment and transference arguments frequently restrict n to arithmetic progressions, so one needs *total equidistribution*: every progression of length at least δN should have approximately the expected average. Quantitative factorization theorems are normally designed to produce this stronger property after passing to controlled residue classes.

Warning C.4 (A metric statement depends on a rational coordinate system). Saying that an element of G/Γ is “ ε -close” or that a polynomial sequence has “complexity at most M ” is not invariant without quantitative rationality data. Mal’cev coordinates provide that data. This is why rational bases and heights, though technically unglamorous, are indispensable in quantitative nilmanifold arguments.

Appendix D

The Green–Tao–Ziegler Inverse Theorem

D.1 The theorem and the proof contract

We now prove the central structural theorem of higher-order Fourier analysis on intervals.

Theorem D.1 (Green–Tao–Ziegler inverse theorem). *Fix $s \geq 1$ and $\delta > 0$. There are constants*

$$c = c(s, \delta) > 0, \quad M = M(s, \delta), \quad N_0 = N_0(s, \delta)$$

such that, for every integer $N \geq N_0$, every 1-bounded $f : [N] \rightarrow \mathbb{C}$ satisfying

$$\|f\|_{U^{s+1}[N]} \geq \delta$$

correlates with an s -step polynomial nilsequence of complexity at most M :

$$\left| \mathbb{E}_{n \leq N} f(n) \overline{F(g(n)\Gamma)} \right| \geq c.$$

Proof location. A complete proof is given later under the heading “Step 8: close the induction.” The present statement is the precise dependency interface used before that proof.

The theorem is due to Green, Tao, and Ziegler [3]. The present text gives the additive-combinatorial and nilcharacter reconstruction rather than citing the theorem as an input. The general nilmanifold infrastructure needed by that reconstruction is also included in the detailed proof appendices of this volume: Mal’cev coordinates and rational subgroups in Appendix L, and quantitative Leibman/factorization at Theorems J.2, K.2, and M.23. They are geometric prerequisites, not substitutes for the inverse-theorem argument. Every step that turns large Gowers norm into coherent derivative structure is proved here or in Appendices F–H.

Proof architecture The proof in one line

$$\begin{aligned} & \boxed{U^{s+1} \text{ large}} \rightarrow \boxed{\text{structured derivatives}} \\ & \rightarrow \boxed{\text{coherent nilcharacters}} \rightarrow \boxed{\text{bracket-linear symbol}} \\ & \rightarrow \boxed{\text{symmetric } s\text{-symbol}} \rightarrow \boxed{\text{integrated } s\text{-step nilsequence}}. \end{aligned}$$

The main issue is the middle arrow. A separate inverse theorem for every derivative is useless until the derivative models are shown to fit together.

D.2 Bias notation

Definition D.2 (Quantified r -bias). Let $r \geq 0$, $M \geq 1$, and $\rho > 0$. A bounded sequence $a : [N] \rightarrow \mathbb{C}$ is (r, M, ρ) -*biased* if there is a degree-at-most- r polynomial nilsequence ψ of complexity at most M and $\|\psi\|_\infty \leq 1$ such that

$$\left| \mathbb{E}_{n \leq N} a(n) \overline{\psi(n)} \right| \geq \rho.$$

When the surrounding argument has fixed parameters (s, δ) , we say simply r -*biased* to mean $(r, M(s, \delta), \rho(s, \delta))$ -biased for some explicitly tracked bounds $M(s, \delta) < \infty$ and $\rho(s, \delta) > 0$ produced by the preceding lemmas. Every load-bearing use below occurs inside such a fixed-parameter induction.

Thus the desired conclusion is that f is s -biased in this quantified sense.

We shall freely use the following elementary principle: a finite chain of restrictions to positive-density subsets, pigeonhole selections among boundedly many alternatives, and losses of a positive correlation produces final constants depending only on the original complexity parameters. Consequently we suppress explicit tower-type formulae and write

$$\gg_{s, \delta} 1, \quad O_{s, \delta}(1).$$

No compactness or ultralimit argument is needed for this bookkeeping.

D.3 Base case: degree one

For $s = 1$ the theorem is ordinary Fourier analysis. On a cyclic group one has

$$\|f\|_{U^2}^4 = \sum_{\xi} |\widehat{f}(\xi)|^4.$$

Since Parseval gives

$$\sum_{\xi} |\widehat{f}(\xi)|^2 \leq 1,$$

a lower bound $\|f\|_{U^2} \geq \delta$ forces

$$\max_{\xi} |\widehat{f}(\xi)| \geq \delta^2.$$

After zero-extension from $[N]$ to a cyclic group of size $O(N)$, this yields correlation with a linear phase, which is a one-step nilsequence. The interval normalization was established in Volume III.

The cases $s = 1$ and $s = 2$ are already proved: $s = 1$ is ordinary Fourier inversion, while $s = 2$ is Corollary B.8, obtained from the complete cyclic U^3 argument of Chapter B. We therefore carry out the genuinely higher-order induction step $\text{GI}(s-1) \Rightarrow \text{GI}(s)$ for $s \geq 3$. The degree-two chapter is not a black box: it is the fully worked base model of the same coherence–symmetry–integration mechanism.

D.4 Step 1: many derivatives have large lower Gowers norm

The recursive identity is the first mechanism to understand.

Lemma D.3 (Derivative selection). *If $|f| \leq 1$ and*

$$\|f\|_{U^{s+1}[N]} \geq \delta,$$

then there is a set $H \subseteq [-N, N]$ with

$$|H| \gg_{s,\delta} N$$

such that

$$\|\Delta_h f\|_{U^s} \gg_{s,\delta} 1 \quad (h \in H),$$

with the norms interpreted on the common overlap of the shifted intervals.

Proof. For a group the identity is exact:

$$\|f\|_{U^{s+1}}^{2^{s+1}} = \mathbb{E}_h \|\Delta_h f\|_{U^s}^{2^s}.$$

For $[N]$, use the zero-extension definition of the interval norm. The same group identity holds in the ambient cyclic group, and shifts with overlap $o(N)$ contribute only a controlled boundary error. Since every derivative norm is at most 1, an average bounded below by $\delta^{2^{s+1}}$ forces a positive proportion of shifts to have

$$\|\Delta_h f\|_{U^s}^{2^s} \geq \frac{1}{2} \delta^{2^{s+1}}.$$

This gives the claim. $\square$

Assume inductively that $\text{GI}(s-1)$ has been proved. For every $h \in H$, the derivative $\Delta_h f$ therefore correlates with a degree- $(s-1)$ nilsequence. Apply the vertical Fejér decomposition of Lemma F.2 and pigeonhole the bounded-complexity data. We obtain degree- $(s-1)$ nilcharacters χ_h and degree-at-most- $(s-2)$ auxiliary nilsequences λ_h such that

$$\left| \mathbb{E}_n \Delta_h f(n) \overline{\chi_h(n) \lambda_h(n)} \right| \gg_{s,\delta} 1 \quad (\text{D.1})$$

for all h in a positive-density subset of H .

Idea Why nilcharacters, not arbitrary nilsequences?

For ordinary Fourier analysis a function is decomposed into pure characters because a single frequency can be added and subtracted. The vertical Fourier decomposition does the same thing at top nilpotent degree. Lower-degree pieces are deliberately discarded into λ_h because the induction can already understand them.

D.5 Steps 2–5: the inner degree–rank linearisation

The derivative character χ_h in (D.1) has ordinary degree at most $s-1$, hence degree–rank at most $(s-1, s-1)$. It is tempting to linearise its entire ordinary top-degree symbol in one pass. That shortcut is not valid. A bracket term of degree–rank $(s-1, r)$ can survive after the $(s-1, r+1)$ part is removed while retaining the same ordinary degree $s-1$. We therefore use an inner induction on r .

The invariant at rank r_* is the following. On a positive-density set of increments,

$$\Delta_h f(n) \text{ is } (s-2)\text{-biased by } \kappa^{(r_*)}(h, n) \chi_h^{(r_*)}(n), \quad (\text{D.2})$$

where

- $\kappa^{(r_*)}$ is one already-constructed two-variable nilcharacter of multidegree at most $(1, s-1)$;
- $\chi_h^{(r_*)}$ has degree–rank at most $(s-1, r_*)$ and is the still unstructured residual character.

At the start take $r_* = s-1$, $\kappa^{(s-1)} = 1$, and $\chi_h^{(s-1)} = \chi_h$ from Step 1.

D.5.1 2A. Cauchy–Schwarz eliminates the function and the part already built

For exact derivatives one has

$$\Delta_{h+k}f(n) = \Delta_h f(n+k) \Delta_k f(n).$$

From the weaker correlations (D.2), Gowers’s Cauchy–Schwarz argument produces many additive quadruples $h_1 + h_2 = h_3 + h_4$. The point that matters for the inner induction is that the fourfold product of the already-built factor $\kappa^{(r_*)}$ disappears at top n -degree. If $t = h_1 - h_4$, its homogeneous $(1, s-1)$ symbol contributes

$$\sigma(h_1) + \sigma(h_2) - \sigma(h_3) - \sigma(h_4) = 0.$$

Lemma G.3 makes this precise. Thus Corollary G.4 leaves an $(s-2)$ -biased fourfold product involving only the residual characters:

$$\chi_{h_1}^{(r_*)}(n) \chi_{h_2}^{(r_*)}(n+t) \overline{\chi_{h_3}^{(r_*)}(n) \chi_{h_4}^{(r_*)}(n+t)}.$$

The degree–rank bias lemma then gives an additive parallelogram law for their top $(s-1, r_*)$ symbols.

D.5.2 2B. Represent and regularise the residual frequencies

Lemma F.8 represents each residual top symbol in a bounded free filtered group by horizontal Taylor frequencies and a top vertical character; all lower degree–rank material is placed in a separate junk factor. Apply the frequency sunflower/regularisation lemma, Lemma G.6, degree by degree at the natural scales N^{-i} .

After a controlled restriction, every residual frequency list is represented using an external core, its own petals, rational frequencies, and $O(N^{-i})$ frequencies. The rational pieces are frozen on boundedly many residue classes and the small pieces are absorbed into smooth or lower degree–rank factors. Crucially, the core is a *representing* core; it need not be a literal subset of every original list.

For a good additive quadruple, quantitatively independent petals generate an equidistributed factor. A nontrivial top vertical character on a commutator involving petals from two different increments would then have Haar integral zero and hence tiny orbit average. This contradicts the fixed bias. Lemmas G.9 and G.10 prove this Furstenberg–Weiss elimination.

D.5.3 2C. Additive combinatorics forces bracket-linearity

What remains at the current layer $(s-1, r_*)$ is common-core data together with a bounded collection of h -dependent scalar frequencies satisfying many additive quadruple relations. Their graphs have large additive energy. The route

$$\text{energy} \rightarrow \text{Balog–Szemerédi–Gowers} \rightarrow \text{small doubling} \rightarrow \text{Freiman coordinates}$$

turns those statistical relations into bracket-linear dependence on h . Theorem G.14 proves this locally rather than citing it. A triangular descent through the Hall coordinates then yields Theorem G.17: the complete $(s-1, r_*)$ residual symbol is bracket-linear.

Proof architecture Where the additive-combinatorial core sits

At each degree–rank, not merely once, the proof uses

$$\begin{array}{c} \boxed{\text{symbol parallelograms}} \longrightarrow \boxed{\text{frequency-graph energy}} \\ \longrightarrow \boxed{\text{BSG/Freiman}} \longrightarrow \boxed{\text{bracket-linear top symbol}}. \end{array}$$

The inner degree–rank induction is therefore a repeated interface between higher-order Fourier analysis and the small-doubling theory developed in Volumes I–II.

D.5.4 2D. Realise the current rank and lower the residual rank

Theorem H.3 realizes the bracket-linear $(s-1, r_*)$ symbol as a two-variable nilcharacter $\kappa_{r_*}(h, n)$ of multidegree at most $(1, s-1)$. Dividing the residual character by this correction kills its total top representation. The rank-lowering lemma, Lemma F.6, then gives a new residual of degree–rank at most $(s-1, r_*-1)$. Replace

$$\kappa^{(r_*-1)} = \kappa^{(r_*)} \kappa_{r_*}$$

and continue.

Starting at $r_* = s-1$ and repeating this finitely many times reaches rank zero. Because $G_{(s-1,0)} = G_{(s-1,1)}$, there is no genuine rank-zero layer of ordinary degree $s-1$; the final residual has ordinary n -degree at most $s-2$ and is absorbed into the bias witness. The result is a single two-variable nilcharacter

$$\chi(h, n) := \prod_{r=1}^{s-1} \kappa_r(h, n) \tag{D.3}$$

of multidegree at most $(1, s-1)$ such that

$$\Delta_h f(n) \text{ is } (s-2)\text{-biased by } \chi(h, n)$$

for a positive-density set of increments. This is Theorem H.4.

Worked Example D.4 (The degree–rank obstruction in one line). At cubic degree,

$$e(\{\alpha n^2\}\beta n) \quad \text{and} \quad e(\gamma n^3)$$

are both cubic, but their top pieces have degree–rank $(3, 2)$ and $(3, 1)$, respectively. Removing the first does not make the second quadratic. The rank loop removes $(3, 2)$ and then $(3, 1)$ before ordinary degree can fall.

D.6 Step 6: symmetry

A priori the multidegree object distinguishes its first variable from the remaining $s-1$ variables. But true derivatives do not:

$$\Delta_h \Delta_k f = \Delta_k \Delta_h f.$$

Polarize the top symbol of χ as

$$\sigma(h; n_1, \dots, n_{s-1}).$$

The n -slots are symmetric by polarization. Apply the Cauchy–Schwarz coherence argument once more to (D.3), now comparing the two orders of differentiation. The quotient of the two model expressions is biased by lower-degree structure. Lemma F.11 therefore forces its top symbol to vanish, giving

$$\sigma(h; k, n_2, \dots, n_{s-1}) = \sigma(k; h, n_2, \dots, n_{s-1}).$$

Lemma H.6 carries out the argument and extends the relation from many pairs to the structured increment set using the bracket-linear parametrisation. Hence the top s -linear symbol is fully symmetric.

D.7 Step 7: integrate the symmetric symbol

For an ordinary symmetric s -linear form T in characteristic zero one writes

$$P(x) = \frac{1}{s!} T(x, \dots, x).$$

The nilpotent construction has the same logic, but the diagonalization creates lower multi-degree commutator terms and the vertical frequency may need a finite cover to divide by s .

Fix $\varepsilon > 0$ much smaller than the bias parameter retained from Step 6. Theorem H.9 gives a degree- s nilcharacter $\Theta(n)$, lower-order multivariable nilsequence factors $\Psi(h, n)$, and a uniform error E with $\|E\|_\infty \leq \varepsilon$ such that

$$\chi(h, n) = \Theta(n + h) \overline{\Theta(n)} \Psi(h, n) + E(h, n) \pmod{n\text{-degree at most } s - 2}. \quad (\text{D.4})$$

Every component of Ψ either has total degree at most $s - 1$ or has n -degree at most $s - 2$. Since every test function in the bias relation is bounded, the contribution of E is at most ε and is absorbed by our choice of ε .

Worked Example D.5 (Why the error multidegrees are inevitable). Let T be symmetric s -linear and set $P(n) = T(n, \dots, n)$. Then

$$P(n + h) - P(n) = sT(h, n, \dots, n) + \binom{s}{2} T(h, h, n, \dots, n) + \dots$$

Only the first term has one h -slot and $s - 1$ n -slots. Terms with two or more copies of h necessarily have lower n -degree. The nilpotent integration theorem records exactly this elementary binomial phenomenon, together with the commutator corrections required by noncommutativity.

D.8 Step 8: close the induction

Insert (D.4) into (D.3). Corollary F.17 proves the required multidegree separation: after vertical Fourier decomposition and pigeonholing, every error term splits into a fixed n -factor and a factor of n -degree at most $s - 2$. Absorbing the latter into the lower-degree test gives, for many h , that $\Delta_h f$ is $(s - 2)$ -biased by an expression

$$\theta(n + h) \theta'(n),$$

where θ, θ' have degree at most s and are independent of h .

By the converse estimate, Proposition F.14, the function

$$f(n+h)\overline{\theta(n+h)}\overline{f(n)}\overline{\theta'(n)}$$

has large U^{s-1} norm for many h . The Cauchy–Schwarz–Gowers closing lemma, Lemma H.10, removes the second factor and yields

$$\|f\bar{\theta}\|_{U^s} \gg_{s,\delta} 1.$$

Apply the inductive inverse theorem GI($s-1$). The function $f\bar{\theta}$ correlates with a degree-at-most- $(s-1)$ nilsequence φ . Therefore f correlates with

$$\theta\varphi,$$

a degree-at-most- s nilsequence. This proves GI(s).

The detailed closing calculation is Theorem H.11.

Proof of Theorem D.1. We induct on s . The Fourier case $s=1$ was proved above, and the global interval case $s=2$ is Corollary B.8. Let $s \geq 3$ and assume GI($s-1$).

Lemma D.3 and the inductive hypothesis give derivative models (D.1). Appendices F and G turn the statistical derivative correlations into coherent bracket-linear top symbols, Theorem G.17. Appendix H realizes those symbols as one multidegree nilcharacter, proves symmetry, integrates it, and closes the induction via Theorem H.11. All restrictions and correlation losses are controlled by functions of (s, δ) , so the final nilmanifold complexity and correlation bound also depend only on (s, δ) . This proves the theorem. $\square$

D.9 What the proof actually taught us

The theorem is often summarized as

$$\|f\|_{U^{s+1}} \gg 1 \Rightarrow f \text{ correlates with an } s\text{-step nilsequence.}$$

The proof reveals more:

1. U^{s+1} detects structure through derivatives, not directly.
2. The derivative models first appear independently and must be synchronized.
3. Cauchy–Schwarz converts weak correlation into additive relations among their top symbols.
4. Equidistribution eliminates independent frequency directions.
5. BSG and Freiman theory turn statistical symbol relations into bracket-linear dependence.
6. Commuting derivatives force symmetry.
7. Symmetric top-order derivative data can be integrated inside a nilpotent category closed under the required commutators and carries.

This is why nilsequences are inevitable: they are the class stable under the repeated operations “differentiate, linearize, symmetrize, integrate.”

D.10 The corrected filtration in the Green–Tao–Ziegler argument

The current arXiv revision of Green–Tao–Ziegler, version 5 dated 23 April 2026, explicitly corrects the filtration specified in the proof of Lemma 13.2, in addition to the 2024 erratum [3].

We therefore check the filtration, rather than only the underlying nilpotent group, at every free-model and factorisation step. In the proof above the free nilpotent objects are equipped with the degree–rank/multidegree filtration used by the symbol being integrated; quotient maps, rational subgroups and factorisation maps are required to be filtered maps; and an (i, j) commutator is placed in the sum layer prescribed by that filtration. This is not cosmetic bookkeeping: choosing the lower-central filtration when the argument requires the finer degree–rank filtration can put a commutator in the wrong degree and invalidate the induction. The multidegree checks in Appendices [F–H](#) prove this point explicitly.

Appendix E

A Heisenberg Nilsequence Laboratory

The Heisenberg group is the smallest example in which nilpotent geometry produces genuinely new phases beyond ordinary characters. We compute its group law, iterates, lattice reduction, and bracket-polynomial coordinates explicitly.

E.1 The group

Let

$$G = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, z \in \mathbb{R} \right\}.$$

Identify the matrix with (x, y, z) . Multiplication is

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy').$$

The identity is $(0, 0, 0)$ and

$$(x, y, z)^{-1} = (-x, -y, -z + xy).$$

A direct multiplication gives

$$[(x, y, z), (x', y', z')] = (0, 0, xy' - x'y).$$

Hence

$$[G, G] = Z(G) = \{(0, 0, z) : z \in \mathbb{R}\},$$

so G is two-step nilpotent.

E.2 Iterating one element

Let $g = (\alpha, \beta, \gamma)$. We claim

$$g^n = \left(n\alpha, n\beta, n\gamma + \binom{n}{2}\alpha\beta \right).$$

The case $n = 1$ is clear. If it holds for n , multiplication by g adds

$$\gamma + (n\alpha)\beta$$

to the central coordinate. Therefore the central coefficient evolves by

$$\binom{n}{2} + n = \binom{n+1}{2},$$

proving the formula.

The quadratic term $\binom{n}{2}\alpha\beta$ is created purely by noncommutativity. Neither horizontal coordinate is nonlinear.

E.3 The integer lattice

Let

$$\Gamma = \{(m, n, k) : m, n, k \in \mathbb{Z}\}.$$

It is a subgroup because the cross term mn' is integral. The quotient G/Γ is compact.

For right cosets, right multiplication by $(m, n, k) \in \Gamma$ sends

$$(x, y, z) \mapsto (x + m, y + n, z + k + xn).$$

Thus changing the integer representative of the y coordinate also changes z . This coupling is the geometric origin of floor functions in a fundamental-domain coordinate system.

E.4 Reduction to a fundamental domain

Choose $m = -\lfloor x \rfloor$ and $n = -\lfloor y \rfloor$. After right multiplication, the first two coordinates become

$$\{x\} = x - \lfloor x \rfloor, \quad \{y\} = y - \lfloor y \rfloor.$$

The central coordinate becomes

$$z - x\lfloor y \rfloor + k.$$

Choosing k to reduce this modulo 1, a convenient vertical coordinate on the fundamental domain is

$$\{z - x\lfloor y \rfloor\}.$$

This formula is discontinuous at the boundary of the fundamental domain, which is unavoidable for a global coordinate chart on a nontrivial compact quotient.

E.5 A bracket-polynomial orbit

Insert the orbit g^n :

$$x_n = n\alpha, \quad y_n = n\beta, \quad z_n = n\gamma + \binom{n}{2}\alpha\beta.$$

The reduced vertical phase is represented by

$$\theta(n) = n\gamma + \binom{n}{2}\alpha\beta - n\alpha\lfloor n\beta \rfloor \pmod{1}.$$

Using

$$\lfloor n\beta \rfloor = n\beta - \{n\beta\},$$

we can rewrite

$$\theta(n) = n\alpha\{n\beta\} + \text{an ordinary quadratic polynomial in } n \pmod{1}.$$

Thus the characteristic bracket term

$$n\alpha\{n\beta\}$$

appears automatically when a polynomial nilpotent orbit is written in Euclidean coordinates.

Worked Example E.1 (Carry discontinuity is coordinate, not geometric). The function $n \mapsto \{n\beta\}$ jumps whenever $n\beta$ crosses an integer. Therefore the bracket expression appears to have singular derivatives. But the underlying orbit $g^n\Gamma$ moves continuously in the compact nilmanifold. The jump records only a change of lattice representative. Localizing to a Bohr set freezes the representative; working on the nilmanifold removes the artificial singularity altogether.

E.6 Vertical characters

The center acts by

$$(0, 0, t) : (x, y, z)\Gamma \mapsto (x, y, z + t)\Gamma.$$

A vertical character of frequency $m \in \mathbb{Z}$ is an observable satisfying

$$F((0, 0, t)x) = e(mt)F(x).$$

A pure vertical frequency detects the genuinely two-step component of the orbit. In a fundamental-domain chart it becomes a bracket-quadratic phase plus lower-order terms.

A globally smooth vertical character must be defined geometrically rather than simply by $e(mz)$ in one coordinate chart, because z itself changes under lattice reduction. This distinction is precisely why nilsequences are a better invariant language than bracket formulas.

E.7 Horizontal characters

Since a horizontal character must annihilate $[G, G]$, it has the form

$$\eta(x, y, z) = kx + \ell y \pmod{1}, \quad k, \ell \in \mathbb{Z}.$$

Applied to g^n ,

$$\eta(g^n) = n(k\alpha + \ell\beta).$$

Thus a horizontal obstruction to equidistribution is an ordinary Diophantine relation among α and β . The nonlinear vertical behavior is controlled indirectly through the way the orbit sits over this horizontal torus.

E.8 A two-variable cube calculation

Take two group increments

$$a = (u, v, w), \quad b = (u', v', w').$$

Their commutator is

$$[a, b] = (0, 0, uv' - u'v).$$

Hence exchanging the order of the two directions in a parallelogram changes only the central coordinate, by a bilinear form in the horizontal increments. This is the nilpotent version of the Hessian of a quadratic phase.

In particular, the two-step cube is not an ordinary affine parallelogram in coordinates: its closing error is central. Host–Kra cube groups are designed exactly to keep track of these higher-order closing errors.

E.9 Polynomial sequences with independent central motion

More generally, let

$$g(n) = (\alpha n, \beta n, \gamma_2 \binom{n}{2} + \gamma_1 n).$$

With the lower-central filtration this is a polynomial sequence of degree at most two. More explicitly, if $\kappa := \gamma_2 - \alpha\beta$, then the first right derivative has horizontal part (α, β) and central part affine with slope κ ; the second derivative is the constant central element $(0, 0, \kappa)$; and the third derivative is trivial. In the special case $\gamma_2 = \alpha\beta$, the sequence reduces (up to the central linear term) to a power sequence and the genuine filtered degree drops to one.

This example shows why a polynomial sequence in a nilpotent group is not simply the power sequence g^n . The central coordinate may contain its own polynomial data compatible with the filtration.

E.10 What generalizes

In an s -step nilpotent group:

- collection produces binomial-coefficient polynomials in deeper commutator coordinates;
- lattice reduction produces nested floor/fractional-part expressions;
- vertical characters on successive central quotients become higher-order frequencies;
- horizontal characters remain the ultimate obstructions to equidistribution.

The Heisenberg calculation is therefore not merely an example of one exotic phase. It is the first nontrivial case of the general nilsequence mechanism.

Appendix F

Nilcharacters, Symbols, and Bias

This appendix develops the part of nilcharacter calculus that is used in the proof of the Green–Tao–Ziegler inverse theorem. The point is to make precise the slogan

“a nilcharacter is a higher-order Fourier mode.”

The only external geometric ingredient is the horizontal-character equidistribution criterion recorded in Chapter C; all Fourier and symbol manipulations are proved here.

F.1 Vertical tori and vertical frequencies

Let $G_\bullet$ be a filtered connected simply connected nilpotent Lie group of degree at most s and let Γ be a compatible lattice. The final filtration group G_s is central. Its compact quotient

$$T_s := G_s / (G_s \cap \Gamma)$$

is a torus acting on G/Γ by left translation.

Definition F.1 (Nilcharacter). A bounded Lipschitz function $F : G/\Gamma \rightarrow \mathbb{C}$ is a *vertical character of frequency $\eta \in \widehat{T}_s$* if

$$F(tx) = e(\eta(t))F(x) \quad (t \in T_s).$$

If $g : \mathbb{Z} \rightarrow G$ is polynomial, the sequence

$$\chi(n) = F(g(n)\Gamma)$$

is called an s -degree nilcharacter.

The definition allows vector-valued functions, but scalar-valued characters suffice for the present exposition because a bounded vector-valued nilsequence can be decomposed into finitely many scalar coordinates.

F.2 Vertical Fourier decomposition

The analogue of selecting one Fourier coefficient is the following elementary fiberwise decomposition.

Lemma F.2 (Vertical Fejér decomposition). *Let $F : G/\Gamma \rightarrow \mathbb{C}$ be Lipschitz and $|F| \leq 1$. For every $\varepsilon > 0$ there is a finite set $\Omega \subset \widehat{T}_s$, with cardinality and frequencies bounded in terms of the complexity of G/Γ , $\|F\|_{\text{Lip}}$, and ε , and vertical-character functions F_η such that*

$$\left\| F - \sum_{\eta \in \Omega} F_\eta \right\|_\infty \leq \varepsilon.$$

Proof. Choose coordinates $T_s \cong \mathbb{R}^d / \mathbb{Z}^d$. Let K_R be the d -dimensional Fejér kernel of order R . Define convolution only in the vertical direction:

$$F_R(x) = \int_{T_s} K_R(t) F(tx) dt.$$

Because F is Lipschitz and K_R is an approximate identity,

$$\|F_R - F\|_\infty \rightarrow 0$$

as $R \rightarrow \infty$, uniformly in x . The Fejér kernel is a finite trigonometric polynomial,

$$K_R(t) = \sum_{\|m\|_\infty < R} a_m e(m \cdot t).$$

Therefore

$$F_R(x) = \sum_{\|m\|_\infty < R} a_m \int_{T_s} e(m \cdot t) F(tx) dt.$$

Set

$$F_m(x) = a_m \int_{T_s} e(m \cdot t) F(tx) dt.$$

Changing x to ux and then t to tu^{-1} shows

$$F_m(ux) = e(-m \cdot u) F_m(x),$$

so F_m is a vertical character. Taking R large enough gives the claimed approximation. $\square$

Corollary F.3 (Selecting a nilcharacter). *Let $F : G/\Gamma \rightarrow \mathbb{C}$ be Lipschitz with $\|F\|_\infty \leq 1$. If a 1-bounded function $f : [N] \rightarrow \mathbb{C}$ satisfies*

$$\left| \mathbb{E}_{n \leq N} f(n) \overline{F(g(n)\Gamma)} \right| \geq \delta,$$

then, after increasing the complexity by a quantity depending only on $(\delta, G/\Gamma, \|F\|_{\text{Lip}})$, there is a 1-bounded nilcharacter χ with

$$\left| \mathbb{E}_{n \leq N} f(n) \overline{\chi(n)} \right| \geq c(\delta) > 0.$$

Proof. Apply Lemma F.2 with error $\delta/2$. The correlation with the finite sum has magnitude at least $\delta/2$. The triangle inequality then forces one summand to have correlation at least $\delta/(2|\Omega|)$. $\square$

F.3 Equivalence modulo lower degree

A higher-order frequency is meaningful only up to lower-degree structure.

Definition F.4 (Symbol equivalence). For degree- s nilcharacters χ, χ' , write

$$\chi \sim_{s-1} \chi'$$

if $\overline{\chi\chi'}$ is a bounded linear combination of degree at most $s-1$ nilsequences. The equivalence class $[\chi]_s$ is the *degree- s symbol* of χ .

Tensor product makes symbols into an abelian group:

$$[\chi]_s + [\psi]_s = [\chi\psi]_s, \quad -[\chi]_s = [\bar{\chi}]_s.$$

The commutativity occurs at symbol level even though the underlying nilpotent groups need not commute: commutator errors land one degree lower.

Lemma F.5 (Symbol operations are well defined). *The relation $\sim_{s-1}$ is an equivalence relation, and the operations*

$$[\chi]_s + [\psi]_s := [\chi\psi]_s, \quad -[\chi]_s := [\bar{\chi}]_s$$

are independent of the chosen representatives.

Proof. Reflexivity and symmetry are immediate. For transitivity, if

$$\chi\bar{\chi}' = L_1, \quad \chi'\bar{\chi}'' = L_2$$

with L_1, L_2 finite linear combinations of lower-degree nilsequences, then

$$\chi\bar{\chi}'' = L_1 L_2,$$

and products of lower-degree nilsequences remain lower degree by Lemma 5.6. The same calculation proves independence of representatives under multiplication and conjugation. $\square$

F.4 Degree–rank symbols and the inner induction

For the general inverse theorem, ordinary degree alone is too coarse. Fix $(d, r) \in \text{DR}$ with $1 \leq r \leq d$. A degree–rank (d, r) nilcharacter is a nilcharacter on a degree–rank filtered nilmanifold whose top vertical action occurs in the layer $G_{(d,r)}$. Two such nilcharacters are equivalent at (d, r) if their quotient is a finite bounded-complexity combination of nilsequences of degree–rank strictly smaller than (d, r) in lexicographic order. We denote the class by

$$[\chi]_{(d,r)}.$$

Tensor product and conjugation again make these symbols an abelian group. The proof is the same as Lemma F.5, because products preserve the maximum of the degree–rank filtrations and commutator errors move to a strictly lower layer.

Lemma F.6 (Rank lowering from a trivial top representation). *Let χ be a bounded-complexity nilcharacter of degree–rank at most (d, r) , where $r \geq 1$. Suppose that, in a total frequency representation of the form constructed in Lemma F.8, the top (d, r) vertical character is trivial. Then χ is a bounded-complexity nilsequence of degree–rank at most $(d, r - 1)$. In particular its (d, r) symbol vanishes. When $r = 1$, the convention $G_{(d,0)} = G_{(d,1)}$ means that the ordinary degree drops below d .*

Proof. Write the total frequency representation as

$$\chi(n) = F(O(n), O_0(n)),$$

where O takes values in a degree–rank at most (d, r) nilmanifold G/Γ , O_0 takes values in a degree–rank at most $(d, r - 1)$ “junk” nilmanifold, and there is a filtered homomorphism

$$\phi : \tilde{G} \longrightarrow G$$

from the universal free filtered group which realizes all horizontal Taylor frequencies of O . Triviality of the top representation says precisely that F is invariant under the subtorus generated by

$$\phi(\tilde{G}_{(d,r)}) \text{ inside } G_{(d,r)}/(G_{(d,r)} \cap \Gamma).$$

Let T be the connected closed subgroup generated by that subtorus. It is rational and central at the top degree–rank layer. Since F is T -invariant, F descends to the quotient G/T and the polynomial orbit O descends with it.

It remains to check the filtration, because this is exactly where ordinary degree bookkeeping is insufficient. On G/T define

$$G'_{(a,b)} := \langle \pi(G_{(a,b+1)}), \pi\phi(\tilde{G}_{(a,b)}) \rangle^\circ,$$

with the convention $G_{(a,b)} = G_{(a+1,0)}$ when $b > a$ and with $\pi : G \rightarrow G/T$ the quotient map. The generators satisfy the degree–rank commutator law because both the original filtration and the universal filtration do; hence $[G'_{(a,b)}, G'_{(a',b')}] \subseteq G'_{(a+a',b+b')}$ in the degree–rank order. The Taylor coefficient formula in the total representation shows that the descended orbit is polynomial for this smaller filtration. At the old top layer (d, r) the universal contribution has been killed by T , while the first generator above is $G_{(d,r+1)}$; therefore the new filtration has degree–rank at most $(d, r-1)$. The factor O_0 already had this bound, so their product does as well.

For $r = 1$, there is no separate rank-zero layer at ordinary degree d because $G_{(d,0)} = G_{(d,1)}$; thus degree–rank at most $(d, 0)$ is the same as ordinary degree at most $d-1$. This proves the claim. $\square$

Idea Why this extra index is indispensable

At degree three, for example, removing a genuinely three-way commutator does not remove a bracket term of the same polynomial degree but lower commutator rank. An induction only on ordinary degree would silently classify that surviving term as “lower order” when it is not. The degree–rank induction removes the top-degree pieces one rank at a time.

F.5 Derivatives lower degree

Let

$$\Delta_h \chi(n) = \chi(n+h) \overline{\chi(n)}.$$

Lemma F.7 (Derivative lemma). *Let $\chi(n) = F(g(n)\Gamma)$ be a degree- s nilcharacter of complexity at most M . Then for every fixed h the sequence*

$$n \mapsto \Delta_h \chi(n) = \chi(n+h) \overline{\chi(n)}$$

is a degree-at-most- $(s-1)$ nilsequence of complexity $O_s(M)$, uniformly in h . More precisely, the two-variable map

$$(h, n) \mapsto \Delta_h \chi(n)$$

is a polynomial nilsequence of multidegree contained in

$$\{(i, j) : i + j \leq s, j \leq s-1\},$$

and its component of multidegree $(1, s-1)$ depends only on $[\chi]_s$.

Proof. After a left translation of the orbit and the corresponding change of the observable, we may assume $g(0) = e$. Put $a = g(1)$ and

$$g_2(n) := g(n)a^{-n}.$$

The relative-square construction of Lemmas M.11 and M.12 is designed exactly for this derivative. For

$$G_i^\square = G_i \times_{G_{i+1}} G_i = \{(x, y) \in G_i^2 : xy^{-1} \in G_{i+1}\}$$

one has a rational filtered subgroup $G^\square \leq G \times G$ whose top layer is G_s^Δ . The quotient

$$\overline{G}^\square := G^\square / G_s^\Delta$$

has degree at most $s-1$; write $\overline{\Gamma}^\square$ for the image of $\Gamma^\square$ in this quotient. Its Mal'cev complexity is bounded in terms of the original complexity.

For each fixed h , write the fundamental-domain decomposition $a^h = \{a^h\}[a^h]$ and use the polynomial orbit

$$g_h^\square(n) = (\{a^h\}^{-1}g_2(n+h)a^n\{a^h\}, g_2(n)a^n) \in G^\square.$$

If ξ is the vertical frequency of F , then

$$F_h^\square(x, y) := F(\{a^h\}x)\overline{F(y)}$$

is invariant under G_s^Δ , because the two copies of the vertical character cancel. Hence it descends to $\overline{G}^\square / \overline{\Gamma}^\square$. Moreover, evaluation gives

$$F_h^\square(g_h^\square(n)\Gamma^\square) = F(g(n+h)\Gamma)\overline{F(g(n)\Gamma)} = \Delta_h\chi(n).$$

Thus, for every fixed h , $\Delta_h\chi$ is a degree-at-most- $(s-1)$ nilsequence. The rationality bounds in Lemma M.11, together with the fact that $\{a^h\}$ remains in one fixed Mal'cev fundamental domain, give complexity $O_s(M)$ uniformly in h .

It remains to record the dependence on h . In binomial Taylor coordinates,

$$g(n) = g_0 g_1^{\binom{n}{1}} \cdots g_s^{\binom{n}{s}}, \quad g_j \in G_j,$$

and

$$\binom{n+h}{j} - \binom{n}{j} = \sum_{a=1}^j \binom{h}{a} \binom{n}{j-a}. \quad (\text{D})$$

Consequently every contribution has total (h, n) -degree at most $j \leq s$ and n -degree at most $j-1 \leq s-1$. The BCH corrections have filtration weight equal to the sum of the weights of their inputs, so they obey the same downset bound. This proves the asserted two-variable multidegree containment.

Finally, multiplying χ by a degree-at-most- $(s-1)$ nilcharacter changes its derivative by a two-variable nilsequence whose n -degree is at most $s-2$. Hence the $(1, s-1)$ component is unchanged and depends only on $[\chi]_s$. $\square$

This lemma is the nilpotent replacement for

$$\Delta_h e(P(n)) = e(P(n+h) - P(n)),$$

which lowers the polynomial degree in n by one.

F.6 Universal horizontal-frequency representations

The coherence argument needs to talk about the “frequencies” of a nilcharacter just as ordinary Fourier analysis talks about the frequency of $e(\xi n)$. The following lemma makes that passage explicit at top order.

For the degree–rank filtration used in the higher-order inverse theorem, write

$$\text{Horiz}_i(G) := G_{(i,1)}/G_{(i,2)}$$

for the i th horizontal group, and quotient by the image of the lattice to obtain the corresponding horizontal torus. Equivalently one may work in the associated Lie-algebra quotient. A polynomial orbit has an i th Taylor coefficient in this torus. We use this notation only in the degree–rank setting; for an ordinary one-parameter filtration the usual horizontal torus is the abelianization appropriate to that filtration.

Lemma F.8 (Universal total frequency representation). *Let $\chi(n)$ be a bounded-complexity nilcharacter of degree–rank at most (d, r_*) . After adjoining a bounded-complexity “junk” nilmanifold of strictly smaller degree–rank, one can write*

$$\chi(n) = \mathcal{F}(O(n), O_0(n)),$$

where O takes values in a bounded-complexity filtered nilmanifold G/Γ , O_0 has degree–rank strictly below (d, r_*) , and the following data are available. There are bounded integers D_i , a free filtered nilpotent Lie group $\tilde{G}$ of degree–rank at most (d, r_*) with formal generators $e_{i,j}$ of filtration degree i , a filtered homomorphism

$$\phi : \tilde{G} \longrightarrow G,$$

frequencies $\xi_{i,j} \in \mathbb{R}/\mathbb{Z}$, and a character

$$\eta : \tilde{G}_{(d, r_*)} \longrightarrow \mathbb{R}/\mathbb{Z}$$

such that:

1. the i th horizontal Taylor coefficient of the free orbit is

$$\text{Taylor}_i(\tilde{O}) = \prod_{j=1}^{D_i} e_{i,j}^{\xi_{i,j}};$$

2. O and $\phi\tilde{O}$ have the same horizontal Taylor data, with their discrepancy absorbed into O_0 ; and
3. on the top free vertical layer the representing function has the exact transformation law

$$\mathcal{F}(\phi(t)x, x_0) = e(\eta(t))\mathcal{F}(x, x_0) \quad (t \in \tilde{G}_{(d, r_*)}). \quad (\text{F.1})$$

Thus the pair consisting of the horizontal frequency vector and the character η represents the complete top (d, r_*) symbol. All dimensions and rationality data are bounded by the complexity of χ .

Moreover, if every old frequency $\xi_{i,j}$ is an integral linear combination of a new list $\xi'_{i,1}, \dots, \xi'_{i,D'_i}$ at the same polynomial degree, then a filtered homomorphism between the corresponding free groups realizes this change of basis, and pulling back η gives a new total representation of the same top symbol.

Proof. For each $1 \leq i \leq d$, use the degree–rank horizontal space

$$\text{Horiz}_i(G) = G_{(i,1)}/G_{(i,2)}$$

(with its lattice quotient) and write the i th Taylor coefficient of g in a bounded rational basis:

$$\text{Taylor}_i(g) = \sum_{j=1}^{D_i} \xi_{i,j} X_{i,j}.$$

The dimensions and rationality constants are bounded by the nilmanifold complexity.

Take the free nilpotent Lie group on formal generators $e_{i,j}$, impose all commutator relations of degree–rank strictly above (d, r_*) , and equip the quotient with the induced degree–rank filtration. By freeness, $e_{i,j} \mapsto X_{i,j}$ extends to a filtered homomorphism

$$\phi : \tilde{G} \longrightarrow G.$$

Let $\tilde{O}$ be the free polynomial orbit with the displayed horizontal Taylor coefficients. The principal orbit $O = g(\cdot)\Gamma$ and $\phi\tilde{O}$ have the same horizontal Taylor data. Collecting their quotient in a Hall basis shows that the discrepancy has degree–rank strictly below (d, r_*) ; store it in a secondary “junk” nilmanifold G_0/Γ_0 and let O_0 denote the corresponding orbit. After vertical Fourier decomposition of the original Lipschitz function, keep the single top vertical mode that carries the nilcharacter. Pulling this character back through ϕ gives

$$\eta : \tilde{G}_{(d, r_*)} \rightarrow \mathbb{R}/\mathbb{Z}.$$

The defining vertical-character identity is then exactly

$$\mathcal{F}(\phi(t)x, x_0) = e(\eta(t))\mathcal{F}(x, x_0),$$

which is (F.1). This identity is important: when η vanishes on a top subgroup, the representing function is genuinely invariant under that subgroup, so quotienting it later is legitimate rather than merely a formal operation on frequency lists. The top degree–rank behaviour is therefore completely determined by the frequency vector and η , while all strictly smaller degree–rank terms are isolated in O_0 .

For the change of basis, write

$$\xi_{i,j} = \sum_k a_{j,k} \xi'_{i,k}, \quad a_{j,k} \in \mathbb{Z}.$$

Send each old free generator to the degree- i word whose horizontalization is $\sum_k a_{j,k} e'_{i,k}$. The universal property gives a filtered homomorphism; BCH commutator corrections automatically lie in the filtration layers prescribed by the degree–rank order. The new free orbit therefore maps to one with the same horizontal Taylor data as the old representation, and the discrepancy again belongs to the junk factor. Pulling back $\tilde{\eta}$ proves invariance of the represented symbol. $\square$

Worked Example F.9 (The Heisenberg frequency pair). For the bracket quadratic phase modeled by the Heisenberg group, the degree-one horizontal Taylor data are two frequencies α, β . The top central character reads the commutator of the two free generators. Replacing $\alpha = 2\alpha' + 3\beta'$ and $\beta = \beta'$ simply sends the old first generator to the horizontal word $2e'_1 + 3e'_2$ and the second to e'_2 ; bilinearity of the commutator then transforms the central frequency automatically. This is the degree-two shadow of the universal construction used later.

F.7 Bias and the top symbol

We formalize the phrase “biased modulo lower degree.”

Definition F.10 (r -bias). A bounded sequence $a : [N] \rightarrow \mathbb{C}$ is r -biased at level ρ if there is a degree at most r nilsequence ψ of bounded complexity such that

$$|\mathbb{E}_{n \leq N} a(n) \overline{\psi(n)}| \geq \rho.$$

When only positivity matters, we say simply r -biased.

Lemma F.11 (A biased nilcharacter has trivial top symbol on its orbit). *Fix a complexity bound M and a bias level $\rho > 0$. There is $N_0 = N_0(s, M, \rho)$ with the following property. Let $\chi(n) = F(g(n)\Gamma)$ be a degree- s nilcharacter of complexity at most M on $[N]$, $N \geq N_0$. If χ is $(s-1)$ -biased at level ρ , then after restricting to one member of a bounded partition into rational residue classes and long subintervals, the top vertical character of χ is trivial on the rational orbit closure. Equivalently, on that piece χ is symbol-equivalent to a degree-at-most $(s-1)$ nilsequence.*

Proof. Let ψ be a degree-at-most- $(s-1)$ nilsequence of complexity at most M' witnessing bias, so that

$$|\mathbb{E}_{n \leq N} \chi(n) \overline{\psi(n)}| \geq \rho.$$

Tensor the two nilmanifolds. The product

$$\Phi(n) := \chi(n) \overline{\psi(n)}$$

has exactly the same degree- s vertical frequency as χ , because the second factor has trivial degree- s vertical action.

Choose a small parameter

$$0 < \kappa < \frac{\rho}{100(1 + \|F\|_{\text{Lip}})(1 + \|\psi\|_{\text{Lip}})}.$$

Apply the quantitative smooth-rational-equidistributed factorization theorem from Chapter C to the product polynomial orbit, with equidistribution parameter κ . It partitions $[N]$ into $O_{s,M,\rho}(1)$ residue-class/interval pieces on each of which

$$g_{\text{prod}}(n) = \varepsilon(n) g'(n) \gamma(n),$$

where ε varies by $O(\kappa)$ on the piece, γ is constant modulo the lattice, and g' is κ -equidistributed in a connected rational subnilmanifold Y . Since the original average has magnitude at least ρ , the triangle inequality forces one piece on which the corresponding normalized average has magnitude at least $\rho/2$ after κ is chosen as above.

Freeze the rational factor and replace the smooth factor by its value at one base point. Lipschitz continuity changes the average by at most $O_M(\kappa) < \rho/10$. We are left with a Lipschitz function F_0 on a translate of Y whose top vertical frequency is the restriction of the top frequency of χ , and whose orbit average has magnitude at least $2\rho/5$.

If that restricted vertical frequency were nontrivial, Haar invariance in the top vertical torus would give

$$\int_Y F_0 d\mu_Y = 0$$

by Lemma G.8. The κ -equidistribution of g' would then imply

$$|\mathbb{E} F_0(g'(n)\Gamma_Y)| \leq \kappa \|F_0\|_{\text{Lip}} < \rho/10,$$

a contradiction. Hence the top vertical frequency is trivial on Y .

Quotient the top vertical subgroup of Y on which F_0 is invariant. In the corresponding filtered quotient the final nontrivial degree is at most $s - 1$; the smooth and rational factors that were frozen contribute only bounded lower-degree nilsequences. Thus the restriction of χ differs from a degree-at-most- $(s - 1)$ nilsequence only by lower-degree factors. This is exactly vanishing of the degree- s symbol on the selected piece. All choices were quantitative in (s, M, ρ) , so the assertion is finite rather than asymptotic. $\square$

The preceding statement is where quantitative equidistribution enters the inverse proof: it converts a nonzero average into an algebraic annihilation relation among frequencies.

Lemma F.12 (Multivariable bias kills the top symbol). *Let $\Xi : \prod_{j=1}^k I_j \rightarrow \mathbb{C}$ be a bounded-complexity degree- d nilcharacter on a product of long intervals, equipped with any fixed multidegree filtration. If Ξ has correlation at least $\rho > 0$ with a bounded-complexity nilsequence whose top multidegree is strictly smaller, then, after a bounded rational/box partition, the top vertical character of Ξ is trivial on the connected rational orbit closure. In particular its top multidegree symbol vanishes on that piece.*

Proof. The proof of Lemma F.11 did not use that the polynomial parameter had one coordinate. Tensor Ξ with the conjugate lower-multidegree test, apply the quantitative smooth-rational-equidistributed factorization to the resulting polynomial map from $\mathbb{Z}^k$, and choose a box carrying a fixed proportion of the correlation. Freeze the smooth and rational factors. If the terminal vertical character were nontrivial on the connected equidistributed orbit closure, Haar integration in that vertical torus would give mean zero, while quantitative equidistribution would make the box average $o_\rho(1)$, a contradiction. Quotienting the invariant terminal vertical subgroup removes the top multidegree. This is exactly the one-variable proof with a product box in place of an interval. $\square$

Lemma F.13 (Degree-rank bias kills the top representation). *Fix (d, r) with $r \geq 1$, a complexity bound M , and $\rho > 0$. Let χ be a degree-rank at most (d, r) nilcharacter of complexity at most M . Suppose χ correlates at level ρ with a nilsequence of degree-rank strictly smaller than (d, r) . Then, after restricting to one member of a bounded rational/interval partition, the top (d, r) character in a total frequency representation of χ is trivial on the connected rational orbit closure. Consequently χ is equivalent there to a nilsequence of degree-rank at most $(d, r - 1)$.*

Proof. The proof is the degree-rank version of Lemma F.11, and we spell out the only point that changes. Tensor χ with the conjugate of the lower-degree-rank test nilsequence and apply the quantitative smooth-rational-equidistributed factorization to the product orbit. The test factor has no vertical action in the layer (d, r) , so the product retains exactly the top (d, r) character of χ . On a piece carrying a definite proportion of the correlation, freeze the smooth and rational factors. If the top character were nontrivial on the connected equidistributed orbit closure, integration over the (d, r) vertical torus would give mean zero; quantitative equidistribution would then make the orbit average smaller than the prescribed bias, a contradiction.

Thus the top character is trivial on the orbit closure. Pass to the associated total frequency representation and quotient the invariant top subtorus. Lemma F.6 then replaces the principal factor by one of degree-rank at most $(d, r - 1)$; all junk factors already have strictly smaller degree-rank. This gives the conclusion. $\square$

F.8 The converse estimate: structured correlation forces Gowers uniformity

The final induction in the inverse theorem also uses the easy direction: correlation with degree- r structure forces a large U^{r+1} norm. The proof below is the finitary interval form of the necessity argument in Green–Tao–Ziegler [2, Appendix G]; the general inverse-theorem paper records the same implication as its converse principle [3, Proposition 5.6].

Proposition F.14 (Converse inverse estimate). *Fix $r \geq 0$, a complexity bound $M \geq 1$, and $\rho > 0$. There exists $c = c(r, M, \rho) > 0$ such that the following holds for every $N \geq 1$. If $f : [N] \rightarrow \mathbb{C}$ is 1-bounded and*

$$|\mathbb{E}_{n \leq N} f(n) \overline{\psi(n)}| \geq \rho$$

for a 1-bounded degree-at-most- r nilsequence ψ of complexity at most M , then

$$\|f\|_{U^{r+1}[N]} \geq c.$$

Proof. We induct on r . When $r = 0$, a degree-zero polynomial orbit is constant, so ψ is constant. The assumed correlation therefore gives $|\mathbb{E}_{n \leq N} f(n)| \geq \rho$, which is exactly the $U^1[N]$ conclusion.

Assume $r \geq 1$ and the result has been proved at degree $r - 1$. Apply Corollary F.3. After changing the constants in terms of (r, M, ρ) , there is a 1-bounded degree- r nilcharacter χ satisfying

$$|\mathbb{E}_{n \leq N} f(n) \overline{\chi(n)}| \geq \rho_1 > 0. \quad (\text{C1})$$

Here ρ_1 depends only on (r, M, ρ) . Set $a_n = f(n) \overline{\chi(n)}$ for $1 \leq n \leq N$. For $-(N - 1) \leq h \leq N - 1$, let

$$I_h := \{n \in [N] : n + h \in [N]\}.$$

The elementary correlation identity

$$\left| \sum_{n=1}^N a_n \right|^2 = \sum_{h=-(N-1)}^{N-1} \sum_{n \in I_h} a_{n+h} \overline{a_n} \quad (\text{C2})$$

and (C1) imply

$$\sum_h \left| \sum_{n \in I_h} \Delta_h f(n) \overline{\Delta_h \chi(n)} \right| \geq \rho_1^2 N^2. \quad (\text{C3})$$

Every inner sum has magnitude at most N . Taking, for example, $c_3 = \rho_1^2/4$ in the next display, (C3) shows that at least $c_2 N$ shifts h satisfy

$$\left| \sum_{n \in I_h} \Delta_h f(n) \overline{\Delta_h \chi(n)} \right| \geq c_3 N, \quad (\text{C4})$$

where $c_2 > 0$ depends only on (r, M, ρ) . For every such shift, $|I_h| \geq c_3 N$, so

$$|I_h| \geq c_3 N \quad \text{and} \quad \left| \mathbb{E}_{n \in I_h} \Delta_h f(n) \overline{\Delta_h \chi(n)} \right| \geq c_5. \quad (\text{C5})$$

This is the point at which a bounded set of shifts would be insufficient: the recursive Gowers identity averages over all shifts, so we need the positive proportion supplied by (C3)–(C5).

By Lemma F.7, $\Delta_h \chi$ is, uniformly in h , a degree-at-most- $(r - 1)$ nilsequence of controlled complexity. The inductive hypothesis applied on the interval I_h therefore gives

$$\|\Delta_h f\|_{U^r[I_h]} \geq c_6 \quad (\text{C6})$$

for every shift satisfying (C4).

It remains to pass from the interval norms in (C6) to one common Gowers average. This bookkeeping is short but important. Put

$$Q := 2^{r+2}N, \quad G := \mathbb{Z}/Q\mathbb{Z},$$

and let $\tilde{f} : G \rightarrow \mathbb{C}$ be the zero extension of f . Recall that our interval convention is

$$\|u\|_{U^d[J]} := \frac{\|u1_J\|_{U^d(G)}}{\|1_J\|_{U^d(G)}}$$

whenever the cyclic ambient group is large enough that the relevant cubes do not wrap. The present choice of Q is large enough simultaneously for $[N]$ at level $r+1$ and for every I_h at level r . Identifying I_h with a translate of $[I_h]$, translation invariance and zero extension give

$$\|\Delta_h \tilde{f}\|_{U^r(G)} = \|1_{I_h}\|_{U^r(G)} \|\Delta_h f\|_{U^r[I_h]}. \quad (\text{C7})$$

Because $|I_h| \geq c_3 N$, the indicator factor in (C7) is bounded below by a positive constant depending only on (r, c_3) . Indeed, choose the base point in the middle half of I_h and each of the r increments to have absolute value at most $|I_h|/(4r)$. When $|I_h| \geq 8r$, every resulting r -cube stays inside I_h , so direct cube counting gives

$$\|1_{I_h}\|_{U^r(G)}^{2^r} \geq c_r \left(\frac{|I_h|}{Q} \right)^{r+1} \geq c_7 > 0. \quad (\text{C8})$$

If $|I_h| < 8r$, then $|I_h| \geq c_3 N$ bounds N in terms of (r, c_3) ; decreasing c_7 handles these finitely many sizes as well. Thus (C6)–(C8) imply

$$\|\Delta_h \tilde{f}\|_{U^r(G)} \geq c_8$$

for at least $c_2 N$ values of h . Since $Q = 2^{r+2}N$, these good shifts form a positive proportion of G . The exact finite-group recursion now yields

$$\|\tilde{f}\|_{U^{r+1}(G)}^{2^{r+1}} = \mathbb{E}_{h \in G} \|\Delta_h \tilde{f}\|_{U^r(G)}^{2^r} \geq c_9. \quad (\text{C9})$$

Finally, by the definition of the interval norm,

$$\|f\|_{U^{r+1}[N]} = \frac{\|\tilde{f}\|_{U^{r+1}(G)}}{\|1_{[N]}\|_{U^{r+1}(G)}} \geq \|\tilde{f}\|_{U^{r+1}(G)},$$

because the denominator is at most 1. Combining this with (C9) proves the claim. $\square$

F.9 Multidegree and polarization

For several variables, filter $\mathbb{Z}^d$ by a downset $J \subset \mathbb{N}^d$. A polynomial nilsequence has multidegree in J if every mixed derivative whose multi-index lies outside J vanishes at the group-polynomial level.

The main example is multidegree $(1, s-1)$ in (h, n) . Its top symbol behaves like a multilinear form with one h slot and $s-1$ n slots. By taking successive derivatives in independent copies of n we obtain a polarized symbol

$$\sigma(h; n_1, \dots, n_{s-1}).$$

The derivative identities above imply additivity in every argument, modulo lower-degree terms.

Lemma F.15 (Polarization). *Let $\chi(h, n)$ be a multidegree $(1, s - 1)$ nilcharacter. There is a well-defined top symbol*

$$\sigma_\chi(h; n_1, \dots, n_{s-1})$$

which is additive in each variable. Two such nilcharacters have the same top symbol if and only if their quotient has multidegree strictly below $(1, s - 1)$ in degree-rank order.

Proof. Take one derivative in h and $s - 1$ independent derivatives in n , retaining only the terminal vertical character. The polynomial-sequence derivative rule shows that the result is independent of the base point and additive in each increment. If the top symbols agree, their quotient has vanishing terminal mixed derivative, hence has lower multidegree by the definition of polynomial degree. The converse is immediate. $\square$

This is the exact higher-order analogue of recovering a symmetric bilinear form from the second derivative of a quadratic phase.

F.10 Splitting multidegree factors

The final induction must separate two different kinds of lower-order error. The following elementary nilmanifold lemma is the precise tool.

Lemma F.16 (Splitting lemma for downsets). *Let $J, J' \subset \mathbb{N}^d$ be finite downsets and let $M \geq 1$. Let $\psi : \mathbb{Z}^d \rightarrow \mathbb{C}$ be a bounded Lipschitz nilsequence of multidegree contained in $J \cup J'$ and complexity at most M . For every $\varepsilon > 0$ there are integers*

$$K = K(J, J', \varepsilon, M), \quad M' = M'(J, J', \varepsilon, M)$$

and nilsequences

$$\psi_k \in \text{Nil}_{\subset J}, \quad \psi'_k \in \text{Nil}_{\subset J'} \quad (1 \leq k \leq K)$$

of complexity at most M' such that

$$\left\| \psi - \sum_{k=1}^K \psi_k \psi'_k \right\|_\infty \leq \varepsilon.$$

Proof. Write $\psi(n) = F(g(n)\Gamma)$. We use the following explicit universal lift. Choose homogeneous Mal'cev generators $e_{a,j}$, with multidegree $a \in J \cup J'$, and replace G by the universal nilpotent Lie group generated by these symbols subject only to the relations that a commutator vanishes whenever the sum of the multidegrees of its entries lies outside $J \cup J'$. The original nilmanifold, orbit and function are factors of this universal model, so proving the decomposition upstairs and pushing it down is sufficient.

Let $G_{>J}$ be the closed normal subgroup generated by the homogeneous generators whose degrees lie in $J' \setminus J$, and define $G_{>J'}$ symmetrically. Then

$$X_J := G/(G_{>J}\Gamma), \quad X_{J'} := G/(G_{>J'}\Gamma)$$

carry filtrations of multidegree contained in J and J' respectively.

The key point is that in the universal model

$$G_{>J} \cap G_{>J'} = \{e\}.$$

Indeed, a Hall commutator belonging to the first normal subgroup contains at least one generator of degree $a \in J' \setminus J$, while one belonging to the second contains at least one generator of degree $b \in J \setminus J'$. If a Hall word contains both, its total multidegree dominates

$a + b$. That degree cannot lie in J : since J is a downset, $a + b \in J$ would force $a \in J$, contrary to the choice of a ; similarly it cannot lie in J' . Hence such a mixed Hall word is killed by the defining relations of the universal group. Distinct surviving Hall words are linearly independent in the free Lie algebra, so no other intersection is possible.

It follows that the product map

$$G/\Gamma \longrightarrow X_J \times X_{J'}$$

is injective (and hence a homeomorphism onto its compact image). Therefore the pullbacks of continuous functions on X_J and on $X_{J'}$ form two self-adjoint unital algebras whose union separates points of G/Γ . By Stone–Weierstrass, finite sums of products

$$F_{J,k}(x_J)F_{J',k}(x_{J'})$$

are uniformly dense in $C(G/\Gamma)$. Approximate the Lipschitz function F to error ε by such a finite sum and compose with the polynomial orbit $g(n)\Gamma$. The first factor in every product is a J -nilsequence and the second a J' -nilsequence. Smooth approximation on the two compact factor nilmanifolds gives the asserted controlled complexity. $\square$

Corollary F.17 (The factorisation needed in the GTZ closing step). *Let $\Psi(h, n)$ have multidegree in*

$$\mathcal{J}_s := \{(i, j) : i + j \leq s - 1\} \cup \{(i, s - i) : 2 \leq i \leq s\}.$$

For every $\varepsilon > 0$ there is a finite sum

$$\sum_{j=1}^K \psi_j(n) \psi'_j(h, n)$$

whose uniform distance from Ψ is at most ε , where each ψ_j is independent of h and has degree at most $s - 1$ in n , while each ψ'_j has n -degree at most $s - 2$.

Proof. Take

$$J = \{(0, j) : 0 \leq j \leq s - 1\}, \quad J' = \{(i, j) : j \leq s - 2, i + j \leq s\}.$$

These are downsets and $\mathcal{J}_s \subseteq J \cup J'$. Apply Lemma F.16. A J -nilsequence has no positive h -degree, hence depends only on n ; a J' -nilsequence has n -degree at most $s - 2$. $\square$

Lemma F.18 (Torsion-freeness of higher-degree symbols). *Let $d \geq 2$ and let χ be a degree- d nilcharacter on an interval. If, for a nonzero integer q of bounded size,*

$$q[\chi]_d = 0,$$

then $[\chi]_d = 0$.

Proof. The hypothesis says that χ^q is equivalent to a degree-at-most- $(d - 1)$ nilsequence. Apply the factorization/equidistribution argument from Lemma F.11 to the polynomial orbit of χ , restricting to one rational residue class so that its orbit closure is connected. If η is the top vertical character of χ , the fact that χ^q drops degree says that $q\eta$ annihilates the connected top vertical subgroup of the orbit closure.

The character group of a connected torus is a free abelian group. Hence $q\eta = 0$ on that torus implies $\eta = 0$ there. Quotienting the annihilated top vertical direction lowers the degree of χ itself. Thus $[\chi]_d = 0$. $\square$

Appendix G

Coherence, Sunflowers, and Bracket Linearisation

The difficult middle of a higher-order inverse theorem is not the first derivative or the final antiderivative. It is the passage from a separate structured model for each derivative

$$h \mapsto \chi_h$$

to one coherent model depending polynomially on h . This appendix proves the combinatorial steps used in Chapter D.

G.1 A Gowers–Cauchy–Schwarz coherence lemma

We begin with the exact analogue of the frequency-graph calculation in the U^3 proof.

Definition G.1 (Correlation modulo lower degree). Let $r \geq 0$, $\rho \in (0, 1]$, and $M \geq 1$. For $h \in [-N, N]$ and a 1-bounded model $a_h : [N] \rightarrow \mathbb{C}$, we say that $\Delta_h f$ *correlates with a_h modulo n -degree at most r at level (ρ, M)* if there is a 1-bounded degree-at-most- r nilsequence ψ_h of complexity at most M such that

$$\left| \mathbb{E}_{n \leq N} \Delta_h f(n) \overline{a_h(n) \psi_h(n)} \right| \geq \rho.$$

As throughout this appendix, all functions are extended by 0 off $[N]$ when a translated argument occurs.

Lemma G.2 (Derivative correlations create additive-quadruple bias). *Let $H \subseteq [-N, N]$ have density at least α in that interval. Suppose $|f| \leq 1$ and for every $h \in H$ there are 1-bounded sequences a_h such that*

$$\left| \mathbb{E}_n \Delta_h f(n) \overline{a_h(n)} \right| \geq \rho,$$

where all functions are extended by 0 outside $[N]$. Then for $\gg_{\alpha, \rho} N^3$ additive quadruples

$$h_1 + h_2 = h_3 + h_4, \quad h_i \in H,$$

one has

$$\left| \mathbb{E}_n a_{h_1}(n) a_{h_2}(n + h_1 - h_4) \overline{a_{h_3}(n) a_{h_4}(n + h_1 - h_4)} \right| \gg_{\alpha, \rho} 1.$$

The same conclusion holds if every a_h is multiplied by an auxiliary sequence from a class closed under translation, multiplication, and conjugation; the surviving auxiliary factor remains in that class.

Proof. Multiply each a_h by a scalar phase so that its correlation has real part at least ρ . Put all variables in one fixed interval Ω of length $O(N)$ containing $[N]$ and all relevant translates, and extend every function by zero. Changing from the normalized averages on $[N]$ and H to uniform averages on Ω only changes constants depending on α .

Introduce the bipartite kernel

$$B(x, y) = 1_H(y - x) \overline{a_{y-x}(x)}.$$

Then the averaged correlation is

$$\left| \mathbb{E}_{x, y \in \Omega} f(y) \overline{f(x)} B(x, y) \right| \gg_{\alpha} \rho \alpha =: \eta.$$

We now perform the two Cauchy–Schwarz steps explicitly. Since $|f| \leq 1$,

$$\begin{aligned} \eta^2 &\leq \mathbb{E}_x |\mathbb{E}_y f(y) B(x, y)|^2 \\ &= \mathbb{E}_{x, y, y'} f(y) \overline{f(y')} B(x, y) \overline{B(x, y')}. \end{aligned}$$

Regard the last expression as an average in (y, y') and use Cauchy–Schwarz once more, again with $|f| \leq 1$:

$$\begin{aligned} \eta^4 &\leq \mathbb{E}_{y, y'} \left| \mathbb{E}_x B(x, y) \overline{B(x, y')} \right|^2 \\ &= \mathbb{E}_{x, x', y, y'} B(x, y) \overline{B(x, y')} \overline{B(x', y)} B(x', y'). \end{aligned}$$

This is the usual bipartite four-cycle inequality; importantly, every copy of f has now disappeared.

Take the complex conjugate of the last integrand and make the change of variables

$$\begin{aligned} n &= x, & t &= x' - x, \\ h_1 &= y - x, & h_2 &= y' - x', & h_3 &= y' - x, & h_4 &= y - x'. \end{aligned}$$

Then

$$h_1 + h_2 = h_3 + h_4, \quad t = h_1 - h_4,$$

and the a -part of the four-cycle is exactly

$$a_{h_1}(n) a_{h_2}(n + t) \overline{a_{h_3}(n) a_{h_4}(n + t)}.$$

Conversely, (n, h_1, h_2, h_3, h_4) satisfying the displayed additive relation determines (x, x', y, y') by

$$x = n, \quad x' = n + h_1 - h_4, \quad y = n + h_1, \quad y' = n + h_3,$$

so no multiplicity has been hidden in the reparametrization.

We have therefore proved a lower bound $\gg_{\alpha, \rho} 1$ for the ambient normalized average, over additive quadruples in H , of the displayed inner n -average (after harmless conjugation). There are $O(N^3)$ additive quadruples in $[-N, N]$ and every inner average has magnitude at most 1. If fewer than $c_{\alpha, \rho} N^3$ quadruples had magnitude at least a sufficiently small $c_{\alpha, \rho} > 0$, the total four-cycle average would be smaller than the lower bound. This proves the claimed positive-proportion statement.

Finally, if $a_h = \chi_h \lambda_h$ and the class containing λ_h is closed under translation, multiplication, and conjugation, the four surviving λ factors remain in that class. No additional Cauchy–Schwarz step is needed. $\square$

Lemma G.3 (The already-built two-variable factor disappears at top order). *Let $d \geq 1$ and let $\kappa(h, n)$ be a multidegree at most $(1, d)$ nilcharacter. For an additive quadruple $h_1 + h_2 = h_3 + h_4$ put $t = h_1 - h_4$. Then*

$$\kappa(h_1, n)\kappa(h_2, n+t)\overline{\kappa(h_3, n)}\overline{\kappa(h_4, n+t)}$$

has n -degree at most $d - 1$ modulo bounded-complexity lower multidegree factors.

Proof. Polarize the top multidegree $(1, d)$ symbol of κ as

$$\sigma(h; n_1, \dots, n_d).$$

Translation $n \mapsto n + t$ does not change the homogeneous n -degree- d symbol: all terms containing t use at most $d - 1$ copies of the n variable. Hence the top n -degree symbol of the displayed fourfold product is

$$\sigma(h_1; n_1, \dots, n_d) + \sigma(h_2; n_1, \dots, n_d) - \sigma(h_3; n_1, \dots, n_d) - \sigma(h_4; n_1, \dots, n_d).$$

The first slot is degree one, so this is

$$\sigma(h_1 + h_2 - h_3 - h_4; n_1, \dots, n_d) = 0.$$

Lemma F.15 therefore lowers the n -degree by one. This is the finitary version of the cancellation that allows the Gowers Cauchy–Schwarz argument to eliminate not only f but also the coherent nilobject already constructed at earlier rank steps. $\square$

Corollary G.4 (Residual rank-symbol parallelogram law). *Fix $s \geq 2$, $1 \leq r_* \leq s - 1$, $\alpha, \rho \in (0, 1]$, and $M \geq 1$. There are constants*

$$c = c(s, r_*, \alpha, \rho, M) > 0, \quad C = C(s, r_*, \alpha, \rho, M) \geq 1,$$

and $N_0 = N_0(s, r_, \alpha, \rho, M)$ with the following property. Let $N \geq N_0$, let $H \subseteq [-N, N]$ satisfy $|H| \geq \alpha(2N + 1)$, and let $|f| \leq 1$. Suppose that for every $h \in H$ the nilcharacters $\kappa(h, \cdot)$ and χ_h have complexity at most M , that κ has multidegree at most $(1, s - 1)$, that χ_h has degree–rank at most $(s - 1, r_*)$, and that $\Delta_h f$ correlates with $\kappa(h, \cdot)\chi_h$ modulo n -degree at most $s - 2$ at level (ρ, M) . Then at least cN^3 additive quadruples $h_1 + h_2 = h_3 + h_4$ in H have the following property. With $t = h_1 - h_4$, the residual product*

$$\chi_{h_1}(n)\chi_{h_2}(n+t)\overline{\chi_{h_3}(n)}\overline{\chi_{h_4}(n+t)}$$

is $(s - 2)$ -biased at level at least c by a nilsequence of complexity at most C . On the rational/interval piece supplied by Lemma F.13, its $(s - 1, r_)$ symbol therefore vanishes.*

Proof. Apply Lemma G.2 to the full models. The surviving fourfold product of κ has n -degree at most $s - 2$ by Lemma G.3; the four lower-degree test factors also remain of degree at most $s - 2$. Absorb both into the bias witness. The residual fourfold product is therefore $(s - 2)$ -biased. Since $(s - 2)$ is strictly below $(s - 1, r_*)$ in degree–rank order, Lemma F.13 kills its top $(s - 1, r_*)$ symbol. $\square$

Corollary G.5 (Top-symbol parallelogram law). *Fix $s \geq 2$, $\alpha, \rho \in (0, 1]$, and $M \geq 1$. There are $c = c(s, \alpha, \rho, M) > 0$ and $N_0 = N_0(s, \alpha, \rho, M)$ such that the following holds for $N \geq N_0$. Let $H \subseteq [-N, N]$ satisfy $|H| \geq \alpha(2N + 1)$ and let $|f| \leq 1$. For every $h \in H$, let χ_h be a degree- $(s - 1)$ nilcharacter of complexity at most M , let λ_h be a 1-bounded degree-at-most- $(s - 2)$ nilsequence of complexity at most M , and assume*

$$\left| \mathbb{E}_n \Delta_h f(n) \overline{\chi_h(n)} \overline{\lambda_h(n)} \right| \geq \rho.$$

Then for at least cN^3 additive quadruples $h_1 + h_2 = h_3 + h_4$ in H , after passing to the bounded rational/interval piece on which the bias is carried, the degree- $(s-1)$ symbols satisfy

$$[\chi_{h_1}]_{s-1} + [T_{h_1-h_4}\chi_{h_2}]_{s-1} - [\chi_{h_3}]_{s-1} - [T_{h_1-h_4}\chi_{h_4}]_{s-1} = 0.$$

Here $T_t\chi(n) = \chi(n+t)$; translation acts on the top symbol through its usual polynomial translation action.

Proof. The inner product supplied by Lemma G.2 says that the product of the four nilcharacters is $(s-2)$ -biased, because the product of the lower-degree λ factors can be included in the test nilsequence. Lemma F.11 therefore forces the top symbol of that product to vanish. Symbol addition under tensor product gives the displayed relation. $\square$

G.2 Frequency regularisation: core, petals, and representation

Nilcharacters of bounded complexity involve only boundedly many horizontal frequencies in Mal'cev coordinates. A point that matters logically is that the “core” frequencies need *not* occur literally in every original frequency list. They are an external list with respect to which every original list can be represented. This is exactly what is needed when a relation involving two fixed increments is used to eliminate one frequency from a third increment.

For $x = (x_1, \dots, x_m) \in \mathbb{T}^m$ write

$$\|x\|_{\mathbb{T}^m} = \max_j \|x_j\|_{\mathbb{R}/\mathbb{Z}}.$$

If $\mathcal{C}$ is a finite list in $\mathbb{T}^m$, say that a list $\mathcal{P}$ is (L, ε) -independent modulo $\mathcal{C}$ if there is no relation

$$\sum_{p \in \mathcal{P}} a_p p + \sum_{c \in \mathcal{C}} b_c c = u \quad \text{in } \mathbb{T}^m, \quad \|u\|_{\mathbb{T}^m} \leq \varepsilon,$$

with $|a_p|, |b_c| \leq L$ and at least one $a_p \neq 0$.

Lemma G.6 (Finite frequency sunflower/regularisation). *Fix $D, L \geq 1$ and $0 < \tau < 1$. There is $c = c(D, L, \tau) > 0$ with the following property. Let H be finite and, for each $h \in H$, let $\mathcal{F}_h$ be a list of at most D frequencies in $\mathbb{T}^m$. Fix a scale $\varepsilon \geq 0$. Then there are*

1. *a set $H' \subseteq H$ with $|H'| \geq c|H|$;*
2. *an external core list $\mathcal{F}_*$ of $O_{D,L,\tau}(1)$ frequencies;*
3. *a bounded list $\mathcal{F}_{\text{rat}}$ of rational frequencies of bounded denominator;*
4. *for each $h \in H'$, a petal list $\mathcal{P}_h$ of a common length $d \leq D$ and a list $\mathcal{S}_h$ of $O_{D,L,\tau}(1)$ frequencies of size $O_{D,L,\tau}(\varepsilon)$,*

such that every frequency in $\mathcal{F}_h$ is an integer linear combination, with bounded coefficients, of

$$\mathcal{F}_* \uplus \mathcal{F}_{\text{rat}} \uplus \mathcal{P}_h \uplus \mathcal{S}_h.$$

Moreover, for all but at most $\tau|H'|^3$ triples $(h_1, h_2, h_3) \in (H')^3$, the union

$$\mathcal{P}_{h_1} \uplus \mathcal{P}_{h_2} \uplus \mathcal{P}_{h_3}$$

is $(L, C\varepsilon)$ -independent modulo $\mathcal{F}_$, where $C = C(D, L, \tau)$.*

Proof. The proof is a finite quantitative version of the minimal-rank argument behind the Green–Tao–Ziegler sunflower lemma. We give it because the distinction between *representation* and literal set intersection is essential later.

Call a *partial representation* any choice of a positive-density subset $H_0 \subseteq H$, a fixed core, bounded rational and small lists, and a petal list $\mathcal{P}_h$ of common length d for $h \in H_0$, such that every original frequency in $\mathcal{F}_h$ is represented by these lists with bounded coefficients. At least one partial representation exists: take $H_0 = H$, empty core/rational/small lists, and $\mathcal{P}_h = \mathcal{F}_h$ after padding the finitely many possible list lengths and pigeonholing one length. Among all partial representations obtainable after finitely many restrictions of density bounded below in terms of (D, L, τ) , choose one for which d is minimal; subject to that, choose one with a minimal core size.

Suppose the desired independence conclusion fails. Then more than $\tau|H_0|^3$ triples admit an $(L, C\varepsilon)$ -relation involving a petal coefficient. There are only finitely many coefficient patterns and positions, depending on D and L . Pigeonholing therefore gives one fixed pattern on $\gg_{D,L,\tau} |H_0|^3$ triples. Relabeling the three slots if necessary, one coefficient of the first petal list, say $a p_{h_1,1}$ with $a \neq 0$, occurs in that pattern.

By averaging, we may fix h_2, h_3 so that the same relation holds for $\gg_{D,L,\tau} |H_0|$ values of h_1 . Restrict H_0 to those h_1 . The torus is divisible, so for every frequency x appearing in the relation choose once and for all an a th root $\tilde{x}$ with $a\tilde{x} = x$. Dividing the relation by a now expresses $p_{h_1,1}$ as

$$p_{h_1,1} = \sum_{j=2}^d u_j \tilde{p}_{h,j} + \sum_{c \in \mathcal{F}_*} v_c \tilde{c} + \sum_{q \in \mathcal{Q}} w_q \tilde{q} + q_0 + s_h,$$

where the coefficients are bounded in terms of (D, L) , the list $\mathcal{Q}$ consists of the finitely many petal frequencies attached to the two fixed indices h_2, h_3 , q_0 is a rational frequency of bounded denominator (the ambiguity in choosing an a th root), and $\|s_h\| = O_{D,L}(\varepsilon)$. Add the fixed roots from $\mathcal{Q}$ to the external core, add q_0 to the rational list and s_h to the small list, and replace the remaining petal coordinates by their chosen a th roots. We have produced a valid partial representation on a still positive-density subset but with petal length $d - 1$, contradicting minimality.

If a violating relation contains no petal coefficient, the same division argument shortens the core instead, after moving the resulting rational and small terms to their designated lists; this contradicts the secondary minimality. Hence the desired independence property must hold. At most D petal eliminations and $O_D(1)$ core eliminations occur, so the product of all density losses and every coefficient/denominator bound depends only on (D, L, τ) . This proves the lemma. $\square$

Remark G.7 (How the scale is chosen). In the nilsequence application the lists occur separately at each filtration degree. We apply the lemma degree by degree with $\varepsilon \asymp N^{-i}$ at degree i . The quantitative Leibman criterion supplies a bound $L = O_{s,\delta}(1)$ for every resonance capable of affecting the fixed correlation threshold. Rational frequencies are frozen after passing to controlled residue classes, and $O(N^{-i})$ frequencies are absorbed into smooth/lower-degree factors. Thus the frequencies that remain as petals are precisely the genuinely irrational directions relevant to top-order cancellation.

G.3 Why independent petals cannot interact

The next two lemmas isolate the exact implication that is often compressed into the phrase “independent frequencies equidistribute and hence cancel.” The first part is purely harmonic analysis on a compact nilmanifold; the second is the contrapositive of the quantitative Leibman criterion.

Lemma G.8 (Vertical cancellation on an orbit closure). *Let $X = G/\Gamma$ be a compact filtered nilmanifold, let $K \leq G$ be a connected rational subgroup, and put $Y = K/(K \cap \Gamma)$. Suppose*

$F : X \rightarrow \mathbb{C}$ is Lipschitz and has vertical frequency η on the top central torus T_s . Let

$$T_Y := (K \cap G_s) / ((K \cap G_s) \cap \Gamma).$$

If $\eta|_{T_Y}$ is nontrivial, then for every $x \in X$

$$\int_Y F(xy) d\mu_Y(y) = 0.$$

Consequently, if a finite polynomial orbit $(g(n)\Gamma)_{n \in P}$ is δ -equidistributed in a translate xY in the Lipschitz sense, then

$$|\mathbb{E}_{n \in P} F(g(n)\Gamma)| \leq \delta \|F\|_{\text{Lip}}$$

whenever $\eta|_{T_Y}$ is nontrivial.

Proof. Let

$$I = \int_Y F(xy) d\mu_Y(y).$$

For $t \in T_Y$, Haar invariance gives

$$I = \int_Y F(xty) d\mu_Y(y) = e(\eta(t))I.$$

Since $\eta|_{T_Y}$ is nontrivial, choose t with $e(\eta(t)) \neq 1$; hence $I = 0$. The quantitative assertion is now exactly the definition of Lipschitz δ -equidistribution, applied to $F|_{xY}$ and the zero Haar integral just proved. $\square$

Lemma G.9 (Independent petals force equidistribution). *Fix a free filtered nilmanifold model of complexity at most M and a correlation threshold $\rho > 0$. There are $L = L(M, \rho) \geq 1$ and $c_0 = c_0(M, \rho) > 0$ with the following property. Suppose the horizontal Taylor coefficients of a polynomial orbit are divided into a fixed core and a finite petal list $\mathcal{P}$, and suppose $\mathcal{P}$ is $(L, c_0 N^{-i})$ -independent modulo the core at every relevant Taylor degree i . After freezing the core coordinates, the orbit is c_0 -equidistributed in the rational subnilmanifold generated by the petal directions. Moreover, every vertical character nontrivial on a top commutator generated by those petal directions has orbit average of modulus $< \rho$.*

Proof. Argue contrapositively. If the petal factor is not c -equidistributed, the quantitative Leibman theorem from Chapter C supplies a nonzero horizontal character of complexity $O_\rho(1)$ whose composition with the polynomial orbit has $C^\infty[N]$ norm $O_\rho(1)$. Writing the composition in binomial Taylor coordinates, this says at some degree i that a bounded nonzero integer combination of the petal Taylor frequencies, together with a bounded combination of the frozen core frequencies, is $O_\rho(N^{-i})$ in $\mathbb{R}/\mathbb{Z}$. Taking L larger than the horizontal-character complexity and c smaller than the implied constant contradicts (L, cN^{-i}) -independence modulo the core.

Thus the petal factor is equidistributed. If the top vertical character is nontrivial on a commutator direction generated by those petals, that direction lies in the top vertical torus of the petal orbit closure. Lemma G.8 gives zero Haar integral and hence, by equidistribution, an orbit average smaller than ρ . $\square$

The next lemma is the higher-order analogue of the familiar assertion that independent Fourier frequencies cannot produce a nonzero constant phase.

Lemma G.10 (No petal–petal top term). *Let an additive-quadruple product of bounded-complexity nilcharacters be $(s-2)$ -biased, and suppose their horizontal frequency lists have been regularised and represented by Lemma G.6, with the rational/small pieces already held*

fixed or absorbed into lower degree. Then the top vertical character of the product contains no commutator monomial involving petal frequencies from two distinct indices. Nor can it contain a frequency component independent of the common core and of the bracket-linear frequencies forced by the additive relation.

Proof. Choose the exceptional parameter τ in Lemma G.6 much smaller than the positive density of biased additive quadruples supplied by Lemma G.2. A bad triple belongs to at most $O(N^0)$ additive quadruples once the three entries and their positions are fixed: the fourth increment is determined by $h_1 + h_2 = h_3 + h_4$. Since every quadruple has four three-element subcollections, at most $4\tau|H|^3$ additive quadruples contain a bad triple. Hence we may choose a biased additive quadruple for which every three of the four petal lists satisfy the independence conclusion.

Represent the four nilcharacters on a common free filtered nilpotent group whose generators correspond to the external core and to the four petal lists. The representation clause of Lemma G.6, not literal inclusion of the core in every original list, is exactly what permits this change of basis. Bounded-complexity nilcharacters are quotients of this free model by rational relations; the construction is given explicitly in Appendix H.

Assume, for contradiction, that the top vertical character is nonzero on a commutator word w containing petal generators from two different indices, say h_i and h_j . Select a third index different from both. Freeze the common-core coordinates and project the joint polynomial orbit to the rational subnilmanifold generated by the petal entries occurring in these three lists. The chosen good-triple independence is exactly the hypothesis of Lemma G.9; hence this petal factor is quantitatively equidistributed.

The commutator word w belongs to the top vertical subgroup of that petal factor, because it is generated by two of its horizontal petal directions. By assumption the vertical character of the additive-quadruple product is nontrivial on w . Lemma G.8 therefore makes the average of the top nilcharacter arbitrarily small compared with the fixed bias threshold. Tensoring with the degree- $(s-2)$ test nilsequence does not alter this top vertical frequency, so the same cancellation applies to the product that witnesses the alleged bias. This contradicts $(s-2)$ -bias.

The same argument applies to an unresolved regular frequency: include that frequency and the common core in the horizontal factor. If it is not forced by a bounded relation, Lemma G.9 gives equidistribution; if its top vertical frequency is nonzero, Lemma G.8 gives cancellation. Thus only core terms and frequency directions whose dependence on h is forced by the additive relations can survive. $\square$

Proof architecture The Furstenberg–Weiss step

This is the conceptual heart of the “no petal–petal” argument:

$$\boxed{\text{independent horizontal motion}} \implies \boxed{\text{equidistribution}} \implies \boxed{\text{vertical cancellation}}.$$

Bias can persist only if algebraic dependence collapses the supposedly independent directions. That collapse is precisely the structure we keep.

G.4 From many symbol relations to a Freiman graph

After the petal elimination, the surviving top data attached to h live in a fixed finite-dimensional torus $\mathbb{T}^d$ modulo lower-order terms. Denote this data by ξ_h .

Lemma G.11 (Balog–Szemerédi extraction for the frequency graph). *Suppose $H \subseteq [-N, N]$ has size at least αN and that for at least ρN^3 additive quadruples*

$$h_1 + h_2 = h_3 + h_4$$

one has

$$\xi_{h_1} + \xi_{h_2} = \xi_{h_3} + \xi_{h_4}$$

in an abelian group W . Then there exists $H_0 \subseteq H$, $|H_0| \geq c(\alpha, \rho)N$, such that the graph

$$\Gamma_0 = \{(h, \xi_h) : h \in H_0\} \subseteq \mathbb{Z} \times W$$

has doubling

$$|\Gamma_0 + \Gamma_0| \leq C(\alpha, \rho)|\Gamma_0|.$$

Proof. The number of additive quadruples in the graph is exactly its additive energy. The hypothesis therefore gives

$$E(\Gamma) \gg_{\alpha, \rho} |\Gamma|^3.$$

Apply the asymmetric Balog–Szemerédi–Gowers theorem (Volume I, Section 2.2, Theorem 2.19 (project tag I.2.015)) to obtain a subset Γ_0 of comparable size and small doubling. Since the first-coordinate projection is injective on the graph, Γ_0 is the graph of the restricted map on some $H_0 \subseteq H$. $\square$

G.5 Freiman homomorphisms become bracket-linear

The exact input needed in the higher-order proof is slightly more precise than the slogan “small doubling implies a progression.” The frequency maps are initially only approximately additive on many quadruples. We now convert that statistical statement into a genuine Freiman homomorphism and then identify every such homomorphism on a dense Bohr cell.

Lemma G.12 (Freiman homomorphisms on a Bohr cell). *Let $B = B(S, \rho) \subseteq \mathbb{Z}/Q\mathbb{Z}$ be a regular Bohr set of bounded rank and radius bounded below, and let*

$$\zeta : B \rightarrow \mathbb{R}/\mathbb{Z}$$

be a Freiman homomorphism of order 2. After shrinking B by a bounded factor and partitioning it into $O(1)$ cells according to the fractional parts $(\{\alpha h\})_{\alpha \in S}$, one has on each cell

$$\zeta(h) = \delta + \lambda h + \sum_{j=1}^r \{\alpha_j h + \theta_j\} \beta_j \pmod{1},$$

with $r = O(|S|)$ and controlled parameters.

Proof. Volume II proves that a bounded-rank Bohr set contains a proper generalized progression

$$P = \left\{ \sum_{j=1}^r n_j v_j : |n_j| \leq L_j \right\}$$

of comparable density after a bounded shrink. Properness makes the coordinates n_j unique. Since ζ is a Freiman homomorphism,

$$\zeta\left(\sum_j n_j v_j\right) = \zeta(0) + \sum_j n_j (\zeta(v_j) - \zeta(0))$$

whenever the relevant partial sums stay in a fixed further shrink of P .

It remains to express the coordinate functions n_j in terms of the ambient integer h . The progression was obtained from the lattice of relations among the Bohr coordinates

$$h \mapsto (h/Q, (\alpha h)_{\alpha \in S}) \in \mathbb{R}^{1+|S|}/\mathbb{Z}^{1+|S|}.$$

Choosing a fundamental parallelepiped for this lattice expresses each n_j as an affine linear function of h plus integer carry functions:

$$n_j = a_j h + b_j + \sum_{\ell} c_{j\ell} [\alpha_{\ell} h + \theta_{\ell}].$$

This is the elementary coordinate calculation behind the Bohr-to-progression lemma of Volume II. Substitute

$$[t] = t - \{t\}.$$

All genuinely linear terms combine into $\lambda h + \delta$, and the remaining terms are finite linear combinations of fractional parts. Finally partition each fractional part into intervals of length $< 1/10$. On such a cell no new carry is created by a two-term additive relation, so the displayed expression retains the Freiman property. The number of cells and all parameters depend only on the Bohr rank and radius. $\square$

Lemma G.13 (Finite vertical-kernel refinement). *Let $G = \mathbb{Z}/Q\mathbb{Z}$, let W be a finite abelian group, and let $\Gamma \subseteq G \times W$ be a graph over its first coordinate. Suppose*

$$|\Gamma| \geq cQ, \quad |9\Gamma - 8\Gamma| \leq KQ.$$

Then there is $\Gamma'' \subseteq \Gamma$ with $|\Gamma''| \geq c'(c, K)Q$ such that $4\Gamma'' - 4\Gamma''$ is a graph.

Proof. Put

$$A = \{w \in W : (0, w) \in 8\Gamma - 8\Gamma\}.$$

For distinct $w \in A$, the translates $(0, w) + \Gamma$ are disjoint: if two met, their difference would be a nonzero vertical element between two points of the graph Γ . All these translates lie in $9\Gamma - 8\Gamma$, and hence

$$|A| |\Gamma| \leq |9\Gamma - 8\Gamma| \leq KQ.$$

Thus $|A| \leq K/c$.

We next separate the nonzero elements of A by characters of W . If $a \neq 0$ has order $q \geq 2$ and χ is uniform in $\widehat{W}$, then $\chi(a)$ is uniform among the q th roots of unity. (Restriction of characters from W to the cyclic subgroup generated by a is surjective; this follows directly from the structure theorem for finite abelian groups proved in Volume I.) Therefore

$$\Pr_{\chi}(\|\chi(a)\|_{\mathbb{R}/\mathbb{Z}} \leq 1/4) \leq 3/4.$$

Choose $m = O(\log(2 + |A|))$ independent characters. The expected number of nonzero $a \in A$ lying in all m corresponding quarter-arcs is less than 1 once the implicit constant is large enough. Hence there are $\chi_1, \dots, \chi_m$ such that

$$a \in A, \quad \|\chi_j(a)\| \leq 1/4 \quad (1 \leq j \leq m) \quad \implies \quad a = 0. \quad (\text{G.1})$$

Partition $\mathbb{T}^m$ into cubes of side $1/64$ and map $(h, w) \in \Gamma$ to $(\chi_1(w), \dots, \chi_m(w))$. One cube contains at least $64^{-m} |\Gamma| \gg_{c, K} Q$ graph points; call that subset Γ'' .

If two elements of $4\Gamma'' - 4\Gamma''$ have the same first coordinate, their difference is $(0, a) \in 8\Gamma'' - 8\Gamma''$, hence $a \in A$. In each character coordinate, a is a signed sum of sixteen points lying in one interval of length $1/64$, so

$$\|\chi_j(a)\| \leq 16/64 = 1/4.$$

Equation (G.1) gives $a = 0$. Thus the second coordinate is uniquely determined by the first, as required. $\square$

Theorem G.14 (Approximate bracket-linearisation). *Fix $\alpha, \rho \in (0, 1]$ and $\varepsilon \in (0, 1/10)$. There are constants $c = c(\alpha, \rho, \varepsilon) > 0$, $C = C(\alpha, \rho, \varepsilon) \geq 1$, $K = K(\alpha, \rho, \varepsilon) \in \mathbb{N}$, and $N_0 = N_0(\alpha, \rho, \varepsilon)$ with the following property. Let $N \geq N_0$, let $H \subseteq [-N, N]$ have $|H| \geq \alpha N$, and let $\xi_i : H \rightarrow \mathbb{R}/\mathbb{Z}$ for $1 \leq i \leq 4$. Suppose that for at least ρN^3 additive quadruples $h_1 + h_2 = h_3 + h_4$ one has*

$$\|\xi_1(h_1) + \xi_2(h_2) + \xi_3(h_3) + \xi_4(h_4) - \omega\|_{\mathbb{R}/\mathbb{Z}} \leq \varepsilon.$$

Then there is $H' \subseteq H$ with $|H'| \geq cN$, a Freiman homomorphism $\xi : H' \rightarrow \mathbb{R}/\mathbb{Z}$, constants $\delta, \lambda \in \mathbb{R}/\mathbb{Z}$, and frequencies $\alpha_j, \beta_j, \theta_j \in \mathbb{R}/\mathbb{Z}$ for $1 \leq j \leq K$ such that

$$\xi(h) = \lambda h + \sum_{j=1}^K \{\alpha_j h + \theta_j\} \beta_j \pmod{1}$$

and the set of $h \in H'$ satisfying

$$\|\xi_1(h) - \xi(h) - \delta\|_{\mathbb{R}/\mathbb{Z}} \leq C\varepsilon$$

has cardinality at least cN .

Proof. Choose an integer $M \asymp \varepsilon^{-1}$ and round every $\xi_i(h)$ to the nearest multiple of $1/M$. This changes the relation by $O(\varepsilon)$. There are only $O(1)$ possible rounded values for its right-hand side inside the original $O(\varepsilon)$ arc, so pigeonhole one value and reduce to an exact relation

$$\xi_1(h_1) + \xi_2(h_2) + \xi_3(h_3) + \xi_4(h_4) = \omega_0$$

for $\gg N^3$ additive quadruples.

Choose a cyclic group $G = \mathbb{Z}/Q\mathbb{Z}$ with $40N < Q < 100N$ and identify $[-N, N]$ with its image in G ; no sum or difference involving at most seventeen copies of H can wrap around. Define the *signed* graphs

$$\begin{aligned} \Gamma_1 &= \{(h, \xi_1(h)) : h \in H\}, & \Gamma_2 &= \{(h, \xi_2(h)) : h \in H\}, \\ \Gamma_3 &= \{(-h, \xi_3(h)) : h \in H\}, & \Gamma_4 &= \{(-h, \xi_4(h)) : h \in H\} \end{aligned}$$

in $G \times (\frac{1}{M}\mathbb{Z}/\mathbb{Z})$. The signs in the first coordinate are essential: the relation $h_1 + h_2 = h_3 + h_4$ now says exactly that every good quadruple is a representation of $(0, \omega_0)$ as

$$\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4, \quad \gamma_i \in \Gamma_i.$$

Thus this representation number is $\gg N^3$.

For arbitrary finite sets $A_1, \dots, A_4$ in an abelian group, Cauchy–Schwarz twice gives

$$r_{A_1+A_2+A_3+A_4}(x) \leq \prod_{i=1}^4 E(A_i)^{1/4}.$$

Indeed, pair (A_1, A_2) against (A_3, A_4) and then use $E(A_i, A_j) \leq E(A_i)^{1/2} E(A_j)^{1/2}$. Hence

$$\prod_{i=1}^4 E(\Gamma_i)^{1/4} \gg N^3.$$

Each energy is at most $|\Gamma_i|^3 \ll N^3$. Consequently *every* Γ_i has energy $\gg N^3$; in particular

$$E(\Gamma_1) \gg N^3.$$

There is no relabelling here, which is important because the conclusion is about the original map ξ_1 .

Apply Balog–Szemerédi–Gowers and then Plünnecke–Ruzsa to Γ_1 . We obtain $\Gamma' \subseteq \Gamma_1$, $|\Gamma'| \gg N$, with

$$|9\Gamma' - 8\Gamma'| \ll N.$$

Lemma G.13 applies to the finite second coordinate group and gives a further subset

$$\Gamma'' = \{(h, \xi_1(h)) : h \in H''\}, \quad |H''| \gg N,$$

for which $4\Gamma'' - 4\Gamma''$ is a graph.

Hence there is a unique Freiman homomorphism

$$\zeta : 2H'' - 2H'' \rightarrow \mathbb{R}/\mathbb{Z}$$

satisfying

$$\xi_1(h_1) + \xi_1(h_2) - \xi_1(h_3) - \xi_1(h_4) = \zeta(h_1 + h_2 - h_3 - h_4)$$

for $h_i \in H''$. Bogolyubov’s lemma puts a regular bounded-rank Bohr set B of positive radius inside $2H'' - 2H''$.

Shrink B once to a regular Bohr set B' with $B' - B' \subseteq B$. Average the translates $x + B'$ over $x \in G$. Because $|H''| \gg Q$ and $|B'| \gg Q$, some translate contains $\gg |B'|$ points of H'' . Choose

$$h_0 \in H'' \cap (x + B')$$

and retain

$$H_1 := H'' \cap (x + B').$$

Then $h - h_0 \in B' - B' \subseteq B$ for every $h \in H_1$.

Apply Lemma G.12 to $\zeta|_B$ and refine H_1 by the bounded Bohr-cell partition occurring there. One cell still contains $\gg N$ retained increments. From the definition of ζ , using the representation

$$h - h_0 = h + h_0 - h_0 - h_0,$$

we have exactly

$$\zeta(h - h_0) = \xi_1(h) - \xi_1(h_0).$$

Thus on the retained set

$$\xi_1(h) = \zeta(h - h_0) + \delta, \quad \delta = \xi_1(h_0).$$

Substituting the bracket formula for ζ yields the claimed expression. A final refinement into fractional-part intervals shorter than $1/10$ ensures that all carries occurring in two-term Freiman relations are fixed, so the displayed bracket expression is a genuine Freiman homomorphism on H' . Undoing the initial rounding changes it by $O(\varepsilon)$. $\square$

Worked Example G.15 (Rank one). If the graph lies in a rank-one progression generated by (q, ω) , then

$$h = h_0 + nq, \quad \xi_h = \xi_0 + n\omega.$$

On one residue class $h \equiv h_0 \pmod{q}$,

$$n = (h - h_0)/q,$$

so the frequency is genuinely affine-linear. Bracket terms appear only when several progression coordinates compete to represent one integer coordinate and a fundamental-domain choice creates carries.

G.6 The degree–rank coherence theorem

The preceding machinery must be used one degree–rank layer at a time. This is not mere bookkeeping. At ordinary degree $s - 1$ there can be bracket terms of several commutator ranks, and removing a top-rank term may leave a term of the same ordinary degree.

Fix $1 \leq r_* \leq s - 1$. Choose a Hall basis for the universal free model of a residual degree–rank $(s - 1, r_*)$ nilcharacter, ordered so that the Hall words in the layer $(s - 1, r_*)$ come first and then by the usual Hall order inside that layer.

Lemma G.16 (One triangular coherence step at fixed degree–rank). *Fix $s \geq 2$, $1 \leq r_* \leq s - 1$, $\alpha, \rho \in (0, 1]$, and $M \geq 1$. There are $c = c(s, r_*, \alpha, \rho, M) > 0$, $C = C(s, r_*, \alpha, \rho, M) \geq 1$, and $N_0 = N_0(s, r_*, \alpha, \rho, M)$ with the following property. Let $N \geq N_0$, $H \subseteq [-N, N]$ with $|H| \geq \alpha(2N + 1)$, and $|f| \leq 1$. Suppose that κ is a multidegree-at-most- $(1, s - 1)$ nilcharacter of complexity at most M , every χ_h has degree–rank at most $(s - 1, r_*)$ and complexity at most M , and $\Delta_h f$ correlates with $\kappa(h, \cdot)\chi_h$ modulo n -degree at most $s - 2$ at level (ρ, M) for every $h \in H$. Put the χ_h in a common total frequency representation and fix a Hall word w in the layer $(s - 1, r_*)$. If every Hall coordinate strictly above w has on H a bracket-linear representation using at most M bracket frequencies and parameters of complexity at most M , then there is $H' \subseteq H$ with $|H'| \geq cN$ on which the coefficient of w has a bracket-linear representation using at most C frequencies and parameters of complexity at most C . Subtracting the associated formal two-variable symbol changes only Hall coordinates at or below w .*

Proof. Corollary G.4 first eliminates both f and the already-built factor κ , leaving the additive-quadruple relation for the $(s - 1, r_*)$ symbols of the residual characters alone. Cancel the coordinates already made coherent. Their four-point second difference vanishes at top degree–rank; any carry produced by a fundamental-domain choice belongs to the fixed core or to smaller degree–rank.

Apply Lemma G.6 to total frequency representations of the remaining residual symbols. Lemma G.10, using the degree–rank bias lemma rather than ordinary-degree cancellation, removes every top Hall word containing independent petals from two different increments. Hence the coefficient of w depends only on the common core and on a bounded family of scalar torus frequencies which satisfy the additive-quadruple relation on a positive proportion of quadruples. Theorem G.14 makes those scalar frequencies bracket-linear on a positive-density subset.

Represent the resulting coefficient by the corresponding formal Hall word with one h -slot. Hall collection is triangular: subtracting this word can create corrections only in lower Hall coordinates within the same degree–rank layer or below that layer. Thus coordinates already treated are not reintroduced. $\square$

Theorem G.17 (Coherence at one degree–rank). *Fix $s \geq 2$, $1 \leq r_* \leq s - 1$, $\alpha, \rho \in (0, 1]$, and $M \geq 1$. There are $c = c(s, r_*, \alpha, \rho, M) > 0$, $C = C(s, r_*, \alpha, \rho, M) \geq 1$, and $N_0 = N_0(s, r_*, \alpha, \rho, M)$ with the following property. Let $N \geq N_0$, let $H \subseteq [-N, N]$ satisfy $|H| \geq \alpha(2N + 1)$, and let $|f| \leq 1$. Suppose κ is a multidegree-at-most- $(1, s - 1)$ nilcharacter of complexity at most M , every χ_h is a nilcharacter of degree–rank at most $(s - 1, r_*)$ and complexity at most M , and for every $h \in H$ the derivative $\Delta_h f$ correlates with $\kappa(h, \cdot)\chi_h$ modulo n -degree at most $s - 2$ at level (ρ, M) . Then there is $H' \subseteq H$ with $|H'| \geq cN$ such that the complete $(s - 1, r_*)$ symbol of χ_h has a bracket-linear description on H' . More precisely, every top Hall coefficient belongs to the algebra generated by*

$$h, \quad \{\alpha_1 h + \theta_1\}, \dots, \{\alpha_R h + \theta_R\}, \quad R \leq C,$$

with all coefficients and auxiliary frequency data of complexity at most C .

Proof. Use Lemma F.8 and bounded complexity to pigeonhole a common free Hall model for the residual characters. There are at most $C_0 = C_0(s, M)$ Hall coordinates in the top layer $(s - 1, r_*)$ of the chosen common model. Starting with the highest unresolved coordinate, apply Lemma G.16; then descend through the Hall order. Triangularity prevents a later subtraction from changing a coordinate already treated. Each step costs only a controlled density factor and introduces only boundedly many bracket frequencies. After finitely many steps every $(s - 1, r_*)$ coefficient is bracket-linear, as claimed. $\square$

Proof architecture Two nested descents

There are two different inductions and they must not be conflated:

inside a fixed $(s - 1, r_*)$: Hall-coordinate descent

followed by

$(s - 1, r_*) \rightarrow (s - 1, r_* - 1)$: degree–rank descent.

The first produces a coherent top symbol. Appendix H realizes that symbol by a two-variable nilcharacter; quotienting it from χ_h then performs the second descent. Only after r_* reaches zero has the residual ordinary degree actually fallen to $s - 2$.

Appendix H

Building, Symmetrising, and Integrating the Nilobject

Appendix G shows that the top symbol of the derivative model varies bracket-linearly with the increment. We now turn that symbolic information into an actual multivariable nilcharacter, prove the symmetry forced by commuting derivatives, and integrate it to a one-variable degree- s nilsequence.

H.1 Free filtered nilpotent groups

Fix a finite list of formal generators $e_{i,j}$, where $i \in \{1, \dots, s\}$ is the assigned filtration weight. Let $\mathfrak{f}$ be the free real Lie algebra on these generators modulo all commutators whose total weight exceeds s , and let

$$\tilde{G} = \exp(\mathfrak{f}).$$

The BCH series terminates because $\mathfrak{f}$ is nilpotent. Give $\tilde{G}$ the filtration generated by commutators of total weight at least r .

Lemma H.1 (Universal property). *If $G_\bullet$ is any filtered s -step nilpotent Lie group and $x_{i,j} \in G_i$, there is a unique filtered homomorphism from $\tilde{G}$ to the connected subgroup generated by the $x_{i,j}$ that sends*

$$e_{i,j} \mapsto x_{i,j}.$$

Proof. On Lie algebras, the universal property of the free Lie algebra gives a homomorphism sending the formal generators to $\log x_{i,j}$. Every commutator of total weight greater than s maps to zero by the filtration rule

$$[G_i, G_j] \subseteq G_{i+j}.$$

Hence the map factors through $\mathfrak{f}$. Exponentiation is a bijection for connected simply connected nilpotent groups and intertwines Lie homomorphisms with group homomorphisms, proving existence and uniqueness. $\square$

A rational lattice is obtained by taking the subgroup generated by sufficiently divisible rational powers of the formal generators. The precise denominator clearing is proved internally in the Mal'cev appendix, Appendix L, through the lattice-rationality and controlled-basis results; only the explicitly stated bounded-complexity compatible-lattice conclusion is used below.

H.2 Realising bracket-linear symbols

A typical bracket-linear frequency has the form

$$\alpha(h) = \alpha_0 + \lambda h + \sum_{j=1}^r \gamma_j \{\eta_j h + \theta_j\}.$$

The fractional part is discontinuous, but the discontinuity is a fundamental-domain artifact. In a free nilpotent group one represents $\eta_j h$ by motion in a horizontal torus, and the carry produced by reducing that motion modulo the lattice is recorded by a higher commutator coordinate.

Lemma H.2 (Bracket monomials are nilcharacters). *Fix finitely many bracket-linear functions $\alpha_1(h), \dots, \alpha_m(h)$ and polynomial phases in n of degrees whose total with the h -weight is at most s . Every finite product of formal top-order terms obtained from them, for example*

$$e(\{\eta h\} \beta n^{s-1}), \quad e(\{\gamma\{\eta h\} n\} \beta n^{s-2}),$$

away from their coordinate discontinuities, is the coordinate expression of a multidegree $(1, s-1)$ nilcharacter on a bounded-complexity filtered nilmanifold. After a smooth partition of unity, the same is true globally as a bounded linear combination of smooth nilcharacters.

Proof. We induct on the number of nested brackets. With no bracket, a monomial

$$e(\alpha h n^r)$$

is realized on a torus by a polynomial orbit; alternatively it is the vertical coordinate of the free group generated by one h -weight-1 direction and r n -weight-1 directions.

Suppose a bracket $\{u(h, n)\}$ has already been realized by reducing a horizontal coordinate u to a fundamental domain. Introduce one new generator v of the required complementary weight and a central commutator coordinate $[u, v]$. The BCH identity

$$\exp(U) \exp(V) = \exp\left(U + V + \frac{1}{2}[U, V] + \dots\right)$$

shows that the commutator coordinate contains the product of the two horizontal coordinates, plus terms of strictly lower degree-rank. Reducing u modulo the lattice replaces it by $\{u\}$ and transfers the integer carry into the central coordinate. A vertical character therefore reads off $e(\{u\}v)$ modulo lower-degree factors.

Repeating this construction realizes any finite nested bracket monomial. The filtration weights add under commutators, so terms of total weight greater than s vanish in the free s -step quotient. Finally cover the finitely many fundamental-domain boundaries by small neighborhoods and use a smooth partition of unity. On each chart the coordinate formula is smooth; summing the chart pieces gives a bounded linear combination of smooth nilcharacters with the same top symbol. $\square$

Theorem H.3 (One degree-rank correction). *Fix $s \geq 2$, $1 \leq r_* \leq s-1$, $\alpha, \rho \in (0, 1]$, and $M \geq 1$. There are $c = c(s, r_*, \alpha, \rho, M) > 0$, $C = C(s, r_*, \alpha, \rho, M) \geq 1$, and $N_0 = N_0(s, r_*, \alpha, \rho, M)$ with the following property. Let $N \geq N_0$, let $H \subseteq [-N, N]$ satisfy $|H| \geq \alpha(2N+1)$, and let $|f| \leq 1$. Suppose κ is a multidegree-at-most- $(1, s-1)$ nilcharacter of complexity at most M , every χ_h has degree-rank at most $(s-1, r_*)$ and complexity at most M , and $\Delta_h f$ correlates with $\kappa(h, \cdot) \chi_h$ modulo n -degree at most $s-2$ at level (ρ, M) for every $h \in H$. Assume moreover that the complete $(s-1, r_*)$ symbol of χ_h is bracket-linear in h using at most M bracket frequencies and data of complexity at most M . Then there are $H' \subseteq H$, $|H'| \geq cN$,*

and a multidegree-at-most- $(1, s-1)$ nilcharacter $\kappa_{r_*}(h, n)$ of complexity at most C such that, for every $h \in H'$,

$$\chi_h(n) \overline{\kappa_{r_*}(h, n)}$$

has degree-rank at most $(s-1, r_*-1)$ after multiplication by a lower-degree-rank nilsequence of complexity at most C . The correlation level retained after this replacement is at least c .

Proof. Choose the common universal free model supplied by Lemma F.8. Theorem G.17 expresses each coefficient of its top $(s-1, r_*)$ vertical character as a finite bracket-linear function of h . For every top Hall word, realize that coefficient by Lemma H.2; take the product over the finitely many top Hall words. The resulting two-variable nilcharacter $\kappa_{r_*}(h, n)$ has exactly the same total frequency representation at layer $(s-1, r_*)$ as χ_h .

Therefore the quotient has trivial top $(s-1, r_*)$ character in the universal representation. Lemma F.6 lowers it to degree-rank at most $(s-1, r_*-1)$. Lower Hall-coordinate corrections produced by BCH collection are already below the removed layer and are part of that residual factor. This is exactly why the construction is performed after the triangular Hall-coordinate descent rather than one monomial at a time without order. $\square$

Theorem H.4 (Construction of the two-variable nilobject). *Fix $s \geq 2$, $\alpha, \rho \in (0, 1]$, and $M \geq 1$. There are $c = c(s, \alpha, \rho, M) > 0$, $C = C(s, \alpha, \rho, M) \geq 1$, and $N_0 = N_0(s, \alpha, \rho, M)$ with the following property. Let $N \geq N_0$, let $H \subseteq [-N, N]$ satisfy $|H| \geq \alpha(2N+1)$, and let $|f| \leq 1$. For every $h \in H$, let χ_h be a degree-at-most- $(s-1)$ nilcharacter of complexity at most M , and assume that $\Delta_h f$ correlates with χ_h modulo n -degree at most $s-2$ at level (ρ, M) . Then there are a set $H' \subseteq H$ with $|H'| \geq cN$ and a single multidegree-at-most- $(1, s-1)$ nilcharacter $\chi(h, n)$ of complexity at most C such that, for every $h \in H'$, $\Delta_h f$ correlates with $\chi(h, \cdot)$ modulo n -degree at most $s-2$ at level (c, C) .*

Proof. An ordinary degree-at-most- $(s-1)$ nilcharacter has degree-rank at most $(s-1, s-1)$. Set

$$\kappa^{(s-1)}(h, n) = 1, \quad \chi_h^{(s-1)} = \chi_h.$$

We induct downward on $r_* = s-1, s-2, \dots, 1$. Assume that on a positive-density increment set

$$\Delta_h f \text{ is } (s-2)\text{-biased by } \kappa^{(r_*)}(h, \cdot) \chi_h^{(r_*)},$$

where $\kappa^{(r_*)}$ has multidegree at most $(1, s-1)$ and $\chi_h^{(r_*)}$ has degree-rank at most $(s-1, r_*)$. Theorem G.17 makes the top $(s-1, r_*)$ residual symbol bracket-linear. Theorem H.3 constructs κ_{r_*} and a new residual $\chi_h^{(r_*-1)}$ of degree-rank at most $(s-1, r_*-1)$. Put

$$\kappa^{(r_*-1)} := \kappa^{(r_*)} \kappa_{r_*}.$$

Products preserve multidegree at most $(1, s-1)$, and all restrictions and bias losses are controlled. This closes the inner induction.

After the step $r_* = 1$, the residual has degree-rank at most $(s-1, 0)$. By the convention $G_{(s-1,0)} = G_{(s-1,1)}$, this means that its ordinary n -degree is at most $s-2$. Absorb that residual into the $(s-2)$ -degree bias witness and set

$$\chi(h, n) := \kappa^{(0)}(h, n).$$

This is the desired single two-variable nilcharacter. $\square$

Worked Example H.5 (Why one pass is insufficient at cubic degree). A cubic bracket term such as

$$e(\{\alpha n^2\} \beta n)$$

has degree-rank $(3, 2)$, while an ordinary cubic phase $e(\gamma n^3)$ has degree-rank $(3, 1)$. Removing the rank-two bracket term can therefore leave a rank-one cubic term. Both have ordinary degree three. The inner induction removes $(3, 2)$ first and $(3, 1)$ second; calling the latter “lower degree” after the first step would be false.

H.3 Symmetry from commuting derivatives

Let

$$\sigma(h; n_1, \dots, n_{s-1})$$

be the polarized top symbol of $\chi(h, n)$. It is symmetric in the n -variables. The difficult point is to exchange the distinguished increment h with an n -variable. For $s = 2$ this is precisely the local symmetry calculation of Chapter B; in the general induction we therefore assume $s \geq 3$.

Choose a polarized degree- $(1, \dots, 1)$ nilcharacter

$$\tilde{\chi}(u_1, \dots, u_s)$$

so that, at symbol level,

$$[\chi(h, n)]_s = s[\tilde{\chi}(h, n, \dots, n)]_s. \quad (\text{H.1})$$

The finite vertical-root lemma below permits this normalization.

Lemma H.6 (Second-derivative symmetry at symbol level). *Fix $s \geq 3$, $\alpha, \rho \in (0, 1]$, and $M \geq 1$. There is $N_0 = N_0(s, \alpha, \rho, M)$ with the following property. Let $N \geq N_0$, $H \subseteq [-N, N]$ satisfy $|H| \geq \alpha(2N + 1)$, let $|f| \leq 1$, and let $\chi(h, n)$ be a multidegree-at-most- $(1, s - 1)$ nilcharacter of complexity at most M . Suppose that, for every $h \in H$, $\Delta_h f$ correlates with $\chi(h, \cdot)$ modulo n -degree at most $s - 2$ at level (ρ, M) . Let $\tilde{\chi}$ be the polarized nilcharacter normalized by (H.1). Then*

$$[\tilde{\chi}(h, k, n_2, \dots, n_{s-1})]_s = [\tilde{\chi}(k, h, n_2, \dots, n_{s-1})]_s.$$

Consequently the polarized top s -symbol is symmetric in all s variables.

Proof. We give the Cauchy–Schwarz elimination rather than replacing it by the phrase “commuting derivatives.”

For each retained h , choose a degree-at-most- $(s - 2)$ nilsequence φ_h so that

$$\left| \mathbb{E}_n \Delta_h f(n) \overline{\chi(h, n) \varphi_h(n)} \right| \geq \rho$$

with $\rho > 0$ controlled. Apply Lemma G.2 and pigeonhole a base increment h_0 . Using the cocycle identity

$$\Delta_{h_0+a+b} f(n) \Delta_{h_0} f(n+a) = \Delta_{h_0+a} f(n) \Delta_{h_0+b} f(n+a)$$

inside the Cauchy–Schwarz expansion, we obtain for many pairs (a, b) a bias relation for

$$\begin{aligned} \tau(a, b, n) &:= \chi(h_0 + a, n) \chi(h_0 + b, n + a) \\ &\quad \times \overline{\chi(h_0 + a + b, n)} \overline{\chi(h_0, n + a)}. \end{aligned} \quad (\text{H.2})$$

All accompanying φ -terms have n -degree at most $s - 2$.

We now compute the top part of τ . Insert (H.1) and use multilinearity of $\tilde{\chi}$. Terms in which a or b occurs fewer than once cancel in the four-term alternating sum; terms in which

the n -variable occurs at most $s - 3$ times are lower order for the present calculation. Modulo symbols of n -degree at most $s - 3$,

$$[\tau(a, b, n)]_s = s(s - 1)[\tilde{\chi}(b, a, n, \dots, n)]_s. \quad (\text{H.3})$$

This is simply the coefficient of abn^{s-2} in the discrete mixed second derivative of the polarized form.

The lower-degree factors in the bias for (H.2) still depend on a and b . We now eliminate them with the full rectangular Cauchy–Schwarz calculation.

First note a simple shift fact. If ϕ is a degree-at-most- $(s - 2)$ nilcharacter, then the shifted sequence $\phi(n + a)$ factors, at top degree, as a fixed n -nilcharacter times a factor of n -degree at most $s - 3$. Indeed, write the polynomial orbit in Taylor coordinates; replacing n by $n + a$ leaves the top n^{s-2} Taylor coefficient unchanged, so the quotient of the two top vertical modes has n -degree at most $s - 3$. For a general bounded-complexity nilsequence, first apply vertical Fejér approximation with uniform error $\varepsilon \ll \rho$; this gives a finite controlled sum of such nilcharacter modes. The total contribution of the uniform error is $O(\varepsilon)$ and can be absorbed into the original correlation. Pigeonholing one retained Fourier mode in each shifted φ -factor therefore gives a controlled constant $\rho_1 > 0$ and lower-degree nilsequences $\psi_{a,b}$ such that

$$|\mathbb{E}_{a,b,n} \tau(a, b, n) A_a(n) B_b(n) C_{a+b}(n) D(n) \psi_{a,b}(n)| \geq \rho_1, \quad (\text{H.4})$$

where A, B, C, D have n -degree at most $s - 2$ and every $\psi_{a,b}$ has n -degree at most $s - 3$.

Regard the average over b in (H.4) as the inner average and apply Cauchy–Schwarz in (a, n) . The factors $A_a D$, which are independent of b , disappear. After expanding the square we obtain

$$\begin{aligned} & \left| \mathbb{E}_{a,b,b',n} \tau(a, b, n) \overline{\tau(a, b', n)} B_b(n) \overline{B_{b'}(n)} \right. \\ & \quad \left. \times C_{a+b}(n) \overline{C_{a+b'}(n)} \psi_{a,b,b'}(n) \right| \geq \rho_1^2, \end{aligned} \quad (\text{H.5})$$

for a controlled degree-at-most- $(s - 3)$ factor $\psi_{a,b,b'}$. Set

$$c = a + b + b'.$$

Changing variables in (H.5) and pigeonholing c gives a value c_0 such that, for

$$\alpha_0(b, b', n) := \tau(c_0 - b - b', b, n) \overline{\tau(c_0 - b - b', b', n)}, \quad (\text{H.6})$$

we have

$$\left| \mathbb{E}_{b,b',n} \alpha_0(b, b', n) U_b(n) V_{b'}(n) \psi'_{b,b'}(n) \right| \gg_\rho 1, \quad (\text{H.7})$$

where $U_b, V_{b'}$ have n -degree at most $s - 2$ and $\psi'_{b,b'}$ has n -degree at most $s - 3$.

We record the top-symbol calculation because it is the algebraic heart of the symmetry step. Using (H.3), additivity in the first two polarized slots, and the fact that translating either parameter contributes only terms independent of the other parameter, one obtains, modulo symbols of n -degree at most $s - 3$,

$$[\alpha_0(b, b', n)]_s \equiv -s(s - 1)[\tilde{\chi}(b + b', b - b', n, \dots, n)]_s + L_b + M_{b'}, \quad (\text{H.8})$$

where L_b is independent of b' and $M_{b'}$ is independent of b . The harmless one-parameter terms are written explicitly because the next rectangular difference is designed to remove them.

Apply Cauchy–Schwarz to (H.7) first in (b', n) and then in (b, n) . The U - and V -factors disappear. Expanding both squares and pigeonholing one pair of base parameters b_0, b'_0 leaves a positive bias, against an n -degree-at-most- $(s - 3)$ nilsequence, for the rectangular product

$$\alpha_1(b, b', n) := \alpha_0(b, b', n) \overline{\alpha_0(b_0, b', n) \alpha_0(b, b'_0, n) \alpha_0(b_0, b'_0, n)} \quad (\text{H.9})$$

for a positive density of pairs (b, b') . Taking the same rectangular difference in (H.8) kills L_b and $M_{b'}$. Bilinearity in the first two polarized slots gives

$$\begin{aligned} [\alpha_1(b, b', n)]_s &\equiv s(s-1) ([\tilde{\chi}(b - b_0, b' - b'_0, n, \dots, n)]_s \\ &\quad - [\tilde{\chi}(b' - b'_0, b - b_0, n, \dots, n)]_s) \end{aligned} \quad (\text{H.10})$$

modulo symbols of n -degree at most $s - 3$. This is the desired antisymmetric part; it has survived precisely because the one-parameter terms have been removed by the rectangle.

Since $\alpha_1(b, b', \cdot)$ is $(s - 3)$ -biased for many (b, b') , Proposition F.14 gives

$$\|\alpha_1(b, b', \cdot)\|_{U^{s-2}[N]} \gg_{s, \rho} 1$$

for a positive density of those pairs. Average the 2^{s-2} -th powers and expand the Gowers norm. With independent increments $t_1, \dots, t_{s-2}$ this produces a multivariable nilcharacter

$$\Sigma(b, b', \mathbf{t}, n) = \prod_{\omega \in \{0,1\}^{s-2}} \mathcal{C}^{|\omega|} \alpha_1(b, b', n + \omega \cdot \mathbf{t}) \quad (\text{H.11})$$

whose average on a fixed product polytope is bounded below by a controlled positive constant. The lower- n -degree error in (H.10) is annihilated by these $s - 2$ derivatives. Repeated polarization therefore gives

$$\begin{aligned} [\Sigma]_s &= s! ([\tilde{\chi}(b - b_0, b' - b'_0, t_1, \dots, t_{s-2})]_s \\ &\quad - [\tilde{\chi}(b' - b'_0, b - b_0, t_1, \dots, t_{s-2})]_s). \end{aligned} \quad (\text{H.12})$$

The average of Σ is a bias against the constant sequence, which has strictly smaller multidegree. Lemma F.12 forces the symbol in (H.12) to vanish on a rational subbox. Additivity of the polarized symbol and the bracket-linear extension from Theorem G.17 extend the identity from that positive-density structured set of increments to the whole generated increment group. Finally, Lemma F.18 lets us cancel the nonzero integer $s!$. After renaming $b - b_0$ and $b' - b'_0$ as h and k , we obtain

$$[\tilde{\chi}(h, k, t_1, \dots, t_{s-2})]_s = [\tilde{\chi}(k, h, t_1, \dots, t_{s-2})]_s.$$

The last $s - 1$ slots were already symmetric by polarization, so exchanging the first slot with the second and then permuting the remaining slots generates the full symmetric group. Hence the complete s -symbol is symmetric. $\square$

Worked Example H.7 (The quadratic shadow). When $s = 2$, the top symbol is a bilinear form $B(h, n)$. The long elimination above collapses to

$$B(h, k) - B(k, h),$$

and Chapter B shows directly, through two Cauchy–Schwarz steps and a Bohr-set annihilator argument, that this antisymmetric part is negligible on a smaller Bohr set. Thus the higher-order calculation is the nilcharacter version of Hessian symmetry.

H.4 Diagonal integration

We now prove the nilpotent counterpart of integrating a symmetric multilinear form.

Lemma H.8 (Taking a finite vertical root). *For every $s, m, M \geq 1$ there is $M' = M'(s, m, M)$ with the following property. Let Θ_0 be a degree- s nilcharacter represented on a filtered nilmanifold of complexity at most M . Then there is a finite vertical cover of degree at most m^{d_s} , where d_s is the dimension of the top vertical torus, whose complexity is at most M' , and on that cover there is a degree- s nilcharacter Θ with*

$$[\Theta^m]_s = [\Theta_0]_s.$$

Proof. The top vertical torus is $T_s \cong \mathbb{R}^d / \mathbb{Z}^d$ and the top frequency of Θ_0 is some $k \in \mathbb{Z}^d$. Replace the lattice in the top central group by the sublattice $m\mathbb{Z}^d$. The resulting quotient is an m^d -fold vertical cover. On the cover, the character k/m is integral relative to the new lattice, so it defines a vertical character whose m -th power has frequency k . Pulling back the remaining data gives the required Θ . $\square$

Theorem H.9 (Symmetric multidegree integration). *Let $\chi(h, n)$ be a multidegree $(1, s-1)$ nilcharacter whose polarized top symbol is symmetric in all s variables. For every $\varepsilon > 0$ there exist a degree- s nilcharacter $\Theta(n)$, a bounded finite linear combination of multivariable nilsequences $\Psi(h, n)$, and an error $E(h, n)$ with $\|E\|_\infty \leq \varepsilon$ such that*

$$\chi(h, n) = \Theta(n+h) \overline{\Theta(n)} \Psi(h, n) + E(h, n) \quad (\text{mod } n\text{-degree at most } s-2),$$

and every component of Ψ either has total degree at most $s-1$ or has n -degree at most $s-2$. All complexity bounds depend only on the original complexity and on ε .

Proof. Polarize the top symbol to a symmetric multidegree $(1, \dots, 1)$ nilcharacter

$$\tilde{\chi}(n_1, \dots, n_s)$$

with symbol $\sigma(n_1, \dots, n_s)$. This is constructed on a free filtered group exactly as in Lemma H.2; polarization replaces repeated n -slots by independent variables.

Restrict to the diagonal and set

$$\Theta_0(n) = \tilde{\chi}(n, \dots, n).$$

Expand the top symbol of its derivative. Multilinearity gives

$$\sigma(n+h, \dots, n+h) - \sigma(n, \dots, n) = s \sigma(h, n, \dots, n) + \sum_{j=2}^s \binom{s}{j} \sigma(\underbrace{h, \dots, h}_j, \underbrace{n, \dots, n}_{s-j}).$$

The first term has multidegree $(1, s-1)$. Every remaining term has h -degree at least 2 and n -degree at most $s-2$; these are exactly the permissible error multidegrees.

By Lemma H.8, after a finite vertical cover we may choose Θ whose top symbol is $1/s$ of that of Θ_0 . Therefore

$$\Theta(n+h) \overline{\Theta(n)}$$

has the same $(1, s-1)$ top symbol as $\chi(h, n)$. Their quotient consequently has no $(1, s-1)$ component. The quotient is a smooth nilsequence on a compact product nilmanifold. Apply the vertical Fejér decomposition of Lemma F.2 on the remaining vertical tori and truncate at frequency R . It gives a finite sum Ψ_R with

$$\|\Psi_R - \Psi_{\text{full}}\|_\infty \leq \varepsilon,$$

where $R = R(\varepsilon)$ and the Lipschitz complexity remains controlled. Every retained vertical mode has either total degree at most $s - 1$ or n -degree at most $s - 2$, because the $(1, s - 1)$ top mode has already been removed. Choose ε smaller than the correlation threshold in the later closing argument and absorb the uniform error there. Renaming the finite approximation Ψ proves the asserted decomposition (with the harmless controlled error understood, as throughout the symbol calculus). $\square$

H.5 The final Gowers closing argument

We isolate the analytic inequality used to close the induction.

Lemma H.10 (Cauchy–Schwarz–Gowers closing inequality). *Let G be a finite abelian group, let $u, v : G \rightarrow \mathbb{C}$ be 1-bounded, and let $r \geq 1$. Then*

$$\mathbb{E}_{h \in G} \|T_h u v\|_{U^r(G)}^{2^r} \leq \|u\|_{U^{r+1}(G)}^{2^r} \|v\|_{U^{r+1}(G)}^{2^r}. \quad (\text{H.13})$$

Consequently, if a set $H \subseteq G$ has density at least η and

$$\|T_h u v\|_{U^r(G)} \geq \rho \quad (h \in H),$$

then

$$\|u\|_{U^{r+1}(G)} \geq \eta^{1/2^r} \rho.$$

The same conclusion, with constants depending only on r and the density of the admissible shifts, holds for the normalized interval norms $U^r[N]$ after the zero-extension embedding of Volume III.

Proof. Expand the left side of (H.13). With $\omega \in \{0, 1\}^r$ and $\mathcal{C}$ denoting complex conjugation,

$$\mathbb{E}_{h, x, t_1, \dots, t_r} \prod_{\omega} \mathcal{C}^{|\omega|} u(x + h + \omega \cdot t) \prod_{\omega} \mathcal{C}^{|\omega|} v(x + \omega \cdot t).$$

This is an $(r + 1)$ -dimensional Gowers multilinear form: u occupies one h -face and v the opposite h -face. Apply Cauchy–Schwarz successively in $t_1, \dots, t_r$ (equivalently, apply the Gowers–Cauchy–Schwarz inequality proved in Volume III). The first face contributes $\|u\|_{U^{r+1}}^{2^r}$ and the opposite face contributes $\|v\|_{U^{r+1}}^{2^r}$, proving (H.13).

If the displayed lower bound holds on H , the left side of (H.13) is at least $\eta \rho^{2^r}$. Since v is 1-bounded, $\|v\|_{U^{r+1}} \leq 1$, and the stated conclusion follows.

For $[N]$, extend u, v by zero to a cyclic group of order larger than $2^{r+2}N$. No relevant cube wraps around. Dividing by the fixed normalizing factors defining $U^j[N]$ changes the preceding inequality only by constants depending on r ; a positive-density set of admissible integer shifts remains positive-density in the auxiliary group. This gives the interval form. $\square$

Theorem H.11 (Closing the inverse-theorem induction). *Fix $s \geq 2$, $\alpha, \rho \in (0, 1]$, and $M \geq 1$, and assume $\text{GI}(s - 1)$. There are $c = c(s, \alpha, \rho, M) > 0$, $C = C(s, \alpha, \rho, M) \geq 1$, and $N_0 = N_0(s, \alpha, \rho, M)$ with the following property. Let $N \geq N_0$, let $|f| \leq 1$, and suppose there is a set $H \subseteq [-N, N]$ with $|H| \geq \alpha(2N + 1)$ such that, for every $h \in H$, $\Delta_h f$ correlates modulo n -degree at most $s - 2$ at level (ρ, M) with*

$$\Theta(n + h) \overline{\Theta(n)} \Psi(h, n),$$

where Θ is a degree- s nilcharacter of complexity at most M and Ψ has complexity at most M and only the error multidegrees listed in Theorem H.9. Then there is a degree-at-most- s nilsequence Φ of complexity at most C such that

$$\left| \mathbb{E}_{n \leq N} f(n) \overline{\Phi(n)} \right| \geq c.$$

Proof. Apply Corollary F.17 to the error factor Ψ and then its finite vertical Fourier expansion. After pigeonholing one summand that retains a controlled proportion of the original bias, we may write the selected error contribution as

$$\psi(n)\psi'(h, n),$$

where ψ is independent of h and has degree at most $s - 1$, while ψ' has n -degree at most $s - 2$. The factor ψ' can therefore be absorbed into the degree- $(s - 2)$ test sequence that witnesses the bias.

Set

$$\theta(n) := \Theta(n)\psi(n).$$

This is a fixed degree-at-most- s nilsequence, independent of h . After one further finite vertical Fourier decomposition of the fixed n -factor created by complex conjugation, and pigeonholing, there is a second fixed degree-at-most- s nilsequence $\theta'(n)$ such that, for a positive-density set of h ,

$$\Delta_h f(n) \text{ is } (s - 2)\text{-biased by } \theta(n + h)\theta'(n). \quad (\text{H.14})$$

All complexity and density losses depend only on the induction parameters.

Define fixed 1-bounded sequences (after dividing the bounded nilsequences by their controlled sup norms)

$$u(n) = f(n)\overline{\theta(n)}, \quad v(n) = \overline{f(n)}\overline{\theta'(n)}.$$

Then (H.14) says that for every good h the product

$$(T_h u)(n)v(n)$$

correlates with a degree-at-most- $(s - 2)$ nilsequence by a controlled amount. Proposition F.14 therefore yields

$$\|T_h u v\|_{U^{s-1}[N]} \gg 1$$

for a positive-density set of admissible shifts h . Lemma H.10, with $r = s - 1$, now gives

$$\|u\|_{U^s[N]} = \|f\bar{\theta}\|_{U^s[N]} \gg 1.$$

Apply the inductive inverse theorem GI($s - 1$) to u . We obtain a degree-at-most- $(s - 1)$ nilsequence φ such that

$$\left| \mathbb{E}_{n \leq N} f(n)\overline{\theta(n)} \overline{\varphi(n)} \right| \gg 1.$$

Equivalently f correlates with $\theta\varphi$. By Lemma 5.6, this product is a degree-at-most- s nilsequence of controlled complexity. This closes the induction. $\square$

Appendix I

Rational Nilmanifolds and Mal'cev Coordinates

I.1 The objects and the convention

Throughout this volume G is a connected simply connected nilpotent Lie group, $\Gamma \leq G$ is a discrete cocompact subgroup, and G/Γ is written with right cosets. The exponential map is a global diffeomorphism because the Lie algebra is nilpotent.

A filtration is a finite descending chain

$$G = G_0 = G_1 \supseteq G_2 \supseteq \cdots \supseteq G_{s+1} = \{e\}, \quad [G_i, G_j] \subseteq G_{i+j}.$$

A subgroup $H \leq G$ is Γ -rational when $H \cap \Gamma$ is cocompact in H .

I.2 Strong Mal'cev bases

Definition I.1 (Mal'cev basis adapted to a filtration). A basis $\mathcal{X} = (X_1, \dots, X_m)$ of $\mathfrak{g}$ is a strong Mal'cev basis for G/Γ if the tails

$$\mathfrak{g}_j = \text{span}(X_j, \dots, X_m)$$

are ideals, $G_j^* = \exp(\mathfrak{g}_j)$ are Γ -rational, and every $g \in G$ has a unique representation

$$g = \exp(t_1 X_1) \cdots \exp(t_m X_m),$$

with $g \in \Gamma$ precisely when the successive reduction algorithm produces integer coordinates. It is adapted to $G_\bullet$ when every $\mathfrak{g}_i = \text{Lie}(G_i)$ is spanned by a terminal block of the basis.

Theorem I.2 (Mal'cev coordinate theorem). *Let G/Γ be a nilmanifold and let $G_\bullet$ be a Γ -rational filtration. There is a strong Mal'cev basis adapted to $G_\bullet$. In coordinates of the second kind:*

1. multiplication and inversion are triangular polynomial maps with rational coefficients;
2. a fundamental domain is obtained by successive reduction of the coordinates to $[0, 1)$;
3. rational subgroups admit adapted rational Mal'cev bases; and
4. after fixing dimension, step, and a height bound for the structure constants, all denominators and polynomial coefficients arising from these operations are bounded by a function of those parameters alone.

Proof. The complete construction is given in Appendix L, Theorems L.7 and L.12. The proof is by induction along a rational central series. The terminating Baker–Campbell–Hausdorff formula supplies the polynomial coordinate laws; rational Gaussian elimination and successive lattice reduction supply the denominator bounds. $\square$

I.3 The metric and Lipschitz control

Write $\psi_{\mathcal{X}}(g) = (t_1, \dots, t_m)$ for second-kind coordinates. On a fixed fundamental box define a coordinate metric by taking the infimum of coordinate lengths over representatives of two cosets. Compactness makes this bi-Lipschitz equivalent to any fixed Riemannian metric, with constants controlled by the rationality data.

Lemma I.3 (Controlled translations). *Fix a bounded-complexity Mal'cev system. If a has coordinates bounded by M , then left and right translation by a are $M^{O(1)}$ -Lipschitz on the fundamental domain, away from its finitely many coordinate faces; after quotient reduction the same bound holds globally for the quotient metric.*

Proof. In Mal'cev coordinates, multiplication is triangular polynomial with coefficients of controlled height by [theorem I.2](#). On a coordinate box with $|\psi(a)| \leq M$, every first derivative of that polynomial is $M^{O(1)}$. The mean-value theorem gives the local bound. Crossing a face merely changes representatives by a lattice element; the quotient metric is defined by an infimum over such representatives, so the same estimate survives reduction. $\square$

Worked Example I.4 (Heisenberg reduction). For

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy'),$$

the standard basis is Mal'cev for the integer lattice. Reducing y by $\lfloor y \rfloor$ changes the central coordinate by $-x\lfloor y \rfloor$. This is the exact source of the bracket term $x\{y\}$ in elementary coordinates; no discontinuity exists on the nilmanifold itself.

Appendix J

Polynomial Orbits, Horizontal Characters, and Equidistribution

J.1 Polynomial sequences

For $g : \mathbb{Z} \rightarrow G$ define

$$\partial_h g(n) = g(n+h)g(n)^{-1}.$$

We write $g \in \text{poly}(\mathbb{Z}, G_\bullet)$ when every j -fold discrete derivative takes values in G_j .

Proposition J.1 (Taylor expansion in a filtered group). *For a polynomial sequence g of degree at most s there are uniquely determined Taylor coefficients $g_j \in G_j$ modulo the natural deeper-filtration ambiguity such that*

$$g(n) = g_0 g_1^{\binom{n}{1}} \cdots g_s^{\binom{n}{s}}$$

after ordering the factors from low to high filtration degree. Conversely every such expression is polynomial.

Proof. Induct on s . The s -fold derivative is constant modulo $G_{s+1} = \{e\}$; call its value g_s . The sequence $g_s^{-\binom{n}{s}} g(n)$ has vanishing s -fold derivative because $\Delta^s \binom{n}{s} = 1$, with commutator errors landing in filtration degree $> s$ and therefore vanishing. Apply the induction hypothesis. The converse follows from the identity $\Delta_h \binom{n}{j}$ being a polynomial of degree $j-1$ together with $[G_i, G_j] \subseteq G_{i+j}$. $\square$

J.2 Horizontal characters

A horizontal character is a continuous homomorphism $\eta : G \rightarrow \mathbb{R}/\mathbb{Z}$ annihilating Γ . It factors through

$$G/[G, G]\Gamma,$$

a torus. In a rational Mal'cev basis it has the form

$$\eta(g) = k \cdot \pi_{\text{hor}} \psi_{\mathcal{X}}(g) \pmod{1}, \quad k \in \mathbb{Z}^r.$$

Its complexity is $|k|$.

For a polynomial phase

$$P(n) = \alpha_0 + \alpha_1 \binom{n}{1} + \cdots + \alpha_s \binom{n}{s} \pmod{1}$$

define

$$\|P\|_{C^\infty[N]} = \max_{1 \leq j \leq s} N^j \|\alpha_j\|_{\mathbb{R}/\mathbb{Z}}.$$

J.3 Quantitative obstruction theorem

Theorem J.2 (Quantitative Leibman theorem). *Fix dimension m , filtration degree s , and rationality bound M_0 . There is $C = C(m, s)$ such that the following holds. Let G/Γ have an M_0 -rational Mal'cev basis adapted to $G_\bullet$, let $g \in \text{poly}(\mathbb{Z}, G_\bullet)$, and suppose the orbit $(g(n)\Gamma)_{1 \leq n \leq N}$ is not δ -equidistributed. Then there is a nontrivial horizontal character η with*

$$0 < |\eta| \leq M_0^C \delta^{-C}, \quad \|\eta \circ g\|_{C^\infty[N]} \leq M_0^C \delta^{-C}.$$

Conversely, there is $C' = C'(m, s)$ such that for every $R \geq 2$, if a nonzero horizontal character satisfies

$$|\eta| \leq R, \quad \|\eta \circ g\|_{C^\infty[N]} \leq R,$$

then the orbit fails total δ_R -equidistribution for

$$\delta_R = M_0^{-C'} R^{-C'}.$$

(Enlarging C' absorbs harmless absolute constants.)

Proof. Appendix M, Sections M.2–M.3, proves the theorem by induction on dimension. The abelian base case is the multidimensional Weyl criterion plus the inverse Weyl estimate of Volume V. In the induction step, vertical Fourier decomposition and van der Corput convert a failure of equidistribution into many structured derivatives; commutator quotients lower dimension, and the Mal'cev rationality theorem converts the resulting resonance into a horizontal character with controlled integer coefficients. All parameter losses are finite compositions of polynomial bounds. $\square$

Corollary J.3 (Total equidistribution criterion). *Under the hypotheses of theorem J.2, there is $C_* = C_*(m, s)$ such that the following implication holds. If every nontrivial horizontal character η with*

$$|\eta| \leq M_0^{C_*} \delta^{-C_*}$$

satisfies

$$\|\eta \circ g\|_{C^\infty[N]} > M_0^{C_*} \delta^{-C_*},$$

then $(g(n)\Gamma)_{n \leq N}$ is totally δ -equidistributed on every arithmetic progression of length at least δN .

Proof. Suppose instead that a progression $P = \{a + qr : 1 \leq r \leq L\} \subset [N]$, $L \geq \delta N$, violates δ -equidistribution. Then $q \leq \delta^{-1}$. Apply theorem J.2 to the polynomial sequence $g_P(r) = g(a + qr)$ on $[L]$. We obtain a nonzero horizontal character η with

$$|\eta|, \|\eta \circ g_P\|_{C^\infty[L]} \leq M_0^C \delta^{-C}.$$

The triangular binomial change of variables for $n = a + qr$ and the extrapolation lemma in Appendix M give an integer $1 \leq r_0 \leq \delta^{-O_s(1)}$ such that

$$|r_0 \eta|, \|r_0 \eta \circ g\|_{C^\infty[N]} \leq M_0^{O_{m,s}(1)} \delta^{-O_{m,s}(1)}.$$

Choosing C_* larger than all exponents in these bounds contradicts the hypothesis, since $r_0 \eta$ is still a nontrivial horizontal character. Hence no such progression P exists. $\square$

Appendix K

Smooth–Equidistributed–Rational Factorization

K.1 Three kinds of motion

A polynomial sequence ε is (M, N) -smooth if

$$d(\varepsilon(n), e) \leq M, \quad d(\varepsilon(n+1), \varepsilon(n)) \leq M/N.$$

A polynomial sequence γ is M -rational if a bounded power of every Taylor coefficient lies in Γ ; then $\gamma(n)\Gamma$ is periodic with period $M^{O(1)}$.

Lemma K.1 (Rational polynomial sequences are periodic). *Fix degree and Mal’cev complexity. If every Taylor coefficient of γ has denominator dividing Q , then $\gamma(n)\Gamma$ is periodic with period dividing $Q^{O(1)}$.*

Proof. In second-kind Mal’cev coordinates, each coordinate of $\gamma(n)$ is a rational polynomial in the binomial coefficients $\binom{n}{j}$ whose denominators divide $Q^{O(1)}$ by [theorem I.2](#). Choosing q divisible by those denominators and by $1, \dots, s!$ makes every coordinate difference $\psi(\gamma(n+q)\gamma(n)^{-1})$ integral after successive triangular reduction. Hence the quotient points agree. $\square$

K.2 The factorization theorem

Theorem K.2 (Quantitative factorization). *Fix bounded nilmanifold complexity, degree s , an initial parameter $M_0 \geq 2$, and $A > 0$. There is M with*

$$M_0 \leq M \leq M_0^{O_{m,s,A}(1)}$$

and a factorization

$$g(n) = \varepsilon(n)g'(n)\gamma(n)$$

such that

1. ε is (M, N) -smooth;
2. g' takes values in a connected M -rational subgroup $G' \leq G$ and is totally M^{-A} -equidistributed in $G'/(G' \cap \Gamma)$;
3. γ is M -rational and $\gamma(n)\Gamma$ is periodic with period at most $M^{O(1)}$.

Proof. The detailed descent is in [Appendix M](#), [Section M.5](#). Briefly, if g is not totally M^{-A} -equidistributed, [theorem J.2](#) produces a bounded horizontal character η whose polynomial

coefficients are almost rational. Dirichlet approximation splits those coefficients into rational and N^{-j} -small pieces. Lift the small pieces to a smooth polynomial ε_1 , the rational pieces to a periodic polynomial γ_1 , and leave a residual polynomial g_1 in the connected kernel of η . The kernel has smaller dimension and is rational with controlled complexity. Iterate. Dimension decreases at every failed-equidistribution step, so after at most m steps the process stops. Multiplying the smooth and rational factors preserves their classes by the filtered polynomial calculus proved in Appendix L. $\square$

Idea Why termination is quantitative

No compactness argument hides in the proof. Every failure produces an explicit integer horizontal character and therefore a proper rational kernel. Dimension drops, while rationality heights grow only by bounded algebraic operations. There are at most m drops.

Appendix L

Mal'cev Theory with Rationality Bounds

This appendix proves the rational nilmanifold facts used in Chapters 1–4. The point is not merely existence of coordinates. The later quantitative arguments need three facts simultaneously: the coordinates must respect a filtration, the lattice must be visible in those coordinates, and every change of coordinates must have controlled rational height.

L.1 The exponential map and rational Lie algebras

Lemma L.1 (Formal BCH theorem and the nilpotent truncation). *Let $\hat{\mathcal{A}} = \mathbb{Q}\langle\langle X, Y \rangle\rangle$ be the completion of the free associative algebra by word length, and put*

$$Z(t) := \log(e^{tX} e^{tY}) = \sum_{n \geq 1} t^n Z_n.$$

Then every Z_n is a homogeneous Lie polynomial of bracket weight n in X, Y . Consequently, if $\mathfrak{g}$ is an s -step nilpotent real Lie algebra, evaluation of this book in $\mathfrak{g}$ gives the finite polynomial

$$\text{BCH}_{\mathfrak{g}}(X, Y) = \sum_{n=1}^s Z_n(X, Y),$$

*and the operation $X * Y = \text{BCH}_{\mathfrak{g}}(X, Y)$ is associative with identity 0 and inverse $-X$.*

Moreover, for every finite-dimensional real Lie group G with Lie algebra $\mathfrak{g}$ there is an open neighbourhood U of 0 such that $\exp_G$ restricts to a diffeomorphism from U onto $\exp_G(U)$ and, whenever $X, Y \in U$ and $\exp_G X \exp_G Y \in \exp_G(U)$,

$$\log_G(\exp_G X \exp_G Y) = \text{BCH}_{\mathfrak{g}}(X, Y). \quad (\text{A.1})$$

Proof. All exponential and logarithm series below are formal series in the word-length completion, so rearranging a fixed homogeneous degree involves only finitely many terms. For a differentiable formal series $W(t)$ with zero constant term, direct differentiation of $e^W = \sum_{m \geq 0} W^m/m!$ and multiplication on the right by e^{-W} gives

$$\frac{d}{dt} e^{W(t)} e^{-W(t)} = \sum_{r \geq 0} \frac{1}{(r+1)!} \text{ad}_{W(t)}^r W'(t) = \frac{e^{\text{ad}_{W(t)}} - 1}{\text{ad}_{W(t)}} W'(t). \quad (\text{A.2})$$

The quotient is notation for its power series; its constant coefficient is 1, so it is formally invertible. If $E(t) = e^{tX} e^{tY}$, then

$$E'(t)E(t)^{-1} = X + e^{tX} Y e^{-tX} = X + e^{t \text{ad}_X} Y.$$

Applying (A.2) with $W = Z = \log E$ yields

$$Z'(t) = \frac{\text{ad}_{Z(t)}}{e^{\text{ad}_{Z(t)}} - 1} (X + e^{t \text{ad}_X} Y). \quad (\text{A.3})$$

Define the rational numbers b_r by the formal identity

$$\frac{z}{e^z - 1} = \sum_{r \geq 0} b_r z^r.$$

The right side of (A.3) is therefore a sum of iterated commutators of $Z(t)$ with the Lie series $X + e^{t \text{ad}_X} Y$. Comparing the coefficient of t^{n-1} gives nZ_n as a rational linear combination of iterated brackets involving $X, Y, Z_1, \dots, Z_{n-1}$. Since $Z_1 = X + Y$, induction shows that every Z_n lies in the Lie subalgebra generated by X, Y . Word length in the associative algebra is preserved by commutators and by (A.3), so Z_n is homogeneous of bracket weight n . The first terms are

$$Z_1 = X + Y, \quad Z_2 = \frac{1}{2}[X, Y], \quad Z_3 = \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[Y, [Y, X]].$$

This proves the formal BCH assertion without assuming in advance that the logarithm is a Lie series.

If $\mathfrak{g}$ is s -step nilpotent, every Lie monomial of weight $> s$ vanishes, so evaluation factors through the finite sum displayed in the statement. Formal exponential and logarithm are inverse under composition in the completion, whence

$$\begin{aligned} \exp(\text{BCH}(\text{BCH}(X, Y), W)) &= e^X e^Y e^W \\ &= \exp(\text{BCH}(X, \text{BCH}(Y, W))). \end{aligned}$$

Applying the formal logarithm and then passing to the step- s nilpotent quotient proves associativity of $*$. The identities $\text{BCH}(0, X) = X$ and $\text{BCH}(X, -X) = 0$ give the identity and inverse.

It remains to justify the comparison with an actual Lie group, since this is what is used in the integration lemma below. We first derive the differential of the exponential map rather than cite it. For fixed $A, V \in \mathfrak{g}$, let $g_u(t) = \exp_G(t(A + uV))$ and right-trivialize the variation by

$$W(t) = \left. \frac{\partial g_u(t)}{\partial u} \right|_{u=0} g_0(t)^{-1}.$$

Since $g'_u(t)g_u(t)^{-1} = A + uV$, differentiation in u gives the linear equation

$$W'(t) = V + [A, W(t)], \quad W(0) = 0.$$

Solving it in the finite-dimensional vector space $\mathfrak{g}$ gives

$$W(1) = \int_0^1 e^{v \text{ad}_A} V dv = \frac{e^{\text{ad}_A} - 1}{\text{ad}_A} V. \quad (\text{A.4})$$

Thus (A.2) is also the right-trivialized differential of the ordinary exponential map.

For small t define $z(t) = \log_G(\exp_G(tX) \exp_G(tY))$. The product curve has right logarithmic derivative $X + e^{t \text{ad}_X} Y$; applying (A.4) to $e^{z(t)}$ shows that z satisfies exactly (A.3) with $z(0) = 0$. Comparing Taylor coefficients recursively therefore gives $z(t) = \sum_{n \geq 1} t^n Z_n(X, Y)$ in a neighbourhood of 0. Setting $t = 1$ whenever X, Y lie in a sufficiently small exponential neighbourhood proves (A.1). $\square$

Worked Example L.2 (Why nilpotence turns BCH into a polynomial). In the Heisenberg Lie algebra take $[X, Y] = Z$ with Z central. Every bracket of weight three vanishes, so Lemma L.1 gives exactly

$$\text{BCH}(xX + yY + zZ, x'X + y'Y + z'Z) = (x + x')X + (y + y')Y + \left(z + z' + \frac{1}{2}(xy' - yx')\right)Z.$$

Thus the nonlinear part of multiplication is not mysterious: it is precisely the weight-two commutator term, and there can be no higher correction because the next lower-central layer is zero.

Lemma L.3 (Integration of a Lie-algebra homomorphism). *Let G_1 be a connected simply connected Lie group and let G_2 be a Lie group. Every Lie-algebra homomorphism $T : \mathfrak{g}_1 \rightarrow \mathfrak{g}_2$ integrates to a unique smooth group homomorphism $\Phi : G_1 \rightarrow G_2$ with $d\Phi_e = T$.*

Proof. Choose exponential neighbourhoods U_i of the identity on which $\exp_i$ is a diffeomorphism. Shrinking U_1 if necessary, define locally

$$\Phi_0(\exp_1 X) = \exp_2(TX).$$

The local BCH identity (A.1) and the fact that T preserves brackets show that $T(\text{BCH}(X, Y)) = \text{BCH}(TX, TY)$. Hence $\Phi_0(xy) = \Phi_0(x)\Phi_0(y)$ whenever x, y, xy lie in that neighbourhood.

For a piecewise smooth path $c : [0, 1] \rightarrow G_1$ beginning at the identity, subdivide so finely that every increment $c(t_j)^{-1}c(t_{j+1})$ lies in the local neighbourhood and multiply their Φ_0 -images. Local multiplicativity shows that the result is unchanged by refinement. It is also unchanged under a homotopy of c with fixed endpoints: subdivide a homotopy square into rectangles small enough that all four edge increments lie in the local neighbourhood; local multiplicativity makes the product around each small rectangle trivial, and the interior edges cancel. Since G_1 is simply connected, the product therefore depends only on $c(1)$. This defines $\Phi : G_1 \rightarrow G_2$. Concatenating paths proves that Φ is a homomorphism; locally it equals Φ_0 , hence it is smooth and has differential T . Uniqueness follows because two such homomorphisms agree on an identity neighbourhood and a connected group is generated by any identity neighbourhood. $\square$

Lemma L.4 (Exponential coordinates for nilpotent groups). *Let G be connected and simply connected with nilpotent Lie algebra $\mathfrak{g}$. Then $\exp : \mathfrak{g} \rightarrow G$ is a global diffeomorphism. Moreover the group law transported to $\mathfrak{g}$ is the finite Baker–Campbell–Hausdorff polynomial*

$$X * Y = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]] + \cdots,$$

where every displayed and omitted term is an iterated commutator and all terms of bracket length exceeding the nilpotency step vanish.

Proof. On the vector space $\mathfrak{g}$ define $X * Y$ by the finite BCH polynomial of Lemma L.1. That lemma proves directly that $*$ is associative, with identity 0 and inverse $-X$. Thus $G_{\text{BCH}} = (\mathfrak{g}, *)$ is a Lie group. Its underlying manifold is the vector space $\mathbb{R}^{\dim \mathfrak{g}}$, so it is connected and simply connected, and differentiating the BCH law at the origin shows that its Lie algebra is precisely $\mathfrak{g}$ with the given bracket.

Apply Lemma L.3 to the identity Lie-algebra map $\mathfrak{g} \rightarrow \text{Lie}(G)$. It gives a homomorphism $\Phi : G_{\text{BCH}} \rightarrow G$ with differential the identity. Applying the same lemma in the opposite direction gives $\Psi : G \rightarrow G_{\text{BCH}}$. Both $\Psi\Phi$ and the identity on G_{BCH} integrate the identity map of $\mathfrak{g}$, so uniqueness gives $\Psi\Phi = \text{id}$; similarly $\Phi\Psi = \text{id}$. Hence Φ is an isomorphism. The one-parameter subgroup $t \mapsto tX$ in G_{BCH} is carried to the one-parameter subgroup $t \mapsto \exp_G(tX)$, so $\Phi(X) = \exp_G X$. Therefore $\exp_G$ is a global diffeomorphism and its transported multiplication is exactly the finite BCH polynomial. $\square$

A *rational structure* on $\mathfrak{g}$ is a $\mathbb{Q}$ -Lie algebra $\mathfrak{g}_{\mathbb{Q}} \subset \mathfrak{g}$ whose real span is $\mathfrak{g}$. A subspace is rational when it is spanned over $\mathbb{R}$ by its intersection with $\mathfrak{g}_{\mathbb{Q}}$.

Lemma L.5 (The lattice determines a rational structure). *Let Γ be a lattice in a connected simply connected nilpotent group G . Then*

$$\mathfrak{g}_{\mathbb{Q}} := \text{span}_{\mathbb{Q}}(\log \Gamma)$$

is a rational Lie algebra whose real span is $\mathfrak{g}$. If $\mathfrak{h} \leq \mathfrak{g}$ is a rational Lie subalgebra and $H = \exp \mathfrak{h}$, then $H \cap \Gamma$ is a lattice in H . Conversely, if H is connected and $H \cap \Gamma$ is cocompact in H , then $\mathfrak{h}$ is rational.

Proof. Step 1: bracket closure and real spanning. For $X = \log \gamma$ and $Y = \log \gamma'$ with $\gamma, \gamma' \in \Gamma$, the commutator

$$C(n, m) = \log[\exp(nX), \exp(mY)]$$

belongs to $\log \Gamma$ for all integers n, m . BCH makes $C(n, m)$ a finite polynomial in (n, m) with rational coefficients and leading bidegree term $nm[X, Y]$; all other terms are iterated brackets of length at least three. Proceed by descending bracket length. At the top nonzero bracket length, finite differences in n, m isolate every coefficient as a rational linear combination of the values $C(n, m)$, hence place those brackets in $\text{span}_{\mathbb{Q}} \log \Gamma$. Subtracting the already recovered higher-bracket terms and repeating downward eventually gives $[X, Y] \in \mathfrak{g}_{\mathbb{Q}}$. Bilinearity then proves that $\mathfrak{g}_{\mathbb{Q}}$ is a $\mathbb{Q}$ -Lie algebra.

Let $V = \text{span}_{\mathbb{R}} \mathfrak{g}_{\mathbb{Q}}$ and $H_V = \exp V$. Since V is a Lie subalgebra, H_V is a closed connected subgroup: in BCH coordinates it is the closed vector subspace V . It contains Γ . The quotient map $G/\Gamma \rightarrow G/H_V$ is therefore surjective, so G/H_V is compact. On the other hand a closed connected subgroup of a BCH nilpotent group admits a linear complement adapted to a central flag; successive projection along that flag identifies G/H_V diffeomorphically with $\mathbb{R}^{\dim G - \dim H_V}$. (Construct the coordinates one one-dimensional central quotient at a time exactly as in the proof of Lemma L.6 below.) Such a quotient is compact only in dimension zero. Hence $V = \mathfrak{g}$.

Step 2: the centre and its quotient lattice. The centre $\mathfrak{z}$ is the kernel of the rational linear map $X \mapsto ([X, Y_j])_j$ for a rational basis (Y_j) of $\mathfrak{g}_{\mathbb{Q}}$, hence is rational. Put $Z = \exp \mathfrak{z}$. We claim that the image $\bar{\Gamma}$ of Γ in G/Z is discrete. If $\gamma_n Z \rightarrow Z$, choose $z_n \in Z$ with $x_n := \gamma_n z_n \rightarrow e$. The lattice Γ is finitely generated: if $K\Gamma = G$ with K compact, a compact identity neighbourhood enlarged slightly beyond $K^{-1}K$ meets only finitely many lattice translates, and the usual connected-neighbourhood chain argument generates Γ from that finite set. Let S be finite generators. For $\delta \in S$,

$$[\gamma_n, \delta] = [x_n, \delta] \longrightarrow e.$$

These commutators lie in the discrete group Γ , so they equal e for all large n , simultaneously for $\delta \in S$. Thus γ_n centralizes Γ . The differential $\text{Ad}(\gamma_n)$ fixes $\log \Gamma$, which spans $\mathfrak{g}$ over $\mathbb{R}$ by Step 1; hence $\text{Ad}(\gamma_n) = 1$. In a connected simply connected nilpotent BCH group, $\ker \text{Ad} = Z$, so $\gamma_n \in Z$. Therefore $\gamma_n Z = Z$ eventually and $\bar{\Gamma}$ is discrete. It is cocompact because $(G/Z)/\bar{\Gamma}$ is the continuous image of the compact space G/Γ . The fibre of

$$G/\Gamma \longrightarrow (G/Z)/\bar{\Gamma}$$

over the identity coset is $Z/(Z \cap \Gamma)$; being a closed fibre of a compact space, it is compact. Thus $Z \cap \Gamma$ is a full lattice in Z .

Step 3: rational subgroups. We prove $H \cap \Gamma$ cocompact in H by induction on $\dim G$; the abelian case is ordinary rational linear algebra for a full lattice in a vector space. Pass to

$\bar{G} = G/Z$. The image $\bar{H} = HZ/Z$ has rational Lie algebra $(\mathfrak{h} + \mathfrak{z})/\mathfrak{z}$, so by induction $\bar{H} \cap \bar{\Gamma}$ contains a lattice in $\bar{H}$. Choose finitely many elements $\bar{\gamma}_j$ of this lattice and lifts $\gamma_j \in \Gamma$. Because $\log \gamma_j \in \mathfrak{g}_{\mathbb{Q}}$ and its image lies in the rational space $(\mathfrak{h} + \mathfrak{z})/\mathfrak{z}$, write

$$\log \gamma_j = H_j + Z_j, \quad H_j \in \mathfrak{h} \cap \mathfrak{g}_{\mathbb{Q}}, \quad Z_j \in \mathfrak{z} \cap \mathfrak{g}_{\mathbb{Q}}.$$

The full central lattice $\log(Z \cap \Gamma)$ spans $\mathfrak{z} \cap \mathfrak{g}_{\mathbb{Q}}$ over $\mathbb{Q}$. Hence for some integer $q_j \geq 1$, $q_j Z_j \in \log(Z \cap \Gamma)$. Since Z_j is central,

$$\gamma_j^{q_j} \exp(-q_j Z_j) = \exp(q_j H_j) \in H \cap \Gamma.$$

Thus the image of $H \cap \Gamma$ in $\bar{H}$ contains finite multiples of a lattice and is itself cocompact. Also $H \cap Z$ is a rational vector subgroup of the abelian group Z , so $(H \cap Z) \cap \Gamma$ is a lattice in $H \cap Z$. A compact fundamental set in the base together with a compact fundamental parallelepiped in this central fibre gives a compact fundamental set for $H \cap \Gamma$ in H .

Conversely, suppose $H/(H \cap \Gamma)$ is compact. Apply Step 1 inside the nilpotent group H with the lattice $H \cap \Gamma$. Its logarithms span $\mathfrak{h}$ over $\mathbb{R}$, while they all lie in $\mathfrak{g}_{\mathbb{Q}}$. Hence $\mathfrak{h}$ is spanned by rational vectors and is rational. This completes all directions. $\square$

L.2 A rational flag and a strong basis

Lemma L.6 (Rational central flag). *Let $(\mathfrak{g}, \mathfrak{g}_{\mathbb{Q}})$ be nilpotent and rational. There is a flag of rational ideals*

$$\mathfrak{g} = \mathfrak{g}^{(1)} \supset \mathfrak{g}^{(2)} \supset \cdots \supset \mathfrak{g}^{(m)} \supset \mathfrak{g}^{(m+1)} = 0, \quad \dim \mathfrak{g}^{(j)} / \mathfrak{g}^{(j+1)} = 1,$$

such that each quotient $\mathfrak{g}^{(j)} / \mathfrak{g}^{(j+1)}$ is central in $\mathfrak{g} / \mathfrak{g}^{(j+1)}$. If finitely many rational ideals $\mathfrak{h}_1, \dots, \mathfrak{h}_r$ are prescribed, the flag may be chosen so that every $\mathfrak{h}_i$ is a union of terminal stages of a refinement.

Proof. The centre of a nonzero nilpotent rational Lie algebra contains a nonzero rational vector: solve the rational homogeneous system $[X, Y_j] = 0$ in a rational basis Y_j . Choose a rational central line L and pass to $\mathfrak{g}/L$. Induction gives a rational central flag in the quotient; take inverse images and append L . To adapt to finitely many rational ideals, at each quotient choose the rational central line inside the first nonzero member of the induced family of ideals. Since intersections and quotients of rational subspaces remain rational, the induction continues and gives a common refinement. $\square$

Theorem L.7 (Strong lattice-adapted Mal'cev basis). *Let G/Γ be a nilmanifold and let $G_{\bullet}$ be a Γ -rational filtration. There is a basis $\mathcal{X} = (X_1, \dots, X_m)$ of $\mathfrak{g}$ such that:*

1. *every tail $\text{span}(X_j, \dots, X_m)$ is a rational ideal;*
2. *each $\text{Lie}(G_i)$ is spanned by a terminal block;*
3. *every $g \in G$ has a unique second-kind expansion*

$$g = \exp(t_1 X_1) \cdots \exp(t_m X_m);$$

4. *the successive reduction map sends every coset $g\Gamma$ to a unique representative with $0 \leq t_j < 1$; and*
5. *with the normalization used in the construction, the lattice is generated successively by $\exp(X_j)$ modulo the later tail group.*

Proof. Apply [theorem L.6](#) to the rational structure from [theorem L.5](#), including the ideals $\text{Lie}(G_i)$ among the prescribed rational ideals. Write $H_j = \exp \mathfrak{g}^{(j)}$ and $\Gamma_j = H_j \cap \Gamma$. By [theorem L.5](#), Γ_j is a lattice in H_j . The quotient H_j/H_{j+1} is one-dimensional central and the image of Γ_j is a nonzero discrete subgroup, hence is $a_j\mathbb{Z}$ in a chosen real coordinate. Choose $X_j \in \mathfrak{g}^{(j)}$ whose image exponentiates to the positive generator of this quotient lattice. This gives (1), (2), and (5).

For existence of (3), work from $j = 1$ downward. Since $G/H_2 \simeq \mathbb{R}$, choose the unique t_1 making $g \exp(-t_1 X_1) \in H_2$ and continue in H_2 . Uniqueness follows by projecting two expansions to the first quotient where their coordinates differ.

For (4), given the expansion of g , choose $n_1 = \lfloor t_1 \rfloor$ and multiply on the right by a lattice element whose image in H_1/H_2 is $\exp(-n_1 X_1)H_2$. Centrality of this quotient changes only later coordinates. Repeat in H_2/H_3 , and so on. This produces a representative with every coordinate in $[0, 1)$. If two such representatives differ by an element of Γ , projection to H_1/H_2 forces equality of their first coordinates; induction forces equality of all coordinates. Thus the representative is unique. $\square$

L.3 Triangular polynomial coordinate laws

Lemma L.8 (First-kind/second-kind conversion). *For a strong Mal'cev basis, if*

$$\exp(u_1 X_1 + \cdots + u_m X_m) = \exp(t_1 X_1) \cdots \exp(t_m X_m),$$

then

$$t_j = u_j + P_j(u_1, \dots, u_{j-1}), \quad u_j = t_j + Q_j(t_1, \dots, t_{j-1}),$$

where P_j, Q_j are rational polynomials of degree bounded in terms of the nilpotency step. If the structure constants have rational height at most H , the coefficients have height at most $H^{O_{m,s}(1)}$.

Proof. Apply BCH successively to the product of exponentials. Because every tail is an ideal and the successive one-dimensional quotient is central, a commutator that contributes to the X_j coordinate can involve only $X_1, \dots, X_{j-1}$ in addition to X_j itself. The coefficient of u_j is exactly one. BCH coefficients are rational numbers whose denominators depend only on the step, and each iterated bracket is obtained by multiplying at most $s - 1$ structure constants. This gives the first triangular system and its height bound. Solve recursively for u_j to obtain the inverse triangular system and the same type of bound. $\square$

Proposition L.9 (Polynomial multiplication and inversion). *In second-kind coordinates there are triangular rational polynomials $P = (P_1, \dots, P_m)$ and $I = (I_1, \dots, I_m)$ such that*

$$\psi_{\mathcal{X}}(gh) = P(\psi_{\mathcal{X}}(g), \psi_{\mathcal{X}}(h)), \quad \psi_{\mathcal{X}}(g^{-1}) = I(\psi_{\mathcal{X}}(g)).$$

Their degrees depend only on the nilpotency step and their coefficient heights are polynomially controlled by the structure-constant height.

Proof. Convert both factors to first-kind coordinates by [theorem L.8](#), apply the finite BCH polynomial to multiply (or replace X by $-X$ to invert), and convert back. A finite composition of triangular rational polynomial maps is triangular and preserves a polynomial height bound when dimension and degree are fixed. $\square$

L.4 Quantitative rational linear algebra

For a rational number a/b in lowest terms, its height is $\max(|a|, |b|)$. A rational vector or matrix has height at most H when all entries do.

Lemma L.10 (Height-controlled kernels and bases). *Fix m . Let A be an $r \times m$ rational matrix of height at most H . Then $\ker A$ has a rational basis of height at most $H^{O_m(1)}$. If $V, W \leq \mathbb{Q}^m$ have bases of height at most H , then $V + W$, $V \cap W$, and the image of V in $\mathbb{Q}^m/W$ have bases of height $H^{O_m(1)}$.*

Proof. Clear denominators row by row; this produces an integer matrix with entries at most $H^{O_m(1)}$. Put it in row-echelon form. Every pivot denominator and every numerator in the resulting nullspace basis is a minor of order at most m . Hadamard's determinant bound is at most $m!H^{O_m(1)}$. This proves the kernel statement. Sums and images are obtained by concatenating basis matrices; intersections are kernels of the map $(v, w) \mapsto v - w$. Applying the kernel bound again proves all assertions. $\square$

Lemma L.11 (Rational subalgebras inherit controlled Mal'cev bases). *Suppose the structure constants of $\mathfrak{g}$ in $\mathcal{X}$ have height at most M , and a rational Lie subalgebra $\mathfrak{h}$ is cut out by rational linear equations of height at most M . Then $H = \exp \mathfrak{h}$ has a strong Mal'cev basis for $H/(H \cap \Gamma)$ whose vectors are rational combinations of the X_i of height $M^{O_m(1)}$. The same is true after intersecting with a fixed rational filtration.*

Proof. Clear denominators in the equations defining $\mathfrak{h}$. By Lemma L.10, the lattice $\mathfrak{h} \cap \text{span}_{\mathbb{Z}}(X_1, \dots, X_m)$ has a basis of height $M^{O_m(1)}$: one may compute it by Hermite elimination, whose pivots and back-substitution denominators are minors of the cleared integer matrix, hence are bounded by Hadamard's inequality.

We now construct the central flag quantitatively rather than invoke an uncontrolled choice. In a controlled rational basis, the centre is the kernel of the matrix of all bracket maps; its entries have height $M^{O_m(1)}$, so Lemma L.10 gives a controlled rational basis for the centre. Choose a primitive lattice vector on a rational central line, quotient by that line, and repeat. At each of at most m stages, kernels, intersections, and quotient maps cost only a polynomial power of the current height. Since m is fixed, every vector in the resulting flag has height $M^{O_m(1)}$.

For a one-dimensional rational quotient V/V' in this flag, the image of the integer lattice is a rank-one subgroup. Its positive generator is obtained by taking the gcd of the finitely many integer coordinates after clearing the same controlled denominators. Bézout (or Hermite normal form) expresses that generator using coefficients of size $M^{O_m(1)}$. Lifting it therefore gives a vector of height $M^{O_m(1)}$. Applying this successively is precisely the construction in Theorem L.7, now with the height bookkeeping included. Intersecting at each stage with the fixed rational filtration uses only the controlled kernel/intersection operations above, so the same polynomial bound persists. $\square$

Theorem L.12 (Quantitative rational-coordinate calculus). *Fix dimension m and step s . Suppose a Mal'cev basis has structure constants of height at most M . Then:*

1. *multiplication, inversion, first/second-kind conversion, and quotient reduction are given on each reduction cell by rational polynomials of degree $O_{m,s}(1)$ and height $M^{O_{m,s}(1)}$;*
2. *kernels of horizontal characters of size at most M and all rational subgroups obtained by at most m successive kernel/intersection operations have adapted Mal'cev bases of rationality $M^{O_{m,s}(1)}$;*
3. *conjugation by an element with rational coordinates of height at most M transforms rational subgroups and their Mal'cev bases with the same polynomial loss; and*

4. on coordinate boxes of radius M , all these maps and their inverses are $M^{O_{m,s}(1)}$ -Lipschitz.

Proof. Part (1) is [theorems L.8](#) and [L.9](#); quotient reduction uses finitely many translations by lattice generators on each cell. Part (2) is [theorems L.10](#) and [L.11](#); a horizontal character is one rational linear equation in the horizontal coordinates. Part (3) follows because $\text{Ad}(\exp X) = e^{\text{ad } X}$ and the exponential is a finite polynomial in the nilpotent operator $\text{ad } X$, followed by the rational linear algebra of part (2). For (4), differentiate the controlled polynomial maps on a bounded box and use the mean-value theorem. Their inverses are again controlled triangular polynomial maps, so the same estimate applies in both directions. $\square$

Appendix M

Quantitative Leibman Theory and Factorization

The purpose of this appendix is to prove the two dynamical inputs that carry the rest of this detailed appendix: quantitative detection of non-equidistribution by a horizontal character, and the smooth–equidistributed–rational factorization. The argument is arranged so that every induction parameter is visible.

M.1 Why filtered polynomial maps form a group

The closure of filtered polynomial maps under multiplication is easy to state and surprisingly easy to underprove. Because the target is nonabelian, repeatedly applying a product rule creates nested conjugations. We therefore give the cube-group proof of the Lazard–Leibman theorem before using polynomial calculus elsewhere in this appendix.

For $k \geq 1$ identify $\{0, 1\}^k$ with subsets of $[k]$. For $\omega_0 \in \{0, 1\}^k$ put

$$F(\omega_0) = \{\omega : \omega \supseteq \omega_0\}, \quad \text{codim } F(\omega_0) = |\omega_0|.$$

If F is such an upper face and $a \in G_{\text{codim } F}$, let $a^F \in G^{\{0,1\}^k}$ equal a on F and e off F . The *Host–Kra cube group* $\text{HK}^k(G_\bullet)$ is the subgroup generated by all these upper-face elements.

Lemma M.1 (Ordered face collection). *Order the upper faces by reverse lexicographic order of their minimal vertices, starting with the singleton face $\{(1, \dots, 1)\}$ and ending with the whole cube. Every element of $\text{HK}^k(G_\bullet)$ can be written as a product of one element from each face group in this order.*

Proof. If F, F' are upper faces, then coordinatewise commutation gives

$$[G(F), G(F')] \subseteq G(F \cap F'),$$

because

$$[G_{\text{codim } F}, G_{\text{codim } F'}] \subseteq G_{\text{codim } F + \text{codim } F'} \subseteq G_{\text{codim}(F \cap F')}.$$

When two adjacent face factors occur in the wrong order, the identity $ab = [a, b]ba$ swaps them and creates only a factor supported on $F \cap F'$. That intersection is at least as deep in the filtration and occurs no earlier than the factors being collected. Repeating this finite “bubble sort” first moves every factor from the first face group to the far left, then every factor from the second face group immediately after it, and so on. This produces the claimed ordered product. We do not need uniqueness. $\square$

For an abelian domain H and $x, h_1, \dots, h_k \in H$, write

$$[x; h_1, \dots, h_k] := (x + \omega_1 h_1 + \dots + \omega_k h_k)_{\omega \in \{0,1\}^k}$$

for the corresponding parallelepiped.

Proposition M.2 (Host–Kra cube characterization of polynomial maps). *Let H be an abelian group and let $G_\bullet$ be a filtration on a nilpotent group G . A map $g : H \rightarrow G$ is polynomial adapted to $G_\bullet$ if and only if, for every k and every parallelepiped $c = [x; h_1, \dots, h_k]$,*

$$(g(c_\omega))_{\omega \in \{0,1\}^k} \in \text{HK}^k(G_\bullet).$$

Consequently the polynomial maps $H \rightarrow G$ form a group under pointwise multiplication and inversion.

Proof. We prove the forward implication by induction on k . The case $k = 0$ is trivial. Split a $(k+1)$ -cube into its lower and upper k -faces. If $c = [x; h_1, \dots, h_k]$, then pointwise

$$(g(c), g(c + h_{k+1})) = (e, (\partial_{h_{k+1}} g)(c)) (g(c), g(c)). \quad (\text{B.LL.1})$$

The derivative $\partial_{h_{k+1}} g$ is polynomial for the shifted filtration $G_i^+ := G_{i+1}$, directly from the definition of iterated derivatives. By induction its k -cube lies in $\text{HK}^k(G_\bullet^+)$, while the cube of g lies in $\text{HK}^k(G_\bullet)$. A generator a^F of the first group becomes, when placed only on the upper half of the $(k+1)$ -cube, an upper-face generator whose codimension has increased by one; a generator of the second becomes a diagonal upper-face generator with the same codimension. Thus both factors on the right of (B.LL.1) lie in $\text{HK}^{k+1}(G_\bullet)$.

Conversely suppose every image cube lies in the Host–Kra cube group. Fix $x, h_1, \dots, h_k$ and write its image cube in the ordered form supplied by Lemma M.1. The first factor is $a^{\{(1, \dots, 1)\}}$ with $a \in G_k$. We claim that a is exactly the multiplicative k -fold derivative $\partial_{h_k} \dots \partial_{h_1} g(x)$, up to multiplication by factors in $G_{k+1} \subseteq G_k$, which is all that is required. To see this without commutative cancellation, peel the cube one coordinate at a time. All face factors not supported on the top vertex occur with the same value in the two halves at the coordinate where they are first peeled, and hence disappear in the quotient defining that derivative. A commutator created while moving such a factor across a factor of codimensions i and j lies in G_{i+j} ; after all k peelings every such correction has total codimension at least k . The only factor that survives all peelings at minimal codimension k is the singleton-top-face coefficient a . Therefore the k -fold derivative belongs to G_k . Since k and the increments were arbitrary, g is polynomial.

Finally, if g and $\tilde{g}$ are polynomial, both image every parallelepiped into the subgroup $\text{HK}^k(G_\bullet)$. Their pointwise product and pointwise inverse do the same because that set is a group. The characterization just proved therefore shows that $g\tilde{g}$ and g^{-1} are polynomial. This is the Lazard–Leibman group theorem in the form needed throughout this book. $\square$

M.2 Polynomial phases on a torus

Lemma M.3 (Fejér detection on a torus). *Let $x_1, \dots, x_N \in \mathbb{T}^r$. If the sequence is not δ -equidistributed, then for some nonzero $k \in \mathbb{Z}^r$ with $|k| \ll_r \delta^{-O_r(1)}$,*

$$\left| \frac{1}{N} \sum_{n \leq N} e(k \cdot x_n) \right| \gg_r \delta^{O_r(1)}.$$

Proof. Choose a 1-Lipschitz mean-zero witness F with $|N^{-1} \sum F(x_n)| > \delta$. Convolve F with the product Fejér kernel of order K . The approximation error is $O_r(K^{-1})$. Taking $K \asymp_r \delta^{-1}$ leaves correlation at least $\delta/2$. The convolution is a trigonometric polynomial supported on $|k|_\infty \leq K$, with total absolute Fourier mass $\ll_r K^r$. Since its zero Fourier coefficient is zero, one nonzero frequency has correlation $\gg_r \delta K^{-r}$. $\square$

Lemma M.4 (Dense near-integer multiples). *There are absolute constants $c, C > 0$ with the following property. Let $0 < \sigma \leq 1/2$, let $H \geq C\sigma^{-1}$, and suppose that $\|h\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \varepsilon$ for at least σH integers $1 \leq h \leq H$, where $0 < \varepsilon \leq c\sigma^2$. Then there is an integer*

$$1 \leq q \leq C\sigma^{-1}$$

such that

$$H\|q\theta\|_{\mathbb{R}/\mathbb{Z}} \leq C\sigma^{-2}\varepsilon. \quad (\text{B.0})$$

Proof. Let A be the set of good h . Put $h_0 = \min A$, $h_1 = \max A$ and $r = h_1 - h_0$. Since A contains at least σH distinct integers, $r \geq \sigma H - 1 \geq \sigma H/2$. We have $\|r\theta\| \leq 2\varepsilon$; write

$$r\theta = m + u, \quad |u| \leq 2\varepsilon.$$

Reduce $m/r = a/q$, so $(a, q) = 1$ and $q \mid r$. For every $h \in A$,

$$\left\| \frac{ha}{q} \right\| \leq \|h\theta\| + \frac{h|u|}{r} \leq \varepsilon + 4\sigma^{-1}\varepsilon \leq 5\sigma^{-1}\varepsilon =: \tau. \quad (\text{B.01})$$

For a reduced fraction a/q , the residues $ha \pmod{q}$ run through all residue classes as h runs through a block of q consecutive integers. Hence the number of $1 \leq h \leq H$ satisfying $\|ha/q\| \leq \tau$ is at most

$$(H/q + 1)(2\tau q + 2) \leq 4\tau H + 2H/q + 2.$$

Choose c small enough that $4\tau \leq \sigma/2$. Comparing with $|A| \geq \sigma H$ and using $H \geq C\sigma^{-1}$ gives $q \leq C\sigma^{-1}$. Finally qm/r is an integer, so

$$\|q\theta\| = \frac{q|u|}{r} \leq \frac{C\sigma^{-1}2\varepsilon}{\sigma H/2} \leq C\sigma^{-2}\varepsilon/H,$$

which is (B.0). $\square$

Lemma M.5 (Dense polynomial coefficient recovery). *Fix $s \geq 1$. There are constants $c_s, C_s > 0$ such that the following holds. Let*

$$Q(h) = \sum_{j=0}^s \beta_j h^j \pmod{1}, \quad 1 \leq h \leq H,$$

and suppose $\|Q(h)\| \leq \varepsilon$ for at least σH values of h , where

$$H \geq \sigma^{-C_s}, \quad 0 < \varepsilon \leq c_s \sigma^{C_s}.$$

Then some $1 \leq q \leq \sigma^{-C_s}$ satisfies

$$H^j \|q\beta_j\| \leq \sigma^{-C_s} \varepsilon \quad (0 \leq j \leq s). \quad (\text{B.02})$$

The same conclusion holds in the binomial basis, after changing C_s .

Proof. We induct on s . The case $s = 0$ follows by evaluating at one good point. Assume $s \geq 1$ and let A be the good set. Counting ordered pairs in A by their difference gives

$$\sum_{|h| < H} |A \cap (A - h)| = |A|^2.$$

After discarding $h = 0$ and the boundary half of the possible shifts, there is a set $\mathcal{H} \subset [1, H/2]$ of at least $\sigma^{O(1)}H$ shifts for which

$$|A \cap (A - h)| \geq \sigma^{O(1)}H. \quad (\text{B.03})$$

For each $h \in \mathcal{H}$, the polynomial $\Delta_h Q(x) = Q(x + h) - Q(x)$ has degree $s - 1$ and is within 2ε of an integer at all $x \in A \cap (A - h)$. By the induction hypothesis there is $q_h \leq \sigma^{-O_s(1)}$ such that every coefficient of $\Delta_h Q$ has the required scale- H bound, with right side $\sigma^{-O_s(1)}\varepsilon$. Pigeonholing the finitely many possible q_h leaves a set $\mathcal{H}_0 \subset \mathcal{H}$ of density $\sigma^{O_s(1)}$ on which one integer q_0 works.

The coefficient of x^{s-1} in $\Delta_h Q$ is $sh\beta_s$. Hence, for all $h \in \mathcal{H}_0$,

$$\|q_0 sh\beta_s\| \leq \sigma^{-O_s(1)}\varepsilon H^{-(s-1)}.$$

Apply Lemma M.4; the hypotheses hold after enlarging C_s (and the case of smaller H was excluded in the statement). We obtain a further factor $q_s \leq \sigma^{-O_s(1)}$ with

$$H^s \|q_s q_0 s \beta_s\| \leq \sigma^{-O_s(1)}\varepsilon.$$

Absorb the fixed factor s into the denominator.

Now descend. Suppose one common integer Q_r , divisible by q_0 , already controls $\beta_{r+2}, \dots, \beta_s$ at their scales. The coefficient of x^r in $\Delta_h Q$ is

$$(r+1)h\beta_{r+1} + \sum_{j=r+2}^s \binom{j}{r} \beta_j h^{j-r}. \quad (\text{B.04})$$

The induction applied to $\Delta_h Q$ controls (B.04), after multiplication by Q_r , to accuracy $\sigma^{-O_s(1)}\varepsilon H^{-r}$. The already controlled higher coefficients contribute at the same scale because $h \leq H$. Thus

$$\|Q_r(r+1)h\beta_{r+1}\| \leq \sigma^{-O_s(1)}\varepsilon H^{-r}$$

for the dense set $\mathcal{H}_0$. A further application of Lemma M.4 gives a bounded factor which controls β_{r+1} at scale $H^{-(r+1)}$. Descending to $r = 0$ controls every nonconstant coefficient. Evaluating qQ at one good point and subtracting the controlled nonconstant terms controls β_0 . Multiplying the finitely many bounded denominators keeps $q \leq \sigma^{-C_s}$.

The change between monomials and $\binom{h}{j}$ is triangular with integer entries in one direction and denominators dividing $s!$ in the other, so a fixed extra factor transfers the result to the binomial basis. $\square$

Lemma M.6 (Multivariate dense coefficient recovery). *Fix d, s . Let*

$$Q(\mathbf{h}) = \sum_{|\mathbf{i}| \leq s} \beta_{\mathbf{i}} \mathbf{h}^{\mathbf{i}} \pmod{1}, \quad \mathbf{h} \in [H]^d.$$

If $\|Q(\mathbf{h})\| \leq \varepsilon$ on at least σH^d points, with $H \geq \sigma^{-C_{d,s}}$ and $\varepsilon \leq c_{d,s} \sigma^{C_{d,s}}$, then some $q \leq \sigma^{-C_{d,s}}$ satisfies

$$H^{|\mathbf{i}|} \|q \beta_{\mathbf{i}}\| \leq \sigma^{-C_{d,s}} \varepsilon \quad (|\mathbf{i}| \leq s). \quad (\text{B.05})$$

Proof. Induct on d . The case $d = 1$ is Lemma M.5. Write $Q(\mathbf{x}, y) = \sum_{j=0}^s B_j(\mathbf{x})y^j$. By averaging, at least $\sigma H/2$ -density worth of base points $\mathbf{x} \in [H]^{d-1}$ have a y -fiber containing at least $\sigma H/2$ good points. Apply the one-variable lemma on each such fiber. There are only $\sigma^{-O_s(1)}$ possible denominators, so one denominator q_0 works for a set of $\sigma^{O_s(1)}H^{d-1}$ base points. Consequently, for each j ,

$$\|q_0 B_j(\mathbf{x})\| \leq \sigma^{-O_s(1)} \varepsilon H^{-j}$$

on that dense set. Apply the $(d-1)$ -variable induction hypothesis to each polynomial $q_0 B_j$ and multiply the finitely many resulting denominators. This yields $H^{|\mathbf{i}|} \|q \beta_{\mathbf{i}}\| \leq \sigma^{-O_{d,s}(1)} \varepsilon$ as required. $\square$

Lemma M.7 (Polynomial Weyl obstruction). *Let*

$$P(n) = \sum_{j=0}^s \alpha_j \binom{n}{j} \pmod{1}.$$

If

$$\left| \frac{1}{N} \sum_{n \leq N} e(P(n)) \right| \geq \rho,$$

then there is an integer $1 \leq q \ll_s \rho^{-O_s(1)}$ such that

$$\max_{1 \leq j \leq s} N^j \|q \alpha_j\|_{\mathbb{R}/\mathbb{Z}} \ll_s \rho^{-O_s(1)}.$$

Proof. We induct on s . For $s = 1$, the geometric-sum estimate proved in Volume V, Chapter 2 gives $N \|\alpha_1\| \ll \rho^{-1}$, so $q = 1$ works. We may assume $N \geq \rho^{-C_s}$, since otherwise the conclusion is trivial after enlarging C_s .

Take $H = \lfloor N/2 \rfloor$. Van der Corput's inequality and the lower bound $|\mathbb{E}e(P)| \geq \rho$ imply that for at least $c\rho^2 H$ shifts $1 \leq h \leq H$,

$$\left| \frac{1}{N-h} \sum_{n \leq N-h} e(P(n+h) - P(n)) \right| \geq c\rho^2. \quad (\text{B.06})$$

For such an h , write

$$P(n+h) - P(n) = \sum_{r=0}^{s-1} B_r(h) \binom{n}{r}, \quad B_r(h) = \sum_{t=1}^{s-r} \binom{h}{t} \alpha_{r+t}. \quad (\text{B.07})$$

The induction hypothesis, with $N-h \asymp N$, supplies $q_h \leq \rho^{-O_s(1)}$ such that $N^r \|q_h B_r(h)\| \leq \rho^{-O_s(1)}$ for $1 \leq r \leq s-1$. Pigeonhole one value q_0 on a set $\mathcal{H}$ of at least $\rho^{O_s(1)} H$ good shifts.

For $r = s-1$, (B.07) says $B_{s-1}(h) = h\alpha_s$. Apply Lemma M.4 to $q_0 \alpha_s$ with tolerance $\rho^{-O_s(1)} N^{-(s-1)}$. We obtain a bounded extra factor and hence a common integer Q with

$$N^s \|Q \alpha_s\| \leq \rho^{-O_s(1)}. \quad (\text{B.08})$$

Proceed downward. Suppose Q already controls $\alpha_{r+2}, \dots, \alpha_s$. Multiply the bound on $B_r(h)$ by Q/q_0 and subtract the terms with $t \geq 2$ in (B.07). By the existing controls and $h \leq N$ these terms contribute $O(\rho^{-O_s(1)} N^{-r})$. Thus on the same dense set $\mathcal{H}$,

$$\|Q h \alpha_{r+1}\| \leq \rho^{-O_s(1)} N^{-r}.$$

Lemma M.4 supplies another bounded factor and gives $N^{r+1} \|Q \alpha_{r+1}\| \leq \rho^{-O_s(1)}$. After finitely many steps one common $Q \leq \rho^{-O_s(1)}$ controls every nonconstant coefficient. This is the claimed obstruction. $\square$

Proposition M.8 (Quantitative Weyl criterion on $\mathbb{T}^r$). *Let $p : \mathbb{Z} \rightarrow \mathbb{T}^r$ be polynomial of degree at most s . If $(p(n))_{n \leq N}$ is not δ -equidistributed, then there is $0 \neq k \in \mathbb{Z}^r$ with $|k| \leq \delta^{-O_{r,s}(1)}$ and*

$$\|k \cdot p\|_{C^\infty[N]} \leq \delta^{-O_{r,s}(1)}.$$

Proof. Use [theorem M.3](#) to find a frequency k . Apply [theorem M.7](#) to the polynomial phase $k \cdot p$ and replace k by qk . $\square$

M.3 Vertical Fourier analysis and dimension descent

Let G_s be the last nontrivial group in the filtration. It is central, and $T_s = G_s/(G_s \cap \Gamma)$ is a torus acting on G/Γ by left translation. A function has *vertical frequency* $\xi \in \widehat{T}_s$ if $F(ux) = e(\xi(u))F(x)$.

Lemma M.9 (Vertical Fourier truncation). *Let F be Lipschitz on a bounded-complexity nilmanifold and have mean zero. For every $K \geq 2$ there are vertical-frequency functions F_ξ , indexed by $|\xi| \leq K$, such that*

$$F = \sum_{|\xi| \leq K} F_\xi + R_K, \quad \|R_K\|_\infty \ll K^{-1} \|F\|_{\text{Lip}},$$

and

$$\sum_{|\xi| \leq K} \|F_\xi\|_{\text{Lip}} \ll K^{O_m(1)} \|F\|_{\text{Lip}}.$$

If $\xi = 0$, then F_0 descends to $G/(G_s\Gamma)$ and still has mean zero.

Proof. For $x \in G/\Gamma$, convolve $u \mapsto F(ux)$ on the compact torus T_s with a product Fejér kernel. Fourier expansion in u gives the F_ξ and the K^{-1} approximation bound follows from the first moment of the kernel and the Lipschitz property. Differentiating the convolution in x and summing the Fourier coefficients gives the polynomial K loss. The zero-frequency term is constant on T_s -orbits, hence descends. Fubini shows that its mean on the quotient equals the mean of F . $\square$

Lemma M.10 (van der Corput for bounded sequences). *If $|a_n| \leq 1$ and $1 \leq H \leq N$, then*

$$\left| \frac{1}{N} \sum_{n \leq N} a_n \right|^2 \leq \frac{2}{H} + \frac{2}{H} \sum_{h=1}^H \left| \frac{1}{N} \sum_{n \leq N-h} a_{n+h} \overline{a_n} \right|.$$

Proof. Write $H \sum_{n \leq N} a_n$ as $\sum_{n \leq N+H} \sum_{1 \leq h \leq H} a_{n-h}$, taking $a_j = 0$ outside $[N]$. Cauchy-Schwarz in n and expansion of the inner square give the diagonal HN and, for each nonzero difference $h - h'$, the corresponding shifted correlation. Divide by $H^2 N^2$ and bound the boundary by $O(H/N)$, which is absorbed by $2/H$ after replacing H by $\min(H, \lfloor N/2 \rfloor)$. $\square$

For a nonzero vertical frequency one must first restrict the shifted pair orbit before quotienting. This is the point at which a naive use of all of $G \times G$ fails: the quotient filtration on the full product does not lose a degree. The correct object is the *relative square*.

For $0 \leq i \leq s$ put

$$G_i^\square := G_i \times_{G_{i+1}} G_i := \{(x, y) \in G_i \times G_i : xy^{-1} \in G_{i+1}\}, \quad G_{s+1} = \{e\}, \quad (\text{B.10})$$

and set $G^\square = G_1^\square$ and $\Gamma^\square = (\Gamma \times \Gamma) \cap G^\square$. We also write $G_i^\Delta = \{(u, u) : u \in G_i\}$.

Lemma M.11 (Relative-square filtration and rationality). *The sequence $(G_i^\square)_{i \geq 1}$ is a filtration of $G^\square$ and*

$$G_i^\square = \langle G_i^\Delta, G_{i+1} \times G_{i+1} \rangle. \quad (\text{B.11})$$

Moreover $G^\square$ and every $G_i^\square$ are rational subgroups of $G \times G$, $\Gamma^\square$ is a lattice in $G^\square$, and $(G_s^\square) = G_s^\Delta$. Hence

$$\overline{G}^\square := G^\square / G_s^\Delta$$

with the quotient filtration $\overline{G}_i^\square = G_i^\square / G_s^\Delta$ for $i < s$ and $\overline{G}_s^\square = \{e\}$ has degree at most $s - 1$. Its Mal'cev rationality is bounded by a fixed power of the rationality of G/Γ .

Proof. If $(x, y) \in G_i^\square$, write $x = uy$ with $u \in G_{i+1}$; then $(x, y) = (u, e)(y, y)$, which proves (B.11), since (u, e) lies in $G_{i+1} \times G_{i+1}$. Conversely both displayed generating subgroups lie in $G_i^\square$.

We verify the commutator condition. By (B.11) it is enough to commute the two types of generators. The filtration law in G gives

$$[G_i^\Delta, G_j^\Delta] \subseteq G_{i+j}^\Delta, \quad [G_i^\Delta, G_{j+1} \times G_{j+1}] \subseteq G_{i+j+1} \times G_{i+j+1},$$

and the same estimate with the factors reversed, while

$$[G_{i+1} \times G_{i+1}, G_{j+1} \times G_{j+1}] \subseteq G_{i+j+2} \times G_{i+j+2}.$$

All four groups lie in $G_{i+j}^\square$, proving that $(G_i^\square)$ is a filtration. Since $G_{s+1} = \{e\}$, the defining condition at level s forces $x = y$, so $G_s^\square = G_s^\Delta$.

For rationality, choose a rational Mal'cev basis adapted to $G_\bullet$. The Lie algebra of $G_i^\square$ is

$$\mathfrak{g}_i^\square = \{(X, Y) \in \mathfrak{g}_i^2 : X - Y \in \mathfrak{g}_{i+1}\} = \mathfrak{g}_i^\Delta + (\mathfrak{g}_{i+1} \times \mathfrak{g}_{i+1}),$$

so it is cut out by rational linear equations of height bounded by a fixed power of the original height. Appendix A's rational-subgroup criterion and controlled-basis construction therefore give a rational Mal'cev basis for $G_i^\square$ and show that $(\Gamma \times \Gamma) \cap G^\square$ is a lattice. The same criterion applies to the central rational subgroup G_s^Δ and to the quotient. Row reduction over $\mathbb{Q}$ shows that every new height is bounded by a fixed power of the old one. $\square$

Lemma M.12 (The shifted pair is polynomial in the relative square). *Assume $g(0) = e$ and put $a = g(1)$ and $g_2(n) = g(n)a^{-n}$. Then $g_2(n) \in G_2$ for all n . If $a^h = \{a^h\}[a^h]$ is the Mal'cev fundamental-domain decomposition, define*

$$g_h^\square(n) = \left(\{a^h\}^{-1} g_2(n+h) a^n \{a^h\}, g_2(n) a^n \right). \quad (\text{B.12})$$

Then $g_h^\square \in \text{poly}(\mathbb{Z}, G_\bullet^\square)$. Furthermore, if F has vertical frequency ξ , then

$$F_h^\square(x, y) := F(\{a^h\}x) \overline{F(y)}$$

is invariant under G_s^Δ and hence descends to $\overline{G}^\square / \overline{\Gamma}^\square$.

Proof. Modulo G_2 , the filtration is abelian of degree one. Since $g(0) = e$ and $g(1) = a$, the second discrete derivative of g vanishes in G/G_2 , whence $g(n)G_2 = a^n G_2$ by induction on n ; thus $g_2(n) \in G_2$.

The class of polynomial sequences in a filtered nilpotent group is closed under products, inverses, shifts, and conjugation by constants. For completeness, this follows by induction on the number of derivatives from

$$\partial_h(uv) = (\partial_h u) u (\partial_h v) u^{-1}, \quad [G_i, G_j] \subseteq G_{i+j}; \quad (\text{B.13})$$

at each differentiation, the conjugation error is a commutator whose filtration index is the sum of the two indices. Hence it suffices to check the Taylor generators. Write a Taylor expansion of g_2 as a product of $c_i^{(n)}$, $c_i \in G_i$, $i \geq 2$. For each j th derivative the two coordinates of $(c_i^{(n+h)}, c_i^{(n)})$ differ by an element of G_{j+1} : if $j < i$, their difference lies already in $G_i \subseteq G_{j+1}$, and if $j \geq i$ both derivatives are trivial. Thus this pair is polynomial for $G_\bullet^\square$. Multiplying by the diagonal polynomial (a^n, a^n) and conjugating the first component by $\{a^h\}$ proves (B.12).

Finally G_s is central, so for $u \in G_s$,

$$F_h^\square(ux, uy) = e(\xi(u))F(\{a^h\}x) e(-\xi(u))\overline{F(y)} = F_h^\square(x, y).$$

□

Lemma M.13 (Characters of the relative square). *Let $\eta : G^\square \rightarrow \mathbb{R}/\mathbb{Z}$ be a horizontal character. There are homomorphisms*

$$\eta_1 : G \rightarrow \mathbb{R}/\mathbb{Z}, \quad \eta_2 : G_2 \rightarrow \mathbb{R}/\mathbb{Z}$$

such that

$$\eta(x', x) = \eta_1(x) + \eta_2(x'x^{-1}) \quad ((x', x) \in G^\square). \quad (\text{B.14})$$

Here η_1 is a horizontal character of G , η_2 annihilates $\Gamma \cap G_2$ and $[G, G_2]$, and their coefficient sizes are bounded by a fixed power of the size of η and of the Mal'cev rationality.

Proof. Set $\eta_1(x) = \eta(x, x)$ and $\eta_2(u) = \eta(u, e)$. Since $(x', x) = (x'x^{-1}, e)(x, x)$, additivity gives (B.14). The lattice $\Gamma^\square$ contains Γ^Δ and $(\Gamma \cap G_2) \times \{e\}$, so the asserted lattice annihilation follows. Also $[G^\Delta, G_2 \times \{e\}] = [G, G_2] \times \{e\}$ lies in $[G^\square, G^\square]$, hence η_2 kills $[G, G_2]$. In rational Mal'cev coordinates the restriction maps $x \mapsto (x, x)$ and $u \mapsto (u, e)$ are rational polynomial maps of bounded height by Lemma M.11; substituting them into the integer coordinate vector of η gives the stated coefficient bounds. □

Lemma M.14 (Polynomial stitching lemma). *Fix degree s . Let $P, Q : \mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$ be polynomial phases of degree at most s , with $P(0) = 0$ and $Q(0) = \Delta Q(0) = 0$, and let $\sigma : [N] \rightarrow \mathbb{R}/\mathbb{Z}$ be arbitrary. Suppose that, for at least ρN shifts h ,*

$$\|P(n) + Q(n+h) - Q(n) + \sigma(h)n\|_{C^\infty[N]} \leq K, \quad (\text{B.15})$$

where $K \leq \rho^{-O_s(1)}$. Then, after multiplication by one integer $q \leq \rho^{-O_s(1)}$, all Taylor coefficients of Q of degree at least three obey their natural scale bounds

$$N^j \|q \Delta^j Q(0)\| \leq \rho^{-O_s(1)} K \quad (j \geq 3), \quad (\text{B.16})$$

and for at least $\rho^{O_s(1)} N$ of the same shifts,

$$N \|qP(1) + q\alpha h + q\sigma(h)\| \leq \rho^{-O_s(1)} K, \quad \alpha = \Delta^2 Q(0). \quad (\text{B.17})$$

Proof. Take the second discrete derivative in n at $n = 0$ in (B.15). Its value is

$$\Delta^2 P(0) + \Delta^2 Q(h) - \Delta^2 Q(0),$$

and is within $O(KN^{-2})$ of an integer on a set of ρN values of h . This is a polynomial in h of degree at most $s - 2$. Apply Lemma M.5, with the harmless rescaling by N^{-2} absorbed into its tolerance. We obtain a common $q_0 \leq \rho^{-O_s(1)}$ for which the coefficient of $\binom{h}{j-2}$ yields

$$N^j \|q_0 \Delta^j Q(0)\| \leq \rho^{-O_s(1)} K \quad (j \geq 3).$$

Multiplying by one further bounded integer, choose real lifts of these coefficients in their nearest-integer classes and write

$$q_0 Q(n) = q_0 \alpha \binom{n}{2} + R(n),$$

where $R(0) = R(1) = R(2) = 0$ modulo 1 and $\|R\|_{C^\infty[N]} \leq \rho^{-O_s(1)} K$. Substitute this into q_0 times (B.15) and take the first derivative in n at zero. Since $\Delta\binom{n+h}{2}|_{n=0} = h$,

$$\|q_0 P(1) + q_0 \alpha h + q_0 \sigma(h) + \Delta R(h)\| \leq \rho^{-O_s(1)} K/N.$$

Every nonconstant coefficient of R is already controlled. Clearing the finitely many denominators supplied by (B.16) makes $N\|q_1 \Delta R(h)\| \leq \rho^{-O_s(1)} K$ uniformly for $h \leq N$. With $q = q_0 q_1$ we obtain (B.17), and (B.16) is preserved. $\square$

Lemma M.15 (Bracket-polynomial dichotomy). *Fix r . Let $0 < \rho < 1/2$, let $\gamma \in \mathbb{T}^r$, $\zeta \in \mathbb{R}^r$, and let $|\zeta| \leq \rho^{-A}$. Suppose $|\alpha| \leq \rho^{-A}/N$ and that for at least ρN integers $h \leq N$,*

$$\|\beta + \alpha h + \zeta \cdot \{\gamma h\}\| \leq \rho^{-A}/N. \quad (\text{B.18})$$

Then, with $C = C(A, r)$, either

$$|\zeta_i| \leq \rho^{-C}/N \quad (1 \leq i \leq r), \quad (\text{B.19})$$

or there is $0 \neq k \in \mathbb{Z}^r$, $|k| \leq \rho^{-C}$, such that

$$N\|k \cdot \gamma\| \leq \rho^{-C}. \quad (\text{B.20})$$

Proof. Put $Z = \max_i |\zeta_i|$. If $Z \leq \rho^{-C}/N$ there is nothing to prove, so assume the contrary. We first localise the harmless linear term. A dense set of values of $\beta + \alpha h$ lies in an interval of length $O_r(Z + \rho^{-A}/N)$ modulo one. Taking pairwise differences and using Lemma M.4 shows, unless $N \leq \rho^{-C}$ (when (B.20) is trivial after Dirichlet approximation), that $|\alpha| \leq \rho^{-C_0} Z/N$ after choosing the representative of α of least absolute value.

Partition $[N]$ into intervals of length $L = \lfloor \rho^{C_1} N \rfloor$, where C_1 is much larger than $A + C_0$. One interval I contains at least $\rho|I|/2$ good shifts. Across I the variation of αh is at most $\rho^{C_1 - C_0} Z$. Thus for some θ and at least $\rho|I|/2$ values $h \in I$,

$$\|\theta + \zeta \cdot \{\gamma h\}\| \leq \rho^{C_2} Z \quad (\text{B.21})$$

with C_2 as large as desired by increasing C_1 . Choose i with $|\zeta_i| = Z$. On every line parallel to the i th coordinate axis, the set

$$\Omega = \{x \in \mathbb{T}^r : \|\theta + \zeta \cdot \{x\}\| \leq \rho^{C_2} Z\}$$

is a union of $O(1 + Z)$ intervals, each of length $O(\rho^{C_2})$. Since $Z \leq \rho^{-A}$, Fubini gives $|\Omega| \ll \rho^{C_2 - A}$. Thicken it by radius $r_0 = \rho^{C_3}$ with $C_3 > A + 3$; the same slicing bound gives $|\Omega_{r_0}| < \rho/4$ when the constants are chosen in descending order. Let $\Phi(x) = \max(1 - d(x, \Omega)/r_0, 0)$. Then $\|\Phi\|_{\text{Lip}} \leq 1 + r_0^{-1} = \rho^{-O(1)}$, $\Phi = 1$ on Ω , and $\int \Phi < \rho/4$, while (B.21) gives an average at least $\rho/2$ along γh , $h \in I$. Thus that torus orbit segment is not $\rho^{O(1)}$ -equidistributed. Proposition M.8 (applied after translating and reparametrising I) yields a nonzero $k \in \mathbb{Z}^r$ with $|k| \leq \rho^{-C}$ and $L\|k \cdot \gamma\| \leq \rho^{-C}$. Since $L = \rho^{O(1)} N$, this is (B.20). $\square$

The proof of quantitative Leibman requires one second induction parameter. Put

$$m_{\text{lin}} = \dim(G/G_2), \quad m_{\text{ab}} = \dim(G/[G, G]), \quad m_* = m_{\text{ab}} - m_{\text{lin}} = \dim(G_2/[G, G]). \quad (\text{B.22})$$

We call m_* the *nonlinearity rank* of the filtration. The induction is lexicographic in (s, m_*) : first on the filtration degree, and at fixed degree on m_* . This is essential; degree descent alone can stop at a reducible filtration.

Lemma M.16 (Finer filtration in the reducible case). *Let $\eta_2 : G_2 \rightarrow \mathbb{R}/\mathbb{Z}$ be nonzero, annihilate $[G, G_2]$, and suppose in addition that it annihilates $[G, G]$. Let $\tilde{\eta}_2 : G_2 \rightarrow \mathbb{R}$ be its differential lift. Define*

$$G'_0 = G'_1 = G, \quad G'_i = G_i \cap \ker \tilde{\eta}_2 \quad (i \geq 2). \quad (\text{B.23})$$

Then $G'_\bullet$ is a rational filtration of degree at most s , and its nonlinearity rank is at most $m_ - 1$.*

Proof. Because G_2 is connected and simply connected, a continuous character on it has a unique real differential lift $\tilde{\eta}_2(\exp X) = d\eta_2(X)$; the assumption that η_2 kills $[G_2, G_2] \subseteq [G, G_2]$ makes this a homomorphism. Its differential has rational coefficients in an adapted Mal'cev basis because η_2 annihilates $\Gamma \cap G_2$. Hence every G'_i is a connected rational subgroup by Appendix A.

For $i, j \geq 2$, commutators lie in $[G_2, G_2]$ and are killed by $\tilde{\eta}_2$, so $[G'_i, G'_j] \subseteq G'_{i+j}$. Also $[G, G'_i] \subseteq [G, G_i] \subseteq [G, G_2] \cap G_{i+1} \subseteq G'_{i+1}$. Thus $G'_\bullet$ is a filtration. Since η_2 is nonzero on $G_2/[G, G]$, intersecting G_2 with its kernel lowers $\dim(G_2/[G, G])$ by at least one, proving the rank assertion. $\square$

Lemma M.17 (Reducible-sequence splitting). *Assume the hypotheses of Lemma M.16 and*

$$\|\eta_2 \circ g_2\|_{C^\infty[N]} \leq K, \quad g_2(n) = g(n)g(1)^{-n}, \quad (\text{B.24})$$

with K polynomially bounded in the ambient complexity. Then

$$g = \varepsilon g' \gamma \quad (\text{B.25})$$

where ε is $(K^{O(1)}, N)$ -smooth, $\gamma(n)\Gamma$ is periodic of period $K^{O(1)}$, and $g' \in \text{poly}(\mathbb{Z}, G'_\bullet)$. Moreover every horizontal character χ of complexity $K^{O(1)}$ satisfies $\|\chi \circ \varepsilon\|_{C^\infty[N]} \leq K^{O(1)}$.

Proof. Choose Mal'cev coordinates adapted to $G_\bullet$. The coordinate Taylor formula (proved by induction from BCH and (B.13)) gives

$$\psi(g_2(n)) = \sum_{j=2}^s \binom{n}{j} t_j, \quad t_j \in \mathfrak{g}_j \quad (\text{B.26})$$

with the coordinates of t_j before the $\mathfrak{g}_j$ block equal to zero. Write $\tilde{\eta}_2(X) = k \cdot X$ in these coordinates. From (B.24), for each $j \geq 2$ choose an integer a_j such that

$$|k \cdot t_j - a_j| \leq KN^{-j}.$$

If k vanishes on $\mathfrak{g}_j$, set $u_j = t_j$ and $v_j = 0$. Otherwise alter t_j in one allowed coordinate on which k is nonzero to obtain $u_j \in \mathfrak{g}_j$ with $k \cdot u_j = a_j$ and $|t_j - u_j| \leq K^{O(1)}N^{-j}$. Choose $v_j \in \mathfrak{g}_j(\mathbb{Q})$ with the same value $k \cdot v_j = a_j$ and rational height $K^{O(1)}$ (put the whole value in one nonzero rational coordinate of k).

Define polynomial sequences through their coordinate Taylor data by

$$\psi(\varepsilon(n)) = \sum_{j=2}^s \binom{n}{j} (t_j - u_j), \quad \psi(\gamma(n)) = \sum_{j=2}^s \binom{n}{j} v_j,$$

and set $g' = \varepsilon^{-1} g \gamma^{-1}$. BCH and the filtration law show that these coordinate polynomials are filtered polynomial sequences. The bounds on $t_j - u_j$ give $d(\varepsilon(n+1), \varepsilon(n)) \leq K^{O(1)}/N$ and bounded coordinates on $[N]$, hence smoothness; the same coefficient estimates give the asserted C^∞ bounds for bounded horizontal characters. Rational Taylor coefficients of bounded common denominator make $\gamma(n)\Gamma$ periodic after multiplying the period by a bounded power accounting for BCH denominators (Appendix A).

It remains to locate g' . The nonlinear factor $\varepsilon^{-1} g_2 \gamma^{-1}$ is annihilated by $\tilde{\eta}_2$ coefficient by coefficient, because $k \cdot (t_j - (t_j - u_j) - v_j) = a_j - a_j = 0$. The linear sequence $g(1)^n$ is polynomial for every filtration beginning with $G'_0 = G'_1 = G$. When it is moved past γ , the commutator correction lies in $[G, G_2] \subseteq \ker \tilde{\eta}_2$ and in the correct deeper filtration level. Closure under products from (B.13) therefore gives $g' \in \text{poly}(\mathbb{Z}, G'_\bullet)$. $\square$

Lemma M.18 (Single-progression extrapolation). *Let $P(n) = \sum_{j=0}^s \alpha_j \binom{n}{j}$ and let $\tilde{P}(n) = P(a + qn)$ with $1 \leq q \leq Q$ and $|a| \leq QN$. Then some $1 \leq r \leq Q^{O_s(1)}$ satisfies*

$$\|rP\|_{C^\infty[N]} \leq Q^{O_s(1)} \|\tilde{P}\|_{C^\infty[N]}. \quad (\text{B.27})$$

Proof. The binomial identity $\binom{a+qn}{j} = \sum_{\ell \leq j} c_{j\ell}(a, q) \binom{n}{\ell}$ has rational coefficients with denominators dividing a fixed power of $j!$ and with $|c_{j\ell}| \leq Q^{O_s(1)} N^{j-\ell}$ after scaling by the relevant powers of N . Its coefficient matrix is triangular with diagonal q^j . Back-substitution from degree s down to degree one, multiplying once by $\prod_{j \leq s} q^j (j!)^{O_s(1)} \leq Q^{O_s(1)}$, yields (B.27). $\square$

Theorem M.19 (Quantitative Leibman theorem, detailed form). *Under the hypotheses of [theorem J.2](#), failure of δ -equidistribution produces a nonzero horizontal character η with*

$$|\eta|, \|\eta \circ g\|_{C^\infty[N]} \leq M_0^{O_{m,s}(1)} \delta^{-O_{m,s}(1)}.$$

Proof. Let K denote a quantity of the form $M_0^{O_{m,s}(1)} \delta^{-O_{m,s}(1)}$; it may increase by a fixed power from line to line. We prove the theorem lexicographically in (s, m_*) . The abelian degree-one case is Proposition M.8. The assertion is trivial when $N \leq K$, by Dirichlet approximation to the finitely many horizontal Taylor coefficients, so assume $N > K$.

Choose a mean-zero Lipschitz witness and apply Lemma M.9 with a polynomially large cutoff. One vertical mode F of frequency $|\xi| \leq K$ has correlation at least K^{-1} . If $\xi = 0$, F descends to G/G_s , whose quotient filtration has degree $s-1$, and the outer induction gives the required horizontal character after pullback.

Assume $\xi \neq 0$. Left translation and multiplication by a lattice-valued linear sequence allow us to normalise $g(0) = e$ and to choose the Mal'cev representative of $a = g(1)$ in a fixed fundamental box; these operations do not change the horizontal Taylor seminorms except by a factor $K^{O(1)}$. Van der Corput, Lemma M.10, gives at least $K^{-1}N$ shifts h for which

$$\left| \mathbb{E}_n F(g(n+h)\Gamma) \overline{F(g(n)\Gamma)} \right| \geq K^{-1}. \quad (\text{B.28})$$

Use Lemma M.12 to rewrite each correlation on the relative square. The descended function has Haar integral zero: indeed $G_s \times \{e\} \subseteq G^\square$, and integration along this fiber multiplies the integrand by the nontrivial character ξ . By Lemma M.11, quotienting by G_s^Δ leaves a filtered nilmanifold of degree at most $s-1$. The outer induction therefore gives, for every good h , a nonzero horizontal character of the quotient with complexity K . There are only $K^{O(1)}$ such

integer characters, so pigeonholing gives one lifted character η of $G^\square$ for a set $\mathcal{H}$ of at least $K^{-1}N$ shifts, with

$$\|\eta \circ g_h^\square\|_{C^\infty[N]} \leq K. \quad (\text{B.29})$$

Decompose η by Lemma M.13. A direct commutator calculation, using that η_2 kills $[G, G_2]$, gives

$$\eta(g_h^\square(n)) = P(n) + Q(n+h) - Q(n) + \sigma(h)n, \quad (\text{B.30})$$

where $P = \eta_1 \circ g$, $Q = \eta_2 \circ g_2$, and $\sigma(h) = \eta_2([a, \{a^h\}])$ up to the fixed commutator convention. To verify (B.30), substitute (B.12) into (B.14), move every G_2 term through a^n , and use $y^{-1}a^ny a^{-n} = [a, y]^n$ modulo $[G, G_2]$; all discarded terms are killed by η_2 . If $\eta_2 = 0$, (B.29) is already the required bound for the nonzero horizontal character η_1 , so suppose $\eta_2 \neq 0$.

Apply Lemma M.14 to (B.29)–(B.30). After multiplying η_1, η_2 by one integer at most K , we obtain the higher coefficient bounds for Q and, on a dense set of shifts,

$$\|\beta + \alpha h + \eta_2([a, \{a^h\}])\| \leq K/N, \quad (\text{B.31})$$

where $\beta = \eta_1(a)$ and α is the quadratic Taylor coefficient of Q (again after a bounded integer multiple). Because η_2 kills $[G, G_2]$, the map $x \mapsto \eta_2([a, x])$ is a homomorphism on G/G_2 . In the first m_{lin} Mal'cev coordinates it has the form $\zeta \cdot \psi_{\text{lin}}(x)$ with $|\zeta| \leq K$. Writing $\gamma = \psi_{\text{lin}}(a)$ and using $\psi_{\text{lin}}(\{a^h\}) = \{h\gamma\}$ turns (B.31) into

$$\|\beta + \alpha h + \zeta \cdot \{h\gamma\}\| \leq K/N. \quad (\text{B.32})$$

We do not yet know that $|\alpha| \ll K/N$. Apply Lemma M.15 first in dimension $m_{\text{lin}} + 1$ to $(\zeta, 1)$ and $(\gamma, q\alpha)$ with zero linear coefficient. It yields either an immediate nonzero integer relation $N\|k \cdot \gamma\| \leq K$, or a bounded integer r for which every coordinate of $r\zeta$ is $O(K/N)$ after subtracting an integer vector. In the latter case multiply (B.32) by r and absorb that integer vector into the torus character; a second application of the bracket dichotomy, now with the small linear coefficient, yields the same horizontal resonance unless

$$|r\zeta_i| \leq K/N \quad (1 \leq i \leq m_{\text{lin}}). \quad (\text{B.33})$$

In the resonance case, the character using the integer vector k on G/G_2 is nonzero and satisfies the required $C^\infty[N]$ bound because $g(n)G_2 = a^n G_2$.

Assume (B.33). For every basis vector X_i define

$$\tau_i(x) = r\eta_2([x, \exp X_i]).$$

The commutator identities and the fact that η_2 kills $[G, G_2]$ show that τ_i is a horizontal character of G annihilating G_2 ; its coordinate size is K . Moreover $\tau_i(g(n)) = nr\zeta_i$, so (B.33) gives the required smoothness bound. If some τ_i is nonzero, we are done.

It remains that every τ_i vanishes. Since the basis generates G and $y \mapsto \eta_2([x, y])$ is a homomorphism modulo $[G, G_2]$, this implies that η_2 annihilates $[G, G]$, hence $\zeta = 0$. Equation (B.32), together with Lemma M.4, controls α after a bounded integer multiple. Combining this with (B.16) gives

$$\|\eta_2 \circ g_2\|_{C^\infty[N]} \leq K. \quad (\text{B.33a})$$

Now apply Lemmas M.16 and M.17: write $g = \varepsilon g' \gamma_0$ with g' polynomial in a rational filtration of the same degree but nonlinearity rank at most $m_* - 1$.

The original correlation survives on one long residue-class progression on which $\gamma_0(n)\Gamma$ is constant and ε changes by at most a small fixed amount. Explicitly, split $[N]$ first by the period $Q \leq K$ of γ_0 and then into blocks of length $N' = \lfloor K^{-C}N \rfloor$. One block has correlation at least K^{-1} ; freezing ε at its left endpoint changes it by at most half that

amount when C is large. Thus a translated/reparametrised polynomial sequence from g' fails K^{-1} -equidistribution on $[N']$. The inner induction on m_* gives a nonzero horizontal character χ of the finer filtration with $\|\chi \circ g'(a + Q \cdot)\|_{C^\infty[N']} \leq K$. Lemma M.18 transfers this to $\|q\chi \circ g'\|_{C^\infty[N]} \leq K$ for a bounded q . Multiply once more by the bounded period denominator so that $\chi \circ \gamma_0$ has zero nonconstant Taylor coefficients. The last property of Lemma M.17 controls $\chi \circ \varepsilon$. Since horizontal characters are homomorphisms,

$$\|\chi \circ g\|_{C^\infty[N]} \leq \|\chi \circ \varepsilon\|_{C^\infty[N]} + \|\chi \circ g'\|_{C^\infty[N]} + \|\chi \circ \gamma_0\|_{C^\infty[N]} \leq K.$$

This closes the inner induction, and hence the lexicographic induction.

For the converse, let $0 \neq \eta$ be horizontal and suppose $\|\eta \circ g\|_{C^\infty[N]} \leq R$. Clearing one denominator bounded by $R^{O_s(1)}$ makes every nonconstant Taylor coefficient at most $R^{O_s(1)}N^{-j}$. On a progression of length $\gg R^{-O_s(1)}N$ the phase $e(\eta(g(n)))$ varies by less than $1/10$, while its Haar integral on G/Γ is zero. This is a quantitative obstruction to total equidistribution. $\square$

M.4 Lifting phase decompositions

Lemma M.20 (Smooth–rational splitting of a polynomial phase). *Let $P(n) = \sum_{j=0}^s \alpha_j \binom{n}{j}$ and suppose $N^j \|q_j \alpha_j\| \leq M$ with $1 \leq q_j \leq M$ for $j \geq 1$. Then*

$$P = P_{\text{sm}} + P_{\text{rat}} + P_0$$

where P_0 is constant, every nonconstant coefficient of P_{sm} is $O_s(M^{O(1)}N^{-j})$, and every coefficient of P_{rat} is rational with a common denominator $Q \leq M^{O_s(1)}$.

Proof. Let Q be the least common multiple of the q_j , bounded by their product. Choose $a_j \in \mathbb{Z}$ with $|\alpha_j - a_j/Q| \leq M^{O(1)}N^{-j}$. Put the rational pieces a_j/Q into P_{rat} and the errors into P_{sm} . The constant coefficient is placed in P_0 . $\square$

Lemma M.21 (Layerwise lifting of a horizontal phase). *Let $\phi : G \rightarrow \mathbb{R}$ be the real homomorphism whose differential is a primitive integral horizontal covector, and let $d \geq 1$. For every j for which $\phi|_{G_j}$ is nonzero there is a rational vector $Y_j \in \mathfrak{g}_j$ and an integer $1 \leq r_j \leq M^{O_m(1)}$ such that*

$$d\phi(\exp Y_j) = r_j. \tag{B.34}$$

The heights of the Y_j are $M^{O_m(1)}$. If $\phi|_{G_j} = 0$, the j th coefficient of $d\phi \circ g$ is zero for every $g \in \text{poly}(\mathbb{Z}, G_\bullet)$.

Proof. Choose an adapted rational Mal'cev basis. The differential of $d\phi$ is an integer row vector k of height $M^{O(1)}$. Its restriction to the rational subspace $\mathfrak{g}_j$ is again a rational row vector of that height. If the restriction is nonzero, Hermite–Smith reduction of a bounded-height rational basis of $\mathfrak{g}_j$ gives a rational vector $Y_j \in \mathfrak{g}_j$ for which $k(Y_j) = r_j \neq 0$, with $|r_j|$ and the height of Y_j bounded by a fixed power of M ; change the sign if necessary. This is (B.34).

For the last assertion, write the group Taylor expansion $g(n) = c_0 c_1^{\binom{n}{1}} \cdots c_s^{\binom{n}{s}}$ with $c_j \in G_j$. A homomorphism sends this to the additive Taylor expansion whose j th coefficient is $d\phi(c_j)$. It vanishes when $\phi|_{G_j} = 0$. $\square$

Lemma M.22 (Horizontal splitting lifts to the group). *Let η be a nonzero horizontal character of rational complexity at most M , let $g \in \text{poly}(\mathbb{Z}, G_\bullet)$, and suppose $\eta \circ g = P_{\text{sm}} + P_{\text{rat}} + P_0$ is*

a splitting supplied by [theorem M.20](#). Then, after increasing M by a fixed power, there are filtered polynomial sequences

$$g(n) = \varepsilon_1(n)g_1(n)\gamma_1(n) \quad (\text{B.35})$$

with:

1. ε_1 ($M^{O(1)}, N$)-smooth;
2. g_1 taking values in a proper connected rational subgroup $H < G$;
3. γ_1 rational of complexity $M^{O(1)}$, hence periodic modulo Γ with period $M^{O(1)}$.

One may take $H = \ker \phi$, where $\phi : G \rightarrow \mathbb{R}$ is the real homomorphism associated with the primitive integer covector in the direction of η . All rationality bounds are $M^{O_{m,s}(1)}$.

Proof. Write the integer horizontal covector of η as $k = dk_0$, where k_0 is primitive. Because G is connected and simply connected nilpotent, the Lie-algebra homomorphism determined by k_0 integrates to a real homomorphism $\phi : G \rightarrow \mathbb{R}$, and

$$\eta(x) = d\phi(x) \pmod{1}. \quad (\text{B.36})$$

The subgroup $H = \ker \phi$ is connected (in exponential coordinates it is the kernel of a nonzero linear functional), rational by Appendix A, and proper.

Write the group Taylor expansion

$$g(n) = c_0 c_1^{\binom{n}{1}} \cdots c_s^{\binom{n}{s}}, \quad c_j \in G_j. \quad (\text{B.37})$$

Put $A_j = d\phi(c_j) \in \mathbb{R}$. Its class modulo one is precisely the j th Taylor coefficient α_j of $\eta \circ g$. For $j \geq 1$ choose, as in [theorem M.20](#),

$$A_j = b_j + \rho_j + \sigma_j, \quad b_j \in \mathbb{Z}, \quad \rho_j \in Q^{-1}\mathbb{Z}, \quad |\sigma_j| \leq M^{O(1)}N^{-j}, \quad (\text{B.38})$$

with one $Q \leq M^{O_s(1)}$. If $d\phi|_{G_j} = 0$, then $A_j = 0$ and all three terms are taken to be zero. Otherwise choose Y_j, r_j from [theorem M.21](#). Enlarge Q by the product of the finitely many r_j .

Define the j th Taylor generator of the smooth factor by

$$e_j = \exp\left(\frac{\sigma_j}{r_j} Y_j\right) \in G_j$$

and that of the rational factor by

$$q_j = \exp\left(\frac{b_j + \rho_j}{r_j} Y_j\right) \in G_j.$$

The constant coefficient is treated in $G_0 = G$ in the same way, after first multiplying the whole decomposition by one fixed element; constants do not affect filtered polynomiality. Set

$$\varepsilon_1(n) = e_0 e_1^{\binom{n}{1}} \cdots e_s^{\binom{n}{s}}, \quad \gamma_1(n) = q_0 q_1^{\binom{n}{1}} \cdots q_s^{\binom{n}{s}}. \quad (\text{B.39})$$

Every e_j, q_j lies in the correct filtration layer G_j . The polynomial closure lemma (B.13), proved above from the commutator filtration law, therefore shows that both sequences are in $\text{poly}(\mathbb{Z}, G_\bullet)$. This layerwise choice is essential: lifting every degree along one generic horizontal direction would not preserve filtered polynomiality.

The logarithmic size of e_j is $M^{O(1)}N^{-j}$. BCH in a bounded Mal'cev box therefore gives $d(\varepsilon_1(n), e) \leq M^{O(1)}$ and $d(\varepsilon_1(n+1), \varepsilon_1(n)) \leq M^{O(1)}/N$ for $n \leq N$; hence ε_1 is smooth. The q_j have rational Mal'cev coordinates with bounded denominators. Their numerators may

be reduced modulo the corresponding rational lattice without changing the orbit modulo Γ . Appendix A and BCH denominator clearing then give a common period $M^{O_{m,s}(1)}$ for $\gamma_1(n)\Gamma$.

Set $h = \varepsilon_1^{-1}g\gamma_1^{-1}$. Applying $d\phi$ and using (B.38)–(B.39), every nonconstant Taylor coefficient of $d\phi \circ h$ is zero. Thus $d\phi(h(n))$ is a constant C . Choose a fixed rational-height element x_0 with $d\phi(x_0) = C$ modulo the lattice period, absorb the bounded fundamental-domain part of x_0 into ε_1 , and absorb its rational lattice-direction part into γ_1 . The remaining middle factor has $\phi = 0$ identically and hence takes values in H .

For clarity about the preceding constant adjustment, in horizontal Mal'cev coordinates it is just Euclidean division in the primitive direction k_0 : the bounded remainder is a constant smooth factor, while the integral/rational quotient is a constant rational factor. No assertion that $\ker \eta / (\ker \eta)^\circ$ is finite is used; that quotient can be infinite before passing modulo the lattice. This proves (B.35). $\square$

M.5 The factorization descent

Theorem M.23 (Factorization by finite rational-kernel descent). *The factorization asserted in [theorem K.2](#) holds, with all parameters bounded by a fixed power tower determined only by the initial rationality, M_0 , the dimension, the degree, and the requested exponent A .*

Proof. We maintain an ordered factorization

$$g = \varepsilon^{(r)} g^{(r)} \gamma^{(r)}, \quad (\text{B.40})$$

where $g^{(r)}$ takes values in a connected rational subgroup $G^{(r)}$ and the outer factors have controlled smooth and rational Taylor data. At stage zero these factors are trivial and $G^{(0)} = G$.

If $g^{(r)}$ is totally M_r^{-A} -equidistributed in $G^{(r)}/(\Gamma \cap G^{(r)})$, stop. Otherwise a progression $P = a + q[L]$, with $q \leq M_r^A$ and $L \geq M_r^{-A}N$, witnesses failure. Apply [theorem M.19](#) to the reparametrized sequence $g^{(r)}(a + qn)$. This gives a nonzero horizontal character $\tilde{\eta}_r$ whose $C^\infty[L]$ norm is $M_r^{O_A(1)}$. Now apply [theorem M.18](#) to each Taylor coefficient. After multiplying by one integer $u_r \leq M_r^{O_{s,A}(1)}$, the character $\eta_r = u_r \tilde{\eta}_r$ satisfies

$$\|\eta_r \circ g^{(r)}\|_{C^\infty[N]} \leq M_r^{O_{m,s,A}(1)}. \quad (\text{B.41})$$

Thus no unproved “restriction-to-progression inverse” is being used.

Apply [theorems M.20](#) and [M.22](#). We obtain

$$g^{(r)} = \varepsilon_r g^{(r+1)} \gamma_r, \quad G^{(r+1)} = \ker \phi_r < G^{(r)}, \quad (\text{B.42})$$

where ϕ_r is the primitive real horizontal homomorphism associated with η_r . The subgroup $G^{(r+1)}$ is connected, rational, and proper, so

$$\dim G^{(r+1)} < \dim G^{(r)}. \quad (\text{B.43})$$

Substituting (B.42) directly into (B.40) gives

$$g = (\varepsilon^{(r)} \varepsilon_r) g^{(r+1)} (\gamma_r \gamma^{(r)}). \quad (\text{B.44})$$

No factor is commuted past another. Products of the smooth factors remain (M_{r+1}, N) -smooth: all factors stay in a bounded Mal'cev coordinate box and multiplication/inversion are polynomial-Lipschitz there, so the first-difference bound follows from the product rule. For the rational factors we do not infer periodicity merely by multiplying two periodic coset

sequences (which would be unsafe in a nonnormal lattice). Instead use their rational group Taylor coefficients. Closure of filtered polynomial sequences and BCH show that $\gamma_r \gamma^{(r)}$ again has rational Taylor coefficients with denominators bounded by a fixed power of the old ones; Appendix A then supplies a new common period modulo Γ .

Choose $M_{r+1} = M_r^{C_{m,s,A}}$, enlarged if necessary to dominate the rationality of the induced Mal'cev basis on $G^{(r+1)}$. Since (B.43) can occur at most $m = \dim G$ times, the descent terminates. At termination the middle sequence is totally M^{-A} -equidistributed, while the accumulated left and right factors have the required smoothness and rational periodicity. Taking M to be the largest parameter encountered proves the factorization theorem. $\square$

Appendix N

The Equal-Sides Multiparameter Counting Theorem

The multiparameter case requires care. A naive proof by changing the preferred coordinate direction after producing a horizontal character is invalid because the character can change with the direction. We therefore prove precisely the equal-sides statement needed in this book: all parameter boxes have side length comparable with one common scale N . The proof keeps one horizontal character fixed while varying many line directions.

N.1 Multivariate smoothness norms

Let $\mathbf{i} = (i_1, \dots, i_d)$ be a multi-index and $\binom{\mathbf{n}}{\mathbf{i}} = \prod_r \binom{n_r}{i_r}$. Every polynomial phase of total degree at most s has a unique expansion

$$P(\mathbf{n}) = \sum_{|\mathbf{i}| \leq s} \alpha_{\mathbf{i}} \binom{\mathbf{n}}{\mathbf{i}} \pmod{1}.$$

On $[N]^d$ set

$$\|P\|_{C^\infty[N]^d} = \max_{0 < |\mathbf{i}| \leq s} N^{|\mathbf{i}|} \|\alpha_{\mathbf{i}}\|_{\mathbb{R}/\mathbb{Z}}.$$

Lemma N.1 (Schwartz–Zippel on an integer box). *If $Q \in \mathbb{R}[x_1, \dots, x_d]$ is nonzero of total degree at most s , then*

$$|\{\mathbf{x} \in [L]^d : Q(\mathbf{x}) = 0\}| \leq C_{d,s} L^{d-1}.$$

Proof. Induct on d . Write $Q = \sum_{j=0}^s Q_j(x_1, \dots, x_{d-1})x_d^j$ and choose a nonzero coefficient Q_{j_0} . By induction, at most CL^{d-2} base points annihilate Q_{j_0} . For every other base point the resulting one-variable polynomial is nonzero and has at most s roots. Hence the number of roots is $O(L^{d-2})L + L^{d-1}s = O(L^{d-1})$. $\square$

Lemma N.2 (Many small polynomial values force small coefficients). *Fix d, s . Let $Q(\mathbf{x}) = \sum_{|\mathbf{i}| \leq s} \beta_{\mathbf{i}} \mathbf{x}^{\mathbf{i}}$ with coefficients in $\mathbb{R}/\mathbb{Z}$. Suppose that for at least ρL^d points $\mathbf{x} \in [L]^d$ one has $\|Q(\mathbf{x})\| \leq \varepsilon$, where*

$$L \geq \rho^{-C_{d,s}}, \quad \varepsilon \leq c_{d,s} \rho^{C_{d,s}}.$$

Then there is $1 \leq q \leq \rho^{-O_{d,s}(1)}$ such that

$$L^{|\mathbf{i}|} \|q\beta_{\mathbf{i}}\| \ll_{d,s} \rho^{-O(1)} \varepsilon \quad \text{for every } \mathbf{i}.$$

Proof. This is exactly Lemma M.6, proved in Appendix B by induction on the number of variables from the one-variable dense coefficient-recovery lemma. That proof preserves the factor ε at every finite-difference stage; this preservation is essential here because the applications below take ε of order N^{-j} . The change between ordinary monomials and multivariate binomial monomials is a fixed triangular rational change of basis and costs only a factor depending on d, s . $\square$

N.2 From bad cubes to one fixed horizontal character

Lemma N.3 (Line covering of a cube). *Let F be 1-Lipschitz and mean zero, and suppose*

$$\left| \mathbb{E}_{\mathbf{n} \in [N]^d} F(g(\mathbf{n})\Gamma) \right| > \delta.$$

Let $L \geq C_d \delta^{-1}$ and $N \geq L^2$. For every direction $\mathbf{q} \in [L]^d$, at least $c_d \delta N^d$ base points $\mathbf{x} \in [N]^d$ have

$$\left| \mathbb{E}_{1 \leq r \leq N/L^2} F(g(\mathbf{x} + r\mathbf{q})\Gamma) \right| > \delta/2.$$

Proof. Average the line averages over $\mathbf{x}$. Every interior point of $[N]^d$ is counted the same number of times after translating $\mathbf{x}$, while points within distance N/L of the boundary contribute an error $O_d(L^{-1})$ because $|\mathbf{q}| \leq dL$ and the line length is N/L^2 . Thus the average over $\mathbf{x}$ differs from the original cube average by at most $C_d/L < \delta/4$. If fewer than $c_d \delta N^d$ base points had line average exceeding $\delta/2$, the total average would be at most $\delta/2 + c\delta + O(1/L) < 3\delta/4$ for c small, contradiction. $\square$

Theorem N.4 (Multiparameter quantitative obstruction, equal sides). *Fix d, m, s and bounded rationality. Let $g \in \text{poly}(\mathbb{Z}^d, G_\bullet)$. If $(g(\mathbf{n})\Gamma)_{\mathbf{n} \in [N]^d}$ is not δ -equidistributed, then there is a nonzero horizontal character η with*

$$|\eta| \leq \delta^{-O_{d,m,s}(1)}, \quad \|\eta \circ g\|_{C^\infty[N]^d} \leq \delta^{-O_{d,m,s}(1)}.$$

The same conclusion holds for boxes whose side lengths are between cN and CN , with constants also depending on c, C .

Proof. Choose $L = \delta^{-C_1}$, with C_1 large, and assume first that $N \geq L^{C_2}$; for smaller N the conclusion follows from simultaneous Dirichlet approximation after enlarging the polynomial loss in δ^{-1} . Apply theorem N.3. For every $\mathbf{q} \in [L]^d$ there are $\gg \delta N^d$ base points $\mathbf{x}$ for which the restriction $r \mapsto g(\mathbf{x} + r\mathbf{q})$ fails quantitative equidistribution on a line of length $R \asymp N/L^2$. The one-dimensional quantitative Leibman theorem assigns to each bad pair $(\mathbf{x}, \mathbf{q})$ a nonzero horizontal character in a coefficient box of cardinality $\delta^{-O(1)}$. Pigeonholing over all bad pairs, and then discarding low-density fibers, produces one fixed character η , a set $\mathcal{Q} \subset [L]^d$ with $|\mathcal{Q}| \gg \delta^{O(1)} L^d$, and for every $\mathbf{q} \in \mathcal{Q}$ a set $X_{\mathbf{q}} \subset [N]^d$ of size $\gg \delta^{O(1)} N^d$ such that

$$P_{\mathbf{x}, \mathbf{q}}(r) = \eta(g(\mathbf{x} + r\mathbf{q}))$$

has $C^\infty[R]$ norm at most $\delta^{-O(1)}$ whenever $\mathbf{x} \in X_{\mathbf{q}}$. Thus the same horizontal character is kept while both the base point and the direction vary.

Write in ordinary multivariate Taylor coordinates

$$\eta(g(\mathbf{n})) = \sum_{|\mathbf{i}| \leq s} \beta_{\mathbf{i}} \mathbf{n}^{\mathbf{i}}.$$

For $1 \leq j \leq s$ the coefficient of r^j in $P_{\mathbf{x}, \mathbf{q}}$ is the polynomial

$$C_j(\mathbf{x}, \mathbf{q}) = \sum_{|\mathbf{i}| \geq j} \beta_{\mathbf{i}} \sum_{|\mathbf{h}|=j} c_{\mathbf{i}, \mathbf{h}} \mathbf{x}^{\mathbf{i}-\mathbf{h}} \mathbf{q}^{\mathbf{h}}. \quad (\text{C.1})$$

For all the selected pairs,

$$\|C_j(\mathbf{x}, \mathbf{q})\| \leq \delta^{-O(1)} R^{-j} = \delta^{-O(1)} (L^2/N)^j. \quad (\text{C.2})$$

We now recover the coefficients by a two-scale fiber argument; this is the step that prevents an invalid change of preferred direction.

Start with $j = s$. Then C_s is independent of $\mathbf{x}$ and homogeneous of degree s in $\mathbf{q}$. Apply [theorem N.2](#) on the dense set $\mathcal{Q}$. Since $L = \delta^{-O(1)}$, (C.2) yields, after multiplying by $q_s \leq \delta^{-O(1)}$,

$$N^s \|q_s \beta_{\mathbf{i}}\| \leq \delta^{-O(1)} \quad (|\mathbf{i}| = s). \quad (\text{C.3})$$

Indeed the recovery first gives an L^{-s} factor, while (C.2) contains $L^{2s} N^{-s}$; all powers of L are absorbed into $\delta^{-O(1)}$.

Suppose inductively that all coefficients of total degree $> t$ have been controlled at their natural N -scales by one common denominator. Multiply (C.1) by that denominator and subtract the already controlled higher-degree contribution. Take $j = t$. The remaining leading part is

$$\sum_{|\mathbf{i}|=t} \beta_{\mathbf{i}} \sum_{|\mathbf{h}|=t} c_{\mathbf{i},\mathbf{h}} \mathbf{q}^{\mathbf{h}},$$

again independent of $\mathbf{x}$, so recovery in $\mathbf{q}$ gives (C.3) with s replaced by t . Equivalently, one may take any $j < t$ first: for each $\mathbf{q}$ the dense set $X_{\mathbf{q}}$ lets [theorem N.2](#) recover the coefficients as polynomials in $\mathbf{x}$; pigeonholing the finitely many denominators and then applying the same lemma in $\mathbf{q}$ gives the identical conclusion. This fiberwise version explains why the unequal scales N and L cause no gap. Descending through $t = s, s-1, \dots, 1$ gives one $q \leq \delta^{-O(1)}$ with

$$N^{|\mathbf{i}|} \|q \beta_{\mathbf{i}}\| \leq \delta^{-O(1)} \quad (0 < |\mathbf{i}| \leq s).$$

Replace η by $q\eta$. The ordinary/binomial change of basis is triangular with coefficients depending only on d, s , so the same bound holds for the multivariate C^∞ norm.

For comparable side lengths, partition into $O_{c,C}(1)$ boxes with one common side scale and choose a subbox carrying a fixed proportion of the discrepancy. The preceding argument applies verbatim there. $\square$

N.3 Multiparameter factorization

Theorem N.5 (Equal-scale multiparameter factorization). *Let $g : \mathbb{Z}^d \rightarrow G$ be polynomial and consider $[N]^d$. Fix $A > 0$ and an initial rationality scale M_0 . There is $M_0 \leq M \leq M_0^{O_{d,m,s,A}(1)}$ and*

$$g(\mathbf{n}) = \varepsilon(\mathbf{n}) g'(\mathbf{n}) \gamma(\mathbf{n})$$

such that:

1. *changing one coordinate of $\mathbf{n}$ by one changes ε by $O(M/N)$, and ε stays in an M -bounded coordinate box;*
2. *g' takes values in a connected M -rational subgroup and is totally M^{-A} -equidistributed on every comparable subbox and residue class of density at least M^{-A} ; and*
3. *$\gamma(\mathbf{n})\Gamma$ is periodic in every coordinate with common period at most $M^{O(1)}$.*

Proof. Run the rational-kernel descent of Appendix B with [theorem N.4](#) in place of the one-parameter obstruction theorem. At a bad stage a fixed horizontal character η has a multivariate binomial phase

$$\eta(g(\mathbf{n})) = \sum_{\mathbf{i}} \alpha_{\mathbf{i}} \binom{\mathbf{n}}{\mathbf{i}}$$

whose coefficients are within $M^{O(1)}N^{-|\mathbf{i}|}$ of rationals with one bounded common denominator. Use the primitive real horizontal homomorphism ϕ associated with η . For every multi-degree $\mathbf{i}$, the group Taylor generator lies in the filtration layer $G_{|\mathbf{i}|}$. If ϕ is nonzero on that layer, the rational linear-algebra lemma [theorem M.21](#) supplies a controlled rational direction $Y_{|\mathbf{i}|} \in \mathfrak{g}_{|\mathbf{i}|}$; if it vanishes, the corresponding phase coefficient is zero. Lift the small and rational pieces along these *layer-correct* directions. Hence the lifted generators remain in $G_{|\mathbf{i}|}$ and define filtered polynomial sequences.

The small coefficient of multi-degree $\mathbf{i}$ is $O(M^{O(1)}N^{-|\mathbf{i}|})$. Taking a unit difference in any coordinate lowers the total degree by one, so on $[N]^d$ the accumulated smooth factor moves by $O(M^{O(1)}/N)$ per coordinate step and stays in a controlled Mal'cev box. The rational Taylor generators have bounded denominators; Appendix A therefore gives a common period in every coordinate. Removing the lifted phase leaves the middle sequence in the proper connected rational kernel $\ker \phi$.

Substitution into the ordered factorization is exactly as in (B.44), so no noncommuting factors are moved past one another. Products of smooth factors are smooth by the bounded-coordinate Lipschitz calculus, while products of rational factors are handled through their rational Taylor coefficients rather than by multiplying coset periods. Every failure of total equidistribution strictly lowers the dimension of the connected middle subgroup; after at most m steps the process stops. Choosing the complexity parameter recursively by fixed powers gives the asserted M and all three properties. $\square$

N.4 Freezing and counting

Theorem N.6 (Polynomial nilsequence counting on equal-scale boxes). *Let $B \subset [-N, N]^d$ be a box whose side lengths are all between cN and CN . Under fixed complexity hypotheses, for every $\delta > 0$ one may partition $B \cap \mathbb{Z}^d$ into $O_{\delta, c, C}(1)$ cells, each the intersection of a residue class of controlled modulus with a box, so that on every cell*

$$g = \varepsilon g' \gamma$$

and

$$\mathbb{E}_{\mathbf{n} \in \mathcal{C}} F(g(\mathbf{n})\Gamma) = \int_{Y_{\mathcal{C}}} F_{\mathcal{C}} dm_{Y_{\mathcal{C}}} + O(\delta \|F\|_{\text{Lip}}).$$

Proof. Apply [theorem N.5](#) with equidistribution scale much smaller than δ . Split into residue classes on which γ is constant modulo Γ . Then partition each coordinate interval into pieces of length $\asymp \delta N/M$; on a cell, telescoping the coordinate increments shows that ε moves by $O_d(\delta)$. Freeze it at one point. The translation changes F by $O(\delta \|F\|_{\text{Lip}})$ using the controlled metric calculus of Appendix A.

The remaining average is a Lipschitz test along a totally equidistributed orbit g' in the rational subnilmanifold $Y_{\mathcal{C}}$. After rescaling the cell to a comparable box, total equidistribution gives its Haar integral with error at most the chosen equidistribution parameter times the Lipschitz norm. Summing the finitely many cells proves the result. $\square$

References

- [1] Vitaly Bergelson, Terence Tao, and Tamar Ziegler. “An inverse theorem for the uniformity seminorms associated with the action of $\mathbb{F}^\omega$ ”. In: *Geom. Funct. Anal.* 19 (2010). arXiv:0901.2602, pp. 1539–1596.
- [2] Ben Green, Terence Tao, and Tamar Ziegler. “An inverse theorem for the Gowers U^4 norm”. In: *Glasgow Mathematical Journal* 53.1 (2011). arXiv:0911.5681, pp. 1–50.
- [3] Ben Green, Terence Tao, and Tamar Ziegler. “An inverse theorem for the Gowers $U^{s+1}[N]$ -norm”. In: *Ann. of Math.* 176 (2012). with later erratum; current arXiv revision v5, April 2026, pp. 1231–1372.
- [4] Shachar Lovett. “The Analytic Rank of Tensors and Its Applications”. In: *Discrete Analysis* 7 (2019). arXiv:1806.09179.
- [5] Frederick Manners. *Quantitative bounds in the inverse theorem for the Gowers U^{s+1} -norms over cyclic groups*. Version 2, 16 February 2024. 2018. arXiv: [1811.00718](#).
- [6] Terence Tao and Tamar Ziegler. “The inverse conjecture for the Gowers norm over finite fields in low characteristic”. In: *Annals of Combinatorics* (2012). arXiv:1101.1469.